# From True-Concurrent Games to a Categorical Logic of Agency

Conor Sheridan
siridconor@gmail.com

November 3, 2026

**Abstract**

Two joint policies can prescribe identical actions on the observed inputs, yet only one may satisfy an institution's rule for execution. When the rule also concerns responses to an unobserved input, an overseer who sees only the current actions cannot decide which policy it faces. We develop a categorical account of agency in which one can specify the evidence such a decision requires and trace how it reaches the component that acts on it.

We construct an agent type whose states represent owner-indexed families of true-concurrent strategies. The correspondence preserves sequential and parallel composition, so the same owner retains control of its policy as components are wired together. Each strategy records which events can occur independently and which must precede others. A completed execution also carries its outcome and accumulated charge in a cost monad. We characterize when evaluating these records gives a well-defined payoff on policies. In a finite authorization protocol, the checker receives the two additional policy entries tested by its rule, and a final action depends causally on receipt of authorization.

The same distinction between policies and executions governs abstraction. We show when public messages preserve strategic and epistemic alternatives, how much memory a controller needs to implement a complete local policy, and which refinements of a visible state preserve specified future alternatives. The calculus is sound, and its first-layer equations are complete for the designated class of models. In the fixed concurrent semantics, the original arenas separate open agent morphisms, and finite retained frames admit characteristic formulas for bisimulation-invariant predicates.

## Contents

**Introduction** **4**

**I Concurrent semantics and first-class agents** **8**

**1 True-Concurrent Games and Dependent Linear Semantics** **8**
1.1 Operational event structures and concurrent interaction . . . . . 8
1.2 Dependent completion of concurrent interfaces . . . . . 18
1.3 The indexed model underlying agent types . . . . . 29

**2 Agents as First-Class Linear Objects** **30**
2.1 Agent types and distinct levels of control . . . . . 30
2.2 Economic vocabulary: policies, outcomes, and information . . . . . 31
2.3 Owned agents and their situated interfaces . . . . . 36

**II Composition, control, and game-theoretic evaluation** **42**

**3 The Compositional Calculus of Agents** **43**
3.1 Linear agent operations and ownerwise composition . . . 43
3.2 Open arenas, ownership and contextual wiring . . . 44

**4 Control, Coalitions, and Delegation** **51**
4.1 Complete-policy alternatives and their composition . . . 51
4.2 Collective control and regrouping of agent interfaces . . . 53
4.3 Delegation and substitution of agent interfaces . . . 55
4.4 Displayed coalition and delegation operations . . . 59

**5 Contextual Evaluation and Strategic Equilibrium** **61**
5.1 Presentations, state contexts, and assertions . . . 62
5.2 Predicate feedback, canonical lenses, and coplay . . . 64
5.3 Observable outcomes . . . 66
5.4 The category of operational contexts . . . 67
5.5 Dependent assertions and semantic testing . . . 69
5.6 Selection and logical equilibrium . . . 75
5.7 Situated agent presentations . . . 88
5.8 Legal feedback, best response and equilibrium . . . 89

**III Information, institution, and epistemic dynamics** **91**

**6 Observation, Perspective, and Knowledge** **91**
6.1 Observation and admissible policies . . . 91
6.2 Observation-based responses of an agent . . . 91
6.3 An individual agent's knowledge from observation . . . 93
6.4 Qualitative belief and common knowledge . . . 94
6.5 Dependent observation descent . . . 97

**7 Informed Choice and Collective Agency** **107**
7.1 Institutionally available agent interfaces . . . 107
7.2 Institutional arrangements and collective agent interfaces . . . 109
7.3 Quantified spans and their action . . . 109
7.4 Informed legal policies as sections of a dependent product . . . 111
7.5 Local states, knowledge, and a uniform ability witness . . . 113
7.6 Legal coalition effectivity and stability . . . 114
7.7 Uniformly informed and legal policies . . . 126

**8 Witnessed Institutions and Dynamic Agency** **128**
8.1 Dependent witnesses and certified realizations . . . 128
8.2 Witnessed institutional executions . . . 129
8.3 Linear realization and dynamic modalities . . . 132

**9 Public Announcements and Epistemic Actions** **137**
9.1 Communication and epistemic updates . . . 137
9.2 Public announcement in the agent calculus . . . 149
9.3 General epistemic action models and delivery . . . 151

**IV Operational and strategic communication** **153**

**10 True-Concurrent Communication Between Agents** **153**
10.1 Agents implemented by causal decision protocols . . . 153
10.2 Causal decision and session protocols . . . 157
10.3 Epistemic action events from completed communication . . . 170

**11 Strategic Communication and the Creation of Knowledge** **171**
11.1 Public inspection and the disclosure of pooled information . . . 172
11.2 Agent-labelled truthful public actions . . . 174
11.3 The full truthful message family and coalition announcement games . . . 176
11.4 Complete truthful policies of communicating agents . . . 178
11.5 Concurrent implementation and retained policy witnesses . . . 182
11.6 Coalition ability through complete communication policies . . . 184
11.7 Messages change observations . . . 185
11.8 Pooling and collective knowledge . . . 185
11.9 Complete truthful communication policies in the agent calculus . . . 187
11.10 Concurrent implementation and coalition announcement ability . . . 188
11.11 Choices, disclosure, and the price of authorization . . . 189
11.12 A concrete finite check of linearity and information . . . 191
11.13 Which evidence enables the authorization gate? . . . 191

**V Continuing agency and exactness** **192**

**12 Continuing Agents and Temporal Agency** **192**
12.1 Fixed points in assertion fibres and their transport . . . 192
12.2 Residual strategies and continuing agents . . . 193

**13 Epistemic Rationality and Strategic Dynamics** **222**
13.1 Improvement dynamics and termination . . . 223
13.2 Epistemic rationality and elimination . . . 225
13.3 Reporting agents, incentive compatibility and truthfulness . . . 234

**14 Reconstruction and Exact Abstraction of Agent Systems** **238**
14.1 Reconstructing the complete agent presentation . . . 238
14.2 Soundness and classifying-model completeness . . . 254

**Conclusion** **262**

**A Natural-Deduction Presentation of the Logic of Agency** **264**
A.1 Ordinary and linear agent type theory . . . 264
A.2 Strategic, epistemic, institutional, and temporal assertions . . . 270

**B Soundness, Reconstruction, and Equational Completeness** **278**
B.1 Reconstruction of semantic presentations . . . 278
B.2 Soundness and reconstruction of the agency calculus . . . 280
B.3 The classifying model and equational completeness . . . 282

## Introduction

Consider two agents that each observe a private bit and submit a decision to an authorization checker. Both agents submit an affirmative decision on the current input. In one pair of policies, the first agent would also act on another input for which the second would decline to act. In another pair, both would decline to act on that input. The institution permits only the second pair in this comparison. A checker that receives only the decisions made now sees the same evidence in both cases. In the finite protocol constructed below, the checker receives two additional entries from the complete policies. It can then distinguish the cases and issue the token required by the executor (Section 11.13).

For AI alignment, this is a question about the design of oversight. A rule can refer to counterfactual behaviour. To implement that rule, a designer must identify evidence that distinguishes permitted from prohibited cases and a way to supply it to the checker. The example gives the checker explicit access to two entries of a finite policy. Applying the argument to a learned policy would require an account of how the corresponding evidence is obtained. For cybernetics, the same example poses a question about control under partial observation. The two decisions can occur independently. The checker must wait for the information on which its verdict depends, and the executor must wait for the checker's token. Which of these dependencies survive when the whole arrangement is represented as a single component?

Strategic games describe an owner's complete contingent choices. True-concurrent strategies describe execution through causal prerequisites, conflict, and independent events. We connect the two by representing each complete policy as a parameter of a concurrent interaction. An owner can occupy several decision positions. Changing that owner's policy can therefore change several responses at once, including responses at positions the current execution never reaches. This is the unit of comparison in the strategic analysis.

Four theorems develop the connection for pure policies and qualitative information. Theorem A gives the representation and its composition law. Theorem B identifies the alternatives available to an owner and evaluates them on the supplied preference and institutional data. A completed execution records both an outcome and a charge. This allows a budget to restrict execution and an owner to price the executions that remain. Theorem C constructs the protocols that deliver observations and the continuations available after a history. Theorem D identifies which data the representation recovers and which tests remain valid after abstraction.

**Theorem A** (Concurrent agents and compositional representation)**.** *Receptive courteous strategies on polarized event structures form a compact closed category $\mathcal{C}$ under secured interaction and hiding. Its restricted Yoneda completion $\mathcal{V}$ is symmetric monoidal closed and supports the indexed dependent linear–nonlinear structure. For every finite owner list $N$, ordinary complete-policy family $\vec{\Sigma} = (\Sigma_i)_{i\in N}$ and original arenas $A, B$, there is a natural bijection*

$$\mathcal{V}(I, [\mathsf{Lin}(\prod_i \Sigma_i) \otimes yA, yB]) \cong \mathbf{Set}(\prod_i \Sigma_i, \mathcal{C}(A, B)).$$

*Under the displayed ownership signature, this representation preserves sequential and parallel composition, ownerwise reparametrizations, and the realization-preserving 2-cells of the owned-open-arena bicategory.*

To represent a complete policy inside an interaction, Theorem A uses the operational category, its Day completion, and the dependent model of Theorems 1.10, 1.31 and 1.42. The agent abstraction in Theorems 2.15, 2.18 and 2.19 recovers the owner-indexed family

from its internal representation. This is why a subsequent change of policy can still be attributed to its owner.

**Theorem B** (Controlled, situated, and institutional agency)**.** *Owner-coordinate projections determine compositional coalition counterfactuals. Owner regrouping and associative delegation preserve the realized concurrent interaction and admit exact deviation comparisons on the implemented policy image. Predicate feedback along realization constructs canonical selection, best response, and equilibrium. Observation quotients construct admissible policy descent and knowledge; dependent universal images construct observation-uniform legal policies. Witnessed record-changing institutions form a symmetric monoidal bicategory, and their dependent sums and products determine proof-relevant dynamic modalities compatible with sequential execution. Owner evaluation of measured records descends to profile utility exactly when it equalizes the completed-witness kernel pair. Pullback of these values to controlled deviations gives the cost-sensitive equilibrium test.*

To test a proposed intervention, first fix the policies of everyone outside the controlling group. Then compare the executions obtained as the group varies its own policy. Theorem B justifies this order of operations for regrouping, delegation, and institutional restrictions. The component comparisons appear in Theorems 4.4, 4.8, 4.10, 5.11, 7.40, 8.4 and 8.10. They also allow a change to be permitted even when every owner ranks it below the reference choice.

**Theorem C** (Communication, epistemic change, and continuation)**.** *Finite owned causal decision protocols and finite binary sessions admit true-concurrent realizations preserving complete off-path policies. Their residual operations satisfy subject reduction, and secured interaction is adequate for the completed communication witnesses. Actual delivery records determine epistemic action-event quotients. For complete truthful message policies, evaluation into the truthful public-message model is a split surjection with the designated strategic and epistemic lifting properties. At the completed delivery boundary, collective disclosure transports distributed knowledge to the stated common-knowledge and announcement-ability predicates. Complete local policies generate their residual controllers, with the constructed criterion for finite-state implementation; witnessed residual transitions support temporal fixed-point agency and exact visible refinement. On finite public game trees with the displayed round meter, backward decomposition computes each child equilibrium at its inherited charge.*

Theorem C identifies the boundary at which a delivered message can stand in for the policy that produced it. The protocols in Theorems 10.4, 10.23 and 11.18 and Lemma 10.20 exhibit actual delivery and retain the responses prescribed elsewhere. The comparisons in Theorems 11.27 and 11.28 identify the information recipients gain at that boundary. Section 12 asks what the agents can still do after the exchange.

**Theorem D** (Soundness, recovery, and finite behavioural completeness)**.** *The ordinary, linear, displayed-agent, and contextual judgments of the agency calculus are sound in the completed true-concurrent semantics. The owner-indexed and enriched situated presentations are recovered by the explicit reconstruction maps, while reductions relative to designated tests retain their exhibited alternative-lifting properties. Retaining the measured record recovers both its charge and outcome; retaining the evaluated continuation tests preserves the corresponding subgame-perfect predicates.*

*The first-layer equational calculus is complete for its class of split indexed linear–nonlinear models by the syntactic classifying model. In the fixed concurrent completion, original arenas separate open agent morphisms. On image-finite retained frames, the exact bisimulation quotient identifies precisely the states satisfying the same designated Boolean modal tests; on*

*finite frames, characteristic formulas define all bisimulation-invariant predicates. Explicit finite causal-agent interpretations have decidable semantic equality.*

Theorem D answers two questions about using the calculus. Section 14 identifies the displayed data that can be reconstructed and the tests that distinguish behaviours in the fixed concurrent semantics. Appendix B proves soundness and characterizes equational derivability by validity across the designated class of models.

**Construction-to-theorem guide.** The middle column identifies the construction on which each main theorem depends. The last column locates the proofs, including the comparisons that connect the imported categorical results to the agent model.

| Claim | Construction and comparison | Proof location |
|---|---|---|
| A | Imported secured interaction and Day convolution; constructed $\mathsf{Lin} \dashv \mathsf{Val}$ and the open-context agent representation. | Theorems 1.10, 1.31, 1.42, 2.18 and 2.19. |
| B | Ownerwise kernel-pair deviations, legal alternative lifting, comparison feedback, and matched institutional witnesses. | Lemma 4.11 and Theorems 4.12, 5.39 and 8.4. |
| C | Finite causal wiring, session-prefix residuals, truthful-message back-lifts, and the greatest-bisimulation quotient. | Theorems 10.4, 10.23, 11.18 and 12.37 and Lemma 11.19. |
| D | Reconstruction and the first-layer classifying model; original-arena separation and finite-state modal characterization. | Theorems 14.26, 14.32, 14.33, 14.35 and 14.36. |

The proof of each operational, strategic, or epistemic comparison follows its statement in Parts I–V. For the rules of the calculus, see Appendix A. For their soundness, reconstruction, and equational completeness, see Appendix B. The compact-closed strategy category and Day convolution supply the categorical foundations. The constructions in the table explain how ownership, legal alternatives, and delivered observations survive passage to this semantics.

**Three semantic levels.** The model has three levels. Ordinary policies and their dependent families live in **Set**. Linear interfaces, interpreted as presheaves over the concurrent strategy category, describe the resources an execution uses. Assertions express properties of states and execution witnesses. At the first level one can consider several policies for a single device. Running several copies of that device would require the corresponding resources at the second level. At the third level one can assert that authorization occurred and, when the witness is retained, identify the event that establishes the assertion.

**What the operational semantics determines.** Execution determines the enabled moves, their causes, the signals carried by the wires, and the residual process. The model specifies who may change a policy and how an owner compares outcomes. A permission rule likewise enters as institutional data. Once a protocol implements that rule, its execution can establish that the required event occurred. This distinction matters whenever an external observer uses the event as evidence of authorization.

**A running finite instance.** The recurring example has two owners and a private bit for each: its world carrier is $W = 2 \times 2$, with observations $o_1(x, y) = x$ and $o_2(x, y) = y$. At separate decision nodes owner 1 applies $\pi_1 : 2 \to 2$ to $x$ and owner 2 applies $\pi_2 : 2 \to 2$ to $y$. Thus $\Sigma_1 = 2^2$, $\Sigma_2 = 2^2$, and the complete profile carrier has 16 elements, including all off-path responses. The operational realization is the tensor of the two response strategies, with their specified private inputs. An institutional rule can couple the policies: we use the concrete legal domain

$$L = \{(\pi_1, \pi_2) : \pi_1(1) \leq \pi_2(1)\} \hookrightarrow \Sigma_1 \times \Sigma_2,$$

which has 12 elements, using the usual order on $2 = \{0, 1\}$. To compare choices at a given world, start with the exact rational payoffs $u_1(a, b) = 2ab - a$ and $u_2(a, b) = 2ab - b$, where $a = \pi_1(x)$ and $b = \pi_2(y)$. A collector later delivers $(x, y)$ to the agents. A separate checker enforces the permission rule for final execution. The calculations in Sections 11.11 and 11.13 show which deviations are profitable, which facts become common knowledge, and what evidence the executor receives.

**Route through the paper.** Parts I and II follow a policy from its representation as a concurrent strategy to the comparison of deviations under composition. Parts III and IV follow a signal from a private observation through a communication protocol to a decision another owner can make with it. Part V asks what remains possible after a history and how much of that history a controller or an abstraction must retain. Appendix A gives the inference rules. Appendix B gives the reconstruction, soundness, and equational-completeness proofs.

## Relation to the literature

Two questions organize the comparison with earlier work. How does a process acquire an owner whose control remains identifiable under composition? Which observations and strategic alternatives survive a change of representation? On the first question, the action calculi of Barber, Gardner, Hasegawa, and Plotkin connect interaction to linear logic (Barber et al., 1998), and resource-sensitive agency logics connect action modalities to resources (Porello and Troquard, 2014). Here the agent itself is a typed resource. Passing it to another process must preserve the designated ownership of its policy.

Parameterized open games distinguish policy coordinates from their evaluation (Capucci et al., 2022b). Categorical cybernetics relates this distinction to selection and the backward transport of comparisons (Capucci et al., 2022a). Cartesian frames describe the division of control between agent and environment (Garrabrant, Herrmann, and Lopez-Wild, 2021). Section 4 recovers that division on unrestricted policy products and computes the available choices when a permission rule couples the two sides. The present construction also assigns each profile a concurrent execution, so one can ask how an observation reaches its recipient. Session calculi relate channel interaction to cut (Caires and Pfenning, 2010). Our finite sessions retain the complete endpoint policies and the matched events that witness delivery.

For the second question, epistemic actions as resources (Baltag, Coecke, and Sadrzadeh, 2007) and multi-type dynamic epistemic logic (Frittella et al., 2016) provide accounts of information change. In this paper a recipient's observation map determines which states it can distinguish. Public restriction changes the space of possible states, and a witnessed transition supplies the event of a general update. This separation lets us compare the information an abstract update prescribes with the signal a protocol actually delivers.

The Hennessy–Milner separation theorem (Hennessy and Milner, 1980, 1985) provides the modal precedent for testing behaviour in finitely branching systems. Our tests observe the

designated epistemic, deviation, and witnessed transition relations. On a finite retained frame, characteristic formulas then distinguish exactly the classes of its behavioural equivalence.

Three comparisons account for the passage from the categorical ingredients to claims about agency. The legal lift of Lemma 4.11 realizes an alternative while keeping the reference outsiders fixed. The protocol of Theorem 10.4 realizes a decision with its causal prerequisites. The back-lifts of Theorem 11.18 recover strategic and epistemic alternatives from a public message at the given source. The underlying copower, internal-hom, and Yoneda identities give the representation. These comparisons establish what an owner can still do or distinguish when that representation changes.

**Use of AI-assisted tools.** Generative AI systems were used during the research and development of this manuscript. Responsibility for the mathematical content, arguments, and conclusions rests with the author.

# Part I
# Concurrent semantics and first-class agents

## 1 True-Concurrent Games and Dependent Linear Semantics

A checker can wait for two decisions without requiring either agent to move first. To describe such a protocol, we need a model that records a dependency where a component waits and leaves independent events unordered. Polarized event structures and the secured composition of Castellan et al. (2017) provide this model. We first build finite value and collection protocols to identify the barriers they impose. We then use restricted Yoneda and Day convolution to introduce the sums, products, and ordinary parameters needed for dependent agent types. The embedding recovers every original concurrent arrow, so these additional operations preserve the causal account of execution.

### 1.1 Operational event structures and concurrent interaction

#### 1.1.1 Concurrent histories

**Definition 1.1** (Event structures and configurations)**.** An **event** is one occurrence of an action or communication, so two occurrences of the same kind of action remain distinct events. **True concurrency** records prerequisites and compatibility directly, without first choosing a total schedule for independent events. In the formal data below, a down-closed set contains every prerequisite of each of its events. An event is **enabled** after a configuration $x$ when $e \notin x$ and $x \cup \{e\}$ is again a configuration. Two distinct events are concurrent when they can occur together and neither causally precedes the other; incompatible collections are in conflict. The consistency family also records incompatibilities involving more than two events.

An **event structure** is a triple $E = (|E|, \leq_E, \mathrm{Con}_E)$. The relation $\leq_E$ is a partial order with finite principal downsets. The family $\mathrm{Con}_E$ consists of finite subsets of $|E|$, contains the empty set and every singleton, is closed under subsets, and satisfies

$$X \in \mathrm{Con}_E, \quad e \leq_E e' \in X \quad \Longrightarrow \quad X \cup \{e\} \in \mathrm{Con}_E\,.$$

A finite **configuration** is a down-closed consistent subset. Write $\mathrm{Conf}(E)$ for the set of finite configurations.

The order records prerequisites. Consistency records which finite collections can occur together. Two events can be consistent without either causing the other. A configuration therefore records a partially ordered history rather than an arbitrary linear ordering of its independent events.

**Construction 1.2** (Enabled-event span)**.**

(1) **Enabled extensions.** For an event structure $E$ put $W_E = \mathrm{Conf}(E)$ and define

$$\mathsf{Step}(E) = \{(x, e) \in W_E \times |E| : e \notin x,\ x \cup \{e\} \in W_E\}.$$

It has projections $p(x, e) = x$ and $\ell(x, e) = e$, and a target $t(x, e) = x \cup \{e\}$.

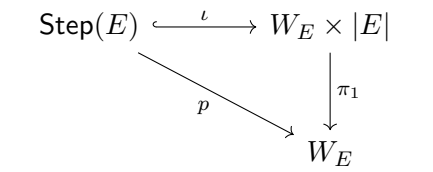
(2) **Source and target.** Thus

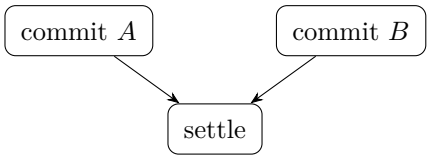
$$W_E \xleftarrow{p} \mathsf{Step}(E) \xrightarrow{t} W_E$$

is a relation span: $(p, t)$ is jointly injective, since $e = t(x, e) \setminus x$ is determined by its two endpoints. The inclusion $\mathsf{Step}(E) \hookrightarrow W_E \times |E|$ is the comprehension of the event-enablement predicate.

(3) **Execution in a configuration.** Every event of a finite configuration can be executed in some linear extension of that configuration's causal order, but the span does not choose a linear extension.

The incidence inclusion and source projection give the enabled-event diagram.




Neither commitment orders the other. Settlement depends on both.

Figure 1: Causal dependence without a scheduler-imposed order.

**Definition 1.3** (Total event-structure map)**.** For event structures $E, F$, a **total map of event structures** is a function $f : |E| \to |F|$ satisfying, for every finite configuration $x \in \mathrm{Conf}(E)$,

$$f[x] \in \mathrm{Conf}(F), \qquad f|_x : x \longrightarrow f[x] \text{ is injective.}$$

It thus preserves configurations and is locally injective. It need not preserve every source causal order as an order relation between event labels: a strategy may add dependencies specifying when it responds.

**Definition 1.4** (Polarized arenas)**.** The polarity is relative to an interaction boundary: negative events are moves the surrounding interaction may supply, and positive events are moves the strategy may make. Reversing an interface reverses these roles. Economic

ownership is recorded by the separate owner-indexed policy signature. For example, several players' chosen messages can all be positive outputs of separate components, while becoming negative inputs of the component that receives them.

An **arena** is an event structure with a polarity $\mathrm{pol} : |A| \to \{-, +\}$. Its dual $A^{\perp}$ reverses polarities. The parallel arena $A \parallel B$ is the disjoint union, with componentwise causality and consistency. The empty arena is denoted $I$.

### 1.1.2 Strategies and their composition

An output and an input at a shared interface can synchronize only if their combined prerequisites admit a causal order. The synchronization pullback records the secured history for each match. Two matches with identical visible labels may therefore remain distinct when their histories are incompatible.

We use the secured-bijection pullback of Castellan et al. (2017). Its universal property supplies the synchronization object subsequently used in interaction and hiding.

**Definition 1.5** (Secured matching)**.** Let $f : S \to A$ and $g : T \to A$ be total maps of unpolarized event structures. A matching is a finite graph $\varphi \subseteq |S| \times |T|$ whose projections are configurations, which is the graph of a bijection between those configurations, and for which $f(s) = g(t)$ whenever $(s, t) \in \varphi$. Order its pairs by the transitive closure of

$$(s, t) \trianglelefteq (s', t') \quad \text{if} \quad s \leq_S s' \text{ or } t \leq_T t'.$$

The matching is secured when this closure is antisymmetric. It is prime when this finite partial order has a greatest pair. Write $\mathcal{M}(f, g)$ and $\mathcal{M}_{\bullet}(f, g)$ for the secured and prime secured matchings.

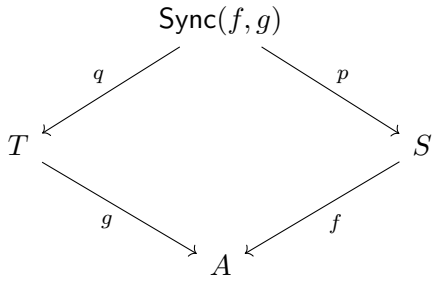
**Construction 1.6** (Synchronization object)**.**

(1) **Events and order.** The events of $\mathsf{Sync}(f, g)$ are $\mathcal{M}_{\bullet}(f, g)$, ordered by inclusion.

(2) **Consistency.** A finite set of primes is consistent precisely when its union is a secured matching.

(3) **Projections.** A prime is projected to the coordinates of its greatest pair, giving maps $p : \mathsf{Sync}(f, g) \to S$ and $q : \mathsf{Sync}(f, g) \to T$.

The pullback compares the two occurrences proposed for synchronization by their images at the shared interface.




**Theorem 1.7** (Synchronization universal property; recalled)**.** *The construction is an event structure, with a bijection*

$$\mathrm{Conf}(\mathsf{Sync}(f, g)) \cong \mathcal{M}(f, g), \qquad X \longmapsto \bigcup X.$$

*The maps $p, q$ exhibit it as the pullback of $f, g$ in the category of event structures and total maps.*

*Source and identification.*

(1) **Secured operational interaction.** This is the secured-bijection pullback theorem of Castellan et al. (2017). In the notation fixed above, a secured matching $\varphi$ determines the configuration

$$X_\varphi = \{\downarrow_\varphi u : u \in \varphi\}$$

of prime secured matchings, and a configuration $X$ of primes recovers the secured matching $\bigcup X$. These operations are inverse.

(2) **Lifting through the comparison.** The maps $p$ and $q$ take a prime to the coordinates of its greatest pair, and the cited pullback theorem identifies the unique lift of a cone with the prime secured histories generated by its configurations. We use that universal property below; no different synchronization construction is introduced here.

□

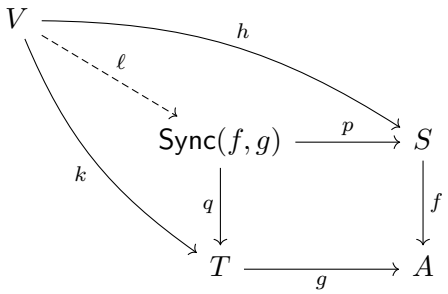



Figure 2: Prime secured histories supply the synchronization pullback.

**Definition 1.8** (Concurrent strategy)**.** Receptivity requires a unique extension in the strategy for the arrival of each enabled negative event. Courtesy permits the strategy to wait for an input before producing an output, but prevents it from inserting arbitrary new dependencies between moves whose ordering is already controlled by the arena. Here $e \lessdot e'$ means an immediate causal dependency, with no event strictly between them. Thus a protocol may wait for both submitted messages before announcing their joint outcome without imposing an order between the two submissions.

A concurrent strategy from $A$ to $B$ is a polarity-preserving total map

$$s : S \longrightarrow A^\perp \parallel B$$

with two conditions. **Receptivity** says that every enabled negative move has a unique extension in the strategy. More explicitly, if $x$ is a configuration of $S$ and $s(x) \cup \{a\}$ is a configuration extending $s(x)$ by a negative event, there is a unique $e$ with $s(e) = a$ and $x \cup \{e\} \in \mathrm{Conf}(S)$. **Courtesy** says that a new immediate dependency may only run from a negative event to a positive event. Thus, if $e \lessdot_S e'$ and either $\mathrm{pol}(e) = +$ or $\mathrm{pol}(e') = -$, then $s(e) \lessdot s(e')$ in the arena.

**Construction 1.9** (Secured interaction and hiding)**.**

(1) **Synchronize the shared interface.** For strategies $S \to A^\perp \parallel B$ and $T \to B^\perp \parallel C$, interaction synchronizes the occurrences on $B$. Only **secured** synchronizations are admitted: the combined causal orders of the matched histories must be acyclic.

(2) **Hide the internal events.** Composition then hides the matched $B$-events while retaining the induced order on visible events. The acyclicity condition prevents a circular collection of unfulfilled prerequisites from being mistaken for a possible interaction.

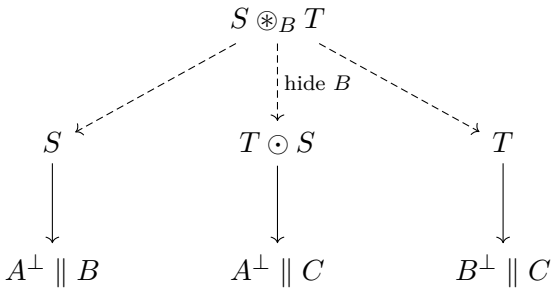



Figure 3: Interaction precedes hiding. Dashed arrows are partial maps of event structures: projection forgets events private to the other component, and hiding forgets internal events. Solid arrows are total strategy maps.

**Theorem 1.10** (Concurrent composition; imported)**.** *Polarized event structures and receptive courteous strategies, taken up to isomorphism over their arenas, form a compact closed category $\mathcal{C}$. Composition is secured interaction followed by hiding, identity is copycat, tensor is parallel composition, and duality reverses polarity.*

*Reference and size verification.*

(1) **Composition.** For composition and compact closure, see Castellan et al. (2017). Parallel composition, finite secured histories, and hiding preserve countability. On a fixed countable carrier, orders, consistency families, polarities, and labellings range over sets.

(2) **Causal execution.** Taking a set of countable carriers and quotienting strategy presentations by isomorphism therefore gives a set $\mathcal{C}(A, B)$ for every $A, B$.

□

**Notation 1.11** (Compact structure)**.** We write $\otimes$ for parallel composition in $\mathcal{C}$ and $I$ for the empty arena. Arena-preserving strategy isomorphisms represent equality of arrows. Put

$$\widetilde{\eta}_A = I \xrightarrow{\eta_A} A^\perp \otimes A \xrightarrow{\gamma} A \otimes A^\perp.$$

Interaction polarities are independent of economic ownership.

### 1.1.3 The linear account of interaction

**Construction 1.12** (Composition and copycat in the chosen convention)**.**

(1) **Form the interaction.** The preceding pullback describes the interaction used by Theorem 1.10. Forget the polarities on the shared interface while forming the pullback of

$$S \parallel C \xrightarrow{s\parallel \mathrm{id}} A^\perp \parallel B \parallel C \xleftarrow{\mathrm{id}\parallel t} A^\perp \parallel T.$$

The additional factors carry the events private to the other component.

(2) **Restrict to the exterior arena.** The pullback has a map to the three-interface arena. Keep just its events labelled in $A^\perp \parallel C$, restrict order and consistency, and restrict the labelling map. This gives the composite. Restricting the transitive order retains the visible consequence of an internal causal chain; it does not simply erase the chain's dependencies.

(3) **Construct copycat.** The copycat strategy of Castellan et al. (2017) is constructed as follows: start with the two interface copies of $A$, add the forwarding edge from the negative occurrence of each arena event to its positive occurrence, take transitive closure, and define consistency through the causal closure in the arena. We use the cited verification that this gives a receptive courteous strategy and that it is the identity for secured interaction followed by hiding. Thus no separate copycat correctness claim is being introduced here; the concrete description only fixes the convention used by the imported compact-closed structure.

**Construction 1.13** (The closed structure of concurrent interaction)**.**

(1) **Tensor and internal hom.** The tensor is $A \otimes B = A \parallel B$, with unit the empty arena. Compact closure constructs the internal hom $A \multimap B = A^{\perp} \parallel B$ and the natural bijection

$$\mathcal{C}(X \otimes A, B) \cong \mathcal{C}(X, A \multimap B).$$

(2) **Evaluation and transposition.** Evaluation and transposition are obtained from the displayed cups, caps, and symmetry. The snake equations prove the inverse laws for this bijection. Identity is copycat, composition is interaction and hiding, and tensor places interactions in parallel.

Compact closure provides a tensor for combining linear interfaces and an internal hom for linear interaction, as in Girard (1987). An agent type also needs an ordinary parameter that can index several uses of a policy description. Section 1.2 constructs such parameters and the linear interfaces that depend on them.

*Remark* 1.14 (Related concurrent semantics). The extension by symmetry and the construction of concurrent Hyland–Ong games are developed in Castellan, Clairambault, and Winskel (2019). Winskel (2023) relates concurrent strategies to functional models, including lenses and optics. These works situate the chosen operational category and its functional fragments; the constructions used here are the ones explicitly specified above and below.

### 1.1.4 Ordinary functions as response protocols

**Construction 1.15** (Finite value arenas and response strategies)**.**

(1) **Form the value arena.** For a finite set $X$, let $D(X)$ be the positive arena with one event for each $x \in X$, no nontrivial causal order, and configurations $\varnothing$ and the singletons. Distinct values conflict.

(2) **Realize a response.** A function $f : X \to Y$ determines a strategy $\widehat{f} : D(X) \to D(Y)$. For each $x$, its corresponding branch contains a negative input event $x$ and a positive output event labelled $f(x)$, with the input preceding the output; different input branches conflict.

**Proposition 1.16.** *The assignments $X \mapsto D(X)$ and $f \mapsto \widehat{f}$ form a faithful functor from finite sets and functions to $\mathcal{C}$.*

*Proof.*

(1) **Secured operational interaction.** The only added dependency is negative-to-positive, so courtesy holds. At the empty configuration every input is accepted once; after an input, all other inputs conflict with it in the arena.

(2) **Receptivity and functional composition.** This proves receptivity. Synchronizing $\widehat{f}$ and $\widehat{g}$ produces, on the branch indexed by $x$, the chain $x \to f(x) \to g(f(x))$. Hiding the middle event gives $\widehat{g \circ f}$.

(3) **Secured operational interaction.** For the identity function the construction is copycat on $D(X)$. Distinct functions differ on an input and therefore on an output label of the corresponding strategy; they cannot be isomorphic over the arena.

□

The point strategies form a natural family with respect to finite response maps.

$$\begin{array}{ccc} I & \xrightarrow{\mathsf{pt}_X(x)} & D(X) \\ \| & & \downarrow \widehat{f} \\ I & \xrightarrow[\mathsf{pt}_Y(f(x))]{} & D(Y) \end{array}$$

**Notation 1.17** (Value states)**.** For $x \in X$, write

$$\mathsf{pt}_X(x) : I \longrightarrow D(X)$$

for the one-event positive strategy selecting $x$. Equivalently, if $\bar{x} : 1 \to X$ is the corresponding function and $\upsilon : I \to D(1)$ is the unique one-event state, then

$$\mathsf{pt}_X(x) = \widehat{\bar{x}} \circ \upsilon.$$

Functoriality gives the naturality equation

$$\widehat{f} \circ \mathsf{pt}_X(x) = \mathsf{pt}_Y(f(x)). \tag{1.1}$$

Equivalently, under $X \cong \mathbf{FinSet}(1, X)$, the functions $\mathsf{pt}_X : X \to \mathcal{C}(I, D(X))$ are the components of the natural transformation

$$\mathsf{pt} : \mathbf{FinSet}(1, -) \Longrightarrow \mathcal{C}(I, D(-)),$$

between functors $\mathbf{FinSet} \to \mathbf{Set}$. Its construction is the hom-functor action of $D$ on $X \cong \mathbf{FinSet}(1, X)$, followed by precomposition with $\upsilon$. The symbol denotes a selected point of a data arena and is separate from both compact coevaluation $\eta_A$ and later quotation.

A recipient of $D(X \times Y)$ receives one report containing both values. A recipient of $D(X) \otimes D(Y)$ can receive either output first. Faithfulness records the distinctions between the original response maps. Collecting independent outputs into a joint report imposes a further requirement. The collector must wait for both, which determines when a later policy may first use both values.

### 1.1.5 Communication of data and synchronization barriers

**Construction 1.18** (Copying and disposal of data)**.**

(1) **Specify the data maps.** For data that may be reported to several recipients there are explicit copying and disposal strategies

$$\Delta_X : D(X) \to D(X) \otimes D(X), \qquad \epsilon_X : D(X) \to I.$$

(2) **Copy an input.** The source of $\Delta_X$ has, for each value $x$, one negative input and two independent positive outputs, each labelled $x$ and each dependent on the input. Different input branches conflict.

(3) **Dispose of an input.** The source of $\epsilon_X$ is just the negative input arena. These are receptive and courteous by direct inspection: every input is accepted uniquely, and the only new dependencies are negative-to-positive.

**Proposition 1.19** (Duplicable data)**.** *The triple* $(D(X), \Delta_X, \epsilon_X)$ *is a commutative comonoid in* $\mathcal{C}$*. For every finite function* $f : X \to Y$*,*

$$(\widehat{f} \otimes \widehat{f})\Delta_X = \Delta_Y \widehat{f}, \qquad \epsilon_Y \widehat{f} = \epsilon_X.$$

*Proof.*

(1) **Associativity.** Both bracketings of the iterated copy strategy have, after interaction and hiding, one negative input event labelled $x$ and three independent positive output events labelled $x$, each with that input as its prerequisite. Matching corresponding events gives an isomorphism over the arena.

(2) **Copying comonoid laws.** This proves coassociativity, including its prescribed associators. Swapping the two outputs is a branchwise isomorphism, proving cocommutativity. Composing either output with disposal leaves one input and its single copied output, exactly copycat. This proves the two counit equations.

(3) **Causal execution.** For the naturality equation, each side has an input branch for $x$ and two independent outputs labelled $f(x)$. Their only visible causal prerequisite is the input. The branchwise identification is again an isomorphism.

(4) **Secured operational interaction.** Discarding the output of $\widehat{f}$ leaves just its receptive input behavior, which is $\epsilon_X$.

□

Naturality of copying is expressed by the following commuting square.

$$\begin{array}{ccc} D(X) & \xrightarrow{\Delta_X} & D(X) \otimes D(X) \\ \Big\downarrow{\scriptstyle \widehat{f}} & & \Big\downarrow{\scriptstyle \widehat{f} \otimes \widehat{f}} \\ D(Y) & \xrightarrow[\Delta_Y]{} & D(Y) \otimes D(Y) \end{array}$$

Copying and disposal are designated operations on these finite data interfaces, not structural rules of the linear syntax. An isomorphism in the ambient concurrent category can transport a comonoid structure, so material nonduplicability cannot be inferred from an object's isomorphism class alone. Material feasibility and authorization are instead tracked by exhibited certificates and witnessed institutional executions, as constructed in Section 8.1; an ambient morphism alone does not establish either property. Thus communication may copy a signal without thereby certifying another authorization or resource occurrence.

**Construction 1.20** (Distribution of a paired value)**.**

(1) **Distribute a paired input.** There is also a distributor $d_{X,Y} : D(X \times Y) \to D(X) \otimes D(Y)$. For each pair $(x, y)$ its source has one negative input labelled $(x, y)$ and two independent positive outputs labelled $x$ and $y$, both following that input. Different paired inputs conflict.

(2) **Verify the strategy conditions.** It is receptive and courteous by the same direct check as the response protocols above.

A collector must accommodate partially received tuples. Completed tuples sharing a channel value therefore share the corresponding negative input event, rather than generating incompatible copies of that input. The canonical collector below makes this sharing explicit.

**Construction 1.21** (Canonical finite collector)**.**

(1) **Form the input and output events.** Let $J$ be a finite index set and let $(X_j)_{j\in J}$ be finite sets. Put $X_J = \prod_{j\in J} X_j$. Define an event structure $C_{\mathbf{X}}$ with negative events

$$e_{j,x} \qquad (j \in J,\ x \in X_j)$$

and positive events

$$o_{\mathbf{x}} \qquad (\mathbf{x} = (x_j)_{j\in J} \in X_J).$$

(2) **Generate order and consistency.** Generate its causal order by

$$e_{j,x_j} < o_{\mathbf{x}} \qquad (j \in J).$$

A finite set of events is consistent precisely when it contains at most one input value on each component, at most one tuple output, and every input that occurs together with $o_{\mathbf{x}}$ has the value prescribed by $\mathbf{x}$.

(3) **Label the collector.** Label $e_{j,x}$ by the input occurrence $x$ on component $j$ and label $o_{\mathbf{x}}$ by the tuple $\mathbf{x}$. Denote the resulting map by

$$c_{\mathbf{X}} : C_{\mathbf{X}} \longrightarrow \left(\bigotimes_{j\in J} D(X_j)\right)^{\perp} \parr D(X_J).$$

For $J = \{1, 2\}$ write $c_{X,Y}$, and for $J = \varnothing$ write $c_0 : I \to D(1)$.

**Proposition 1.22** (Finite collectors)**.** *The map $c_{\mathbf{X}}$ is a receptive courteous strategy*

$$c_{X_j \,|\, j\in J} : \bigotimes_{j\in J} D(X_j) \longrightarrow D\left(\prod_{j\in J} X_j\right).$$

*A configuration containing the tuple output $o_{\mathbf{x}}$ contains exactly its required input events $e_{j,x_j}$ in their causal past. Every iterated binary collector, followed by the canonical reassociation of the Cartesian product, is isomorphic to this strategy over its arena. The nullary collector $c_0$ is the one-event positive state selecting the empty tuple.*

*Proof.*

(1) **Causal execution.** A principal downset of an input event is a singleton, while

$$\downarrow o_{\mathbf{x}} = \{o_{\mathbf{x}}\} \cup \{e_{j,x_j} : j \in J\},$$

so principal downsets are finite. Consistency is downward closed. For the cause-extension axiom, the only nontrivial case is $e_{j,x_j} \le o_{\mathbf{x}}$ with $o_{\mathbf{x}}$ already in a consistent set.

(2) **Causal execution: second comparison.** Such a set cannot contain a different input value on component $j$; therefore adjoining $e_{j,x_j}$ preserves consistency. Thus $C_{\mathbf{X}}$ is an event structure.

(3) **Causal execution.** A configuration containing $o_{\mathbf{x}}$ is down-closed, hence contains all $e_{j,x_j}$, and its image is a configuration of the arrow arena. A configuration without an output contains at most one value on each input component. The labelling is injective on either kind of configuration, so it is a total map of event structures.

(4) **Secured operational interaction.** For receptivity, let $z$ be a source configuration and suppose that the arrow arena enables a negative input $x \in X_j$. Then $z$ contains no input on component $j$. It cannot contain a tuple output either, because down-closure of such an output would already contain an input on component $j$.

(5) **Collector receptivity and courtesy.** Hence $z \cup \{e_{j,x}\}$ is a configuration. The source has exactly one event with that arena label, so the lift is unique. Every added causal edge is of the form $e_{j,x_j} < o_{\mathbf{x}}$, hence is negative-to-positive; courtesy follows.

(6) **Secured operational interaction.** Now expand any iterated binary collector before hiding its intermediate tuple events. On a completed tuple $\mathbf{x}$, the visible source events are the original inputs $e_{j,x_j}$ and the final tuple output. The internal tuple events form a directed tree from those inputs to the final output, so secured interaction is acyclic.

(7) **Secured operational interaction: second comparison.** Hiding the internal tuple events leaves $e_{j,x_j} < o_{\mathbf{x}}$ for every $j$ and adds no order between distinct inputs. Its visible consistency condition is exactly the one in Construction 1.21. Matching equally labelled visible events therefore gives an isomorphism over the arena.

(8) **Causal execution.** Cartesian coherence identifies every parenthesization with the same product tuple. The nullary case has no negative events and one positive event labelled by the unique element of 1.

□

**Proposition 1.23** (Lax symmetric-monoidal data interface)**.** *The faithful functor* $D : \mathbf{FinSet} \to \mathcal{C}$ *of Proposition 1.16, together with*

$$c_{X,Y} : D(X) \otimes D(Y) \to D(X \times Y), \qquad c_0 : I \to D(1),$$

*is lax symmetric monoidal. Its canonical* $J$*-ary comparison maps are the finite collectors of Proposition 1.22.*

*Proof.*

(1) **Coherence of finite collectors.** The associativity and unit diagrams compare two iterated collectors with the same finite collector; Proposition 1.22 supplies the required isomorphisms, and their coherence is the coherence of Cartesian reassociation together with symmetric-monoidal coherence in $\mathcal{C}$. Symmetry exchanges the input components and sends a tuple to its permuted tuple, giving the symmetry diagram branchwise.

(2) **Causal execution.** For functions $f_j : X_j \to Y_j$, both composites

$$\widehat{\prod_j f_j} \circ c_{X_j \,|\, j \in J} \quad \text{and} \quad c_{Y_j \,|\, j \in J} \circ \bigotimes_j \widehat{f_j}$$

have one input event $e_{j,x}$ for each supplied coordinate and one final output labelled $(f_j(x_j))_j$, depending on all of those inputs. After hiding the intermediate response events they are isomorphic over the arena. This proves naturality and the lax symmetric-monoidal laws.

□

The lax monoidal structure records the naturality of finite collection.

$$\begin{array}{ccc} D(X)\otimes D(Y) & \xrightarrow{c_{X,Y}} & D(X\times Y) \\ {\scriptstyle \widehat{f}\otimes\widehat{g}}\downarrow & & \downarrow{\scriptstyle \widehat{f\times g}} \\ D(X')\otimes D(Y') & \xrightarrow[c_{X',Y'}]{} & D(X'\times Y') \end{array}$$

**Proposition 1.24** (The synchronization barrier)**.** *One has*

$$c_{X,Y}\circ d_{X,Y} = \mathrm{id}_{D(X\times Y)}\,.$$

*For nonempty $X, Y$, the reverse composite is not the identity on $D(X)\otimes D(Y)$: each of its outputs depends on both inputs.*

*Proof.*

(1) **Secured operational interaction.** In the first composite there is one negative input pair and, after hiding its two distributed occurrences, one positive output carrying the same pair. Its source is copycat on $D(X\times Y)$.

(2) **Reverse implication.** In the reverse composite the collector's tuple output requires both independent input events. Each distributed output therefore inherits both prerequisites after hiding. In identity copycat on $D(X)\otimes D(Y)$, the first positive output can occur after its own negative input alone, before the second input.

(3) **Secured operational interaction.** Choosing any values in the two nonempty sets exhibits a configuration present in copycat and absent from $d_{X,Y}\circ c_{X,Y}$, so the two strategies are not isomorphic over their arena.

□

Collection and distribution determine a retraction onto paired data.

$$\begin{array}{ccc} D(X\times Y) & \xrightarrow{d_{X,Y}} & D(X)\otimes D(Y) \\ & {\scriptstyle \mathrm{id}}\searrow & \downarrow{\scriptstyle c_{X,Y}} \\ & & D(X\times Y) \end{array}$$

The retraction identifies one paired report with the completed collection of two channels. Its reverse retains the synchronization barrier: neither output can be forwarded before both inputs have arrived. This causal difference persists in protocols that make decisions between individual submissions and completion of a joint report.

## 1.2 Dependent completion of concurrent interfaces

An owned interaction starts from the policy sets $\Sigma_i$ and a realization $g : \prod_i \Sigma_i \to \mathcal{C}(A, B)$ in the concurrent category $\mathcal{C}$. These form its policy family and concurrent realization, but these alone do not interpret a policy-dependent boundary or quantified alternatives. The completion below adds those interfaces without changing what an arrow between original arenas means. We first index the operational category by ordinary parameters and then embed it fully faithfully into a presheaf category with Day tensor. The resulting indexed linear–nonlinear model will supply agent abstraction and the logical operations used later.

### 1.2.1 Indexed families of true-concurrent interactions

**Construction 1.25** (Indexed operational category)**.**

(1) **Index the operational arrows.** The ordinary parameters of an economic model are kept separate from its linear interfaces. For a set $\Gamma$, let $\mathcal{C}_\Gamma$ have the same objects as $\mathcal{C}$ and hom-sets

$$\mathcal{C}_\Gamma(A, B) = \mathbf{Set}(\Gamma, \mathcal{C}(A, B)).$$

(2) **Lift composition and tensor.** Composition, tensor, identity, symmetry, cups, and caps are defined pointwise.

(3) **Reindex the parameters.** A function $f : \Delta \to \Gamma$ induces reindexing

$$f^* : \mathcal{C}_\Gamma \longrightarrow \mathcal{C}_\Delta, \qquad (f^* t)_\delta = t_{f(\delta)}.$$

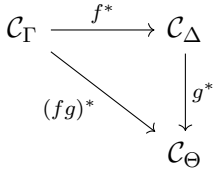
**Proposition 1.26** (Indexed operational semantics)**.** *The assignment $\Gamma \mapsto \mathcal{C}_\Gamma$ and $f \mapsto f^*$ is contravariantly functorial. Every fiber $\mathcal{C}_\Gamma$ is compact closed with the pointwise structure inherited from $\mathcal{C}$, and every reindexing functor preserves that structure. Thus ordinary parameter substitution and linear interaction form an indexed compact-closed semantics over* **Set***.*

*Proof.*

(1) **Monoidal comparison.** Identity and successive reindexing are the identity and associativity laws for function composition. Every categorical operation is evaluated at one parameter value at a time, so reindexing commutes with composition and tensor and sends the constant identity, symmetry, cup, and cap families to the same constant families. The compact-closed equations therefore hold pointwise in every fiber because they hold in $\mathcal{C}$.

□

Successive parameter substitutions commute with indexed realization.




**Definition 1.27** (Owner-indexed parameter category)**.** Fix a finite set $N$ of economic agents and put

$$\mathbf{Par}_N = \mathbf{Set}^N, \qquad \Pi_N(\vec{\Sigma}) = \prod_{i \in N} \Sigma_i.$$

The product functor $\Pi_N : \mathbf{Par}_N \to \mathbf{Set}$ forgets which factor is controlled by which owner while retaining the total profile space. Pulling the indexed operational semantics back along $\Pi_N$ leaves the true-concurrent operational category unchanged and records exactly the factorization of ordinary alternatives that will determine deviations.

### 1.2.2 The canonical completion of concurrent interfaces

Economically, the completion supplies spaces in which policy families, their dependent alternatives, and predicates can coexist with the original operational interfaces. Its new objects organize those families; they do not add new economic decisions to a given arena. Full faithfulness of Yoneda will ensure that an arrow between original interfaces still denotes exactly one original concurrent strategy.

**Lemma 1.28** (Scalars and the obstruction to empty linear limits)**.** *The scalar set $\mathcal{C}(I, I)$ is a singleton. Every arena $A$ has a state $I \to A$, but $\mathcal{C}$ has neither an initial nor a terminal object.*

*Proof.*

(1) **Causal execution.** A total event-structure map to the empty arena has empty source, proving the scalar assertion. In $A$ take the induced substructure

$$A_- = \{e \in A : d \leq_A e \implies \mathrm{pol}(d) = -\}.$$

It is downward closed. Its inclusion into $A$ is a total event-structure map and is courteous: an immediate dependency in this downward-closed substructure is immediate in $A$.

(2) **Lifting through the comparison.** If a negative event is enabled after an image configuration, all its strict predecessors belong to that configuration, so it belongs to $A_-$ and has a unique lift. The inclusion is therefore a receptive courteous strategy $I \to A$.

(3) **Causal execution.** If $O$ were initial, its state $s : I \to O$ and the unique $r : O \to I$ would satisfy $rs = 1_I$ by scalar uniqueness and $sr = 1_O$ by initiality. Then $I$ would be initial. The one-positive-event arena, however, has distinct empty-source and one-event-source strategies from $I$.

(4) **Secured operational interaction.** Neither violates receptivity, and their sources are not isomorphic. This is a contradiction. Compact duality exchanges initial and terminal objects, so a terminal object is impossible as well.

□

**Convention 1.29** (Base, size, and finite ownership)**. Set** denotes the ordinary set-theoretic parameter category. The ambient logic is classical. Predicate calculations use its subobject doctrine, written with Heyting operations; arguments that use classical order or finite maximization are identified as such. Set-indexed meets, joins, and power objects are taken in this base. The operational category $\mathcal{C}$ has arbitrary countable arena carriers. For each fixed pair of arenas, its arrows form the set of countable strategy presentations whose source carriers are subsets of $\mathbb{N}$, modulo arena-preserving isomorphism. Every countable source admits such a presentation, so this convention retains every operational arrow. Let $\mathcal{C}_0$ be the full subcategory on arenas whose event carriers are subsets of $\mathbb{N}$, and write $J : \mathcal{C}_0 \hookrightarrow \mathcal{C}$ for the inclusion. The category $\mathcal{C}_0$ is small. Fixed bijections for finite tagged carriers equip it with a tensor $\otimes_0$ and coherent comparisons making $J$ strong symmetric monoidal. The empty arena is its unit $I$. Every countable arena is isomorphic to some $Jc$ by transport of its structure along an injection into $\mathbb{N}$. This is an objectwise existence statement; no simultaneous choice of representatives is made. The owner set is a fixed finite list $N$. A coalition $C \subseteq N$ inherits its order from this list, and $-C$ denotes the complementary list. Thus $\Sigma_C = \prod_{i \in C} \Sigma_i$ retains exactly its members' controlled coordinates. Empty products are singletons and empty conjunctions are true; the empty coalition changes no coordinate. Collective improvement predicates below explicitly quantify over nonempty coalitions.

An empty dependent linear sum or product requires limits and colimits not present in the original arena category. We use the small monoidal category $\mathcal{C}_0$ and the fully faithful, essentially surjective inclusion $J : \mathcal{C}_0 \hookrightarrow \mathcal{C}$ from Convention 1.29. Presentations with carriers in $\mathbb{N}$ retain finite data arenas $D(X)$ and residual arenas as objects of $\mathcal{C}$. For each original arena, restricted Yoneda recovers its concurrent arrows. The new colimits and limits permit dependent interfaces that the operational arena category itself lacks. Thus the completion

enlarges the language of interfaces, not the possible executions of any already specified original strategy.

**Construction 1.30** (Presheaves and Day convolution)**.**

(1) **Form the presheaves and their tensor.** Put $\widehat{\mathcal{C}} = [\mathcal{C}_0^{\mathrm{op}}, \mathbf{Set}]$ and write $yA = \mathcal{C}(J(-), A)$ for restricted Yoneda. For presheaves $P, Q$ and $c \in \mathcal{C}_0$ define

$$(P \otimes_D Q)(c) = \int^{a,b \in \mathcal{C}_0} \mathcal{C}_0(c, a \otimes_0 b) \times P(a) \times Q(b), \qquad \widehat{I} = yI,$$

$$[P, Q]_D(c) = \mathrm{Nat}(P, Q(c \otimes_0 -)).$$

(2) **Identify coend representatives.** The coend is the quotient of the disjoint union over pairs $(a, b)$ by the relations induced by arrows in either variable. For example, for $u : a' \to a$, it identifies $(k, P(u)x, z)$ with $((u \otimes_0 1)k, x, z)$; the other variable is treated identically.

(3) **Reindex and take limits.** Reindexing in $c$ is precomposition. Limits and colimits of presheaves are formed objectwise.

**Theorem 1.31** (Completion and exact recovery of concurrent arrows)**.** *The category $\widehat{\mathcal{C}}$ is complete, cocomplete, and symmetric monoidal closed under the displayed operations. Tensor preserves colimits separately. Yoneda is fully faithful and strong symmetric monoidal, and preserves the original internal hom:*

$$\begin{aligned} \widehat{\mathcal{C}}(yA, yB) &\cong \mathcal{C}(A, B), \\ yA \otimes_D yB &\cong y(A \otimes B), \\ [yA, yB]_D &\cong y(A \multimap B). \end{aligned}$$

*It preserves the compact duals of representable objects. A natural transformation $\alpha : yA \to yB$ recovers the unique original strategy*

$$\alpha_c(u) \circ u^{-1} : A \longrightarrow B, \qquad u : Jc \xrightarrow{\cong} A.$$

*The result is independent of the chosen isomorphism $u$.*

*Proof.*

(1) **Day-convolution calculation.** Day's convolution theorem supplies the monoidal coherence and closed structure (Day, 1970). In the present formulas, the coend universal property gives directly

$$\widehat{\mathcal{C}}(R \otimes_D P, Q) \cong \widehat{\mathcal{C}}(R, [P, Q]_D).$$

Products with fixed sets and coends preserve colimits, proving the separate cocontinuity of tensor. Objectwise limits and colimits prove completeness and cocompleteness.

(2) **Representation and Yoneda decoding.** We first verify restricted Yoneda directly. Choose, for this argument only, an isomorphism $u : Jc \to A$. Put $f = \alpha_c(u)u^{-1}$.

(3) **Functoriality and substitution.** For any $t : Jd \to A$, fullness of $J$ gives the unique arrow $v : d \to c$ with $Jv = u^{-1}t$. Naturality yields $\alpha_d(t) = \alpha_c(u)Jv = ft$. Thus $\alpha = yf$, and evaluation at $u$ proves uniqueness. The recovered $f$ is therefore independent of the local witness $u$.

(4) **Day-convolution calculation.** The strong monoidal comparison of $J$ induces canonical maps $yA \otimes_D yB \to y(A \otimes B)$. For $A = Ja$ and $B = Jb$, co-Yoneda makes these maps isomorphisms. Every $A, B$ is locally isomorphic to such a pair, so the same canonical maps are isomorphisms for all $A, B$.

(5) **Unit and identity laws.** The unit comparison is likewise an isomorphism; coherence follows from these canonical maps and that of $J$. For the closed structure the same local argument gives

$$[yA, yB]_D(c) \cong \mathcal{C}(Jc \otimes A, B) \cong \mathcal{C}(Jc, A \multimap B).$$

The comparisons are natural and require no global choice of arena presentations. They carry cups and caps to cups and caps, preserving their snake equations.

(6) **Equation reflection under Yoneda.** Full faithfulness proves exact recovery and reflection of equations between original strategies.

□

The Yoneda comparison intertwines the original tensor product and Day convolution.

$$\begin{array}{ccc} \mathcal{C}_0 \times \mathcal{C}_0 & \xrightarrow{y \times y} & \widehat{\mathcal{C}} \times \widehat{\mathcal{C}} \\ {\scriptstyle \otimes_0}\downarrow & & \downarrow{\scriptstyle \otimes_D} \\ \mathcal{C}_0 & \xrightarrow[y]{} & \widehat{\mathcal{C}} \end{array}$$

Every presheaf is canonically a colimit of representables: $P \cong \int^{c \in \mathcal{C}_0} P(c) \cdot y(Jc)$, where the dot denotes a copower. Thus the new interfaces are generated from the concurrent ones by specified categorical constructions. They need not themselves be single event structures. General presheaves have the proved symmetric monoidal closed structure; compact duality remains available on the original arena fragment. The theorem is a preservation and recovery result for semantic arrows.

### 1.2.3 Dependent ordinary and linear fibres

Dependence in this model is on ordinary parameters. The discrete index category $\Gamma_{\mathrm{disc}}$ indexes a family of presheaf interfaces, so substitution changes the parameter at which a linear object is read. The linear context still accounts for resource occurrences. This division allows dependent quantification without turning a linear resource into an ordinary variable.

**Construction 1.32** (The two indexed categories)**.**

(1) **Define the two fibres.** For a set $\Gamma$ define

$$\mathsf{T}(\Gamma) = \mathbf{Set}^{\Gamma_{\mathrm{disc}}}, \qquad \mathsf{L}(\Gamma) = \widehat{\mathcal{C}}^{\Gamma_{\mathrm{disc}}}.$$

Objects of $\mathsf{T}(\Gamma)$ are ordinary dependent families; objects of $\mathsf{L}(\Gamma)$ are linear dependent families. Their morphisms are families of functions and natural transformations, respectively.

(2) **Reindex the families.** For $f : \Delta \to \Gamma$ put $(f^*A)_\delta = A_{f(\delta)}$. Composition, tensor, internal hom, and their comparisons in the linear fibres are pointwise. The unit is the constant family $\widehat{I}$.

(3) **Extend the ordinary context.** An ordinary family $X$ has context extension $p_X : \Gamma.X = \coprod_\gamma X_\gamma \to \Gamma$.

**Proposition 1.33** (Substitution and the original indexed fragment)**.** *These assignments are contravariantly functorial. Ordinary reindexing preserves the cartesian closed structure, and linear reindexing preserves the symmetric monoidal closed structure. Pointwise Yoneda embeds $\mathcal{C}^{\Gamma_{\mathrm{disc}}}$ fully faithfully into $\mathsf{L}(\Gamma)$, compatibly with substitution. Its constant-arena part is exactly $\mathcal{C}_\Gamma$.*

*Proof.*

(1) **Functoriality and substitution.** Both identity and composite substitution are identities of index precomposition. Every listed structure is evaluated at a single index, so reindexing preserves its defining universal property and comparison.

(2) **Representation and Yoneda decoding.** The hom-set of constant representables $yA, yB$ is

$$\prod_{\gamma\in\Gamma} \widehat{\mathcal{C}}(yA, yB) \cong \mathbf{Set}(\Gamma, \mathcal{C}(A, B)).$$

Theorem 1.31, applied at each index, proves the remaining preservation and full-faithfulness assertions.

□

**Proposition 1.34** (Indexed operational equations)**.** *Composition, tensor, and currying of parameterized concurrent arrows commute with parameter substitution. They satisfy associativity, units, interchange, and the closed-structure equations*

$$\mathrm{ev} \circ (\Lambda_A t \otimes 1_A) = t, \qquad \Lambda_A(\mathrm{ev} \circ (u \otimes 1_A)) = u.$$

*The same equations hold in every completed linear fibre.*

*Proof.*

(1) **Adjunction and transposition.** In the original fragment each equation holds at every parameter by compact closure, with the two displayed equations reducing to the snake identities. In the completion they are the triangle identities for the closed adjunction. Reindexing evaluates the identical map at the substituted index, proving compatibility.

□

Ordinary parameters may be reused to describe or compare different interactions; a linear occurrence must instead be consumed according to its process. The functor $\mathsf{Val}$ extracts monoidal states of a linear interface, not economic values, while $\mathsf{Lin}$ makes an ordinary family of chosen states available linearly. Their adjunction is the required bridge between policy descriptions and interactive resources; it introduces no copying operation on arbitrary resources.

**Construction 1.35** (Values and linearization)**.**

(1) **Form the two functors.** Define functors between the two fibres by

$$(\mathsf{Val}_\Gamma A)_\gamma = \widehat{\mathcal{C}}(\widehat{I}, A_\gamma) = A_\gamma(I), \qquad (\mathsf{Lin}_\Gamma X)_\gamma = \coprod_{x\in X_\gamma} \widehat{I}.$$

(2) **Construct comprehension.** For a linear family its comprehension is

$$p_A : \Gamma.A = \coprod_{\gamma\in\Gamma} (\mathsf{Val}_\Gamma A)_\gamma \longrightarrow \Gamma.$$

The generic element at $(\gamma, a)$ is the state $a : \widehat{I} \to A_\gamma$.

**Theorem 1.36** (Constructed comprehension and linear–nonlinear adjunction)**.** *There is a substitution-stable adjunction* $\mathsf{Lin}_\Gamma \dashv \mathsf{Val}_\Gamma$ *with* $\mathsf{Lin}_\Gamma$ *strong symmetric monoidal. Linear comprehension has the representing bijection*

$$\mathsf{L}(\Delta)(\widehat{I}, f^*A) \cong \mathbf{Set}/\Gamma(f, p_A).$$

*Moreover* $\mathsf{Val}_\Gamma\mathsf{Lin}_\Gamma X \cong X$*, so* $\mathsf{Lin}_\Gamma$ *is fully faithful and* $\Gamma.\mathsf{Lin}_\Gamma X \cong \Gamma.X$ *over* $\Gamma$*.*

Comprehension transports linear states into ordinary sections of the displayed family.

$$\begin{array}{ccc} \mathsf{L}(\Delta)(\widehat{I}, f^*A) & \xrightarrow{\cong} & \mathbf{Set}/\Gamma(f, p_A) \\ \Big\downarrow{\scriptstyle g^*} & & \Big\downarrow{\scriptstyle (-)\circ g} \\ \mathsf{L}(\Theta)(\widehat{I}, (fg)^*A) & \xrightarrow[\cong]{} & \mathbf{Set}/\Gamma(fg, p_A) \end{array}$$

*Proof.*

(1) **Frobenius and coproduct comparison.** The coproduct universal property gives

$$\mathsf{L}(\Gamma)(\mathsf{Lin}_\Gamma X, A) \cong \prod_\gamma \mathbf{Set}(X_\gamma, \widehat{\mathcal{C}}(\widehat{I}, A_\gamma)) = \mathsf{T}(\Gamma)(X, \mathsf{Val}_\Gamma A).$$

Separate cocontinuity of tensor gives $\mathsf{Lin}(X \times Y) \cong \mathsf{Lin}(X) \otimes_D \mathsf{Lin}(Y)$ and $\mathsf{Lin}(1) \cong \widehat{I}$.

(2) **Monoidal comparison.** On summands the comparisons are unitors and the bijection between pairs of indices; their coherence follows on each summand from monoidal coherence. Evaluation at $I$ yields

$$(\mathsf{ValLin}X)_\gamma = \coprod_{x\in X_\gamma} \mathcal{C}(I, I) \cong X_\gamma$$

by Lemma 1.28. This is the adjunction unit, proving full faithfulness.

(3) **Secured operational interaction.** For comprehension, a family of states $a_\delta : \widehat{I} \to A_{f(\delta)}$ corresponds to the map $\delta \mapsto (f(\delta), a_\delta)$; projection recovers the family. Pullback reindexes these same pairs. All the constructions commute with index precomposition, establishing their substitution comparisons and coherence.

□

**Construction 1.37** (The exponential)**.**

(1) **Define the comonad.** Put $!_\Gamma = \mathsf{Lin}_\Gamma\mathsf{Val}_\Gamma$. The counit $\epsilon_A : !A \to A$ evaluates the state labelling each summand; the comultiplication is $\mathsf{Lin}(\eta_{\mathsf{Val}A})$. The adjunction triangles prove the comonad identities.

(2) **Transport the structural maps.** The diagonal and terminal map of $\mathsf{Val}A$, transported through the strong monoidal functor $\mathsf{Lin}$, give $!A \to !A \otimes_D !A$ and $!A \to \widehat{I}$ and their commutative-comonoid laws. Thus these structural maps are constructed, and are stable under substitution. Here ! is idempotent because $\eta$ is invertible. This is the monoidal-adjunction account of linear–nonlinear semantics (Benton, 1995).

(3) **Construct the monoidal comparison.** For completeness, the lax monoidal comparison on values sends a pair of states $a : \widehat{I} \to A$, $b : \widehat{I} \to B$ to

$$\widehat{I} \cong \widehat{I} \otimes_D \widehat{I} \xrightarrow{a \otimes_D b} A \otimes_D B;$$

its unit is the identity state of $\widehat{I}$. Applying $\mathsf{Lin}$ gives the monoidal comparison $!A \otimes_D !B \to !(A \otimes_D B)$. Its coherence, and compatibility with the comonad maps, can be checked on each pair of state-labelled summands: both sides evaluate the same tensor of states. Also

$$!(A \,\&\, B) \cong !A \otimes_D !B, \qquad !\top \cong \widehat{I},$$

because values preserve products and $\mathsf{Lin}$ is strong monoidal.

The exponential renders named states reusable through the ordinary interface. It does not turn the physical response arena $D(X)$ into a copower: $\mathsf{Lin}(1) = \widehat{I}$, whereas $yD(1)$ has more than one state and is not isomorphic to $\widehat{I}$. The copying and collection strategies on finite data therefore retain the causal behaviour proved above, independently of the abstract linear–nonlinear adjunction.

### 1.2.4 Dependent adjoints, additives, and equality

**Construction 1.38** (Dependent sums and products)**.**

(1) **Form sums and products on fibres.** For $f : \Delta \to \Gamma$ define

$$(\Sigma^{\ell}_f A)_\gamma = \coprod_{\delta:\, f(\delta)=\gamma} A_\delta, \qquad (\Pi^{\ell}_f A)_\gamma = \prod_{\delta:\, f(\delta)=\gamma} A_\delta.$$

They act on arrows componentwise using coproduct injections and product projections.

(2) **Recover the ordinary adjoints.** The identical set formulas define ordinary functors $\Sigma^{\mathrm{c}}_f, \Pi^{\mathrm{c}}_f$ on $\mathsf{T}$.

**Theorem 1.39** (Adjunctions, base change, and Frobenius)**.** *For every $f$, $\Sigma^{\ell}_f \dashv f^* \dashv \Pi^{\ell}_f$. Both adjoints satisfy Beck–Chevalley for pullback squares. There is a natural Frobenius isomorphism*

$$\Sigma^{\ell}_f(f^*B \otimes_D A) \cong B \otimes_D \Sigma^{\ell}_f A.$$

*The comparisons are coherent with identity and composite substitution. The ordinary adjoints have the corresponding properties, with cartesian product in place of tensor. All empty fibres are included.*

*Proof.*

(1) **Frobenius and coproduct comparison.** The coproduct universal property gives

$$\mathsf{L}(\Gamma)(\Sigma^{\ell}_f A, B) \cong \prod_{\gamma} \prod_{f(\delta)=\gamma} \widehat{\mathcal{C}}(A_\delta, B_\gamma) \cong \mathsf{L}(\Delta)(A, f^*B).$$

Replacing the coproduct universal property by the product universal property gives the right adjunction.

(2) **Base-change comparison.** For a pullback square $\Delta' \xrightarrow{a} \Delta$, $\Gamma' \xrightarrow{b} \Gamma$ with vertical arrows $f', f$, the map $a$ is a bijection between the fibre of $f'$ at $\gamma'$ and the fibre of $f$ at $b\gamma'$. It therefore induces $b^*\Sigma^{\ell}_f \cong \Sigma^{\ell}_{f'} a^*$ and $b^*\Pi^{\ell}_f \cong \Pi^{\ell}_{f'} a^*$. Successive pullback comparisons compose the same index bijections, so their coherence follows.

(3) **Base-change comparison: completion.** Frobenius is, at each $\gamma$, the distribution of $B_\gamma \otimes_D -$ over the coproduct indexed by that fibre. Its comparison on each summand is the identity; this also proves compatibility with base change. The set-valued proof is identical.

□

The pullback square supplies Beck–Chevalley comparisons for both dependent adjoints.

$$\begin{array}{ccc} \Delta' & \xrightarrow{a} & \Delta \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ \Gamma' & \xrightarrow[b]{} & \Gamma \end{array}$$

**Construction 1.40** (Additives and equality)**.**

(1) **Form the additive types.** Finite products and coproducts in each linear fibre give $A \,\&\, B$, $\top$, $A \oplus B$, and $0$. The object $\top$ is the terminal presheaf; the tensor unit is $\widehat{I} = yI$.

(2) **Construct the equality family.** For an ordinary family $X$ let $d : \Gamma.X \to \Gamma.X.X$ be its diagonal and put

$$\mathrm{Id}^\ell_X = \Sigma^\ell_d \widehat{I}.$$

Its fibre at $(\gamma, x, x')$ is $\widehat{I}$ if $x = x'$ and $0$ otherwise. It equals $\mathsf{Lin}(\mathrm{Eq}_X)$, where $\mathrm{Eq}_X$ is the ordinary equality family. Reflexivity is the identity of $\widehat{I}$ on the diagonal.

(3) **Obtain equality elimination.** For any linear $E, D$ over $\Gamma.X.X$, Frobenius and adjunction give

$$\mathsf{L}(\Gamma.X.X)(E \otimes_D \mathrm{Id}^\ell_X, D) \cong \mathsf{L}(\Gamma.X)(d^*E, d^*D).$$

This is its equality-elimination universal property; the adjunction identities give evaluation and uniqueness. All these constructions are stable under substitution by Theorem 1.39.

**Proposition 1.41** (Recovery of ordinary dependent structure)**.** *Every ordinary context extension is the comprehension of a linearized family. Moreover*

$$\mathsf{Val}(\Sigma^\ell_f \mathsf{Lin} X) \cong \Sigma^{\mathrm{c}}_f X, \qquad \mathsf{Val}(\Pi^\ell_f \mathsf{Lin} X) \cong \Pi^{\mathrm{c}}_f X.$$

*For $p : \Gamma.X \to \Gamma$ and $B \in \mathsf{L}(\Gamma)$,*

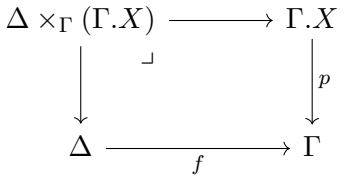
$$\Sigma^\ell_p p^* B \cong \mathsf{Lin} X \otimes_D B, \qquad \Pi^\ell_p p^* B \cong [\mathsf{Lin} X, B]_D.$$

*Proof.*

(1) **Frobenius and coproduct comparison.** Evaluation at $I$ preserves objectwise products and coproducts, and $\mathsf{Val}\mathsf{Lin} X \cong X$. This proves the first two comparisons and, by Theorem 1.36, the context claim. The constant-family sum distributes $B$ over one copy of $\widehat{I}$ per $x$; the product comparison follows by applying closedness to that copower. These calculations are natural in all parameters.

□

Dependent ordinary families are transported along the pullback of context extension.




**Theorem 1.42** (The categorical dependent linear model)**.** *The constructed ordinary and linear indexed categories, comprehension, and substitution-stable monoidal adjunction form a dependent linear–nonlinear model over* **Set***. They carry ordinary dependent sums, products, and extensional equality; linear multiplicatives and additives; the exponential of Construction 1.37; and the dependent linear sums, products, and equality just constructed. Dependence is on ordinary parameters. Original concurrent arena families embed fully faithfully, preserving their operational structure, and all original ordinary families are recovered by* ValLin*.*

*Proof.*

(1) **Adjunction and transposition.** Proposition 1.33 supplies the indexed monoidal structure and its substitution laws. Theorem 1.36 constructs the comprehension representing property and the monoidal adjunction. Theorem 1.39 proves the adjoints, Beck–Chevalley, and Frobenius, including their coherence.

(2) **Assembly of dependent linear structure.** Construction 1.40 gives equality and additives. These are the structural requirements in the indexed comprehension account of dependent linear semantics (Vákár, 2015, Sections 3–4). The preservation assertions are Theorem 1.31 and Proposition 1.41.

□

**Proposition 1.43** (Transport of dependent linear realizations)**.** *For $A, B \in \mathsf{L}(W)$ and any $t : U \to W$, an ordinary section $\rho : 1_W \to \mathsf{Val}_W[A, B]_D$ determines a section*

$$t^*\rho : 1_U \to \mathsf{Val}_U[t^*A, t^*B]_D.$$

*Under the linear–nonlinear adjunction, its linear transpose is $t^*\widetilde{\rho} : t^*A \to t^*B$. These comparisons commute with composition of state maps, and, for representable fibres, recover the original concurrent strategies pointwise by Yoneda.*

*Proof.*

(1) **Fibrewise calculation.** At $u \in U$ the transported state is the arrow $\rho_{t(u)} : A_{t(u)} \to B_{t(u)}$. The adjunction is pointwise, so transposing before or after reindexing gives that same component. Successive substitutions select the same component. For representables, $\widehat{\mathcal{C}}(yA, yB) \cong \mathcal{C}(A, B)$ proves recovery.

□

The transport result will be needed when an epistemic action changes the ordinary state index. Its reindexed family still denotes the same pointwise concurrent arrow; later residualization can therefore track an interaction across the update without replacing it by an abstract scalar.

### 1.2.5 Policy-parameterized interactions and their composition

**Construction 1.44** (Dependent owned realization)**.**

(1) **Specify the policy family and its realization.** Fix $A, B \in \mathsf{L}(\Gamma)$ and ordinary policy families $\vec{\Sigma} = (\Sigma_i)_{i \in N}$; put $\Sigma = \prod_i \Sigma_i$. A realization is an ordinary morphism

$$g : \Sigma \longrightarrow \mathsf{Val}_\Gamma[A, B]_D.$$

Since $\mathsf{Val}[A, B]_D \cong (\widehat{\mathcal{C}}(A_\gamma, B_\gamma))_\gamma$, it specifies $g_{\gamma,\sigma} : A_\gamma \to B_\gamma$.

(2) **Transpose the realization.** Adjunction and closedness give the equivalent single linear morphism

$$\widetilde{g} : \mathsf{Lin}_\Gamma \Sigma \otimes_D A \longrightarrow B.$$

(3) **Change policy coordinates.** An ownerwise map $r_i : \Theta_i \to \Sigma_i$ reparametrizes $g$ by $\prod_i r_i$; its linear form is precomposition with $\mathsf{Lin}(\prod_i r_i) \otimes_D 1_A$.

**Proposition 1.45** (Categorical construction of dependent game composition)**.** *Dependent owned realizations have sequential and parallel composition, identity, and ownerwise reparametrizations. With the canonical ownerwise product comparisons they form a symmetric monoidal bicategory at each $\Gamma$, stable under substitution of $\Gamma$. On original representable interfaces the construction recovers precisely the concurrent open arenas and their compositions of Section 3.2.*

The realization of a composite is computed by composing the policy-indexed arrows.

$$\begin{array}{ccc} \Sigma \times T & \xrightarrow{g \times h} & \mathcal{C}(A, B) \times \mathcal{C}(B, C) \\ \cong \downarrow & & \downarrow \circ \\ \prod_i (\Sigma_i \times T_i) & \xrightarrow[h \circ g]{} & \mathcal{C}(A, C) \end{array}$$

*Proof.*

(1) **Composition through linearized policy sums.** For $\widetilde{g} : \mathsf{Lin}\Sigma \otimes_D A \to B$ and $\widetilde{h} : \mathsf{Lin}\mathcal{T} \otimes_D B \to C$, form

$$\mathsf{Lin}(\Sigma \times \mathcal{T}) \otimes_D A \cong \mathsf{Lin}\mathcal{T} \otimes_D \mathsf{Lin}\Sigma \otimes_D A \xrightarrow{1 \otimes \widetilde{g}} \mathsf{Lin}\mathcal{T} \otimes_D B \xrightarrow{\widetilde{h}} C.$$

Its policy component is $h_{\gamma,\tau} g_{\gamma,\sigma}$.

(2) **Ownerwise alternatives.** Parallel composition uses the strong monoidal comparison for $\mathsf{Lin}$ and symmetry to apply the two realized arrows to their respective inputs; its component is $g_{\gamma,\sigma} \otimes_D k_{\gamma,\rho}$. For each owner use the paired policy family, so the total-profile comparison is the canonical rearrangement of those pairs. Identity uses singleton policies.

(3) **Ownerwise alternatives: second comparison.** A 2-cell is an ownerwise map preserving the realization, and horizontal and parallel composition take products of these maps. Associators, unitors, and interchange rearrange the same named policy coordinates; their realization equations are the category and monoidal equations in $\mathsf{L}(\Gamma)$. Thus every coherence diagram has identical coordinate maps and identical realizations.

(4) **Representation and Yoneda decoding.** Substitution evaluates these formulas at the substituted index. Finally $\mathsf{Val}[yA, yB]_D \cong \mathcal{C}(A, B)$ and Yoneda preserves composition and tensor, proving exact recovery.

□

The owner-indexed factorization remains displayed after the dependent realization is curried. A diagonal can require two components to use one policy description, but changing the reference parameter does not restrict which alternative policies exist. The next section makes agent types from these policy families; subsequent state-context and comparison constructions will distinguish reference substitution from changes of controlled alternatives.

## 1.3 The indexed model underlying agent types

### 1.3.1 The indexed model used by the agent calculus

The operational category, its restricted Yoneda completion, and its substitution-stable dependent linear–nonlinear structure have already been constructed in Theorems 1.10, 1.31 and 1.42. In the present notation the canonical comparisons are

$$\mathcal{V}(yA, yB) \cong \mathcal{C}(A, B), \qquad yA \otimes yB \cong y(A \otimes B). \tag{1.2}$$

Put $\mathcal{T}(W) = \mathbf{Set}^{W_{\mathrm{disc}}}$, $\mathcal{L}(W) = \mathcal{V}^{W_{\mathrm{disc}}}$ and $\mathcal{P}(W) = \mathrm{Sub}(W)$. The functors $\mathsf{Lin}X = \coprod_{x \in X} \mathsf{I}$ and $\mathsf{Val}A = \mathcal{V}(\mathsf{I}, A)$, and their indexed forms, are those of Construction 1.35; in particular $\mathsf{Lin}(X \times Y) \cong \mathsf{Lin}X \otimes \mathsf{Lin}Y$ and $\mathsf{ValLin}X \cong X$ by Theorem 1.36. For $f : V \to W$ the previously constructed linear adjoints have fibres

$$(\Sigma^{\ell}_f A)_w = \coprod_{f(v)=w} A_v, \qquad (\Pi^{\ell}_f A)_w = \prod_{f(v)=w} A_v. \tag{1.3}$$

The ordinary adjoints use set sums and products; the predicate adjoints use existential and universal quantification along the same fibres.

**Theorem 1.46** (Indexed adjunctions and substitution)**.** *There are adjunctions $\Sigma^{\ell}_f \dashv f^* \dashv \Pi^{\ell}_f$, Beck–Chevalley isomorphisms for pullback squares, and a coherent Frobenius isomorphism*

$$\Sigma^{\ell}_f(f^* B \otimes A) \cong B \otimes \Sigma^{\ell}_f A.$$

*These include empty fibres and commute with successive substitutions. Ordinary and predicate versions hold as well.*

*Proof.*

(1) **Base-change comparison.** This is Theorem 1.39, applied to the same pointwise fibres. Its proof constructs the adjoints from their universal properties, Beck–Chevalley by the pullback-induced bijections of index fibres, and Frobenius by separate cocontinuity of Day tensor. The ordinary comparisons use the same fibrewise set formulas.

(2) **Base-change comparison: completion.** For predicates, universal quantification is the product of subterminal families, whereas existential quantification is the support (image) of their indexed coproduct; their Beck–Chevalley and Frobenius laws follow from the same fibre bijections and the predicate adjunctions proved in Theorem 5.22.

□

**Proposition 1.47** (Linearization as a dependent sum of units)**.** *For an ordinary family $X$ over $\Gamma$ and its comprehension $p : \Gamma.X \to \Gamma$ there are substitution-natural isomorphisms*

$$\mathsf{Lin}_{\Gamma} X \cong \Sigma^{\ell}_p \mathsf{I}_{\Gamma.X}, \qquad \mathsf{Lin}_{\Gamma} X \otimes B \cong \Sigma^{\ell}_p p^* B.$$

*They interpret* Lin*-elimination with an independent linear continuation.*

*Proof.*

(1) **Frobenius and coproduct comparison.** The first fibre is the coproduct of one tensor unit for each $x \in X_\gamma$, by the definitions of $\mathsf{Lin}$ and $\Sigma_p^\ell$; the second comparison is the Frobenius isomorphism of Theorem 1.39. Both comparisons are natural in $\gamma$ and preserve substitution by Beck–Chevalley.

□

### 1.3.2 Closed reindexing and dependent-product coherence

**Lemma 1.48** (Closed reindexing in the concrete indexed model)**.** *For every ordinary substitution $f : V \to W$, the canonical comparison $f^*[A, B] \to [f^*A, f^*B]$ is a natural isomorphism, coherent under successive substitutions.*

*Proof.*

(1) **Fibrewise calculation.** At $v \in V$ both sides are canonically the same internal hom:

$$(f^*[A, B])_v = [A, B]_{f(v)} = [A_{f(v)}, B_{f(v)}] = [(f^*A)_v, (f^*B)_v].$$

The comparison and its inverse are the identities on these fibres. Both orders of successive reindexing select $A_{f(g(u))}$ and $B_{f(g(u))}$, proving naturality and coherence, including empty indexing sets.

□

Reindexing commutes with the internal hom by the identity on each fibre.

$$\begin{array}{ccc} f^*[A, B] & \xrightarrow{\cong} & [f^*A, f^*B] \\ {\scriptstyle g^*}\downarrow & & \downarrow{\scriptstyle g^*} \\ (fg)^*[A, B] & \xrightarrow[\cong]{} & [(fg)^*A, (fg)^*B] \end{array}$$

For the dependent-product Beck–Chevalley comparison in a pullback $V' = W' \times_W V$, both sides at $w' \in W'$ are products over the same fibre $\{v : f(v) = g(w')\}$. The comparison is uniquely specified by its product projections; repeated base change selects the same factors. The corresponding sum comparison uses the same bijection and coproduct injections. Thus the explicit indexed formulas verify the coherence already established in Theorem 1.39.

# 2 Agents as First-Class Linear Objects

## 2.1 Agent types and distinct levels of control

An owner may make decisions at several sites in a process. To identify which decisions can change together, we represent an agent by a linear interaction indexed by its owner's complete policy, following the distinction between parameter and realization in Capucci et al. (2022b). The owner determines which policy may change. The ports determine how the process interacts, the observation map determines which signals a response may use, and the comparison interface determines whether an alternative is preferred.

**Four distinctions in the agent calculus.** An *owner identity* indexes control and can occur at several decision positions. A *linear occurrence* of an agent is a resource in a typing context; its use must be accounted for. A *complete policy* is an ordinary parameter specifying responses at all relevant decision positions, including positions not reached during a particular execution. A *situated agent* is an agent presentation together with a legal domain, a reference-state space, observation maps, and comparison data.

If one owner makes two decisions, both entries belong to its complete policy. Replacing that policy can change both decisions together. The same device can admit several alternative policies. Running two copies of it would require two linear occurrences in the execution.

**Constructed and supplied interfaces.** An institution supplies its permission rule, and an owner supplies its comparison of outcomes. The calculus represents these inputs by typed maps or subobjects. Once a protocol is wired, its signals and certificates have causal histories. Images, pullbacks, and dependent quantification then determine which observations and legal executions those histories support.

**The agent-type construction.**

$$\left|\mathsf{Ag}_{\vec{\Sigma}}(A,B)\right| = [\mathsf{Lin}\Sigma \otimes A, B]. \tag{2.1}$$

The signature keeps ownership displayed rather than attempting to recover it from the underlying linear object. In this sense "first-class" concerns the linear occurrence and higher-order use of the agent resource; ownership is a displayed index of its typing judgment, not a property of its underlying object alone. Theorems 2.15 and 2.19 identify dependent abstraction and owned presentations; Theorems 3.2, 4.15 and 4.17 then compare composition, regrouping, and implemented choices. The subsequent contextual and operational sections use precisely this representation rather than adding an independent meaning for an agent.

Thus an agent is not merely an available policy set or an operational strategy. Its linear type can be supplied to another process, while the displayed signature records which owner controls each choice. Polarity continues to describe interface behaviour; rationality belongs to the later comparison predicate. The representation theorem establishes the exact relation between states of this type and owned concurrent realizations.

## 2.2 Economic vocabulary: policies, outcomes, and information

To read the type of an agent economically, we must distinguish a complete contingent policy from the concurrent arrow implementing it. Ownership identifies who may replace a policy coordinate; comparison and observation specify, separately, how that replacement is evaluated and what information the owner possesses. The definitions below fix this vocabulary before Section 2.3 packages the resulting data into a formal agent interface.

**Definition 2.1** (Agent identities, policies, and profiles)**.** An **economic game** in ordinary strategic form consists of a finite owner list $N$, sets $\Sigma_i$ of available policies, an outcome set $Z$, a realization, and strict comparison relations:

$$\Sigma = \prod_{i\in N} \Sigma_i, \qquad z : \Sigma \longrightarrow Z, \qquad \prec_i \hookrightarrow\ Z \times Z \quad (i \in N).$$

An **agent identity** $i$ controls the coordinate $\Sigma_i$. The term **owner** describes this coordinate control; **player** is the customary game-theoretic name for the agent when its incentives are considered. The same identity may own several decisions in one process. A **pure policy**

$\sigma_i \in \Sigma_i$ is a complete contingent plan for the decisions of $i$; a **profile** is $\sigma = (\sigma_i)_{i \in N} \in \Sigma$. Write $\sigma_{-i}$ for the profile of the other owners.

More explicitly, at a decision node $v$ with effective observation set $O_v$ and action set $A_v$, a decision policy has type $\pi_v : O_v \to A_v$ and chooses the **action** $\pi_v(o) \in A_v$ upon observation $o$. For a protocol with node set $V$ and ownership $i(v)$, the complete owner policy space is

$$\Sigma_i = \prod_{v \in V:\, i(v)=i} A_v^{O_v},$$

including decisions at nodes not reached by realized play. The general open-arena interface takes the sets $\Sigma_i$ as parameters; its operational realization has type $g : \Sigma \to \mathcal{C}(A, B)$. An economic policy is an element of $\Sigma_i$, whereas an **operational concurrent strategy** is an event-structure morphism representing an arrow $g(\sigma)$ of $\mathcal{C}$. These are distinct kinds of objects.

**Definition 2.2** (Coalitions and deviations)**.** A **coalition** is a subset $C \subseteq N$ of owners, with complement $-C = N \setminus C$ and policy carriers $\Sigma_C = \prod_{i \in C} \Sigma_i$ and $\Sigma_{-C} = \prod_{i \notin C} \Sigma_i$. Its **deviation relation** is the kernel pair of the outsider projection:

$$\Sigma \xleftarrow{p_1} D_C = \Sigma \times_{\Sigma_{-C}} \Sigma \xrightarrow{p_2} \Sigma, \qquad (\sigma, \tau) \in D_C \iff \sigma_{-C} = \tau_{-C}.$$

A coalition deviation from a reference $\sigma$ is an element of the fibre $\mathsf{Alt}_C(\sigma) = p_1^{-1}(\sigma)$; for the ordinary outcome realization $z$ its paired outcome is $\delta_C^z = (zp_1, zp_2) : D_C \to Z^2$. A **unilateral deviation** has $C = \{i\}$; the **grand coalition** has $C = N$. For $C = \varnothing$, $D_C$ is the diagonal, so no policy changes. The word deviation denotes a comparison of available alternatives, not a temporal adjustment.

Joint control of these coordinates is distinct from a collective preference, pooled access to observations, or common resource endowments. Joint choice and guaranteed outcomes against arbitrary outsider choices therefore have different quantifier structures. Control of material resources is represented separately through occurrence-preserving processes and their certificates.

**Definition 2.3** (Preferences, utilities, and valuations)**.** For an outcome set $Z$, an agent's **strict preference** is a relation $\prec_i \hookrightarrow Z \times Z$; $z \prec_i z'$ means that $z'$ is strictly better than $z$ for owner $i$. For exact numerical evaluations write $R_+ = \mathbb{Q}_{\geq 0}$ and $R_\pm = \mathrm{Gr}(R_+) \cong \mathbb{Q}$, with inclusion $\eta : R_+ \to R_\pm$. Explicitly, the pair $(a, b)$ represents the signed rational $a - b$, and $(a, b)$ and $(c, d)$ represent the same value precisely when $a + d = b + c$. In numerical game statements $\mathbb{K}$ denotes $R_\pm$ in the exact presentation and $\mathbb{R}$ in the classical real-valued presentation. Finite numerical examples use the former unless otherwise indicated. A numerical **utility** or **payoff** for completed plays evaluates the measured result via $e_i : T_C Z \to \mathbb{K}$ (Constructions 2.7 and 2.8). An outcome-only payoff $u_i : Z \to \mathbb{K}$ represents the given relation on $Z$ when

$$z \prec_i z' \quad \iff \quad u_i(z) < u_i(z').$$

For an allocation carrier $X$, a numerical **valuation** is a map $v_i : X \to \mathbb{K}$ assigning the value of an allocation before separately specified payments. Alternatively, when owner $i$ receives a bundle in $X_i$ through $a_i : X \to X_i$, use $v_i : X_i \to \mathbb{K}$ and the composite $v_i a_i : X \to \mathbb{K}$. The resulting **social welfare** is the map

$$\mathrm{SW} : X \longrightarrow \mathbb{K}, \qquad \mathrm{SW}(x) = \sum_{i \in N} v_i(a_i(x)),$$

with $a_i = \mathrm{id}_X$ in the common-allocation presentation. A valuation need not be an agent's final utility after payments, and an allocation's resource feasibility is separately witnessed.

The general theory uses the supplied comparison relation directly, including when no numerical utility representation is available. When numerical utilities are supplied, their differences measure incentive gains. The displayed welfare map records the specified sum of valuations, not an additional rationality law on agents.

**Definition 2.4** (Best responses and Nash equilibrium)**.** Let $z : \Sigma \to Z$ realize the outcome of a policy profile. Against fixed outsider policies $\sigma_{-i}$, a policy $a_i \in \Sigma_i$ is a **best response** if no available $b_i \in \Sigma_i$ satisfies

$$z(a_i, \sigma_{-i}) \prec_i z(b_i, \sigma_{-i}).$$

For a numerical utility this is equivalent to $u_i(z(a_i, \sigma_{-i})) \geq u_i(z(b_i, \sigma_{-i}))$ for every $b_i$. A **Nash equilibrium** is a profile $\sigma$ at which each $\sigma_i$ is a best response to $\sigma_{-i}$. With a legal domain, the alternatives are restricted to those that complete the fixed outsiders to a legal profile.

Equilibrium expresses stability of a profile under the designated deviations. Coalitional stability tests joint changes using an explicitly stated improvement criterion. The selection construction below recovers these predicates and retains all ties.

**Example 2.5** (Individual and joint incentives)**.** Two agents choose whether to contribute to a project, with actions $a_1, a_2 \in \{0, 1\}$. Let

$$u_1(a_1, a_2) = 2a_1a_2 - a_1, \qquad u_2(a_1, a_2) = 2a_1a_2 - a_2.$$

At $(0, 0)$ each receives 0. Either unilateral contribution gives its contributor $-1$, so $(0, 0)$ is Nash. The joint change to $(1, 1)$ gives both agents 1, so the same profile admits a strict improvement by the grand coalition. At $(1, 1)$ either unilateral withdrawal reduces the withdrawing agent's payoff to 0, so that profile is Nash as well. This calculation separates individual stability from joint improvement. The joint improvement is assessed in each agent's own utility; no collective utility is needed.

**Definition 2.6** (Legality, observational admissibility, and resource feasibility)**. Legality** of a policy profile is a subobject $m : L \hookrightarrow \Sigma$, with exactly the elements of $L$ permitted. For a history set $H$, an effective observation quotient $o : H \twoheadrightarrow O$, and an action set $A$, a response $f : H \to A$ is **observationally admissible** precisely when there is a map $\pi : O \to A$ with $f = \pi o$. Equivalently, for the kernel-pair projections $p_1, p_2 : H \times_O H \rightrightarrows H$, one has

$$fp_1 = fp_2;$$

the descended $\pi$ is unique by effective descent in **Set**.

A **resource feasibility certificate** is retained in an exhibited carrier $K$ together with its realization $w : K \to Z$ into the specified allocation or outcome carrier. Its existential feasibility predicate is

$$\mathrm{Feas}_w = \mathrm{im}(w) \hookrightarrow Z.$$

The image says which outcomes admit a certificate, while $K$ retains the certificates themselves. Institutional legality, information-dependent admissibility, and physical feasibility therefore have different carriers and need not coincide.

**Construction 2.7** (The writer of measured outcomes)**.**

(1) **Form the ordinary writer.** For a commutative charge monoid $(C, +, 0)$ put $T_C Z = C \times Z$ and $T_C f = 1_C \times f$. Its maps

$$\eta^T_Z(z) = (0, z), \qquad \mu^T_Z(c, (d, z)) = (c + d, z)$$

insert a result at zero cost and combine nested records. For $f : X \to T_C Y$ its extension is $f^\dagger = \mu^T_Y T_C f$: if $f(x) = (d, y)$, then $f^\dagger(c, x) = (c+d, y)$. This is the writer interpretation of cost in Niu et al. (2022, Section 1.1). Here its input cost is obtained from a displayed execution witness.

(2) **Verify the sequential and parallel laws.** Naturality of $\eta^T, \mu^T$ follows by applying the same outcome map on both sides. The two unit laws reduce to $0 + c = c = c + 0$; the two multiplications on $(b, (c, (d, z)))$ agree by associativity. Thus $(g^\dagger f)^\dagger = g^\dagger f^\dagger$ for consecutive measured operations. The strength sends $(x, (c, y))$ to $(c, (x, y))$. Running two independent branches in either order yields $(c + d, (x, y))$ from $(c, x), (d, y)$, by commutativity. For a triple the two bracketings give the sum of the same three charges; a unit branch adds zero, and symmetry permutes the paired outcomes. These calculations give the symmetric monoidal comparison, fibrewise also in ordinary parameter contexts.

(3) **Erase, regrade and evaluate.** The projection $\epsilon_Z(c, z) = z$ is a monad morphism to the identity. A monoid map $j : C \to C'$ induces the monad morphism $T_j(c, z) = (j(c), z)$, by $j(0) = 0$ and $j(c + d) = j(c) + j(d)$. Each owner supplies an evaluator $e_i : T_C Z \to V_i$ with a comparison on $V_i$. An ordinary utility $u_i$ gives $e_i = u_i \epsilon_Z$, also on nonzero charges. An ordinary game with outcome map $z$ embeds by $\eta^T_Z z$; $\epsilon_Z \eta^T_Z = \mathrm{id}_Z$ recovers its outcomes. More generally, $e_i(c, z) = u_i(z) - \ell_i(c)$ for an additive price $\ell_i : C \to (\mathbb{K}, +)$. A product charge monoid allows each owner to price its own coordinates. An arbitrary evaluator can also assign different rankings at different accumulated charges.

**Construction 2.8** (Measured outcomes of completed executions)**.**

(1) **Retain the completed witness.** For a specified result interface, let $p : K \to \Sigma$ send each completed execution witness to its policy profile and let $a : K \to Z$ read its outcome. A measurement with values in a commutative monoid $(C, +, 0)$ gives an ordinary term $m : K \to C$. The measured outcome and its erasure are

$$r = z^{\mathrm{m}} = \langle m, a \rangle : K \longrightarrow T_C Z, \qquad \epsilon_Z r = a.$$

The fibre $K_\sigma$ can be empty or contain several executions. A witness of zero cost is still an element of this fibre.

(2) **Compare measured outcomes.** Give owner $i$ a strict improvement relation $P_i \hookrightarrow (T_C Z)^2$. A value presentation takes $P_i = (e_i \times e_i)^* \prec_i$, so the actual witness value is $v_i = e_i r$. The paired records of two witnesses are tested by $(z^{\mathrm{m}} \times z^{\mathrm{m}})^* P_i$. Taking $P_i = (\epsilon_Z \times \epsilon_Z)^* \prec_i$ recovers the original outcome comparison by composition of pullbacks. In the exact numerical instance, use $u_i : Z \to R_\pm$, a charge $m_i : K \to R_+$ read from a coordinate of $m$, and a price $\lambda_i \in R_+$. The net value is

$$v_i(k) = u_i(a(k)) - \eta(\lambda_i m_i(k)) \in R_\pm,$$

where $\eta : R_+ \hookrightarrow R_\pm$ is the canonical inclusion. An owner can compare records directly even when the charge has no price.

(3) **Restrict to a budget.** For $B \hookrightarrow C$ retain $K_B = K \times_C B$. The image of $a|_{K_B}$ consists of outcomes with an execution meeting the budget. Taking this image after the pullback permits two witnesses for the same outcome to have different costs. The budget is a predicate on the measurement and can be used independently of the owner's comparison.

**Words with more than one role.** The following distinctions will be maintained when the constructions meet.

| Term | Meaning in this paper |
|---|---|
| Agent | A persistent owner identity with a complete policy interface in an interaction; its situated record includes observations and comparison feedback. |
| Policy | An ordinary complete contingent plan. Its realized concurrent strategy is the corresponding arrow of the operational category. |
| State | A monoidal state is an arrow from the tensor unit. An economic reference state indexes a policy profile; a public state records a stage of a protocol. |
| Context | A state context indexes references, an operational context surrounds an interaction by wiring, and a comparison context supplies the outcome comparisons used for incentives. |
| Type | A dependent type is a family in the semantic model. An economic type indexes an agent's valuation or utility. A resource type names a kind of material occurrence. |
| Value | $\mathsf{Val}$ extracts states of a linear object. An economic valuation or utility instead evaluates desirability. |
| History | A concurrent history is a configuration. A tree history records a path of decisions or completed public rounds. |

**Definition 2.9** (Mixed strategies and probabilistic information)**.** For a finite set $S$ write

$$\mathrm{Dist}(S) = \left\{ \mu : S \to [0,1] : \sum_{s \in S} \mu(s) = 1 \right\}.$$

When $\Sigma_i$ is finite, a **mixed strategy** of owner $i$ has type $\mu_i \in \mathrm{Dist}(\Sigma_i)$, as opposed to a pure policy $\sigma_i \in \Sigma_i$. A finite **prior** on a type space $T$ is $\mu \in \mathrm{Dist}(T)$. For an observation $o : T \to O$ and $b \in O$ with $\sum_{o(t)=b} \mu(t) > 0$, its **posterior** has type $\mu_b \in \mathrm{Dist}(o^{-1}(b))$ and is given by

$$\mu_b(t) = \frac{\mu(t)}{\sum_{o(t')=b} \mu(t')} \quad (t \in o^{-1}(b)).$$

For $u : T \to \mathbb{R}$, the **expected utility** is $\mathbb{E}_\mu[u] = \sum_{t \in T} \mu(t) u(t)$. A finite **chance kernel** is a map $k : W \times \prod_i A_i \to \mathrm{Dist}(W)$. A **Bayesian game** additionally indexes agents' information and utilities by private types and a specified probability distribution; a **stochastic game** uses a state-dependent chance kernel for its transitions. For general measurable carriers the corresponding data are probability measures and measurable kernels, not the displayed finite sums. These probabilistic notions describe the boundary of the present pure, qualitative model; no probability structure is used in the constructions below.

The paper's policies are pure, and its belief models record possibilities without probabilities. The probabilistic structures in Definition 2.9 are vocabulary for the theory's boundary, not additional data assumed by the constructions. The carrier presentations in Convention 1.29 establish smallness of the presheaf indexing category. Policy parameters retain their ordinary set-theoretic meaning.

**Notation 2.10** (Operational and ordinary structure)**.** The symbol $A \multimap B$ denotes the internal hom of concurrent interaction interfaces. Ordinary predicates have implication $P \Rightarrow Q$. Policy parameters are ordinary objects; resource occurrences belong to the linear interfaces. Their categorical relationship is constructed in Section 1.2.

## 2.3 Owned agents and their situated interfaces

The policy-parameterized concurrent realization is now available, but an agent must also retain an identity across all of its controlled decisions. This section packages those decisions into one complete policy interface and connects it to the canonical selection relation. Legal choices and observations are displayed with their state contexts, so that neither evaluation nor information is mistaken for part of operational polarity.

### 2.3.1 Agent identities, complete policies, and situated interfaces

Selection enters as a comparison interface, not as a property of a concurrent event. The canonical relation below accepts a proposed policy when no available alternative is designated as an improvement. It retains all ties; the later feedback construction supplies the actual improvement predicate.

**Definition 2.11** (Selection at a comparison interface)**.** For a set $S$, a **selection relation** has the type

$$\varepsilon_S \hookrightarrow S \times \big(S \to \mathsf{Pred}(S)\big), \qquad \varepsilon_S(a,q) \iff \forall b \in S\ \neg q(a)(b).$$

Here $a$ is a state of $\mathbb{L}(S)$, and $q : S \to \mathsf{Pred}(S)$ is a comparison costate: $q(a)$ is the predicate of improving alternatives to $a$. The selection relation retains ties and does not choose a representative; for numerical utilities it expresses maximization among all available choices.

In the terminology of Definition 3.4, a **concurrent open game** consists of the typed owned arena $G = (\vec{\Sigma}, g) : A \to B$ with the canonical tuple $(\varepsilon_{\Sigma_i^G})_{i\in N}$ of such owner interfaces. An **evaluated concurrent open game** is $\big(G, (\varepsilon_{\Sigma_i^G})_i, P\big)$ with $P \in \mathsf{Cmp}_N(A,B)$. Connecting the agent interfaces to the pulled-back comparison costates constructs $\mathsf{BR}_G(P) \hookrightarrow (\Sigma^G)^2$ in Section 5.6. A numerical evaluation retains both its value maps and their induced comparison relations. This is the selection-relation interface of Capucci et al. (2022a) specialized to the paper's canonical qualitative feedback.

**Definition 2.12** (Agent interface and situated agent)**.** In an owned open game $G = (\vec{\Sigma}, g) : A \to B$, the **agent interface** of identity $i \in N$ is

$$\mathsf{Ag}_i(G) = (i, \Sigma_i^G, \varepsilon_{\Sigma_i^G}).$$

The complete policy carrier belongs to this occurrence of the interface; the identity $i$ is retained when occurrences are composed. Its contextual realization is

$$\alpha_i^G : \Sigma_{-i}^G \times \Sigma_i^G \longrightarrow \mathcal{C}(A,B), \qquad \alpha_i^G(u,a) = g_{(u,a)}.$$

An **agent process** is the owned open game connecting these interfaces through $g$. A **situated, evaluated agent presentation** consists of the process together with a legal comprehension $m : L \hookrightarrow \Sigma^G$, a reference map $s : W \to L$, observation maps $o_i : W \to O_i$, and a comparison context $P_i \hookrightarrow \mathcal{C}(A,B)^2$. The unrestricted case uses $L = \Sigma^G$. The ordinary reference context of the presentation is $ms$. The corresponding situated agent is the record

$$\big(\mathsf{Ag}_i(G), \alpha_i^G, o_i, P_i, m, s\big).$$

Its legal interface at $\sigma \in L$ is the dependent carrier

$$S_i^L(\sigma_{-i}) = \{a \in \Sigma_i^G : (\sigma_{-i}, a) \in L\},$$

with its own canonical selection relation and the restriction of $\alpha_i^G$. The outsider tuple indexes available alternatives; the observation map specifies the information available to the agent.

The owned policy parameter and the concurrent arrow $g_\sigma$ have different types and different roles. The former supplies an alternative; the latter executes that alternative. Selection, contextual comparison, and execution are consequently interpreted by different constructions, which are connected only after the evaluation context has been specified.

The triple $\mathsf{Ag}_i(G)$ records control and the canonical selection interface, but it does not by itself determine observation, objective, legality, or implementation. The situated presentation retains those fields so that knowledge, rationality, ability, and commitment can be posed as distinct predicates. In particular, failing a rationality or knowledge test does not remove the agent's control interface.

Over an ordinary parameter context $\Gamma$, the dependent version uses the same record in each fibre. Its implementation is the linear map $\widetilde{g} : \mathsf{Lin}_\Gamma\Sigma \otimes_D A \to B$; world-indexed observations and comparisons remain ordinary dependent terms. This makes the situated interface compatible with substitution and with the original representable arena fragment.

**Proposition 2.13** (Recovery of the owned presentation from agent interfaces)**.** *Bundling an owned open game and its displayed state, observation, comparison, and legal data into the preceding agent presentation, and then unpacking it, recovers the original data. Conversely, unpacking and rebundling recovers the agent presentation. These constructions preserve ownerwise maps, sequential and parallel composition of the underlying processes, and substitution of the ordinary base. Displayed state, observation, and comparison data are retained under the same context transport.*

*Proof.*

(1) **Secured operational interaction.** The interfaces retain the same indexed carriers $\Sigma_i$, and the process retains $g$. Its contextual realization is obtained by the canonical rearrangement $\Sigma_{-i} \times \Sigma_i \cong \Sigma$; evaluating it on the corresponding coordinates recovers $g$. The selection relation is the displayed predicate constructed from $\Sigma_i$.

(2) **Ownerwise alternatives.** All additional data remain the same maps $m, s, o_i, P_i$. Thus the two constructions are inverse in indexed family notation. An ownerwise map remains the same family of maps and satisfies the same realization equation.

(3) **Policy-level comparison.** Products of its coordinates give the same composition comparisons, and substitution selects the same fibre. In slice notation the comparison is the canonical equivalence between a displayed map and the disjoint union of its fibres. No policy representative is chosen.

□

### 2.3.2 Displayed ownership and the underlying linear object

**Definition 2.14** (Displayed agent signatures and judgments)**.** An agent signature over $\Gamma$ is a tuple $\mathfrak{s} = (N, \vec{\Sigma}, A, B)$ comprising a finite owner list, ordinary families $\vec{\Sigma} = (\Sigma_i)_{i\in N}$, and linear boundaries $A, B$. Write $\Sigma = \prod_i \Sigma_i$. The signature is part of the judgment:

$$\Gamma \vdash \mathfrak{s}\ \mathsf{AgSig}, \qquad \Gamma; \Delta \vdash a : \mathsf{Ag}_{\mathfrak{s}}.$$

Its underlying linear type, obtained by forgetting the display, is

$$\left|\mathsf{Ag}_{\mathfrak{s}}\right| = H_{\mathfrak{s}} := [\mathsf{Lin}\Sigma \otimes A, B].$$

Formally a displayed agent term is a pair $(\mathfrak{s}, t)$ with $\Gamma; |\Delta| \vdash t : H_{\mathfrak{s}}$. The notation $\mathsf{Ag}_{\vec{\Sigma}}(A, B)$ suppresses the other entries of $\mathfrak{s}$. Unpacking forgets the tag and returns $t$; packing $t$ with a

specified signature returns $(\mathfrak{s}, t)$. Packing and unpacking are definitionally inverse, and their equations are those of the underlying linear calculus. Every linear constructor acts on the underlying terms; a resulting agent is tagged by an explicitly constructed signature.

Ordinary substitution transports both an agent signature and its term, whereas linear substitution transports the term with its signature fixed. Consequently, conversion of displayed agent types must respect the owner factors. An isomorphism of underlying hom objects cannot by itself say whether two coordinates are controlled by one owner or by two: the signatures $(2 \times 2)$ and $(2, 2)$ have isomorphic total profiles but different unilateral-deviation kernel pairs. Thus ownership cannot be recovered from the underlying object $H_{\mathfrak{s}}$.

An agent variable $a : \mathsf{Ag}_{\mathfrak{s}}$ occurs linearly in $\Delta$ and can be packed with its specified signature. Because an agent term is linear, another process may receive it as an input without acquiring permission to duplicate its occurrences. The owner-indexed display has a different purpose: it specifies which reparametrizations count as strategic maps. An underlying arrow between agent objects need not preserve those coordinates, and repacking a term with a different display is not a strategic morphism. The category of elements in Theorem 2.19 characterizes the realization-preserving ownerwise maps. Regrouping and delegation then give explicit changes of display, while higher-order linear terms use agents as resources.

### 2.3.3 Introduction, execution, and extensionality

The derived introduction and execution rules are

$$\frac{\Gamma, \sigma : \Sigma; \Delta, x : A \vdash t : B}{\Gamma; \Delta \vdash \Lambda\sigma.\lambda x.t : \mathsf{Ag}_{\vec{\Sigma}}(A, B)},$$

$$\frac{\Gamma; \Delta \vdash a : \mathsf{Ag}_{\vec{\Sigma}}(A, B) \quad \Gamma \vdash \sigma : \Sigma \quad \Gamma; \Theta \vdash x : A}{\Gamma; \Delta, \Theta \vdash a[\sigma](x) : B}.$$

In the first rule, $\Delta, A, B$ are pulled back along the policy parameter. The introduction first forms the underlying linear term

$$\lambda z.\, \mathsf{let}\ u \otimes x = z\ \mathsf{in}\ \big(\mathsf{let}\ \mathsf{return}(\sigma) = u\ \ \mathsf{in}\ t\big),$$

where $u : \mathsf{Lin}\Sigma$ and the ordinary variable $\sigma$ is bound in the second elimination. The specified signature tag is then attached. Execution unpacks the agent and supplies $\mathsf{return}(\sigma) \otimes x$ to the internal hom. Neither operation changes the ownership tag implicitly. The equations are

$$(\Lambda\sigma.\lambda x.t)[\tau](u) = t[\tau/\sigma, u/x], \tag{2.2}$$

$$\Lambda\sigma.\lambda x.a[\sigma](x) = a. \tag{2.3}$$

The second equation follows from the uniqueness equations for internal hom, tensor, and $\mathsf{Lin}$ elimination applied to the displayed nested term. The $\beta$-equation is elimination after introduction and the $\eta$-equation is the corresponding universal-property uniqueness. Both are equations of open terms, so no separation of arrows by global points is invoked.

**Theorem 2.15** (Agent abstraction is a dependent linear product)**.** *Let $p : \Gamma.\Sigma \to \Gamma$ be the ordinary comprehension of the complete policy family. In the indexed linear–nonlinear semantics of Section 1.3, there is a substitution-natural isomorphism*

$$[\mathsf{Lin}_{\Gamma}\Sigma \otimes A, B] \cong \Pi^{\ell}_{p}[p^{*}A, p^{*}B]. \tag{2.4}$$

*Its transposition is precisely agent introduction, and its inverse is policy-indexed application. Hence the displayed agent $\beta\eta$ equations hold on open linear contexts, without using global points to separate arrows or postulating an independent agent abstraction rule.*

*Proof.*

(1) **Frobenius and coproduct comparison.** For every test object $X$ of $\mathcal{L}(\Gamma)$, closure, Proposition 1.47, Frobenius, the left indexed adjunction, fibrewise closure, and the right indexed adjunction give a natural chain

$$\begin{aligned}
&\mathcal{L}(\Gamma)(X, [\mathsf{Lin}\Sigma \otimes A, B]) \\
&\cong \mathcal{L}(\Gamma)(X \otimes \mathsf{Lin}\Sigma \otimes A, B) \\
&\cong \mathcal{L}(\Gamma)(\Sigma^{\ell}_{p} p^{*}(X \otimes A), B) \\
&\cong \mathcal{L}(\Gamma.\Sigma)(p^{*}(X \otimes A), p^{*}B) \\
&\cong \mathcal{L}(\Gamma.\Sigma)(p^{*}X, [p^{*}A, p^{*}B]) \\
&\cong \mathcal{L}(\Gamma)(X, \Pi^{\ell}_{p}[p^{*}A, p^{*}B]).
\end{aligned}$$

Reindexing preserves tensor by construction and internal hom by Lemma 1.48.

(2) **Representation and Yoneda decoding.** Yoneda therefore gives the displayed isomorphism. The first and last transpositions send a body $\Gamma, \sigma : \Sigma; \Delta, x : A \vdash t : B$ to its agent abstraction and back to its application. The triangle identities and the closed-structure $\beta\eta$ equations prove (2.2)–(2.3) even for open terms.

(3) **Base-change comparison.** Each comparison is an indexed adjunction comparison, so Beck–Chevalley makes the isomorphism natural under ordinary substitution.

□

The indexed adjunctions identify agent abstraction with dependent currying.

$$\begin{array}{ccc}
\mathsf{L}(\Gamma)(X, [\mathsf{Lin}\Sigma \otimes A, B]) & \xrightarrow{\cong} & \mathsf{L}(\Gamma)(X \otimes \mathsf{Lin}\Sigma \otimes A, B) \\
\downarrow{\scriptstyle \cong} & & \downarrow{\scriptstyle \cong} \\
\mathsf{L}(\Gamma)(X, \Pi^{\ell}_{p}[p^{*}A, p^{*}B]) & \xrightarrow[\cong]{} & \mathsf{L}(\Gamma.\Sigma)(p^{*}X, [p^{*}A, p^{*}B])
\end{array}$$

**Proposition 2.16** (Agents as additive families of interactions)**.** *The following is the copower–closedness comparison specialized to the canonical model, rather than a property peculiar to event structures. In the canonical model constructed in Section 1.3, there is a natural isomorphism*

$$[\mathsf{Lin}\Sigma \otimes A, B] \cong \prod_{\sigma \in \Sigma} [A, B].$$

*The product is an additive product in the linear category.*

*Proof.*

(1) **Fibrewise additive-product representation.** By Theorem 2.15, the left-hand object is $\Pi^{\ell}_{p}[p^{*}A, p^{*}B]$. At $\gamma$ its fibre is, by the constructed indexed product formula, precisely $\prod_{\sigma \in \Sigma_{\gamma}} [A_{\gamma}, B_{\gamma}]$. Naturality and substitution stability are those of the dependent-product comparison, so the result is a natural isomorphism of linear families.

□

*Remark* 2.17 (Agent transformations and strategic maps). A linear arrow between agent types is a transformation of agent resources. An ownerwise reparametrization is a more specific piece of displayed data: it changes the policy parameters in a way that identifies which owner controls each change. The category of underlying linear types alone does not recover this distinction. In particular, an isomorphism of total policy sets does not identify their coalition structures.

### 2.3.4 Open agents

An open term $\Gamma; \Delta \vdash a : \mathsf{Ag}_{\vec{\Sigma}}(A, B)$ denotes, at each ordinary parameter $\gamma$, an arrow

$$\bigotimes \llbracket \Delta \rrbracket_\gamma \longrightarrow [\mathsf{Lin}\Sigma_\gamma \otimes A_\gamma, B_\gamma].$$

Its uncurried form is

$$\bigotimes \llbracket \Delta \rrbracket_\gamma \otimes \mathsf{Lin}\Sigma_\gamma \otimes A_\gamma \longrightarrow B_\gamma.$$

The uncurried form accounts for the resources needed to supply an agent separately from the policy and interface supplied when it executes. Thus higher-order application does not erase the linear inputs on which an open agent term depends.

### 2.3.5 Exact representation of owned presentations

**Theorem 2.18** (Representation of agents)**.** *There is a natural bijection*

$$\mathcal{V}(\mathsf{I}, |\mathsf{Ag}_{\vec{\Sigma}}(yA, yB)|) \cong \mathbf{Set}(\Sigma, \mathcal{C}(A, B)).$$

*More generally, for every linear test object $T \in \mathcal{V}$ there is a natural open-context bijection*

$$\mathcal{V}(T, [\mathsf{Lin}\Sigma \otimes yA, yB]) \cong \prod_{\sigma \in \Sigma} \mathcal{V}(T \otimes yA, yB). \tag{2.5}$$

*These comparisons identify execution, sequential and parallel composition, ownerwise reparametrization, regrouping, and delegation with the corresponding constructions on owned concurrent realizations.*

*Proof.*

(1) **Frobenius and coproduct comparison.** Closedness and the coproduct description of $\mathsf{Lin}$ give, naturally in $T$ and in the policy set $\Sigma$,

$$\begin{aligned} \mathcal{V}(T, [\mathsf{Lin}\Sigma \otimes yA, yB]) &\cong \mathcal{V}(T \otimes \mathsf{Lin}\Sigma \otimes yA, yB) \\ &\cong \mathcal{V}\Big(\coprod_{\sigma \in \Sigma} (T \otimes yA), yB\Big) \\ &\cong \prod_{\sigma \in \Sigma} \mathcal{V}(T \otimes yA, yB). \end{aligned}$$

The middle comparison is an isomorphism because Day tensor preserves coproducts separately; the last is the coproduct universal property.

(2) **Representation and Yoneda decoding.** For $T = \mathsf{I}$, restricted Yoneda gives $\mathcal{V}(yA, yB) \cong \mathcal{C}(A, B)$ and hence the first assertion. For $r : \Theta \to \Sigma$, precomposition by $\mathsf{Lin}r \otimes 1$ sends the component at $\theta$ to the component at $r(\theta)$. For closed agents, sequential and parallel composition therefore have components $h_\tau g_\sigma$ and $g_\sigma \otimes h_\tau$.

(3) **Composition in open linear contexts.** For agents supplied from $T$ and $U$, the decoded components for sequential composition are $g_\sigma : T \otimes yA \to yB$ and $h_\tau : U \otimes yB \to yC$. Evaluation of the sequential composite on the $(\sigma, \tau)$-summand is the arrow

$$(U \otimes T) \otimes yA \xrightarrow{\alpha} U \otimes (T \otimes yA) \xrightarrow{1_U \otimes g_\sigma} U \otimes yB \xrightarrow{h_\tau} yC.$$

For parallel composition take $k_\tau : U \otimes yA' \to yB'$. The component on the same summand rearranges $(T \otimes U) \otimes (yA \otimes yA')$ into $(T \otimes yA) \otimes (U \otimes yA')$ and applies $g_\sigma \otimes k_\tau$. These formulas result by precomposing the defining evaluation maps with the two policy injections. The coproduct universal property recovers the entire open map from these components. Precomposition in either test object commutes with the displayed rearrangements by tensor naturality. Taking $U = T = \mathsf{I}$ recovers the closed formulas through the unitors.

(4) **Ownerwise alternatives.** Ownerwise regrouping and delegation apply the same named-coordinate substitutions on both sides; the comparison maps obey coherence by ordinary-product and symmetric-monoidal coherence.

(5) **Policy-level comparison.** The equality $g = h \prod_i r_i$ defining a realization-preserving 2-cell becomes exactly the equality between the corresponding reparametrized agent states.

□

Policy indexing, currying, and Yoneda give an exact representation of realized agents.

$$\begin{array}{ccc} \mathcal{V}(\mathsf{I}, [\mathsf{Lin}\Sigma \otimes yA, yB]) & \xrightarrow{\cong} & \mathcal{V}(\mathsf{Lin}\Sigma \otimes yA, yB) \\ \downarrow{\scriptstyle \cong} & & \downarrow{\scriptstyle \cong} \\ \mathbf{Set}(\Sigma, \mathcal{C}(A, B)) & \xrightarrow[\cong]{} & \prod_{\sigma \in \Sigma} \mathcal{V}(yA, yB) \end{array}$$

**Theorem 2.19** (Equivalence on owned presentations)**.** *For fixed original arenas $A, B$ and owner list $N$, form the presheaf of agent states on $\mathbf{Set}^N$,*

$$F_{A,B}(\vec{\Sigma}) = \mathcal{V}(\mathsf{I}, [\mathsf{Lin}(\prod_i \Sigma_i) \otimes yA, yB]), \qquad F_{A,B}(r)(b) = b[r].$$

*Let $\mathsf{StateAg}_N(A, B) = \int F_{A,B}$ be its category of elements. Its objects are $(\vec{\Sigma}, a)$ with the retained owner display; an arrow $(\vec{\Sigma}, a) \to (\vec{T}, b)$ is an ownerwise map $r_i : \Sigma_i \to T_i$ satisfying $a = b[r]$. Then execution and restricted Yoneda define an isomorphism of categories*

$$\mathsf{StateAg}_N(A, B) \;\cong\; \mathbf{Arena}_N(\mathcal{C})(A, B).$$

*These isomorphisms are compatible with sequential and parallel composition, identity presentations, and their ownerwise associativity and symmetry comparisons. They also commute with ordinary base substitution.*

*Proof.*

(1) **Representation and Yoneda decoding.** For $g : \Sigma \to \mathcal{C}(A, B)$, copair the arrows $yg_\sigma : yA \to yB$:

$$\mathsf{Lin}\Sigma \otimes yA \cong \coprod_{\sigma \in \Sigma} yA \xrightarrow{[yg_\sigma]_\sigma} yB.$$

Currying and tagging this map constructs the agent state $a_g$.

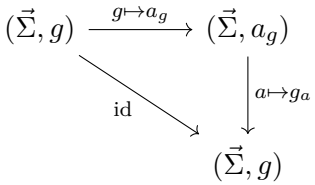
(2) **Reverse implication.** Conversely, evaluate an agent state $a$ on every coproduct injection, then decode the resulting map $yA \to yB$ by Yoneda; call the family $g_a$. The coproduct and closed-structure $\beta\eta$ equations give $g_{a_g} = g$ and $a_{g_a} = a$, without relying on global-point extensionality.

(3) **Ownerwise alternatives.** For an ownerwise map $r$, the equation $a = b[r]$ holds exactly when $g_a(\sigma) = g_b(r(\sigma))$ for every profile, which is precisely the original presentation 2-cell condition and the category-of-elements arrow condition. In particular arbitrary arrows of the underlying linear category are not thereby declared strategic maps. The execution equations (3.1)–(3.2) then identify horizontal and tensor composition.

(4) **Ownerwise alternatives: completion.** Each higher coherence map has the same ownerwise coordinate projections and the same concurrent realizations on both sides. The ordinary and symmetric-monoidal coherence laws prove compatibility with all comparison maps. Pointwise execution also commutes with ordinary substitution.

□

Copairing and componentwise execution are inverse correspondences on owned presentations.




**Proposition 2.20** (Interpretation of open terms)**.** *Every core derivation determines a morphism* $\bigotimes[\![\Delta]\!] \to [\![A]\!]$ *in* $\mathcal{L}([\![\Gamma]\!])$*. Definitional equality gives equality of morphisms. Ordinary and linear substitution are interpreted by reindexing and composition.*

*Proof.*

(1) **Frobenius and coproduct comparison.** Interpret context splitting by tensor, abstraction and application by closedness, additives by products and coproducts, the ordinary–linear rules by $\mathsf{Lin} \dashv \mathsf{Val}$ together with Proposition 1.47, and indexed binders by (1.3). Theorem 1.46 makes ordinary substitution commute with these operations.

(2) **Fixed-point argument.** The induction of Lemma A.1 interprets linear substitution as composition. Each $\beta$ equation is an evaluation equation of one of these universal properties; each $\eta$ equation is its uniqueness equation. Commuting conversions express naturality. Agent constructors are the specified composites of these operations.

□

*Remark* 2.21 (Linearization and physical data). The functor $\mathsf{Lin}$ organizes ordinary alternatives in the completion. The physical finite-data functor is $D : \mathbf{FinSet} \to \mathcal{C}$. Its collector $D(X) \otimes D(Y) \to D(X \times Y)$ waits for both inputs and belongs to a lax monoidal structure; it is generally not an isomorphism. We retain $D$ for communication implementations and $\mathsf{Lin}$ for the ordinary–linear adjunction. Identifying them would erase a real synchronization barrier.

# Part II
# Composition, control, and game-theoretic evaluation

## 3 The Compositional Calculus of Agents

When two agents are wired together, which causal process results, and which decisions may the same owner change in it? Composition and tensor answer the first question. Ordinary products collect each owner's contingent plans, as in parameterized open games (Ghani et al., 2018; Atkey et al., 2021). Their compatibility gives a symmetric monoidal calculus and a bicategory of owned presentations. Retaining the owner factors matters even in a two-decision process, where a single owner may gain by changing both decisions together.

### 3.1 Linear agent operations and ownerwise composition

#### 3.1.1 The two composition operations

For signatures on a common owner set, define $(\vec{\Sigma}\times\vec{T})_i = \Sigma_i \times T_i$. The canonical rearrangement identifies its total profile with $\Sigma \times T$. Unused owners have singleton policy carriers; disjoint owner lists can therefore be combined without identifying owners.

**Construction 3.1** (Agent composition)**.**
(1) **Form the linear operations.** The following are linear operations:

$$\mathsf{seq} : \mathsf{Ag}_{\vec{\Sigma}}(A,B) \otimes \mathsf{Ag}_{\vec{T}}(B,C) \longrightarrow \mathsf{Ag}_{\vec{\Sigma}\times\vec{T}}(A,C),$$
$$\mathsf{par} : \mathsf{Ag}_{\vec{\Sigma}}(A,B) \otimes \mathsf{Ag}_{\vec{T}}(A',B') \longrightarrow \mathsf{Ag}_{\vec{\Sigma}\times\vec{T}}(A\otimes A', B\otimes B').$$

(2) **Compute their executions.** Their defining executions are

$$(a\,;b)[\sigma,\tau](x) = b[\tau](a[\sigma](x)), \tag{3.1}$$
$$(a \parallel b)[\sigma,\tau](x\otimes y) = a[\sigma](x)\otimes b[\tau](y). \tag{3.2}$$

(3) **Supply the identity.** The identity agent has singleton policies and executes $x \mapsto x$.

The first operation consumes each supplied agent once. The second splits both the agent resources and their inputs. These are comparison maps between types; they are not asserted to be isomorphisms. An arbitrary composite agent need not admit a decomposition into two independently supplied agents.

**Theorem 3.2** (Coherence of agent composition)**.** *Sequential composition is associative and unital under the canonical ownerwise product bijections. Parallel composition is coherently symmetric monoidal. Sequential and parallel composition satisfy interchange:*

$$(a \parallel b)\,;(c \parallel d) \cong (a\,;c) \parallel (b\,;d),$$

*where the comparison regroups the complete policy coordinates.*

*Proof.*

(1) **Associativity.** For associativity, agent abstraction followed by (2.2) reduces both sides, at the same named policy coordinates, to

$$c[v](b[\tau](a[\sigma](x))).$$

The unit calculation reduces to $a[\sigma](x)$. The tensor comparison is obtained by reassociating and exchanging the linear inputs and the ordinary profile factors.

(2) **Interchange and higher coherence.** For interchange, both executions reduce to

$$c[v](a[\sigma](x)) \otimes d[\omega](b[\tau](y)).$$

Agent $\eta$ then gives the open-term equations. In every higher coherence diagram, the ordinary maps have the same coordinate projections and the linear maps are the coherence maps of the symmetric monoidal calculus. The product and monoidal coherence laws therefore give the required commuting diagrams.

□

### 3.1.2 Ownerwise policy substitution

For ordinary maps $r_i : \Theta_i \to \Sigma_i$, write $r = \prod_i r_i$ and set

$$a[r][\theta](x) = a[r(\theta)](x).$$

This defines a linear map $\mathsf{Ag}_{\vec{\Sigma}}(A, B) \to \mathsf{Ag}_{\vec{\Theta}}(A, B)$, implemented by precomposition with $\mathsf{Lin}r \otimes 1_A$. Computation proves

$$a[\mathrm{id}] = a, \qquad a[r][s] = a[r \circ s],$$

and

$$(a \,;\, b)[r \times q] = a[r] \,;\, b[q], \qquad (a \parallel b)[r \times q] = a[r] \parallel b[q].$$

Here $r \times q$ is formed ownerwise and then flattened.

At the level of declared presentations, an ownerwise map $r : a \Rightarrow b$ has components $r_i : \Sigma_i^a \to \Sigma_i^b$ satisfying $a[\sigma] = b[r(\sigma)]$. Vertical composition is function composition; horizontal and parallel composition take ownerwise products. Together with the preceding coherence maps these presentations form the owned symmetric monoidal bicategory recovered in Section 1.3.

**Example 3.3** (Shared policy and duplicated execution)**.** Two occurrences controlled by one owner initially have policy carrier $\Sigma \times \Sigma$. Precomposition with the ordinary diagonal $\Sigma \to \Sigma \times \Sigma$ makes them use the same policy description. Both linear occurrences must still be supplied. The diagonal ties parameters; it does not construct a linear diagonal on the agent type.

## 3.2 Open arenas, ownership and contextual wiring

The internal-hom construction defines composition on agent types. We now compute that operation on the original concurrent arenas. Full faithfulness of Yoneda identifies each component with a concurrent strategy there. This gives a wiring map along which a later comparison can evaluate the complete policy of each owner.

A comparison can depend on several ports at once. Taking its inverse image along the complete wiring map preserves that dependence. The owner coordinates then determine which joint changes count as unilateral deviations. This applies the contextual treatment of open games in Ghani et al. (2018); Atkey et al. (2021); Capucci et al. (2022b); Hedges (2020) to the concurrent category.

### 3.2.1 The bicategory of open arenas

**Definition 3.4** (Open arenas and open games)**.** An **owned open arena** on a linear interaction boundary $(A, B)$ is the data

$$G = (\vec{\Sigma}, g) : A \to B, \qquad \vec{\Sigma} \in \mathbf{Par}_N, \qquad g : \Sigma^G = \prod_{i \in N} \Sigma_i^G \longrightarrow \mathcal{C}(A, B).$$

Its owned policy families parameterize the concurrent realizations; at this stage it carries no preference. A **concurrent open game** adds the prescribed owner-selection interfaces

$$\varepsilon_{\Sigma_i^G} \hookrightarrow \Sigma_i^G \times (\Sigma_i^G \to \mathsf{Pred}(\Sigma_i^G)), \qquad \varepsilon_S(a, q) \iff \forall b \in S\ \neg q(a)(b),$$

whose construction is developed in Definition 2.11. An **evaluated concurrent open game** has in addition a comparison context $P = (P_i)_{i \in N}$ of type

$$P \in \mathsf{Cmp}_N(A, B) = \prod_{i \in N} \mathsf{Pred}(\mathcal{C}(A, B)^2).$$

Connecting this context to the owner interfaces determines the best-response relation $\mathsf{BR}_G(P) \hookrightarrow (\Sigma^G)^2$. The outcome interface can retain the entire exposed concurrent interaction, with scalar evaluations and material allocations as particular observations.

**Definition 3.5** (Reparametrisations of owned open arenas)**.** For owned open arenas $G = (\vec{\Sigma}^G, g)$ and $G' = (\vec{\Sigma}^{G'}, g')$ on the same linear interaction boundary $(A, B)$, a **reparametrisation** or 2-cell $f : G \Rightarrow G'$ is an ownerwise map $f_i : \Sigma_i^G \to \Sigma_i^{G'}$ satisfying $g = g' \Pi_N f$. An element of $\Sigma_i^G$ is a complete policy of owner $i$. A presentation is **owned by** $i$ if the other policy sets are singletons, and **nonstrategic** if every policy set is a singleton. The backward interface of an owned open arena is the canonical predicate feedback constructed from its realization below, not additional unspecified data.

**Construction 3.6** (Horizontal and parallel composition)**.**

(1) **Compose in sequence.** For $G : A \to B$ and $H : B \to C$, put

$$\Sigma_i^{H \circ G} = \Sigma_i^G \times \Sigma_i^H, \qquad (h \circ g)_{(\sigma,\tau)} = h_\tau \circ g_\sigma.$$

(2) **Compose in parallel.** For $G : A \to B$ and $K : A' \to B'$, put

$$\Sigma_i^{G \otimes K} = \Sigma_i^G \times \Sigma_i^K, \qquad (g \otimes k)_{(\sigma,\rho)} = g_\sigma \otimes k_\rho.$$

Use the canonical product rearrangements throughout.

(3) **Act on maps and identities.** Both operations on 2-cells are ownerwise products. The identity at $A$ has singleton policy sets and realizes $\mathrm{id}_A$.

**Theorem 3.7** (Bicategory of concurrent open arenas)**.** *These operations define a symmetric monoidal bicategory* $\mathbf{Arena}_N(\mathcal{C})$*. Its hom-category at* $(A, B)$ *is the category of elements of the presheaf*

$$F_{A,B} : \mathbf{Par}_N^{\mathrm{op}} \to \mathbf{Set}, \qquad F_{A,B}(\vec{\Sigma}) = \mathbf{Set}(\Pi_N \vec{\Sigma}, \mathcal{C}(A, B)).$$

*The quotient by invertible* 2*-cells is a symmetric monoidal category* $\mathsf{Arenalso}_N(\mathcal{C})$*.*

*Proof.*

(1) **Functoriality and substitution.** The category-of-elements description is exactly the equation defining a 2-cell. The comparison $\Pi_N(\vec{\Sigma} \times \vec{\Theta}) \cong \Pi_N \vec{\Sigma} \times \Pi_N \vec{\Theta}$ is natural. It therefore makes the displayed horizontal operations functorial on hom-categories.

(2) **Ownerwise alternatives.** The associator and unitors are the canonical ownerwise product bijections; their realization equations are the associativity and unit equations in $\mathcal{C}$. Their pentagon and triangle hold coordinatewise.

(3) **Ownerwise alternatives: second comparison.** The tensor compositor is the ownerwise permutation between the parameter orders of $(H \otimes H') \circ (G \otimes G')$ and $(H \circ G) \otimes (H' \circ G')$. Its realization equation is interchange in $\mathcal{C}$. The monoidal associator, unitors, and symmetry have singleton parameters and realize those of $\mathcal{C}$; their naturality 2-cells are the corresponding product rearrangements.

(4) **Monoidal comparison.** In each coherence diagram both parameter maps send every named coordinate to the same coordinate. The realized arrows agree by symmetric monoidal coherence in $\mathcal{C}$ (Selinger, 2011). Since a 2-cell is determined by its parameter maps, this verifies the coherence diagrams themselves.

(5) **Passage to isomorphism classes.** Finally, invertible 2-cells are exactly ownerwise bijections; the coherence maps become equalities on their isomorphism classes.

□

Ownerwise reparametrization commutes with evaluation of the composite realization.

$$\begin{array}{ccc} \Sigma^G \times \Sigma^H & \xrightarrow{r \times s} & \Sigma^{G'} \times \Sigma^{H'} \\ \Big\downarrow{\scriptstyle h \circ g} & & \Big\downarrow{\scriptstyle h' \circ g'} \\ \mathcal{C}(A, C) & = & \mathcal{C}(A, C) \end{array}$$

$$\begin{array}{ccc} \Sigma^G \times \Sigma^H & \xrightarrow{g \times h} & \mathcal{C}(A, B) \times \mathcal{C}(B, C) \\ \Big\downarrow{\scriptstyle \cong} & & \Big\downarrow{\scriptstyle \circ} \\ \Sigma^{H \circ G} & \xrightarrow[h \circ g]{} & \mathcal{C}(A, C) \end{array}$$

**Construction 3.8** (Reindexing and the nonstrategic embedding)**.**

(1) **Reindex an owned arena.** For an ownerwise map $f : \vec{\Theta} \to \vec{\Sigma}$ define

$$f^* G = (\vec{\Theta}, g \Pi_N f). \tag{3.3}$$

It carries the canonical 2-cell $f : f^*G \Rightarrow G$.

(2) **Embed an ordinary arrow.** Define

$$\mathbf{j}_N : \mathcal{C} \to \mathbf{Arena}_N(\mathcal{C}), \qquad u \longmapsto (\vec{1}, \mathrm{const}_u). \tag{3.4}$$

Its structure maps are the ownerwise bijections $1 \times 1 \cong 1$.

**Proposition 3.9** (Reindexing and nonstrategic recovery)**.** *Reindexing is functorial and gives the cartesian arrows of each category-of-elements projection to* $\mathbf{Par}_N$*. The functor* $\mathbf{j}_N$ *is locally fully faithful and its image is biequivalent to the nonstrategic sub-bicategory.*

*Proof.*

(1) **Ownerwise alternatives.** The cartesian universal property is substitution of the equation $g = g' \Pi_N f$. Between two singleton-parameter presentations there is one possible ownerwise map, and it is a 2-cell exactly when their realized arrows are equal. Every family of singleton policy sets is uniquely isomorphic to $\vec{1}$.

□

*Remark* 3.10 (Reference sharing and policy restriction). Reindexing a family of reference states to a diagonal does not change its space of alternative policies. To enforce reuse of a policy in two components, use the ownerwise diagonal $\Sigma_i \to \Sigma_i \times \Sigma_i$ in Construction 3.8, or construct the corresponding legal subobject. These are changes to the available policy presentation; ordinary substitution of the reference state alone is not such a change.

### 3.2.2 Joint feedback for concurrent wiring

When the same owner controls two components, a change in one can alter the value of a change in the other. Pulling the comparison back along the entire wiring function tests both choices in the same composite execution.

**Construction 3.11** (Multi-input coplay)**.**

(1) **Wire the interaction maps.** Let $H_j = \mathcal{C}(A_j, B_j)$, and let $m : \prod_{j=1}^{r} H_j \to H$ be a wiring function constructed using composition, tensor, fixed concurrent arrows, and their coherent reassociations. Apply $\mathbb{L}$ to $m$.

(2) **Transport comparisons.** For a comparison $P \hookrightarrow H^2$ its transported costate is the comparison

$$m^\sharp P((f_j)_j, (f'_j)_j) \quad \Longleftrightarrow \quad P\big(m((f_j)_j), m((f'_j)_j)\big). \tag{3.5}$$

The input continuation interface is $\mathbb{L}(\prod_j H_j)$: feedback is a predicate on joint alternatives.

(3) **Compute the coplay of the composite.** For component arenas $G_j$ with plays $g_j$, their wired play is $m \circ \prod_j g_j$, followed by the ownerwise parameter rearrangement. Its coplay is the canonical coplay of that composite function.

**Theorem 3.12** (Canonical coplay and open-arena coherence)**.** *Wiring substitution commutes with backward feedback. In particular,*

$$(mn)^\sharp = n^\sharp m^\sharp, \qquad (m \circ \textstyle\prod_j g_j)^\sharp = (\prod_j g_j)^\sharp m^\sharp.$$

*The sequential, parallel, unit, symmetry, and reparametrisation laws of open arenas hold for both play and coplay. Every arena retains its owner-indexed policy sets and its complete concurrent play map.*

*Proof.*

(1) **Equality of the two constructions.** The displayed equations are $((mn) \times (mn))^* = (n \times n)^*(m \times m)^*$. For sequential composition the two arguments of a transported comparison are $(h_\tau g_\sigma, h_{\tau'} g_{\sigma'})$. For a triple composite the two bracketings give

$$P(k_\upsilon(h_\tau g_\sigma), k_{\upsilon'}(h_{\tau'} g_{\sigma'})) = P((k_\upsilon h_\tau) g_\sigma, (k_{\upsilon'} h_{\tau'}) g_{\sigma'}).$$

Associativity in $\mathcal{C}$ proves equality.

(2) **Ownerwise alternatives.** Interchange, units, and symmetry are handled by their equations in $\mathcal{C}$ on both arguments. The ownerwise permutations are exactly those of Theorem 3.7. For a reparametrisation $f$, the equation $g = g'\Pi_N f$ yields

$$(g \times g)^* P = (\Pi_N f \times \Pi_N f)^* (g' \times g')^* P.$$

Thus all coherence diagrams commute also on costates.

(3) **Recovery of canonical coplay.** Finally, forgetting the canonical backward map recovers the same $g$ and the same policy coordinates, while Theorem 5.11 reconstructs that backward map uniquely by the displayed construction.

□

Ownerwise reparametrization commutes with joint wiring.

$$\begin{array}{ccc} \prod_j \Sigma_j & \xrightarrow{\prod_j r_j} & \prod_j \Theta_j \\ {\scriptstyle m\circ\prod_j g_j}\downarrow & & \downarrow{\scriptstyle m\circ\prod_j h_j} \\ H & = & H \end{array}$$

$$\begin{array}{ccc} (\Sigma^G \times \Sigma^H)^2 & \xrightarrow{(g\times h)^2} & (H_{A,B} \times H_{B,C})^2 \\ {\scriptstyle (h\circ g)^2}\downarrow & & \downarrow{\scriptstyle m^2} \\ H_{A,C}^2 & = & H_{A,C}^2 \end{array}$$

The owner grading records precisely the policy factors preserved by composition. When an owner acts in more than one component, its composite coordinate is the product of those decisions. The graded-arrow structure below packages parameter-indexed hom-families with their unit, composition, strength, and reindexing; its coherence says that these operations commute with substitution and concurrent tensor.

**Proposition 3.13** (The owner grading)**.** *The family*

$$\mathcal{A}_{\vec{\Sigma}}(A, B) = \mathbf{Set}(\Pi_N \vec{\Sigma}, \mathcal{C}(A, B))$$

*is a commutative* $\mathbf{Par}_N^{\mathrm{op}}$*-graded arrow over* $\mathcal{C}$*: its unit is the singleton family, composition uses ownerwise products, reparametrisation is precomposition, and strength sends* $g_\sigma$ *to* $g_\sigma \otimes \mathrm{id}_Z$*.*

*Proof.*

(1) **Naturality.** Naturality in the grade is substitution. Graded associativity and the unit equations are the corresponding equations of $\mathcal{C}$ after the canonical product comparisons of $\Pi_N$. The strength laws are functoriality and associativity of tensor.

(2) **Ownerwise alternatives.** The two ways of forming parallel composition agree by interchange. These are precisely the graded-arrow equations of Atkey et al. (2021), verified here for the constructed owner grading. Taking its category of elements retains the reparametrisations and yields the bicategory already constructed.

□

Owner grading respects the two ways of inserting a fixed interaction.

$$\begin{array}{ccc} \mathcal{A}_{\vec{\Sigma}}(A, B) & \xrightarrow{-\otimes \mathrm{id}_Z} & \mathcal{A}_{\vec{\Sigma}}(A \otimes Z, B \otimes Z) \\ {\scriptstyle r^*}\downarrow & & \downarrow{\scriptstyle r^*} \\ \mathcal{A}_{\vec{\Theta}}(A, B) & \xrightarrow[-\otimes \mathrm{id}_Z]{} & \mathcal{A}_{\vec{\Theta}}(A \otimes Z, B \otimes Z) \end{array}$$

The grading extends the agency and contextual-wiring constructions of Capucci et al. (2022b,a); Hedges (2020). It matters because feedback must evaluate a repeated owner's *complete* policy. For an owner acting in $G$ and $H$, the forward carrier is $\Sigma_i^G \times \Sigma_i^H$ and the backward carrier is $\mathsf{Pred}(\Sigma_i^G \times \Sigma_i^H)$. The next step fixes the other owners' coordinates to obtain the individual costate, as in Section 5.6.

### 3.2.3 Composition of contexts and their logical interfaces

**Construction 3.14** (Sequential and parallel context composition)**.**

(1) **Compose the state contexts.** For $G : A \to B$, $H : B \to C$, use the canonical owner rearrangement $\Sigma^{H \circ G} \cong \Sigma^G \times \Sigma^H$ to define

$$\mathsf{St}(G) \times \mathsf{St}(H) \longrightarrow \mathsf{St}(H \circ G), \qquad (W \xrightarrow{s} \Sigma^G, V \xrightarrow{t} \Sigma^H) \longmapsto (W \times V \xrightarrow{s \times t} \Sigma^{H \circ G}).$$

Its realization is $(w, v) \mapsto h_{t(v)} g_{s(w)}$.

(2) **Form the parallel context.** For parallel arenas use the same product context and realization $(w, v) \mapsto g_{s(w)} \otimes k_{t(v)}$.

(3) **Combine its predicates.** Predicates have an external conjunction $P \boxtimes Q = \pi_W^* P \wedge \pi_V^* Q$, while the complete fibre is $\mathsf{Pred}(W \times V)$, including arbitrary joint predicates.

**Theorem 3.15** (Fibred coherence of concurrent wiring)**.** *The context-composition functors preserve cartesian lifts jointly in their inputs. External conjunction commutes with these lifts. Their associativity, units, symmetry, and interchange comparisons satisfy the corresponding bicategorical coherence laws. For any concurrent wiring $m : \prod_j H_j \to H$ and realizations $a_j : W_j \to H_j$,*

$$(m \circ \textstyle\prod_j a_j)^* Q = (\prod_j a_j)^* m^* Q, \qquad (m \circ \prod_j a_j)^\sharp P = (\prod_j a_j)^\sharp m^\sharp P.$$

*Proof.*

(1) **Monoidal comparison.** For maps to $X$ and $Y$ there is a natural bijection

$$(S \times_X W) \times (T \times_Y V) \cong (S \times T) \times_{X \times Y} (W \times V).$$

Both sides consist of tuples $(s, t, w, v)$ satisfying the same two equations. It identifies the product of the cartesian lifts in Theorem 5.3 with the cartesian lift of their composite. Inverse image commutes with conjunction and projections, proving the predicate assertion.

(2) **Coherence.** The coherence maps rearrange the same named coordinates. Their realizations agree by associativity and interchange in $\mathcal{C}$, so the coherence diagrams commute in the tower itself. The final equations are functoriality of inverse image, applied once to the realization and once to its square.

□

Cartesian lifts are compatible with products of independently wired state contexts.

$$\begin{array}{ccc} (S \times_X W) \times (T \times_Y V) & \xrightarrow{\cong} & (S \times T) \times_{X \times Y} (W \times V) \\ \downarrow & & \downarrow \\ S \times T & = & S \times T \end{array}$$

*Remark* 3.16 (Whole joint feedback). The source interface for a product of carriers is $\mathbb{L}(X \times Y) = (X \times Y, \mathsf{Pred}(X \times Y))$. The ordinary lens tensor instead has backward carrier $\mathsf{Pred}(X) \times \mathsf{Pred}(Y)$; it is not identified with the former. In particular $\mathbb{L}$ is faithful but not strong monoidal for the cartesian tensor on sets and the Cartesian-product tensor on lenses. The constructed context fibre keeps whole joint feedback until ownership determines which alternatives are varied. This implements the distinction between joint and partially evaluated feedback described by Capucci (2023). The graded-arrow organization remains the one proved in Proposition 3.13; fibrational lenses and related variants are developed in Atkey et al. (2021, Section 6).

### 3.2.4 Composition of complete agent interfaces

**Proposition 3.17** (The complete-policy composition law)**.** *For composable agent processes $G : A \to B$ and $H : B \to C$, and for $K : A' \to B'$, the agent of identity $i$ has interfaces*

$$\mathsf{Ag}_i(H \circ G) = \big(i, \Sigma_i^G \times \Sigma_i^H, \varepsilon_{\Sigma_i^G \times \Sigma_i^H}\big),$$
$$\mathsf{Ag}_i(G \otimes K) = \big(i, \Sigma_i^G \times \Sigma_i^K, \varepsilon_{\Sigma_i^G \times \Sigma_i^K}\big).$$

*The realizations are $h_\tau g_\sigma$ and $g_\sigma \otimes k_\rho$. The identity process has singleton agent policies. These operations and ownerwise realization-preserving reparametrizations form the agent calculus of $\mathbf{Arena}_N(\mathcal{C})$, with its symmetric monoidal coherence. They are stable under substitution in the dependent model.*

*Proof.*

(1) **Ownerwise alternatives.** The parameter family of a composite is the ownerwise product by Proposition 1.45. Applying Definition 2.12 gives the displayed interfaces. Three sequential components have the same flattened policy tuple $(\sigma_i, \tau_i, \upsilon_i)$ for each $i$, and both bracketings realize $k_\upsilon h_\tau g_\sigma$.

(2) **Ownerwise alternatives: second comparison.** Unit comparisons remove singleton coordinates; tensor comparisons reassociate and permute named coordinates. Interchange holds because it holds in $\mathcal{C}$. Products of ownerwise maps preserve these equations.

(3) **Dependent substitution of the composite.** Thus every coherence diagram is the corresponding product coordinate diagram together with the same realized concurrent arrow. The dependent construction evaluates these formulas at each base index, so substitution commutes with all comparisons.

□

The composite realization is obtained by joining the two complete-policy profiles.

$$\begin{array}{ccc} \Sigma^{H\circ G} & \xrightarrow{\ \cong\ } & \Sigma^G \times \Sigma^H \\ {\scriptstyle g^{H\circ G}}\downarrow & & \downarrow{\scriptstyle g\times h} \\ \mathcal{C}(A,C) & \xleftarrow[\circ]{} & \mathcal{C}(A,B) \times \mathcal{C}(B,C) \end{array}$$

Section 5.6 uses the comparison on the whole composite to calculate best responses. For a single owner with payoff $u(x, y) = xy$, neither bit alone can improve on $(0, 0)$. Changing the complete policy to $(1, 1)$ raises the payoff. The ownerwise product makes this a unilateral deviation.

Reparametrization and reference substitution must remain distinct. The map $r_i : \Theta_i \to \Sigma_i$ changes the implemented policy presentation by precomposition; substitution of a state

reference changes which policy is currently being tested but leaves the alternative carrier intact. The transport laws in Theorem 5.65 make this difference precise.

### 3.2.5 The semantic composition map on open agent resources

Let $G = [\mathsf{Lin}\Sigma \otimes A, B]$ and $H = [\mathsf{Lin}T \otimes B, C]$. The uncurried sequential operation is

$$G \otimes H \otimes \mathsf{Lin}(\Sigma \times T) \otimes A \cong H \otimes \mathsf{Lin}T \otimes G \otimes \mathsf{Lin}\Sigma \otimes A \xrightarrow{1\otimes 1\otimes \mathrm{ev}} H \otimes \mathsf{Lin}T \otimes B \xrightarrow{\mathrm{ev}} C.$$

Currying gives the map of Section 3.1. For the parallel operation, rearrange

$$G \otimes H \otimes \mathsf{Lin}(\Sigma \times T) \otimes A \otimes A'$$

into the tensor of $G \otimes \mathsf{Lin}\Sigma \otimes A$ and $H \otimes \mathsf{Lin}T \otimes A'$, evaluate each, and curry. These descriptions prove naturality on arbitrary test objects and open linear contexts. They do not reduce equality of morphisms to equality only on global states.

**Policy substitution for a measured presentation.** Over $p : K \to \Sigma$, an ownerwise reparametrization with total map $\theta : \Theta \to \Sigma$ gives the witness carrier $K_\theta = \Theta \times_\Sigma K$. Its measured record and owner value are $r\pi_K$ and $e_i r\pi_K$. At $\vartheta$ the projection identifies its entire witness fibre with $K_{\theta(\vartheta)}$. Successive reparametrizations compose by flattening these pullback tuples. In particular values that descend to policies pull back by $\theta$. For a legal deviation, the further comparison in Lemma 4.11 lifts the alternative at its specified source; the fibre identification then transports its measured records. This construction reindexes the given execution. A physically different implementation has its own secured witnesses.

# 4 Control, Coalitions, and Delegation

Which decisions can a group change while outsiders keep their policies fixed? Project a profile onto the outsiders' coordinates. Its kernel pair consists of the profile pairs that agree there. We follow these pairs through regrouping and delegation. Regrouping assigns existing decisions to a common controller. Delegation supplies subordinate policies that implement a parent's choices, and the comparison is exact on the implemented image. The relation to multiplicative and additive control in Cartesian frames (Garrabrant, Herrmann, and Lopez-Wild, 2021) becomes explicit below.

## 4.1 Complete-policy alternatives and their composition

**Definition 4.1** (Counterfactual policy alternatives)**.** For a coalition $C \subseteq N$ of an open arena $G = (\vec{\Sigma}, g) : A \to B$, its **counterfactual alternative span** is the kernel pair of the outsider projection:

$$\Sigma^G \xleftarrow{p_1} D_C = \Sigma^G \times_{\Sigma^G_{-C}} \Sigma^G \xrightarrow{p_2} \Sigma^G, \qquad \mathsf{Alt}_C(\sigma) = p_1^{-1}(\sigma).$$

The counterfactual realization and its comparison tests have types

$$\delta^G_C = (gp_1, gp_2) : D_C \to \mathcal{C}(A, B)^2, \qquad (\delta^G_C)^* P_i \in \mathsf{Pred}(D_C).$$

Thus a reference and its alternative retain identical outsider policies. The deviation relates a reference policy to its available counterfactual alternatives. The following construction gives the projections in ownerwise coordinates and their compositional comparison maps.

These kernel pairs support both existential and universal deviation modalities. In either case the reference policy remains fixed while the quantifier ranges over alternatives obtained by changing the coalition's coordinates. No temporal move is implied by the word "counterfactual."

**Construction 4.2** (Deviation kernel pairs)**.**

(1) **Form the kernel pair.** For $\Sigma = \prod_i \Sigma_i$, write $\Sigma_C = \prod_{i \in C} \Sigma_i$. Define

$$D_C = \Sigma \times_{\Sigma_{-C}} \Sigma, \qquad p_1, p_2 : D_C \rightrightarrows \Sigma, \tag{4.1}$$

using the outsider projection $\pi_{-C}$. Put $D_i = D_{\{i\}}$.

(2) **Identify its coordinates.** The coordinate bijection is

$$\chi_C : \Sigma_{-C} \times \Sigma_C \times \Sigma_C \xrightarrow{\cong} D_C, \quad (u, a, b) \longmapsto ((u, a), (u, b)). \tag{4.2}$$

(3) **Realize both alternatives.** For $G = (\vec{\Sigma}, g)$ its counterfactual realization is

$$\delta_C^G = (gp_1, gp_2) : D_C \to \mathcal{C}(A, B)^2. \tag{4.3}$$

The first leg retains the reference profile and the second its alternative.

$$\begin{array}{ccc} D_C & \xrightarrow{p_2} & \Sigma \\ {\scriptstyle p_1}\downarrow & \lrcorner & \downarrow{\scriptstyle \pi_{-C}} \\ \Sigma & \xrightarrow[\pi_{-C}]{} & \Sigma_{-C} \end{array} \qquad \begin{array}{ccc} D_C & \xrightarrow{(p_1,p_2)} & \Sigma^2 \\ & {\scriptstyle \delta_C^G}\searrow & \downarrow{\scriptstyle g\times g} \\ & & \mathcal{C}(A,B)^2 \end{array}$$

**Construction 4.3** (Counterfactualization)**.**

(1) **Duplicate the controlled coordinates.** Define $\mathsf{Dup}_C : \mathbf{Par}_N \to \mathbf{Par}_N$ by squaring the coordinates in $C$, leaving the others unchanged, and acting on maps by $f_i \times f_i$ or $f_i$ respectively. Identify $\Pi_N \mathsf{Dup}_C \vec{\Sigma}$ with $D_C$ through $\chi_C$.

(2) **Act on arenas and maps.** On objects put $\mathsf{CF}_C(A) = (A, A)$; on open arenas put

$$\mathsf{CF}_C(G) = (\mathsf{Dup}_C \vec{\Sigma}, \delta_C^G) \quad \text{in } \mathbf{Arena}_N(\mathcal{C} \times \mathcal{C}).$$

On 2-cells use $\mathsf{Dup}_C f$.

(3) **Compose the two realizations.** Write $\mathrm{Comp}((f_0, f_1), (h_0, h_1)) = (h_0 f_0, h_1 f_1)$ and define Ten coordinatewise.

**Theorem 4.4** (Compositional counterfactualization)**.** $\mathsf{CF}_C$ *is a symmetric monoidal pseudofunctor. In particular there is a canonical bijection* $\rho_C : D_C^{H \circ G} \cong D_C^G \times D_C^H$ *such that*

$$\delta_C^{H \circ G} = \mathrm{Comp}(\delta_C^G \times \delta_C^H) \rho_C. \tag{4.4}$$

*The tensor equation uses* Ten*.*

*Proof.*

(1) **Ownerwise alternatives.** The strong product comparison of $\mathsf{Dup}_C$ at a coalition coordinate is the bijection

$$((a, b), (a', b')) \longmapsto ((a, a'), (b, b')).$$

Outside the coalition it is the identity product comparison. These comparisons and $1 \times 1 \cong 1$ are natural and satisfy symmetric monoidal coherence, since every composite has the same action on named coordinates. They induce $\rho_C$.

(2) **Policy-level comparison.** Both realizations in the displayed equation send a tuple of reference and alternative policies to $(h_\tau g_\sigma, h_{\tau'} g_{\sigma'})$. A realization-preserving 2-cell stays realization-preserving after duplication. The pseudofunctor and monoidal coherence equations reduce to these product comparisons and the equations in $\mathcal{C} \times \mathcal{C}$.

□

$$\begin{array}{ccc} D_C^{H\circ G} & \xrightarrow[\cong]{\rho_C} & D_C^G \times D_C^H \\ {\scriptstyle \delta_C^{H\circ G}}\downarrow & & \downarrow{\scriptstyle \delta_C^G \times \delta_C^H} \\ \mathcal{C}(A,C)^2 & \xleftarrow[\text{Comp}]{} & \mathcal{C}(A,B)^2 \times \mathcal{C}(B,C)^2 \end{array}$$

**Corollary 4.5** (Compositional counterfactual tests)**.** *For a predicate $P \hookrightarrow \mathcal{C}(A,C)^2$,*

$$(\delta_C^{H\circ G})^* P = \rho_C^* (\delta_C^G \times \delta_C^H)^* \,\mathrm{Comp}^*\, P.$$

*Proof.*

(1) **Functoriality and substitution.** Apply functoriality of pullback to Equation (4.4).

□

The dependent reading of $D_C$ takes $p_1$ as the reference map and the second projection as the alternative. Its paired realization is a term into a constant comparison type. Thus the composition result above is also a substitution law in the state-context doctrine of Theorem 3.15: changing a reference preserves the entire dependent family of alternative policies.

## 4.2 Collective control and regrouping of agent interfaces

A coalition may be treated as one controller without altering the component policies it controls. Regrouping collects its coordinates into a single owner factor, leaving the realized interaction unchanged. This changes the deviation relation, not the members' preferences: collective evaluation is still supplied by a separate comparison. The construction therefore distinguishes common control from a common objective.

The coalition-deviation relation $D_C$ was defined in Equation (4.1) as the kernel pair of $\pi_{-C} : \Sigma \to \Sigma_{-C}$. Regrouping will identify it with the unilateral-deviation relation of the new collective owner, without changing the underlying profile tuple.

**Proposition 4.6.** *On a full product profile space,*

$$D_C ; D_D = D_{C \cup D}.$$

*Proof.*

(1) **Composition of coalition changes.** Two successive changes in $C$ and $D$ preserve every coordinate outside $C \cup D$. Conversely, if $\sigma$ and $\tau$ agree outside $C \cup D$, let $\rho$ have $\tau$'s coordinates on $C$ and $\sigma$'s coordinates elsewhere. Then $\sigma D_C \rho$ and $\rho D_D \tau$. The construction also covers overlap between $C$ and $D$.

□

**Construction 4.7** (Regrouping by right Kan extension)**.**

(1) **Take the right Kan extension.** Regard an owner family $\vec{\Sigma}$ as a functor

$$\Sigma : N_{\mathrm{disc}} \longrightarrow \mathbf{Set}.$$

For a surjection of owner sets $c : N \twoheadrightarrow J$, regarded as a functor between the corresponding discrete categories, define

$$c_*\Sigma = \mathrm{Ran}_c\, \Sigma.$$

Because the categories are discrete, the right Kan extension is the finite product over each fiber:

$$(c_*\Sigma)_j = \prod_{i \in c^{-1}(j)} \Sigma_i. \tag{4.5}$$

(2) **Flatten its fibres.** There is a canonical flattening bijection

$$\kappa_\Sigma : \prod_{j \in J} (c_*\Sigma)_j \xrightarrow{\cong} \prod_{i \in N} \Sigma_i. \tag{4.6}$$

**Theorem 4.8** (Owner regrouping)**.** *Every surjection $c : N \twoheadrightarrow J$ induces a symmetric-monoidal pseudofunctor*

$$c_* : \mathbf{Arena}_N(\mathcal{C}) \longrightarrow \mathbf{Arena}_J(\mathcal{C})$$

*which is the identity on arenas. For an open arena $G = (\vec{\Sigma}, g) : A \to B$ its regrouping has parameter family $c_*\vec{\Sigma}$ and realization*

$$g^c = g \circ \kappa_\Sigma.$$

*For a 2-cell $f : G \Rightarrow G'$, its $j$th component is*

$$(c_*f)_j = \prod_{i \in c^{-1}(j)} f_i.$$

*Under the flattening bijection, deviation by the new owner $j$ is exactly deviation by the old coalition $c^{-1}(j)$:*

$$D_j^{c_*G} \cong D^G_{c^{-1}(j)}.$$

*For another surjection $d : J \twoheadrightarrow K$ there is a canonical pseudonatural isomorphism*

$$d_*c_* \cong (d \circ c)_*.$$

*Proof.*

(1) **Naturality.** Naturality of the flattening bijections gives

$$\kappa_{\Sigma'} \circ \prod_j (c_*f)_j = \left( \prod_i f_i \right) \circ \kappa_\Sigma.$$

Hence, if $g = g' \prod_i f_i$, then

$$g\kappa_\Sigma = g'\kappa_{\Sigma'} \prod_j (c_*f)_j,$$

so $c_*f$ is realization-preserving.

(2) **Monoidal comparison.** The right Kan extension is strong monoidal for the pointwise Cartesian products used in owned composition and tensor. Its component is the canonical bijection

$$\prod_{i \in c^{-1}(j)} (\Sigma_i \times \Theta_i) \cong \left( \prod_{i \in c^{-1}(j)} \Sigma_i \right) \times \left( \prod_{i \in c^{-1}(j)} \Theta_i \right),$$

and it sends the singleton owner family to the singleton owner family.

(3) **Monoidal comparison: second comparison.** These maps supply the compositor and unitor of the induced pseudofunctor. Their coherence is Cartesian-product coherence. Since composition and tensor of realized interactions are evaluated after flattening the same parameter tuple, the required realization equations are respectively associativity and interchange in $\mathcal{C}$. The same strong-monoidal maps give compatibility with tensor and symmetry.

(4) **Ownerwise alternatives.** For the deviation statement, forgetting the new $j$-coordinate from $\prod_{k\in J}(c_*\Sigma)_k$ forgets, after $\kappa_\Sigma$, precisely the old coordinates indexed by $c^{-1}(j)$. The two forgetful maps therefore commute up to the canonical bijections, and their kernel pairs are canonically isomorphic. This yields $D_j^{c_*G} \cong D^G_{c^{-1}(j)}$.

(5) **Iterated right Kan extension.** Finally, the component at $k \in K$ of $d_*c_*\Sigma$ is

$$\prod_{j\in d^{-1}(k)} \prod_{i\in c^{-1}(j)} \Sigma_i,$$

canonically isomorphic to

$$\prod_{i\in(d\circ c)^{-1}(k)} \Sigma_i = ((d\circ c)_*\Sigma)_k.$$

These reassociations are natural and satisfy the pseudofunctorial coherence because every composite is the same flattening of the same owner-indexed tuple.

□

Flattening preserves realized arrows while identifying a regrouped owner with its original coalition.

$$\begin{array}{ccc} \prod_{j\in J}(c_*\Sigma)_j & \xrightarrow[\cong]{\kappa_\Sigma} & \prod_{i\in N}\Sigma_i \\ & \searrow^{g^c} & \downarrow g \\ & & \mathcal{C}(A,B) \end{array}$$

Regrouping gives the coalition authority to change a block of policies. To rank the changes, it still needs a collective comparison, such as a welfare order or a Pareto rule. Delegation poses a further question. Which subordinate decisions actually implement the parent's choice?

### 4.3 Delegation and substitution of agent interfaces

Regrouping reallocates control over existing coordinates; delegation must additionally say how subordinate policies implement each parent choice. The implementation maps therefore precede any comparison of preferences. This follows the separation of agency from selection in Capucci et al. (2022b) and makes the distinction from Cartesian frame subagents explicit (Garrabrant, Herrmann, and Lopez-Wild, 2021). The main issue is which parent alternatives remain available after the subordinate implementation is fixed.

**Construction 4.9** (Substitution of controlled policies)**.**

(1) **Specify the implementation maps.** Let $G = (\vec{\Sigma}, g) : A \to B$ have finite owner set $N$. A delegation presentation consists of a finite set $J$, a surjection $c : J \twoheadrightarrow N$, subordinate policy carriers $\Theta_j$, and maps

$$h_i : \prod_{j\in c^{-1}(i)} \Theta_j \longrightarrow \Sigma_i.$$

Write $\Theta = \prod_{j \in J} \Theta_j$.

(2) **Compose the controlled realizations.** Regrouping and the product of the $h_i$ give $h : \Theta \to \Sigma$. The substituted process

$$G[h] = (\vec{\Theta}, gh) : A \to B$$

is owned by $J$.

(3) **Pull back the legal domain.** If $L \hookrightarrow \Sigma$ is the legal domain, its substituted legal domain is $L_h = h^{-1}(L)$. The construction retains the implementation maps $h_i$: an unimplemented parent policy is not silently made available to the delegates.

**Theorem 4.10** (Associative delegation and compatibility with composition)**.** *For $d : K \twoheadrightarrow J$ and $k_j : \prod_{\ell \in d^{-1}(j)} \Xi_\ell \to \Theta_j$, define*

$$(h \star k)_i = h_i \circ \left( \prod_{j \in c^{-1}(i)} k_j \right) \circ \kappa_i,$$

*where $\kappa_i$ groups the $\Xi$-coordinates over $(cd)^{-1}(i)$ by their intermediate owner $j$. Then*

$$(G[h])[k] \cong G[h \star k], \qquad (L_h)_k = L_{h \star k}.$$

*Identity implementation maps are units, and three successive substitutions agree under the canonical product reassociations.*

*Suppose also that $H : B \to C$ has owner set $N$, and that $r_i : \prod_{j \in c^{-1}(i)} \Lambda_j \to \Sigma_i^H$ implements its policies using the same owner map $c$. Define $h \boxtimes r$ at owner $i$ by the canonical map*

$$\prod_{j \in c^{-1}(i)} (\Theta_j \times \Lambda_j) \longrightarrow \left( \prod_{j \in c^{-1}(i)} \Theta_j \right) \times \left( \prod_{j \in c^{-1}(i)} \Lambda_j \right) \xrightarrow{h_i \times r_i} \Sigma_i^G \times \Sigma_i^H.$$

*There are canonical realization-preserving comparisons*

$$(H \circ G)[h \boxtimes r] \cong H[r] \circ G[h].$$

*The same assertion holds for tensor whenever the boundaries are those of a tensor product. In particular, delegation respects the complete-policy composition law without identifying the subordinate owners.*

*Proof.*

(1) **Monoidal comparison.** On a subordinate tuple $\xi$, the $i$th parent coordinate of either iterated implementation is $h_i((k_j((\xi_\ell)_{\ell \in d^{-1}(j)}))_{j \in c^{-1}(i)})$. Thus the realizations agree after the displayed product bijection. Inverse images compose, giving the legal-domain equality.

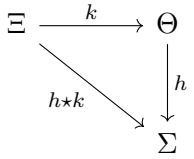
(2) **Monoidal comparison: second comparison.** For three levels, both bracketings evaluate the same nested coordinate functions; the product reassociations are coherent because they have identical coordinate projections. Identity maps leave every coordinate unchanged.

(3) **Composition.** For sequential composition, both sides have, after rearrangement, the $J$-indexed carrier $(\Theta_j \times \Lambda_j)_j$. Their value at $(\theta, \lambda)$ is $r_H(\lambda) \circ r_G(\theta)$, where $r_G = gh$ and $r_H$ is the realization of $H$ composed with $r$. For tensor the common value is $r_G(\theta) \otimes r_H(\lambda)$.

(4) **Witness and legality comparison.** The comparison maps are products of coordinate rearrangements, so preserve realizations, identities, and the associativity and symmetry comparisons. If legal domains on a composite are specified by an additional comprehension, pulling that comprehension back through either equal implementation map gives the same legal domain. No independence of that additional rule is required.

□

Associative delegation expresses equality of the iterated implementation maps.




**Lemma 4.11** (Ownerwise lifting of legal alternatives)**.** *Let $r_i : T_i \twoheadrightarrow S_i$ be surjections over a finite owner set $N$, put $r = \prod_i r_i$, and let $L \hookrightarrow S = \prod_i S_i$ be any legal domain. With $K = r^{-1}(L)$, for every coalition $C \subseteq N$ the canonical map over $K$*

$$\lambda_C : D_C^K \longrightarrow K \times_{L,p_1} D_C^L, \qquad (x, x') \longmapsto (x, (r(x), r(x')))$$

*is surjective. For any $g : S \to H$, it preserves both realized endpoints under the factorized realization $g \circ r$. Writing $g_L = g|_L : L \to H$ and $(gr)_K = (gr)|_K : K \to H$, and setting $\delta_C^{g,L} = (g_L p_1, g_L p_2)$ and $\delta_C^{gr,K} = ((gr)_K p_1, (gr)_K p_2)$, for every comparison $Q \hookrightarrow H^2$ one consequently has an equality of predicates on $K$:*

$$r|_K^* \exists_{p_1 : D_C^L \to L} (\delta_C^{g,L})^* Q = \exists_{p_1 : D_C^K \to K} (\delta_C^{gr,K})^* Q. \tag{4.7}$$

*Proof.*

(1) **Lifting along the surjection.** A point of the target is $x \in K$ and $y \in L$ agreeing with $r(x)$ outside $C$. Keep $x'_i = x_i$ for $i \notin C$. For each $i \in C$, surjectivity of $r_i$ provides $x'_i \in T_i$ with $r_i(x'_i) = y_i$.

(2) **Witness and legality comparison.** The finitely many coordinate witnesses assemble to $x' \in T$. Then $r(x') = y \in L$, so $x' \in K$, and $x'$ agrees with $x$ outside $C$. Hence $(x, x') \in D_C^K$ maps to $(x, (r(x), y))$.

(3) **Ownerwise alternatives.** The case $C = \varnothing$ uses $x' = x$, including empty owner lists. Both endpoints are preserved because the delegated realization is $g \circ r$ by construction. To prove (4.7), fix $x \in K$.

(4) **Witness and legality comparison.** The predicate on the right has a witness $x'$ exactly when $x D_C^K x'$ and $Q(grx, grx')$. Such a witness maps through $\lambda_C$ to the target alternative $(rx, rx')$, giving the left predicate.

(5) **Reverse implication.** Conversely a witness $y \in L$ for the left predicate is a target point $(x, (rx, y))$ of $\lambda_C$; the explicit coordinate construction above lifts it to $x'$ at that *same* reference $x$. The two realized endpoints agree, so $Q(grx, grx')$ holds. This gives equality of the existential predicates; applying negation and intersecting the specified coalitional tests gives exact stability transport.

□

Starting from the given subordinate profile, the lift realizes the proposed legal parent alternative with every outsider block fixed.

$$\begin{array}{ccc} D_C^K & \xrightarrow{\lambda_C} & K \times_L D_C^L \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle \pi_1} \\ K & = & K \end{array}$$

**Theorem 4.12** (Exact strategic comparison with implemented parent policies)**.** *For the preceding delegation, put $T_i = \mathrm{im}(h_i)$, $T = \prod_i T_i$, and $L_T = L \cap T$. Restrict the parent process to $T$ and regroup the subordinate process by $c$. Write $\Theta^c = \prod_{i \in N} \prod_{j \in c^{-1}(i)} \Theta_j$ for the regrouped profile carrier, identified with $\Theta$ by flattening. The induced map*

$$\bar{h} : \Theta^c \longrightarrow T$$

*is ownerwise surjective and takes $L_h$ onto $L_T$. For every coalition $I \subseteq N$, every $\theta \in L_h$, and every $t' \in L_T$ agreeing with $\bar{h}(\theta)$ outside $I$, there is $\theta' \in L_h$ agreeing with $\theta$ outside the blocks indexed by $I$ and satisfying $\bar{h}(\theta') = t'$. Realizations of both the reference and the alternative are preserved.*

*Consequently, writing $\bar{h}_L : L_h \twoheadrightarrow L_T$ for its legal restriction and pulling the same parent comparisons back through realization,*

$$\mathsf{NE}_{c_* G[h],\, L_h} = \bar{h}_L^* \mathsf{NE}_{G|_T,\, L_T}.$$

*The same equality holds for stability against any specified family of parent coalitions, using the corresponding whole blocks of delegates and the same coalition comparisons.*

*Proof.*

(1) **Lifting along the surjection.** Corestricting each implementation map $h_i$ to its image $T_i$ produces an ownerwise surjection $\bar{h} : \Theta^c \twoheadrightarrow T$. Its inverse-image legal domain is exactly $L_h = \bar{h}^{-1}(L_T)$. Lemma 4.11 now gives the required lift of every legal $I$-alternative, keeping the entire outsider blocks fixed and preserving both concurrent realizations.

(2) **Reverse implication.** Conversely, every legal subordinate block deviation maps to a legal parent deviation. Thus an improving deviation exists at a subordinate reference if and only if one exists at its implemented parent reference. Universal negation over singleton parent owners proves the displayed Nash equality; the same comparison for each designated coalition proves the final stability statement.

□

The image factorization retains the exact implemented parent policies.

$$\begin{array}{ccc} L_h & \xrightarrow{\bar{h}_L} & L_T \\ \downarrow & & \downarrow \\ \Theta^c & \xrightarrow{\bar{h}} & T \end{array}$$

The image factorization answers that issue even when the delegation maps $h_i$ are not surjective onto the original parent carriers: every implemented parent alternative has a subordinate lift with the outsider blocks fixed. Collective authority is nevertheless distinct from a collective preference. If delegates share their parent's comparison, whole-block stability implies individual stability, but not conversely. For two delegate bits $x, y \in \{0,1\}$, with $h(x,y) = (x,y)$ and common value $xy$, the profile $(0,0)$ has no profitable isolated change, while the block improves to $(1,1)$. Thus exact parent comparison requires whole-block deviations.

**Proposition 4.13** (The Cartesian frame underlying a control interface)**.** *On an unrestricted profile carrier, every coalition $C$ determines the Cartesian frame*

$$\mathsf{Frame}_C(G) = \Big(\Sigma_C, \Sigma_{-C}, (a,e) \longmapsto g_{(a,e)}\Big)$$

*with outcome set $\mathcal{C}(A,B)$. If $C \subseteq D$, write $X = \Sigma_C$, $Y = \Sigma_{D\setminus C}$, and $Z = \Sigma_{-D}$. The two frames have the form*

$$(X, Y \times Z, p), \qquad (X \times Y, Z, p)$$

*for the same three-variable outcome function. Thus their relation is exactly the multiplicative redistribution of control and environment used in Cartesian frames. Restricting the controlled carrier to $K \subseteq \Sigma_C$ instead gives the corresponding additive restriction.*

*Proof.*

(1) **Policy-level comparison.** Reassociate the profile product as $X \times Y \times Z$. For $C$, the $Y$-coordinates belong to the environment; for $D$, they belong to the controlled coordinates. Both evaluation maps are the same $g$ after this reassociation. Restricting the first coordinate of the evaluation to $K$ gives its additive restriction.

□

A permission rule may couple a coalition's decisions to an outsider's choice. The available actions then form a family $S_C^L(e)$ indexed by that choice. This is where the comparison with a rectangular Cartesian frame reaches its limit. A Chu-style map also reverses the environment coordinate, whereas an owned map here preserves a specified implementation. To pass from joint control to a corporate judgement or commitment requires the corresponding aggregation or update data, as in List and Pettit (2011).

Legal comprehension turns this distinction into a dependent interface and supplies the legal context $(L, m)$. For $m : L \hookrightarrow \Sigma$, the kernel pair of the outsider map gives $D_C^L = L \times_{L_{-C}} L$; its two legs select a legal reference and a legal alternative term in $\Delta_G L$. The following dependent choice spaces are the fibres of that span, so coupled legal opportunities are not replaced by unrestricted products.

## 4.4 Displayed coalition and delegation operations

### 4.4.1 Counterfactual alternatives in agent syntax

For $C \subseteq N$, the unrestricted deviation is $D_C = \Sigma \times_{\Sigma_{-C}} \Sigma$, as constructed in Definition 4.1 and Theorem 4.4. On a legal domain $L \hookrightarrow \Sigma$, its actual alternatives are instead

$$D_C^L = L \times_{\Sigma_{-C}} L. \tag{4.8}$$

Their paired realization is $(a[\sigma], a[\tau])$. The coordinate comparison in Theorem 4.4 gives $D_C^{\vec{\Sigma}\times\vec{T}} \cong D_C^{\vec{\Sigma}} \times D_C^{\vec{T}}$, including the sequential and parallel agent presentations. Likewise Proposition 4.6 proves $D_C; D_D = D_{C\cup D}$ on the unrestricted profile product. For legal carriers the intermediate profile of that proof need not be legal; legal relations compose by their actual pullback and image, not by replacing them with unrestricted products.

### 4.4.2 Collective agents by regrouping

**Construction 4.14** (Collective control)**.**

(1) **Regroup the policy factors.** Let $c : N \twoheadrightarrow J$ be a surjection. Define

$$(c_*\vec{\Sigma})_j = \prod_{i \in c^{-1}(j)} \Sigma_i, \qquad \kappa_{\vec{\Sigma}} : \prod_{j \in J} (c_*\vec{\Sigma})_j \xrightarrow{\cong} \Sigma.$$

(2) **Execute the regrouped agent.** The regrouped agent $c_*a$ has execution $(c_*a)[\theta](x) = a[\kappa_{\vec{\Sigma}}(\theta)](x)$. Its underlying linear type is canonically isomorphic to that of $a$, while its displayed control interface has owner set $J$.

**Theorem 4.15** (Collective composition and coalition deviations)**.** *Regrouping is compatible with sequential and parallel agent composition and with ownerwise maps. It satisfies*

$$d_*c_*a \cong (d \circ c)_*a, \qquad D_j^{c_*a} \cong D^a_{c^{-1}(j)}.$$

*The deviation comparison also holds on the transported legal domain.*

*Proof.*

(1) **Ownerwise alternatives.** Apply the regrouping pseudofunctor of Theorem 4.8 through the owned-presentation isomorphism of Theorem 2.19. Its compositor at new owner $j$ is explicitly

$$\prod_{ci=j} (\Sigma_i \times T_i) \cong \left( \prod_{ci=j} \Sigma_i \right) \times \left( \prod_{ci=j} T_i \right).$$

Both routes have identical policy projections and concurrent realizations.

(2) **Descent along the kernel pair.** Forgetting $j$ forgets exactly the old $c^{-1}(j)$ coordinates; the kernel-pair comparisons restrict along any transported legal inclusion.

□

The kernel-pair comparison preserves the deviations controlled by each regrouped owner.

$$\begin{array}{ccc} D_j^{c_*a} & \xrightarrow{\cong} & D^a_{c^{-1}(j)} \\ {\scriptstyle p_2}\downarrow\downarrow{\scriptstyle p_1} & & {\scriptstyle p_2}\downarrow\downarrow{\scriptstyle p_1} \\ \Sigma^{c_*a} & \xrightarrow[\cong]{\kappa_\Sigma} & \Sigma^a \end{array}$$

A coalition $C$ is represented by the partition with block $C$ and singleton outsiders. The empty coalition has unit policy carrier and identity alternatives. Overlapping coalitions are represented by their different projections $\Sigma \to \Sigma_{-C}$; they need not be blocks of a single partition. Thus the calculus retains both nested collective agents and overlapping coalitional tests.

Regrouping determines joint control. Collective observation is constructed in Sections 6.5.4 and 9.3; collective evaluation is constructed in Section 5.8. The original owner list and regrouping map remain displayed, so members' separate observations and comparisons are not lost.

### 4.4.3 Delegation as implementation of policies

For a parent owner set $N$, take subordinate owners $J$, a surjection $c : J \twoheadrightarrow N$, and maps

$$h_i : \prod_{cj=i} \Theta_j \longrightarrow \Sigma_i.$$

Flattening gives $h : \Theta \to \Sigma$. The delegated agent is $a[h]$, owned by $J$, with execution $a[h(\theta)]$ and legal domain $h^{-1}L$. For a further implementation $k$, define $h \star k$ by composing the coordinate maps after regrouping their subordinate inputs.

**Proposition 4.16** (Associative implementation)**.** *There are coherent comparisons*

$$(a[h])[k] \cong a[h \star k], \qquad (h^{-1}L) \text{ pulled back along } k = (h \star k)^{-1}L.$$

*Delegation commutes with sequential and parallel composition through the ownerwise product of implementation maps.*

*Proof.*

(1) **Functoriality and substitution.** The implementation maps $h \star k$ and the natural composition comparisons are those constructed in Theorem 4.10. The owned-presentation equivalence Theorem 2.19 preserves their explicit flattening maps and realized arrows. Legal domains agree by functoriality of inverse image, $(h \circ k)^{-1}L = k^{-1}(h^{-1}L)$.

□

**Theorem 4.17** (Exact implemented alternatives)**.** *Put $T_i = \mathrm{im}(h_i)$, $T = \prod_i T_i$, and $L_T = L \cap T$. The regrouped delegated legal carrier maps onto $L_T$. Every legal parent alternative in a coalition $C \subseteq N$ lifts to a change of exactly the subordinate blocks over $C$, with all other subordinate coordinates fixed.*

*Proof.*

(1) **Lifting along the surjection.** After regrouping delegates into parent blocks, each implementation corestricts to its image. The resulting ownerwise surjection has inverse-image legal domain $h^{-1}L$. The explicit lift in Lemma 4.11 fixes the outsider blocks and realizes the requested legal parent alternative.

(2) **Ownerwise alternatives.** Its converse takes the image of any legal block deviation, as in Theorem 4.12.

□

**Example 4.18** (Delegated choices and collective authority)**.** Two delegates choose bits $x, y$, with common outcome value $xy$. At $(0, 0)$ neither isolated delegate has a strictly improving change, but the whole block can change to $(1, 1)$. Consequently the exact strategic comparison in Theorem 4.17 is between parent coalitions and whole delegate blocks. The distinction is produced by the explicit projections defining alternatives.

# 5 Contextual Evaluation and Strategic Equilibrium

Knowing which policies an owner can change leaves open which changes it prefers. We compare two realizations, then pull that comparison back to their policies. Here *feedback* means this backward transport of an evaluation. Delivering a signal to a controller is a separate operation with an event and a recipient. An institution can therefore price a measured execution even when the owner never observes its charge. Applying a descended

comparison to the owner's legal alternatives gives the equilibrium test and recovers pure Nash equilibrium in a finite normal-form game. This use of feedback specializes the selection interface of categorical cybernetics (Capucci et al., 2022a).

### 5.1 Presentations, state contexts, and assertions

A policy specifies what to do. A reference state specifies the circumstances in which we evaluate that policy and may carry a history or an observation. The fibration below lets us change the reference and pull back assertions. It keeps the owner's policies available for the counterfactual comparison that defines coplay.

**Definition 5.1** (Economic state contexts)**.** A state context $s : W \to \Sigma^G$ is a family of reference profiles indexed by $W$.

Its points need not be policies themselves: an epistemic world may also carry information about which other worlds an agent considers possible. Several points of $W$ may consequently have the same profile. A predicate on $W$ asks a question at those references, while the full policy space still supplies alternatives. This use of state is distinct from a monoidal state $I \to A$.

**Definition 5.2** (The tower over a concurrent interface)**.** Put $\mathcal{A}_{A,B} = \mathbf{Arena}_N(\mathcal{C})(A, B)$. The profile functor $U_{A,B} : \mathcal{A}_{A,B} \to \mathbf{Set}$ sends $G$ to $\Sigma^G$ and an ownerwise reparametrisation $r$ to $\bar{r} = \Pi_N r$. The category $\mathsf{St}_{A,B}$ has objects $(G, s : W \to \Sigma^G)$ and arrows

$$(r, u) : (G, s : W \to \Sigma^G) \longrightarrow (H, t : V \to \Sigma^H), \qquad tu = \bar{r}s.$$

An object of $\mathsf{Assn}_{A,B}$ additionally carries a predicate $P \hookrightarrow W$. An arrow over $(r, u)$ from $P$ to $Q$ exists exactly when $P \leq u^*Q$, and is then unique over that arrow. Here $\mathsf{Pred}(W) = \mathrm{Sub}_{\mathbf{Set}}(W)$ and $u^*Q = u^{-1}(Q)$. Composition uses composition of the two displayed maps and functoriality of inverse image. The projections form the tower

$$\mathsf{Assn}_{A,B} \xrightarrow{\ p\ } \mathsf{St}_{A,B} \xrightarrow{\ q\ } \mathcal{A}_{A,B} \xrightarrow{\ \pi\ } \mathbf{Par}_N.$$

The category $\mathsf{St}(G) = q^{-1}(G)$ is the category of **state contexts** of $G$. Its objects retain their reference map to the full policy space.

**Theorem 5.3** (Cartesian lifts and the logical fibres)**.** *All three projections in Definition 5.2 are fibrations. There are canonical identifications*

$$\mathsf{St}(G) = \mathbf{Set}/\Sigma^G, \qquad p^{-1}(G, s : W \to \Sigma^G) = \mathsf{Pred}(W).$$

*More precisely, $p$ is a* posetal *fibration, obtained by change of base of the subobject fibration along the domain functor $\mathsf{St}_{A,B} \to \mathbf{Set}$; $q$ is the change of base of the codomain fibration along $U_{A,B}$; and $\pi$ is a* discrete *fibration, as the category-of-elements projection of the presheaf $F_{A,B}$.*

*Proof.*

(1) **Lifting through the comparison.** For $\pi$, lifting an owner map $r : \vec{\Theta} \to \vec{\Sigma}$ at $g$ gives $(\vec{\Theta}, g\Pi_N r)$, and the factorization equation is unique. This is the category-of-elements construction of Theorem 3.7.

(2) **Lifting along the surjection: second comparison.** For $q$, lift $t : V \to \Sigma^H$ along $r : G \to H$ by the square

$$\begin{array}{ccc} \Sigma^G \times_{\Sigma^H} V & \xrightarrow{a} & V \\ {\scriptstyle s}\downarrow & \lrcorner & \downarrow{\scriptstyle t} \\ \Sigma^G & \xrightarrow[\bar{r}]{} & \Sigma^H. \end{array}$$

If $(r\ell, v) : (K, z) \to (H, t)$ lies over $r\ell$, its unique factorization through this square sends $w$ to $(\bar{\ell}z(w), v(w))$.

(3) **Cartesian lifting of assertion predicates.** This proves cartesianness. Finally the lift of $Q \hookrightarrow V$ along $(r, u)$ is $u^*Q \hookrightarrow W$. A factorization into it exists exactly when its predicate is below the inverse image of $u^*Q$; functoriality of inverse image gives precisely the required condition. The two change-of-base descriptions and the fibre identifications follow from the definitions.

□

The state-context fibration is obtained by changing base along the policy projection.

$$\begin{array}{ccc} \Sigma^G \times_{\Sigma^H} V & \longrightarrow & V \\ \downarrow & \lrcorner & \downarrow{\scriptstyle t} \\ \Sigma^G & \xrightarrow[\bar{r}]{} & \Sigma^H \end{array}$$

**Construction 5.4** (Indexed presentation and constant types)**.**

(1) **Index the state presentation.** Write $\mathsf{P}_G : \mathsf{St}(G)^{\mathrm{op}} \to \mathbf{Pos}$ for the indexed predicate assignment $(W, s) \mapsto \mathsf{Pred}(W)$. The terminal context is $1_G = (\Sigma^G \xrightarrow{\mathrm{id}} \Sigma^G)$.

(2) **Interpret constants and their tests.** For an ordinary set $X$, its constant type is

$$\Delta_G X = (\Sigma^G \times X \xrightarrow{\pi_1} \Sigma^G).$$

For $s : W \to \Sigma^G$, a term $a : W \to X$ is represented by $(s, a) : (W, s) \to \Delta_G X$. A predicate $Q \hookrightarrow X$ is represented by $\Delta_G Q = \Sigma^G \times Q \hookrightarrow \Sigma^G \times X$. Consequently

$$(s, a)^* \Delta_G Q = a^* Q. \tag{5.1}$$

(3) **Represent play and ownership.** In particular play is the section $\widehat{g} = (\mathrm{id}, g) : 1_G \to \Delta_G \mathcal{C}(A, B)$. Each owner projection likewise gives a section of $\Delta_G \Sigma_i^G$.

**Proposition 5.5** (Change of game and constant types)**.** *A reparametrisation $r : G \to H$ induces pullback $\bar{r}^* : \mathsf{St}(H) \to \mathsf{St}(G)$ and, on each predicate, inverse image along the top map of its cartesian lift. There are natural isomorphisms*

$$\bar{r}^* \Delta_H X \cong \Delta_G X, \qquad \overline{tr}^* \cong \bar{r}^* \bar{t}^*.$$

*They satisfy the identity and associativity coherence equations. Under the first isomorphism the pullback of $\widehat{h}$ is $\widehat{g}$.*

*Proof.*

(1) **Pullback comparison isomorphisms.** The first map sends $(\sigma, (\bar{r}\sigma, x))$ to $(\sigma, x)$. The second forgets the repeated intermediate coordinate in an iterated pullback. Their inverses insert the displayed coordinates.

(2) **Coherence.** Every coherence composite therefore inserts or deletes the same coordinates. The final assertion is the realization equation $g = h\bar{r}$. Equivalently, represent a context as its family of fibres $(W_\sigma)$; reindexing is $(W_\eta) \mapsto (W_{\bar{r}\sigma})$, with strict substitution.

(3) **Equivalence with split families.** Passing between this split presentation and slices uses the canonical bijection $W \cong \coprod_\sigma W_\sigma$.

□

**Construction 5.6** (Dependent fibres over state contexts)**.**

(1) **Pull back the two fibres.** Pull back the indexed categories $\mathsf{T}$ and $\mathsf{L}$ along $\mathrm{Dom} : \mathsf{St}_{A,B} \to \mathbf{Set}$, which sends $(G, W, s)$ to $W$ and $(r, u)$ to $u$. Their fibres are $\mathsf{T}(W) \simeq \mathbf{Set}/W$ and $\mathsf{L}(W)$, with reindexing $u^*$.

(2) **Construct the fibrations.** Their Grothendieck constructions are fibrations: the cartesian arrow above $u$ is the identity $u^*E \to u^*E$ in the source fibre, and every factorization is forced by composition of index maps. The comparison, comprehension, and adjunction maps of Theorem 1.42 pull back to these fibrations. The original $\mathcal{C}_W$ is their constant representable linear fragment, by Proposition 1.33.

The original assertion fibre is recovered as $\mathsf{Pred}(W) = \mathrm{Sub}_{\mathsf{T}(W)}(1)$: a subterminal family has a singleton or empty fibre at each state. These are the ordinary subobject structures of Jacobs (1999, Chapters 1, 4, and 10), now situated inside the constructed dependent linear–nonlinear model. For the original boundary, the realization section is equivalently a section of $\mathsf{Val}_W[yA, yB]_D$, since this family is constant with value $\mathcal{C}(A, B)$.

## 5.2 Predicate feedback, canonical lenses, and coplay

A forward map already determines how to carry a test backward by taking its inverse image. Power objects represent this operation by the canonical lens used here for *coplay*. Its forward value can be an entire concurrent strategy. Its backward value compares alternatives. This gives the particular continuation semantics used here, within the broader account of Capucci (2023).

**Definition 5.7** (Play, predicate feedback, and canonical coplay)**.** For a forward realization $f : X \to Y$, a **predicate continuation** is $Q \in \mathsf{Pred}(Y)$. The subobject fibration supplies its **predicate feedback** by reindexing. The **canonical coplay** of the induced lens is this same map:

$$\mathsf{coplay}_f = f^* : \mathsf{Pred}(Y) \longrightarrow \mathsf{Pred}(X), \qquad Q \longmapsto f^{-1}(Q).$$

A numerical or other $V$-valued continuation has type $k : Y \to V$, and its backward transport has type $f_V^- : V^Y \to V^X$, $k \mapsto kf$. For a comparison predicate $P \hookrightarrow Y^2$, its transported comparison is $(f \times f)^* P \hookrightarrow X^2$. These are the backward components of the canonical lenses constructed below. Play returns a result; coplay tests which alternatives produce desired results, and does not execute a physical protocol backwards.

For example, let $X$ encode keeping or exchanging a resource and let $f$ return the resulting bundle. Pulling back the goal of possessing a specified resource selects exactly the choices achieving it. The calculation evaluates forward possibilities; it does not run the protocol backwards.

**Definition 5.8** (Lenses and comparison interfaces)**.** Write $\mathsf{Pred}(X) = \mathrm{Sub}_{\mathbf{Set}}(X)$. A lens $(X, S) \to (Y, R)$ is a pair of maps $f : X \to Y$ and $f^- : X \times R \to S$. Its composite with $(h, h^-) : (Y, R) \to (Z, T)$ is

$$(hf,\ (x, t) \longmapsto f^-(x, h^-(f(x), t))).$$

The identity has backward map $(x, s) \mapsto s$. The tensor is the componentwise product of forward and backward maps, with unit $(1, 1)$. These operations define $\mathsf{Lens}(\mathbf{Set})$. The **comparison interface** on $X$ is

$$\mathbb{L}(X) = (X, \mathsf{Pred}(X)).$$

A comparison $P \hookrightarrow X \times X$ is represented by the costate $q_P : X \to \mathsf{Pred}(X)$, where $q_P(x) = \{x' : P(x, x')\}$. Thus its backward output at a reference $x$ is the predicate of alternatives compared favorably with that reference.

**Definition 5.9** (Lens states and comparison costates)**.** A lens state is an arrow $(1, 1) \to \mathbb{L}(X)$, hence an element $x \in X$. A lens costate is an arrow $\mathbb{L}(X) \to (1, 1)$, hence a map $q : X \to \mathsf{Pred}(X)$. The two arguments of $q(x)(x')$ have distinct roles: $x$ is the reference and $x'$ is its candidate alternative.

To test improvement one must know both the reference outcome and the proposed replacement. The costate on $X$ retains this pair, so it can distinguish two changes that start from the same acceptable outcome.

**Construction 5.10** (Canonical relational and numerical feedback)**.**

(1) **Pull back a predicate.** The cartesian lift of $Q \hookrightarrow Y$ along $f : X \to Y$ is $f^*Q \hookrightarrow X$. Using the membership predicates of the power objects to represent this operation, define

$$\mathbb{L}(f) = (f, f^-), \qquad f^-(x, Q) = f^*Q.$$

(2) **Pull back a value map.** For a value object $V$, put $\mathbb{L}_V(X) = (X, V^X)$ and define $\mathbb{L}_V(f)$ with backward map $(x, k) \mapsto kf$. Numerical values, their differences, and arbitrary comparison predicates can consequently be transported along the same forward map. A value presentation consists of the actual map $k : X \to V$ and its comparison relation; the numerical map remains part of the evaluation data.

**Theorem 5.11** (Functorial feedback and comparison costates)**.** *The assignments* $\mathbb{L}$ *and* $\mathbb{L}_V$ *are faithful functors from* **Set** *to* $\mathsf{Lens}(\mathbf{Set})$*. Transporting a comparison costate along* $\mathbb{L}(f)$ *gives*

$$(\mathbb{L}(f)^* q_P)(x)(x') \quad \Longleftrightarrow \quad P(f(x), f(x')), \tag{5.2}$$

*so its uncurried comparison is* $(f \times f)^* P$*.*

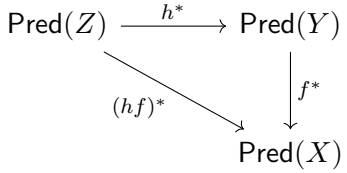
*Proof.*

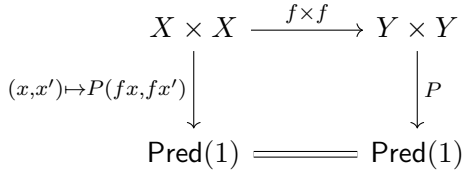
(1) **Functoriality of predicate feedback.** For $f : X \to Y$ and $h : Y \to Z$, the composite backward map sends $Q$ to $f^*(h^*Q) = (hf)^*Q$; identities give identity pullback. For numerical continuation interfaces the same calculation is $(kh)f = k(hf)$. The forward component recovers the original map, which proves faithfulness.

(2) **Pullback of comparison costates.** A costate of $(Y, \mathsf{Pred}(Y))$ sends the reference $f(x)$ to $q_P(f(x))$. Pullback along $f$ then tests $P(f(x), f(x'))$, proving the displayed equivalence.

□

The canonical coplay is functorial under composition of forward realizations.






Here a predicate is displayed as a truth-value map only to draw its classification square; all subsequent constructions use subobjects and pullback, as developed in Section 5.5.

The forward carriers used below are hom-sets $H_{A,B} = \mathcal{C}(A, B)$. Therefore a value of the play map is the entire concurrent strategy of Section 1.1. Applying $\mathbb{L}$ to an ordinary map into such a hom-set constructs feedback on that strategy; the linear interaction ports are still the original objects $A, B$. For families $u, v : \Gamma \to H_{A,B}$, write $\mathrm{Test}_P(u, v) = \langle u, v \rangle^* P$. Section 5.5 derives the logical structure of this testing operation.

Applied to the realization $g : \Sigma^G \to \mathcal{C}(A, B)$, this construction equips the arena with $\mathbb{L}(g)$ and backward map $g^-(\sigma, Q) = g^* Q$. The same operation is substitution along $\widehat{g}$ into a constant predicate type in its state doctrine. The equation $g = h\bar{r}$ for a reparametrisation therefore gives both the forward equation and $\mathbb{L}(g) = \mathbb{L}(h)\mathbb{L}(\bar{r})$.

When an operation incurs a charge, its continuation receives that charge along with the output. Pulling back an evaluator must therefore account for the cost already incurred before it compares what could happen next. The writer extension below gives precisely this backward action under sequential composition.

**Proposition 5.12** (Feedback for a measured operation)**.** *For $f : X \to T_C Y$, define $f_C^\sharp(e) = e f^\dagger$ on evaluators $e : T_C Y \to V$, and act on paired comparison predicates by inverse image along $f^\dagger \times f^\dagger$. For $g : Y \to T_C Z$ and $h = g^\dagger f$, one has $h_C^\sharp = f_C^\sharp g_C^\sharp$ and $(\eta^T)_C^\sharp = 1$.*

*Proof.*

(1) **Compute the continuation.** If $f(x) = (d, y)$ then $f_C^\sharp(e)(c, x) = e(c + d, y)$. A further operation $g(y) = (b, z)$ gives $e((c + d) + b, z) = e(c + (d + b), z)$; the unit inserts zero.

(2) **Transport comparisons.** Both legs of each paired test compose by the same extension law. Apply Theorem 5.11 to the ordinary functions $f^\dagger, g^\dagger$ on the measured carriers. For a witness-indexed game the forward measured term is defined on its witness carrier; pull the test back along that term before taking any profile image.

□

## 5.3 Observable outcomes

**Proposition 5.13** (Economic closure retains an outcome interface)**.** *$\mathcal{C}(I, I)$ is a singleton. Therefore an evaluation depending only on a fully hidden operational scalar is constant.*

*Proof.*

(1) **Constancy of scalar evaluation.** This is the scalar calculation in Lemma 1.28. Any function on that singleton is constant.

□

**Definition 5.14** (Economically closed realization)**.** An economically closed realization has type $I \to O$, with $O$ a retained outcome interface. Economic closure supplies the decision inputs; it does not hide $O$ before evaluation.

## 5.4 The category of operational contexts

**Definition 5.15** (Operational contexts, comparison contexts, and environments)**.** An **operational context** from $(A, B)$ to $(A', B')$ is an allowed wiring-induced function

$$T : \mathcal{C}(A, B) \longrightarrow \mathcal{C}(A', B'), \qquad T \in \mathsf{Ctx}(\mathcal{C})((A, B), (A', B')),$$

where the generators of $\mathsf{Ctx}(\mathcal{C})$ are defined below. A **comparison context** at $(A', B')$ is a tuple $P \in \mathsf{Cmp}_N(A', B')$ of predicates on pairs of interactions. An **environment** for $(A, B)$ is a typed pair $(T, P)$; its induced comparison has type $T^\sharp P \in \mathsf{Cmp}_N(A, B)$, with

$$(T^\sharp P)_i = (T \times T)^* P_i.$$

A value-presented environment in particular may give $a : I \to A$, $b : B \to O$, value maps $u_i : \mathcal{C}(I, O) \to V_i$, and $\prec_i \hookrightarrow V_i^2$. It induces

$$k_i : \mathcal{C}(A, B) \to V_i, \qquad k_i(f) = u_i(bfa), \qquad (k_i \times k_i)^* \prec_i \hookrightarrow \mathcal{C}(A, B)^2.$$

A **state context** $s : W \to \Sigma^G$ instead indexes reference profiles; it is distinct from these operational and comparison contexts.

**Definition 5.16** (Operational context)**.** An interface pair is an ordered pair $(A, B)$ of objects of $\mathcal{C}$. The category $\mathsf{Ctx}(\mathcal{C})$ has interface pairs as objects. Its arrows $(A, B) \to (A', B')$ are functions

$$T : \mathcal{C}(A, B) \longrightarrow \mathcal{C}(A', B')$$

generated under composition by identities, whiskering $f \mapsto bfa$ for $a : A' \to A$, $b : B \to B'$, and tensor insertion $f \mapsto f \otimes r$ or $f \mapsto r \otimes f$ for a fixed arrow $r$. Generated functions are identified when they are equal. A parameterized context is an ordinary family of such functions.

**Construction 5.17** (Backward action on tests)**.**

(1) **Form the comparison space.** Put

$$\mathsf{Cmp}_N(A, B) = \prod_{i \in N} \mathsf{Pred}(\mathcal{C}(A, B)^2).$$

(2) **Transport the tests.** For a context $T$, define

$$T^\sharp : \mathsf{Cmp}_N(A', B') \to \mathsf{Cmp}_N(A, B), \qquad (T^\sharp P)_i = (T \times T)^* P_i.$$

(3) **Transport the values.** For value maps $k_i : \mathcal{C}(A', B') \to V_i$, the corresponding action is $k_i \mapsto k_i T$.

**Theorem 5.18** (Contextual evaluation and coherence)**.** $\mathsf{Cmp}_N$ *with this action is a contravariant functor* $\mathsf{Ctx}(\mathcal{C})^{\mathrm{op}} \to \mathbf{Pos}$*. It preserves the pointwise predicate operations under reindexing, and*

$$\mathrm{Test}_{T^\sharp P}(u, v) = \mathrm{Test}_P(Tu, Tv)$$

*for families* $u, v : \Gamma \to \mathcal{C}(A, B)$*. The action respects every equation between operational contexts.*

*Proof.*

(1) **Composition of contextual predicate pullback.** The costate action is the comparison part of the canonical lens functor in Theorem 5.11. In particular, for contexts $T, U$, $((UT) \times (UT))^* = (T \times T)^*(U \times U)^*$. Testing follows by the same pullback identity: $\langle u, v\rangle^*(T \times T)^* P = \langle Tu, Tv\rangle^* P$.

(2) **Functoriality and substitution.** All context equations are equalities of the generated functions, so their backward actions agree by this functoriality.

□

The external test and its contextual pullback coincide.

$$\begin{array}{ccc} \Gamma & \xrightarrow{\langle u,v\rangle} & \mathcal{C}(A,B)^2 \\ \| & & \downarrow{\scriptstyle T\times T} \\ \Gamma & \xrightarrow[\langle Tu,Tv\rangle]{} & \mathcal{C}(A',B')^2 \end{array}$$

$$\begin{array}{ccc} \Gamma & \xrightarrow{(u,v)} & \mathcal{C}(A,B)^2 \\ & {\scriptstyle (Tu,Tv)}\searrow & \downarrow{\scriptstyle T\times T} \\ & & \mathcal{C}(A',B')^2 \end{array}$$

$$\mathsf{Cmp}_N(A,B) \xleftarrow{T^\sharp} \mathsf{Cmp}_N(A',B') \xleftarrow{U^\sharp} \mathsf{Cmp}_N(A'',B''), \qquad (UT)^\sharp : \mathsf{Cmp}_N(A,B) \leftarrow \mathsf{Cmp}_N(A'',B'')$$

**Construction 5.19** (Evaluation supplied by an environment)**.**

(1) **Close an interaction in its environment.** For $a : I \to A$, $b : B \to O$, and $u_i : \mathcal{C}(I, O) \to V_i$, define

$$k_i(f) = u_i(bfa).$$

More generally a comparison context is an element $P \in \mathsf{Cmp}_N(A, B)$; it need not have a numerical representation. If $\prec_i \subseteq V_i^2$, define $P_i = (k_i \times k_i)^* \prec_i$.

(2) **Specialize the comparison to each component.** For open arenas $G : A \to B$ and $H : B \to C$, the contexts obtained by fixing $\tau \in \Sigma^H$ or $\sigma \in \Sigma^G$ are

$$T_\tau(f) = h_\tau f, \qquad S_\sigma(h) = h g_\sigma.$$

Write $P^{G|\tau} = T_\tau^\sharp P$ and $P^{H|\sigma} = S_\sigma^\sharp P$.

**Proposition 5.20** (Context action for sequential and parallel composition)**.** *For fixed $h : B \to C$, $j : C \to D$, $T^\sharp_{jh} = T^\sharp_h T^\sharp_j$. Whiskering on opposite sides commutes. For fixed composable $r, r'$, tensor insertion is compatible with composition by interchange. All these equations hold also for parameterized contexts, pointwise in their parameters.*

*Proof.*

(1) **Monoidal comparison.** The context functions satisfy $(jh)f = j(hf)$ and $b(fa) = (bf)a$. The tensor assertion follows from $(h \otimes r') \circ (f \otimes r) = (hf) \otimes (r'r)$. Apply Theorem 5.18.

□

The backward context action is the costate action of the canonical lenses of Construction 5.10. Its inputs and outputs are complete concurrent strategies. Closing an evaluation context specifies the comparison received by each agency interface; the retained output arena continues to expose the economic outcome.

## 5.5 Dependent assertions and semantic testing

Comparisons of policies become assertions only after a reference context has been chosen. The state-context fibration supplies ordinary dependent objects for policies and witnesses and subterminal objects for predicates. Comprehension restricts contexts to satisfying states; quantifier adjoints express selection and existence of witnesses. We now identify these operations with the standard dependent semantics so that subsequent strategic and epistemic tests use the same logic.

### 5.5.1 The predicate fragment and comprehension

A feasibility assertion answers whether at least one allocation certificate exists. An institution that later uses the certificate must also retain a witness. An ordinary dependent family stores those witnesses, and its existential image answers the feasibility question. Comprehension selects the states satisfying a predicate. Its subterminal fragment supplies truth conditions for the assertions about agency.

**Definition 5.21** (The assertion hyperdoctrine)**.** For an arena $G$, use the indexed assignment $\mathsf{P}_G$ of Construction 5.4. For $s : W \to \Sigma^G$, an assertion is a subobject of $W$; entailment is inclusion. Its conjunction, disjunction, implication, and constants are those of $\mathsf{Pred}(W)$, with $\neg P = (P \Rightarrow \bot)$. An arrow $u : (W, s) \to (V, t)$ substitutes predicates by $u^*$. The ambient notation $\mathsf{Pred}(X)$ permits the same calculation before a reference context is chosen; constant types insert it into the game's doctrine by Equation (5.1).

**Theorem 5.22** (Quantifiers, equality, and comprehension)**.** *The indexed assignment $\mathsf{P}_G$ is the subobject hyperdoctrine of $\mathbf{Set}/\Sigma^G$. Every context map has adjoints $\exists_u \dashv u^* \dashv \forall_u$, with*

$$(\exists_u P)(v) \iff \exists w\,(u(w) = v \wedge P(w)), \qquad (\forall_u P)(v) \iff \forall w\,(u(w) = v \Rightarrow P(w)).$$

*Both quantifiers satisfy Beck–Chevalley for pullback squares. Moreover $\exists_u(P \wedge u^*Q) = \exists_u P \wedge Q$. Equality on a context $W \to \Sigma^G$ is its diagonal in $W \times_{\Sigma^G} W$.*

*Comprehension of $P \hookrightarrow W$ is $\{P\} = (P \hookrightarrow W \xrightarrow{s} \Sigma^G)$. For the projection $p_G$ of the predicate fibration, the truth functor $\top_G$ and comprehension satisfy $p_G \dashv \top_G \dashv \{-\}$. The same comprehension construction works in the total state-context category and is stable under its cartesian substitutions.*

Comprehension represents the predicates by inclusions of their satisfying contexts.

$$\begin{array}{ccc} \{P\} & \xhookrightarrow{m_P} & W \\ {\scriptstyle s|_P}\downarrow & & \downarrow{\scriptstyle s} \\ \Sigma^G & = & \Sigma^G \end{array}$$

*Proof.*

(1) **Predicate calculation.** A subobject of $(W, s)$ in the slice is exactly a subset of $W$ equipped with the restricted map $s$. Slice pullbacks have the underlying pullbacks of sets, proving the asserted identification of doctrines.

(2) **Base-change comparison.** The elementwise formulas give

$$\exists_u P \le Q \iff P \le u^*Q, \qquad u^*Q \le P \iff Q \le \forall_u P.$$

In a pullback square

$$\begin{array}{ccc} W' & \xrightarrow{a} & W \\ {\scriptstyle u'}\downarrow & \lrcorner & \downarrow{\scriptstyle u} \\ V' & \xrightarrow[b]{} & V \end{array}$$

the fibre of $u'$ at $v'$ is canonically the fibre of $u$ at $b(v')$.

(3) **Frobenius and coproduct comparison.** The quantifier formulas therefore give $b^*\exists_u = \exists_{u'}a^*$ and $b^*\forall_u = \forall_{u'}a^*$. Frobenius follows because $Q(u(w))$ is constant on a fibre of $u$. The diagonal classifies equality in the slice product.

(4) **Predicate calculation.** An arrow from a predicate to truth is specified by its underlying context map, proving $p_G \dashv \top_G$. An arrow from a true context to $P$ is a context map whose image lies in $P$; it factors uniquely through $\{P\}$, proving $\top_G \dashv \{-\}$. In the total category the same factorization keeps its arena reparametrisation unchanged. Inverse image of a subset is its pullback, giving stability of comprehension.

□

**Construction 5.23** (Dependent types and predicate types)**.**

(1) **Identify the ordinary fibre.** Over a state context $(W, s)$, a dependent type is a map $E \to W$; its terms are sections. The slice $(\mathbf{Set}/\Sigma^G)/(W, s)$ is canonically $\mathbf{Set}/W$. These are the ordinary fibres of Construction 1.32.

(2) **Form dependent sums and products.** For $u : W \to V$, their dependent sums and products have fibres

$$(\Sigma_u E)_v = \coprod_{u(w)=v} E_w, \qquad (\Pi_u E)_v = \prod_{u(w)=v} E_w.$$

We write $\Sigma_u, \Pi_u$ for the ordinary adjoints $\Sigma_u^{\mathrm{c}}, \Pi_u^{\mathrm{c}}$ of Theorem 1.39; their adjunctions follow from the same sum and product universal properties. The image of the dependent sum of a predicate gives $\exists_u$; its dependent product gives $\forall_u$.

(3) **Construct predicate types.** For $W \to \Sigma^G$, the power object has fibre $\mathcal{P}_{\Sigma^G}(W)_\sigma = \mathcal{P}(W_\sigma)$, with fibrewise membership. A subobject of a slice product is uniquely classified by the map sending a parameter to its corresponding subset of $W_\sigma$. The truth-value object is $\Sigma^G \times \{0, 1\} \to \Sigma^G$. Thus the construction also supplies higher-order predicate types.

**Proposition 5.24** (Propositional reflection and recovery of quantifiers)**.** *For $E \in \mathsf{T}(W)$ define $\|E\|$ by the image of its projection $\coprod_w E_w \to W$. This is left adjoint to the inclusion of subterminal families into ordinary families. It commutes with substitution. For a subterminal $P$ over $W$ and $u : W \to V$,*

$$\exists_u P = \|\Sigma_u^{\mathrm{c}} P\|, \qquad \forall_u P = \Pi_u^{\mathrm{c}} P.$$

*Thus the existing predicate doctrine and its quantifiers are recovered inside the constructed dependent semantics.*

*Proof.*

(1) **Characterization of propositional reflection.** The fibre $\|E\|_w$ is a singleton exactly when $E_w$ is inhabited. A map from $E_w$ to a subterminal $P_w$ exists exactly when this inhabitation implies inhabitation of $P_w$, and is then unique.

(2) **Naturality and quantified reflection.** This proves the adjunction without choosing an element of $E_w$. Reindexing retains the same fibres, proving substitution stability. The coproduct is inhabited exactly when some summand is inhabited.

(3) **Monoidal comparison.** A product of subterminal sets is a singleton exactly when every factor is inhabited: its element is the uniquely specified family of their unique elements. Otherwise it is empty.

(4) **Quantifiers on empty and inhabited fibres.** This proves the quantifier formulas, including empty indexing fibres, without a choice of witnesses from arbitrary inhabited types.

□

**Proposition 5.25** (Logical change of base)**.** *The change of game in Proposition 5.5 preserves the constructed finite-limit, comprehension, and predicate-type structure up to its canonical comparisons. Quantifiers commute with this change of base along the associated pullback squares. These comparisons satisfy substitution coherence.*

*Proof.*

(1) **Monoidal comparison.** In family notation a context has fibres $W_\eta$ and a dependent type has fibres $E_{\eta,w}$. Pullback along $\bar{r}$ replaces $\eta$ by $\bar{r}\sigma$ in these formulas. Products, equalities, subsets, and power objects are consequently the identical constructions on identical fibres.

(2) **Monoidal comparison: completion.** Sums and products indexed by a dependent context map range over the same sets after this substitution. This proves their comparisons and the quantifier equations; repeated substitution of the index proves coherence. The set-theoretic constructions implement the fibred logical semantics of Jacobs (1999).

□

The doctrine lives over state contexts, not over a Cartesian structure on the category $\mathcal{A}_{A,B}$ of agent presentations itself. The latter indexes the contexts and their changes of base; operational contexts in $\mathsf{Ctx}(\mathcal{C})$ separately act on hom-set values as in Section 3.2. Keeping these roles apart is necessary for treating owned reparametrizations and logical substitutions correctly.

### 5.5.2 Algebras and coalgebras of predicate adjunctions

**Definition 5.26** (Posetal monads and comonads)**.** On a poset, a monad is a monotone endomap $M$ with $\mathrm{id} \leq M$ and $M^2 \leq M$; a comonad is a monotone endomap $K$ with $K \leq \mathrm{id}$ and $K \leq K^2$. These inequalities imply idempotence. Their algebra and coalgebra posets are respectively the full subposets $MP = P$ and $KP = P$.

**Theorem 5.27** (The two universal approximations to invariant assertions)**.** *For a surjection $q : X \twoheadrightarrow Q$, let $\mathsf{Inv}(q)$ be the full subposet of predicates constant on the fibers of $q$, with inclusion $j : \mathsf{Inv}(q) \hookrightarrow \mathsf{Pred}(X)$. Then*

$$M_q = q^*\exists_q \dashv j \dashv K_q = q^*\forall_q, \qquad \mathrm{Alg}(M_q) \cong \mathsf{Inv}(q) \cong \mathrm{Coalg}(K_q) \cong \mathsf{Pred}(Q).$$

*In the adjunction display, $M_q$ and $K_q$ are corestricted to $\mathsf{Inv}(q)$; as endomaps they are the resulting idempotent monad and comonad.*

*Proof.*

(1) **Verification of the universal property.** Surjectivity gives $\exists_q q^* = \forall_q q^* = \mathrm{id}$. Thus $M_q$ and $K_q$ have invariant values, are idempotent, and the adjunction units and counits give $P \le M_q P$ and $K_q P \le P$. If $Q' = q^* \bar{Q}$ is invariant, then

$$M_q P \le Q' \iff P \le Q', \qquad Q' \le K_q P \iff Q' \le P$$

by the two quantifier adjunctions.

(2) **Predicate calculation.** These are the claimed universal properties. Their fixed predicates are precisely those in the image of $q^*$, and effective predicate descent identifies this image with $\mathsf{Pred}(Q)$.

□

Invariant predicates descend uniquely along the observation quotient.

$$\begin{array}{ccc} \mathsf{Pred}(Q) & \xrightarrow[\cong]{q^*} & \mathsf{Inv}(q) \\ \| & & \downarrow j \\ \mathsf{Pred}(Q) & \xrightarrow[q^*]{} & \mathsf{Pred}(X) \end{array}$$

$$\mathsf{Pred}(X) \underset{j}{\overset{M_q}{\rightleftarrows}} \mathsf{Inv}(q) \xrightarrow{\simeq} \mathsf{Pred}(Q), \qquad K_q : \mathsf{Pred}(X) \to \mathsf{Inv}(q)$$

*Remark* 5.28 (One construction, several economic meanings). Observation quotients give epistemic invariant assertions. Owner-coordinate quotients give assertions invariant under the corresponding deviations. Quotients of redundant presentations give representation-independent assertions. The adjunction is the same; the constructed quotient specifies its economic meaning.

### 5.5.3 Realization terms and their assertion predicates

**Construction 5.29** (Semantic testing)**.**

(1) **Pull back the designated predicate.** For $t : \Gamma \to X$ and $P \hookrightarrow X$, put $\mathrm{Test}_P(t) = t^* P$. In a state context $s : \Gamma \to \Sigma^G$, its typed expression is $(s,t)^* \Delta_G P = t^* P$, by Equation (5.1). Thus the same test is an internal assertion on the reference context.

(2) **Test a tuple.** For a finite tuple of indexed arrows use $X = \prod_k \mathcal{C}(A_k, B_k)$ and

$$\mathrm{Test}_P(t_1, \ldots, t_n) = \langle t_1, \ldots, t_n \rangle^* P.$$

**Theorem 5.30** (Classification of semantic tests)**.** *Predicates on $X$ correspond bijectively to assignments $\tau_\Gamma(t) \in \mathsf{Pred}(\Gamma)$ for $t : \Gamma \to X$ that are natural under substitution. The inverse sends $\tau$ to $\tau_X(\mathrm{id}_X)$. In particular,*

$$f^* \,\mathrm{Test}_P(t) = \mathrm{Test}_P(tf). \tag{5.3}$$

*For $F : X \to Y$ one has $\mathrm{Test}_{F^* Q}(t) = \mathrm{Test}_Q(Ft)$.*

*Proof.*

(1) **Naturality.** Naturality at $t : \Gamma \to X$ forces $\tau_\Gamma(t) = t^* \tau_X(\mathrm{id}_X)$. Conversely pullback is functorial, so every predicate gives such an assignment. These operations are inverse. This is the Yoneda argument in its elementwise form.

□

For $f : \Delta \to \Gamma$ and $t : \Gamma \to X$, both squares in the following diagram are pullbacks.

$$\begin{array}{ccccc} \mathrm{Test}_P(tf) & \longrightarrow & \mathrm{Test}_P(t) & \longrightarrow & P \\ \downarrow & & \downarrow & & \downarrow \\ \Delta & \xrightarrow[f]{} & \Gamma & \xrightarrow[t]{} & X \end{array}$$

Indeed, $\delta$ lies over the middle comprehension exactly when $t(f(\delta)) \in P$. This identifies its pullback along $f$ with the left comprehension and gives (5.3). The classification concerns tests of the specified semantic values. Ownership, observations, and legality supply additional data; it does not assert that every assertion about them depends only on the realized arrow.

### 5.5.4 State contexts, legal comprehension, and operational substitution

**Definition 5.31** (State realization)**.** A state realization is an object $(W, s)$ of $\mathsf{St}(G)$. Its operational family is $gs : W \to \mathcal{C}(A, B)$, equivalently a morphism $yA \to yB$ in $\mathsf{L}(W)$. Its realization section $(s, gs)$ takes values in the ordinary family $\mathsf{Val}_W[yA, yB]_D$, identified with the constant concurrent-arrow family by Theorem 1.31. For a legal predicate with comprehension $m : L \hookrightarrow \Sigma^G$, a legal state realization is a context over that comprehension, $s : W \to L$, with operational family $gms$. For $f : W' \to W$, substitution is $s \mapsto sf$ and $P \mapsto f^*P$.

**Proposition 5.32** (Synchronized substitution)**.** *For every semantic predicate $Q$,*

$$f^* \,\mathrm{Test}_Q(gms) = \mathrm{Test}_Q(gmsf).$$

*The intermediate state space need not contain representatives of every available alternative policy.*

*Proof.*

(1) **Representation and Yoneda decoding.** The equality is Theorem 5.30. The definitions require only the displayed map $s$ and impose no surjectivity condition on it.

□

**Theorem 5.33** (Fibred compatibility of dependent realization and assertions)**.** *Over $\mathsf{St}_{A,B}$ the indexed ordinary and linear categories of Construction 5.6, their comprehension, monoidal adjunction, and dependent adjoints share coherent substitution. The assertion fibration is their subterminal ordinary fragment. The original indexed operational category embeds fully faithfully in the linear fibration, preserving its concurrent structure.*

*If $t : W \to \prod_j \mathcal{C}(A_j, B_j)$ is a tuple of realized families, $m$ is composition, tensor, or another exhibited operation on these arrow carriers, and $Q$ is a predicate on its target, then*

$$u^* \,\mathrm{Test}_Q(mt) = \mathrm{Test}_Q(m(tu)) = \mathrm{Test}_{m^*Q}(tu)$$

*for every $u : V \to W$. The same equations hold for comparison predicates on paired carriers. For dependent boundaries, replace the constant carriers by $\mathsf{Val}[A, B]_D$ and use their fibrewise operations.*

*Proof.*

(1) **Representation and Yoneda decoding.** The categorical assertions follow by change of base of Theorem 1.42 along Dom. Proposition 5.24 identifies the assertion fibration, and pointwise Yoneda proves the embedding assertion. Each cartesian lift reindexes the same ordinary parameter, so all the comparison maps are those already proved coherent in the model theorem.

(2) **Secured operational interaction.** For the displayed equalities, $(mt)u = m(tu)$ and $(m(tu))^*Q = (tu)^*m^*Q$ are inverse-image functoriality. The paired and dependent cases are the identical calculation on the corresponding fibres.

□

### 5.5.5 Effective descent of terms and predicates

Observation descent gives a concrete example of this logical architecture. If $q : H \to O$ identifies histories an agent cannot distinguish, an implementable response $f : H \to A$ must be constant on the fibres of $q$ and hence factor through $O$. It cannot prescribe different actions at histories producing the same signal. The quotient thus has an operational meaning before knowledge modalities are introduced, and its dependent extension will retain complete families of such responses.

**Proposition 5.34** (Descent and finite products of quotients)**.** *Let $q : X \twoheadrightarrow Y$ have kernel pair $p_1, p_2 : X \times_Y X \rightrightarrows X$. A map $F : X \to Z$ descends uniquely to $Y$ exactly when $Fp_1 = Fp_2$. A predicate $P \hookrightarrow X$ descends uniquely exactly when $p_1^*P = p_2^*P$; its descent is*

$$\bar{P}(y) \iff \exists x\, (q(x) = y \wedge P(x)).$$

*Surjections are effective regular epimorphisms and are stable under pullbacks and finite products, including the nullary product.*

*Proof.*

(1) **Predicate calculation.** These are the effective-descent properties of sets (Jacobs, 1999, Chapters 4–5). Explicitly, the relation $\bar{F}(y) = z \iff \exists x\, (q(x) = y \wedge F(x) = z)$ is total and single-valued precisely when $F$ is fiber-constant. The displayed formula for $\bar{P}$ pulls back to $P$ precisely when $P$ is fiber-invariant.

(2) **Verification of the universal property.** Surjectivity gives uniqueness in both cases. A pullback has the same fibers as the original surjection. Products of two surjections are composites of two such pullbacks; finite induction includes the identity on 1.

□

$$X \times_Y X \underset{p_2}{\overset{p_1}{\rightrightarrows}} X \xrightarrow{q} Y, \quad F : X \to Z, \quad \bar{F} : Y \to Z$$

### 5.5.6 Substituting the reference state

Consider an owner with $\Sigma_i = \{a, b\}$. Selecting $a$ as the current policy leaves $b$ available as a deviation. Restricting the owner's carrier to $\{a\}$ removes it. The following comparison of spans shows why changing the reference state preserves the deviation fibre even when the two reference maps look alike.

**Construction 5.35** (Reference pullback)**.**

(1) **Pull back the source leg.** For $X \xleftarrow{p} R \xrightarrow{q} Y$ and $f : U \to X$, put $R_f = U \times_X R$, with legs $p_f : R_f \to U$ and $q_f = q \,\mathrm{pr}_R : R_f \to Y$.

(2) **Recover coalition alternatives.** For $s : W \to \Sigma$, the available $C$-alternatives are $\mathsf{Alt}_C(s) = W \times_{\Sigma, p_1} D_C$. Their paired realization sends $(w, \tau)$ to $(gs(w), g\tau)$.

**Proposition 5.36** (Substitution of relational assertions)**.** *$f^*[R]P = [R_f]P$ and $f^*\langle R\rangle P = \langle R_f\rangle P$. More generally, for a predicate $Q$ on the relation itself, $f^*\forall_p Q = \forall_{p_f} \mathrm{pr}_R^* Q$, and similarly for $\exists$. No target alternative has to lie in the image of $f$.*

*Proof.*

(1) **Base-change comparison.** Apply both Beck–Chevalley equations of Theorem 5.22 to the displayed cartesian lift. Proposition 7.6 types its target as a term into $\Delta_G Y$, so pulling back the reference context retains every target alternative. This proves both modal identities and their quantified form.

□

$$\begin{array}{ccccc} R_f & \xrightarrow{\mathrm{pr}_R} & R & \xrightarrow{q} & Y \\ {\scriptstyle p_f}\downarrow & \lrcorner & \downarrow{\scriptstyle p} & & \\ U & \xrightarrow[f]{} & X & & \end{array}$$

## 5.6 Selection and logical equilibrium

With a comparison costate in place, we can ask whether one owner's response improves on another with the outsiders fixed. Pullback supplies the comparison for that owner's complete policy. Universal quantification excludes each profitable replacement. Restricting best response to the diagonal tests equilibrium at the current profile. The off-diagonal relation still records which replacements were compared and how both endpoints were realized.

### 5.6.1 Connecting agent interfaces to joint comparison feedback

The complete owner interface, constructed in Section 2.3, now receives the comparison on the wired interaction. Fixing outsider coordinates turns that joint comparison into a costate on the owner's own policy set. The selection relation then classifies best responses, and its diagonal classifies equilibrium; the same calculation is later restricted to legal alternatives.

**Construction 5.37** (Connecting an agent to joint feedback)**.**

(1) **Fix the reference and vary one policy.** Fix an open arena $G$, comparison context $P = (P_i)_i$, and reference profile $\sigma$. For owner $i$ put

$$j_{i,\sigma} : \Sigma_i \to \Sigma, \qquad j_{i,\sigma}(a) = (\sigma_{-i}, a).$$

(2) **Transport the costate.** The agent's costate is the costate transported through $\mathbb{L}(gj_{i,\sigma})$:

$$q_{i,\sigma}(a)(b) \iff P_i(g_{(\sigma_{-i},a)}, g_{(\sigma_{-i},b)}). \tag{5.4}$$

(3) **Form best responses.** Define the best-response relation by

$$\mathsf{BR}_G(P)(\sigma, \tau) \iff \bigwedge_i \varepsilon_{\Sigma_i}(\tau_i, q_{i,\sigma}).$$

For arbitrary comparisons this means absence of an improving alternative; for exact rational and classical real-valued maximization it is the best-response condition of ordinary game theory.

In $\mathsf{BR}_G(P)(\sigma,\tau)$, the first profile $\sigma$ determines the outsiders against whom responses are evaluated. The second supplies the candidate choices $\tau_i$, each compared with alternatives for that same $\tau_i$ while $\sigma_{-i}$ remains fixed. Evaluating against $\tau_{-i}$ would instead change the reference game. Only on the diagonal $\tau = \sigma$ does the relation say that each current choice is a best response to all the other current choices.

**Proposition 5.38** (Selection and best response as internal predicates)**.** *For $T_S = S \times (S \to \mathsf{Pred}(S))$, let $B_S \hookrightarrow T_S \times S$ be the predicate $B_S(a,q,b) \iff q(a)(b)$ and let $\rho : T_S \times S \to T_S$. Then the agent's selection predicate is*

$$\varepsilon_S = \forall_\rho \neg B_S \in \mathsf{Pred}(T_S).$$

*For an evaluated arena $G$, define*

$$\kappa_i : \Sigma^G \times \Sigma^G \to T_{\Sigma_i^G}, \qquad \kappa_i(\sigma,\tau) = (\tau_i, q_{i,\sigma}).$$

*Its complete best-response relation and equilibrium predicate are*

$$\mathsf{BR}_G(P) = \bigwedge_i \kappa_i^* \varepsilon_{\Sigma_i^G}, \qquad \mathsf{NE}_G = \Delta^*_{\Sigma^G} \mathsf{BR}_G(P).$$

*All these expressions are terms and predicates of $\mathsf{P}_G$ using the constant types of Construction 5.4.*

*Proof.*

(1) **Strategic comparison.** The fibre of $\rho$ at $(a,q)$ is $S$, so its universal quantifier is exactly $\forall b\, \neg q(a)(b)$. Substitution along $\kappa_i$ gives the agent condition defining best response, and substitution along the diagonal makes its candidate response equal to the reference. The constant-type interpretation preserves each pullback by Equation (5.1).

(2) **Strategic comparison: completion.** This retains the off-diagonal relation; passage to equilibrium is the displayed additional pullback.

□

The selection-relation approach of Capucci et al. (2022a,b) organizes selection covariantly over lenses, with the Nash product giving a monoidal opfibration. Here the corresponding predicates live in a contravariant assertion fibration. Connecting feedback to selection is the specified substitution, not an identification of those two variances. A repeated owner retains one complete policy carrier before this test is performed.

**Theorem 5.39** (Selection, counterfactual feedback, and logical equilibrium)**.** *For every comparison context, the following predicates on $\Sigma$ agree:*

(i) *the diagonal $\mathsf{BR}_G(P)(\sigma,\sigma)$;*

(ii) *satisfaction of every agent selection relation after connecting the agent interfaces to the arena;*

(iii) $\bigwedge_i \forall_{p_1 : D_i \to \Sigma} \neg (\delta_i^G)^* P_i$.

*The equality retains the off-diagonal best-response relation and all paired deviation realizations as separate, constructed semantic data.*

*Proof.*

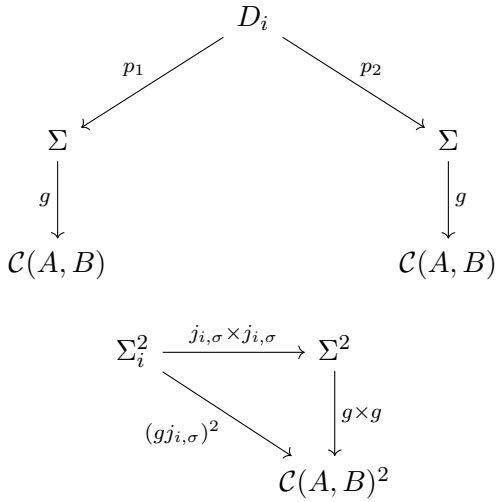
(1) **Policy-level comparison.** The first two assertions both expand to $\bigwedge_i \forall b \in \Sigma_i \neg P_i(g_\sigma, g_{(\sigma_{-i},b)})$. The fiber of $p_1 : D_i \to \Sigma$ at $\sigma$ is precisely that set of alternative policies, and $\delta_i^G$ realizes its two endpoints. The adjunction $p_1^* \dashv \forall_{p_1}$ therefore proves equality with the third assertion.

(2) **Typing of paired deviation tests.** The paired realization is typed into a constant comparison type by Proposition 7.6, so this is an equality in the assertion fibre over $1_G$. The proof uses no order property of $P_i$. If $P_i(f, f')$ is $k_i(f) < k_i(f')$ for classical real values, its negation is $k_i(f) \geq k_i(f')$, giving the argmax interpretation with every tie retained.

(3) **Evaluate a measured comparison.** Given profile values descended from completed records, use the two endpoint values on $D_i$ in the selection test. Its diagonal again excludes every profitable deviation, because its fibre quantifies the same policies. The precise descent conditions and the restriction to legal, completed alternatives are given in Proposition 5.43.

□

The two projections of a unilateral deviation determine the pair of realized interactions.




Independent owners recover the relational Nash product: every owner receives the comparison obtained by fixing the other owners' policies. A repeated owner has one larger carrier $\Sigma_i$ containing all its decisions, and therefore must be tested against changes to that entire carrier. This is the distinction on which equilibrium under composition depends.

**Construction 5.40** (Maximum of a finite numerical family)**.**

(1) **Compute the maximum.** For scalars in $\mathbb{K}$, a nonempty finite list $(a_1, \ldots, a_m)$ has a maximum constructed by $b_1 = a_1$ and $b_{k+1} = \max(b_k, a_{k+1})$.

(2) **Retain a maximizing record.** To retain a maximizing record from a displayed finite ordered domain, retain its first record at a tie. Induction proves that $b_k$ is the maximum of the first $k$ values and that the retained record has that value. Thus it provides both a maximum value and a specified first maximizer. For exact rational entries, comparison by cross multiplication of positive denominators makes this recursion a finite algorithm. For classical real entries its correctness uses the usual linear order.

**Definition 5.41** (Gain on a deviation)**.** For a numerical utility $u_i : \Sigma \to \mathbb{K}$ and a deviation $(\sigma, \tau) \in D_i$, put $\Delta_i u(\sigma, \tau) = u_i(\tau) - u_i(\sigma)$. Strict profitable deviation is the inverse image of $\mathbb{K}_{>0}$. For composable deviations of the same owner these gains add, and reversing a deviation negates its gain, by cancellation of the intermediate utility.

### 5.6.2 Comparison contexts and equilibrium

**Definition 5.42** (Preferences and value presentations)**.** A strict comparison records an improvement, while a weak comparison allows equally valued alternatives. For a preorder, two alternatives are **indifferent** when each is weakly preferred to the other and **incomparable** when neither is. Both fail to be strict improvements over one another, for different reasons. The general comparison context can be an arbitrary relation; preference-order terminology applies when the indicated preorder or numerical presentation is supplied. The second argument of $\mathsf{Better}_i$ is always the candidate improvement.

A comparison context for $G : A \to B$ is $\mathsf{Better} \in \mathsf{Cmp}_N(A, B)$, with the second argument denoting the improved interaction. No order law is part of this type. A value presentation is a family $k_i : \mathcal{C}(A, B) \to V_i$ with relations $\prec_i \subseteq V_i^2$ such that $\mathsf{Better}_i = (k_i \times k_i)^* \prec_i$. This gives $v_i^G = k_i g$ for an arrow-level value context. For a measured completion the numerical input is $e_i r$ on witnesses; Proposition 5.43 gives its profile-level form, including values that distinguish implementations of the same hidden arrow. When $V_i$ is preordered, its strict part is $x \prec_i y \iff x \preceq_i y \wedge \neg(y \preceq_i x)$. Every comparison has the tautological relational presentation $V_i = \mathcal{C}(A, B)$, $k_i = \mathrm{id}$. For an arbitrary comparison, the selection predicate expresses *stability against the specified relation*; it has the Nash interpretation when that relation is strict improvement. In particular, reflexivity or irreflexivity is not inferred from the bare type $\mathsf{Cmp}_N(A, B)$.

A policy can have several completed executions. An analyst may assign it a single payoff when the owner gives every execution over that policy the same value. An auditor who needs the actual expense asks more of the record. The dashed arrow in the proof exists precisely in the first case, and the same argument applied to the whole record tests the second.

**Proposition 5.43** (Descent of measured comparisons to policy profiles)**.** *Write $q : K \twoheadrightarrow D = \mathrm{im}(p)$ and $r = \langle m, a \rangle : K \to T_C Z$. The record descends to $D$ exactly when $m$ and $a$ each equalize the kernel pair of $q$. For a fixed owner the value $v_i = e_i r$ descends exactly when $v_i$ equalizes that kernel pair. Either descended map is unique. In particular, a unique completed witness at each profile gives total descended records and values. For a legal completed carrier $L \hookrightarrow D$, the descended values determine equilibrium by*

$$\mathsf{NE}_L(\sigma) \iff \bigwedge_i \forall \tau \in L \ (\sigma_{-i} = \tau_{-i} \ \Rightarrow \ \neg(\bar{v}_i(\sigma) \prec_i \bar{v}_i(\tau))).$$

*Proof.*

(1) **Descend exactly the observed quantity.** For any $b : K \to V$ constant on fibres, define $\bar{b}(\sigma)$ as the unique value $b(k)$ for $q(k) = \sigma$. Surjectivity supplies a witness and constancy gives uniqueness. Conversely $b = \bar{b} q$ equalizes the kernel pair. Apply this once to the pair $(m, a)$ and once to $e_i r$. For example, erasure can identify different charges on the same outcome. The value square is

$$\begin{array}{ccc} K & \xrightarrow{\ r\ } & T_C Z \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle e_i} \\ D & \xrightarrow[\bar{v}_i]{} & V_i \end{array}$$

To represent $\bar{v}_i$ on the image of the hidden concurrent realizer $g|_D$, apply the same argument to its image surjection. Profile descent and factorization through the hidden arrow are distinct tests.

(2) **Pull the comparison back to legal alternatives.** Restrict $K$ to $K_L = q^{-1}(L)$. Put $D_i^L = \{(\sigma, \tau) \in L^2 : \sigma_{-i} = \tau_{-i}\}$ and $\widetilde{D}_i = (K_L \times K_L) \times_{L^2} D_i^L$. The projection $\widetilde{D}_i \to D_i^L$ is onto because each of the two endpoints has a witness. A predicate on witness pairs descends exactly when it agrees on the fibres of this projection, by the same image argument applied to its characteristic function. When values descend, the improving-pair test is the inverse image of $\bar{v}_i p_1 \prec_i \bar{v}_i p_2$. Universal exclusion of those alternatives is the displayed Nash formula, by the coordinate argument of Theorem 5.39.

(3) **Keep unresolved outcomes on witnesses.** Without paired-test descent, existential and universal images along $\widetilde{D}_i \to D_i^L$ give the respective tests that some or every completing pair improves. An empty execution fibre contributes no point to $D$. For outcome-only evaluation $e_i = u_i \epsilon_Z$ this calculation recovers the usual numerical predicate whenever $u_i a$ descends.

□

**Construction 5.44** (Profitable deviations and equilibrium)**.**

(1) **Exclude profitable deviations.** On $D_i$ put $\mathsf{Prof}_i = (\delta_i^G)^* \mathsf{Better}_i$ and define

$$\mathsf{NE}_G = \bigwedge_{i \in N} \forall_{p_1 : D_i \to \Sigma} \neg \mathsf{Prof}_i. \tag{5.5}$$

(2) **Name the resulting predicate.** Write $\mathrm{NE}(G, \mathsf{Better})$ for this predicate, and $\mathrm{NE}(G, k)$ when the value presentation is understood.

**Theorem 5.45** (Derivation of pure strategic equilibrium)**.** *For every profile $\sigma$,*

$$\mathsf{NE}_G(\sigma) \iff \bigwedge_i \forall a_i \in \Sigma_i \; \neg \mathsf{Better}_i(g_\sigma, g_{(\sigma_{-i}, a_i)}). \tag{5.6}$$

*For utilities valued in $\mathbb{K}$ this is exactly $v_i(\sigma) \geq v_i(\sigma_{-i}, a_i)$ for all $i, a_i$. For measured play on a legal completed domain, the same derivation uses $\bar{v}_i(\sigma) \geq \bar{v}_i(\tau)$ on $D_i^L$ with $\bar{v}_i q = e_i r$. The arrow presentation in (5.6) applies when the paired comparison factors through the two realizations. For an incomplete preorder it expresses absence of a strictly better alternative, without asserting comparability of all alternatives.*

*Proof.*

(1) **Strategic comparison.** The fiber of $p_1$ at $\sigma$ is the set of profiles agreeing with $\sigma$ outside $i$. Use the coordinate bijection (4.2), then expand the quantifier and test. In a classical linear order, absence of a strict improvement is the displayed weak inequality.

□

**Construction 5.46** (The ordinary dependent object of equilibrium proofs)**.**

(1) **Form the dependent product.** Regard each $\mathsf{Prof}_i(\sigma, \tau)$ as its subterminal ordinary family, and write $D_i(\sigma)$ for the fibre of the deviation projection. Construct in $\mathsf{T}(\Sigma)$

$$\mathsf{Eqm}_G(\sigma) = \prod_{i \in N} \prod_{\tau \in D_i(\sigma)} (\mathsf{Prof}_i(\sigma, \tau) \to 0).$$

(2) **Identify its fibres.** Each factor and therefore the product is subterminal. Its fibre is inhabited precisely when there is no profitable deviation of any owner, so it is the subterminal object classifying $\mathsf{NE}_G$. This follows from Proposition 5.24, applied to the displayed dependent products.

(3) **Substitute the reference.** Reference substitution pulls back the complete deviation fibres, as in Theorem 5.33.

**Construction 5.47** (Improvement relation)**.**

(1) **Restrict comparison to controlled alternatives.** Put $\mathsf{Cmp}_i^G = (g \times g)^* \mathsf{Better}_i$ on $\Sigma^2$ and $\mathsf{PDev}_i = D_i \cap \mathsf{Cmp}_i^G$. For descended measured values take $\mathsf{Cmp}_i^G = (\bar{v}_i \times \bar{v}_i)^* \prec_i$ on the completed carrier. Intersecting it with the deviation span then gives the same modal safety formula below.

**Proposition 5.48** (Equilibrium as modal safety)**.**

$$\mathsf{NE}_G = \bigwedge_i \neg\langle \mathsf{PDev}_i \rangle \top. \tag{5.7}$$

*For a state map $s : W \to \Sigma$, testing and quantifying the reference pullback $\mathsf{Alt}_i(s)$ yields exactly $s^* \mathsf{NE}_G$.*

*Proof.*

(1) **Strategic comparison.** An existential successor of $\mathsf{PDev}_i$ is precisely a profitable $i$-deviation. The state-indexed assertion follows from Proposition 5.36 with the predicate $\neg \mathsf{Prof}_i$ on $D_i$.

□

**Proposition 5.49** (Preferences are not determined by interaction)**.** *No predicate depending only on the concurrent open-arena realization defines equilibrium for all comparison contexts.*

*Proof.*

(1) **Strategic comparison.** Use one owner and two distinct value-state realizations. Rank the first strictly above the second, and then reverse the ranking. The two equilibrium sets are different singletons, while the open-arena realization is unchanged.

□

### 5.6.3 Contextual decisions

**Proposition 5.50** (The open-decision calculation)**.** *For a decision presentation $\mathsf{Dec}_i(X, Y)$, fix an observed input $x \in X$ and a continuation $k : Y \to \mathbb{K}$. Define the comparison by*

$$P_{x,k}(f, f') \iff \exists y, y' \, (f\mathsf{pt}(x) = \mathsf{pt}(y) \wedge f'\mathsf{pt}(x) = \mathsf{pt}(y') \wedge k(y) < k(y')).$$

*Its equilibrium condition is*

$$\pi \in \mathrm{NE}(x, k) \iff \pi(x) \in \arg\max_{y \in Y} k(y).$$

*Proof.*

(1) **Policy-level comparison.** The response-state equation gives $\widehat{\pi}\mathsf{pt}(x) = \mathsf{pt}(\pi(x))$. Consequently the comparison on response policies is $k(\pi(x)) < k(\pi'(x))$. Every action $y$ occurs at $x$ under the constant policy with value $y$.

(2) **Policy-level comparison: completion.** Quantifying over policies is therefore equivalent to quantifying over actions. This recovers the contextual utility-maximizing decision of Ghani et al. (2018).

□

### 5.6.4 Exact recovery of pure normal-form games

**Definition 5.51** (Normal-form games)**.** A **normal-form game** consists of an agent set $N$, a policy set $\Sigma_i$ for each agent, and payoff maps $u_i : \prod_{j \in N} \Sigma_j \to \mathbb{K}$. It specifies a payoff for each joint policy profile without displaying the internal order of decisions. A policy in this representation may itself be a contingent plan from a larger protocol.

For a finite normal-form game, each pure action profile produces a singleton outcome. On these singletons the Hoare preorder agrees with the ordinary numerical payoff order, although it also compares other strategy states. The specialization below therefore tests the general selection construction against the usual pure Nash inequalities.

**Construction 5.52** (Realizing a finite normal-form game)**.**

(1) **Realize independent actions.** Let $A_i$ be finite nonempty action sets and $u_i : \prod_j A_j \to \mathbb{K}$ utilities. Player $i$ has a singly owned open game $G_i : I \to D(A_i)$ whose parameter $a_i$ realizes the one-event strategy $\mathsf{pt}_{A_i}(a_i)$ selecting $a_i$. Form

$$G = \bigotimes_{i \in N} G_i : I \longrightarrow O, \qquad O = \bigotimes_{i \in N} D(A_i).$$

At a profile $a$, the realization has one event for each agent, all independent. Its unique maximal configuration contains the tuple $a$.

(2) **Evaluate outcome sets.** The following evaluation is defined on the full hom-set $\mathcal{C}(I, O)$. Let $\mathrm{Out}(f)$ be the set of tuples labelling configurations of $f$ with one event on every output component. Define

$$k_i(f) = \{u_i(a) : a \in \mathrm{Out}(f)\}.$$

(3) **Order the sets of values.** Order subsets of $\mathbb{K}$ by the Hoare preorder

$$U \preceq_H V \quad \Longleftrightarrow \quad \forall u \in U \; \exists v \in V \; (u \leq v).$$

This is reflexive and transitive: reflexivity uses $v = u$, and transitivity composes the two witnesses with transitivity of $\leq$. On singleton sets it restricts to the order on $\mathbb{K}$. This particular extension to nondeterministic or incomplete outcomes is used only to make the context total; the extension provides a total comparison context for this construction.

**Theorem 5.53.** *The constructed concurrent open game has profile space $\prod_i A_i$, the ordinary unilateral deviations, and exactly the pure Nash equilibria of $(A_i, u_i)$. Its equilibrium assertion in the extended linear language is the ordinary Nash inequality.*

*Proof.*

(1) **Recovery of normal-form payoffs.** All factors except agent $i$'s factor in $G_i$ are singletons; consequently the composite has $\Sigma_i = A_i$. Its realized outcome set at $a$ is $\{a\}$. The measured presentation has $K = \prod_i A_i$, $p = \mathrm{id}$, $r(a) = \eta^T(a) = (0, a)$ and evaluator $e_i = u_i \epsilon$. Consequently $\bar{v}_i(a) = e_i(r(a)) = u_i(a)$ and $k_i(g_a) = \{\bar{v}_i(a)\}$. Strict comparison of these singleton values is strict comparison of the utilities.

(2) **Descent along the kernel pair.** The kernel pair $D_i$ fixes exactly the other agents' actions. Substituting these facts into Equation (5.6) gives precisely the ordinary Nash inequalities.

□

The diagram shows the classical case $\mathbb{K} = \mathbb{R}$. The pure action profile and its realized concurrent strategy determine identical payoffs.

$$\begin{array}{ccc} \prod_i A_i & \xrightarrow{\;g\;} & \mathcal{C}(I,O) \\ {\scriptstyle u_i}\downarrow & & \downarrow{\scriptstyle k_i} \\ \mathbb{R} & \xrightarrow[r\mapsto\{r\}]{} & \mathcal{P}(\mathbb{R}) \end{array}$$

The recovered pure-equilibrium predicate can be empty. In **matching pennies**, both agents choose $H$ or $T$. Player 1 receives 1 when the choices agree and $-1$ otherwise; agent 2 receives the opposite payoff. At an agreeing profile agent 2 improves by switching, and at a disagreeing profile agent 1 improves by switching. Thus every pure profile has a profitable unilateral deviation. A mixed strategy would instead be a probability distribution over $H, T$; existence results using such strategies belong to the probabilistic extension.

### 5.6.5 Boolean games

**Definition 5.54** (Boolean games and truth assignments)**.** A **finite Boolean game** consists of a finite partition $V = \coprod_{i\in N} V_i$ of propositional variables, with policy sets

$$\Sigma_i = 2^{V_i}, \qquad \Sigma = \prod_i \Sigma_i \cong 2^V,$$

and a propositional goal $\gamma_i \in \mathrm{Form}(V)$ for each agent. Here a **valuation** is a propositional truth assignment $\sigma : V \to 2$. Interpret the formula by its satisfaction subobject $[\![\gamma_i]\!] \hookrightarrow \Sigma$ and give agent $i$ the payoff map

$$u_i : \Sigma \longrightarrow \{0,1\}, \qquad u_i(\sigma) = \begin{cases} 1, & \sigma \models \gamma_i, \\ 0, & \sigma \not\models \gamma_i. \end{cases}$$

Thus satisfying a goal is strictly preferred to failing it. The constructed game is the Boolean-game instance of Bonzon et al. (2006), realized by the concurrent action protocol.

Truth of a finite propositional formula at a valuation is decidable by structural recursion, and quantification over the finitely many assignments to $V_i$ is likewise decidable. The case distinction in the proof below therefore does not invoke a general excluded-middle principle for the assertion language.

**Corollary 5.55.** *For a joint valuation $a$,*

$$a \in \mathrm{NE} \quad\Longleftrightarrow\quad a \models \bigwedge_{i\in N} (\gamma_i \vee \neg \exists V_i\, \gamma_i),$$

*where the quantified variables are changed while all other variables retain their values at $a$.*

*Proof.*

(1) **Strategic comparison.** If $\gamma_i$ already holds, the utility is 1 and cannot increase. If it fails, a profitable deviation exists precisely when some assignment of $V_i$ makes it true while the other variables remain fixed. Taking the conjunction over agents gives the result.

□

The Boolean-game formula is a predicate on complete valuations, not a linear operation on the move realizing a valuation. Consequently logical goals may be tested against resource-sensitive interactions without turning a truth value into a consumable resource.

### 5.6.6 Equilibrium under composition and reparameterization

**Theorem 5.56** (Selection after sequential and parallel wiring)**.** *Let $G : A \to B$ and $H : B \to C$ be concurrent open arenas. Fix a comparison context $P$ on $\mathcal{C}(A, C)$ and a reference profile $(\sigma, \tau)$. The feedback at the composed interface of owner $i$ is*

$$q_i^{H\circ G}((a,b))((a',b')) \iff P_i\big(h_{(\tau_{-i},b)}g_{(\sigma_{-i},a)},\ h_{(\tau_{-i},b')}g_{(\sigma_{-i},a')}\big).$$

*For every candidate response $(\sigma', \tau')$,*

$$\mathsf{BR}_{H\circ G}(P)((\sigma,\tau),(\sigma',\tau')) \iff \bigwedge_i \varepsilon_{\Sigma_i^G\times\Sigma_i^H}((\sigma_i',\tau_i'), q_i^{H\circ G}).$$

*The same construction for parallel wiring replaces composition of concurrent arrows by their tensor. Both laws respect reassociation, unit insertion, and interchange. They hold for arbitrary comparisons and repeated ownership of components.*

*Proof.*

(1) **Ownerwise alternatives.** The ownerwise product in the composite arena gives the agency carrier $\Sigma_i^G \times \Sigma_i^H$. Insert the fixed outsider coordinates in each component, apply the two plays, and then apply the concurrent composition map. Pulling $P_i$ back along both arguments of this function gives the displayed $q_i^{H\circ G}$ by Theorem 5.11.

(2) **Strategic comparison.** Applying the agent selection relation to that costate is precisely the definition of the composite best-response relation. This calculation uses all alternatives $(a', b')$ at once. Equivalently, Theorem 4.4 identifies their paired realizations with the component deviation pairs followed by concurrent composition.

(3) **Base-change comparison.** The tensor calculation uses the same product of policy coordinates and the tensor wiring function. Construction 3.14 gives the composite reference context, and Theorem 3.15 gives coherence of its cartesian lifts and feedback. Pulling back the internal selection predicates of Proposition 5.38 along equal costate terms therefore preserves those equations.

□

**Example 5.57** (One owner with complementary decisions)**.** Equilibrium must compare the complete policy of each owner, including the owner's coordinates in separate components. Consider one agent choosing bits $x$ and $y$ in two components, with value $xy$. At $(0, 0)$, changing either coordinate alone gives no improvement, while the joint change $(0, 0) \mapsto (1, 1)$ is profitable. Componentwise stationarity is therefore strictly weaker than the correct unilateral-deviation condition.

**Construction 5.58** (Component deviation subrelations)**.**

(1) **Separate the component deviations.** Under the canonical isomorphism of Theorem 4.4, define the subrelations

$$E_i^G = D_i^G \times \Delta_{\Sigma^H},$$
$$E_i^H = \Delta_{\Sigma^G} \times D_i^H$$

of $D_i^{H\circ G}$.

(2) **Identify the changing coordinate.** The first consists of deviations that change only the $G$-policy of owner $i$, and the second of deviations that change only its $H$-policy.

**Theorem 5.59** (Deviation-cover factorization)**.** *Suppose that for every owner $i$ the two component-deviation subrelations cover the composite deviation relation,*

$$D_i^{H\circ G} = E_i^G \cup E_i^H. \tag{5.8}$$

*Then, for every composite profile $(\sigma, \tau)$,*

$$(\sigma,\tau) \in \mathrm{NE}(H \circ G, \mathsf{Better}) \quad \Longleftrightarrow \quad \begin{cases} \sigma \in \mathrm{NE}(G, \mathsf{Better}^{G|\tau}), \\ \tau \in \mathrm{NE}(H, \mathsf{Better}^{H|\sigma}). \end{cases}$$

*The analogous statement holds for tensor and its pullback comparison contexts.*

*Proof.*

(1) **Strategic comparison.** If the composite profile is an equilibrium, every $G$-only deviation is a composite deviation with the $H$-profile held fixed. Its two realized interactions are

$$h_\tau \circ g_\sigma \quad \text{and} \quad h_\tau \circ g_{\sigma'},$$

so its comparison is exactly $\mathsf{Better}_i^{G|\tau}(g_\sigma, g_{\sigma'})$. Hence no profitable $G$-deviation exists. The same argument for $H$-only deviations gives the second component equilibrium.

(2) **Reverse implication.** Conversely suppose both component equilibrium predicates hold and take any composite $i$-deviation. By Equation (5.8) it is either in $E_i^G$ or in $E_i^H$. In the first case its $H$-coordinate is unchanged, so profitability would contradict the $G$-equilibrium with comparison $\mathsf{Better}^{G|\tau}$.

(3) **Strategic comparison.** In the second case profitability would contradict the $H$-equilibrium with comparison $\mathsf{Better}^{H|\sigma}$. Thus no profitable composite $i$-deviation exists. The tensor proof is identical with whiskering by composition replaced by tensoring with the fixed other component.

□

Disjoint active ownership is a classical sufficient criterion for the cover equation.

**Corollary 5.60** (Disjoint active owners; classical specialization)**.** *In ordinary classical* **Set***, let*

$$\mathrm{act}(G) = \{i : |\Sigma_i^G| > 1\}.$$

*If* $\mathrm{act}(G) \cap \mathrm{act}(H) = \varnothing$*, then*

$$(\sigma,\tau) \in \mathrm{NE}(H \circ G, \mathsf{Better}) \quad \Longleftrightarrow \quad \begin{cases} \sigma \in \mathrm{NE}(G, \mathsf{Better}^{G|\tau}), \\ \tau \in \mathrm{NE}(H, \mathsf{Better}^{H|\sigma}). \end{cases}$$

*The analogous equivalence holds for tensor.*

*Proof.*

(1) **Construction on policy coordinates.** Fix $i$. If the composite profile space is empty there is nothing to prove for its deviation relation. Otherwise the two policy factors of owner $i$ are inhabited.

(2) **Ownerwise alternatives.** Classical disjointness of the active-owner sets implies that at least one of $\Sigma_i^G, \Sigma_i^H$ has at most one element; being inhabited, that factor is a singleton. Hence every composite $i$-deviation changes only the other factor, so $D_i^{H\circ G} = E_i^G \cup E_i^H$. Apply Theorem 5.59.

□

**Corollary 5.61** (Selection by one agent across a composite interaction)**.** *For $G : A \to B$, $H : B \to C$, and a comparison context $P$ on $\mathcal{C}(A, C)$, the equilibrium condition at $(\sigma, \tau)$ is*

$$\bigwedge_i \ \forall a \in \Sigma_i^G \ \forall b \in \Sigma_i^H \ \neg P_i(h_\tau g_\sigma, h_{\tau[i\leftarrow b]} g_{\sigma[i\leftarrow a]}).$$

*For tensor composition, replace the two sequential composites by the corresponding tensor products. These are selections on the complete agent interfaces of Proposition 3.17.*

*Proof.*

(1) **Strategic comparison.** The alternatives of agent $i$ in the composite are exactly the pairs $(a, b) \in \Sigma_i^G \times \Sigma_i^H$. Substitute its composite realization in Theorem 5.39. Its universal quantifier over that product is the displayed pair of quantifiers. The tensor case is the same substitution using the tensor realization.

□

### 5.6.7 Additive reward composition

The deviation-cover condition is one route to equilibrium factorization. Another is a specified separable reward: if the evaluation is additive across the two components, a joint change improves the sum only if the component best-response conditions fail somewhere. This argument applies even when one owner controls both components, because its complete paired policy remains the unit of deviation.

**Construction 5.62** (Additive reward composition)**.**

(1) **Add the component rewards.** For two games with separate retained outcome records, form their parallel interaction and evaluate agent $i$ by $v_i(s, t) = u_i^G(s) + u_i^H(t)$. For disjoint measured branches with respective records $(c, z), (d, z')$, take additive intrinsic rewards and an additive price $\ell_i$. The writer pairs the outcomes at charge $c + d$, so

$$u_i^G(z) + u_i^H(z') - \ell_i(c + d) = (u_i^G(z) - \ell_i(c)) + (u_i^H(z') - \ell_i(d)).$$

This gives the displayed factorization for the evaluated component utilities. In general an evaluator of $c + d$ can couple the two choices.

(2) **Retain joint ownership.** This defines the additive reward composition, with owner $i$ controlling both $s_i$ and $t_i$.

**Proposition 5.63** (Additive equilibrium factorization)**.**

$$\mathrm{NE}(G \boxplus H) = \mathrm{NE}(G) \times \mathrm{NE}(H).$$

*Proof.*

(1) **Strategic comparison.** A profitable component deviation gives a profitable joint deviation by keeping the other coordinate fixed. Conversely, at two component equilibria each component's utility weakly decreases under its corresponding deviation, so their sum cannot increase under any joint deviation. This is a specified economic composition with a proof; it does not impose separability on an arbitrary pre-existing institutional context.

□

### 5.6.8 Fibred substitution and changes of strategic power

**Construction 5.64** (The deviation comparison of a reparametrisation)**.**

(1) **Construct the comparison map.** For $r : G \to H$ and $i \in N$, put

$$E_i(r) = \Sigma^G \times_{\Sigma^H, p_1} D_i^H, \qquad \lambda_i : D_i^G \to E_i(r), \quad (\sigma, \tau) \mapsto (\sigma, (\bar{r}\sigma, \bar{r}\tau)).$$

It is a map over $\Sigma^G$.

(2) **Compare its endpoints.** Its paired realization in $H$ equals the paired realization in $G$, because $g = h\bar{r}$. These maps commute with composition of reparametrisations: factoring through successive reference pullbacks or through their composite sends $(\sigma, \tau)$ to the same pair of images.

**Theorem 5.65** (Strategic transport and exact quotient transport)**.** *For every ownerwise reparametrisation $r : G \to H$ and common comparison context on their concurrent-arrow sort,*

$$\bar{r}^* \mathsf{NE}_H \leq \mathsf{NE}_G. \tag{5.9}$$

*For the ownerwise quotient presentations of Theorem 5.67, the comparison maps $\lambda_i$ are surjective and the displayed inclusion is equality. For a reference context $s : W \to \Sigma^G$, by contrast, substitution always gives the exact predicate*

$$s^* \mathsf{NE}_G = \bigwedge_i \forall_{d_i^s} \neg (\delta_i^G a_i)^* P_i, \quad D_i^s = W \times_{\Sigma^G, p_1} D_i^G, \tag{5.10}$$

*where $d_i^s$ and $a_i$ are the two pullback projections.*

A reparametrization carries each source deviation to a deviation at the same target reference.

$$\begin{array}{ccc} D_i^G & \xrightarrow{\lambda_i} & \Sigma^G \times_{\Sigma^H} D_i^H \\ {\scriptstyle p_1}\downarrow & & \downarrow \\ \Sigma^G & = & \Sigma^G \end{array}$$

*Proof.*

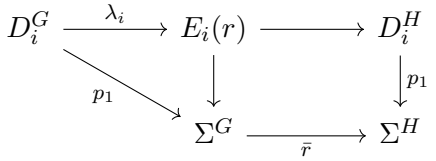
(1) **Strategic comparison.** An alternative in $D_i^G$ maps under $\lambda_i$ to an alternative of $H$ at the image reference, with identical realized interactions. If none of the latter alternatives improves the comparison, neither does an original alternative. Conjunction over $i$ proves Equation (5.9).

(2) **Lifting along the surjection.** For an ownerwise quotient $r$, take an element $(\sigma, (\bar{r}\sigma, \eta))$ of $E_i(r)$. Its outsiders already agree with those of $\bar{r}\sigma$. Surjectivity of $r_i$ supplies $b \in \Sigma_i^G$ with $r_i(b) = \eta_i$.

(3) **Lifting along the surjection: second comparison.** The profile $\tau = (\sigma_{-i}, b)$ is a lift. Thus every alternative in the reference pullback has an original representative with the same comparison. Universal quantification over these fibres gives equality. Only the displayed single alternative is lifted; no section of the quotient is selected.

(4) **Beck–Chevalley for reference reindexing.** Finally, $D_i^s$ is the cartesian lift of the deviation context along $s$. Beck–Chevalley from Theorem 5.22 gives $s^*\forall_{p_1} = \forall_{d_i^s} a_i^*$. Inverse image preserves negation and conjunction, proving Equation (5.10).

□




**Example 5.66** (A restriction is a change of game)**.** One owner has policies $a, b$ with values $0, 1$. The larger game's equilibrium predicate is $\{b\}$. Reparametrising along $\{a\} \hookrightarrow \{a, b\}$ constructs the one-policy game, whose equilibrium predicate is $\{a\}$; the inverse image of $\{b\}$ is empty. Reference substitution along the same underlying map instead retains the alternative $b$ in $D_i^s$, and correctly returns the empty predicate. The two operations have the same reference map but different deviation contexts.

### 5.6.9 Changing the presentation of available strategies

**Theorem 5.67.** *Let $f_i : \Sigma_i^G \twoheadrightarrow \Sigma_i^H$ be surjective maps with $g_\sigma = h_{f(\sigma)}$. In the same comparison context,*

$$\mathrm{NE}(G, \mathsf{Better}) = f^{-1}(\mathrm{NE}(H, \mathsf{Better})).$$

*In a value-presented context the same equation is written with $k$ in place of* Better*.*

*Proof.*

(1) **Lifting deviations along surjective maps.** Apply Lemma 4.11 to the ownerwise surjection $f$, with full legal domain. It lifts every unilateral target deviation at a given reference while fixing all outsider policies. Since both realized endpoints agree under $g = hf$, the predicates of profitable deviations agree.

(2) **Lifting along the surjection.** Universal negation gives the stated equality of Nash predicates. The lift chooses preimages only for the finitely many coordinates varied by the deviation.

□

*Remark* 5.68 (Restriction of available policies). For injective parameter maps the useful implication is different. A profile that is an equilibrium in a larger strategy space remains an equilibrium when the available deviations are restricted and that profile is retained. The converse fails because enlarging a strategy space can create a profitable deviation. The 2-cell language therefore distinguishes redundant descriptions from genuine changes in strategic power.

**Example 5.69** (Complementary commitments)**.** Two owners independently choose 0 or 1. Tensor their value-state presentations and evaluate the completed record by

$$u(0,0) = (0,0), \quad u(0,1) = (0,-1), \quad u(1,0) = (-1,0), \quad u(1,1) = (1,1).$$

The profile $(0,0)$ has only unilateral deviations giving the deviator $-1$; at $(1,1)$ a unilateral withdrawal gives 0. Both are equilibria. At either off-diagonal profile the contributor improves by withdrawing. Hence the equilibrium set is exactly $\{(0,0),(1,1)\}$.

## 5.7 Situated agent presentations

### 5.7.1 Declared presentations and reference states

**Definition 5.70** (Situated presentation)**.** A situated presentation is displayed data

$$\mathsf{Ctx} = (a, L \hookrightarrow \Sigma, s : W \to L, (o_i : W \to O_i)_{i\in N}, (P_i)_{i\in N}).$$

Here $a$ is an owned agent presentation, $L$ is its legal policy domain, $s$ selects the reference profile, $o_i$ specifies observation, and $P_i$ is its comparison interface, typed in Section 5.8. The state space may retain histories, private inputs, policies, and institutional records. Each observation is corestricted to its effective image whenever local states are quantified.

**Construction 5.71** (Situated and legal presentations generated by a protocol)**.**

(1) **Realize the finite protocol.** Let $U$ be the product of the external-input carriers of a finite causal decision protocol, $V$ its node set, and $\Sigma$ its complete owner profile space. The causal-height recursion gives the unique map

$$r : U \times \Sigma \longrightarrow \prod_{v\in V} M_v, \qquad W_{\mathrm{prot}} = \{(u,\sigma,a) : a = r(u,\sigma)\} \cong U \times \Sigma.$$

Let $s : W_{\mathrm{prot}} \to \Sigma$ be projection and $z : W_{\mathrm{prot}} \to Z$ the specified outcome-record map.

(2) **Read the observations.** Nodewise observations are the maps of Theorem 10.11: they are evaluated on their *own causal prefix* and use only actually available incoming wires. They need not coincide with one owner's later, accumulated observation record. Any final owner observation used in a situated presentation is the image of the specified actually delivered end-of-protocol signal; it is not the unrestricted product of all its earlier and later signals.

(3) **Form the acceptance and policy domains.** For an exhibited institutional acceptance predicate $Q \hookrightarrow Z$, first form the accepted-*state* subobject $P = z^*Q \hookrightarrow W_{\mathrm{prot}}$. There are canonical, distinct policy-domain constructions:

$$L_\exists = \exists_s P = \mathrm{im}(s|_P) \hookrightarrow \Sigma,$$
$$L_\forall = \forall_s P \hookrightarrow \Sigma.$$

The restricted map $s|_P$ factors canonically into a surjection $P \twoheadrightarrow L_\exists$ and its image inclusion; this gives the situated presentation of *accepted executions*. Pulling back $s$ along $L_\forall \hookrightarrow \Sigma$ gives $W_\forall \to L_\forall$, the presentation of policies whose *every* external-input execution passes the acceptance test.

(4) **Retain certificates.** For exhibited certificates $k : K \to W_{\mathrm{prot}}$, the policy domain is $L_K = \mathrm{im}(sk)$, retaining $K$ as its witness carrier. None of these subobjects is confused with the accepted-state predicate $P$. Preference comparisons and institutional acceptance rules remain separately supplied economic and normative inputs.

**Proposition 5.72** (Universal characterization of protocol legality)**.** *$L_\exists$ consists exactly of profiles having at least one accepted execution. The subobject $L_\forall$ is the greatest $J \hookrightarrow \Sigma$ such that all protocol states with reference in $J$ lie in $P$. These constructions are stable under substitution of ordinary protocol parameters along cartesian squares.*

*Proof.*

(1) **Adjunction and transposition.** The first assertion is the image formula $\sigma \in L_\exists \iff \exists w\,(s(w) = \sigma \wedge P(w))$. For the second, the indexed adjunction gives

$$s^*J \subseteq P \iff J \subseteq \forall_s P.$$

Taking $J = L_\forall$ yields its desired acceptance and greatestness. Beck–Chevalley for the image/existential and the right dependent adjoint supplies substitution, with the same source pullback as the protocol's reference map. No execution representative is selected.

□

The accepted execution space factors onto precisely the existentially legal profiles.

$$\begin{array}{ccc} P & \twoheadrightarrow & L_\exists \\ \downarrow & & \downarrow \\ W_{\mathrm{prot}} & \xrightarrow[s]{} & \Sigma \end{array}$$

A situated presentation depends on declared owner and policy data. It does not extract a new reusable owner presentation from an arbitrary linear agent variable. Instead, an agent executes within a state context, whose displayed signature determines which control and observation tests are meaningful.

The need for the economic fields can be checked independently of the categorical construction. Reverse the preference on the same two-choice concurrent process: equilibrium changes, although realization does not. Change the permitted subobject from $E$ to $\varnothing$ for an unchanged execution span: permission changes without changing the underlying process. Replace a constant observation by an injective observation of the same worlds: knowledge changes without changing execution. Preferences, permissions, and nondelivered signals therefore cannot be reconstructed from a bare operational arrow. Where the information is generated by a protocol, Construction 5.71 instead derives its associated maps by image, pullback, and quotient. No additional agency equation is needed.

These constructions occupy three judgments: ordinary terms in $\mathcal{T}(W)$, linear terms in $\mathcal{L}(W)$, and assertions in $\mathcal{P}(W)$. The entailment $W \mid \Phi \vdash \varphi$ means $\bigcap_{\psi\in\Phi}[\![\psi]\!] \subseteq [\![\varphi]\!]$. Logical conjunction, implication, and ordinary quantification act on assertions; they do not supply a tensor diagonal to a linear resource. The concrete predicate fibres are Boolean, while proofs using only the adjunctions and Heyting operations remain constructive.

## 5.8 Legal feedback, best response and equilibrium

### 5.8.1 Comparison feedback and its pullback

The legal version of selection must compare only legal alternatives. Let $H$ denote the retained outcome or interaction carrier, let $g : L \to H$ be its realization, and let $Q_i \hookrightarrow H \times H$ classify strict improvement in the second coordinate. Pulling this relation back along the paired realization supplies the improvement predicate on legal profiles.

$$P_i(\sigma, \tau) \iff Q_i(g\sigma, g\tau).$$

The realization $g$ and comparison $Q_i$ may be indexed by outsider policies or by an ordinary state context; the calculation is the same in each fibre. Evaluating a component inside a larger interaction first supplies its contextual realization into $H$. Inverse image of the joint comparison then gives the component feedback, compatibly with further wiring.

**Proposition 5.73** (Compositional comparison feedback)**.** *If $k : H \to Z$ is a further contextual realization and $Q \subseteq Z^2$, then*

$$(g \times g)^*(k \times k)^* Q = ((kg) \times (kg))^* Q.$$

*Hence evaluating a component in a composite context agrees with successive contextual evaluations, retaining the entire joint predicate.*

*Proof.*

(1) **Equality of the two constructions.** Both sides are $((kg)\times(kg))^*Q$ by the functoriality of inverse image proved in Theorems 3.12 and 5.11. Their common test at $(\sigma, \tau)$ is $Q(kg\sigma, kg\tau)$, including all joint alternatives.

□

The backward comparison must remain joint: separate marginal predicates would lose improvements that depend on complementary changes at decisions controlled by the same owner.

### 5.8.2 Selection and best response

For a legal own-choice fibre $S_i^L(u) = \{v : \Sigma_i \mid (u, v) \in L\}$, define the canonical selection predicate

$$\mathrm{Sel}_i(u, v) \iff v \in S_i^L(u) \ \wedge \ \forall v' \in S_i^L(u)\, \neg Q_i(g(u, v), g(u, v')).$$

The predicate therefore retains exactly the legal choices admitting no strictly improving legal alternative. Its construction uses only comprehension, comparison feedback, and dependent universal quantification, so it is an assertion in the established doctrine.

**Definition 5.74** (Best response and equilibrium)**.** For a legal reference profile $\sigma$, define

$$\begin{aligned} \mathsf{BR}_i(\sigma, v) &\iff v \in S_i^L(\sigma_{-i}) \ \wedge \ \forall u \in S_i^L(\sigma_{-i})\, \neg Q_i(g(\sigma[i := v]), g(\sigma[i := u])), \\ \mathsf{NE}(\sigma) &\iff \bigwedge_{i \in N} \mathsf{BR}_i(\sigma, \sigma_i). \end{aligned}$$

The same predicates at a situated state are obtained by substituting $\sigma = s(w)$.

**Proposition 5.75** (Equilibrium from owned alternatives)**.**

$$\mathsf{NE}(\sigma) \iff \bigwedge_{i \in N} \forall \tau\, (\sigma D_i^L \tau \Rightarrow \neg P_i(\sigma, \tau)).$$

*It is therefore an assertion constructed from the agent's displayed control, legality, realization, and comparison interfaces.*

*Proof.*

(1) **Predicate calculation.** For fixed $\sigma$, the map $u \mapsto \sigma[i := u]$ is a bijection from $S_i^L(\sigma_{-i})$ to the target fibre of $D_i^L$ at $\sigma$. Under this bijection the two comparisons are the same pullback of $Q_i$. Dependent universal quantification along the bijection proves the equality.

□

Write $\Pi = \mathsf{Pol}_{o_i}(F_i)$ for the uniformly legal informed policies. For $\pi, \rho \in \Pi$ let

$$Q_w(\pi, \rho) = Q_i(g(s(w)[i := \pi(o_i w)]), g(s(w)[i := \rho(o_i w)])).$$

Two selection predicates on this same carrier have different meanings:

$$\mathsf{EverywhereBest}(\pi) \iff \forall w \in W\ \forall \rho \in \Pi\ \neg Q_w(\pi, \rho),$$
$$\mathsf{StrictlyUndominated}(\pi) \iff \neg\exists \rho \in \Pi\ \forall w \in W\ Q_w(\pi, \rho).$$

The first excludes improvement at each reference. The second excludes a single informed competitor that strictly improves at every reference. For two indistinguishable worlds and two always legal constant responses with payoff vectors $(1, 0)$ and $(0, 1)$, both responses satisfy the second predicate and neither satisfies the first. The two references can realize different outsider coordinates while giving $i$ the same observation.

**Example 5.76** (Jointly controlled decisions)**.** One owner controls two bits $x, y$, and values the outcome by $xy$. At $(0, 0)$, changing either coordinate in isolation yields no improvement; changing the complete owned policy to $(1, 1)$ does. The product law $\Sigma_i^{a;b} = \Sigma_i^a \times \Sigma_i^b$ makes the latter an ordinary unilateral deviation by that owner. Consequently equilibrium of the composite is not established by testing isolated decision positions.

# Part III
# Information, institution, and epistemic dynamics

## 6 Observation, Perspective, and Knowledge

A controller can distinguish histories through the observations supplied at its decision interface. Following the local-state account of Halpern and Fagin (1989), we group histories with the same observation. A response factors through this quotient exactly when the controller can choose it from the signal alone. The same quotient determines what the owner knows, and its action on dependent linear families keeps track of resources available from that perspective. For several owners, the distinction between pooling signals and actually delivering the pooled result will become decisive.

### 6.1 Observation and admissible policies

First take the image of the raw signal map. Two histories lie over the same point of this image precisely when the decision interface receives the same signal. Kernel-pair descent identifies the responses implementable at that interface. Section A.2.14 uses the same fibres to decide which assertions the owner knows.

### 6.2 Observation-based responses of an agent

**Construction 6.1** (Observation quotient and admissible policies)**.**

(1) **Form the observation quotient.** Let $H$ be a set of histories available at a decision and let $\widetilde{o} : H \to Y$ be the raw signal supplied by the protocol. Construct the observation object by the image factorization in **Set**,

$$H \xrightarrow{o} O \hookrightarrow Y, \qquad O = \widetilde{o}[H],$$

so $o$ is the effective quotient obtained by corestricting the raw signal to its image. Equivalently, if indistinguishability is supplied directly as an equivalence relation $R \rightrightarrows H$, one may take its coequalizer $H \twoheadrightarrow H/R$.

(2) **Describe its policies.** Available policies with action set $A$ are then functions $\pi : O \to A$. When $H, O, A$ are finite, these functions have the concurrent realization of Proposition 1.16.

For a concrete example take histories $H = \{h_1, h_2, h_3\}$, observations $O = \{o, o'\}$ with $o(h_1) = o(h_2) = o$ and $o(h_3) = o'$, and actions $A = \{0, 1\}$. An admissible history response is determined by two choices, one for $o$ and one for $o'$, so there are four such policies. The response assigning 0 at $h_1$ and 1 at $h_2$ is inadmissible for this observation. The raw signal's image removes unused signal labels without changing any decision the agent can make.

**Proposition 6.2** (Admissibility as effective descent)**.** *Let*

$$H \times_O H \underset{p_1}{\overset{p_2}{\rightrightarrows}} H \xrightarrow{o} O$$

*be the kernel pair of the constructed observation quotient. For every action set $A$, precomposition gives an equalizer*

$$A^O \xrightarrow{o^*} A^H \underset{p_1^*}{\overset{p_2^*}{\rightrightarrows}} A^{H \times_O H}.$$

*Consequently a response $f : H \to A$ is implementable using only the observation $o$ exactly when it equalizes the observation kernel pair, equivalently when there is a unique $\pi : O \to A$ with $f = \pi o$.*

*Proof.*

(1) **Descent along the kernel pair.** The image factorization makes $o$ surjective. By effective regularity of **Set** it is the coequalizer of its kernel pair. A map $f : H \to A$ equalizes the two projections precisely when

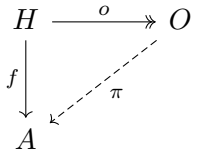
$$o(h) = o(h') \Longrightarrow f(h) = f(h').$$

The coequalizer universal property therefore gives a unique factor $\pi : O \to A$ with $f = \pi o$.

(2) **Representation and Yoneda decoding.** Applying the contravariant representable $\mathbf{Set}(-, A)$ to this coequalizer is exactly the displayed equalizer. Thus "admissible" introduces no further policy datum: it names this descent property.

□

Admissible response strategies factor through the effective observation quotient.




The left square expresses the information restriction; for finite data, Proposition 1.16 realizes the same factorization by concurrent response strategies. The policy responds after its actual observation event. Other strategies on the arenas may stall or behave differently; effective descent classifies the specified total response family. In particular, a decision cannot depend on a signal whose event has not occurred in its causal past.

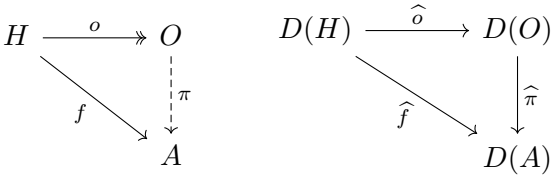



Figure 4: An information restriction is effective descent through the constructed observation quotient; finite response protocols realize the factorization concurrently.

### 6.3 An individual agent's knowledge from observation

**Definition 6.3** (Observations, indistinguishability, and knowledge)**.** An **observation** is information actually made available to an agent. States yielding the same observation are **indistinguishable** to that agent. It knows a predicate when the predicate holds at every indistinguishable state.

Let $W$ be a set of economic states and $o_i : W \to O_i$ an observation map. Its epistemic relation is the kernel-pair span

$$W \xleftarrow{p_1} W \times_{O_i} W \xrightarrow{p_2} W.$$

Knowledge is the corresponding universal span modality:

$$K_i P = [\ker(o_i)]P = \forall_{p_1} p_2^* P. \tag{6.1}$$

For example, observing whether a public total is zero need not reveal the individual contributions that produced it. The resulting knowledge concerns everything forced by that observation, not a probability assigned to the unobserved states.

**Definition 6.4** (Factivity and introspection)**.** A knowledge operator $K_i$ is **factive** when $K_i P \leq P$ for every predicate $P$. It has **positive introspection** when $K_i P \leq K_i K_i P$, and **negative introspection** when $\neg K_i P \leq K_i \neg K_i P$. These conditions say, respectively, that knowledge implies truth, that an agent knows what it knows, and that it knows when it does not know.

Because an observation kernel pair is an equivalence relation, it also determines the familiar factivity and introspection conditions. The same reasoning applies after epistemic action update when the updated relation arises from the effective observation quotient of Proposition 9.22.

**Proposition 6.5** (Factivity and introspection)**.** *For an observation map, $K_i P \leq P$, $K_i P \leq K_i K_i P$, and $\neg K_i P \leq K_i \neg K_i P$.*

*Proof.*

(1) **Descent along the kernel pair.** The observation kernel pair is reflexive, so the reference state is an alternative. Every related state has the same equivalence class. Truth of $P$ throughout that class is therefore invariant throughout it, as is its negation.

(2) **Predicate calculation.** These arguments use the implication and negation operations of the predicate doctrine.

□

Two kernel pairs appear in strategic reasoning. Fixing outsider policies compares changes under a coalition's control. Fixing an observation compares states the owner cannot distinguish. The modalities for intervention and for knowledge therefore quantify over different pairs, even when they are predicates on the same state carrier.

**Corollary 6.6** (Observation quotients classify invariant predicates)**.** *For a constructed observation quotient $o : W \twoheadrightarrow O$, put $K_o = o^*\forall_o$ and $M_o = o^*\exists_o$. Knowledge is the idempotent comonad $K_o$, and possibility is the idempotent monad $M_o$. For a predicate $P$, the following are equivalent: it is the pullback of a unique predicate on $O$; it is invariant on* $\ker(o)$*; $K_oP = P$; and $M_oP = P$. Moreover $K_oP \leq P \leq M_oP$ and*

$$\mathrm{Coalg}(K_o) \cong \mathsf{Pred}(O) \cong \mathrm{Alg}(M_o).$$

*Proof.*

(1) **Adjunction and transposition.** The relation factorization identifies the knowledge and possibility operators with the displayed composites. Apply Theorem 5.27 and effective predicate descent.

□

The quotient plays two roles that are now visibly linked. Its maps into action sets classify implementable responses by Proposition 6.2; its descended predicates classify observation-invariant assertions by Corollary 6.6. The operational and epistemic interpretations therefore arise from one effective quotient, not from separate information primitives.

**Proposition 6.7** (An agent's responses and observation-local assertions)**.** *For an effective observation quotient $o_i : W \twoheadrightarrow \Omega_i$ and an action set $M$, there are canonical identifications*

$$M^{\Omega_i} \cong \{f : W \to M : \ker(o_i) \subseteq \ker(f)\}, \qquad \mathsf{Pred}(\Omega_i) \cong \{P \in \mathsf{Pred}(W) : K_iP = P\}.$$

*The first identifies admissible responses; the second identifies assertions local to the agent's current observation.*

*Proof.*

(1) **Observation and knowledge.** The first bijection is precomposition with $o_i$ and its inverse is the unique factorization of Proposition 6.2.

(2) **Lifting along the surjection.** For the second, $o_i^*B$ is constant on every observation fibre, hence equals its own knowledge predicate. If $K_iP = P$, every fibre meeting $P$ lies entirely in $P$, so $P = o_i^{-1}(o_i[P])$. Surjectivity makes the representing subset unique.

(3) **Observation and knowledge.** These maps are exactly the predicate descent and knowledge identifications already constructed.

□

Observation-local assertions live on the common situated carrier $W$ alongside assertions about other agents and later events. At a particular decision, however, the relevant quotient is that decision's actual history-to-signal map. Applying the same factorization at the correct causal stage yields precisely the responses available there.

## 6.4 Qualitative belief and common knowledge

**Definition 6.8** (Possibility model and belief)**.** A **possibility model** records which states each agent considers possible. An agent **believes** a predicate when it holds at every state it considers possible. The actual state need not be among them, so belief need not imply truth. If the possibility set is empty, universal belief is vacuous; no nonemptiness condition is implicit.

A possibility model over a profile space $\Sigma$ consists of a set $W$, a map $s : W \to \Sigma$, and relations $R_i \hookrightarrow W^2$. Put $P_i(w) = \{w' : wR_iw'\}$, $B_i = [R_i]$, and $E(P) = \bigwedge_i B_iP$. No relational laws are implicit. Observation maps construct the special case $R_i = \ker(o_i)$.

**Definition 6.9** (Common belief and common knowledge)**. Common belief** requires belief by every agent, belief by every agent that all agents believe, and all further finite iterations. For observation relations, **common knowledge** also includes truth at the current state. The two fixed points below distinguish iterated belief from truth along all finite chains, including the chain of length zero.

Define

$$\mathsf{CB}(P) = \nu X\, E(P \wedge X), \qquad C_P = \nu X\, (P \wedge E(X)). \tag{6.2}$$

When the relations are observation kernel pairs, call $C_P$ the **common-knowledge predicate** $\mathsf{CK}(P)$. Let $R = \bigcup_i R_i$ and let $R^*$ be the reflexive transitive closure constructed from finite $R$-paths, including paths of length zero.

**Proposition 6.10** (Truth, common belief, and finite paths)**.** *For every finite owner list, including the empty list,*

$$C_P = P \wedge \mathsf{CB}(P) = [R^*]P.$$

*In particular* $\mathsf{CK}(P) \leq P$*.*

*Proof.*

(1) **Coinduction and bisimulation.** The fixed-point equation for $C_P$ gives $C_P \leq P$ and $C_P \leq E(C_P) = E(P \wedge C_P)$, so coinduction gives $C_P \leq P \wedge \mathsf{CB}(P)$.

(2) **Reverse implication.** Conversely, with $B = \mathsf{CB}(P)$, the predicate $P \wedge B$ equals $P \wedge E(P \wedge B)$ and is fixed for the operator defining $C_P$. For the path description, any post-fixed point of $X \mapsto P \wedge E(X)$ satisfies $P$ after every finite path by induction on path length.

(3) **Reverse implication: completion.** Conversely $[R^*]P \leq P$ by the zero-length path and is closed under one further $R$-step, so it is post-fixed. Coinduction proves equality. For $N = \varnothing$, these formulas give $\mathsf{CB}(P) = \top$ and $C_P = P$, as required.

□

**Theorem 6.11** (Common knowledge as a coreflection)**.** *For observation relations* $R_i = \ker(o_i)$*, the relation* $E = (\bigcup_i R_i)^*$ *is an equivalence relation. Its quotient* $c : W \twoheadrightarrow W/E$ *satisfies*

$$\mathsf{CK} = c^* \forall_c, \qquad \mathrm{Coalg}(\mathsf{CK}) = \bigcap_i \mathrm{Coalg}(K_i).$$

*Thus common knowledge is the greatest jointly observation-invariant predicate below the given predicate.*

*Proof.*

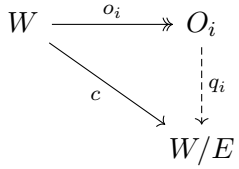
(1) **Fixed-point argument.** Reflexivity and transitivity hold by the path construction. Reversing each edge of a finite path proves symmetry. A predicate is $E$-invariant iff it is invariant under every $R_i$: one direction uses the inclusions and the other is induction on path length.

(2) **Adjunction and transposition.** The finite-path characterization gives $\mathsf{CK} = [E] = c^* \forall_c$. Apply Theorem 5.27 to $c$ and to the image factorization of each $o_i$. Their fixed predicates are exactly the invariant predicates just identified.

(3) **Ownerwise alternatives.** If the owner list is empty, $E$ is equality and the same proof gives the identity coreflection.

□

Each observation quotient factors through the equivalence quotient defining common knowledge.




**Corollary 6.12** (Common knowledge after epistemic action update)**.** *Let $R_i = \ker(o_i)$ and $S_i = \ker(b_i)$ in the action model of Section 9.1.3. On the updated carrier $U$, put $c_{\mathcal{E}} : U \twoheadrightarrow U/{\sim}$, where $\sim$ is generated by the union of its updated knowledge relations. Then*

$$\mathsf{CK}_{\mathcal{E}}(Q) = \left[\left(\bigcup_i R_i^{\mathcal{E}}\right)^*\right] Q = c_{\mathcal{E}}^* \forall_{c_{\mathcal{E}}} Q.$$

*For the pointed action $e$, the corresponding assertion at the old state sort is $[\mathcal{E}, e]\mathsf{CK}_{\mathcal{E}}(Q)$. For a singleton public action, this recovers the common-knowledge specialization of Section 9.1.1.*

*Proof.*

(1) **Observation and knowledge.** Proposition 9.22 constructs each updated kernel relation. Their reflexive transitive union is an equivalence relation, and Theorem 6.11 identifies its box with the displayed quotient coreflection. Equation (9.10) transports the result through the chosen action component; the singleton case is Theorem 9.16.

□

**Theorem 6.13** (Categorical validity of agency operations)**.** *The ordinary dependent fibres and their subterminal fragment recover $\mathsf{P}_G$, its quantifier adjunctions, and comprehension. The relation-action and comprehension-update equations hold in these fibres, as do the order-theoretic induction and coinduction properties of positive fixed-point operators. For external parameters $\Gamma$ and a fixed state sort $X$, pointwise parameterized fixed points commute with ordinary substitution in $\Gamma$.*

*Proof.*

(1) **Base-change comparison.** Theorem 5.22 supplies the quantifier adjunctions, equality, comprehension, and Beck–Chevalley. Construction 5.23 supplies dependent types; Proposition 7.6 supplies the internal modal typing.

(2) **Fixed-point argument.** The modal and update equations were proved using these operations, and Theorem 5.33 identifies their operational tests with substitution of concurrent realization terms. For fixed relations the ordinary polarity induction proves monotonicity. For the extended contextual grammar, Lemma A.5 gives the corresponding induction after expanding the changing domains. The fixed-point theorem then applies.

(3) **Fixed-point argument: second comparison.** For a family $F_\gamma$ on $\mathsf{Pred}(X)$, the operator on $\mathsf{Pred}(\Gamma \times X)$ acting fiberwise has least fixed point with fiber $\mu F_\gamma$: it is fixed fiberwise and lies below every fixed predicate fiberwise. The greatest case is dual. Substitution in $\Gamma$ merely selects these same fibers.

(4) **Functoriality and substitution.** Changes to the state carrier are handled by the separate transport results, not by this parameter-substitution assertion.

□

*Remark* 6.14 (Relation to strategy logics). Explicit quantification over complete policies places these assertions in contact with strategy logics such as Mogavero et al. (2012). Here the interpretation is supplied by concurrent open-arena realizations and the predicate doctrine. The soundness theorem concerns this interpretation; it does not import model-checking results for a different class of models.

A completed charge $m : K \to C$ can enter the institutional evaluator even when an owner receives only $o_i : K \to O_i$. Corestrict $o_i$ to its effective image. The owner can read the charge from that observation exactly when $m$ is constant on $\ker(o_i)$, by effective descent. A public or private charge report instead gives a new observation $\langle o_i, m\rangle$ to the recipients and the corresponding kernel relation. Preference evaluation alone supplies no such report.

## 6.5 Dependent observation descent

The quotient already classifies individual observable responses and predicates. To compare different presentations and information refinements, we must also transport whole dependent families and their witnesses. Effective descent provides this transport; Morita comparisons identify presentations of the same observations, while kernel inclusions order observation strength. The subsequent pooled-policy comparison separates what multiple agents could compute from a delivered joint signal from what they can compute independently.

### 6.5.1 Descent in the dependent-type fibration

**Construction 6.15** (The slice adjunctions)**.**

(1) **Define the fibrewise functors.** For a map $q : X \to Q$, regard an object of $\mathbf{Set}/X$ as a family $(E_x)_{x\in X}$. Define

$$(q_!E)_u = \coprod_{q(x)=u} E_x, \qquad (q^*F)_x = F_{q(x)}, \qquad (q_*E)_u = \prod_{q(x)=u} E_x.$$

(2) **Construct their adjunctions.** The universal properties of disjoint union and product give natural bijections proving $q_! \dashv q^* \dashv q_*$. The empty sum and empty product are included. On maps these functors act componentwise.

(3) **Form the monad and comonad.** Write $T_q = q^*q_!$ and $G_q = q^*q_*$ for the induced monad and comonad on $\mathbf{Set}/X$.

**Definition 6.16** (Kernel groupoid and its actions)**.** For a surjection $q : X \twoheadrightarrow Q$, the **presentation groupoid** $R_q$ has objects $X$ and an arrow $(x, y) : y \to x$ exactly when $q(x) = q(y)$. Its identity, inverse, and composition are $(x, x)$, $(x, y)^{-1} = (y, x)$, and $(x, y)(y, z) = (x, z)$. An action of $R_q$ is a family $E_x$ with maps $\tau_{xy} : E_y \to E_x$ satisfying

$$\tau_{xx} = \mathrm{id}, \qquad \tau_{xy}\tau_{yz} = \tau_{xz}.$$

An equivariant map is a family of functions commuting with all these maps. The resulting category is denoted $\mathrm{Act}(R_q)$.

**Theorem 6.17** (Effective descent as algebra and coalgebra)**.** *There are canonical equivalences*

$$\mathrm{Alg}(T_q) \simeq \mathrm{Act}(R_q) \simeq \mathbf{Set}/Q \simeq \mathrm{Coalg}(G_q).$$

*The comparison from* $\mathbf{Set}/Q$ *is pullback with its canonical transport. It requires no chosen section of* $q$*.*

*Proof.*

(1) **Unit and identity laws.** The unit of $T_q$ includes $E_x$ in the $x$-summand, and its multiplication forgets the middle index in a twice-indexed sum. Consequently a $T_q$-algebra is exactly a family $\tau_{xy}$ with the displayed identity and composition equations. These equations imply that $\tau_{xy}$ and $\tau_{yx}$ are inverse. Algebra morphisms are exactly equivariant maps.

(2) **Construction on states and events.** Given an action, define

$$\bar{E}_u = \left( \coprod_{q(x)=u} E_x \right) / \sim, \qquad (x,e) \sim (y,e') \iff e = \tau_{xy}(e').$$

The action laws prove that this is an equivalence relation. For each $x$, the map $E_x \to \bar{E}_{q(x)}$, $e \mapsto [(x,e)]$, is injective by $\tau_{xx} = \mathrm{id}$ and surjective because $[(y,e')] = [(x, \tau_{xy}(e'))]$.

(3) **Descent along the kernel pair.** These bijections intertwine the action with the identity transports on the pullback of $\bar{E}$. Equivariant maps descend by $[(x,e)] \mapsto [(x, f_x(e))]$.

(4) **Reverse implication.** Conversely, pulling back a family on $Q$ and applying this quotient returns the original family: the map $[(x,e)] \mapsto e$ is a bijection on each fiber, with surjectivity supplied by $q$. The displayed maps are natural, establishing the equivalence.

(5) **Structure of perspective coalgebras.** A $G_q$-coalgebra is a family $c_x : E_x \to \prod_{q(y)=q(x)} E_y$. Its counit law says that the $x$-component of $c_x$ is the identity. Its coassociativity says that the component maps from $E_x$ to $E_y$ and then $E_z$ compose to the component from $E_x$ to $E_z$.

(6) **Equivalence with groupoid actions.** Thus its components are precisely the transports $\tau_{yx}$ above; coalgebra morphisms are again equivariant maps.

(7) **Conclusion of the calculation: completion.** This proves the remaining equivalence. The slice adjunctions follow the categorical-logical account of Jacobs (1999); the fiber formulas give the descent used here.

□

$$\mathbf{Set}/Q \underset{q_*}{\overset{q^*}{\underset{q_!}{\rightleftarrows}}} \mathbf{Set}/X \qquad \mathrm{Alg}(T_q) \xrightarrow{\simeq} \mathrm{Act}(R_q) \xrightarrow{\simeq} \mathbf{Set}/Q$$

The same observable information can have different raw-history presentations. Suppose $X \twoheadrightarrow Q$ and $Y \twoheadrightarrow Q$ present the quotient $Q$. The pullback of pairs $(x,y)$ with the same image gives the comparison bibundle. Its principal actions ensure that policies and dependent families descended from $Q$ do not depend on a chosen history presentation; no additional signal becomes available to an agent.

**Definition 6.18** (Principal comparison and Morita equivalence)**.** Fix groupoids $G$ and $H$ in **Set**. A $(G,H)$-**bibundle** consists of anchor maps $G_0 \xleftarrow{\alpha} P \xrightarrow{\beta} H_0$ and commuting left and right actions, typed as

$$G_1 \times_{s,\alpha} P \longrightarrow P, \quad (g,p) \mapsto gp, \qquad P \times_{\beta,t} H_1 \longrightarrow P, \quad (p,h) \mapsto ph.$$

Each action satisfies its groupoid unit and composition laws and preserves the opposite anchor. It is **biprincipal** when both anchors are surjective and both canonical maps

$$G_1 \times_{s,\alpha} P \xrightarrow{\cong} P \times_\beta P, \quad (g,p) \longmapsto (gp,p),$$
$$P \times_{\beta,t} H_1 \xrightarrow{\cong} P \times_\alpha P, \quad (p,h) \longmapsto (p,ph)$$

are bijective. Thus, over either anchor, two points differ by a unique arrow of the other groupoid. **Morita equivalence** here means an equivalence implemented by such a bibundle. The balanced product identifies $(p\cdot h, p')$ with $(p, h\cdot p')$ wherever defined. All these groupoids are set-based; compare Landsman (2001).

**Theorem 6.19** (Morita invariance of effective presentations)**.** *For surjections $X \xrightarrow{q} Q \xleftarrow{r} Y$, the set $P = X \times_Q Y$ is a biprincipal $(R_q, R_r)$-bibundle. Moreover*

$$P \otimes_{R_r} P^{\mathrm{op}} \cong R_q, \qquad P^{\mathrm{op}} \otimes_{R_q} P \cong R_r, \qquad \mathrm{Act}(R_q) \simeq \mathbf{Set}/Q \simeq \mathrm{Act}(R_r).$$

*These equivalences preserve data descended from $Q$.*

*Proof.*

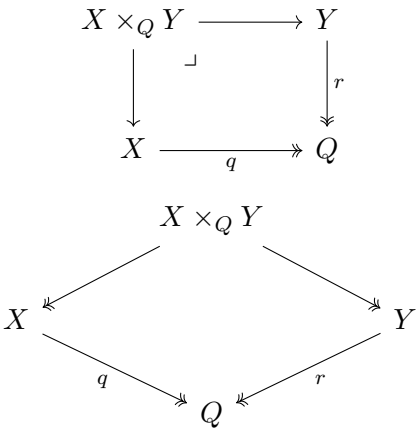
(1) **Lifting along the surjection.** The anchors are the two projections. The actions replace the $X$ or $Y$ coordinate along the corresponding kernel arrow and therefore commute. The anchors are surjective by pullback stability of surjections.

(2) **Monoidal comparison.** Over a fixed $y$, the unique arrow from $(x,y)$ to $(x',y)$ is $(x',x)$; the other principality equation is identical with the roles reversed. The first balanced product sends the class of $((x,y),(y,x'))$ to $(x,x')$. Each admissible pair $(x,x')$ has such a representative because $r$ is surjective.

(3) **Monoidal comparison: completion.** Any two choices of $y$ differ by the unique kernel arrow and hence give the same balanced class. This proves the first bijection, including equivariance; the second is symmetric. The equivalences of action categories follow from Theorem 6.17. Both identify pulled-back data with the same family on $Q$, which proves the last assertion.

□

The comparison bibundle is the pullback of the two quotient presentations.




*Remark* 6.20 (What the comparison retains). A semantic realization $Q \to \mathcal{C}(A,B)$, an observation on $Q$, or a predicate on $Q$ pulls back to both presentations. Data not constant on the quotient fibers remain on the larger presentation. In particular, economic invariance also requires the deviation comparisons constructed in Section 14.1; it does not follow from equivalence of unadorned groupoids alone.

For a response term $a : W \to A$ and observation $o : W \to O$, descent along $o$ says that $a$ is constant on the kernel of $o$. Its unique factor is $\bar{a}$, and tests satisfy $a^*P = o^*\bar{a}^*P$. This factorization through $o$ also holds for dependent policies in the corresponding slice; constant types place it in the state context of an owned arena. This identifies the mathematical meaning of a policy's dependence on available information.

**Proposition 6.21** (Effective descent of dependent linear interfaces)**.** *Let $q : X \twoheadrightarrow Y$. A family $A_x \in \widehat{\mathcal{C}}$ with natural isomorphisms $\theta_{x'x} : A_x \to A_{x'}$ whenever $q(x) = q(x')$, satisfying identity and composition, descends to a family $\bar{A}_y$, uniquely up to its canonical isomorphism. Equivariant families of arrows descend as well.*

*Proof.*

(1) **Predicate calculation.** For each $y, c$ form the set

$$\bar{A}_y(c) = \left( \coprod_{q(x)=y} A_x(c) \right) / ((x, a) \sim (x', \theta_{x'x,c}(a))).$$

The identity and composition equations make this an equivalence relation. For $v : d \to c$ define restriction by $[x, a] \mapsto [x, A_x(v)a]$; naturality of $\theta$ makes it well-defined. The presheaf identities hold on representatives.

(2) **Functoriality and substitution.** For each $x$ the map $a \mapsto [x, a]$ is bijective onto $\bar{A}_{q(x)}(c)$: its inverse sends $[x', a']$ to $\theta_{xx',c}(a')$. This formula is independent of the representative and is natural in $c$. An equivariant arrow sends $[x, a]$ to the class of its image.

(3) **Descent along the kernel pair.** The same formula proves uniqueness of descent and of its canonical comparison with any other descended family. No representative of a fibre of $q$ is selected. Tensor and the other fibre constructions transport along these natural isomorphisms; their substitution comparisons are those of Theorem 1.42.

□

### 6.5.2 Observation quotients and local agreement

**Theorem 6.22** (Testing the order of information)**.** *For two observation maps $o_j : H \twoheadrightarrow O_j$, the following are equivalent:*

(i) *there is a map $q : O_2 \to O_1$ with $o_1 = qo_2$;*

(ii) $\ker(o_2) \subseteq \ker(o_1)$*;*

(iii) *for every action set $A$, every response implementable using $o_1$ is implementable using $o_2$;*

(iv) *the response $o_1 : H \to O_1$, induced by the identity policy on $O_1$, is implementable using $o_2$.*

*The factor $q$ is unique and surjective.*

*Proof.*

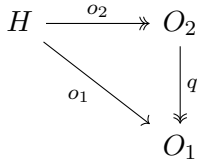
(1) **Descent along the kernel pair.** Factorization implies kernel inclusion. Conversely, if $\ker(o_2) \subseteq \ker(o_1)$, then $o_1$ equalizes the kernel pair of $o_2$. Since $o_2$ is the coequalizer of its kernel pair, there is a unique $q$ with $o_1 = qo_2$.

(2) **Verification of the universal property.** Surjectivity of $q$ follows from surjectivity of $o_1$: for $z \in O_1$, choose $h$ with $o_1(h) = z$ and observe that $z = q(o_2(h))$.

(3) **Policy-level comparison.** If $o_1 = qo_2$ and $\pi : O_1 \to A$, then the same response is implemented using $o_2$ by the policy $\pi q$, proving the third clause. The third clause applied to $A = O_1$ and $\pi = \mathrm{id}_{O_1}$ gives the fourth. The fourth clause is exactly the existence of a map $q$ with $o_1 = qo_2$, so the cycle of implications closes.

□

Information refinement is witnessed by factorization of effective observation maps.




**Corollary 6.23** (Binary tests in the classical finite case)**.** *If $H$ is finite and the finite observation sets are interpreted in ordinary classical* **Set***, the preceding conditions are also equivalent to: every binary-valued policy based on $o_1$ is implementable using $o_2$.*

*Proof.*

(1) **Observation and knowledge.** Factorization gives the binary condition by precomposition. Conversely, if the kernel inclusion fails, take $h, h'$ with equal $o_2$-observations and distinct $o_1$-observations. The characteristic function of the singleton $\{o_1(h)\}$ separates the two $o_1$-observations and therefore cannot factor through $o_2$.

□

**Proposition 6.24** (Gluing observation-based policies)**.** *Let $H = \bigcup_\alpha H_\alpha$, and suppose local policies $\pi_\alpha : o[H_\alpha] \to A$ are given. They glue to one admissible policy iff*

$$o(h) = o(h'),\ h \in H_\alpha,\ h' \in H_\beta \quad \Longrightarrow \quad \pi_\alpha(o(h)) = \pi_\beta(o(h')).$$

*Proof.*

(1) **Observation and knowledge.** Necessity follows by restriction of a common function. For sufficiency, fix an observation $z \in O$. Because the patches cover $H$ and $o$ is onto, there is a pair $(\alpha, h)$ with $h \in H_\alpha$ and $o(h) = z$.

(2) **Gluing observation-based policies.** The displayed agreement shows that all such pairs determine the same value $\pi_\alpha(z)$. Thus there exists a unique value $a_z \in A$ with this property; define $\pi(z) = a_z$. This pointwise unique-value construction gives the global function without choosing representatives, and proves uniqueness.

(3) **Observation and knowledge.** Agreement only on the literal intersections $H_\alpha \cap H_\beta$ would be insufficient: disjoint history patches can still contain observationally indistinguishable histories.

□

### 6.5.3 Information refinement and complete agent policies

**Construction 6.25** (Information refinement and its presentation map)**.**

(1) **Embed the old policies.** Suppose $o_1 = q \circ o_2$ for observation quotients $o_j : H \twoheadrightarrow O_j$. Surjectivity of $o_1$ implies that $q : O_2 \twoheadrightarrow O_1$ is surjective. Precomposition gives

$$q^* : A^{O_1} \hookrightarrow A^{O_2}, \qquad \pi \longmapsto \pi \circ q.$$

It is injective because equal composites with a surjection are equal. The resulting responses on $H$ are unchanged. Thus old policies embed in the more informed policy space, preserving their realized behavior.

When $H, O_j, A$ are finite this embedding is itself a 2-cell of open arenas.

(2) **Realize the change of observation.** Define the single-owner observed decision arena

$$\mathsf{ObsDec}_i(o, A) : D(H) \longrightarrow D(A)$$

by

$$\Sigma_i = A^O, \qquad \Sigma_j = 1 \ (j \neq i), \qquad \pi \longmapsto \widehat{\pi} \circ \widehat{o} = \widehat{\pi o}.$$

If $o_1 = qo_2$, functoriality of the response protocols gives

$$\widehat{\pi o_1} = \widehat{(\pi q) o_2},$$

so the injective map $q^* : A^{O_1} \hookrightarrow A^{O_2}$ is a realization-preserving 2-cell

$$\mathsf{ObsDec}_i(o_1, A) \Longrightarrow \mathsf{ObsDec}_i(o_2, A).$$

Thus refinement of information is represented by ordinary parameter reindexing in $\mathbf{Arena}_N(\mathcal{C})$, while the observation wire itself is the explicitly constructed map $\widehat{o}$.

A finer signal lets a policy prescribe different actions where previously it had to give the same response. The injection above embeds the old policies into this enlarged carrier. Stability against the enlarged set implies stability against the old choices at a retained profile. An old equilibrium can fail the new test because the signal makes a profitable deviation implementable.

**Construction 6.26** (Pooled observation)**.**

(1) **Take the joint observation image.** For a coalition $C$ of observers on a common history space, form the product map

$$\langle o_i \rangle_{i \in C} : H \longrightarrow \prod_{i \in C} O_i$$

and take its image factorization

$$H \twoheadrightarrow O_C \hookrightarrow \prod_{i \in C} O_i.$$

This is the pooled observation quotient.

(2) **Compute its kernel pair.** Its kernel pair is

$$\ker(o_C) = \bigcap_{i \in C} \ker(o_i),$$

because equality in the product is coordinatewise equality. Thus pooled information is constructed as the effective quotient of the joint signal, not assumed to fill the whole Cartesian product of observation values. Pooling information remains distinct from allowing a coalition to coordinate its choices; either operation may occur without the other.

**Proposition 6.27** (Decentralized policies inside the pooled-information interface)**.** *For a nonempty coalition $C$, let $o_C : H \twoheadrightarrow O_C$ be the pooled observation quotient and $p_i : O_C \twoheadrightarrow O_i$ its coordinate maps. For action sets $M_i$ there is a canonical injection*

$$J_C : \prod_{i\in C} M_i^{O_i} \hookrightarrow \left(\prod_{i\in C} M_i\right)^{O_C}, \qquad J_C((\pi_i)_i)(o) = (\pi_i(p_i(o)))_i.$$

*Its image consists exactly of joint policies whose $i$th output is constant on $\ker(p_i)$ for each $i$. Its composite with the observation $o_C$ is the original tuple of decentralized responses on $H$.*

*Proof.*

(1) **Lifting along the surjection.** Each $p_i$ is surjective because $p_i o_C = o_i$ and $o_i$ is surjective. Consequently equality after precomposition with every $p_i$ implies equality of every $\pi_i$, proving injectivity. If a joint policy is in the image, its $i$th component factors through $p_i$ and is constant on its kernel pair.

(2) **Reverse implication.** Conversely such constancy gives a unique factor $\pi_i : O_i \to M_i$ by effective descent. Their tuple is the unique preimage under $J_C$. Finally $p_i o_C = o_i$ gives $J_C((\pi_i)_i)o_C = (\pi_i o_i)_i$.

□

Decentralized responses coincide with evaluation of the induced pooled policy.

$$\begin{array}{ccc} H & \xrightarrow{\;o_C\;} & O_C \\ {\scriptstyle (\pi_i o_i)_i}\downarrow & & \downarrow{\scriptstyle J_C((\pi_i)_i)} \\ \prod_i M_i & = & \prod_i M_i \end{array}$$

Placing two decisions under joint control leaves their information channels as they were. Delivering the pooled signal changes which policies can be executed. The injection above identifies the pooled policies already realizable with separate observations. The finite collector supplies the missing channel for the others, and its synchronization barrier specifies when the joint observation becomes available.

**Observation and implementable dependence.** The quotient equalizer of Proposition 6.2 characterizes observation-based responses; Theorem 6.17 and Proposition 6.21 extend it to dependent families. Consequently the response $\pi_i o_i$ uses only the effective signal actually displayed. No separate admissibility predicate needs to be added to this response construction.

### 6.5.4 Local linear reasoning

Knowledge is a predicate on the states compatible with an observation. A linear argument may instead carry resources that vary across those states. For an effective observation $o : W \twoheadrightarrow O$, strong monoidal reindexing $o^* : \mathcal{L}(O) \to \mathcal{L}(W)$ has right adjoint $\Pi^\ell_o$. A local family $X$ is indexed by observations, whereas a world family $A$ is indexed by states. Their adjunction is

$$\mathcal{L}(W)(o^*X, A) \cong \mathcal{L}(O)(X, \Pi^\ell_o A), \qquad (\Pi^\ell_o A)_z = \prod_{o(w)=z} A_w. \tag{6.3}$$

The right adjoint packages one compatible linear argument for every world in an observation fibre. In judgment form it transposes $O; \Delta \vdash u : \Pi^\ell_o A$ and $W; o^*\Delta \vdash \widetilde{u} : A$. This is proof-relevant perspective: the product retains the resources available across indistinguishable worlds, rather than only the assertion that all such worlds satisfy a predicate.

**Definition 6.28** (Proof-relevant perspective)**.** Define $K^\ell_o = o^*\Pi^\ell_o : \mathcal{L}(W) \to \mathcal{L}(W)$. Its counit is projection at the actual state:

$$\epsilon_{A,w} : \prod_{v:o(v)=o(w)} A_v \longrightarrow A_w.$$

Its comultiplication has, for every $v$ in that fibre, the component identity into the same product:

$$\delta_{A,w} : \prod_{u:o(u)=o(w)} A_u \longrightarrow \prod_{v:o(v)=o(w)} \prod_{u:o(u)=o(v)} A_u.$$

**Proposition 6.29** (Perspective comonad)**.** *The displayed counit and comultiplication make $K^\ell_o$ a comonad.*

*Proof.*

(1) **Adjunction and transposition.** They are the counit and $o^*\eta\Pi^\ell_o$ of the adjunction $o^* \dashv \Pi^\ell_o$. Explicitly, either counit composite projects back to the original indexed family. Both coassociativity composites have, at an index $(v, u, t)$, the projection to $A_t$. Product uniqueness proves the equations, including empty products where applicable.

□

Unlike the predicate knowledge operator, the linear perspective comonad need not be idempotent. Reapplying it can retain multiple additive copies of world-indexed witness products, so its comultiplication need not be invertible. In particular, the construction introduces no contraction map $A \to A \otimes A$.

For a concrete distinction in this very model, take two indistinguishable worlds and the constant family $\mathsf{Lin2}$. Evaluation at the original unit gives $(\mathsf{Lin2})(I) = 2$, since $\mathcal{C}(I, I)$ is a singleton. The first perspective product has four such values, whereas its iterated product has sixteen. Thus its comultiplication cannot be an isomorphism. Evaluation at the tensor unit distinguishes the two presheaf objects directly.

### 6.5.5 Compatibility with linear composition

**Theorem 6.30** (Monoidal perspective comparison)**.** *There are coherent natural maps*

$$\Pi^\ell_o A \otimes \Pi^\ell_o B \longrightarrow \Pi^\ell_o(A \otimes B), \qquad \mathsf{I}_O \longrightarrow \Pi^\ell_o \mathsf{I}_W.$$

*Consequently $K^\ell_o$ is lax symmetric monoidal. For world-indexed agents there is, in particular, a natural operation*

$$K^\ell_o \mathsf{Ag}_{\vec{\Sigma}}(A, B) \otimes K^\ell_o \mathsf{Ag}_{\vec{T}}(B, C) \longrightarrow K^\ell_o \mathsf{Ag}_{\vec{\Sigma}\times\vec{T}}(A, C),$$

*and a corresponding parallel-composition operation.*

*Proof.*

(1) **Observation and knowledge.** At observation $z$, for each $w$ with $o(w) = z$, tensor the two projections to obtain a map into $A_w \otimes B_w$. Product uniqueness gives the comparison. The unit comparison is the family of identity maps on $\mathsf{I}$.

(2) **Associativity.** For associativity, each composite followed by the projection at $w$ is the tensor of the same three projections, reassociated coherently. Units and symmetry are checked by the same argument. Naturality follows by postcomposing the projections with the given component maps.

(3) **Composition under linear perspective.** Applying $o^*$ gives the comparisons for $K_o^\ell$. Compose the first comparison with $K_o^\ell(\mathsf{seq})$ to obtain agent composition; replace $\mathsf{seq}$ by $\mathsf{par}$ for the parallel operation. Their coherence follows from the comparison coherence and Theorem 3.2.

□

The perspective comparison is characterized by its projections at each observationally possible world.

$$\begin{array}{ccc} \Pi_o^\ell A \otimes \Pi_o^\ell B & \xrightarrow{m_{A,B}} & \Pi_o^\ell(A \otimes B) \\ \downarrow{\scriptstyle \pi_w^A \otimes \pi_w^B} & & \downarrow{\scriptstyle \pi_w^{A\otimes B}} \\ A_w \otimes B_w & = & A_w \otimes B_w \end{array}$$

The lax monoidal map combines two resources that are each available from the same perspective; it does not create two copies of one agent or pool observations from different owners. Pooling requires its own joint observation map and, when implemented, a communication protocol. This separates composability of perspective-sensitive resources from access to shared information.

### 6.5.6 Perspective coherence and its unit

The comparison $m_{A,B} : \Pi_o A \otimes \Pi_o B \to \Pi_o(A \otimes B)$ has component $\pi_w^A \otimes \pi_w^B$. Either associativity route, followed by the $w$-projection, is $\pi_w^A \otimes \pi_w^B \otimes \pi_w^C$ under tensor reassociation. The unit route uses $\mathrm{id}_{\mathsf{I}}$ at every $w$ and then the tensor unitor. This proves lax monoidal coherence directly. The counit is monoidal because its projection at the actual world selects precisely these tensor projections. The comultiplication is compatible because all its repeated components are the same indexed-family map. These checks explain why perspective and agent composition are compatible even though the lax comparison need not be invertible.

### 6.5.7 Knowledge as the predicate fragment

For $P \subseteq W$, define

$$K_o P = o^* \forall_o P, \qquad K_o P(w) \iff \forall v\, (o(v) = o(w) \Rightarrow P(v)).$$

The accessibility relation is $R_o = \ker(o)$. Since it is an equivalence relation, the following properties are immediate consequences of its fibres.

**Proposition 6.31** (Epistemic laws)**.** *$K_o$ preserves finite conjunction and truth, is monotone and factive, and satisfies $K_o K_o P = K_o P$ and $\neg K_o P \subseteq K_o \neg K_o P$. It validates the normal modal distribution rule.*

*Proof.*

(1) **Adjunction and transposition.** The kernel pair $R_o = \ker(o)$ is an equivalence relation. Corollary 6.6 and Theorem 5.27 identify $K_o = o^* \forall_o$ with the universal kernel-pair modality and prove its factivity, idempotence, and invariant-predicate characterization. Preservation of conjunction and truth follows from the right-adjoint universal quantifier.

(2) **Predicate calculation.** The negation of a fibre-invariant predicate is fibre-invariant, giving negative introspection. The normal modal implication rule follows from meet preservation: since $K_o((P \Rightarrow Q) \wedge P) \leq K_o Q$, one has $K_o(P \Rightarrow Q) \wedge K_o P \leq K_o Q$, and hence $K_o(P \Rightarrow Q) \leq (K_o P \Rightarrow K_o Q)$ by the Heyting implication adjunction. No preservation-of-implication equality is asserted.

(3) **Restriction to the predicate fragment.** These are the same S5 equations on the ordinary subterminal fragment, not claims about idempotence of $K_o^\ell$.

□

**Proposition 6.32** (Predicate reflection of linear perspective)**.** *For $P \subseteq W$, let $\widehat{P} \in \mathcal{T}(W)$ be the subterminal ordinary family with fibre 1 on $P$ and 0 elsewhere. In the canonical completed concurrent model there is a natural isomorphism*

$$\mathsf{Val}_W K_o^\ell \mathsf{Lin}_W \widehat{P} \cong \widehat{K_o P}.$$

*Proof.*

(1) **Adjunction and transposition.** The scalar calculation $\mathcal{C}(I, I) = 1$ gives $\mathsf{ValLin}X \cong X$ in this canonical model. As a right adjoint, $\mathsf{Val}$ preserves products. Therefore at $w$ the left side is

$$\mathsf{Val}\left(\prod_{o(v)=o(w)} \mathsf{Lin}\widehat{P}_v\right) \cong \prod_{o(v)=o(w)} \mathsf{ValLin}\widehat{P}_v \cong \prod_{o(v)=o(w)} \widehat{P}_v.$$

This subterminal product is inhabited exactly when every world in the observation fibre satisfies $P$, which is $\widehat{K_o P}_w$.

(2) **Functoriality and substitution.** The maps preserve all reindexings and predicate inclusions, proving naturality. The comparison uses the proved $\mathsf{ValLin} \cong 1$ for the chosen concurrent completion, not an equation in every abstract LNL model.

□

The predicate reflection keeps exactly the truth information that is constant across an observation fibre. The proof-relevant linear perspective instead retains the associated product of witness objects. Hence its non-idempotence is compatible with the idempotent S5 operator on assertions.

### 6.5.8 Distributed and common knowledge

For nonempty $C \subseteq N$, factor the tuple of observations through its image:

$$W \xrightarrow{q_C} O_C \hookrightarrow \prod_{i \in C} O_i.$$

Its kernel is $R_C^\cap = \bigcap_{i \in C} R_i$. Define $\mathsf{DK}_C P = [R_C^\cap]P$. This is knowledge of a joint signal. A decentralized coalition does not automatically possess a policy on $O_C$; Section 7.7 constructs the exact comparison.

Let $R_C^*$ be the relation generated by finite paths whose steps belong to some $R_i$, $i \in C$. It is an equivalence relation. Let $q_C^* : W \to W/R_C^*$ be its quotient, and define $\mathsf{CK}_C P = (q_C^*)^* \forall_{q_C^*} P$.

**Proposition 6.33** (Common knowledge as a greatest fixed point)**.**

$$\mathsf{CK}_C P = \nu X. \left(P \wedge \bigwedge_{i \in C} K_i X\right).$$

*Proof.*

(1) **Finite-path characterization of knowledge.** Apply the finite-path construction in Proposition 6.10 and Theorem 6.11 to the subfamily $(\ker(o_i))_{i\in C}$. Its equivalence closure consists exactly of finite $C$-paths, including paths of length zero. The proof there identifies its box with the greatest post-fixed point of $X \mapsto P \wedge \bigwedge_{i\in C} K_i X$. For $C = \varnothing$, the path relation is equality and both sides are $P$.

□

### 6.5.9 Perspective and contextual adjunction rules

For $f : V \to W$, the dependent-adjoint rules are the transpositions

$$\frac{V;\Delta \vdash t : f^*B}{W;\Sigma^\ell_f\Delta \vdash \bar{t} : B}, \qquad \frac{V;f^*\Delta \vdash t : A}{W;\Delta \vdash \widehat{t} : \Pi^\ell_f A}.$$

In the first rule $\Delta$ denotes its tensor object, and the premise is the corresponding arrow from that object; this notation does not distribute $\Sigma^\ell_f$ separately over every context entry. In the second, $f^*$ preserves the tensor context. Both transpositions have inverse elimination rules and their adjunction $\beta\eta$ equations. Perspective uses the second rule at an observation $o$.

At the predicate level, for $P \subseteq V$, $Q \subseteq W$,

$$\exists_f P \subseteq Q \iff P \subseteq f^*Q, \qquad f^*Q \subseteq P \iff Q \subseteq \forall_f P.$$

These supply existential and universal introduction and elimination. Knowledge is the definition $K_o = o^*\forall_o$, and action modalities are the definitions $\exists_s t^*$ and $\forall_s t^*$. Their monotonicity rules follow from functoriality. Factivity, introspection, and reduction equations are the derived results of the main text, not operations acting on arbitrary linear resources.

Common knowledge has the unfolding equation $\mathsf{CK}_C P = P \wedge \bigwedge_{i\in C} K_i \mathsf{CK}_C P$ and the rule

$$\frac{X \subseteq P \qquad X \subseteq K_i X \text{ for all } i \in C}{X \subseteq \mathsf{CK}_C P}.$$

The finite-path construction proves both. Under epistemic update the relations in these rules are those of the updated model.

# 7 Informed Choice and Collective Agency

An owner who sees the same signal in two possible worlds must prescribe one response that is legal in both. A legal action in each world, chosen separately, may fail this test. Taking a dependent universal image computes the menu available at the signal. We use these menus to compare decentralized choice with choice after pooling and to determine which goals a coalition can enforce against every legal outsider response. This connects information-constrained control with coalition effectivity (Pauly, 2002).

## 7.1 Institutionally available agent interfaces

For the legal inclusion $m : L \hookrightarrow \Sigma$, write $L_S = \mathrm{im}(\pi_S m)$ and $e_S : L \twoheadrightarrow L_S$. These image-factorized coordinate maps specify the available outsider indices used here; their full functorial construction is given in Section 7.6.

**Definition 7.1** (Legal domains and legal choice fibres)**.** A **legal domain** is a specified subobject $L \hookrightarrow \Sigma$ of profiles permitted by the institution. Against outsiders $\sigma_{-i}$, the **legal choice fibre** is

$$S_i^L(\sigma_{-i}) = \{a_i \in \Sigma_i : (a_i, \sigma_{-i}) \in L\}.$$

Legality can couple owners' choices even when their unrestricted policy space is a product. The set available to one owner may therefore depend on the outsiders' coordinates. Pulling back the legal subobject along a fixed outsider profile constructs this dependent menu, rather than mistaking its coordinate image for an independently feasible choice.

**Construction 7.2** (Legal selection fibers)**.**

(1) **Form the legal fibre.** For a reference $\sigma \in L$ and owner $i$, put

$$S_i^L(\sigma_{-i}) = \{a \in \Sigma_i : (\sigma_{-i}, a) \in L\}.$$

Let $j_{i,\sigma}^L : S_i^L(\sigma_{-i}) \to L$ insert the fixed outsider coordinates.

(2) **Transport the comparison.** If $m : L \hookrightarrow \Sigma$ is the inclusion, the agent's legal feedback is the costate transported through $\mathbb{L}(gmj_{i,\sigma}^L)$. Connect the agent selection relation $\varepsilon_{S_i^L(\sigma_{-i})}$ to this dependent interface.

(3) **Extend to a coalition.** For a coalition $C$, replace $i$ by $C$ and use the corresponding joint choice fiber. All legally admissible changes of its members are then made at one coalition interface.

**Proposition 7.3** (Legal agency and logical equilibrium)**.** *The conjunction of legal agent selections at a profile $\sigma \in L$ is exactly the legal Nash predicate constructed below. More generally, the coalition interface has paired realization $\delta_C^L : D_C^L \to \mathcal{C}(A, B)^2$, where $D_C^L = \ker(e_{-C} : L \to L_{-C})$ and $L_{-C}$ is the image of the outsider-coordinate map. Its feedback is $(\delta_C^L)^* P$.*

*Proof.*

(1) **Ownerwise alternatives.** The fiber of the first projection $D_C^L \to L$ at $\sigma$ consists exactly of the legal profiles with those outsider coordinates. Inserting the alternative coalition coordinates is a bijection from $S_C^L(\sigma_{-C})$ to that fiber. Both descriptions consequently test $P(g_\sigma, g_\tau)$ on the same pairs.

(2) **Strategic comparison.** Universal quantification of its negation is precisely the selection relation. For $C = \{i\}$, conjoin over $i$ to obtain legal Nash equilibrium.

□

Legal choice fibres are pullbacks of the realizable outsider-coordinate map.

$$\begin{array}{ccc} S_C^L(u) & \xrightarrow{j_{C,u}^L} & L \\ \downarrow & \lrcorner & \downarrow{\scriptstyle e_{-C}} \\ 1 & \xrightarrow[u]{} & L_{-C} \end{array}$$

$$\begin{array}{ccc} S_C^L(\sigma_{-C}) & \xrightarrow{j_{C,\sigma}^L} & L \\ \downarrow & \lrcorner & \downarrow{\scriptstyle e_{-C}} \\ 1 & \xrightarrow[\sigma_{-C}]{} & L_{-C} \end{array}$$

The pullback fixes outsiders and retains exactly the legal alternative choices at that reference. Effectivity has a different quantifier order: a coalition first chooses a realizable coordinate and then requires the goal at every legal completion. The same legal map supports both operations, but they answer different agency questions.

Coalitional power pulls an outcome goal back to legal profiles, quantifies universally over completions of a proposed coalition choice, and then existentially over realizable choices. The base of the final quantifier is the image $L_C$, which removes choices without any legal completion. Thus a formally vacuous universal predicate cannot be mistaken for an enforceable goal.

## 7.2 Institutional arrangements and collective agent interfaces

**Definition 7.4** (Institutional arrangement on an agent process)**.** An **institutional arrangement** on a situated agent process is its displayed legal subobject, observation interfaces, and comparison or outcome terms, together with the concurrent wiring that implements its specified communication and outcome rules. A **collective agent interface** for a nonempty coalition $C \subseteq N$ is obtained by regrouping its controlled policy coordinates. Its available legal carrier against outsiders $u$ is $S_C^L(u)$. When the arrangement supplies a collective observation and comparison, these are attached as the situated and evaluated interfaces of that collective agent. The constituent agents' interfaces remain recorded by the original indexing and the regrouping map. The empty coalition has the unit control carrier $\Sigma_\varnothing = 1$ and identity alternatives; at a legal reference its legal choice fibre is a singleton. This is the nullary control interface, requiring no nonempty block in a regrouping partition.

For nonempty $C$, the partition with block $C$ and singleton outsiders has control given by the regrouping of Theorem 4.8. A particular collective comparison constructed from the members is

$$P_C(f, f') = \bigwedge_{i \in C} P_i(f, f').$$

The collective selection interface tests legal joint deviations against the displayed conjunction of members' comparisons. Its construction uses the choice-fibre bijection of Proposition 7.3; different rules for collective preference are not inferred from regrouping. This is the comparison later used to distinguish collective stability from ability to secure an outcome.

For independent arrangements, the legal domain of the composite is $L_G \times L_H \hookrightarrow \Sigma^G \times \Sigma^H$. Additional institutional rules are further comprehensions, and simultaneous rules intersect. These operations remain coherent with product reassociation. They also reveal why the legal domain of a composite cannot in general be read off from its separate marginal projections: a joint rule may couple otherwise available decisions.

## 7.3 Quantified spans and their action

**Definition 7.5** (Relation modalities)**.** For a relation presented by its jointly monic span $X \xleftarrow{p} R \xrightarrow{q} Y$, put

$$[R] = \forall_p q^*, \qquad \langle R \rangle = \exists_p q^*. \tag{7.1}$$

The converse is denoted $R^\smile$. Relational composition $R; S$ first forms $R \times_Y S$ and then takes its image in the two external factors.

**Proposition 7.6** (Internal typing of an alternative span)**.** *Let $s : X \to \Sigma^G$ be a reference context and $X \xleftarrow{p} R \xrightarrow{q} Y$ a span of alternatives. Its source leg is the context map $p : (R, sp) \to (X, s)$; its target is the term $(sp, q) : (R, sp) \to \Delta_G Y$. For $P \hookrightarrow Y$,*

$$[R]P = \forall_p (sp, q)^* \Delta_G P, \qquad \langle R \rangle P = \exists_p (sp, q)^* \Delta_G P.$$

*Thus both modalities are operations of* $\mathsf{P}_G$ *even when the target alternative changes the reference strategy.*

*Proof.*

(1) **Predicate calculation.** Projection of $(sp, q)$ to $\Sigma^G$ is $sp$, so it is a slice arrow. Equation (5.1) identifies its predicate pullback with $q^*P$. The two quantifiers along $p$ then give precisely Equation (7.1). No equality between the reference of an alternative and the original reference is required.

□

**Theorem 7.7** (The adjunction of a relation)**.** *For* $R \subseteq X \times Y$*, put* $R_! = \exists_q p^* : \mathsf{Pred}(X) \to \mathsf{Pred}(Y)$*. Then*

$$R_! = \langle R^\smile \rangle \dashv [R] = \forall_p q^*.$$

*Consequently* $[R]R_!$ *is an idempotent monad on* $\mathsf{Pred}(X)$ *and* $R_![R]$ *is an idempotent comonad on* $\mathsf{Pred}(Y)$*.*

*Proof.*

(1) **Relational adjunction and idempotence.** For $P \in \mathsf{Pred}(X)$ and $Q \in \mathsf{Pred}(Y)$,

$$\exists_q p^* P \leq Q \iff p^* P \leq q^* Q \iff P \leq \forall_p q^* Q.$$

The unit and counit supply the monad and comonad inequalities; monotonicity gives the reverse idempotence inequalities. The converse relation interchanges $p$ and $q$, proving the first identity.

□

*Remark* 7.8 (Direction of adjunction and qualitative belief)*.* The left adjoint of $[R]$ is the diamond of the converse relation. When $R$ is symmetric this is $\langle R \rangle$ itself. An arbitrary belief operator $[R]$ is not thereby a comonad: for a nonempty state set and the empty relation, $[R]\bot = \top$, so no counit $[R] \leq \mathrm{id}$ exists. The universally available comonad is the composite $R_![R]$ on the indicated carrier.

**Proposition 7.9** (Relation action and graph modalities)**.** *One has* $[R;S] = [R][S]$ *and* $\langle R;S \rangle = \langle R \rangle \langle S \rangle$*. For the graph* $\Gamma_f$ *of* $f : X \to Y$*,*

$$[\Gamma_f] = \langle \Gamma_f \rangle = f^*, \qquad [\Gamma_f^\smile] = \forall_f, \qquad \langle \Gamma_f^\smile \rangle = \exists_f.$$

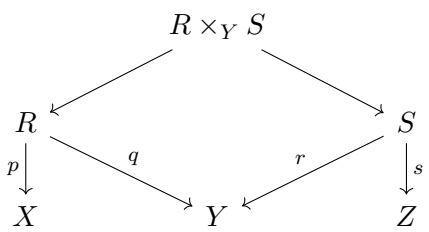
*In particular,*

$$[\ker f] = f^* \forall_f, \qquad \langle \ker f \rangle = f^* \exists_f. \tag{7.2}$$

*Proof.*

(1) **Witness and legality comparison.** Expanding a composite box gives $\forall y\,(R(x,y) \Rightarrow \forall z\,(S(y,z) \Rightarrow P(z)))$, which is equivalent to the box for the image relation. The existential calculation combines the two witnesses. For a graph the sole forward candidate is $f(x)$; its converse has exactly the fiber $f^{-1}(y)$. Finally $\ker f = \Gamma_f ; \Gamma_f^\smile$.

□



## 7.4 Informed legal policies as sections of a dependent product

The individual choice fibre is not yet an informed policy. At the same observation, several worlds may have different feasible menus, so a single response must lie in their intersection. The right adjoint to inverse image constructs that universally feasible menu over each observation. Taking its sections then produces precisely the policies that are simultaneously legal and observation-based.

**Construction 7.10** (The informed legal interface)**.**

(1) **Transport legality along observation.** Let $o : W \twoheadrightarrow O$ be an effective observation quotient, $M$ an action carrier, and $F \hookrightarrow W \times M$ the specified feasibility predicate. Put $u = o \times \mathrm{id}_M$ and define

$$A_o = \forall_u F \hookrightarrow O \times M, \qquad A_o(z) = \{a \in M : \forall w \in o^{-1}(z)\ (w, a) \in F\}.$$

(2) **Take sections of the resulting family.** The informed legal policy carrier is the set of sections

$$\mathsf{Pol}_o(F) = \mathrm{Sect}_O(A_o \to O) = \{\pi : O \to M : \forall w \in W\ (w, \pi(o(w))) \in F\}.$$

Thus ordinary descent imposes dependence on observations, and dependent universal quantification imposes legality at every world with that observation. The predicate $F$ may describe physical availability, institutional permission, or their intersection; these interpretations remain explicit in the supplied data.

**Proposition 7.11** (Universal characterization and information refinement)**.** *For every $\pi : O \to M$, writing $\Gamma_\pi$ for its graph,*

$$\Gamma_\pi \subseteq A_o \quad \Longleftrightarrow \quad u^*\Gamma_\pi \subseteq F.$$

*In particular, $\mathsf{Pol}_o(F)$ is the maximal subobject of $M^O$ whose evaluated responses satisfy $F$. If $o = q\,o'$ for surjective observation maps $o' : W \twoheadrightarrow O'$ and $q : O' \twoheadrightarrow O$, then*

$$A_o = \forall_{q \times \mathrm{id}_M} A_{o'}, \qquad q^* : \mathsf{Pol}_o(F) \hookrightarrow \mathsf{Pol}_{o'}(F), \quad \pi \longmapsto \pi q.$$

*Its image consists exactly of the fine-observation legal policies constant on the fibres of $q$, and it preserves their responses on $W$.*

*Proof.*

(1) **Adjunction and transposition.** The first equivalence is the adjunction $u^* \dashv \forall_u$. The inverse-image graph consists of exactly $(w, \pi(o(w)))$, so it is contained in $F$ exactly when that response is feasible at every $w$. This proves both the section formula and maximality.

(2) **Adjunction and transposition: second comparison.** Since $u = (q \times \mathrm{id}_M)(o' \times \mathrm{id}_M)$, composition of right adjoints gives the displayed equality for $A_o$. Equivalently, the intersection of feasible menus over $o^{-1}(z)$ is the intersection of the corresponding fine-observation menus over $q^{-1}(z)$. If $\pi$ is legal, $\pi q$ has exactly the same world response, so is legal.

(3) **Verification of the universal property.** Surjectivity of $q$ makes precomposition injective. A fine policy constant on $\ker q$ descends uniquely to $\pi$; its identical world response satisfies $F$, proving the image claim.

□

**Proposition 7.12** (Compatibility of admissibility, legality, and knowledge)**.** *For the effective observation $o : W \twoheadrightarrow O$ and $F \hookrightarrow W \times M$ above, let $F_a = \{w \in W : (w, a) \in F\}$. The following square of sets is a pullback, in which both horizontal arrows and the right vertical arrow are inclusions or injections:*

$$\begin{array}{ccc} \mathsf{Pol}_o(F) & \hookrightarrow & M^O \\ {\scriptstyle o^*}\downarrow & & \downarrow{\scriptstyle o^*} \\ \{f : W \to M : \forall w\ F(w, f(w))\} & \hookrightarrow & M^W. \end{array}$$

*Furthermore, for each $a \in M$ the predicate obtained by pulling the universally legal menu back to the history carrier is knowledge of feasibility of that action:*

$$(o \times \underset{M}{\mathrm{id}})^* A_o(w, a) \quad \Longleftrightarrow \quad K_o(F_a)(w).$$

*Thus the same effective observation quotient simultaneously determines which responses descend, which descended responses are uniformly legal, and which individual-action feasibility predicates are known.*

*Proof.*

(1) **Descent of responses.** By Proposition 6.2, precomposition with $o$ embeds $M^O$ into $M^W$ and its image is the set of functions constant on the kernel pair of $o$. This is the right vertical map.

(2) **Legal pullback.** The inverse image under this embedding of the lower-left set consists exactly of those $\pi$ for which $F(w, \pi(o(w)))$ holds for every $w$. By Proposition 7.11, this set is $\mathsf{Pol}_o(F)$; hence the square is cartesian.

(3) **Knowledge of the universally legal menu.** The definition of $A_o$ gives

$$A_o(o(w), a) \iff \forall v\,(o(v) = o(w) \Rightarrow F(v, a)),$$

which is exactly the kernel-pair knowledge predicate $K_o F_a(w)$ of Definition 6.3. All identities are over the original observation and therefore commute with substitution of a state reference by the relevant pullback square.

□

The dependent universal image identifies the observation-uniform legal actions.

$$\begin{array}{ccc} W \times M & \xrightarrow{o \times \mathrm{id}} & O \times M \\ {\scriptstyle \pi_W}\downarrow & & \downarrow{\scriptstyle \pi_O} \\ W & \xrightarrow[o]{} & O \end{array}$$

For example, let $W = \{w_0, w_1\}$ have one observation and $M = \{a, b\}$, with feasible menus $\{a\}$ and $\{b\}$ at the two worlds. Then $A_o$ has empty fibre and there is no informed legal policy. Under the fully revealing observation, the policy $w_0 \mapsto a$, $w_1 \mapsto b$ is legal. The empty universal image exhibits the obstruction to a coarse-observation legal policy.

The construction applies directly to a situated agent. For $s : W \to L$, take $M = \Sigma_i$ and

$$F_i(w, a) \iff (s(w)_{-i}, a) \in L.$$

The sections are complete-policy deviations that depend only on the agent's observation and remain legal at every compatible reference state. When the decision is a local action rather than a complete policy, its own action set and enabling predicate must be used instead. The distinction is fixed by the causal stage of the protocol.

**Proposition 7.13** (Decentralized legal coalition policies)**.** *For observers $o_i : W \twoheadrightarrow O_i$, a nonempty coalition $C$, and a joint feasibility predicate $F_C \hookrightarrow W \times \prod_{i\in C} M_i$, let $o_C : W \twoheadrightarrow O_C$ be the pooled quotient. Then*

$$\mathsf{Pol}_C^{\mathrm{dec}}(F_C) = J_C^{-1}\big(\mathrm{Sect}_{O_C}\big(\forall_{o_C\times \mathrm{id}} F_C \to O_C\big)\big) \hookrightarrow \prod_{i\in C} M_i^{O_i},$$

*where $J_C$ is Proposition 6.27. The inclusion into pooled legal policies is injective, and its image is exactly those pooled legal policies whose $i$th output descends along $p_i : O_C \to O_i$ for every $i$.*

*Proof.*

(1) **Policy-level comparison.** The response at $w$ of $J_C((\pi_i)_i)$ is $(\pi_i(o_i(w)))_i$. By the section characterization, this pooled policy is legal exactly when that tuple lies in $F_C(w)$ at every $w$. This is precisely decentralized joint feasibility.

(2) **Characterization of decentralized legal policies.** The image characterization and injectivity follow by restricting the injection and image characterization of $J_C$ to this comprehension.

□

Decentralized legal policies embed in the globally pooled admissible-policy space.

$$\begin{array}{ccc} \mathsf{Pol}_C^{\mathrm{dec}}(F_C) & \hookrightarrow & \mathsf{Pol}_{o_C}(F_C) \\ \downarrow & & \downarrow \\ \prod_i M_i^{O_i} & \xhookrightarrow[J_C]{} & (\prod_i M_i)^{O_C} \end{array}$$

Separate marginal feasibility does not imply joint legal feasibility. If two binary actions must agree, the joint possibilities $(0,0)$ and $(1,1)$ give both owners nonempty marginal menus, but the product also admits the illegal tuple $(0,1)$. The dependent joint predicate retains this coupling and is therefore needed for informed coalition choice.

The resulting interface is the deterministic policy structure of team decision problems: each agent chooses from its own information, while a joint evaluation is attached to the combined response. The comparison with Radner (1962) concerns these information and decision interfaces, not a probability or optimality principle added to the present qualitative construction.

## 7.5 Local states, knowledge, and a uniform ability witness

**Proposition 7.14** (Local-state presentation of an observation frame)**.** *The map*

$$\ell : W \longrightarrow W \times \prod_{i\in N} O_i, \qquad \ell(w) = (w, (o_i(w))_i)$$

*is injective. On its image, equality of agent $i$'s local coordinate pulls back to $\ker o_i$. An observation policy $\pi_i : O_i \to M_i$ evaluates at the local coordinate exactly as the original response $\pi_i o_i$.*

*Proof.*

(1) **Functoriality and substitution.** The first coordinate recovers $w$, proving injectivity. Equality of the $i$th local coordinates is the equation $o_i(w) = o_i(v)$, and evaluation is substitution in $\pi_i$.

□

This yields the static local-state interface of interpreted-systems semantics (Halpern and Fagin, 1989). The carrier $W$ may consist of histories or points of runs supplied by a protocol; transition relations arise separately from executions or epistemic actions. Knowing one's own currently selected policy, when modeled, is an additional observation fact, not a consequence of ownership alone.

To distinguish knowledge of pointwise availability from possession of one informed ability witness, let $\Pi$ be a specified policy carrier and $S \hookrightarrow W \times \Pi$ a success predicate. For example, $S(w, \pi)$ may already universally quantify legal outsiders' responses. At a current world $w$, compare

$$\forall v\ (o_i(v) = o_i(w) \Rightarrow \exists \pi \in \Pi\ S(v, \pi))$$

with

$$\exists \pi \in \Pi\ \forall v\ (o_i(v) = o_i(w) \Rightarrow S(v, \pi)).$$

The second formula exhibits one policy succeeding across the whole information class; the first only says that each possible world admits some successful policy. The implication cannot be reversed. For two indistinguishable worlds, take policies $\pi_0, \pi_1$ with $S(w_j, \pi_k)$ precisely when $j = k$. Each world has a successful policy, but neither policy succeeds at both. The distinction between knowledge that success is possible and knowing how to secure it is the point of the comparison with Jamroga and van der Hoek (2004).

## 7.6 Legal coalition effectivity and stability

Regrouping gives coalitions a joint control interface, and legal comprehension gives each controlled choice its permitted completions. To determine power, we must now place the quantifiers in the order that corresponds to a coalition choosing before its outsiders. Image factorization removes unrealizable choices, while the same kernel-pair relations continue to express legal deviations. We then compare coalitional guarantees with preference-based stability.

The collective-control map is Theorem 4.8; legal selection is Proposition 7.3; and the comparison of decentralized with pooled information is Proposition 6.27. These supply separate ingredients of the effectivity calculation. The resulting relationship between dependence, preference, and coalitional power also connects with Chen, Shi, and Wang (2022).

### 7.6.1 Institutional choices and power to secure a goal

**Definition 7.15** (Coalition effectivity and enforcement)**.** Let $m : L \hookrightarrow \Sigma$ be a legal profile domain, $e_C : L \twoheadrightarrow L_C$ the image-factorized projection to the coalition's coordinates, $q : L \to Z$ an outcome, and $P \hookrightarrow Z$ a goal. **Coalition effectivity** is the predicate-valued operator

$$[C]_{L,q} : \mathsf{Pred}(Z) \longrightarrow \mathsf{Pred}(1), \qquad [C]_{L,q}(P) = \exists_{!:L_C \to 1}\, \forall_{e_C : L \to L_C}\, q^* P.$$

Equivalently, it holds just when some $c \in L_C$ satisfies $\forall \ell \in L\,(e_C(\ell) = c \Rightarrow q(\ell) \in P)$. A coalition **enforces** $P$ by such a realizable coordinate; every legal completion must satisfy the

goal. For fixed $q$, the outcome subscript is suppressed, giving the notation $[C]_L$ used below. The image $L_C$ excludes unrealizable choices and therefore excludes vacuous enforcement by them. Equilibrium and institutional sanctions are evaluated through their respective additional interfaces. Its state-dependent version is constructed by dependent quantification in the public-state and announcement sections.

Here "enforce" refers to a quantified guarantee, not to rationality or preference satisfaction. A coalition enforces a goal when one realizable choice yields it under every legal completion by outsiders. The enforcing policy need not be an equilibrium choice, since the ability predicate does not compare its desirability with other policies.

**Construction 7.16** (Legal profiles and realizable coordinate images)**.**

(1) **Specify the legal profiles.** Let

$$m : L \hookrightarrow \Sigma = \prod_{i \in N} \Sigma_i$$

be a specified subobject of legal profiles. The ordinary unconstrained case is $m = \mathrm{id}_\Sigma$.

(2) **Take the coordinate images.** For every $S \subseteq N$, factor the coordinate map through its image,

$$L \xrightarrow{e_S} L_S \xhookrightarrow{j_S} \Sigma_S, \qquad \pi_S m = j_S e_S. \tag{7.3}$$

Thus $L_S$ is the set of realizable $S$-coordinates and $e_S$ is a surjection by construction. No legal choice is introduced by selecting a representative of an equivalence class.

The coordinate images inherit all coordinate restrictions by effective descent.

**Proposition 7.17** (The realizable-coordinate functor)**.** *For every inclusion $S \subseteq T \subseteq N$ there is a unique surjection*

$$\rho_S^T : L_T \twoheadrightarrow L_S$$

*satisfying*

$$\rho_S^T e_T = e_S. \tag{7.4}$$

*These maps obey*

$$\rho_S^S = \mathrm{id}_{L_S}, \qquad \rho_R^S \rho_S^T = \rho_R^T \quad (R \subseteq S \subseteq T).$$

*Hence realizable coordinates form a contravariant functor*

$$L_{(-)} : \mathcal{P}(N)^{\mathrm{op}} \longrightarrow \mathbf{Set}, \qquad S \longmapsto L_S,$$

*where the powerset is ordered by inclusion.*

*Proof.*

(1) **Construction of the comparison.** Let $\pi_S^T : \Sigma_T \to \Sigma_S$ be the ordinary coordinate projection. If $e_T(\ell) = e_T(\ell')$, monicity of the image inclusion $L_T \hookrightarrow \Sigma_T$ gives $\pi_T m(\ell) = \pi_T m(\ell')$. Applying $\pi_S^T$ gives $\pi_S m(\ell) = \pi_S m(\ell')$, and monicity of $L_S \hookrightarrow \Sigma_S$ therefore gives $e_S(\ell) = e_S(\ell')$.

(2) **Construction by quotient descent.** Thus $e_S$ equalizes the kernel pair of the surjection $e_T$. Proposition 5.34 supplies a unique map $\rho_S^T$ satisfying Equation (7.4).

(3) **Lifting along the surjection.** It is surjective because every $s \in L_S$ has the form $s = e_S(\ell)$ and then

$$s = \rho_S^T(e_T(\ell)).$$

For $R \subseteq S \subseteq T$, both $\rho_R^S \rho_S^T$ and $\rho_R^T$ become $e_R$ after precomposition with the surjection $e_T$; cancellation of that epimorphism gives equality. The identity equation is the same argument with $S = T$.

□

**Construction 7.18** (Legal deviation kernel pairs)**.**

(1) **Form the legal kernel pair.** The corresponding legal deviation relation is the kernel pair

$$D_C^L = L \times_{L_{-C}} L \overset{p_2}{\underset{p_1}{\rightrightarrows}} L \tag{7.5}$$

of $e_{-C}$.

(2) **Identify the outsider fibre.** It relates exactly those legal profiles that agree on their realizable outsider coordinate. When $L = \Sigma$ this is the deviation relation $D_C$ constructed in Equation (4.1).

**Proposition 7.19** (Functional dependence of legal policy coordinates)**.** *For $S \subseteq N$ and $j \in N$, form the kernel pair $K_S = L \times_{L_S} L$ of $e_S$, with projections $p_1, p_2$. The assertion that the $S$-coordinates locally determine owner $j$'s coordinate is the predicate*

$$\mathsf{FD}_{S \Rightarrow j} = \forall_{p_1 : K_S \to L} \operatorname{Eq}\big((\pi_j m)p_1, (\pi_j m)p_2\big) \quad \in \mathsf{Pred}(L).$$

*It holds at $\ell$ exactly when every legal profile agreeing with $\ell$ on $S$ also agrees on coordinate $j$. Moreover $\mathsf{FD}_{S \Rightarrow j} = \top_L$ exactly when $\pi_j m$ factors uniquely through $e_S : L \twoheadrightarrow L_S$. This is the functional-dependence test associated with the preference-and- dependence interpretation of Chen, Shi, and Wang (2022), expressed in the existing assertion doctrine.*

*Proof.*

(1) **Descent along the kernel pair.** The fibre of $p_1$ at $\ell$ consists of all legal profiles having $S$-coordinate $e_S(\ell)$; universal quantification of equality proves the pointwise claim. Global truth says precisely that $(\pi_j m)p_1 = (\pi_j m)p_2$. Since $e_S$ is the coequalizer of its kernel pair, Proposition 5.34 gives the unique factor.

□

**Proposition 7.20** (Legal deviation is dynamic restriction)**.** *For every coalition $C$,*

$$D_C^L \cong (m \times m)^* D_C. \tag{7.6}$$

*Thus the legal deviation relation is exactly the ordinary coalition-deviation relation restricted to the legal state space. The image object $L_{-C}$ does not introduce a new accessibility relation; it presents the realizable outsider coordinate through which that restricted relation is the kernel pair.*

*Proof.*

(1) **Descent along the kernel pair.** By definition $D_C^L = \ker(e_{-C})$. Since

$$\pi_{-C} m = j_{-C} e_{-C}$$

and $j_{-C}$ is monic,

$$e_{-C}(\ell) = e_{-C}(\ell') \Longleftrightarrow \pi_{-C}m(\ell) = \pi_{-C}m(\ell').$$

Hence $\ker(e_{-C}) = \ker(\pi_{-C}m)$. Kernel pairs are pullbacks of a map along itself and therefore are stable under change of base.

(2) **Pullback of deviation kernel pairs.** Pulling $D_C = \ker(\pi_{-C})$ back along $m \times m$ gives $\ker(\pi_{-C}m)$, proving the claimed isomorphism.

□

**Construction 7.21** (Legal domains from institutional data)**.**

(1) **Impose institutional constraints.** A legal subobject need not be introduced by an uninterpreted legality predicate when lower-level institutional data are available. For a set-indexed family, if $c_\alpha : \Sigma \to Y_\alpha$ are typed constraint maps and $A_\alpha \hookrightarrow Y_\alpha$ are their permitted-value subobjects, define

$$L = \bigwedge_\alpha c_\alpha^* A_\alpha \hookrightarrow \Sigma.$$

(2) **Take certificate images.** If legality instead means that a profile admits an institutional witness, start from a witness object $W$ and a map $w : W \to \Sigma$ and define

$$L = \mathrm{im}(w) \hookrightarrow \Sigma.$$

(3) **Combine the conditions.** Finite combinations of such conditions are obtained by pullback and image factorization. Thus the generic subobject $L \hookrightarrow \Sigma$ below is supplied institutional data only when no lower-level construction has been specified; whenever constraints or witnesses are supplied, legality is the subobject constructed from them.

For an open-arena realization $g : \Sigma \to \mathcal{C}(A, B)$, the legal counterfactual semantics of coalition $C$ is

$$\delta_C^{G,L} = (g \circ m \circ p_1, g \circ m \circ p_2) : D_C^L \longrightarrow \mathcal{C}(A, B)^2, \tag{7.7}$$

**Construction 7.22** (Legal equilibrium)**.**

(1) **Form profitable legal alternatives.** For a singleton $i$, put

$$\mathsf{Prof}_i^L = (\delta_i^{G,L})^* \mathsf{Better}_i.$$

When the comparison comes directly from measured records, take $\mathsf{Prof}_i^L = (\bar{v}_i p_1, \bar{v}_i p_2)^* \prec_i$, or use the descended pair predicate of Proposition 5.43.

(2) **Quantify over deviations.** The legal Nash predicate is then

$$\mathsf{NE}_{G,L} = \bigwedge_{i \in N} \forall_{p_1 : D_i^L \to L} \neg \mathsf{Prof}_i^L. \tag{7.8}$$

When $L = \Sigma$, these constructions are exactly the unrestricted counterfactual semantics and Nash predicate already defined.

**Proposition 7.23** (Legal equilibrium as restricted modal stability)**.** *Let*

$$\mathsf{PDev}_i = D_i \cap (g \times g)^* \mathsf{Better}_i \hookrightarrow \Sigma \times \Sigma.$$

*Then its legal restriction is*

$$\mathsf{PDev}_i^L = (m \times m)^* \mathsf{PDev}_i,$$

*and*

$$\mathsf{NE}_{G,L} = \bigwedge_i \neg\langle \mathsf{PDev}_i^L \rangle \top. \tag{7.9}$$

*Viewed again as a predicate on the ambient profile space through the dynamic box* $[!m] = \forall_m$*,*

$$[!m]\mathsf{NE}_{G,L} = L \Rightarrow \bigwedge_i \neg\langle \mathsf{PDev}_i \rangle L. \tag{7.10}$$

*Thus restricting legality can create equilibria exactly by removing profitable alternatives; legal equilibrium is not generally the pullback of unrestricted equilibrium.*

*Proof.*

(1) **Ownerwise alternatives.** Proposition 7.20 gives $D_i^L = (m \times m)^* D_i$, while functoriality of pullback gives

$$((gm) \times (gm))^* \mathsf{Better}_i = (m \times m)^* (g \times g)^* \mathsf{Better}_i.$$

Pullback preserves finite meets, hence the profitable-deviation relation restricts as displayed. Proposition 5.48 applied in the state space $L$ gives Equation (7.9).

(2) **Witness and legality comparison.** Write $A = \exists_m \top$ for the ambient predicate represented by the legal subobject and put

$$S = \bigwedge_i \neg\langle \mathsf{PDev}_i \rangle A \in \mathsf{Pred}(\Sigma).$$

For every relation $R \hookrightarrow \Sigma \times \Sigma$, change of base along $m$ gives

$$m^* \langle R \rangle A = \langle (m \times m)^* R \rangle \top. \tag{7.11}$$

To see this entirely in the relation calculus, let $\Gamma_m$ be the graph of $m$.

(3) **Predicate calculation.** Proposition 7.9 gives

$$m^* = \langle \Gamma_m \rangle, \qquad A = \exists_m \top = \langle \Gamma_m^\smile \rangle \top.$$

Hence

$$\begin{aligned} m^* \langle R \rangle A &= \langle \Gamma_m ; R ; \Gamma_m^\smile \rangle \top \\ &= \langle (m \times m)^* R \rangle \top, \end{aligned}$$

because $\Gamma_m ; R ; \Gamma_m^\smile$ is exactly the restriction of $R$ to the subobject $L$.

(4) **Legal Nash reduction by implication.** Therefore

$$m^* S = \bigwedge_i \neg\langle \mathsf{PDev}_i^L \rangle \top = \mathsf{NE}_{G,L}.$$

For the monomorphism $m : L \hookrightarrow \Sigma$, each fibre has one element over a point of $A = \exists_m \top$ and no elements outside $A$. The universal and existential fibre formulas therefore give

$\forall_m Q = A \Rightarrow \exists_m Q$ for every predicate $Q$ on $L$: outside $A$ the universal clause is vacuous, and on $A$ its unique preimage determines both quantifiers. Frobenius gives

$$\begin{aligned}[!m]\mathsf{NE}_{G,L} &= A \Rightarrow \exists_m \mathsf{NE}_{G,L} \\ &= A \Rightarrow \exists_m m^* S \\ &= A \Rightarrow (A \wedge S) \\ &= A \Rightarrow S.\end{aligned}$$

Identifying the image predicate $A$ with the displayed legal predicate $L$ gives Equation (7.10).

□

**Definition 7.24** (Amalgamation of legal choices)**.** For disjoint coalitions $C, D \subseteq N$, use the restriction maps of Proposition 7.17 to form the **amalgamation comparison**

$$\lambda_{C,D} = \langle \rho_C^{C\cup D}, \rho_D^{C\cup D} \rangle : L_{C\cup D} \longrightarrow L_C \times L_D.$$

Two separately realizable coordinates $c \in L_C$ and $d \in L_D$ **amalgamate** precisely when the dependent witness fibre

$$\mathsf{Amal}_{C,D}(c,d) = \{u \in L_{C\cup D} : \lambda_{C,D}(u) = (c,d)\}$$

is inhabited. All such pairs amalgamate precisely when $\lambda_{C,D}$ is surjective. A witness is a single realizable joint coordinate simultaneously extending both choices.

Suppose either agent may request a resource, but the institution cannot grant both requests together. Each request has a legal completion when considered on its own. Their pair has none. The amalgamation map below tests exactly this failure of compatibility. On an unrestricted product domain, taking the pair of choices supplies the desired completion.

**Definition 7.25** (Coalition effectivity specialized to an evaluated arena)**.** For an open-arena realization $g : \Sigma \to \mathcal{C}(A,B)$ with an observable evaluation $o : \mathcal{C}(A,B) \to Z$, the outcome map in Definition 7.15 is $q = o \circ g \circ m : L \to Z$; it may also be supplied directly. The already constructed effectivity operator, now written $[C]_L$, is

$$[C]_L P = \exists_{!:L_C \to 1} \forall_{e_C : L \to L_C} q^* P. \tag{7.12}$$

This agrees with the elementwise enforcing criterion of Definition 7.15: a realizable coalition coordinate $c \in L_C$ guarantees the goal under every legal completion. Because $e_C$ is surjective, every quantified coalition choice has at least one legal completion. Hence ability cannot be obtained vacuously from an unrealizable coordinate. If $L = \Sigma$ and the profile space is inhabited, then every factor is inhabited, $L_C \cong \Sigma_C$, and the fiber of $e_C$ is canonically $\Sigma_{-C}$; Equation (7.12) becomes

$$[C]P \quad \Longleftrightarrow \quad \exists s_C \in \Sigma_C \; \forall s_{-C} \in \Sigma_{-C},\; q(s_C, s_{-C}) \in P. \tag{7.13}$$

For an empty profile space the image construction leaves no realizable coalition choice and therefore introduces no vacuous power.

The coalition must choose a single action that succeeds against every legal outsider response. Reversing the quantifiers would let it tailor its action to the outsider's choice even though the two act simultaneously. The coordinate image specifies which actions have a

legal completion, and the order of the dependent quantifiers specifies who can rely on whose choice.

For disjoint coalitions $C, D$, the realizable-coordinate functor constructs the comparison

$$\lambda_{C,D} = \langle \rho_C^{C\cup D}, \rho_D^{C\cup D} \rangle : L_{C\cup D} \longrightarrow L_C \times L_D. \tag{7.14}$$

It measures exactly whether separately realizable choices of $C$ and $D$ can be amalgamated into one legal joint choice.

**Theorem 7.26** (Coalition ability and legal amalgamation)**.** *Let $C \cap D = \varnothing$. If $\lambda_{C,D}$ is surjective, then for every outcome map $q : L \to Z$ and predicates $P, Q \hookrightarrow Z$,*

$$[C]_L P \ \wedge \ [D]_L Q \quad \Longrightarrow \quad [C \cup D]_L (P \cap Q).$$

*Conversely, if this implication holds for every $q, P, Q$, then $\lambda_{C,D}$ is surjective. In particular the implication holds for the ordinary full-product game.*

*Proof.*

(1) **Choice of representatives and quotients.** Let $c \in L_C$ and $d \in L_D$ witness the two ability assertions. If $\lambda_{C,D}$ is surjective, take $u \in L_{C\cup D}$ with $\lambda_{C,D}(u) = (c, d)$. For any legal profile $\ell$ with $e_{C\cup D}(\ell) = u$, functoriality of coordinate restriction gives $e_C(\ell) = c$ and $e_D(\ell) = d$.

(2) **Joint coalition witness.** Hence $q(\ell) \in P$ by the first witness and $q(\ell) \in Q$ by the second. Thus $u$ witnesses $[C \cup D]_L (P \cap Q)$.

(3) **Reverse implication.** Conversely, take an arbitrary pair $(c, d) \in L_C \times L_D$. Let

$$q = (e_C, e_D) : L \longrightarrow L_C \times L_D, \qquad P = \{c\} \times L_D, \qquad Q = L_C \times \{d\}.$$

Then $c$ witnesses $[C]_L P$ and $d$ witnesses $[D]_L Q$. By the assumed superadditivity there is $u \in L_{C\cup D}$ such that every legal completion of $u$ is sent to $(c, d)$.

(4) **Verification of the universal property.** Surjectivity of $e_{C\cup D} : L \twoheadrightarrow L_{C\cup D}$ supplies such a completion $\ell$, and

$$\lambda_{C,D}(u) = \lambda_{C,D} e_{C\cup D}(\ell) = (e_C(\ell), e_D(\ell)) = (c, d).$$

Hence $\lambda_{C,D}$ is surjective.

(5) **Lifting along the surjection.** For $L = \Sigma$ a nonempty full product has $L_S \cong \Sigma_S$, and for disjoint $C, D$ the map $\lambda_{C,D}$ is the canonical product comparison and is surjective. If $\Sigma$ is empty, every $L_S$ is empty and the same map is the unique surjection $\varnothing \to \varnothing$. Thus the ordinary full-product implication holds in either case.

□

For product choices the resulting implication recovers the familiar superadditivity pattern of coalition effectivity Pauly (2002), while the quantifier order parallels the coalition modalities of alternating-time logic (Alur, Henzinger, and Kupferman, 2002). Here the image comparison identifies the exact amalgamation condition needed when legality couples the choices.

**Construction 7.27** (Outsider-coordinate amalgamation map)**.**

(1) **Project to common outsider coordinates.** There is a parallel exact criterion for composing legal deviation relations. Put $O = N \setminus (C \cup D)$. Proposition 7.17 constructs the restriction maps

$$\rho_O^{-C} : L_{-C} \to L_O, \qquad \rho_O^{-D} : L_{-D} \to L_O.$$

(2) **Form the matching pullback.** Form their pullback

$$P_{C,D} = L_{-C} \times_{L_O} L_{-D}$$

and the canonical comparison

$$a_{C,D} : L \longrightarrow P_{C,D}, \qquad \ell \longmapsto (e_{-C}(\ell), e_{-D}(\ell)). \tag{7.15}$$

**Proposition 7.28** (Legal deviation amalgamation)**.** *For arbitrary $C, D \subseteq N$,*

$$D_C^L; D_D^L = D_{C\cup D}^L \quad \Longleftrightarrow \quad a_{C,D} \text{ is surjective.}$$

*Proof.*

(1) **Inclusion of successive legal deviations.** A successive $C$-change and $D$-change preserves every coordinate outside $C \cup D$, so $D_C^L; D_D^L \subseteq D_{C\cup D}^L$ always.

(2) **Forward implication.** Suppose first that $a_{C,D}$ is surjective and $\sigma D_{C\cup D}^L \tau$. Then the pair

$$(e_{-C}(\sigma), e_{-D}(\tau))$$

has equal images in $L_O$, because $\sigma$ and $\tau$ agree on the realizable coordinate outside $C \cup D$. It therefore lies in $P_{C,D}$.

(3) **Choice of representatives and quotients.** Choose $\rho \in L$ above this point under the surjection $a_{C,D}$. Then

$$e_{-C}(\rho) = e_{-C}(\sigma), \qquad e_{-D}(\rho) = e_{-D}(\tau),$$

so $\sigma D_C^L \rho D_D^L \tau$.

(4) **Reverse implication.** Conversely assume equality of the relations and take $(x, y) \in P_{C,D}$. Since the image maps $e_{-C}$ and $e_{-D}$ are surjective, choose legal profiles $\sigma, \tau$ with $e_{-C}(\sigma) = x$ and $e_{-D}(\tau) = y$. Compatibility of $(x, y)$ over $L_O$ says $\sigma D_{C\cup D}^L \tau$.

(5) **Lifting along the surjection.** By the assumed equality there is $\rho$ with $\sigma D_C^L \rho D_D^L \tau$. Hence $a_{C,D}(\rho) = (x, y)$, proving surjectivity.

□

The comparison for successive legal deviations is induced by the pullback of outsider-coordinate images.

$$\begin{array}{ccc} L_{-C} \times_{L_O} L_{-D} & \xrightarrow{\mathrm{pr}_2} & L_{-D} \\ \downarrow{\scriptstyle \mathrm{pr}_1} & \lrcorner & \downarrow{\scriptstyle \rho_O^{-D}} \\ L_{-C} & \xrightarrow[\rho_O^{-C}]{} & L_O \end{array}$$

For the concrete dependent domain

$$L = \{(0,0),(1,1)\} \subseteq \{0,1\}^2,$$

one has $L_{\{1\}} = L_{\{2\}} = \{0,1\}$, while $\lambda_{\{1\},\{2\}} : L \to \{0,1\}^2$ is the diagonal inclusion and is not surjective. Taking $q = \mathrm{id}_L$, $P = \{(0,0)\}$, and $Q = \{(1,1)\}$, agent 1 can force $P$ with its realizable choice 0 and agent 2 can force $Q$ with its realizable choice 1, but their witnesses have no legal amalgamation and $P \cap Q = \varnothing$. Likewise both singleton deviation relations are identities whereas the two-player relation is $L \times L$. The example separates legal dependence from the full-product case.

### 7.6.2 Choice cells and the one-stage logic of agency

Effectivity asks whether some coalition choice guarantees a goal. A related one-stage operation asks what is guaranteed by the coalition's already selected choice. This distinction makes contact with the seeing-to-it-that interpretation of agency (Horty and Belnap, 1995): choice and ability quantify over different fibres even when they use the same legal domain.

**Proposition 7.29** (Choice-cell guarantees and effectivity)**.** *For $e_C : L \twoheadrightarrow L_C$, put*

$$Q_C = \ker e_C = D^L_{N\setminus C}.$$

*For a predicate $R \hookrightarrow L$, the guarantee of the current coalition choice is*

$$[Q_C]R = e_C^* \forall_{e_C} R.$$

*Some realizable coalition choice guarantees $R$ exactly when $\exists_{!:L_C \to 1} \forall_{e_C} R$ holds. At a legal reference $\ell$, the predicate*

$$[Q_C]R \ \wedge \ \neg[L \times L]R$$

*says that the current choice guarantees $R$ while $R$ is not already inevitable among all legal profiles.*

*Proof.*

(1) **Policy-level comparison.** The $Q_C$-successors of $\ell$ are exactly the legal profiles $\ell'$ with $e_C(\ell') = e_C(\ell)$, which proves the first equation by the fibre formula for $\forall_{e_C}$. Existentially quantifying over its image $L_C$ gives the second assertion and excludes unrealizable choices. The universal relation $L \times L$ tests $R$ at every legal profile, giving the final conjunction.

(2) **Complementary choice-cell relation.** Finally, fixing the $C$-coordinates is exactly allowing changes of their complement, which proves the equality with $D^L_{N\setminus C}$.

□

The current-choice modality ranges over outsiders' legal completions; profitable-deviation modalities instead range over changes to the owner's own policy with outsiders fixed. Interchanging these relations would reverse the control interpretation. Their conjunction in the one-stage deliberative clause therefore records two genuinely different tests.

For unrestricted products, independently fixed choices of different owners always have a common realization, given by their coordinate tuple. Coupled legality need not have this property: with $L = \{(0,0),(1,1)\}$, choosing 0 for one owner and 1 for the other produces no joint legal profile. Compatibility of choices must therefore be established from the legal domain, not from ownership alone.

### 7.6.3 Collective improvements and stability

**Definition 7.30** (Blocking and cores)**.** Let $L$ be a reference carrier. For each nonempty coalition $C \subseteq N$, a specified **blocking-witness family** is a map $p_C : B_C \to L$, whose fibre $B_C(\ell) = p_C^{-1}(\ell)$ contains the exhibited $C$-blocks at $\ell$. The **blocking predicate** and its corresponding **core** are

$$\mathsf{Block}_C = \exists_{p_C} \top \hookrightarrow L, \qquad \mathsf{Core}_B = \bigwedge_{\varnothing \neq C \subseteq N} \neg \mathsf{Block}_C \hookrightarrow L.$$

For strict strong-Nash blocks, $B_C$ is the comprehension of $\mathsf{Improve}_C^L \hookrightarrow D_C^L$, with $p_C$ induced by the first deviation projection. For effectivity blocks it is the family of enforcing coalition coordinates; for endowment blocks it is the family of certified reallocations of the coalition's resources. These are different cores because their witness and alternative carriers are different.

The distinctions used here are:

| Stability notion | Alternative ruled out |
|---|---|
| Nash equilibrium | One agent strictly improves while outsiders retain their policies. |
| Strong Nash stability | A nonempty coalition changes its policies, with outsiders fixed, and every member strictly improves. |
| Strong-Pareto stability | A coalition changes its policies, with outsiders fixed; all members weakly improve and at least one strictly improves. |
| Effectivity core | A coalition chooses a coordinate whose every legal completion strictly improves all its members. |
| Endowment core | A coalition reallocates its own initial resources so that every member strictly improves, using a certified reallocation of its endowment. |

In the endowment interpretation, the outsider comparison concerns exclusion of outsiders' resources, leaving their policy coordinates. These are consequently different stability predicates.

**Definition 7.31** (Strict coalitional improvement and strong Nash stability)**.** For a nonempty coalition $C$, put

$$\mathsf{Improve}_C = \bigwedge_{i \in C} (\delta_C^G)^* \mathsf{Better}_i^{A,B} \hookrightarrow D_C. \tag{7.16}$$

Define strong Nash stability by

$$\mathsf{SNE}_G = \bigwedge_{\varnothing \neq C \subseteq N} \forall_{p_1 : D_C \to \Sigma} \neg \mathsf{Improve}_C. \tag{7.17}$$

For the legal domain $m : L \hookrightarrow \Sigma$, put

$$\mathsf{Improve}_C^L = \bigwedge_{i \in C} (\delta_C^{G,L})^* \mathsf{Better}_i^{A,B} \hookrightarrow D_C^L$$

and

$$\mathsf{SNE}_{G,L} = \bigwedge_{\varnothing \neq C \subseteq N} \forall_{p_1 : D_C^L \to L} \neg \mathsf{Improve}_C^L. \tag{7.18}$$

For measured profile utilities replace each arrow comparison on $D_C^L$ by $(\bar{v}_i p_1, \bar{v}_i p_2)^* \prec_i$, keeping the same outsider-fixed projection and universal quantifier. The Pareto-improvement variant is made into an explicit optimality predicate in Construction 7.32 below.

**Construction 7.32** (Weak and strong Pareto optimality)**.**

(1) **Evaluate individual improvements.** On the legal completed profile space $L$ take $v_i$ to be the descent of the numerical evaluation $e_i r$ and form each strict improvement by comparing these two endpoint values, as in Proposition 5.43. An arrow-level presentation gives the instance $v_i(\ell) = k_i(g(m\ell))$. For every nonempty coalition $C$, the existing strict coalition-improvement predicate is $\mathsf{Improve}_C^L \hookrightarrow D_C^L$.

(2) **Form Pareto improvement.** Construct the Pareto-improvement predicate on that same dependent alternative space by

$$\mathsf{ParetoGain}_C^L(\ell, \ell') \iff \left(\bigwedge_{i \in C} v_i(\ell') \geq v_i(\ell)\right) \wedge \left(\bigvee_{i \in C} v_i(\ell') > v_i(\ell)\right).$$

(3) **Test the legal alternatives.** Define the weak and strong Pareto-optimality assertions on $L$ by

$$\mathsf{WPO}_{C,L} = \forall_{p_1 : D_C^L \to L} \neg \mathsf{Improve}_C^L, \qquad \mathsf{SPO}_{C,L} = \forall_{p_1 : D_C^L \to L} \neg \mathsf{ParetoGain}_C^L.$$

For the empty coalition both predicates are $\top_L$. Weak Pareto optimality rules out an alternative strictly improving every member; strong Pareto optimality rules out an alternative weakly improving every member and strictly improving at least one. For $C = N$ these are the ordinary global Pareto predicates relative to $L$; for other coalitions outsider coordinates remain fixed. Their fibres are subterminal ordinary dependent objects. For preorder-valued preferences, the construction evaluates the supplied weak preorder and its strict part (Chen, Shi, and Wang, 2022).

**Proposition 7.33** (Pareto optimality and strategic stability)**.** *For either numerical presentation, for every coalition $C$, $\mathsf{SPO}_{C,L} \leq \mathsf{WPO}_{C,L}$. Moreover*

$$\mathsf{NE}_{G,L} = \bigwedge_{i \in N} \mathsf{WPO}_{\{i\},L} = \bigwedge_{i \in N} \mathsf{SPO}_{\{i\},L}, \qquad \mathsf{SNE}_{G,L} = \bigwedge_{\varnothing \neq C \subseteq N} \mathsf{WPO}_{C,L},$$

*and the all-coalition strong-Pareto condition satisfies*

$$\bigwedge_{\varnothing \neq C \subseteq N} \mathsf{SPO}_{C,L} \;\leq\; \mathsf{SNE}_{G,L}.$$

*All these assertions commute with substitution of a reference context; legal strategic quotients preserve and reflect them when the utility comparisons descend.*

*Proof.*

(1) **Strategic comparison.** A strict gain for every member is a weak gain for every member with at least one strict gain; taking complements and universally quantifying gives $\mathsf{SPO}_{C,L} \leq \mathsf{WPO}_{C,L}$. For a singleton coalition the two improvement predicates coincide and are exactly the profitable unilateral-deviation predicate. Their conjunction is legal Nash equilibrium.

(2) **Base-change comparison.** Conjoining the weak Pareto predicates over all nonempty coalitions gives the definition of strong Nash stability. The final inequality follows pointwise from the first. Substitution retains the entire alternative fibre by Beck–Chevalley, and the quotient claim follows from alternative lifting in Theorem 14.18 and preservation of the descended comparisons.

□

**Proposition 7.34** (Coalitional interpretation and quotient invariance)**.** *For $\mathbb{K}$-valued utilities, $\mathsf{SNE}_G(\sigma)$ holds exactly when, for every nonempty $C$, there is no $\tau_C$ such that $v_i(\tau_C, \sigma_{-C}) > v_i(\sigma)$ for all $i \in C$. On $L$ the same assertion ranges over legal alternatives with the actual outsider coordinates fixed. Legality-sensitive strategic reduction preserves and reflects these predicates, and also the Pareto variant when its utility comparisons descend.*

*Proof.*

(1) **Lifting through the comparison.** The fiber of $p_1$ over the reference consists precisely of the stated alternatives. Expand the right quantifier and the pulled-back comparisons. For quotient invariance, Theorem 14.18 supplies both forward preservation and lifting of every reduced legal alternative.

(2) **Strategic comparison.** The commuting counterfactual realizations preserve each comparison. A profitable alternative therefore exists on one side exactly when it exists on the other. Finite conjunction over the coalitions gives the result.

□

*Remark* 7.35 (Carriers of collective alternatives). Strong Nash stability fixes actual outsiders. Effectivity quantifies over all legal completions of a coalition choice. Exchange economies use another carrier: certificates for reallocating a coalition's own initial endowment. Its core predicate uses the same assertion doctrine with that distinct form of collective control.

**Definition 7.36** (Enforceable blocks and the effectivity core)**.** For $\mathbb{K}$-valued utilities, $\ell \in L$, and nonempty $C$, define

$$P_{\ell,C}(\ell') \quad \Longleftrightarrow \quad \bigwedge_{i \in C} v_i(\ell') > v_i(\ell).$$

An enforceable block at $\ell$ is a realizable coalition coordinate $c \in L_C$ such that every legal completion $\ell'$ of $c$ satisfies $P_{\ell,C}$. Thus existence of such a block is exactly $[C]_L P_{\ell,C}$, using the identity outcome map of $L$. Define the effectivity core by

$$\mathsf{Core}^{\mathrm{eff}}_{G,L}(\ell) = \bigwedge_{\varnothing \neq C \subseteq N} \neg [C]_L P_{\ell,C}.$$

This is a notion of stability defined by effectivity. Characteristic-function games and cores with transferable utility require their own specified data.

**Proposition 7.37** (Amalgamation turns enforceable blocks into deviations)**.** *An enforcing coordinate $c \in L_C$ has a candidate with the actual outsiders of $\ell$ exactly when*

$$(c, e_{-C}(\ell)) \in \mathrm{im}(\lambda_{C,-C} : L \to L_C \times L_{-C}).$$

*Whenever this condition holds, the candidate is a strict strong-Nash block. In particular surjectivity of $\lambda_{C,-C}$ makes every enforceable block a strong-Nash block. The condition holds on a full product.*

*Proof.*

(1) **Image characterization of deviations.** Membership in the displayed image says exactly that there is an $\ell' \in L$ with $e_C(\ell') = c$ and $e_{-C}(\ell') = e_{-C}(\ell)$. The latter equality places $(\ell, \ell')$ in $D_C^L$; the former invokes the enforcing property, so every member strictly benefits. Surjectivity provides that witness for every pair. On a full product the witness is the tuple $(c, \ell_{-C})$.

□

**Example 7.38** (Authority and information in complementary commitments)**.** For Example 5.69, the coalition of both owners can move from $(0,0)$ to $(1,1)$ and strictly benefit both members. Thus $(0,0)$ fails strong Nash stability while $(1,1)$ satisfies it. Regrouping the owners turns this joint deviation into a unilateral deviation of the new owner; pooling their observations alone does not change its ownership. A collective preference for a regrouped owner remains separate from that structural change.

### 7.6.4 Substitution and three different changes of context

**Proposition 7.39** (Stability of agent formation)**.** *For $f : V \to W$ there are coherent isomorphisms*

$$f^*\mathsf{Ag}_{\vec{\Sigma}}(A,B) \cong \mathsf{Ag}_{f^*\vec{\Sigma}}(f^*A, f^*B).$$

*They preserve sequential and parallel composition, execution, and ownerwise reparametrization.*

*Proof.*

(1) **Fibrewise calculation.** At $v$, both sides are $[\mathsf{Lin}\Sigma_{f(v)} \otimes A_{f(v)}, B_{f(v)}]$. All the named operations are defined by these pointwise tensor, evaluation, and ordinary profile maps. Identity and composition of substitutions give the same evaluations.

□

Three substitutions that can look similar on state carriers have different agency meanings. First, changing the reference state pulls back only the source of an alternative span. For $W \xleftarrow{s} E \xrightarrow{t} Y$ and $f : V \to W$, the new source is the corresponding pullback.

$$V \leftarrow V \times_W E \xrightarrow{t\pi_E} Y.$$

The target remains all of $Y$, so the original alternatives stay available. Beck–Chevalley then gives exact substitution of existential and universal tests along the reference map.

Second, a policy reparametrization changes the available controlled alternatives themselves. Its comparison is the realization-preserving map and alternative-lifting construction of Section 4.4, not a source-only pullback.

Third, epistemic restriction changes both endpoints of accessibility. For $j : U \hookrightarrow W$, the relation $R$ becomes $(j \times j)^*R$, deleting excluded alternatives as well as excluded references. Public announcements in Section 9.2 use this operation, which is not interchangeable with the first two.

## 7.7 Uniformly informed and legal policies

### 7.7.1 The legal local menu in agent syntax

Let $o : W \twoheadrightarrow O$ be an effective observation and $F \hookrightarrow W \times M$ a feasibility or permission relation. The construction in Construction 7.10 gives $A_o = \forall_{o \times 1_M} F$. Its agent-calculus dependent type is

$$\mathsf{Pol}_o(F) = \prod_{z:O} \{m : M \mid A_o(z,m)\}, \tag{7.19}$$

canonically identified with the sections of $A_o \to O$.

**Theorem 7.40** (Uniform legality)**.** *For $\pi : O \to M$, membership in $\mathsf{Pol}_o(F)$ is equivalent to $F(w, \pi(o(w)))$ for every $w$. Thus it represents exactly the observation-based responses legal at all compatible worlds.*

*Proof.*

(1) **Adjunction and transposition.** Under the sections isomorphism, membership means $\Gamma_\pi \subseteq A_o$. The universal-image adjunction proves $\Gamma_\pi \subseteq \forall_{o\times 1} F$ if and only if $(o \times 1)^* \Gamma_\pi \subseteq F$; this is exactly the worldwise criterion of Proposition 7.11.

□

For complete-policy changes of owner $i$, take $M = \Sigma_i$ and $F_i(w,u) \iff (s(w)_{-i}, u) \in L$. The empty-menu counterexample and its quantifier-order interpretation are given after Proposition 7.11.

### 7.7.2 Refinement and decentralized coalitions

**Proposition 7.41** (Refinement of information)**.** *If $o = r \circ o'$ with both observations effective, precomposition $\pi \mapsto \pi r$ injects $\mathsf{Pol}_o(F)$ into $\mathsf{Pol}_{o'}(F)$. Its image is precisely the refined policies constant on the fibres of $r$.*

*Proof.*

(1) **Lifting along the surjection.** This is the section-level specialization of the refinement and effective-descent comparisons in Proposition 7.11: surjectivity of $r$ makes the map injective, and unchanged worldwise responses preserve legality in both directions.

□

For joint legality $F_C \hookrightarrow W \times \prod_{i\in C} M_i$, the decentralized type is

$$\mathsf{Pol}_C^{\mathrm{dec}}(F_C) = \left\{ (\pi_i) \in \prod_{i\in C} M_i^{O_i} : \forall w \; F_C(w, (\pi_i(o_i(w)))_i) \right\}.$$

The pooled observation image $O_C$ defines the map $(\pi_i)_i \mapsto (z \mapsto (\pi_i(p_i z))_i)$.

**Proposition 7.42** (Exact decentralized comparison)**.** *This map into pooled legal policies is injective and has as image exactly those pooled policies whose $i$th output descends along $p_i : O_C \to O_i$, for each member of $C$.*

*Proof.*

(1) **Policy-level comparison.** Propositions 7.13 and 6.27 give the injectivity, coordinate-wise descent, and unchanged joint feasibility test. Their explicit map is the one displayed above, so this is an identification of dependent policy types, not a new independence or marginal-legality principle.

□

Pooling can strictly enlarge the policy type, but executable access to the pooled signal still requires its specified communication wiring.

### 7.7.3 Collective evaluation and institutional power

For a nonempty coalition $C$, one constructed collective comparison is

$$P_C^{\mathrm{all}}(\sigma,\tau) = \bigwedge_{i\in C} P_i(\sigma,\tau).$$

The corresponding stability predicate says that no legal $C$-deviation strictly improves every member. Weak Pareto improvement is a different predicate, formed from the members'

weak comparisons and at least one strict comparison. A designated corporate comparison can also be supplied as the institution's evaluation interface. Regrouping controls does not choose between these evaluations.

**Corollary 7.43** (Delegation preserves the implemented stability test)**.** *For the image comparison of Theorem 4.17, pull back the same parent comparison predicates. Stability against a family of parent coalitions pulls back exactly to stability against the corresponding whole delegate blocks. In particular parent Nash equilibrium pulls back to Nash equilibrium after regrouping each delegate block as one owner.*

*Proof.*

(1) **Lifting through the comparison.** Every subordinate deviation maps to a parent deviation with the same comparison value. Every parent deviation lifts at the given reference while retaining outsider blocks. Existence of an improving alternative is therefore equivalent on the two sides. Negating and conjoining the specified tests gives the result.

□

Institutional ability has a different quantifier pattern. Let $e_C : L \twoheadrightarrow L_C = \mathrm{im}(L \to \Sigma_C)$ be the realizable coalition-coordinate map, and let $z : L \to Z$ record outcomes. Define

$$\mathsf{Can}_C(P) \iff \exists u \in L_C \; \forall \ell \in L \, (e_C(\ell) = u \Rightarrow P(z\ell)). \tag{7.20}$$

Each quantified coordinate has an actual legal completion because $e_C$ is surjective. Thus the formula gives no vacuous power to impossible choices. When choices must be informed, the existential witness is a member of the decentralized or pooled legal policy type, and the universal quantification is taken over its compatible worlds and legal completions. This constructs uniform ability directly, with the information constraint in its witness.

A designer who wants to give a coalition the power to act must therefore check three things. Its choice must have a legal completion, must depend only on information available to its members, and must achieve the goal against every relevant outsider response. Preference enters a different test, which asks whether an owner would choose a feasible deviation. The next section follows the witness for an execution through a change of institutional state.

# 8 Witnessed Institutions and Dynamic Agency

An authorization may change what another component can do. A record that the system moved between two states can establish reachability, yet omit the certificate the next component needs. We therefore keep a witness for each institutional execution together with its concurrent realization. Composition matches these witnesses at their common boundary. Dependent sums and products then express what some or every matching execution permits us to conclude.

## 8.1 Dependent witnesses and certified realizations

Consider the difference between learning that an authorization exists and receiving the authorization needed for the next action. An existential image answers the first question. The dependent family retains the certificate and its concurrent realization for the second. These are the proof-irrelevant and proof-relevant readings of the same institutional data.

**Construction 8.1** (Dependent witness families)**.**

(1) **Form the dependent witness context.** Let $A, B \in \mathsf{L}(\Gamma)$ and let $K \in \mathsf{T}(\Gamma)$. Its context extension is $p_K : \Gamma.K = \coprod_{\gamma\in\Gamma} K_\gamma \to \Gamma$.

(2) **Realize an individual witness.** A witness realization is an ordinary morphism

$$\rho : K \longrightarrow \mathsf{Val}_\Gamma[A, B]_D.$$

Thus each witness $k \in K_\gamma$ exhibits a linear arrow $\rho_\gamma(k) : A_\gamma \to B_\gamma$. A certified family is a section $\kappa : 1_\Gamma \to K$; its realized family is $g = \rho\kappa$.

(3) **Take its support.** The predicate that a witness exists is

$$\|K\| = \exists_{p_K} \top \hookrightarrow \Gamma.$$

The witness object $K$, its realization $\rho$, and any exhibited section remain available separately from this predicate.

(4) **Reindex the family.** For $f : \Delta \to \Gamma$, pullback gives $f^*K$, $f^*\rho$, and $f^*\kappa$, and

$$f^*(\rho\kappa) = (f^*\rho)(f^*\kappa), \qquad f^*\|K\| = \|f^*K\|.$$

The first equation is functoriality of substitution; the second is Proposition 5.24. Consequently realization and existential feasibility share the constructed substitution while retaining their different carriers. Allocation and production witnesses are particular instances of this construction.

For a constant interaction boundary and a displayed certificate carrier, the compatibility of its tests with operational tests has the following form. Here $K$ denotes the total certificate carrier and $U$ its realization map.

**Proposition 8.2** (Certificate tests and operational tests)**.** *Let $K$ be a displayed certificate carrier, $U : K \to \mathcal{C}(A, B)$ its concurrent realization, and $\kappa : \Sigma \to K$ a certified arena with $g = U\kappa$. For $Q \hookrightarrow \mathcal{C}(A, B)$ and $P \hookrightarrow K$,*

$$g^*Q = \kappa^*U^*Q, \qquad \mathrm{Test}_P(\kappa) = \kappa^*P.$$

*Thus operational and certificate predicates belong to the same assertion fibre, and the second class may retain distinctions unseen by $U$.*

*Proof.*

(1) **Witness and legality comparison.** The first equality is functoriality of predicate substitution; the second is semantic testing of the certificate term. Both have typed expressions using $\Delta_G K$ and $\Delta_G \mathcal{C}(A, B)$. No inverse to $U$ is needed: the carrier $K$ and the term $\kappa$ are retained.

□

## 8.2 Witnessed institutional executions

To compose two institutional actions we must connect the record produced by the first to the record expected by the second. A witness carries both endpoints and the concurrent process that relates them. This gives a record-based interpretation of institutional power in the sense of Jones and Sergot (1996). If two executions reach the same record with different certificates, keeping both witnesses allows the next institution to distinguish them.

**Definition 8.3** (A witnessed institutional execution)**.** An institutional boundary is a pair $(X, A)$ of an ordinary record set $X$ and a linear boundary $A \in \mathcal{C}$. A witnessed execution from $(X, A)$ to $(Y, B)$ is

$$\mathcal{E} = (X \xleftarrow{s} E \xrightarrow{t} Y,\ \rho : E \to \mathcal{C}(A, B),\ P \hookrightarrow E).$$

Here $E$ retains execution witnesses, $\rho(e)$ is the realized concurrent interaction, and $P$ consists of permitted witnesses. A map $\mathcal{E} \Rightarrow \mathcal{E}'$ is a function $f : E \to E'$ with $s'f = s$, $t'f = t$, $\rho' f = \rho$, and $f[P] \subseteq P'$. The last condition preserves permission; reflection of permission is a stronger property and is not included in every map.

An owned arena $G = (\vec{\Sigma}, g)$ yields such a witness presentation when execution data include a policy map $\lambda : E \to \Sigma$, record maps, and a permission predicate $P \subseteq \lambda^{-1}(L)$. Its realizer is $\rho = g\lambda$. The witness may retain a policy tuple, a certificate, or hidden communication. Restricting to permitted witnesses changes the institutional action but does not identify distinct executions.

**Theorem 8.4** (Composition of witnessed institutions)**.** *Witnessed executions and their maps form a symmetric monoidal bicategory* $\mathbf{Inst}(\mathcal{C})$*. For consecutive executions* $\mathcal{E}_1 : (X, A) \to (Y, B)$ *and* $\mathcal{E}_2 : (Y, B) \to (Z, C)$*, composition is*

$$E_{21} = E_1 \times_Y E_2, \qquad \rho_{21}(e_1, e_2) = \rho_2(e_2) \circ \rho_1(e_1), \qquad P_{21} = P_1 \times_Y P_2.$$

*The source and target are* $s_1(e_1)$ *and* $t_2(e_2)$*. The identity at* $(X, A)$ *has witness carrier* $X$*, both endpoint maps equal to* $\mathrm{id}_X$*, constant realizer* $\mathrm{id}_A$*, and every witness permitted. The tensor has*

$$(X, A) \otimes (X', A') = (X \times X', A \otimes A'), \qquad E_{\mathcal{E}\otimes\mathcal{F}} = E \times F, \qquad P_{\mathcal{E}\otimes\mathcal{F}} = P_{\mathcal{E}} \times P_{\mathcal{F}},$$

*with coordinatewise endpoints and realizer* $\rho_{\mathcal{E}}(e) \otimes \rho_{\mathcal{F}}(f)$*.*

A pair contributes to the composite span exactly when the record produced by the first execution is the input record of the second.

$$\begin{array}{ccc} E_1 \times_Y E_2 & \xrightarrow{\mathrm{pr}_2} & E_2 \\ {\scriptstyle \mathrm{pr}_1}\downarrow & \lrcorner & \downarrow{\scriptstyle s_2} \\ E_1 & \xrightarrow[t_1]{} & Y \end{array}$$

*Proof.*

(1) **Secured operational interaction.** For composable witnessed executions, the pullback has the explicit fibrewise description

$$(E_1 \times_Y E_2)(x, z) \cong \coprod_{y \in Y} E_1(x, y) \times E_2(y, z), \tag{8.1}$$

which sends a matching pair to its uniquely determined intermediate record and its two constituent witnesses. Its inverse assembles that pair, so it retains both realizer labels and both permissions.

(2) **Naturality of witnessed composition.** If $f : E_1 \to E_1'$ and $g : E_2 \to E_2'$ preserve the endpoints, $(e_1, e_2) \mapsto (f(e_1), g(e_2))$ is a well-defined map of the pullbacks. It preserves the composite realizer by the two realization equations and sends permitted pairs to permitted pairs. This defines horizontal composition of maps; vertical composition is composition of functions.

(3) **Interchange of witnessed executions.** The interchange equation is pointwise the equality of the two componentwise composite functions.

(4) **Secured operational interaction.** The associator sends $((e_1, e_2), e_3)$ to $(e_1, (e_2, e_3))$. Both sides are exactly the triples with matching consecutive records. Their realizers agree by associativity in $\mathcal{C}$, and their permission predicates are both the conjunction of the three permission predicates.

(5) **Witness and legality comparison.** The unitors delete the unique intermediate identity witness, whose record is already fixed by the adjacent endpoint. Pentagon and triangle coherence follow because each route has the same tuple projections.

(6) **Parallel compatibility.** For tensor and horizontal composition, the comparison

$$(E_1 \times E_1') \times_{Y \times Y'} (E_2 \times E_2') \cong (E_1 \times_Y E_2) \times (E_1' \times_{Y'} E_2')$$

reorders a matching quadruple. The interchange law in $\mathcal{C}$ preserves its realizer, and the four permission tests agree on both sides.

(7) **Secured operational interaction.** The monoidal associativity, unit, and symmetry components are the graph spans of the corresponding bijections of ordinary records, labelled by the corresponding structural isomorphisms of $\mathcal{C}$, with all witnesses permitted. Their naturality and coherence reduce to those bijections and the monoidal coherence of $\mathcal{C}$. This supplies the stated symmetric monoidal bicategory structure.

□

Products here combine ordinary institutional records, not copies of arbitrary linear resources. The realized concurrent interactions continue to compose by secured interaction and hiding. Two distinct witnesses may therefore have identical source and target records while remaining distinguishable by their strategies or certifications.

**Proposition 8.5** (Witness retention, permission, and institutional modalities)**.** *Write* $E(x, y) = \{e \in E : s(e) = x,\ t(e) = y\}$. *Composition retains the dependent witness family*

$$E_{21}(x, z) \cong \coprod_{y \in Y} E_1(x, y) \times E_2(y, z).$$

*Taking inhabitedness gives ordinary relational composition. For* $Q \hookrightarrow Y$, *the source predicates*

$$\langle \mathcal{E} \rangle Q = \exists_s t^* Q, \qquad [\mathcal{E}] Q = \forall_s t^* Q$$

*express, respectively, that some witnessed execution reaches* $Q$ *and that every witnessed execution reaches* $Q$. *They obey*

$$\langle \mathcal{E}_2 \circ \mathcal{E}_1 \rangle Q = \langle \mathcal{E}_1 \rangle \langle \mathcal{E}_2 \rangle Q, \qquad [\mathcal{E}_2 \circ \mathcal{E}_1] Q = [\mathcal{E}_1][\mathcal{E}_2] Q.$$

*The same equations hold when each execution is restricted to its permitted subobject. A guarantee requiring an actual execution is*

$$(\exists_s \top) \wedge \forall_s t^* Q.$$

*Finally the violating witnesses of a composite are exactly*

$$\begin{aligned}(E_1 \times_Y E_2) \setminus (P_1 \times_Y P_2) = {} & ((E_1 \setminus P_1) \times_Y E_2) \\ & \cup (E_1 \times_Y (E_2 \setminus P_2)).\end{aligned}$$

*Proof.*

(1) **Witness and legality comparison.** A composite witness is a pair with one common intermediate record; recording that value gives the displayed disjoint-union bijection. It has an inhabitant exactly when some intermediate record has an inhabitant in each of the two fibres.

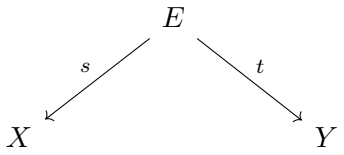
(2) **Fibrewise calculation.** At $x$, the composite diamond says that there are $e_1, e_2$ with $s_1(e_1) = x$, $s_2(e_2) = t_1(e_1)$, and $Q(t_2(e_2))$. This is the nested existential formula on the right. The box replaces these existential tests by the two universal tests, giving its equation, including intermediate records with no outgoing witness.

(3) **Predicate calculation.** Categorically these are the dependent quantifier composition and pullback equations. The restricted permitted spans compose by the same pullback, so the argument applies to them as well. The extra conjunct $\exists_s \top$ excludes a source with no execution, where the box alone is vacuously true.

(4) **De Morgan law for matched execution witnesses.** The final equality is the pointwise equivalence $\neg(P_1(e_1) \wedge P_2(e_2))$ with $\neg P_1(e_1) \vee \neg P_2(e_2)$ on the pullback.

□

An institutional execution witness carries a source record and a target record.




The enabling comprehension and target record describe how an authorization changes the institution. The predicate $P$ states whether the execution is permitted. Keeping the execution witness independent of that judgement allows a later audit or sanction to refer to the very occurrence that violated the rule.

## 8.3 Linear realization and dynamic modalities

The witnessed executions of Definition 8.3 compose in the symmetric monoidal bicategory of Theorem 8.4. We now construct their linear execution maps and the dependent type operations induced by their witness spans. These constructions retain both the realizer and the permission predicate.

### 8.3.1 Measuring secured witnesses before hiding

Suppose a resource meter charges once for delivering a message. The sender and the receiver each record a local occurrence. Adding their logs before matching the occurrences would charge twice for one delivery. The next construction identifies the matched class first, assigns its weight once, and retains it when the shared interface is hidden.

**Construction 8.6** (Additive charges on a finite secured witness)**.**

(1) **Identify matched occurrences.** In a finite interaction retain the component configurations with their tagged occurrences and the actual matches on shared interfaces, using the secured interaction of Castellan et al. (2017). For a finite chain use the full tuple of component configurations with all its matches. On the disjoint union of occurrences take the equivalence relation generated by these matches. Close the component causal orders

transitively on the classes. Global securedness means this relation is antisymmetric, so it gives the causal order of the retained witness $k$. Retain this finite poset $H(k)$ before hiding any internal interface.

(2) **Record each class once.** Use the displayed protocol labels to give each singleton or matched class a signature in $G$. A matched send and receive can carry a single signature recording both ports. With $\mathsf{M}(G) = \mathbb{N}^{(G)}$, the finite multisets of signatures, put

$$U(k) = \sum_{H \in H(k)} [g(H)], \qquad m_w(k) = \bar{w}(U(k)), \qquad \bar{w}(n) = \sum_{g \in \mathrm{supp}(n)} n(g)\, w(g).$$

Here $w : G \to C$ gives the weight of one class and $n(g)\, w(g)$ denotes repeated monoid addition. The empty history has multiset zero. The formula for $\bar{w}$ is additive and is forced on every finite multiset by its values on the generators. Thus it is the unique monoid map extending $w$. In the cost-action interpretation of Niu et al. (2022), successive charges add. Secured matching here determines which occurrences incur one charge.

(3) **Transport the measurement.** An isomorphism transporting the labels and matching induces a bijection on classes and preserves their interpreted weights. Rebracketing a fixed global witness leaves its generating matches and hence its class sum unchanged. Disjoint parallel witnesses give disjoint unions of classes. For a compatible prefix with inherited signatures, partitioning the classes into old and new ones gives $m_w(k') = m_w(k) + m_w(k' \setminus k)$, where the last term is the sum over new classes. A change $j : C \to C'$ of commutative monoids gives $j\bar{w} = \overline{jw}$ by agreement on generators. These comparisons use the original matching even when a hidden class has no visible representative.

(4) **Read span from the order.** For natural durations $d(H)$ compute $t(H) = d(H) + \max(\{0\} \cup \{t(J) : J < H\})$ along the finite order. Induction on strict predecessors identifies $t(H)$ with the maximum duration of a chain ending at $H$. The maximum over all $t(H)$ and 0 gives the span. Two equally labelled unit events have the same multiset in a chain and an antichain, but spans two and one, respectively.

**Proposition 8.7** (Secured prefixes compose in the cost writer)**.** *In a fixed unhidden interaction with its displayed class signatures, let $x \subseteq y$ range over compatible prefixes, inheriting the signatures of the retained occurrences. The charge of an extension is*

$$\delta(x, y) = \sum_{H \in y \setminus x} w(g(H)).$$

*Here each index $H$ denotes a synchronized class in that fixed history. The prefix-inclusion category has a functor to the one-object category of $(C, +, 0)$ sending $x \subseteq y$ to $\delta(x, y)$. Following a prefix by a continuation accumulates charges by writer multiplication.*

*Proof.*

(1) **Partition classes.** The empty difference has charge zero. For $x \subseteq y \subseteq z$, the disjoint union of $y \setminus x$ and $z \setminus y$ is $z \setminus x$. The inherited signatures agree on these sets, hence $\delta(x, z) = \delta(x, y) + \delta(y, z)$. This proves identity and composition, including in monoids without subtraction.

(2) **Compute the writer record.** Give $x \subseteq y$ output $y$ and charge $\delta(x, y)$. Concatenating with $y \subseteq z$ at incoming charge $c$ gives $(c + \delta(x, y) + \delta(y, z), z)$. This is the record of the full extension at $c$ by the first step and Construction 2.7. Parallel independent histories give disjoint unions of classes, and their charges add in either order.

(3) **Identify the measurement boundary.** A pair of matching send and receive events is already one class at the cut where it occurs. Thus the above partition counts it once. The computation is over the retained globally matched history; its two independent component logs can contain representatives of that same class. Critical-path duration in Construction 8.6 still uses the partial order in addition to these classes.

□

**From matched records to a measured execution.** Consider two consecutive institutional executions with shared linear boundary $B$ and intermediate ordinary record set $Y$. Write $D_{21} = E_1 \times_Y E_2$ for the record pairs of Theorem 8.4. At a pair $(e, f) \in D_{21}$, use the two displayed strategy presentations $S_e, T_f$ of its realizer labels in the secured interaction construction of Section 1.1. Write $U_{e,f} = S_e \circledast_B T_f$ for the unhidden interaction. For its finite configurations let $\mathsf{Done}_{e,f}(z)$ be the completion test of the specified result interface. For a displayed completion port it tests whether its output occurs in $z$; for the finite session of Theorem 10.23 it tests whether the matched history has reached $\mathsf{End}$, including a session ended at the root. If the record $e$ or $f$ designates a particular local history, retain only configurations whose component projection equals that marked history. The resulting ordinary dependent family and its total carrier are

$$\begin{aligned} H(e,f) &= \{z \in \mathrm{Conf}(U_{e,f}) : \mathsf{Done}_{e,f}(z) \text{ and } z \text{ has the designated projections}\}, \\ K_{21} &= \coprod_{(e,f)\in D_{21}} H(e,f), \qquad d : K_{21} \to D_{21}, \quad (e,f,z) \longmapsto (e,f). \end{aligned} \tag{8.2}$$

A separately displayed family of local marks is included by taking its dependent sum before forming $H$. The underlying institution supplies $d$'s target and the two realizer labels. Its records can be used without any marks. Measuring occurrences uses the displayed executions and result interface of Constructions 2.8 and 8.6.

To verify (8.2), form the augmented synchronization pullback with the exterior events carried by identity maps, as in the definition of composition. By Theorem 1.7, a configuration of this pullback corresponds to the union of its prime secured matchings. Conversely, the prime pasts of the pairs of a secured matching recover that configuration. The two operations preserve its component histories. Restricting this bijection by the two projection tests and the same completion test on both sides identifies $H(e, f)$ with exactly the completed matching witnesses over $(e, f)$. A change of marked presentations transports configurations, matches and their finite class posets bijectively. Thus the construction is natural under the displayed presentation isomorphisms. With presentations recorded only up to isomorphism one may use their marked isomorphism classes, as in Construction 12.19; the label-preserving class sum descends since it is invariant under every such isomorphism.

For example, take $Y = 1$ and boundary $D(\{0, 1\})$. One component can output either bit, and the other accepts either bit. Retain each of their two completed local bit histories as a separate record. All four pairs match on $Y$. Only $(0, 0)$ and $(1, 1)$ have a matching bit occurrence, and each has one completed secured matching. The projection $d$ therefore accounts for enabling as well as the recorded endpoints.

The source and target on $K_{21}$ are $s_1(e)$ and $t_2(f)$. Its permission predicate is $d^*(P_1 \times_Y P_2)$, and its charge is $m_{21}(e, f, z) = m_w(z)$ from Construction 8.6. For several stages, keep the tuple of local histories, all interface matches and the globally secured order until the class sum has been formed. Reassociation of this retained tuple has inverse given by the opposite reassociation: it preserves every component, generating match and order, hence also the completion test transported along it and the charge. The unmeasured record composite continues to be $E_1 \times_Y E_2$.

**Proposition 8.8** (Budget restriction and change of measurement)**.** *For a witnessed institution $X \xleftarrow{s} E \xrightarrow{t} Y$, let $E_{\mathrm{ret}}$ be its displayed completed witnesses and let $m : E_{\mathrm{ret}} \to C$ measure them. A budget $B \hookrightarrow C$ defines the subspan on $E_B = m^{-1}B$. If $j : C \to C'$ regrades the measurement and $B' \hookrightarrow C'$, the budgets $B'$ for $jm$ and $j^*B'$ for $m$ give the same witness subspan. Its existential and universal modalities, and their dependent linear versions, agree.*

*Proof.*

(1) **Compare fibres.** For a completed witness $k$ the two membership tests are $B'(j(m(k)))$ and $(j^*B')(m(k))$. These are the same predicate by substitution. Intersecting with the institution's permission predicate gives the same permitted subspan as well. The comparison can be displayed on the composite witness carrier:

$$\begin{array}{ccc} K_B & \longrightarrow & B \\ \big\downarrow & \lrcorner & \big\downarrow \\ K_{21} & \xrightarrow[m_{21}]{} & C \end{array}$$

Thus feasibility on witnesses can be tested before taking an image in the policy carrier. A hard budget can remove a deviation; pricing the same execution compares values while retaining that deviation.

(2) **Transport the modalities.** Both restrictions have the same source and target maps $s_B, t_B$. Thus $\exists_{s_B} t_B^*$ and $\forall_{s_B} t_B^*$ agree on predicates, and $\Sigma^\ell_{s_B} t_B^*$ and $\Pi^\ell_{s_B} t_B^*$ have the same indexed summands and factors, including their witness multiplicities. For a composite with measured histories, take $E_{\mathrm{ret}} = K_{21}$ of (8.2). Its witnesses under a regraded budget are the triples $(e, f, z)$ with $j(m_{21}(e, f, z)) \in B'$. This is exactly the comprehension of $m_{21}(e, f, z) \in j^{-1}(B')$ on the same fibre $H(e, f)$.

□

### 8.3.2 The linear execution induced by witnesses

**Proposition 8.9** (Realized witnessed execution)**.** *For a witnessed execution $\mathcal{E} = (X \xleftarrow{s} E \xrightarrow{t} Y, \rho, P)$, let $\underline{yA}$ and $\underline{yB}$ denote the constant linear families on $X$ and $Y$, and put $J_E = \Sigma^\ell_s \mathsf{I}_E$. The concurrent realizer constructs a canonical linear map over $X$:*

$$\mathrm{run}_\rho : \underline{yA} \otimes J_E \longrightarrow \Sigma^\ell_s t^* \underline{yB}. \tag{8.3}$$

*These maps compose along the canonical pullback and coproduct-flattening comparisons, with the realizer of a matching pair $(e, f)$ equal to $y\rho_F(f)\, y\rho_E(e)$. Restricting to $P$ gives the corresponding permitted execution map.*

*Proof.*

(1) **Fibrewise calculation.** Pointwise Yoneda gives the family morphism $\rho^\sharp : s^*\underline{yA} \to t^*\underline{yB}$ with component $y\rho(e)$ at $e$. By Frobenius and the tensor unitor, define

$$\underline{yA} \otimes \Sigma^\ell_s \mathsf{I}_E \xrightarrow{\cong} \Sigma^\ell_s(s^*\underline{yA} \otimes \mathsf{I}_E) \xrightarrow{\Sigma^\ell_s(\rho^\sharp \otimes 1)} \Sigma^\ell_s(t^*\underline{yB} \otimes \mathsf{I}_E) \xrightarrow{\Sigma^\ell_s(\text{right unitor})} \Sigma^\ell_s t^*\underline{yB}.$$

At $x \in X$, this is the copairing of the maps $y\rho(e) : yA \to yB$ over $e$ with $s(e) = x$, retaining each summand's target-record index.

(2) **Base-change comparison.** Under a second execution, Beck–Chevalley and successive dependent sums flatten the summands indexed by matching $(e, f)$; the component map is $y\rho_F(f) \circ y\rho_E(e) = y(\rho_F(f)\rho_E(e))$. The composite execution has exactly that realizer. Coproduct uniqueness proves the comparison equation, and flattening triples proves associativity coherence. Replacing $E$ by its permitted comprehension gives the last assertion.

□

### 8.3.3 Dynamic type and predicate operations

For the underlying span $X \xleftarrow{s} E \xrightarrow{t} Y$, define

$$\langle E\rangle^{\ell} = \Sigma_s^{\ell} t^*, \qquad [E]^{\ell} = \Pi_s^{\ell} t^*.$$

At $x$, the first retains a choice of execution and an output resource; the second supplies an additive family indexed by every execution from $x$. The predicate operations are

$$\langle E\rangle P = \exists_s t^* P, \qquad [E]P = \forall_s t^* P.$$

They quantify over witnessed executions, including distinct executions with the same source and target.

**Theorem 8.10** (Compositional dynamic modalities)**.** *For consecutive witnessed spans $E, F$,*

$$[F \circ E]^{\ell} \cong [E]^{\ell}[F]^{\ell}, \qquad \langle F \circ E\rangle^{\ell} \cong \langle E\rangle^{\ell}\langle F\rangle^{\ell}.$$

*The predicate versions are equalities. Identity spans give identity operations. The comparisons are associative and unital.*

*Proof.*

(1) **Fibrewise calculation.** At $x$, the box on the composite is

$$\prod_{\substack{e : s_E(e) = x \\ f : s_F(f) = t_E(e)}} A_{t_F(f)}.$$

Flattening a nested product gives exactly $\prod_{e:s_E(e)=x} \prod_{f:s_F(f)=t_E(e)} A_{t_F(f)}$. The sum proof flattens the corresponding nested coproduct. Both include empty fibres.

(2) **Representation and Yoneda decoding.** Three successive comparisons flatten the same triples of witnesses, so product projections and coproduct injections prove coherence. The identity fibre has its unique record witness. Predicate quantifiers obey the identical nested quantification formulas.

□

For a relation $R \subseteq X \times Y$, the dynamic diamond $\exists_s t^*$ is left adjoint to $\forall_t s^*$, the box associated with the converse direction. The forward diamond and forward box are not generally an adjoint pair. Recording this variance prevents an incorrect transport rule when sequential execution spans are composed.

Passing from a witness span $E$ to its relational image retains existential and universal truth tests on target predicates. It discards proof-relevant multiplicity: distinct executions at one pair of endpoints still contribute separate dependent summands or factors and may retain different strategies or certificates. The institutional type interpretation therefore uses the full span.

### 8.3.4 Permission and collective institutional agency

Restricting $E$ to $P$ gives permitted modalities. A guarantee of $Q$ with an actual permitted execution is

$$(\exists_{s|P} \top) \wedge \forall_{s|P} (t|P)^* Q.$$

The first conjunct excludes the vacuous truth of a universal test at a record with no permitted execution. On a composite, violation is precisely

$$(\neg P_1 \wedge \top) \ \vee \ (\top \wedge \neg P_2)$$

as a predicate on $E_1 \times_Y E_2$. This records the stage at which permission failed.

An institution can be exposed as a collective agent by retaining its owner-indexed presentation and applying Section 4.4. Its observation and evaluation are then situated collective interfaces. This does not make the institution's normative rules into a preference: who may act, which actions are authorized, and which outcomes the collective prefers remain separately displayed. The preceding bicategory makes their witnessed executions compositional.

# 9 Public Announcements and Epistemic Actions

What does an owner learn when an institutional action occurs? A public announcement restricts the state space and both endpoints of each accessibility relation to states satisfying the announced predicate. An owner who observes only part of an event may face several possibilities even after it occurs. A general action model therefore retains the event and the recipient's observation of it. We construct both updates by comprehension and identify the lifting property that lets an implemented communication support the same modal tests.

## 9.1 Communication and epistemic updates

An agent may announce information, but a stated formula is not yet a delivered message. At the logical level, public update restricts the situated state space and reindexes the epistemic relations. At the operational level, a communicating agent must have a complete policy, an enabled message, and a causal realization of its delivery. We first establish the logical update and then compare it with the complete communication construction.

The later policy-to-message comparison keeps each owner's pre-existing observation and policy record alongside the public message tuple. Its forward map and source-fixed lifts will transport knowledge, distributed knowledge, coalition deviations, and the specified fixed-point assertions. This is stronger than agreement on current messages alone: modal formulas quantify over alternatives that must also be preserved.

### 9.1.1 Public restriction by comprehension

Public announcement is the comprehensional update of public communication logic, originating with Plaza and represented here by Plaza (2007). Its operation on the state set is simple, but its epistemic meaning depends on restricting both arguments of accessibility. We construct this restriction using the predicate doctrine rather than adding an independent update primitive.

**Definition 9.1** (Public announcement)**.** Fix a relation frame $M$ with state set $W$ and relations $(R_i)_{i \in N}$. For a predicate $P \hookrightarrow W$, let $a : U = \{P\} \hookrightarrow W$ be its comprehension. The **public announcement** of $P$ updates the entire frame to

$$M|_P = \big(U, (R_i^P)_{i \in N}\big), \qquad R_i^P = (a \times a)^* R_i \hookrightarrow U^2.$$

Its box and diamond are the dependent quantifiers

$$[!P] = \forall_a : \mathsf{Pred}(U) \to \mathsf{Pred}(W), \qquad \langle !P\rangle = \exists_a : \mathsf{Pred}(U) \to \mathsf{Pred}(W).$$

For a fixed predicate $Q \hookrightarrow W$, $[!P](a^*Q) = P \Rightarrow Q$. Writing $[!a]$ and $\langle !a\rangle$ for the same operators when the inclusion $a$ is specified, epistemic formulas in the updated frame are evaluated using the restricted relations $R_i^P$ on both endpoints. Merely substituting reference states would instead retain the original target alternatives and would not be this update.

This changes the epistemic model, not merely the chosen reference state. Selecting only references satisfying $P$ would leave excluded states available as epistemic alternatives; restricting both endpoints removes them. The corresponding reindexed observation fibres represent the information available after the announcement.

**Construction 9.2** (Public update)**.**

(1) **Restrict the context.** For a monomorphism $a : U \hookrightarrow W$, put $A = \exists_a \top$. Restrict an operational family $t$ to $ta$ and an assertion to $a^*P$.

(2) **Construct the modalities.** Define $[!a] = \forall_a$ and $\langle !a\rangle = \exists_a$.

(3) **Restrict both relation endpoints.** For a relation $R$ on $W$, its restricted relation is $R|_U = (a \times a)^* R$. This restriction changes both endpoints, unlike a reference pullback.

**Proposition 9.3** (Monic change of base and its nucleus)**.** *Put $L = \exists_a$, $\rho = a^*$, and $J = \forall_a$. Then $\rho L = \rho J = \mathrm{id}$ and $L\rho(P) = A \wedge P$. Moreover*

$$JQ = A \Rightarrow LQ, \tag{9.1}$$

*and*

$$J\rho(P) = j_A(P) = A \Rightarrow P. \tag{9.2}$$

*Thus $L$ identifies $\mathsf{Pred}(U)$ with $\downarrow A$, and $J$ identifies it with the fixed predicates of the nucleus $j_A$.*

*Proof.*

(1) **Base-change comparison.** The self-pullback of a monomorphism has identity projections, so Beck–Chevalley gives $\rho L = \rho J = \mathrm{id}$. Frobenius gives $L\rho(P) = A \wedge P$. Consequently $P \le A \Rightarrow LQ$ iff $L\rho P \le LQ$ iff $\rho P \le Q$.

(2) **Verification of the universal property.** Uniqueness of right adjoints proves the formula for $J$. Substitute $Q = \rho P$. The unit of $\rho \dashv J$ gives inflationarity of $J\rho$; $\rho J = \mathrm{id}$ gives idempotence, and both factors preserve finite meets. The two claimed identifications follow from the same identities.

□

The inclusion of an announced submodel and its existential and universal predicate extensions form the update comparison.

$$\begin{array}{ccc} \mathsf{Pred}(U) & \xrightarrow{\exists_m} & \mathsf{Pred}(W) \\ {\scriptstyle \forall_m}\downarrow & & \downarrow{\scriptstyle m^*} \\ \mathsf{Pred}(W) & \xrightarrow[m^*]{} & \mathsf{Pred}(U) \end{array}$$

**Corollary 9.4** (Update monad, support comonad, and restricted knowledge)**.** *For the inclusion $a : U \hookrightarrow W$, the endomaps $L\rho = A \wedge (-)$ and $J\rho = A \Rightarrow (-)$ are respectively an idempotent comonad and monad. Their coalgebra and algebra posets are equivalent to* $\mathsf{Pred}(U)$ *via $L$ and $J$, respectively. For observational knowledge $K = [\ker(o)]$ and $K^U = [\ker(oa)]$,*

$$K^U = \rho KJ, \qquad JK^U = j_A KJ.$$

*Proof.*

(1) **Adjunction and transposition.** The preceding monic adjunction identities identify both fixed-point posets and give their units and counits. For $u \in U$, the assertion $(\rho KJQ)(u)$ says that every $w$ with $o(w) = o(a(u))$ satisfies $JQ$. Outside $U$ this is true; inside $U$ it is exactly $Q$.

(2) **Observation and knowledge.** This is knowledge for the restricted observation $oa$. Applying $J$ and using $J\rho = j_A$ gives the second equation.

□

**Example 9.5** (Restriction and knowledge need not commute)**.** Let $W = \{0, 1\}$ have one observation class and let $A = P = \{0\}$. Then $Kj_AP = W$, whereas $j_AKP = \{1\}$. The restriction formula is $K^U = \rho KJ$; no commutation of $K$ with $j_A$ is used.

**Theorem 9.6** (Public-update reduction and composition)**.** *For predicates $P$ on $W$ and $Q, Q'$ on $U$,*

$$[!a]a^*P = A \Rightarrow P, \qquad [!a](Q \wedge Q') = [!a]Q \wedge [!a]Q',$$
$$[!a](Q \Rightarrow Q') = A \Rightarrow ([!a]Q \Rightarrow [!a]Q'), \qquad [!a]\neg Q = A \Rightarrow \neg[!a]Q.$$

*For a relation $R$ on $W$,*

$$[!a][R|_U]Q = A \Rightarrow [R]([!a]Q). \tag{9.3}$$

*For $R = \ker o$, the restricted relation is* $\ker(oa)$*; hence*

$$[!a]K^UQ = A \Rightarrow K([!a]Q), \tag{9.4}$$

*and*

$$[!a]K^U(a^*P) = A \Rightarrow K(A \Rightarrow P). \tag{9.5}$$

*For $V \overset{b}{\hookrightarrow} U \overset{a}{\hookrightarrow} W$, $[!a][!b] = [!(ab)]$. If $B = \exists_b \top$ in $U$, then*

$$\exists_a B = A \wedge [!a]B. \tag{9.6}$$

*Proof.*

(1) **Reindexed implication and negation.** The first equations follow from $J\rho = j_A$, preservation of meets by $J$, and preservation of implication by $\rho$. For implication use $Q \Rightarrow Q' = \rho(JQ \Rightarrow JQ')$; the negation case also uses $J\bot = \neg A$ and the Heyting implication identities. Let $\Gamma_a$ be the graph of $a$.

(2) **Descent along the kernel pair.** Then $R|_U = \Gamma_a; R; \Gamma_a^\smile$ and $\Gamma_a^\smile; \Gamma_a$ is the partial identity on $A$. Its box is $A \Rightarrow (-)$, proving the modal equation by relation composition. Kernel pairs restrict as stated.

(3) **Frobenius law for repeated announcements.** Finally right adjoints compose, and Frobenius with $a^*\forall_a = \mathrm{id}$ gives $\exists_a B = A \wedge \forall_a B$.

□

Repeated public restriction is computed by a composite of comprehension inclusions.

$$\begin{array}{ccc} U \cap Q & \hookrightarrow & U \\ \downarrow & & \downarrow{\scriptstyle m_U} \\ Q & \hookrightarrow & W \end{array}$$

*Remark* 9.7 (Fixed predicates and re-evaluated epistemic assertions). For an inclusion $a : A \hookrightarrow W$, the equation $\forall_a a^* P = (A \Rightarrow P)$ is an equation about a fixed predicate. It does not imply that an epistemic formula remains true when its own truth set is announced.

**Example 9.8** (A truthful announcement that ceases to be true)**.** Take $W = \{x, y\}$, one indiscriminate agent, and $p = \{x\}$. The formula $\varphi = p \wedge \neg Kp$ holds at $x$. Announcing its truth set leaves $\{x\}$, where $Kp$ holds and $\varphi$ is false. The factual predicate $p$ has not changed; its epistemic context has.

**Corollary 9.9** (Dynamic semantic testing)**.** *For* $\Phi = \mathrm{Test}_P(t_1, \ldots, t_n)$*,*

$$[!a]\,\mathrm{Test}_P(t_1 a, \ldots, t_n a) = A \Rightarrow \Phi, \tag{9.7}$$

*and*

$$[!a] K_i^U\,\mathrm{Test}_P(t_1 a, \ldots, t_n a) = A \Rightarrow K_i(A \Rightarrow \Phi). \tag{9.8}$$

*Proof.*

(1) **Functoriality and substitution.** Apply substitution of semantic tests and the preceding reduction equations.

□

**Corollary 9.10** (Fixed points under public restriction)**.** *For* $a : U \hookrightarrow W$ *and monotone* $F$ *on* $\mathsf{Pred}(U)$*,*

$$\forall_a(\mu F) = \mu(\forall_a F a^*), \qquad \forall_a(\nu F) = \nu(\forall_a F a^*).$$

*Proof.*

(1) **Repeated-announcement fixed-point calculation.** Use $a^* \forall_a = \mathrm{id}$ in Lemma 12.4.

□

### 9.1.2 Epistemic frames and categorical update

The comprehension inclusion for public restriction is a map of state carriers equipped with an accessibility comparison. General action updates have the same shape, but carry an event coordinate as well. To relate their assertions before and after update, we must distinguish ordinary reindexing of dependent families from exact lifting of accessible alternatives.

**Definition 9.11** (Relation frames and their maps)**.** An object $M = (W, (R_i)_{i \in N})$ of $\mathbf{Fr}_N$ is a set with an $N$-indexed family of binary relations. A morphism $f : (U, (T_i)) \to (W, (R_i))$ is a function satisfying

$$T_i \subseteq (f \times f)^* R_i \qquad (i \in N).$$

Write $|M| = W$ for the underlying set. S5 frames form the full subcategory whose relations are equivalence relations. The definitions also permit the paper's non-S5 belief relations.

**Proposition 9.12** (Pullbacks and the existing indexed semantics)**.** *The category* $\mathbf{Fr}_N$ *has finite limits. For maps* $f : M \to K$ *and* $g : L \to K$*, its pullback has underlying set* $|M| \times_{|K|} |L|$ *and relations*

$$R_i^{M \times_K L} = (\pi_M \times \pi_M)^* R_i^M \cap (\pi_L \times \pi_L)^* R_i^L.$$

*These limits restrict to S5 frames. An epistemic refinement update of* $M$ *is an object* $p : U \to M$ *of* $\mathbf{Fr}_N/M$*. Its ordinary and linear families are transported by the existing functors*

$$|p|^* : \mathsf{T}(|M|) \to \mathsf{T}(|U|), \qquad |p|^* : \mathsf{L}(|M|) \to \mathsf{L}(|U|).$$

*Successive updates compose as morphisms. Pullback gives base change between the slice categories, coherently up to their canonical isomorphisms.*

*Proof.*

(1) **Predicate calculation.** The set-theoretic pullback carries the displayed relations so that both projections preserve relations. Any compatible cone has a unique underlying map to that pullback. Since both legs preserve each relation, this map preserves their intersection.

(2) **Finite limits of equivalence frames.** This proves the universal property. A singleton with reflexive relations is terminal; products and equalizers follow. Inverse images and intersections preserve equivalence relations.

(3) **Coherence of epistemic pullbacks.** The remaining statements follow by composition of the underlying functions and the universal property of pullback. The canonical reassociation maps preserve the projected tuples, so their associativity and unit diagrams commute. The same uniqueness gives the coherence of iterated base change.

(4) **Base-change comparison.** The indexed categories of Section 1.2 and Beck–Chevalley comparisons apply to these underlying set pullbacks.

□

A public announcement can delete accessible alternatives, so its update projection is treated as a frame map with the separately stated boundedness criteria. For a forward frame map $p$ one has the comparison

$$|p|^*[R_i]\, Q \;\subseteq\; [T_i]|p|^* Q.$$

The forward frame-map comparison gives an inclusion for each predicate $Q$: every $T_i$-successor maps to an $R_i$-successor. Equality requires lifting abstract successors as well, which is the bounded-morphism condition. The exact equations in Section 9.1.4 make this distinction explicit, in line with the categorical treatment of epistemic updates in (Kishida, 2017).

### 9.1.3 Epistemic action-model updates as dependent comprehension

Public restriction is the singleton-event case of epistemic product update Baltag, Moss, and Solecki (1998). With several possible events, an agent may distinguish some occurrences but not others, and the events may change ordinary observables. The update therefore retains enabled world–event pairs and constructs their new accessibility from both the old state observations and the observations of the action event.

**Definition 9.13** (Semantic epistemic action model)**.** Let $W$ be a state set with relations $R_i \hookrightarrow W^2$ for $i \in N$. An **epistemic action model over** $W$ consists of an action-occurrence set $E$, relations $S_i \hookrightarrow E^2$, and an enabling subobject $P \hookrightarrow W \times E$. Its fibre at $e \in E$ is $P_e = \{w : P(w, e)\} \hookrightarrow W$. The relations $R_i$ and $S_i$ may be arbitrary qualitative possibility relations. If they are observation kernel pairs, they describe knowledge. An updated observable of type $Z$ is a term $v^+ : \{P\} \to Z$; the unchanged-value case is $v^+ = vp$ for $v : W \to Z$, where $p : \{P\} \to W$ is the projection. A state-transition presentation may instead provide $t : \{P\} \to W'$ and set $v^+ = v't$ for $v' : W' \to Z$.

**Construction 9.14** (Epistemic product update)**.**

(1) **Take the enabling comprehension.** Take comprehension and its two projections,

$$j : U = \{P\} \hookrightarrow W \times E, \qquad p : U \to W, \quad q : U \to E.$$

(2) **Construct the updated relations.** Write $R_i \boxtimes S_i$ for the product relation on $W \times E$,

$$(R_i \boxtimes S_i)((w, e), (w', e')) \iff R_i(w, w') \wedge S_i(e, e').$$

Define the updated relation on $U$ by

$$R_i^{\mathcal{E}} = (j \times j)^*(R_i \boxtimes S_i) = (p \times p)^* R_i \wedge (q \times q)^* S_i. \tag{9.9}$$

(3) **Interpret pointed updates.** For each $e$ write $a_e : P_e \hookrightarrow W$ for comprehension and $j_e : P_e \to U$ for $w \mapsto (w, e)$. For a predicate $Q \hookrightarrow U$ define its dynamic box and diamond on $W$ by

$$[\mathcal{E}, e]Q = \forall_{a_e} j_e^* Q, \qquad \langle \mathcal{E}, e \rangle Q = \exists_{a_e} j_e^* Q. \tag{9.10}$$

The action model, its comprehension, and the updated relations are ordinary dependent objects and subobjects in the model of Theorem 1.42. They can equally be constructed in the uncompleted **Set** assertion doctrine.

Epistemic product update is a comprehension of enabled world–event pairs.

$$\begin{array}{ccc} U & \hookrightarrow & W \times E \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \pi_1} \\ W & = & W \end{array}$$

**Proposition 9.15** (Action models in an economic state context)**.** *For $s : W \to \Sigma^G$, the canonical state context after the action update is $sp : U \to \Sigma^G$. Its updated relations, action-indexed predicates, and postcondition terms belong to the ordinary dependent fibres above $U$. For a linear family $A$ over $W$, its unchanged interface after the action is $p^*A$; a transition-indexed residual interface instead uses its exhibited post-state map $t : U \to W'$ and the dependent family over $W'$. These reindexings commute with the ordinary comparison tests under the substitution laws of Theorem 5.33. If the action changes the economic reference profile, the changed profile is the specified post-update term $s^+ : U \to \Sigma^G$; the composite $sp$ denotes the unchanged reference.*

*Proof.*

(1) **Composition.** The composite $sp$ is an object of $\mathbf{Set}/\Sigma^G$. Every relation and postcondition on $U$ is a subobject or term in $\mathsf{T}(U)$; the linear reindexings are index precomposition. Theorem 5.33 gives the asserted comparison-test equations.

(2) **Policy-level comparison.** A post-update profile is simply a term of the constant profile family on $U$, which supplies its typed state context without imposing any equality with $sp$.

□

**Theorem 9.16** (Action-model semantics, base change, and public specialization)**.** *For every $Q \hookrightarrow U$ and $w \in W$,*

$$[\mathcal{E}, e]Q(w) \iff P_e(w) \Rightarrow Q(w, e),$$
$$\langle \mathcal{E}, e\rangle Q(w) \iff P_e(w) \wedge Q(w, e).$$

*If $E = 1$, $S_i = 1^2$, and $P_e = A \hookrightarrow W$, the updated relation and modalities are respectively $R_i|_A$, $[!a]$, and $\langle !a\rangle$ of Section 9.1.1. For $f : W' \to W$, pull back $R_i$ along $f \times f$ and $P$ along $f \times 1_E$. The resulting updated carrier $U'$ is canonically $W' \times_W U$, its relations are inverse images of $R_i^{\mathcal{E}}$, and the action boxes and diamonds commute with this pullback. These comparisons are coherent under successive substitutions.*

*Proof.*

(1) **Predicate calculation.** The fibre of $a_e$ at $w$ is a singleton if $P_e(w)$ holds and empty otherwise. The quantifier formulas of Theorem 5.22 give the two equivalences. For $E = 1$ comprehension identifies $U$ with $A$, and Equation (9.9) reduces to the ordinary restriction of $R_i$.

(2) **Base-change comparison.** The dynamic modalities are exactly the two adjoints along $a$. The universal property of pullback identifies $U'$ with $W' \times_W U$, while relation substitution and both Beck–Chevalley comparisons give the remaining equations. Repeated substitution identifies the same indexed tuples, proving coherence.

□

Changing the economic state context commutes with epistemic product update.

$$\begin{array}{ccc} U' & \longrightarrow & U \\ \downarrow & \lrcorner & \downarrow{\scriptstyle p} \\ W' & \xrightarrow[f]{} & W \end{array}$$

**Theorem 9.17** (Presentation of every semantic refinement by an action model)**.** *Every object $p : (U, (T_i)) \to M = (W, (R_i))$ of $\mathbf{Fr}_N/M$ is isomorphic over $M$ to a semantic action-model update of $M$. Conversely, every action-model update has such a projection.*

*Proof.*

(1) **Causal execution.** For the forward direction take the event set to be $E = U$, its uncertainty relations to be $S_i = T_i$, and its enabling subobject to be the graph

$$P = \{(w, u) \in W \times U : w = p(u)\}.$$

The updated carrier is $P$, with canonical bijection $u \mapsto (p(u), u)$. Under this bijection its $i$-relation is

$$(p \times p)^* R_i \cap T_i = T_i,$$

because $p$ preserves relations. The first projection becomes $p$.

(2) **Reverse implication.** Conversely, the product-update relation $(\pi_W \times \pi_W)^* R_i \cap (\pi_E \times \pi_E)^* S_i$ is contained in the first factor, so its projection is a frame map.

□

The presentation result is relative to a specified base model; its action-event set need not be finite. For finite $W, U$, the construction in Construction 9.19 realizes its enabling relation and the map $P \to U$ by finite response protocols. For every accessibility relation $T_i$, its successor map $b_i : U \to \mathcal{P}(U)$, $b_i(u) = \{v : uT_iv\}$, recovers the relation by $uT_iv \iff v \in b_i(u)$. Equivalently, $T_i$ is the pullback of membership along $(u, v) \mapsto (v, b_i(u))$. When $T_i$ is an observation equivalence, the quotient $U \twoheadrightarrow U/T_i$ recovers $T_i$ as its kernel pair. Indeed two states have the same class precisely when they are $T_i$-related. Conversely, the kernel pair of any observation map is an equivalence relation. The successor-map presentation applies also to directed and nonreflexive belief relations. The witness data retain the underlying execution, so the same update can be compared with an operational protocol when one is supplied.

**Construction 9.18** (Updates with state-dependent edge conditions)**.**

(1) **Combine the two edge tests.** Let $P \subseteq W \times E$ be an enabling carrier and let $G_i \subseteq P \times P$ be specified edge conditions. Put

$$T_i = (\pi_W \times \pi_W)^* R_i \cap G_i.$$

This is another object of $\mathbf{Fr}_N/M$, so Theorem 9.17 gives an action-model presentation.

(2) **Recover ordinary product update.** Ordinary product update is the case $G_i = (\pi_E \times \pi_E)^* S_i$. State-dependent edge conditions provide a semantic counterpart of arrow updates. The syntactic arrow-update literature has its own expressivity and synthesis results (van Ditmarsch et al., 2020); those results are not being inferred from this set-based presentation.

**Construction 9.19** (Finite operational realization of action enablement)**.**

(1) **Form the enabling strategy.** For finite $W, E$ and an action-enabling predicate $P \hookrightarrow W \times E$, let $U = \{P\}$, regarded as its finite set of enabled pairs. Define a strategy

$$a_P : D(W) \longrightarrow D(U)$$

with one negative input event $n_w$ for every $w \in W$ and one positive event $o_{w,e}$ for every $(w, e) \in U$. Generate the causal edges $n_w < o_{w,e}$. Different input branches conflict; within a branch the positive events conflict, so at most one enabled pair is emitted.

(2) **Label its events.** Label the negative event by $w$ at the input boundary and the positive event by $(w, e)$ at the output boundary. Thus the completed outputs following an input $w$ are exactly the enabled actions $(w, e) \in U$.

(3) **Realize the updated observations.** For an updated value term $v^+ : U \to Z$, corestrict to its finite image $F = v^+[U]$ and compose $a_P$ with $\widehat{v^+} : D(U) \to D(F)$. If updated S5 observations are supplied, their finite image quotients likewise have the response strategies of Proposition 1.16.

**Proposition 9.20** (Sound operational realization of finite action models)**.** *The map $a_P$ is a receptive courteous strategy. Its enabled output relation is precisely $P$ and its composition with any finite-image updated observable returns that observable on each completed updated pair. Hence every finite semantic action-model enabling relation and its specified finite-image postconditions admit explicit concurrent operational realizations, while the action uncertainty*

*relations $S_i$ remain ordinary epistemic data until observation protocols are specified. Countable action presentations admit the same carrier-based construction; uncountable* **Set***-based action models belong to the ordinary dependent semantics and are not claimed to be single countable operational arenas.*

*Proof.*

(1) **Causal execution.** The source has finite principal pasts. Its consistency relation is hereditary and closed under adjoining causes, so it is an event structure. Every source configuration maps injectively to a configuration of $D(W)^{\perp} \parallel D(U)$.

(2) **Fibrewise calculation.** At the empty source configuration every negative input $w$ has exactly the lift $n_w$; after one input all other negative inputs conflict in the arena. All new immediate causal edges are negative-to-positive. Thus the map is receptive and courteous.

(3) **Causal execution.** Every source configuration containing an output has $n_w$ and precisely one event $o_{w,e}$, with $P(w,e)$; conversely, each $(w,e) \in P$ exhibits such a configuration. If no action is enabled at $w$, the maximal source configuration $\{n_w\}$ has no positive output.

(4) **Composition.** Composing with the response strategy for the corestriction of $v^+$ gives the value $v^+(w,e)$ by Equation (1.1) and composition of finite response strategies. The same proof uses countably many branches and outputs when the carriers are countable; every individual causal past remains finite.

□

**Proposition 9.21** (Knowledge reduction for epistemic actions)**.** *Put $K_i = [R_i]$ on $W$ and $K_i^{\mathcal{E}} = [R_i^{\mathcal{E}}]$ on $U$. For every $Q \hookrightarrow U$,*

$$[\mathcal{E}, e]K_i^{\mathcal{E}} Q = P_e \Rightarrow \bigwedge_{f\colon S_i(e,f)} K_i([\mathcal{E}, f]Q). \tag{9.11}$$

*The empty meet is $\top$. Ordinary conjunction and unchanged atomic predicates have the reductions*

$$[\mathcal{E}, e](Q \wedge Q') = [\mathcal{E}, e]Q \wedge [\mathcal{E}, e]Q',$$
$$[\mathcal{E}, e]p^*V = P_e \Rightarrow V.$$

*The same predicate doctrine interprets belief modalities for arbitrary $R_i$, without asserting factivity.*

*Proof.*

(1) **Fibrewise calculation.** At a state $w \in P_e$, the left side requires $Q(w', f)$ for every $(w', f) \in U$ satisfying $wR_iw'$ and $eS_if$. Reordering the quantifiers yields

$$\forall f\ (eS_if \Rightarrow \forall w'\ (wR_iw' \Rightarrow (P_f(w') \Rightarrow Q(w', f)))).$$

The inner assertion is $K_i([\mathcal{E}, f]Q)(w)$. At states outside $P_e$ both sides are true. Preservation of meets by the right adjoint and $a_e^* p^* V = a_e^* V$ prove the other equations.

□

**Proposition 9.22** (Observation quotients after action update)**.** *If $R_i = \ker(o_i)$ and $S_i = \ker(b_i)$ for maps $o_i : W \to O_i$ and $b_i : E \to L_i$, then*

$$R_i^{\mathcal{E}} = \ker\big(\langle o_i p, b_i q\rangle : U \to O_i \times L_i\big).$$

*Its image factorization is the effective updated observation quotient. In particular S5 knowledge, the invariant-predicate adjunctions, and the common-knowledge quotient construction remain available after the update, including when $U$ is empty.*

*Proof.*

(1) **Descent along the kernel pair.** Two updated pairs have the same displayed observation exactly when their old observations and action observations agree. This is the conjunction of their two kernel-pair relations, which is Equation (9.9). Image factorization and Corollary 6.6 give the effective quotient.

(2) **Observation and knowledge.** Equivalence relations and their generated common-knowledge relation restrict to the new carrier as constructed, with the empty case interpreted in the ordinary predicate doctrine.

□

**Example 9.23** (Private epistemic information cannot be reduced to announcement)**.** Let $W = \{w_0, w_1\}$, let two owners initially have the indiscrete observation equivalence, and take $E = \{e_0, e_1\}$ with $P_{e_k} = \{w_k\}$. Give owner 1 the diagonal action relation and owner 2 the indiscrete action relation. Then $U = \{(w_0, e_0), (w_1, e_1)\}$. Owner 1's updated relation is equality, while owner 2's still relates both points. Neither original state is removed. Hence this is a genuine private-information update, not the public restriction of a subset of $W$.

**Construction 9.24** (Sequential semantic action update)**.**

(1) **Specify the two stages.** Let $\mathcal{E} = (E, (S_i), P)$ update a relation model on $W$ to $U = \{P\}$. Let $\mathcal{F} = (F, (T_i), Q)$ be an action model over $U$, with $Q \hookrightarrow U \times F$.

(2) **Construct the composite update.** Define the composite $\mathcal{E} \diamond \mathcal{F}$ over $W$ with action set $E \times F$, action relations $S_i \boxtimes T_i$, and enabling predicate

$$P \diamond Q\,(w, e, f) \iff P(w, e) \wedge Q((w, e), f). \tag{9.12}$$

The expression involving $Q$ is evaluated only in the fibre where $P(w, e)$ holds, equivalently by the image under comprehension of the second enabling object. Updated value terms on the two-stage carrier are transported by the same canonical reassociation.

**Theorem 9.25** (Coherent composition of epistemic actions)**.** *There is a canonical isomorphism of relation models and their retained updated ordinary terms*

$$(W \odot \mathcal{E}) \odot \mathcal{F} \cong W \odot (\mathcal{E} \diamond \mathcal{F}), \qquad ((w, e), f) \longmapsto (w, (e, f)). \tag{9.13}$$

*It satisfies associativity and the singleton everywhere-enabled unit laws, including coherent substitution in the reference state. The corresponding dynamic modalities satisfy*

$$[\mathcal{E}, e][\mathcal{F}, f]Q = [\mathcal{E} \diamond \mathcal{F}, (e, f)]Q$$

*under the indicated identification of final predicates.*

*Proof.*

(1) **Predicate calculation.** The two carriers are defined by the identical conjunction of the two enablement tests after reassociation. The two-stage relation between $((w, e), f)$ and $((w', e'), f')$ is

$$R_i(w, w') \wedge S_i(e, e') \wedge T_i(f, f'),$$

which is the relation of the combined action model. Updated terms are the same functions on these identified points.

(2) **Adjunction and transposition.** Reassociation of triples and quadruples obeys Cartesian coherence, including the unit comparison. For the dynamic box, two successive right adjoints along the inclusions of the action components agree with the right adjoint along their composite by uniqueness of right adjoints; the elementwise precondition is Equation (9.12).

□

Two consecutive epistemic action updates associate by matching the intermediate state.

$$\begin{array}{ccc} (W \odot \mathcal{E}) \odot \mathcal{F} & \xrightarrow{\cong} & W \odot (\mathcal{E} \diamond \mathcal{F}) \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle p} \\ W \odot \mathcal{E} & \xrightarrow[p]{} & W \end{array}$$

*Remark* 9.26 (Scope of epistemic action models). The construction is fully semantic: $P$, $R_i$, $S_i$, and updated observable terms are ordinary typed data. Syntactic precondition or postcondition languages are not required. The update changes the state carrier as well as both endpoints of each accessibility relation; it is not the reference pullback of Proposition 5.36. A state map that forgets accessible alternatives need not preserve knowledge formulas merely because it preserves the underlying updated carrier. The bounded morphism and alternative-lifting results of Section 9.1.4 address that distinction.

**Corollary 9.27** (Action models transport across bounded state quotients)**.** *Let $f : X \twoheadrightarrow Y$ be a surjective bounded morphism from relations $R_i$ to $\bar{R}_i$. Give $Y$ an epistemic action model with action set $E$, action relations $S_i$, and enabling subobject $\bar{P} \hookrightarrow Y \times E$. Pull it back along $f \times 1_E$ to form the action model over $X$. The induced map on updated carriers*

$$f^+ : X \odot \mathcal{E} \longrightarrow Y \odot \bar{\mathcal{E}}, \qquad (x, e) \longmapsto (f(x), e)$$

*is a surjective bounded morphism for the updated relations. For every $e \in E$ and $\bar{Q} \hookrightarrow Y \odot \bar{\mathcal{E}}$,*

$$f^*[\bar{\mathcal{E}}, e]\bar{Q} = [\mathcal{E}, e](f^+)^*\bar{Q},$$
$$f^*\langle\bar{\mathcal{E}}, e\rangle\bar{Q} = \langle\mathcal{E}, e\rangle(f^+)^*\bar{Q}.$$

*Hence it preserves and reflects every modal fixed-point assertion with atomic tests descended through $f^+$, including updated knowledge assertions and their dynamic action tests.*

*Proof.*

(1) **Lifting along the surjection.** For an enabled $(y, e)$, surjectivity of $f$ supplies $x$ with $f(x) = y$; the pullback enabling predicate makes $(x, e)$ enabled. Thus $f^+$ is surjective. Forward preservation of accessibility is inherited from $f$, with the unchanged action relation $S_i$.

(2) **Construction of the comparison.** Given a reduced updated alternative $(y', e')$ accessible from $(f(x), e)$, the back clause of $f$ supplies $x'$ with $xR_ix'$ and $f(x') = y'$. Since $\bar{P}(y', e')$, its pullback $P(x', e')$ holds. Thus $(x', e')$ is the required updated alternative.

(3) **Base-change comparison.** The action-component comprehensions of the two models form pullback squares along $f$; both Beck–Chevalley identities of Theorem 5.22 give the dynamic box and diamond equations. Apply Theorem 9.29 to $f^+$.

□

### 9.1.4 Lifting alternatives and preserving fixed points

A map between epistemic models may preserve actual states and forward accessibility while still losing alternatives quantified over by a modality. Exact modal transport therefore needs two comparisons: every concrete successor has an abstract image, and every abstract successor at an image state lifts to a concrete successor at the same reference. These are the forward and back conditions of a bounded morphism. With both, the modal and fixed-point predicates descend without deleting relevant alternatives.

**Definition 9.28** (Bounded morphism)**.** A map $q : X \to Y$ between families of relation frames is **bounded** if, for every relation index $a$,

$$xR_ax' \Longrightarrow q(x)\bar{R}_aq(x'),$$

and every $q(x)\bar{R}_ay'$ has a lift $xR_ax'$ with $q(x') = y'$. For interpreted atomic predicates require $P_X = q^*P_Y$. The second clause is equivalently fibrewise coverage by the canonical comparison to the cartesian lift,

$$R_a \longrightarrow X \times_{Y,\bar{p}_1} \bar{R}_a, \qquad (x, x') \longmapsto (x, (q(x), q(x'))).$$

$$\begin{array}{ccccc} R_a & \twoheadrightarrow & X \times_Y \bar{R}_a & \longrightarrow & \bar{R}_a \\ & \searrow^{p_1} & \downarrow & & \downarrow{\scriptstyle \bar{p}_1} \\ & & X & \xrightarrow[q]{} & Y \end{array}$$

**Theorem 9.29** (Modal fixed-point adequacy)**.** *Let $q : X \to Y$ be a bounded morphism. Interpret every free predicate variable by a valuation $\rho$ on $Y$ and by its componentwise pullback $q^*\rho$ on $X$; interpret each designated atomic predicate on $X$ as the pullback of its interpretation on $Y$. Then every formula of the positive modal fixed-point language satisfies the naturality equation*

$$[\![\varphi]\!]_X(q^*\rho) = q^*[\![\varphi]\!]_Y(\rho).$$

*This includes nested least and greatest fixed points. Validity is preserved; if $q$ is surjective it is also reflected.*

Forward preservation of accessibility is a commuting square on relation witnesses.

$$\begin{array}{ccc} R_i & \xrightarrow{f\times f} & S_i \\ \downarrow & & \downarrow \\ X \times X & \xrightarrow[f\times f]{} & Y \times Y \end{array}$$

*Proof.*

(1) **Construction of the comparison.** Fix the valuation $\rho$ on the target and its pulled-back valuation $q^*\rho$ on the source. Induct on formulas in the assertion fibres. The variable and atomic cases follow from these interpretations by definition.

(2) **Lifting through the comparison.** Reindexing preserves the remaining logical operations by Theorem 5.22. The comparison to the cartesian alternative context has the displayed fibrewise lifts; quantifying those fibres gives $[R_a]q^* = q^*[\bar{R}_a]$ and $\langle R_a\rangle q^* = q^*\langle\bar{R}_a\rangle$ by respectively universal and existential quantification over the corresponding fibers.

(3) **Fixed-point argument.** For a fixed-point body the induction hypothesis gives $F_X q^* = q^* F_Y$. Since $q^* \mu F_Y$ is fixed, $\mu F_X \leq q^* \mu F_Y$. Put $Q = \forall_q \mu F_X$.

(4) **Adjunction and transposition.** Then $q^* Q \leq \mu F_X$, whence $q^* F_Y Q = F_X q^* Q \leq \mu F_X$. Adjunction gives $F_Y Q \leq Q$; leastness and the counit give $q^* \mu F_Y \leq q^* Q \leq \mu F_X$. For greatest fixed points put $Q = \exists_q \nu F_X$.

(5) **Unit and identity laws.** The unit gives $\nu F_X \leq q^* Q$, hence $\nu F_X = F_X \nu F_X \leq q^* F_Y Q$. Adjunction gives $Q \leq F_Y Q$, and coinduction gives $\nu F_X \leq q^* Q \leq q^* \nu F_Y$. The reverse inequality holds because $q^* \nu F_Y$ is fixed.

(6) **Fixed-point argument.** For nested fixed points, first apply structural induction to the body with a designated free variable $p$: for every target candidate $U \in \mathsf{Pred}(Y)$ its source instance uses $q^* U$, giving $F_X(q^* U) = q^* F_Y(U)$ with all other free variables interpreted by $q^* \rho$. The two adjunction arguments above then establish the fixed-point comparison for the binder of $p$.

(7) **Descent along the kernel pair.** Inducting on binding depth proves the displayed equation for nested least and greatest fixed points, without treating arbitrary source valuations as descending. Finally, a surjective $q$ reflects the top predicate.

□

## 9.2 Public announcement in the agent calculus

### 9.2.1 Restricting a situated model

Let $M$ be a situated model on $W$, and let $\varphi$ be an assertion interpreted in $M$. Form its comprehension

$$j : U = \{w \in W \mid M, w \models \varphi\} \hookrightarrow W.$$

The updated model $M|\varphi$ is carried by $U$. Its valuations and linear families are restricted, and observations become $o_i j$, corestricted to their images when appropriate. Accessibility is $R_i^U = (j \times j)^* R_i$, so excluded worlds disappear from both positions of the relation. Strong monoidality of linear restriction preserves agent types and their composition; only the available epistemic context changes.

The dynamic type operation is $\Pi_j^\ell$. At a state removed by the announcement it is the empty additive product $\top$; at a retained state it is the post-update fibre. The tensor unit $\mathsf{I}$ cannot be substituted for $\top$: these are different limits and resource interpretations.

**Theorem 9.30** (Perspective after announcement)**.** *For a linear family $A$ over $U$, there is a natural isomorphism*

$$K_{oj}^\ell A \cong j^* K_o^\ell \Pi_j^\ell A.$$

*It is coherent with successive public restrictions and the monoidal perspective comparisons.*

*Proof.*

(1) **Adjunction and transposition.** Right adjoints compose: $\Pi_{oj}^\ell \cong \Pi_o^\ell \Pi_j^\ell$. Also $(oj)^* = j^* o^*$. Their composite gives the displayed isomorphism.

(2) **Fibrewise calculation.** At $u \in U$, the right side takes a product over all old alternatives, with terminal factors for alternatives outside $U$. Its nonterminal factors are exactly $A_v$ for $v \in U$ with $o(v) = o(u)$. Product projections prove the comparison and its coherence. Tensor comparisons on either side are obtained by tensoring these same remaining projections.

□

### 9.2.2 Dynamic formulas and reduction

The interpretation of a public-announcement formula is

$$[\![[!\varphi]\psi]\!]_M = \forall_j [\![\psi]\!]_{M|\varphi}. \tag{9.14}$$

The formula $\psi$ is always interpreted in the updated model, using its restricted accessibility relations. This rule continues to apply when the postcondition contains knowledge, another announcement, or a common-knowledge fixed point. Otherwise the reduction would silently evaluate an epistemic operator against the old alternatives.

**Theorem 9.31** (Public-announcement reduction laws)**.** *For an atom $p$ whose valuation is restricted by the update,*

$$\begin{aligned}
[!\varphi]p &\leftrightarrow (\varphi \Rightarrow p),\\
[!\varphi](\psi \wedge \chi) &\leftrightarrow [!\varphi]\psi \wedge [!\varphi]\chi,\\
[!\varphi]\neg\psi &\leftrightarrow (\varphi \Rightarrow \neg[!\varphi]\psi),\\
[!\varphi]K_i\psi &\leftrightarrow (\varphi \Rightarrow K_i([!\varphi]\psi)),\\
[!\varphi][!\psi]\chi &\leftrightarrow [!(\varphi \wedge [!\varphi]\psi)]\chi.
\end{aligned}$$

*Proof.*

(1) **Fibrewise calculation.** At each stage interpret the postformula in the restricted model and let $j : U \hookrightarrow W$ be the current announcement comprehension. The atomic, Boolean, and knowledge equations follow, respectively, from inverse image, the monic adjunction identities, and (9.3) of Theorem 9.6.

(2) **Iteration of announcement reduction.** Thus they apply inductively to postformulas involving knowledge as well as fixed pre-update predicates. For successive announcements the second comprehension inside $U$ composes with $j$ to the inclusion classified in $W$ by $\varphi \wedge [!\varphi]\psi$; the right adjoints of the two inclusions compose. This gives the final equation and retains the re-evaluation of epistemic formulas after the first update.

□

The reduction equations follow from comprehension and the right-adjoint interpretation of announcement; compare Plaza (2007); Baltag, Moss, and Solecki (1998). They express the logical operation independently of a physical test or message that might implement it.

**Example 9.32** (Re-evaluating an epistemic announcement)**.** Take two indistinguishable worlds, with $p$ true at just one. At that world $p \wedge \neg K_i p$ is true, so its announcement is enabled. Afterward only the $p$-world remains and $K_i p$ holds. Thus the announced epistemic formula itself becomes false. The update recomputes epistemic formulas in the restricted model; the example shows the possible failure of $[!\varphi]\varphi$.

### 9.2.3 The effect on policies and resources

Public restriction can change which observation-based policies are uniformly legal. For $F \subseteq W \times M$, restricting to $U$ removes some worlds from surviving observation fibres. With $O_U = \mathrm{im}(oj)$ and $r : O_U \hookrightarrow O$, every former legal policy restricts to one over the new quotient, giving the map below.

$$\mathsf{Pol}_o(F) \longrightarrow \mathsf{Pol}_{oj}((j \times 1)^* F), \qquad \pi \longmapsto \pi r.$$

The restriction map need not be injective: policies distinguished only by observations absent from $U$ become indistinguishable. New uniformly legal actions may also appear after a counterexample world is removed. By contrast, linear execution resources are merely reindexed; a public update does not create a resource or authorization token.

So far the update is logical. To regard it as a physical public announcement, an enabled test or message must have execution witnesses whose successful image is exactly $U$. The finite communication protocols in the next part provide such realizations and make it possible to compare the epistemic update with the complete policies that produced it.

**Proposition 9.33** (Observation-based implementability of a public test)**.** *Let $o : W \twoheadrightarrow O$ be an effective observation and $P \subseteq W$. The following are equivalent:*

*(i) $P$ is the inverse image $o^{-1}(B)$ of a predicate $B \subseteq O$;*

*(ii) $P$ is constant on the kernel pair of $o$;*

*(iii) its characteristic function factors as $1_P = \chi o$ for $\chi : O \to 2$.*

*For finite $O$, these equivalent descriptions give a receptive courteous protocol accepting the observed input and emitting a success event precisely at worlds in $P$. Its successful-execution witness image is the comprehension $P \hookrightarrow W$.*

*Proof.*

(1) **Descent along the kernel pair.** The first and third descriptions are equivalent by taking the characteristic predicate. Either makes $P$ constant on the kernel pair.

(2) **Reverse implication.** Conversely kernel-pair constancy lets $1_P$ descend uniquely along the effective quotient $o$, producing $\chi$ and $B = \chi^{-1}(\{1\})$. For finite $O$, realize $\chi$ by $\widehat{\chi} : D(O) \to D(2)$. Connect it to the gate $D(2) \to D(1)$ whose source accepts both negative bit inputs but emits its sole positive success output only following input 1.

(3) **Lifting through the comparison.** It is receptive because both inputs lift uniquely, and courteous because its only added dependency is negative-to-positive. The composite emits success exactly when $\chi(o(w)) = 1$; the image of successful witnesses is therefore $P$. The absent success event is not reinterpreted as a false operational scalar.

(4) **Observation and knowledge.** A test of a predicate not constant on the available observation fibre requires a different actually delivered signal, not an implicit truth oracle.

□

## 9.3 General epistemic action models and delivery

### 9.3.1 General action models

Let an action presentation have event set $E$, event relations $S_i$, and an enabling predicate $\mathsf{pre} \subseteq W \times E$. Its updated carrier and relations are

$$U = \{(w, e) : \mathsf{pre}(w, e)\}, \tag{9.15}$$

$$(w, e)R_i^+(v, f) \iff wR_iv \ \wedge \ eS_if. \tag{9.16}$$

The updated relation is the standard dynamic-epistemic product update (Baltag, Moss, and Solecki, 1998). In the observation-generated case, $R_i = \ker(o_i)$ and $S_i = \ker(b_i)$, so the effective updated signal is the image of $(w, e) \mapsto (o_i(w), b_i(e))$. This construction uses the information actually delivered by the event.

The projection $r : U \to W$ forgets the event, and $r^*$ transports linear families and agent types across that change of state. An implemented event also retains a witness family and

concurrent realizer, as in Section 8.3. Neither an abstract event relation nor a semantic update alone supplies an operational communication wire.

For an event $e$, let $j_e : W_e \hookrightarrow W$ be its enabling inclusion, and $k_e : W_e \to U$ send $w$ to $(w, e)$. Define

$$[\mathcal{A}, e]\psi = \forall_{j_e} k_e^* [\![\psi]\!]_{M \otimes \mathcal{A}}.$$

Its type-level version is $\Pi^{\ell}_{j_e} k_e^*$.

**Proposition 9.34** (Action reduction and composition)**.** *For a finite event presentation,*

$$[\mathcal{A}, e]K_i\psi \leftrightarrow \mathsf{pre}_e \Rightarrow \bigwedge_{f : eS_i f} K_i([\mathcal{A}, f]\psi).$$

*Successive action updates compose by retaining pairs of enabled events, and this composition is associative up to the canonical tuple bijection.*

*Proof.*

(1) **Fibrewise calculation.** At an enabled $(w, e)$, knowledge quantifies over exactly the pairs $(v, f)$ with $wR_iv$, $eS_if$, and $\mathsf{pre}(v, f)$. Grouping first by $f$ gives the displayed conjunction and its guarded boxes. At a disabled $w$ both sides are true.

(2) **Causal execution.** For a second action with precondition $Q$ on $U$, the sequential carrier consists of triples $(w, e, f)$ with $\mathsf{pre}(w, e) \wedge Q((w, e), f)$. Its relation is the conjunction of the original world relation and the two event relations. Reassociating triples preserves these tests, the reindexed linear families, and, for implemented actions, the composite concurrent realizer. Fourfold tuple reassociation proves the coherence diagrams.

□

**Proposition 9.35** (Public restriction as singleton action update)**.** *For a state predicate $P \hookrightarrow W$, let the action carrier be $E = 1$, its sole event have enabling predicate $P$, and every action relation be equality on 1. Its product update is canonically the comprehension $j : \{P\} \hookrightarrow W$, with relations $R_i^+ = (j \times j)^* R_i$. The pointed action box and diamond are $\forall_j$ and $\exists_j$; in particular the corresponding knowledge-reduction law is*

$$[!P]K_i\psi \iff P \Rightarrow K_i([!P]\psi).$$

*Proof.*

(1) **Monoidal comparison.** The enabled carrier is $\{(w, *) : P(w)\} \cong \{w : P(w)\}$ by projection. Because both action coordinates are $*$, the product-update relation reduces to $wR_iv$ on the retained endpoints.

(2) **Communication and delivery.** Pointed action quantification along its enabling inclusion is therefore exactly the comprehension quantification of Definition 9.1; the displayed reduction is the singleton instance of Proposition 9.34 (equivalently Theorem 9.6).

□

### 9.3.2 Institutional and epistemic composition coherence

For three execution spans, both bracketings have witness carrier

$$\{(e, f, g) : t_E(e) = s_F(f),\ t_F(f) = s_G(g)\}.$$

For three composed executions, both bracketings have the same matched triple of witnesses. Permission is $P_E(e) \wedge P_F(f) \wedge P_G(g)$, and the concurrent realizer is $\rho_G(g)\rho_F(f)\rho_E(e)$.

Reassociating the witnesses preserves these fields. The epistemic comparison replaces record matching by enabled event matching and uses the same dependent-sum and -product coherence. Thus repeated updates do not require a new composition law. Thus the extraction of action events from execution witnesses is completed only after the finite causal protocol and session adequacy results of Section 10; see Construction 10.26.

# Part IV
# Operational and strategic communication

## 10 True-Concurrent Communication Between Agents

An epistemic update specifies what each recipient can distinguish after an exchange. A protocol must also explain how the information arrives. We wire finite response strategies to data ports and use secured interaction to identify the sends and receives that match. The event structure allows independent decisions to proceed before the collector has all its inputs. Complete policies retain what each owner would do on branches that the exchange leaves unvisited. Binary sessions describe the same obligations at a pair of dual endpoints. Both constructions supply the delivery witnesses used in the next section.

### 10.1 Agents implemented by causal decision protocols

**Definition 10.1** (Decision generators and generated presentations)**.** Use the locally fully faithful nonstrategic embedding $\mathbf{j}_N$ of Equation (3.4). For finite $O, A$ and $i \in N$, define the singly owned decision presentation

$$\mathsf{Dec}_i(O, A) : D(O) \longrightarrow D(A)$$

by

$$\Sigma_i = A^O, \qquad \Sigma_j = 1 \ (j \neq i), \qquad \pi \longmapsto \widehat{\pi}.$$

Let $\mathbf{DecArena}_N(\mathcal{C})$ be the smallest locally full symmetric monoidal sub-bicategory of $\mathbf{Arena}_N(\mathcal{C})$ that contains the image of $\mathbf{j}_N$ and all finite decision presentations, is closed under horizontal composition and tensor, is closed under the cartesian parameter reindexings of Equation (3.3), and contains every presentation invertibly 2-isomorphic to one already generated. Thus a restriction or duplication of policy descriptions remains in the generated class only through an exhibited parameter map; it is not silently absorbed into the word "decision." This gives an explicit meaning to "generated by singly owned decisions and nonstrategic wiring."

**Definition 10.2** (Finite causal decision protocol)**.** A **finite causal decision protocol** consists of a finite acyclic directed graph of decision nodes $V$, an ownership map $i : V \to N$, finite nonempty action sets $A_v$ for $v \in V$, a finite set $P_0$ of external ports with finite value sets $(X_p)_{p \in P_0}$, and specified typed input wires. Each incoming wire $e$ of a node $v$ has source either an external port $p$ (carrier $X_e = X_p$) or an earlier decision node $u$ (carrier $X_e = A_u$); the wires from nodes respect the directed acyclic graph. If $I_v$ is the finite set of incoming wires to $v$, its input carrier and raw observation map are

$$H_v = \prod_{e \in I_v} X_e, \qquad \widetilde{o}_v : H_v \to Y_v, \qquad H_v \xrightarrow{o_v} O_v \hookrightarrow Y_v, \quad O_v = \mathrm{im}(\widetilde{o}_v).$$

The effective observation quotient $o_v$ determines the node policies $\pi_v : O_v \to A_v$ and the complete owner policy spaces

$$\Sigma_i = \prod_{v:\, i(v)=i} A_v^{O_v}.$$

The protocol also specifies an outcome function on its full institutional record (external data together with all node actions),

$$z : \left(\prod_{p \in P_0} X_p\right) \times \left(\prod_{v \in V} A_v\right) \longrightarrow Z.$$

The value of $z$ can be corestricted to its finite image for the concurrent response protocol. An action at $v$ is $a_v = \pi_v(o_v((x_e)_{e \in I_v}))$. The graph determines which observations are available at each decision; recording an action for an institution does not create an additional observation wire. The generated arena is constructed and its scheduling independence proved below.

A finite causal decision protocol makes the origin of each observation explicit. Nodes respond to the values arriving along their incoming wires; independent nodes remain concurrent. The owned realization assigns a complete contingent function to every decision node, so a policy includes responses at nodes not encountered during a particular execution. The ensuing construction connects these policies to secured event-structure strategies.

**Lemma 10.3** (Decision-event path reflection in a finite protocol)**.** *Fix the external inputs and a complete owner-policy profile of a finite causal decision protocol. Form its secured interaction with all wired occurrences still displayed, before hiding the internal channels. In a completed configuration let $d_v$ be the positive action-output occurrence at decision node $v$. If $u \neq v$ and $d_u < d_v$ in the induced causal order, then the protocol DAG contains a nonempty directed path from $u$ to $v$. Consequently the chosen action outputs of incomparable nodes are concurrent in this completed interaction. Hiding preserves this path restriction on any distinct decision-output occurrences retained as visible events.*

*Proof.*

(1) **Local causal generators.** Expand each response and collector in the protocol into its source event structure. On a completed input branch, a response has only its negative-input to positive-output dependency. A collector has exactly its individual input-to-tuple-output dependencies, with no dependency between its distinct inputs. A copying strategy has an input-to-each-output dependency, with no order between copied outputs. Disposal has no output. These descriptions are the explicit constructions of Propositions 1.16, 1.19 and 1.22.

(2) **Paths through secured matching.** In secured synchronization, each matched internal event is identified with its input/output occurrence on the adjoining components. The secured order is the transitive closure of the component orders, and is acyclic by the secured-matching condition. Hence every strict causal comparison in the unhidden completed interaction is witnessed by a finite chain of local generator edges and matched wire occurrences. The other occurrence of a matched channel cannot insert an edge from its output backwards to its input.

(3) **Reflection into the protocol DAG.** Mark each generator edge that crosses from a decision output towards the next decision input by its specified protocol wire. Edges inside copying or collection boxes only progress towards their outputs. Traversing a

chain from $d_u$ to $d_v$, each passage through a decision component other than the starting one must enter that component through one of its specified incoming wires and leave through its response output. Removing the segments within components therefore gives a directed walk from $u$ to $v$ in the protocol DAG, which is a directed path because the graph is acyclic. External inputs are sources and the final institutional collector is a sink, so neither can introduce a return path between decision nodes.

(4) **Incomparability and hiding.** If $u$ and $v$ are incomparable, neither $d_u < d_v$ nor $d_v < d_u$ can hold. Both occurrences belong to the completed configuration and are therefore consistent, so they are concurrent. Hiding restricts the transitive secured order to visible occurrences; a newly visible comparison between retained decision outputs would already be a comparison in the expanded interaction, which has just been excluded.

□

**Theorem 10.4** (Coherence and scheduling independence of finite causal decision protocols)**.** *Every finite causal decision protocol determines an open arena in* $\mathbf{DecArena}_N(\mathcal{C})$*. For each external input and policy profile it has a unique completed action assignment satisfying*

$$a_v = \pi_v\big(o_v((a_u)_{u\to v}, \textit{incoming external data})\big).$$

*Every evaluation along a linear extension of the causal order yields that assignment. In the concurrent realization, incomparable decision components remain unordered; a linear extension is only a reading of the same symmetric-monoidal diagram.*

Independent nodes can each produce an output. The collector can emit the pair only after both outputs have arrived.

$$\begin{array}{ccc} D(M_v)\otimes D(M_w) & \xrightarrow{c_{M_v,M_w}} & D(M_v\times M_w) \\ \Big\downarrow{\scriptstyle \widehat{f}\otimes\widehat{g}} & & \Big\downarrow{\scriptstyle \widehat{f\times g}} \\ D(N_v)\otimes D(N_w) & \xrightarrow[c_{N_v,N_w}]{} & D(N_v\times N_w) \end{array}$$

*Proof.*

(1) **Monoidal comparison.** Construct a progressive symmetric-monoidal diagram directly from the protocol DAG. A data value with $k$ recipients is routed through the iterated comonoid of Proposition 1.19; zero recipients uses disposal. At a node $v$, put $H_v = \prod_{e\in I_v} X_e$.

(2) **Ownerwise alternatives.** Its implemented component, evaluated at the owner's complete policy coordinate $\pi_v$, is the composite

$$\bigotimes_{e\in I_v} D(X_e) \xrightarrow{c_v} D(H_v) \xrightarrow{\widehat{o_v}} D(O_v) \xrightarrow{\widehat{\pi_v}} D(A_v).$$

Here $c_v$ is the finite collector, including $c_0$ when $I_v$ is empty; the last arrow is the response realized by the singly owned decision generator.

(3) **Causal execution.** Copy the resulting action to exactly its prescribed successor and institutional-record wires. After all required records are available, use a final collector and the response strategy for the corestriction of the outcome function to its finite image. Nodes without incoming wires use the nullary collector $c_0 : I \to D(1)$.

(4) **Ownerwise alternatives.** Every box in this diagram is either nonstrategic wiring from $\mathcal{C}$ or a singly owned decision generator, so its interpretation lies in $\mathbf{DecArena}_N(\mathcal{C})$; grouping the node parameters by owner gives exactly the displayed $\Sigma_i$.

(5) **Fixed-point argument.** Because the graph is finite and acyclic, causal height is well founded. At a minimal node all arguments of its policy equation are external inputs, so its action is uniquely determined. Inductively, once all nodes of smaller height have unique actions, the right side of the equation at $v$ has a unique value.

(6) **Independence from execution ordering.** This constructs a unique completed assignment. The argument mentions no choice of linear extension, so evaluation along every such extension yields the same assignment.

(7) **Causal execution.** The causal-height filtration used above gives a layered factorization of the progressive monoidal diagram: all boxes at one height are tensored, and the successive layers are composed.

(8) **Monoidal comparison.** Within one finite layer, any enumeration is identified with any other by symmetry; reassociation and exchanges of incomparable boxes are identified by symmetric-monoidal coherence, while the constructed data comonoids identify the alternative copying trees by coassociativity and cocommutativity. Hence every reading along a linear extension of the causal order denotes the same layered arrow.

(9) **Monoidal comparison: second comparison.** Any two linear extensions of a finite partial order differ by adjacent swaps of incomparable nodes: move the first node of one reading past its incomparable predecessors in the other, and iterate. Each such swap is justified by monoidal interchange, not by imposing an edge in the event structure. Incomparable decision boxes remain in parallel parts of the diagram, so scheduling independence inserts no causal edge between them. More precisely, Lemma 10.3 shows directly on the secured, unhidden event-level interaction that a comparison of two decision outputs would require a directed path of dependency wires. This proves the stronger event-order assertion independently of the coherence argument for completed values.

(10) **Causal execution.** A linear extension is therefore only an external reading of the constructed arrow, not a scheduler included in the strategy.

□

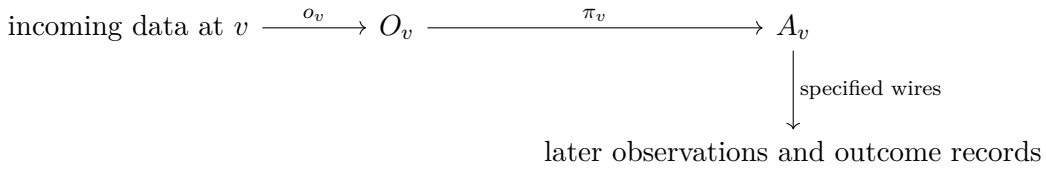



Figure 5: A policy receives its constructed observation quotient. Wiring can retain additional records without making them available to that policy.

The protocol construction now identifies complete economic policies with families of operational response functions. Each function uses the effective observation at its own node, and its realization is receptive and courteous. When several nodes belong to one owner, the resulting policy is their product; ownership neither duplicates the process nor orders otherwise independent events. This provides a concrete implementation of the agent representation.

*Remark* 10.5 (Protocol coherence and incentives). The linear-logical account of multiparty protocol compatibility in Carbone et al. (2015, 2017) concerns the operational layer. In the present construction, the causal protocol specifies the agents' implemented interactions, while preference comparisons and profitable deviations determine incentive properties. The latter are constructed in the following sections.

## 10.2 Causal decision and session protocols

### 10.2.1 Decision generators and causal diagrams

For finite $O, M$, the singly owned decision generator is

$$\mathsf{Dec}_i(O, M) : D(O) \to D(M), \qquad \Sigma_i = M^O, \quad \Sigma_j = 1 \ (j \neq i),$$

with policy $\pi$ realized by $\widehat{\pi}$. The response strategy has an input occurrence for each $o$ and an output labelled $\pi(o)$ dependent on that input. The finite-data construction proves $\widehat{\mathrm{id}} = \mathrm{id}$ and $\widehat{g}\,\widehat{f} = \widehat{gf}$. It supplies data copying and disposal on $D(O)$ and collectors

$$c_{(X_e)} : \bigotimes_e D(X_e) \to D\left(\prod_e X_e\right).$$

A collector output requires all its inputs, without ordering their arrivals.

**Definition 10.6** (Finite causal decision syntax)**.** A finite causal decision diagram is a finite directed acyclic graph of nodes $v$, owned by $i(v)$. Each incoming wire has a finite carrier $X_e$. A node has an effective observation map $o_v : \prod_{e \to v} X_e \twoheadrightarrow O_v$ and a finite output carrier $M_v$. External ports and outcome-record functions are specified as typed wires. Its complete owner policy family is

$$\Sigma_i = \prod_{v : i(v) = i} M_v^{O_v}.$$

Terms are formed by composing and tensoring decision generators, nonstrategic data wiring, and exhibited ownerwise parameter maps.

**Construction 10.7** (The interpreted finite-data generator signature)**.**

(1) **Interpret finite data.** For each finite $X$, take the representable linear object $d_X = yD(X)$. Section 1 constructs, for every finite function $f : X \to Y$, the arrow $y\widehat{f} : d_X \to d_Y$, as well as the copying, disposal, distributor, and collector arrows with their concurrent realizations.

(2) **Represent the policy families.** Via Theorem 2.18, the family $\pi \mapsto y\widehat{\pi}$ determines a state

$$\delta_i : \mathsf{I} \longrightarrow \mathsf{Ag}_{\vec{\Sigma}}(d_O, d_M), \qquad \Sigma_i = M^O, \quad \Sigma_j = 1 \ (j \neq i).$$

These are named generators with *exhibited interpretations*, not arrows obtained from $\mathsf{Lin}$, tensor, and the other abstract LNL constructors alone.

(3) **Transport the generator equations.** Their event-structure equations, including data comonoid coherence and collector synchronization, were proved in Section 1 and are used below in that interpretation. Equality of concurrent interpretations is not automatically a definitional equality between freely declared generator names.

**Lemma 10.8** (A physically token-guarded finite response)**.** *For finite $O, M$ and $f : O \to M$, the composite*

$$D(1) \otimes D(O) \xrightarrow{c_{1,O}} D(1 \times O) \xrightarrow{\widehat{f}} D(M)$$

*(where $f$ is regarded as a map $1 \times O \to M$) is receptive and courteous. Every source configuration containing its positive output labelled $f(o)$ also contains the authorization input event and the input event labelled $o$ in the causal past of that output. Thus the execution cannot deliver its result without receiving both inputs.*

*Proof.*

(1) **Secured operational interaction.** In the finite collector the tuple output $(*, o)$ requires the negative inputs on both components. The response strategy adds the negative-to-positive dependency from this tuple input to its output $f(o)$. Secured composition synchronizes the tuple events; hiding them retains the induced dependencies from both original inputs to the final output.

(2) **Secured operational interaction: completion.** Downward closure of configurations proves the claim. Receptivity and courtesy hold for both factors by the preceding constructions, and are preserved under their secured composition.

□

*Remark* 10.9 (Typing, issuance, and unforgeability). A single linear variable of type $D(1)$ cannot be contracted by the core rules; however the full concurrent category contains the state $\mathsf{pt}_1(*) : I \to D(1)$. Thus this arena alone is not an unforgeable capability. An institutional issuance process is constructed by composing an actual finite observation-based permission gate (Proposition 9.33) with the token input of the above guarded response and retaining the issuance witness. Its output is causally dependent on successful issuance. The issued event's causal predecessor and its execution witness record which gate produced it. Authenticity of issuers is a property of the institution's exhibited certificate and permission interface, not of the mere type $D(1)$: the independent state $\mathsf{pt}_1(*) : I \to D(1)$ is available in the full operational category. The bearer can therefore prove provenance only when the provenance witness or its verified interface is retained.

**Theorem 10.10** (Definability and scheduling independence)**.** *Every finite causal decision diagram has an agent term over the interpreted generator signature of Construction 10.7, with its displayed complete policy family and the concurrent interpretation constructed in the preceding constructions. Each input and profile determine a unique completed assignment of outputs. Different linear readings have the same interpreted concurrent interaction and add no order between independent decisions. Equality of these interpretations is not silently made a definitional equality between unrelated generator terms in the freely presented syntax.*

*Proof.*

(1) **Causal execution.** The generator signature maps each displayed $\mathsf{Dec}_i(O_v, M_v)$ to the same singly owned decision presentation used in the constructive DAG factorization in the proof of Theorem 10.4. Its routing maps, nullary collectors, finite copying/disposal, and ownerwise product coordinates are exactly those of that construction.

(2) **Causal execution: completion.** The causal-height recursion therefore gives the same unique output assignment, while adjacent incomparable-node swaps and collector coherence give the same concurrent interaction for every linear reading. This establishes the interpretation of the agent term, without identifying freely declared primitive terms merely because their interpretations happen to agree.

□

**Theorem 10.11** (Prefix realizations and non-anticipation)**.** *For a downward-closed set $D \subseteq V$ of decision nodes, let $U_D$ be the product of external-input coordinates feeding nodes of*

*$D$, and $\Sigma_D = \prod_{v\in D} M_v^{O_v}$ (regrouped by owner when required). The causal-height recursion constructs a unique prefix realization*

$$r_D : U_D \times \Sigma_D \longrightarrow \prod_{v\in D} M_v.$$

*For downward-closed $D \subseteq E$, these maps obey*

$$\mathrm{pr}_D\, r_E = r_D(\mathrm{pr}_{U_D} \times \mathrm{pr}_{\Sigma_D}). \tag{10.1}$$

*Every node's output therefore depends only on external inputs and policy coordinates at its causal predecessors and itself. For a node $v$, the actual incoming tuple and declared effective observation determine*

$$U_v \times \Sigma_{D_v^-} \xrightarrow{\mathrm{in}_v} H_v := \prod_{e\to v} X_e \xrightarrow{o_v} O_v, \qquad U_v \times \Sigma_{D_v^-} \twoheadrightarrow O_v^{\mathrm{reach}} \hookrightarrow O_v,$$

*where the second factorization is the image of $o_v\,\mathrm{in}_v$, $D_v^-$ is the strict causal past, and $U_v$ includes the external ports feeding it or $v$. Execution evaluates the existing complete policy $\pi_v : O_v \to M_v$ on that image; it does not replace the full policy carrier by one containing only reachable observations.*

*Proof.*

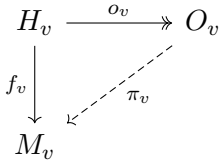
(1) **Fibrewise calculation.** At a minimal node of $D$, both sides of (10.1) evaluate the same external inputs under its same policy coordinate. For a nonminimal $v \in D$, each source of an incoming node wire is already in $D$, because $D$ is downward closed. By induction on causal height all those values agree between the full and prefix realizations.

(2) **Observation and knowledge.** Applying the same raw observation $o_v$ and the same policy $\pi_v$ gives the same output. Extensionality of finite products gives the equation. The map $\mathrm{in}_v$ is formed by projecting the required external ports and the already recursively calculated predecessor outputs.

(3) **Monoidal comparison.** Image factorization of its composite with the declared $o_v$ gives $O_v^{\mathrm{reach}} \hookrightarrow O_v$ directly. All maps are built by recursion, products, and image factorization in **Set**, so the comparisons commute with pullback substitution of ordinary parameters.

□

The factorization identifies the signal available at this prefix. Histories giving the same signal must receive the same response.




**Example 10.12** (One agent informs another)**.** Compose $\mathsf{Dec}_i(O, M)$ with $\mathsf{Dec}_j(M, A)$. At policies $\alpha : O \to M$ and $\beta : M \to A$, the first agent sends $\alpha(o)$ and the second responds with $\beta(\alpha(o))$. Their composite is $\widehat{\beta}\,\widehat{\alpha} = \widehat{\beta\alpha}$. Hiding the message boundary retains both owners' policy coordinates. Recording or forwarding the message uses an explicit output wire.

### 10.2.2 Finite session boundaries

Finite session types provide a second view of these communications: they prescribe how an endpoint can send, receive, and continue after a message. The treatment of channel references in Harper (2016) supplies programming-language context; the present construction makes the linear protocol discipline and its concurrent realization explicit. We use the dependent finite grammar below.

$$S ::= \mathsf{End} \mid !_{m:M} S_m \mid ?_{m:M} S_m, \tag{10.2}$$

where $M$ is a finite ordinary message type and each continuation is already a finite session. The nondependent notations $!M.S$ and $?M.S$ are special cases. Internal choice $\oplus\{\ell : S_\ell\}$ is a send of a finite label; external choice $\&\{\ell : S_\ell\}$ is its receive dual. The prefix ! here is not the exponential $\mathsf{LinVal}$. Duality is defined recursively by

$$\overline{\mathsf{End}} = \mathsf{End}, \quad \overline{!_{m:M} S_m} = ?_{m:M} \overline{S_m}, \quad \overline{?_{m:M} S_m} = !_{m:M} \overline{S_m}.$$

Sending $m$ consumes the current endpoint of type $!_{m:M} S_m$ and returns its residual endpoint $S_m$. Its partner consumes the corresponding receive endpoint and continues at $\overline{S_m}$. An ordinary message may be stored, but the old endpoint does not remain available as a second linear resource.

**Construction 10.13** (Session arenas)**.**

(1) **Form the history tree.** Let $H_S$ be the finite tree of legal message histories of $S$, with empty history at its root. A nonterminal history $h$ has an extension $hm$ for each element of its current message carrier $M_h$.

(2) **Define events and consistency.** Events are nonempty histories, ordered by prefix. A finite event set is consistent when its elements lie on a single branch.

(3) **Assign polarities.** The event $hm$ is positive when the protocol at $h$ sends, and negative when it receives. Write $\mathsf{A}_S$ for this polarized event structure.

Predecessor sets are finite, consistency is downward closed, and adjoining a predecessor preserves consistency. Thus the construction is an event structure. Directly, $\mathsf{A}_{\overline{S}} = \mathsf{A}_S^{\perp}$ and $\mathsf{A}_{\mathsf{End}} = I$. The continuation guard is an actual arena order. In particular $\mathsf{A}_{!M.S}$ is not defined as the unguarded tensor $D(M) \otimes \mathsf{A}_S$: that tensor alone would allow independent activity in the two components.

### 10.2.3 Complete policies of communicating agents

Fix endpoint owners $a, b$. Let $H(S)$ be the prefix-closed set of session histories, including the empty history $\epsilon$. A *decision position* is a nonterminal history, whereas an event of $\mathsf{A}_S$ is a *nonempty extension* of a history. These are different indexing sets. Put

$$\mathcal{D}_a(S) = \{h \in H(S) : S|h \neq \mathsf{End} \text{ and } S|h \text{ begins with a send}\},$$

and define $\mathcal{D}_b(S)$ by replacing send with receive; the send moves of $a$ are exactly the receive moves of $b$. At $h$ let $M_h$ be the current finite message carrier. Define the complete policy spaces

$$\Sigma_i^S = \prod_{h \in \mathcal{D}_i(S)} M_h, \qquad i \in \{a, b\}. \tag{10.3}$$

These products retain every legal decision position, including the positions inconsistent with the choices made at other positions by the same policy. The empty product is the

unit policy type. For an ordinary private value $u_i : O_i$, the corresponding policy carrier is $\prod_{h \in \mathcal{D}_i(S)} M_h^{O_i}$, and its executed choice is $\alpha_h(u_i)$. When the private value is delivered physically, the source strategy additionally records its negative input event, as constructed below.

For $\alpha \in \Sigma_a^S$, recursively construct the prefix-closed subtree $P_\alpha$ by retaining the uniquely selected outgoing event at an $a$-send position and every outgoing event at a $b$-send position. The inclusion $P_\alpha \hookrightarrow \mathsf{A}_S$ denotes a strategy $p_\alpha : I \to \mathsf{A}_S$. Dually, the policy $\beta$ of owner $b$ prunes its positive events in the opposite-polarity arena, giving $q_\beta : \mathsf{A}_S \to I$.

**Lemma 10.14** (Endpoint realization)**.** *The two inclusions are receptive courteous strategies. Finite prefix recursion, followed by abstraction of the complete policy, constructs closed displayed agent terms in the interpreted finite session language*

$$p : \mathsf{Ag}_{(\Sigma_a^S, 1)}(yI, y\mathsf{A}_S), \qquad q : \mathsf{Ag}_{(1, \Sigma_b^S)}(y\mathsf{A}_S, yI)$$

*with owner list* $(a, b)$*.*

*Proof.*

(1) **Causal execution.** Each pruned subtree is down-closed, locally injective, and preserves single-branch consistency and polarity. Given an enabled negative arena event after a retained source configuration, it belongs to the next receive position and was retained, uniquely, by the recursive construction. All causal edges of a pruned tree are inherited prefix edges, so courtesy holds.

(2) **Construct the endpoint expression.** Work in the ordinary context $\alpha : \Sigma_a^S$. For every legal history $h$, define a prefix expression $P_h(\alpha)$ at the suffix type $S|h$ by recursion on its remaining height. At $\mathsf{End}$ use the ended endpoint (the empty tree). At $!_{m:M_h} S_{hm}$, read $m = \alpha_h$ and form $\mathsf{Send}(m); P_{hm}(\alpha)$, using finite case analysis on that coordinate. At $?_{m:M_h} S_{hm}$, form $\mathsf{Recv}(m); P_{hm}(\alpha)$ with one branch for each $m \in M_h$. These expressions are the finite guarded trees of Construction 10.13 and Equation (10.5). At a send each case retains one specified first edge above the root and grafts its continuation below that edge. At a receive the branches have all the negative first edges, and events in distinct branches conflict. An empty receive menu gives the empty family of branches. In an empty send menu the selected coordinate eliminates from the empty ordinary type. All subexpressions retain the same ordinary register. Each recursive call is at a strict child; hence the construction terminates also for dependent continuations and supplies the indicated suffix type in every branch.

(3) **Compute its denotation and abstract.** Induction on that same height identifies the interpreted events with the retained nonempty suffixes of $h$. The first edge carries $m$, and a suffix $k$ in its continuation carries the concatenated history $hmk$. This bijection preserves and reflects prefix order, single-branch consistency, polarity and boundary labels. At $h = \epsilon$ the denotation is therefore $p_\alpha$. In the dual recursion for $\beta : \Sigma_b^S$, sends and receives exchange roles and the denotation on the input boundary is $q_\beta$. Abstract the ordinary registers by the agent introduction of Theorem 2.15, using the tensor unitor at the empty boundary. This gives the displayed terms. Their evaluations at every complete register are the inclusions just constructed, so the representation bijection identifies their denotations with the stated families. The term construction is the finite prefix recursion; the bijection verifies its denotation.

□

### 10.2.4 Physically received private information

**Construction 10.15** (Receptive endpoint with a private input)**.**

(1) **Fix the policy and session interface.** For finite $O_a$, take the complete policy $\alpha \in \prod_{h \in \mathcal{D}_a(S)} M_h^{O_a}$. We construct

$$p_\alpha^{O_a} : D(O_a) \longrightarrow \mathsf{A}_S.$$

(2) **Separate shared and private histories.** Write $B_-$ for the set of nonempty session-event histories whose entire prefix contains only negative moves of $\mathsf{A}_S$. For each $u \in O_a$, let $P_{\alpha,u}$ be the prefix-closed session subtree obtained by applying $\alpha_h(u)$ at every positive decision position and retaining all negative branches. Its events in $B_-$ are shared between all input values.

(3) **Build the source events.** The source event set has: a negative event $e_u$ for each $u \in O_a$; a common event $b_k$ for each $k \in B_-$; and an event $(u, k)$ for each $k \in P_{\alpha,u} \setminus B_-$. The $e_u$ are mutually conflicting.

(4) **Generate causal order.** Order common events by their session prefix relation. Order the tagged events, and the common-to-tagged dependencies, by the same prefix relation. Every first tagged event (which is necessarily positive) has the additional predecessor $e_u$.

(5) **Impose consistency.** Distinct tagged values are inconsistent. A finite set is consistent exactly when its session labels belong to one branch, it uses at most one input value and one tag, and the input value, if present with a tag, equals that tag. The source's finite-consistency family includes down-closures of all these compatible finite sets.

(6) **Label the endpoint.** Label $e_u$ in $D(O_a)^\perp$ and label $b_k, (u, k)$ by the session event $k$ in $\mathsf{A}_S$. The added prerequisite makes the private input causally prior to the first positive session event without imposing an order between the private input and an initial negative-only session prefix.

Apply the identical construction with exchanged session polarities for owner $b$ to obtain

$$q_\beta^{O_b} : \mathsf{A}_S \otimes D(O_b) \longrightarrow I, \qquad \Sigma_b = \prod_{h \in \mathcal{D}_b(S)} M_h^{O_b}.$$

The private-input event is negative in both arrow arenas.

**Lemma 10.16** (Input causality, receptivity, and courtesy)**.** *These are receptive courteous strategies. Session inputs available before the endpoint's first send can precede the private-input event; each input-dependent positive response has that event in its causal past. Every complete policy coordinate remains present in the policy carrier, including off-path coordinates.*

*Proof.*

(1) **Causal execution.** Principal causal downsets are finite because $S$ is finite and a tagged suffix acquires only the one extra private predecessor. Cause extension preserves the stated consistency: predecessors of a tagged event have the same tag or are common prefix events, and its input predecessor is $e_u$. Hence the source is an event structure.

(2) **Causal execution: second comparison.** Its labelling preserves configurations and is locally injective: conflicting input values or tags cannot occur in one configuration and the common and tagged event sets have disjoint session labels.

(3) **Lifting through the comparison.** For receptivity, an enabled negative private-input move has a unique $e_u$ lift. It is available after any common negative-only prefix, and no local positive move can have occurred without some $e_u$. Before the first local positive move, every enabled negative session move has its unique common-event lift, whether the private input is already present or not.

(4) **Lifting along the surjection: second comparison.** After the first local positive move its value tag $u$ is fixed, and the corresponding negative continuation has a unique tagged lift. These cases exhaust negative arena moves.

(5) **Secured operational interaction.** An immediate source edge inherited from the session prefix tree is an arena dependency. The only newly inserted immediate dependencies go from the negative input $e_u$ to a first tagged positive event. Thus courtesy holds.

(6) **Communication and delivery.** The dual endpoint is proved by the same argument with the session polarities reversed. Neither construction restricts the original product of contingent response maps.

□

**A factorization through fixed operational wiring.** The same two agents admit a compilation in the generated presentation of Definition 10.1. For the provider, put $P = \Sigma_a^S$ and specialize Construction 10.15 to the private input $u : P$ and the fixed coordinate evaluations $\alpha_h(u) = u_h$. This gives one fixed receptive courteous strategy $v_a : D(P) \to \mathsf{A}_S$. Produce its input using the nullary collector and the singly owned decision $\mathsf{Dec}_a(1, P)$, reparametrized by $u \mapsto (* \mapsto u)$. At policy $u$ the resulting point is $\mathsf{pt}_P(u)$, and

$$v_a \, \mathsf{pt}_P(u) \cong p_u.$$

Indeed the matched input selects the tagged $u$-suffixes. Hiding it retains every common negative prefix and precisely the selected tagged continuations. Removing the tag gives an isomorphism over the session arena, since the only extra cause was the private input. This also covers a path that has no local send. For the client, use the dual fixed wiring $v_b : \mathsf{A}_S \otimes D(\Sigma_b^S) \to I$ and tensor the policy point with identity on the session input. The same label-preserving bijection gives $q_\beta$. Each $v_i$ is an exhibited nonstrategic wiring in $\mathbf{j}_N(\mathcal{C})$, so these factorizations use the declared closure under composition, tensor and parameter reindexing. The guarded prefix terms and these compiled presentations have the same concurrent denotations. A free signature of flat finite-data generators alone does not declare the general mixed session boundary.

### 10.2.5 Connection, reduction, and fidelity

Connection is linear composition. In provider–client notation its rule is

$$\frac{\Gamma; \Delta \vdash P :: c : S \qquad \Gamma; \Theta, c : S \vdash Q :: z : T}{\Gamma; \Delta, \Theta \vdash (\nu c)(P \mid Q) :: z : T}. \tag{10.4}$$

Here the session boundary is interpreted by $[\![S]\!]_\ell = y\mathsf{A}_S$ and its dual by $[\![\overline{S}]\!]_\ell = y\mathsf{A}_S^\perp$. The first premise denotes a linear arrow $[\![\Delta]\!] \to y\mathsf{A}_S$; the second denotes $[\![\Theta]\!] \otimes y\mathsf{A}_S \to [\![T]\!]_\ell$. The latter occurrence is an *input* linear variable: it is already polarity-reversed when interpreted in the arrow arena. Thus the endpoint duality $\mathsf{A}_{\overline{S}} = \mathsf{A}_S^\perp$ is enforced by the input/output placement, and one does not replace that input variable by a variable of the distinct linear

object $y\mathsf{A}_S^{\perp}$ in this derived cut rule. The conclusion is their tensor-and-composition composite in $\mathcal{V}$. This gives a derived cut judgment from the core linear rules, not an unrestricted process language. The finite operational language consists only of the constructed endpoint prefix trees, complete-policy registers, and a current history. Its $\mathsf{Send}$ and $\mathsf{Recv}$ notations name the corresponding selected positive branch and receptive negative branch of these trees, not independent freely asserted primitive processes. Matching prefixes reduce by

$$(\nu c)(\mathsf{Send}_c(m); P \mid \mathsf{Recv}_c(x); Q) \longrightarrow (\nu c)(P \mid Q[m/x]). \tag{10.5}$$

Dual endpoints synchronize on matching communication events. Hiding the shared interface yields the composite, while the matched history records which conversation took place. The residual theorem identifies what each endpoint can do next by the message actually exchanged.

### 10.2.6 Operational configurations and residual session endpoints

A runtime configuration is $(S, h, \alpha, \beta)$, where $h$ is a legal history compatible with the complete policies and $S|h$ is its residual session. At a nonterminal history the next sender is uniquely determined by the polarity. Its register gives the next label $m_h$, and the transition is

$$(S, h, \alpha, \beta) \longrightarrow (S, hm_h, \alpha, \beta). \tag{10.6}$$

There is no transition from $\mathsf{End}$. At a send node of $a$ one has $m_h = \alpha_h$ and at a send node of $b$ one has $m_h = \beta_h$. For physically received private values $u_a, u_b$, evaluate the corresponding register entries as $\alpha_h(u_a)$ or $\beta_h(u_b)$; the causal input events are prerequisites of the first relevant sends. Local prefix notation projects this communication transition to the matching sender and receiver reduction (10.5). Neither transition alters the complete register.

**Lemma 10.17** (Residual session arena)**.** *Let $h$ be a history of $S$, viewed as a configuration of $\mathsf{A}_S$. The residual arena consisting of events compatible with and strictly after $h$, with the events of $h$ removed and the inherited causal order, is canonically isomorphic to $\mathsf{A}_{S|h}$. Residual formation commutes with duality.*

*Proof.*

(1) **Causal execution.** Every remaining event is a nonempty extension $hk$ of $h$, and $hk \mapsto k$ is a bijection to the nonempty histories of the syntactic continuation $S|h$. Prefix order, single-branch consistency, and the send/receive polarity at a continuation node are unchanged by deleting $h$. Prepending $h$ is the inverse.

(2) **Causal execution: completion.** Reversing polarities before or after this bijection gives the same event labels, proving the duality equation.

□

**Construction 10.18** (Restriction of complete session policies)**.**

(1) **Identify continuation coordinates.** For a legal session history $h$, concatenation gives a bijection

$$\mathcal{D}_i(S|h) \xrightarrow{k \mapsto hk} \{hk \in \mathcal{D}_i(S) : k \in H(S|h)\}.$$

In contrast to the residual *event* bijection, the residual *decision* positions include $k = \epsilon$ when $S|h$ begins with a decision by $i$.

(2) **Project the complete policy.** Define the ordinary product projection

$$\boxed{\rho_{i,h} : \Sigma_i^S \longrightarrow \Sigma_i^{S|h}, \qquad \rho_{i,h}(\alpha)_k = \alpha_{hk}.}$$

This supplies the coordinates needed by the residual endpoint; the original complete policy register remains unchanged and still contains its earlier and counterfactual coordinates.

(3) **Iterate restriction.** For a further legal history $k$ of $S|h$, product extensionality gives

$$\rho_{i,\epsilon} = \mathrm{id}, \qquad \rho_{i,hk} = \rho_{i,k}^{S|h} \rho_{i,h}. \tag{10.7}$$

Indeed both sides have component $\alpha_{hkl}$ at a residual decision position $l$ of $S|hk$.

**Construction 10.19** (Residual status of a physical private input)**.**

(1) **Record the input status.** For owner $i$ define the ordinary local input-status carrier $\mathsf{InputStatus}_i = 1 + O_i$. Its left summand means the physical input is still outstanding; a right-summand value $u$ means its negative input occurrence belongs to the completed source history.

(2) **Residualize an outstanding input.** Let $x$ be a configuration of the strategy of Construction 10.15 and let $h$ be its session projection, so $h$ is a prefix of $S$.

When the input is outstanding, no local positive session event has occurred, because every first such event requires the missing private input. Residualizing at $x$ leaves the private-input strategy associated with $\rho_{i,h}(\alpha)$ and the external input interface $D(O_i)$.

(3) **Residualize a received input.** When $e_u \in x$, the input has already occurred. Define the fixed-value residual session policy by

$$\alpha_k^{u,h} = (\rho_{i,h}\alpha)_k(u) = \alpha_{hk}(u).$$

Residualizing at $x$ and removing its completed private-input occurrence leaves the fixed-value endpoint strategy over $\mathsf{A}_{S|h}$, with no second private-input port. For owner $b$ use the identical construction in the dual session arena.

**Lemma 10.20** (Coherence of input-status residuals)**.** *The two residual descriptions of Construction 10.19 are isomorphisms of strategies over their respective residual session arenas. Receiving a private input changes the status from outstanding to the delivered value $u$; receiving a session message leaves the status unchanged; a local send is possible only in the received-value status. These residual isomorphisms commute with successive residual formation and session duality.*

*Proof.*

(1) **Causal execution.** An outstanding-input configuration has only common negative-prefix session events. Deleting that prefix identifies the remaining common negative events and tagged suffixes, by $hk \mapsto k$, with the source constructed from $\rho_{i,h}\alpha$; its private input events $e_u$ remain present. If the input $e_u$ has arrived, delete it and the completed session prefix.

(2) **Causal execution: second comparison.** The remaining tagged events $(u, hk)$ are identified with the fixed-$u$ pruning of $S|h$, while any common negative-prefix suffix has the identical image under this bijection. The identifications preserve order, consistency, polarity, and labelling, and the additional private-input cause has already been discharged.

(3) **Causal execution.** A transition receiving $e_u$ selects the second description without changing $h$. A negative session transition extends $h$ through the same common or tagged branch and leaves the input status unchanged. Every positive session event is tagged and therefore can occur only in the received-value case.

(4) **Policy-level comparison.** For successive session prefixes the resulting policy components are $\alpha_{hkl}(u)$ by Equation (10.7); the input is removed exactly once. Thus both orders of residualization yield the same source event structure over the suffix arena. Reversal of session polarities exchanges the two endpoint arguments without changing the input-status construction.

□

**Lemma 10.21** (Residual endpoint strategy isomorphisms)**.** *For a history $h$ compatible with the complete policy profile, residual formation of its two fixed-policy endpoint strategies satisfies*

$$\mathrm{Res}_h(p_\alpha) \cong p_{\rho_{a,h}(\alpha)}, \qquad \mathrm{Res}_h(q_\beta) \cong q_{\rho_{b,h}(\beta)},$$

*using the identification $\mathsf{A}_S/h \cong \mathsf{A}_{S|h}$ of Lemma 10.17. These isomorphisms commute with further residual formation and with polarity reversal.*

*Proof.*

(1) **Causal execution.** A remaining session event is exactly an extension $hk$, $k \neq \varnothing$. The bijection $hk \mapsto k$ preserves and reflects its prefix order, branchwise consistency, and polarity. At a positive endpoint decision position $k$ (including $k = \epsilon$ when the residual begins with a local send) both prunings retain the same chosen output, since the residual policy has value $\rho_{i,h}(\alpha)_k = \alpha_{hk}$.

(2) **Fibrewise calculation.** At a negative endpoint node both retain every enabled input. The resulting restriction is therefore an isomorphism of the two strategy event structures over the residual arena. The indexed equation (10.7) proves compatibility of successive residuals, and the event bijection is unchanged by polarity reversal.

□

**Proposition 10.22** (Derived finite-prefix subject reduction)**.** *For the above finite protocol-prefix notation and its closed complete policy registers, a matching send and receive at history $h$ preserves the external linear typing judgment, changes both connected endpoint types to the dual pair $S|hm$ and $\overline{S|hm}$, and preserves the full registers. The reduction (10.5) is interpreted by residual formation of the secured interaction, followed by the canonical residual endpoint isomorphisms.*

*Proof.*

(1) **Fibrewise calculation.** At the nonterminal history $h$, the fixed sender entry selects $m \in M_h$; the receptive endpoint uniquely accepts that same input. The session syntax selects the continuation $S|hm$, and polarity reversal selects its dual. The preceding lemma identifies the resulting endpoint strategies with the respective prunings of these continuations.

(2) **Functoriality and substitution.** Substitution of $m$ into a dependent continuation is ordinary substitution, so it cannot duplicate or drop unrelated linear variables. The core linear substitution lemma validates the cut before and after that step. The projections $\rho_{i,hm}$ supply the first post-prefix decision as well as all later decisions.

(3) **Causal execution.** Its full register is unchanged; only those projections determine its residual use. For endpoints with physical private inputs, the additional input-status isomorphisms of Lemma 10.20 verify that an input is neither used before arrival nor requested a second time.

(4) **Secured operational interaction.** Secured interaction of the matching prefix and hiding give precisely the residual composed strategies, by the interaction and hiding construction of Section 1.

(5) **Session subject reduction.** This proves subject reduction for the specified finite prefix language without claiming a process calculus with arbitrary channel creation or mobility.

□

**Theorem 10.23** (Finite session preservation, progress, and adequacy)**.** *For every* $(\alpha, \beta) \in \Sigma_a^S \times \Sigma_b^S$*:*

*(i) every communication step leaves dual residual endpoint types and preserves the complete-policy profile;*

*(ii) every nonterminal runtime configuration has a unique next communication, and execution reaches* End *in finitely many steps;*

*(iii) the configurations of the secured interaction of* $p_\alpha$ *and* $q_\beta$ *are exactly the prefixes of this execution. Its unique maximal configuration records the completed transcript.*

*Proof.*

(1) **Fibrewise calculation.** At the current history $h$, exactly one endpoint sends. Its complete-policy coordinate selects $m \in M_h$. The other endpoint retains every enabled input and therefore receives this event uniquely.

(2) **Causal execution.** The continuations are indexed by the same $m$ and are dual by Lemma 10.17. Their typed reduction and the unchanged complete register follow from Proposition 10.22. The finite height of the residual syntax tree strictly decreases.

(3) **Policy-level comparison.** A reached node with an empty message carrier would require a component of the given complete profile in the empty set, which is impossible. Hence execution terminates specifically at End.

(4) **Secured operational interaction.** A matching configuration of the endpoint maps identifies session events bearing the same history label. The maps are locally injective, and each endpoint's configurations lie on one root-to-leaf branch. At each nonterminal node the fixed sender pruning retains precisely the event selected by its complete policy, while receptivity gives its unique matching occurrence on the receiver.

(5) **Fixed-point argument.** Induction on the prefix length therefore identifies a matching with exactly one initial segment of the calculated run; conversely, each calculated initial segment gives such a matching. Both endpoint causal orders restrict to prefix order, so their union has no oriented cycle and every such matching is secured.

(6) **Secured operational interaction.** The secured-matching configuration bijection of Theorem 1.7 now identifies the interaction configurations with these prefixes, *before hiding*. The complete run gives the unique maximal secured witness. Hiding can erase the internal transcript events from the composite scalar; the completed witness or the explicit recorder of Lemma 10.25 retains them for subsequent evaluation.

□

**Corollary 10.24** (Input-indexed session adequacy)**.** *For fixed complete input-indexed policies $\alpha, \beta$ and supplied private values $u_a \in O_a, u_b \in O_b$, compose the strategies of Construction 10.15 with the value states $\mathsf{pt}_{O_a}(u_a)$ and $\mathsf{pt}_{O_b}(u_b)$. The projection of their secured interaction to the shared session arena has exactly the prefixes of the execution of (10.6), with next labels $\alpha_h(u_a)$ and $\beta_h(u_b)$. Its maximal session transcript is unique. The physical input events may be concurrent with earlier received session moves; no ordering between the two private inputs is added.*

*Proof.*

(1) **Fibrewise calculation.** At the first local send of each owner its private input is already in its source causal past, by Lemma 10.16. Lemma 10.20 then identifies the remaining endpoint with the fixed-value pruning at each later history, without reissuing the private input. Every negative session continuation remains uniquely receivable.

(2) **Fixed-point argument.** Induction on the current session prefix therefore gives the same unique next synchronized label as the closed case, evaluated at the supplied input. After forgetting private-input occurrences, its session matchings are exactly the calculated prefixes. The input dependencies cannot form a cycle: their edges point into the relevant positive moves and never back towards an input.

(3) **Securedness of synchronized messages.** Hence the matchings are secured. The two input events have no dependency between them, and finite-height termination is unchanged.

□

The theorem establishes adequacy at the level of witnesses. After hiding the entire session, $q_\beta p_\alpha$ has boundary $I \to I$, and the concurrent category has a single scalar. Its transcript must therefore be retained as an execution witness, or delivered on an explicit output wire, when it determines later observations, records, or evaluation.

**Lemma 10.25** (A physical transcript output)**.** *The finite closed session has a recorded implementation with boundary $I \to D(T_S)$, where $T_S$ is its finite set of terminating transcripts. For the input-indexed session there is a recorded implementation with boundary $D(O_a) \otimes D(O_b) \to D(T_S)$. Its complete policy carriers are canonically the original ones. At each complete profile, its visible output is the calculated transcript, and a completed secured matching retains the internal communication history witnessing that output.*

*Proof.*

(1) **Fibrewise calculation.** At each terminating history $h$, adjoin to the session arena a positive record event $r_h$ whose immediate cause is the final session event of $h$ (or no cause if $h = \epsilon$). Its full causal past is therefore exactly the events of $h$. Record events at different leaves remain inconsistent, and their labels distinguish the corresponding leaves.

(2) **Ownerwise alternatives.** Call the extended arena $\mathsf{A}_S^r$. These new final positive moves have a prescribed singleton choice; they contribute unit factors rather than new owner policies.

(3) **Policy-level comparison.** Extend the first endpoint by producing $r_h$ at every leaf consistent with its policy. Extend the dual endpoint by accepting every enabled negative occurrence $r_h$ and then producing a positive output $d_h$ labelled by $h \in T_S$. Its only new immediate dependency is the negative-to-positive edge $r_h < d_h$.

(4) **Secured operational interaction.** The two source maps preserve prefix order and conflict, are locally injective, and accept every enabled negative move uniquely. Their new dependencies satisfy courtesy; thus both extended endpoints are receptive courteous strategies.

(5) **Policy-level comparison.** For a fixed complete policy profile, the secured interaction of the extended endpoints has precisely the session prefixes established by Theorem 10.23, together with a final matched $r_h$ and visible output $d_h$ on the unique completed branch. Before hiding, the completed matching explicitly records the internal session events and the final record occurrence.

(6) **Secured operational interaction.** After hiding the $\mathsf{A}_S^r$ interface, the output event $d_h$ remains with the induced visible causal order, but the matched session events themselves are no longer events of the composite strategy. This distinction proves the claim about the retained execution witness and the visible output separately.

(7) **Causal execution.** For physical private inputs use the endpoints of Construction 10.15, extending their session components by the same terminal recorder. Their reception events remain independent until the protocol requires them, and the extra causal predecessors of input-dependent positive events persist in the secured interaction.

(8) **Secured operational interaction.** For the input-indexed version, first copy each physical private input by its designated data comonoid $\Delta_{O_a}$ or $\Delta_{O_b}$. Send one copy to the corresponding private-input endpoint and the other two copies to $c_{O_a,O_b} : D(O_a) \otimes D(O_b) \to D(O_a \times O_b)$.

(9) **Communication and delivery.** After the session recorder supplies $D(T_S)$, form the final output by

$$D(O_a \times O_b) \otimes D(T_S) \xrightarrow{c_{O_a \times O_b, T_S}} D((O_a \times O_b) \times T_S) \xrightarrow{\widehat{\mathrm{pr}_{T_S}}} D(T_S).$$

This makes the final visible record causally dependent on both physical private inputs and the completed transcript, without ordering the two private input occurrences.

(10) **Ownerwise alternatives.** The entire construction uses only the existing copying, collector, and finite response arrows. Its fixed data wiring preserves the displayed owner policy factors. On composing with the selected value states, Lemma 10.20 and Corollary 10.24 identify the same unique completed transcript.

(11) **Policy-level comparison.** No policy coordinate is created by the recorder: its outputs are functions of the existing terminal history.

□

### 10.2.7 Composing conversations

Sequential continuation is tree grafting. Given a continuation $T_h$ at each terminating leaf $h$ of $S$, its decision nodes are those of $S$ and the tagged nodes of every $T_h$. The complete owner carriers satisfy

$$\Sigma_i^{S[T]} \cong \Sigma_i^S \times \prod_{h \in \mathrm{Leaves}(S)} \Sigma_i^{T_h}.$$

Substituting a continuation at each completed leaf gives the complete policy space of the composite conversation. The policy at an off-path leaf remains part of that product, even

when the chosen execution never visits it. Consequently sequential composition retains the counterfactual choices needed by later strategic evaluation.

This compositional account of finite sessions is the operational basis for the communication witnesses used below: typing records which messages can be exchanged, while secured histories record which messages were actually delivered. Neither alone determines a truthful announcement or the recipient's epistemic update.

These finite sessions implement request and reply, labelled choice, private delivery, and conversations with several rounds. The general connection between session typing and cut reduction is developed in Caires and Pfenning (2010). Here the direct operational construction also identifies the owner of each choice and retains its off-path responses. The syntax covers finite sessions with the specified endpoints.

## 10.3 Epistemic action events from completed communication

The completed secured matchings and the transcript-output wires from this section now provide the event and observation data required by the abstract epistemic action semantics of Section 9. The construction keeps distinct execution witnesses even when their observable action-event labels coincide.

**Construction 10.26** (Action data from completed execution witnesses)**.**

(1) **Retain the completed executions.** For a finite implemented protocol, construct $\Xi$ as the dependent carrier of its completed secured matchings, retaining the source world, profile, and completed labelled causal history. Concretely, over input and complete-policy space $W \times \Sigma$, it is the dependent sum of the sets of completed secured matching configurations associated with the specified protocol realizers. The deterministic protocol and session adequacy theorems just proved construct those fibres; for general witnessed finite interactions they retain all completed matchings together with their individual witnesses. Its source projection is $s : \Xi \to W$.

(2) **Read the delivered ports.** Each recipient's specified actual input ports determine a local-record set $V_i$ of finite labelled causal subhistories and a restriction map

$$d_i : \Xi \longrightarrow V_i.$$

Here the local record includes exactly the events delivered to $i$ and any designated retained local policy register, with their labels and induced causal order; independent receptions are not assigned a total schedule. The finite protocol has finitely many event occurrences and finite message carriers, so its collections of completed labelled local subhistories form finite sets. If an externally visible public record is required, use the final visible output to construct its record map $d_0 : \Xi \to V_0$; otherwise take $V_0 = 1$ and its unique record map.

(3) **Form the joint action image.** Construct the tuple and its image factorization

$$\widetilde{d} = \langle d_0, (d_i)_{i \in N} \rangle : \Xi \xrightarrow{q} E \hookrightarrow V_0 \times \prod_{i \in N} V_i.$$

Here $q$ is the surjective image map and $\iota$ is its inclusion. For each owner define $b_i = \mathrm{pr}_{V_i} \iota : E \to V_i$, so that $d_i = b_i q$. Thus the event labels and their recipient observations are *constructed from delivery*, not separately specified event equivalence relations. Explicitly,

$$q(\xi) = q(\xi') \quad \Longleftrightarrow \quad d_0(\xi) = d_0(\xi') \text{ and } d_i(\xi) = d_i(\xi') \text{ for all } i.$$

This is the coarsest event-label identification preserving the retained public record and every actually delivered recipient record.

(4) **Compute enabling and updated observations.** The enabling predicate is the image

$$U = \mathrm{im}\langle s, q\rangle \hookrightarrow W \times E, \qquad (w, e) \in U \iff \exists \xi \in \Xi\ (s\xi = w \ \wedge\ q\xi = e).$$

The effective updated observation of owner $i$ is the image of

$$o_i^+ : U \to O_i \times V_i, \qquad o_i^+(w, e) = (o_i(w), b_i(e)).$$

Write $R_i = \ker(o_i)$ and construct $S_i = \ker(b_i)$. The witness carrier $\Xi$ remains displayed above $U$ together with each witness's concurrent realization and any institutional certificate. The image is used for the epistemic event label, not to identify distinct execution witnesses.

*Proof.*

(1) **Secured operational interaction.** The local record restrictions are functions because every completed secured history has a determined finite labelled subhistory at each designated recipient port. The image construction has the universal property that every map out of $\Xi$ constant on $\ker(q)$ descends uniquely through the surjection $q$. The coordinate maps $b_i$ exist by factorization through the image and satisfy $b_i q = d_i$ by definition.

(2) **Monoidal comparison.** The product-update kernel is calculated on the enabled carrier:

$$\begin{aligned}(w, e)\ \ker(o_i^+)\ (v, f) &\iff o_i(w) = o_i(v) \ \wedge\ b_i(e) = b_i(f)\\ &\iff wR_iv \ \wedge\ eS_if.\end{aligned}$$

Corestriction to the effective image does not change this kernel pair. The first line is equality in a Cartesian product, and the second is the definition of the two component kernels.

(3) **Witness and legality comparison.** This is exactly Equation (9.16). Nothing in the factorization forgets the separate elements of $\Xi$: each remains an exhibited witness in the institution's dependent execution carrier.

□

# 11 Strategic Communication and the Creation of Knowledge

Communication can give several agents information that none possessed alone. In the two-bit institution, each owner first sees a private bit. The collector already holds the pair before either recipient has necessarily received it. We therefore evaluate what the recipients know at the completed-delivery boundary. The image of the completed execution map gives the event carrier, and each recipient's projection gives its observation. To compare strategic alternatives, we also keep the complete policy behind each current message. The split-surjection and lifting results specify which claims about choice and knowledge can then be read from the public messages.

## 11.1 Public inspection and the disclosure of pooled information

The image in Construction 6.26 is a joint signal. Disclosing its value is an epistemic action that can be described simultaneously at every reference state. For observation relations $R_i = \ker(o_i)$ on $W$ and a coalition $C$, write

$$R_C = \bigcap_{i \in C} R_i, \qquad \mathsf{DK}_C(P) = [R_C]P, \qquad \mathsf{CK}_C(P) = \left[\left(\bigcup_{i \in C} R_i\right)^*\right] P.$$

The first predicate is distributed knowledge. The second is common knowledge within $C$; for $C = N$ it is the previously defined $\mathsf{CK}$. Empty intersection is $W^2$, while the reflexive transitive closure of the empty union is equality. These conventions also define both operations for the empty coalition.

**Construction 11.1** (Public inspection of a signal)**.**

(1) **Construct the inspection update.** For $q : W \to O$, let $E = \mathrm{im}(q)$, enable precisely the pair $(w, q(w))$, and equip every agent's action set with equality. The updated carrier is canonically $W$ and

$$R_i^q = R_i \cap \ker(q).$$

If $R_i = \ker(o_i)$, the updated observation is $\langle o_i, q \rangle$.

(2) **Restrict to the designated signal fibre.** At a designated world $w$, the component $q^{-1}(q(w))$ is the announcement of the observed value. This fibre is closed under all updated accessibility relations. Its inclusion is bounded, so Theorem 9.29 identifies its modal interpretation with that of the full-carrier update at $w$. Retaining every possible outcome at once gives the full-carrier presentation of public inspection (van Eijck, Gattinger, and Wang, 2017).

**Proposition 11.2** (Inspection is an idempotent coreflection)**.** *In the poset of relation families on a fixed carrier $W$, ordered by componentwise inclusion, let $\mathcal{R}_q$ be the subposet with $T_i \subseteq \ker(q)$ for every $i$. Its inclusion $\iota$ has right adjoint*

$$\mathsf{Inspect}_q(R)_i = R_i \cap \ker(q), \qquad \iota \dashv \mathsf{Inspect}_q.$$

*Write $\mathsf{I}_q = \iota\mathsf{Inspect}_q$ for the associated idempotent comonad on the full relation poset. For two fixed signals $q, r$ on $W$, these endofunctors satisfy*

$$\mathsf{I}_r\mathsf{I}_q = \mathsf{I}_{\langle q,r \rangle} = \mathsf{I}_q\mathsf{I}_r.$$

*These statements also hold in the poset of S5 relation families.*

*Proof.*

(1) **Adjunction and transposition.** For $T \in \mathcal{R}_q$, $T \subseteq R$ if and only if $T \subseteq R \cap \ker(q)$. This is the adjunction. Its associated endofunctor $\mathsf{I}_q$ has counit $R \cap \ker(q) \subseteq R$ and is idempotent.

(2) **Predicate calculation.** The composition law follows from $\ker\langle q, r \rangle = \ker(q) \cap \ker(r)$. Meets preserve equivalence relations, proving the S5 version.

□

**Theorem 11.3** (Public pooling turns distributed knowledge into common knowledge)**.** *Let $R_i = \ker(o_i)$ and let $C \subseteq N$ be nonempty. Form the image factorization of the tuple of coalition observations,*

$$W \xrightarrow{q_C} O_C \hookrightarrow \prod_{i \in C} O_i, \qquad R_C = \ker(q_C) = \bigcap_{i \in C} R_i.$$

*After public inspection of $q_C$, the common-knowledge relation of all owners is exactly $R_C$. Consequently, for every fixed factual predicate $A \subseteq W$,*

$$\mathsf{CK}_N^{q_C}(A) = \mathsf{DK}_C(A), \qquad \mathsf{DK}_C(A) := [R_C]A.$$

*Proof.*

(1) **Predicate calculation.** Each updated relation is $R_i^+ = R_i \cap R_C$. If $i \in C$ this is $R_C$; for other $i$ it is contained in $R_C$. Since $C$ is nonempty,

$$\bigcup_{i \in N} R_i^+ = R_C.$$

The relation $R_C$ is an equivalence relation, hence equals its own reflexive transitive closure. Taking relational boxes proves the equation.

□

**Corollary 11.4** (Private pooling within a coalition)**.** *Use the same event set and enabling relation, but take $S_i$ to be equality for $i \in C$ and the universal relation for $i \notin C$. Then members have updated relation $R_C$ and outsiders retain $R_i$. For fixed $A$, common knowledge within $C$ after this update equals $\mathsf{DK}_C(A)$ before it.*

*Proof.*

(1) **Ownerwise alternatives.** Product update intersects member relations with $\ker(q_C)$ and leaves outsider relations unchanged. The union of the member relations is $R_C$.

□

This private construction is the resolution of distributed knowledge studied by Ågotnes and Wáng (2016). Public inspection additionally reveals the pooled value to outsiders. In both equations $A$ is retained as a fixed subobject. An epistemic formula re-evaluated after the update may denote a different subobject; Remark 9.7 makes this explicit.

**Proposition 11.5** (A connectivity obstruction for general signals)**.** *Let $E = (\bigcup_i R_i)^*$ and $E^q = (\bigcup_i (R_i \cap \ker(q)))^*$. Then*

$$E^q \subseteq E \cap \ker(q).$$

*For the quotient $c : W \to W/E$, there is a canonical surjection*

$$W/E^q \longrightarrow \mathrm{im}\langle c, q\rangle.$$

*It is an isomorphism exactly when every nonempty intersection of an $E$-class with a $q$-fibre is connected by the original $R_i$-edges that stay inside that intersection.*

*Proof.*

(1) **Lifting along the surjection.** Every generating updated edge preserves both the $E$-class and $q$. Thus both are constant on $E^q$-classes. This gives the factor map, which is surjective because its target is the image.

(2) **Injectivity of the event quotient.** It is injective exactly when equality of those two retained values implies $E^q$ relatedness. The definition of reflexive transitive closure identifies this with the stated connectivity condition.

□

**Example 11.6.** On $W = \{a, b, c\}$ let $R_1$ have classes $\{a, b\}, \{c\}$ and $R_2$ have classes $\{a\}, \{b, c\}$. Put $q(a) = q(c) = 0$ and $q(b) = 1$. Then $E = W^2$, but every updated individual relation is equality. Hence $E^q$ is equality, whereas $E \cap \ker(q)$ still relates $a$ to $c$. Common knowledge cannot generally be updated by intersecting the old common-knowledge relation with the signal kernel.

**Construction 11.7** (Finite operational disclosure)**.**

(1) **Collect the observed tuple.** For the observation maps $o_i : W \to O_i$, take their image factorizations $W \twoheadrightarrow F_i \hookrightarrow O_i$. When these images are finite and $C$ is nonempty, the collector and iterated data comultiplication form the strategy

$$\bigotimes_{i \in C} D(F_i) \xrightarrow{c_C} D\left(\prod_{i \in C} F_i\right) \xrightarrow{\Delta^{(N)}} \bigotimes_{j \in N} D\left(\prod_{i \in C} F_i\right).$$

The collector law of Proposition 1.22 sends the supplied value states to their tuple state.

(2) **Deliver the tuple.** The comonoid law of Proposition 1.19 places that tuple at every recipient's output. Equality of tuples has kernel $R_C$, so this realizes the disclosed signal of Theorem 11.3 on the supplied observation values.

(3) **Update the recipient observations.** Retaining each recipient's old observation gives the updated map $\langle o_j, q_C \rangle$. The collector waits for all coalition inputs; a decision that uses the disclosure is wired after the corresponding output occurrence.

The signal map identifies the information to be disclosed. The strategy exhibits the collecting and delivery events for those signals. A decision downstream can depend on the disclosure once the relevant delivery has occurred.

## 11.2 Agent-labelled truthful public actions

**Construction 11.8** (The public announcement performed by an agent)**.**

(1) **Form the authored action interface.** Fix an observed agent presentation on $W$, with effective quotients $\bar{o}_j : W \twoheadrightarrow \Omega_j$. For a sender $i$, take the action set $E_i = \mathcal{P}(\Omega_i)$ and enable $B \in E_i$ at $w$ precisely when $\bar{o}_i(w) \in B$. Every observer's action relation is equality on $E_i$.

(2) **Compute its content.** The content and updated carrier are

$$A_{i,B} = \bar{o}_i^{-1}(B), \qquad \mathcal{U}_i = \{(w, B) : w \in A_{i,B}\}.$$

The sender identity is part of the action interface.

(3) **Interpret the pointed announcement.** Write $\mathsf{say}_i(B)$ for the corresponding pointed public action. If $a_{i,B} : A_{i,B} \hookrightarrow W$ and $j_{i,B} : A_{i,B} \to \mathcal{U}_i$ are the inclusion and $w \mapsto (w, B)$, its modalities are

$$[\mathsf{say}_i(B)]Q = \forall_{a_{i,B}} j_{i,B}^* Q, \qquad \langle \mathsf{say}_i(B) \rangle Q = \exists_{a_{i,B}} j_{i,B}^* Q.$$

**Proposition 11.9** (Authored announcement and public restriction)**.** *The content $A_{i,B}$ is invariant under the sender's observation relation, so $K_i A_{i,B} = A_{i,B}$. The component of $\mathcal{U}_i$ at event $B$ is isomorphic as an observed frame to $M|_{A_{i,B}}$. Consequently an agent-labelled announcement has exactly the public-restriction semantics of its content, with epistemic formulas evaluated in that restricted frame.*

*Proof.*

(1) **Observation and knowledge.** If $\bar{o}_i(v) = \bar{o}_i(w)$, membership of that common value in $B$ is the same at $v$ and $w$. This proves observation invariance and the knowledge equation. The product-update relation at $B$ is

$$(w, B)R_j^+(v, B) \iff wR_jv \text{ and } w, v \in A_{i,B},$$

because the action relation is equality.

(2) **Isomorphism of restricted action models.** Thus $w \mapsto (w, B)$ is a frame isomorphism from the restriction onto that component, preserving the restricted factual terms. The displayed dynamic modalities are precisely the inclusion quantifiers of the public restriction. Frame isomorphism and the defining logical operations give the assertion for re-evaluated epistemic formulas and their fixed points.

□

This realizes a sender-labelled public test for the observation-invariant message family. Joint communication consists of one such component per owner, but its strategic interpretation requires complete policies specifying messages at every observation. We construct those carriers next.

**Construction 11.10** (A public test as an institutional agent action)**.**

(1) **Supply the test and its owner.** For any predicate $P \hookrightarrow W$, form its characteristic term $\chi_P : W \to 2$ in the paper's ordinary base and corestrict it to $\Omega_P = \mathrm{im}(\chi_P)$. Extend the owner list by a fresh institutional identity $*$, retaining all original world observations. Give this agent observation $\chi_P : W \twoheadrightarrow \Omega_P$ and the truthful message interface $\mathcal{P}(\Omega_P)$.

(2) **Realize the truthful disclosure.** The complete policy $\eta_P(o) = \{o\}$ discloses the test result. Its successful public event is $B_P = \Omega_P \cap \{1\}$; if $P$ is empty, that event has empty enabling carrier. The original agents publicly receive the disclosed result.

(3) **Factor the implementation.** The ordinary test and its concurrent realization have distinct types:

$$W \xrightarrow{\chi_P} 2, \qquad D(\Omega_P) \xrightarrow{\widehat{\chi}} D(2) \xrightarrow{\text{success gate}} D(1),$$

where $\chi : \Omega_P \to 2$ is the inclusion of the image of $\chi_P$. The first arrow is the *specified ordinary signal input* and not an event-structure sensor inferred from an arbitrary predicate on $W$. The data strategy implements the second, finite-stage operation after that input is actually supplied.

**Proposition 11.11** (Public comprehension realized by a test agent)**.** *The successful branch of this institutional action has enabling carrier $P$, and its frame on the original agent indices is exactly $M|_P$. Its dynamic modalities on retained predicates are the paper's public restriction modalities. The disclosure component has a finite concurrent realization, with its input supplied by the ordinary term $\chi_P$.*

*Proof.*

(1) **Policy-level comparison.** The event $B_P$ is enabled precisely when $\chi_P(w) \in \Omega_P \cap \{1\}$, equivalently $w \in P$. At such a world the complete policy $\eta_P$ outputs exactly $B_P = \{1\}$; if no such world exists both enabling carriers are empty. Equality of public events restricts each original relation $R_j$ to $P \times P$.

(2) **Predicate calculation.** Projection of the successful component to $P$ is therefore a frame isomorphism for those indices and the retained factual terms. Its inclusion into $W$ is the same comprehension map, proving both dynamic quantifier equations.

(3) **Observation and knowledge.** The observation image has at most two elements. Its singleton response and public broadcast are the finite data constructions of the operational semantics. The term $\chi_P$ supplies their ordinary reference input; the implementation realizes the finite disclosure interface.

(4) **Observation and knowledge: completion.** Predicate evaluation remains the ordinary semantic test that constructed $\chi_P$. The original agents' pre-action observations are retained, and the extra institutional observation is explicitly part of this extended presentation.

□

The implementation records who sends the test, which result is delivered, and which event succeeds. Thus the logical update has an observable cause rather than an unexplained restriction of worlds. The finite disclosure protocol acts on the test-result interface; the underlying world remains an ordinary parameter.

## 11.3 The full truthful message family and coalition announcement games

A truthful public message is controlled by its sender, but its content is constrained by that sender's observation. Power objects and comprehension generate the entire semantic family of such messages. The coalition effectivity quantifiers then test whether a group can secure a post-announcement goal against truthful outsiders. This relates strategic control of disclosure to the joint-signal operation of Section 11.1.

**Construction 11.12** (Truthful message families)**.**

(1) **Form the observed message carriers.** Let $M = (W, (\ker(o_i))_{i\in N})$. Factor each observation as $W \xrightarrow{\bar{o}_i} \Omega_i \hookrightarrow O_i$, where $\Omega_i = \mathrm{im}(o_i)$. Define the ordinary message object $\mathsf{Msg}_i = \mathcal{P}(\Omega_i)$ and its admissible fibre

$$\mathsf{Msg}_i(w) = \{B_i \subseteq \Omega_i : \bar{o}_i(w) \in B_i\}.$$

(2) **Interpret message contents.** The message $B_i$ announces the predicate $A_i = \bar{o}_i^{-1}(B_i)$. This predicate is observation-invariant; when $B_i$ is admissible at $w$, $K_i A_i(w)$ holds. Conversely every observation-invariant predicate is the inverse image of a unique subset of $\Omega_i$, by Proposition 5.34.

(3) **Construct the updated model.** For $\mathsf{Msg} = \prod_i \mathsf{Msg}_i$ and $B \in \mathsf{Msg}$, put

$$A_B = \bigcap_i \bar{o}_i^{-1}(B_i), \qquad M_B = M|_{A_B}.$$

The updated carriers form the ordinary dependent family $\pi : \mathcal{U} = \{(B, v) : v \in A_B\} \to \mathsf{Msg}$. Its $i$-relation relates $(B, v)$ and $(B', v')$ precisely when $B = B'$ and $v \ker(o_i) v'$. Thus all epistemic operations in a fibre are those of $M_B$.

(4) **Specify goals and legal messages.** A goal is a specified subobject $\Psi \hookrightarrow \mathcal{U}$.

The legal message family over the reference state is

$$L = \{(w, B) : B_i \in \mathsf{Msg}_i(w) \text{ for all } i\} \longrightarrow W.$$

For $C \subseteq N$, let $L_C = \{(w, B_C) : B_i \in \mathsf{Msg}_i(w)\ (i \in C)\}$, and write $e_C : L \to L_C$ and $u_C : L_C \to W$ for the projections. These are the realizable coalition-coordinate objects in each fibre: every partial tuple has the canonical completion $B_j = \Omega_j$ for $j \notin C$. The map $g : L \to \mathcal{U}$, $(w, B) \mapsto (B, w)$, evaluates the goal in the announced context. Define

$$\mathsf{AnnAbility}_C(\Psi) = \exists_{u_C} \forall_{e_C} g^* \Psi \quad \in \mathsf{Pred}(W). \tag{11.1}$$

In particular, at $w$ this asserts that some truthful coalition message tuple succeeds against every simultaneous truthful outsider tuple.

The construction uses power objects, products, projection, and comprehension. Equation (11.1) is the fibrewise instance of Equation (7.12); its quantifiers follow the coalition-announcement pattern. Standard announcement logics restrict messages to a specified epistemic syntax (Ågotnes et al., 2010; Galimullin and Alechina, 2018); our full semantic family instead contains all observation-invariant subobjects. Group announcement logics with distributed knowledge are discussed by Ågotnes, Alechina, and Galimullin (2022). The results below state their own factual-goal and full-family scope rather than identifying these different languages.

**Theorem 11.13** (Exact announcement ability for factual common knowledge)**.** *For $P \subseteq W$, let $P^+ = \{(B, v) \in \mathcal{U} : v \in P\}$ and take $\Psi_P = \mathsf{CK}_N^{\mathcal{U}}(P^+)$, using the fibre-preserving relations of Construction 11.12. For every nonempty coalition $C$,*

$$\mathsf{AnnAbility}_C(\Psi_P) = \mathsf{DK}_C(P). \tag{11.2}$$

*The sufficient coalition policy is the uniform observation-admissible map $o \mapsto \{o\}$ for each member. For $C = \varnothing$, $\mathsf{AnnAbility}_\varnothing(\Psi_P) = \mathsf{CK}_N^M(P)$.*

A truthful announcement is represented as an informed section over the observation quotient.

$$\begin{array}{ccc} W & \xrightarrow{\ o_C\ } & O_C \\ {\scriptstyle m}\downarrow & & \downarrow{\scriptstyle \pi} \\ M & = & M \end{array}$$

*Proof.*

(1) **Fibrewise calculation.** At $w \in \mathsf{DK}_C(P)$, let every member announce $B_i = \{\bar{o}_i(w)\}$. For every admissible outsider tuple,

$$w \in A_B \subseteq R_C[w] \subseteq P.$$

The fact therefore holds throughout the updated carrier and is common knowledge there. Each chosen message is a function of the owner's observation alone, proving uniform observation-admissibility by Proposition 6.2.

(2) **Reverse implication.** Conversely, let a coalition tuple establish the displayed common knowledge against every outsider tuple. Take $B_j = \Omega_j$ for outsiders. Truthfulness and observation invariance give $R_C[w] \subseteq A_B$.

(3) **Observation and knowledge.** For any $i \in C$, each $v \in R_C[w]$ is $R_i$-accessible from $w$ inside $A_B$. Common knowledge in this updated model therefore gives $v \in P$. This is $\mathsf{DK}_C(P)(w)$.

(4) **Ownerwise alternatives.** For the empty coalition, the outsider tuple of all tautological messages leaves the original model unchanged, proving necessity.

(5) **Reverse implication.** Conversely every path in a restriction containing $w$ is an original path. Hence common knowledge of the fixed fact is preserved under all these restrictions. For $N = \varnothing$, the common-knowledge relation is equality and the same argument gives the predicate $P$.

□

**Proposition 11.14** (Finite message realizations)**.** *When all observation images $\Omega_i$ are finite, the full message objects, singleton disclosure maps, and truthfulness tests in Construction 11.12 have finite data realizations.*

*Proof.*

(1) **Message-policy and membership protocols.** Each $\mathcal{P}(\Omega_i)$ is finite. The map $\Omega_i \to \mathcal{P}(\Omega_i)$, $o \mapsto \{o\}$, and the membership function $\Omega_i \times \mathcal{P}(\Omega_i) \to 2$ have response strategies by Proposition 1.16. Collectors form joint messages, and the data comonoids broadcast them.

(2) **Observation and knowledge.** The public enabling subobject is the conjunction of the membership tests, so its finite presentation is Construction 9.19 on these signal and message carriers. Observational distinctions on an arbitrary original state carrier remain in the ordinary dependent family over that carrier.

□

The construction specifies a full semantic message family. A chosen communication language is represented by its own message object and content map; its effectivity is then obtained by the same dependent quantifiers. The equality in Theorem 11.13 is established for the constructed full family, whose singleton policy is explicitly available.

## 11.4 Complete truthful policies of communicating agents

The message actually sent fixes only one entry of an announcement policy. To ask what the sender could have disclosed at another observation, we retain the entire policy. This follows the contingent announcements of Ågotnes and van Ditmarsch (2011). For the observation-invariant message family constructed above, we compare complete truthful policies with their current public messages and keep both the private policy records and the execution witnesses. The comparison concerns this qualitative family. A restricted message language changes the available responses.

**Construction 11.15** (The complete truthful announcement interface)**.**

(1) **Form complete truthful policies.** Use the effective observation images and message carriers of Construction 11.12. Define

$$\mathsf{AnnPol}_i = \prod_{o \in \Omega_i} \{B \subseteq \Omega_i : o \in B\}, \qquad \mathsf{AnnPol} = \prod_{i \in N} \mathsf{AnnPol}_i.$$

A policy $\pi_i$ is a truthful message response at every possible observation, including observations different from the current one. It is an ordinary subobject of $\mathsf{Msg}_i^{\Omega_i}$.

(2) **Evaluate at the current observation.** Its evaluation is

$$e_i : \Omega_i \times \mathsf{AnnPol}_i \to \mathsf{Msg}_i, \qquad e_i(o, \pi_i) = \pi_i(o).$$

At $w$, the legal output of the agent is $\pi_i(\bar{o}_i(w))$. The agent's policy carrier is $\mathsf{AnnPol}_i$, with its canonical selection interface; a goal or comparison is connected to the resulting message or updated state as in the preceding agent calculus.

**Lemma 11.16** (Explicit extension of a truthful message)**.** *For every $o \in \Omega_i$, the evaluation map*

$$e_{i,o} : \mathsf{AnnPol}_i \twoheadrightarrow \{B \subseteq \Omega_i : o \in B\}$$

*has the explicit section $B \mapsto E_{i,B}$, where*

$$E_{i,B}(x) = B \cup \{x\}.$$

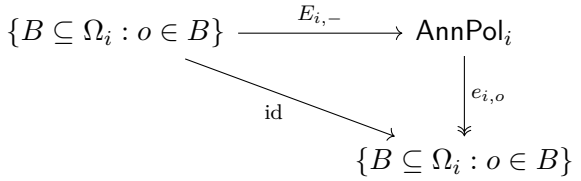
*In particular every legal message at $w$ is the current output of an exhibited complete truthful policy.*

*Proof.*

(1) **Split evaluation of public messages.** Every value of $E_{i,B}$ contains $x$, so the function belongs to $\mathsf{AnnPol}_i$. Since $o \in B$, evaluation at $o$ is $E_{i,B}(o) = B \cup \{o\} = B$. Hence $e_{i,o}(E_{i,B}) = B$. The formula is defined for every indicated pair and uses no choice of a representative.

□

Truthful messages extend to complete policies that evaluate to the chosen message at the given observation.




**Construction 11.17** (The policy-state model of public communication)**.**

(1) **Form policy states and their message map.** Put $\mathcal{X} = W \times \mathsf{AnnPol}$ and define the joint evaluation

$$b(w, \pi) = (\pi_i(\bar{o}_i(w)))_{i \in N}.$$

The map to the paper's public-message carrier, retaining its original coordinate order, is

$$F : \mathcal{X} \to \mathcal{U}, \qquad F(w, \pi) = (b(w, \pi), w).$$

(2) **Update the observations.** Give agent $j$ the pre-communication observation $\widetilde{o}^0_j(w, \pi) = (\bar{o}_j(w), \pi_j)$. Use the public action set $\mathsf{Msg}$ with equality as every action relation and enabling predicate $B = b(w, \pi)$. Its updated carrier is canonically $\mathcal{X}$, since each source has exactly one enabled joint message. Its post-communication observation is

$$\widetilde{o}_j(w, \pi) = (\bar{o}_j(w), \pi_j, b(w, \pi)), \qquad R^{\mathcal{X}}_j = \ker(\widetilde{o}_j).$$

Thus each agent retains its own complete policy, its original observation, and the delivered public messages. This is an explicit product-update instance of the earlier action-model construction.

(3) **Define strategic alternatives.** The strategic relations hold the reference world fixed:

$$(w, \pi) D^{\mathcal{X}}_C (v, \pi') \iff w = v \text{ and } \pi_{-C} = \pi'_{-C},$$
$$(B, w) D^{\mathcal{U}}_C (B', v) \iff w = v \text{ and } B_{-C} = B'_{-C}.$$

The latter are the legal message deviations in each fibre over $W$.

**Theorem 11.18** (Exact policy-to-message comparison)**.** *The map $F : \mathcal{X} \to \mathcal{U}$ is a split surjection and a bounded morphism for every individual epistemic relation, every nonempty intersection of those relations, and every coalition-deviation relation. More explicitly, for $\varnothing \neq J \subseteq N$ it is bounded from $\bigcap_{j\in J} R_j^{\mathcal{X}}$ to $\bigcap_{j\in J} R_j^{\mathcal{U}}$, and for every $C \subseteq N$ it is bounded from $D_C^{\mathcal{X}}$ to $D_C^{\mathcal{U}}$.*

*Proof.*

(1) **Ownerwise alternatives.** A section sends $(B, w) \in \mathcal{U}$ to

$$(w, (E_{i,B_i})_{i\in N}).$$

It is well-defined by truthfulness of $B$ at $w$ and is a section by Lemma 11.16. This includes the empty owner list, when both carriers are canonically $W$.

(2) **Observation and knowledge.** Forward preservation of the epistemic relations follows from their observations: equality of $\widetilde{o}_j$ implies equality of the original $j$-observation and of the public tuple. For the back condition, let $x = (w, \pi)$ and suppose

$$F(x)\Big(\bigcap_{j\in J} R_j^{\mathcal{U}}\Big)(B, v), \qquad J \neq \varnothing.$$

Then $B = b(w, \pi)$ and $\bar{o}_j(v) = \bar{o}_j(w)$ for every $j \in J$.

(3) **Construction on policy coordinates.** Define a profile at $v$ by

$$\pi'_j = \pi_j \quad (j \in J), \qquad \pi'_k = E_{k,B_k} \quad (k \notin J).$$

For a retained coordinate, $\pi_j(\bar{o}_j(v)) = \pi_j(\bar{o}_j(w)) = B_j$; for every other coordinate the extension lemma gives the same output. Hence $F(v, \pi') = (B, v)$. For each $j \in J$ the old observation, its own policy, and the public tuple are unchanged, so $xR_j^{\mathcal{X}}(v, \pi')$.

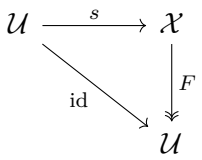
(4) **Lifting intersections along the report map.** This proves the simultaneous back condition and, in particular, its individual case. For an empty intersection the universal relation has the same lifting property by surjectivity alone.

(5) **Ownerwise alternatives.** For strategic relations, retaining the outsider policies at fixed $w$ retains their evaluated messages, which proves the forward clause. Suppose $F(w, \pi)D_C^{\mathcal{U}}(B', w)$. Keep $\pi'_j = \pi_j$ for $j \notin C$ and set $\pi'_i = E_{i,B'_i}$ for $i \in C$.

(6) **Ownerwise alternatives: completion.** The outsider outputs are already $B'_{-C}$, and the changed outputs are $B'_C$ by the extension lemma. Therefore $F(w, \pi') = (B', w)$ and $(w, \pi)D_C^{\mathcal{X}}(w, \pi')$. This proves the back clause for every coalition, including the empty and grand coalitions.

□

The complete-policy comparison is a split surjection with an explicit section.



**Lemma 11.19** (Dependent successor-fibre exactness of truthful communication)**.** *For each nonempty observer set $J \subseteq N$, write $K_J = \bigcap_{j\in J} R_j$ for the joint epistemic relation; for each coalition $C \subseteq N$, retain $D_C$ for strategic deviations. For either $R = K_J$ or $R = D_C$, with the corresponding superscripts $\mathcal{X}$ and $\mathcal{U}$, the canonical map of dependent successor families*

$$\lambda_R : R^{\mathcal{X}} \longrightarrow \mathcal{X} \times_{\mathcal{U},p_1} R^{\mathcal{U}}, \qquad (x, x') \longmapsto (x, (Fx, Fx'))$$

*is surjective. Consequently for every $Q \hookrightarrow \mathcal{U}$,*

$$F^*[R^{\mathcal{U}}]Q = [R^{\mathcal{X}}]F^*Q, \qquad F^*\langle R^{\mathcal{U}}\rangle Q = \langle R^{\mathcal{X}}\rangle F^*Q.$$

*The same statement for an empty epistemic intersection follows from the split surjection $F$.*

*Proof.*

(1) **Fibrewise calculation.** At a fixed $x \in \mathcal{X}$, a point of the displayed pullback is a specified $R^{\mathcal{U}}$-successor of $Fx$. The epistemic and strategic back-lifts constructed in Theorem 11.18 provide an $R^{\mathcal{X}}$-successor of that very $x$ mapping to it. Thus each fibre of $\lambda_R$ is inhabited, which is surjectivity in **Set**.

(2) **Predicate calculation.** Forward preservation ensures well-definedness. The source and target existential successor predicates therefore agree on inverse-image tests. Their universal successor predicates agree as well: a target test is true on every successor if and only if its pullback is true on every source successor, because the successor map is onto.

(3) **Modal preservation for empty coalitions.** This proves both displayed modal equations. For an empty epistemic intersection the relation is universal; the split surjection supplies the requisite lift from any fixed source state.

□

Alternative lifting is performed over the fixed policy reference.

$$\begin{array}{ccc} D_C^{\mathcal{X}} & \xrightarrow{\lambda_C} & \mathcal{X} \times_{\mathcal{U}} D_C^{\mathcal{U}} \\ {\scriptstyle p_1}\downarrow & & \downarrow \\ \mathcal{X} & = & \mathcal{X} \end{array}$$

**Corollary 11.20** (Knowledge, strategic predicates, and dynamic transport)**.** *For predicates and comparisons retained on $\mathcal{U}$, inverse image along $F$ preserves the individual knowledge, distributed knowledge, common-knowledge, coalition-deviation, and positive modal fixed-point operations. It preserves and reflects validity. For a comparison $Q_i \hookrightarrow \mathcal{U}^2$, use $(F \times F)^*Q_i$ on $\mathcal{X}^2$. The Nash predicates defined using the displayed deviation relations satisfy*

$$\mathsf{NE}_{\mathcal{X}} = F^*\mathsf{NE}_{\mathcal{U}},$$

*and the corresponding strict coalitional stability predicates satisfy the same equality. Public restrictions by retained assertions and action models pulled back from $\mathcal{U}$ have the same exact dynamic comparison.*

*Proof.*

(1) **Communication and delivery.** The explicit comparison $\lambda_R : R^{\mathcal{X}} \twoheadrightarrow \mathcal{X} \times_{\mathcal{U},p_1} R^{\mathcal{U}}$ of Lemma 11.19 gives the two inverse-image equations

$$F^*[R^{\mathcal{U}}]Q = [R^{\mathcal{X}}]F^*Q, \qquad F^*\langle R^{\mathcal{U}}\rangle Q = \langle R^{\mathcal{X}}\rangle F^*Q.$$

Here the source reference is not chosen afresh: it is fixed in the pullback, and every target successor lifts over that same reference.

(2) **Lifting through the comparison.** Theorem 9.29 proves the positive fixed-point comparison. The intersection lifting in the theorem supplies the distributed knowledge case directly. Common knowledge uses the same individual relations and its previously constructed greatest fixed point.

(3) **Fibrewise calculation.** At a source $x$, every concrete $i$-deviation maps to a target deviation with comparison $Q_i(Fx, Fx')$, and every target deviation lifts at that same source. Universal absence of improving alternatives is therefore equivalent on the two carriers. For a coalition use its lifting clause and the specified collective comparison; conjunction over coalitions gives the claimed stability equality.

(4) **Base-change comparison.** For $V \hookrightarrow \mathcal{U}$, the restriction $F^{-1}(V) \to V$ is surjective. A lift of an alternative in $V$ belongs to $F^{-1}(V)$ by its image, so it remains bounded. Comprehension and Beck–Chevalley give the public-restriction comparison.

(5) **Lifting along the surjection.** For a pulled-back action model apply Corollary 9.27. Surjectivity reflects validity in each case.

□

The retained language has its predicates on $\mathcal{U}$, pulled back to $\mathcal{X}$. Predicates concerning additional private policy records are also available on $\mathcal{X}$ and remain attached to that presentation. The comparison theorem relates the specified interfaces and their retained tests; it keeps the whole policy family needed to lift alternatives.

**Example 11.21** (A public policy record carries additional information)**.** Take two worlds $0, 1$. Agent 1 observes the world, while agent 2 has one observation. The truthful policy $\pi_1(0) = \{0, 1\}$, $\pi_1(1) = \{1\}$ sends the tautological message at 0. Its content alone leaves both worlds possible to agent 2. If the complete policy $\pi_1$ is also publicly delivered, observing that message identifies world 0. In the policy-state model of Construction 11.17, only agent 1 retains its own policy as a private record. At world 1 the alternative policy $E_{1,\{0,1\}}$ gives the same public message, supplying the lift required for agent 2. The example exhibits the informational effect of an additional policy-record wire and explains the role of the full policy family in $F$.

## 11.5 Concurrent implementation and retained policy witnesses

**Proposition 11.22** (Concurrent realization of complete announcing agents)**.** *When the observation images $\Omega_i$ are finite, the complete truthful policy presentation and its post-communication observations have an implementation in* $\mathbf{DecArena}_N(\mathcal{C})$*. Its complete owner policy carriers are* $\mathsf{AnnPol}_i$*. At reference input $(\bar{o}_i(w))_i$ and policy profile $\pi$, its delivered observation to agent $j$ is*

$$(\bar{o}_j(w), \pi_j, b(w, \pi)).$$

*The report decisions are concurrent, and the joint message is delivered after its component messages are collected.*

*Proof.*

(1) **Realization of announcement policies.** Each $\mathsf{Msg}_i = \mathcal{P}(\Omega_i)$ and $\mathsf{AnnPol}_i$ is finite. Restrict the decision generator $\mathsf{Dec}_i(\Omega_i, \mathsf{Msg}_i)$ along the exhibited inclusion

$$\mathsf{AnnPol}_i \hookrightarrow \mathsf{Msg}_i^{\Omega_i}.$$

It realizes policy $\pi_i$ by the response strategy $\widehat{\pi}_i$. Tensor these components and apply the finite collector to their message outputs.

(2) **Broadcast of the joint report.** The resulting joint output is $b(w, \pi)$. Broadcast this output using the iterated data comonoid, one copy for each agent.

(3) **Observation and knowledge.** Copy the original input observation to its reporting component and its owner's retained-observation wire. For a private policy record, use the singly owned nullary decision generator with response carrier $\mathsf{AnnPol}_i$; at a fixed policy it realizes the constant $\ulcorner \pi_i \urcorner : I \to D(\mathsf{AnnPol}_i)$, using the nullary collector and the response from 1.

(4) **Ownerwise alternatives.** Compose these two occurrences of owner $i$ and reparametrize their product carrier by the diagonal assigning the same $\pi_i$ to the reporting and record components. This is a permitted ownerwise reparametrization in the generated calculus. It leaves exactly $\mathsf{AnnPol}_i$ as the complete policy carrier.

(5) **Collection of observation outputs.** Finally collect, at each owner's output, its retained observation, private policy record, and public tuple. The full boundaries can be taken as

$$A_{\mathrm{ann}} = \bigotimes_i D(\Omega_i), \qquad B_{\mathrm{ann}} = \bigotimes_j D(\Omega_j \times \mathsf{AnnPol}_j \times \mathsf{Msg}).$$

Every component is a decision generator, nonstrategic data wiring, or an exhibited parameter reindexing, so the process lies in $\mathbf{DecArena}_N(\mathcal{C})$.

(6) **Causal execution.** Evaluating each response and collector gives the displayed record. The reporting components are tensored; the message collector is their explicit delivery barrier. Coherence and scheduling independence follow from Theorem 10.4.

(7) **Observation and knowledge.** The original carrier $W$ remains an ordinary reference index: its tuple of observations supplies the finite input, so no finite realization of $W$ is required.

□

**Construction 11.23** (Policy witnesses over a public message state)**.**

(1) **Form the fibre of policy witnesses.** For $u \in \mathcal{U}$ retain the ordinary dependent family

$$K_u = F^{-1}(u).$$

Its total carrier has the canonical bijection

$$\coprod_{u \in \mathcal{U}} K_u \cong \mathcal{X}, \qquad (u, x) \longmapsto x,$$

whose inverse is $x \mapsto (F(x), x)$.

(2) **Realize a witness.** In the finite realization, let $g_{\mathrm{ann}} : \mathsf{AnnPol} \to \mathcal{C}(A_{\mathrm{ann}}, B_{\mathrm{ann}})$ be the process just constructed. Associate to $x = (w, \pi) \in K_u$ the realized arrow $g_{\mathrm{ann}}(\pi)$. This is a witness realization of Construction 8.1; its dependent linear transpose exists by the ordinary–linear adjunction.

(3) **Recover the policy record.** The family retains every implementing policy and its private records, while $F$ gives the exact logical comparison on retained message tests. Reconstructing its total carrier recovers all of $\mathcal{X}$ and the attached realizations.

## 11.6 Coalition ability through complete communication policies

**Theorem 11.24** (Complete-policy announcement ability)**.** *For every goal $\Psi \hookrightarrow \mathcal{U}$ and coalition $C \subseteq N$,*

$$\mathsf{AnnAbility}_C(\Psi)(w) \iff \exists \pi_C \in \prod_{i \in C} \mathsf{AnnPol}_i \; \forall \pi_{-C} \in \prod_{j \notin C} \mathsf{AnnPol}_j \; \Psi(F(w, (\pi_C, \pi_{-C}))).$$

*This is the paper's pointwise effectivity predicate, expressed through the complete policy interfaces of the announcing agents.*

*Proof.*

(1) **Ownerwise alternatives.** If $B_C$ is a winning legal coalition message at $w$, choose the explicit policies $E_{i,B_i}$ for its members. Every outsider policy evaluates to a legal outsider message, so every resulting joint message satisfies $\Psi$.

(2) **Reverse implication.** Conversely, let $\pi_C$ satisfy the right-hand side and let $B_C = (\pi_i(\bar{o}_i(w)))_{i \in C}$. Given any legal outsider tuple $B_{-C}$, extend it by $\pi_j = E_{j,B_j}$ for $j \notin C$. The resulting policy tuple evaluates to $(B_C, B_{-C})$, so that legal joint message satisfies $\Psi$.

(3) **Monoidal comparison.** This is exactly the fibrewise existential–universal quantifier defining $\mathsf{AnnAbility}_C$. The formulas also cover empty products.

□

**Corollary 11.25** (Uniform disclosure by complete agent policies)**.** *For $P \hookrightarrow W$, put $\Psi_P = \mathsf{CK}_N^{\mathcal{U}}(P^+)$ as in Theorem 11.13. For every nonempty coalition $C$, define the fixed complete policies*

$$\eta_i(o) = \{o\} \quad (i \in C).$$

*Then, at every $w$,*

$$\begin{aligned} \mathsf{DK}_C(P)(w) &\iff \forall \pi_{-C} \; \Psi_P(F(w, (\eta_C, \pi_{-C}))) \\ &\iff \exists \pi_C \; \forall \pi_{-C} \; \Psi_P(F(w, (\pi_C, \pi_{-C}))). \end{aligned}$$

*The same $\eta_C$ is used at every world. For $C = \varnothing$, complete-policy announcement ability for this goal is $\mathsf{CK}_N^M(P)$.*

*Proof.*

(1) **Observation and knowledge.** If $R_C[w] \subseteq P$, singleton coalition messages give $w \in A_B \subseteq R_C[w] \subseteq P$ against every truthful outsider output. Hence the fixed fact holds throughout the updated carrier and is common knowledge.

(2) **Comparison of existential witnesses.** This proves the first implication with the exhibited complete policies. Those policies are a particular existential witness, giving the second implication.

(3) **Reverse implication.** Conversely, take any successful complete coalition policies and give each outsider the constant tautological policy $o \mapsto \Omega_j$. At $w$ their joint output $B$ satisfies $R_C[w] \subseteq A_B$ by truthfulness of the coalition messages. For $i \in C$, every $v \in R_C[w]$ is $R_i$-accessible from $w$ inside that updated carrier.

(4) **Observation and knowledge.** Common knowledge of $P^+$ gives $v \in P$, so $R_C[w] \subseteq P$. This proves the converse.

(5) **Observation and knowledge: second comparison.** For the empty coalition, tautological outsider policies leave the original model unchanged and establish necessity of original common knowledge.

(6) **Reverse implication.** Conversely every path in a public restriction containing $w$ is an original path, so a fixed fact that was common knowledge remains common knowledge. If $N$ is empty, the common-knowledge relation is equality and the same statement reduces to $P$.

□

The general ability predicate in Theorem 11.24 is evaluated pointwise at $w$. For the factual common-knowledge goal, the corollary constructs a stronger witness: one coalition policy independent of $w$ that succeeds at every world satisfying distributed knowledge. The distinction is between a response chosen at the current reference and a single complete policy valid throughout the specified information class.

## 11.7 Messages change observations

To turn a delivered message into epistemic information, the state carrier must retain enough evidence to identify what was sent. Let $X$ contain the reference world, the complete policy profile, and any execution witness determining $m : X \to M$. A recipient $j$ keeps its previous observation and receives the message through the updated observation map below.

$$o_j^+(x) = (o_j(x), m(x)), \qquad R_j^+ = R_j \cap \ker(m).$$

For deterministic private delivery, each recipient's event observation $b_i$ records the delivered value, whereas a nonrecipient's $b_i$ is constant on event labels. Outsiders therefore keep their prior relations. Public delivery makes the event relation equality for every recipient. These are instances of (9.15)–(9.16). In finite sessions, the retained transcript of Theorem 10.23 determines the message $m$, while its recipient projection specifies exactly which part of the transcript enters a particular agent's observation.

An endpoint type specifies which messages a recipient may receive next. The enabling predicate tests whether the proposition carried by a message holds at the source world. A correctly ordered exchange therefore supplies evidence of reception. The additional predicate is what licenses the inference from reception to truthful disclosure.

## 11.8 Pooling and collective knowledge

Let $q_C : W \twoheadrightarrow O_C$ be the effective joint signal of a nonempty coalition. In the finite protocol of Lemma 11.26, a public delivery uses $J = N$ and a private delivery to the coalition uses $J = C$. At the *completed execution boundary*, the state carrier is the graph of the deterministic completed transcript, canonically isomorphic to $W$. A designated recipient's new observation is the pair of its earlier observation and the actually received tuple. Thus for public delivery $R_i^+ = R_i \cap \ker(q_C)$ for every owner; private pooling makes that same change

for members of $C$ and leaves outsiders' individual relations unchanged. These statements concern the completed state carrier, not an intermediate configuration where some recipients have not yet received the value.

**Lemma 11.26** (Physical completed-delivery boundary)**.** *Let $C \neq \varnothing$ be finite with finite effective observation carriers $\Omega_i$, and let $J$ be the designated recipient set. Put $T_C = \prod_{i \in C} \Omega_i$. The previously constructed true-concurrent data strategies give a receptive courteous arrow*

$$\bigotimes_{i \in C} D(\Omega_i) \longrightarrow D(T_C)$$

*with a final visible tuple record. In its* secured interaction witness before hiding*, the final record causally follows each recipient's reception of the tuple, while distinct recipients' reception events remain unordered. After hiding, the external input events remain causal predecessors of the final output. At supplied observations the final value is $\widetilde{q}_C(w) = (o_i(w))_{i \in C}$; its effective observation quotient is the image $W \twoheadrightarrow O_C \hookrightarrow T_C$.*

*Proof.*

(1) **Observation and knowledge.** First use the finite collector $c_C$ to form the tuple from the independent observation inputs. Apply the designated data comonoid to supply one copy to each recipient and retain one for the final output. The recipient $j$ is connected to the receptive response strategy $\widehat{!} : D(T_C) \to D(1)$ for the unique function $! : T_C \to 1$.

(2) **Observation and knowledge: second comparison.** Its negative tuple input is the recipient's actual reception event; its positive acknowledgement follows that input. Collect the retained tuple and all the acknowledgements with

$$c_{T_C,(1)_{j \in J}} : D(T_C) \otimes \bigotimes_{j \in J} D(1) \longrightarrow D(T_C \times 1^J),$$

and then apply the response for the projection $T_C \times 1^J \to T_C$. The empty recipient case uses the nullary product $1^{\varnothing} = 1$.

(3) **Observation and knowledge.** To verify the actual causal claim, inspect the secured interaction *before hiding*. Write $e_i$ for original input occurrences, $r_j$ for each matched recipient reception, $a_j$ for its acknowledgement, and $f$ for the final record.

(4) **Observation and knowledge: second comparison.** The collector and copying dependencies followed by acknowledgement and final collection supply, for every $i \in C$ and $j \in J$,

$$e_i < r_j < a_j < f.$$

There is no path from $r_j$ to $r_k$ or conversely for distinct recipients: the copies share predecessors, but their only later common dependency is the final collector output. The causal graph progresses from original inputs through collection, copying, acknowledgements, and the final output.

(5) **Secured operational interaction.** It contains no directed cycle; every matching used is therefore secured. The components are receptive courteous, and their secured composite is receptive courteous by the closure theorem of Section 1.

(6) **Observation and knowledge.** Upon hiding the internal copy, reception, and acknowledgement events, the induced visible causal order still contains $e_i < f$ for every $i$, but the internal receptions themselves are recorded by the completed secured witness, not by events in the hidden composite. Data copying, collectors, and functoriality of the finite response maps ensure that the visible label is exactly the input tuple.

(7) **Quotient for pooled observation.** Image factorization of the tuple map on worlds gives $O_C$, without selecting a retraction of its inclusion into $T_C$.

□

**Theorem 11.27** (Communication realizes pooled information)**.** *For a fixed factual predicate $P \subseteq W$, public inspection satisfies*

$$\mathsf{CK}_N^+ P = \mathsf{DK}_C P.$$

*Private pooling satisfies $\mathsf{CK}_C^+ P = \mathsf{DK}_C P$ and leaves each outsider's individual relation unchanged.*

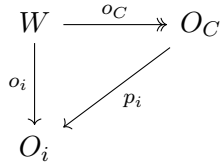
*Proof.*

(1) **Ownerwise alternatives.** Write $R_C = \ker(q_C) = \bigcap_{i \in C} R_i$. For $i \in C$ the updated relation is exactly $R_C$, and for every other owner under public inspection it is contained in $R_C$. Because $C$ is nonempty, the union of all updated relations is $R_C$.

(2) **Ownerwise alternatives: completion.** This is already an equivalence relation, so its finite path closure is $R_C$. Taking boxes gives the first equality. Under private pooling the members again all have relation $R_C$, while outsiders have $R_i$ by their universal event relation. The same path calculation within $C$ proves the second equality.

□

Follow either delivery path at completion. Each designated recipient can recover the same joint signal from its own observation.




Lemma 11.26 supplies the physical finite implementation and its completion witness. The final collector is a barrier for completed delivery, while the independent recipient events remain unordered. The member policies can then use the actual delivered tuple. Owner regrouping separately exposes their combined control; composition coherence for the policy parameters and data wiring makes the two operations compatible. At intermediate asynchronous histories, knowledge must be evaluated on the corresponding history-indexed observations determined by the actual communication history.

## 11.9 Complete truthful communication policies in the agent calculus

The complete response type and the policy-to-message comparison of Constructions 11.15 and 11.17 have the following ordinary-family expressions in the agent calculus:

$$\mathsf{AnnPol}_i = \prod_{z \in \Omega_i} \{B \subseteq \Omega_i : z \in B\},$$
$$X = W \times \prod_i \mathsf{AnnPol}_i,$$
$$U = \{(B, w) : \forall i \ \bar{o}_i(w) \in B_i\}.$$

Evaluation is $F(w, \pi) = (b(w, \pi), w)$. The post-communication observations are $o_j^X(w, \pi) = (\bar{o}_j(w), \pi_j, b(w, \pi))$ and $o_j^U(B, w) = (\bar{o}_j(w), B)$. Each sender retains its own complete policy as a private record, and deviations keep the world and outsider policy coordinates fixed.

For enumerated finite observations the message power set has the ordinary finite function type $2^{\Omega_i}$, with $2 = 1 + 1$. The complete truthful policy carrier is represented by

$$\prod_{z:\Omega_i} \sum_{b:2^{\Omega_i}} \mathrm{Id}_2(b(z), 1).$$

Finite enumeration, equality, and membership give the displayed policy semantics. For infinite sets, $\mathcal{P}(\Omega_i)$ denotes the power set in the set-based semantics.

By Lemma 11.16, the term $E_{i,B}(x) = B \cup \{x\}$ is truthful at every observation and satisfies $E_{i,B}(z) = B$ whenever $z \in B$. It supplies the section $(B, w) \mapsto (w, (E_{i,B_i})_i)$ of $F$. The simultaneous epistemic back-lifts and the outsider-preserving strategic back-lifts of Theorem 11.18 apply to these same families. Thus inverse image along $F$ commutes with the boxes and diamonds of the designated relations, their epistemic intersections, distributed knowledge, and common knowledge, as proved in Lemma 11.19 and Corollary 11.20. The lifts keep the reference complete policy fixed.

For comparisons on $U$ pulled back along $F \times F$, Corollary 11.20 gives

$$\mathsf{NE}_X = F^* \mathsf{NE}_U,$$

together with the analogous equalities for the designated coalitional stability tests. Public restrictions by predicates on $U$ and action updates pulled back from $U$ retain this exact comparison. Assertions about private policy registers remain available on $X$; the comparison applies to the predicates retained on $U$.

## 11.10 Concurrent implementation and coalition announcement ability

For finite $\Omega_i$, the implementation of Proposition 11.22 realizes the internal families above by decision generators, policy reparametrization, collectors, and data copying. At $(w, \pi)$ the delivered observation to owner $j$ is $(\bar{o}_j(w), \pi_j, b(w, \pi))$. The reporting decisions are concurrent, and the joint public delivery follows their collection. The possibly infinite reference carrier $W$ remains an ordinary index; the physical interfaces encode the displayed finite observations.

For a goal $\Psi \subseteq U$, coalition announcement ability at $w$ is the existence of a truthful coalition message tuple that achieves $\Psi$ against every truthful outsider tuple. The extension lemma gives exactly

$$\mathsf{AnnCan}_C(\Psi)(w) \iff \exists \pi_C \, \forall \pi_{-C} \, \Psi(F(w, (\pi_C, \pi_{-C}))). \tag{11.3}$$

Indeed, extend any winning coalition message explicitly to policies; every outsider policy evaluates to a legal outsider message. Conversely extend each possible outsider message while retaining the proposed coalition policies. These two constructions prove both implications.

**Theorem 11.28** (Uniform disclosure of distributed knowledge)**.** *For nonempty $C$ and a fixed factual predicate $P \subseteq W$, let $P^+(B, w) = P(w)$ and $\eta_i(z) = \{z\}$. Then*

$$\begin{aligned} \mathsf{DK}_C P(w) &\iff \forall \pi_{-C} \, \mathsf{CK}_N^U(P^+)(F(w, (\eta_C, \pi_{-C}))) \\ &\iff \exists \pi_C \, \forall \pi_{-C} \, \mathsf{CK}_N^U(P^+)(F(w, (\pi_C, \pi_{-C}))). \end{aligned}$$

*The same complete disclosure policies $\eta_C$ work at every world.*

*Proof.*

(1) **Ownerwise alternatives.** The policies $\eta_i(z) = \{z\}$ are complete and truthful. At $w$ their messages cut the compatible component down to a subset of $R_C[w] = \bigcap_{i\in C} \ker(o_i)[w]$, proving the forward direction from $\mathsf{DK}_C P(w)$ independently of outsider policies.

(2) **Reverse implication.** Conversely, let outsiders use the complete tautological policies $z \mapsto \Omega_j$. Every $v \in R_C[w]$ survives the coalition's truthful messages and is reachable from $w$ by an individual $i \in C$ epistemic step. Common knowledge implies $P(v)$, hence $\mathsf{DK}_C P(w)$.

(3) **Policy-level comparison.** This is the complete-policy realization of the exact disclosure result Corollary 11.25, with evaluation justified by Theorem 11.18. The nonempty condition on $C$ is used only for the final individual-step argument.

□

Policy evaluation retains the underlying world while updating public messages.

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{F} & \mathcal{U} \\ {\scriptstyle \pi_W}\downarrow & & \downarrow{\scriptstyle \pi_W} \\ W & = & W \end{array}$$

For measured communication, Theorem 11.18 applies to the same designated epistemic and strategic relations. If a profile comparison is pulled back from the public message carrier, each forward image and back-lift has identical truth value for that test. Hence the stability equivalence applies, for example to a fee computed from the public message. A fee computed from a private policy register may vary within an evaluation fibre. Test the value (or the paired comparison) for descent as in Proposition 5.43. For a fee that distinguishes such points, retain the relevant record and use the exact refinement of Theorem 12.37 with the specified relations before transporting its tests.

## 11.11 Choices, disclosure, and the price of authorization

Return to the two private bits $x, y$ and the complete policy carriers $\Sigma_i = 2^2$ of the introduction. A policy profile specifies the functions $\pi_1, \pi_2$ independently of the current world, and at $w = (x, y)$ its response realization produces the action tuple

$$(a, b) = (\pi_1(x), \pi_2(y)).$$

The legal coupling $\pi_1(1) \le \pi_2(1)$ has 12 profiles: for the entries at observation 1 it allows $(0,0), (0,1), (1,1)$, while both observation-0 responses vary freely. At the world $w = (0,0)$, let $\pi_1^0$ and $\pi_2^0$ be the constant-zero policies. The reference $(\pi_1^0, \pi_2^0)$ is legal and yields $u_1 = u_2 = 0$. Holding either outsider policy fixed at zero, a change of the current action from 0 to 1 gives the deviator payoff $-1$; changes only at the off-path observation 1 give payoff 0. Thus there is no strictly profitable unilateral legal deviation at this reference. By contrast, the joint deviation to policies $\pi_i^*(0) = 1$, $\pi_i^*(1) = 0$ is legal and yields the action tuple $(1, 1)$, with both utilities equal to 1. This calculates the unilateral and grand-coalition deviation fibres using the same owner-indexed complete-policy space. It also illustrates why changing two decisions under one regrouped owner requires a whole-block deviation test.

Initially each owner sees only its own bit, while the joint image distinguishes the pair. A collector receives both signals and delivers the tuple to the two owners. Theorem 11.27 identifies facts about $(x, y)$ that thereby become common knowledge. For example, put

$P = \{(0,0),(1,1)\} \hookrightarrow W$, the assertion $x = y$. Before delivery, $\ker(o_1) \cap \ker(o_2)$ is equality on $W$, so $\mathsf{DK}_{\{1,2\}}P(0,0)$ holds; but the union of the two observation relations connects all four worlds, so $\mathsf{CK}(P)(0,0)$ fails. After both owners actually receive $(x,y)$, each updated observation is injective and $\mathsf{CK}^+(P) = P$. In particular, $\mathsf{CK}^+(P)(0,0)$ holds. This equality is asserted only at the completed-delivery boundary, not while either copy of the public tuple remains in transit. The recipients' new response policies may depend on the delivered pair; the exact enlargement is given by Propositions 7.41 and 7.42.

To measure one specified link of this collector, give weight 1 to the synchronization of its tuple output with the chosen recipient's input, and weight 0 to every other class. On a completed delivery witness the selected send and its matched receive form one class, so $m(k) = 1$ in $\mathbb{N}$. A budget $m \leq 1$ admits that witness and $m \leq 0$ excludes it. After hiding the link, its internal occurrences no longer appear at the exterior boundary; the retained matching still exhibits the charge. With the cost-insensitive comparison above the action payoffs remain 0 and 1 at the two displayed profiles. A priced evaluation can instead use $v_i = u_i - \eta(\lambda_i m_i)$ on the measured records, with natural counts embedded in $R_+$. On alternatives that all incur this fixed fee, its deduction is the same on both sides of each comparison. A contingent fee gives a different equilibrium calculation.

**Example 11.29** (The price of institutional authorization)**.**

(1) **Complete both branches.** At $w = (0,0)$ use the legal profile domain $L$. Retain the completed accept/reject bit of the finite checker in Section 11.13. In the accepting branch forward the authorization and acknowledge its receipt; in the rejecting branch return a completed rejection record. The result interface thus records a completed decision also on rejection. Give weight one to the designated matched authorization and zero to the remaining classes. With current actions $a, b$ the measured record is $(ab,(a,b))$, since the accepting branch occurs precisely at $(1,1)$.

(2) **Evaluate and compare.** At common nonnegative rational price $\lambda$, the two owner evaluations of this record are $(2ab - a - \lambda ab,\ 2ab - b - \lambda ab)$. They give

| $(a,b)$ | $m$ | $(\bar{v}_1, \bar{v}_2)$ |
|---|---|---|
| $(0,0)$ | 0 | $(0,0)$ |
| $(1,0)$ | 0 | $(-1,0)$ |
| $(0,1)$ | 0 | $(0,-1)$ |
| $(1,1)$ | 1 | $(1-\lambda, 1-\lambda)$. |

(3) **Count complete legal equilibria.** At observation zero each owner can change its present action while keeping its entry at one fixed. At $(0,0)$ a unilateral contribution loses one; at either off-diagonal profile the contributor gains by withdrawing. At $(1,1)$ withdrawal yields zero, so $(1,1)$ is Nash exactly for $\lambda \leq 1$. For each current action pair the off-path entries range over the three pairs $(0,0),(0,1),(1,1)$ allowed by $L$. There are six legal Nash profiles for $\lambda \leq 1$ and three for $\lambda > 1$. Moving jointly from $(0,0)$ to $(1,1)$ strictly improves both payoffs precisely for $\lambda < 1$. At equality the distinction is a tie in each owner's writer-evaluated comparison.

Regroup the owners to form one control interface while retaining their individual comparisons; a chosen collective comparison then evaluates joint decisions. Let the institution record their messages and commitment. Its execution witness retains the policy profile and transcript, its target map records the commitment, and its permitted subobject enforces the rule. The token-guarded executor of Lemma 10.8 receives an actual issuer-produced authorization event. Final execution depends causally on that event; the bare type $D(1)$ does

not certify the token's authority. Composition of witnessed spans and of their concurrent realizations now describes the institutional transaction. A public announcement of the commitment is a further information-changing action.

## 11.12 A concrete finite check of linearity and information

The finite-data barrier can be tested on two independent bits. The collector $c : D(2)\otimes D(2) \to D(2 \times 2)$ waits for both; the distributor $d : D(2 \times 2) \to D(2) \otimes D(2)$ sends the completed pair. Although $cd = \mathrm{id}$ on the paired interface, $dc$ is not the identity on two independent channels: each output inherits both input prerequisites. The distinction is operational, not merely one of value types.

Initially the two observation kernel pairs $R_1$ and $R_2$ distinguish the first and second bit, respectively. Their intersection is equality on the four possible worlds, so private pooling makes facts about the entire pair common knowledge once delivery is complete. For a material resource $R$, this logical change does not supply a linear contraction $R \to R \otimes R$. The example thereby separates information sharing from resource duplication.

## 11.13 Which evidence enables the authorization gate?

Return to the oversight question from the introduction. The two owners observe $x$ and $y$ privately. The issuer must decide whether their present actions and the relevant entries of their policies permit execution. On the legal domain, let $K$ be the carrier of accepted completed executions

$$K = \{(x, y, \pi_1, \pi_2) \in W \times L : \pi_1(x) = 1 \ \wedge \ \pi_2(y) = 1\}.$$

Its projection to the full institutional record retains the source world and complete owner-policy profile. On the legal input domain $W \times L$, let $r_L(w, \pi) = (\pi_1(x), \pi_2(y))$. Then $K = r_L^{-1}\{(1, 1)\}$. A collector followed by the finite success gate for $(1, 1)$ issues the authorization event $t$ exactly on $K$.

There is also an issuer on the ambient input domain $W \times \Sigma$. For each owner $i$, pair its current response component with the nullary response whose output is $\pi_i(1)$. Reparametrize ownerwise by $\pi_i \mapsto (\pi_i, (* \mapsto \pi_i(1)))$, as in the policy-record construction of Proposition 11.22. Route the two current actions and the two additional bits to the issuer. The resulting ordinary value map and acceptance set are

$$d(w, \pi) = (a, b, c_1, c_2) = (\pi_1(x), \pi_2(y), \pi_1(1), \pi_2(1)),$$
$$B = \{(a, b, c_1, c_2) \in 2^4 : a = b = 1, \ c_1 \le c_2\}.$$

The collector, the finite test for $B$, and the success gate form a receptive courteous issuer. The definitions give $d^{-1}(B) = K$ by the rules for $L$ and $r_L$, with $W \times L$ included in $W \times \Sigma$. Only the checker receives the two extra bits. To see what the extra entries establish, take $(x, y) = (0, 0)$ and write each policy in observation order. The profiles $((1, 0), (1, 0))$ and $((1, 1), (1, 0))$ both produce $(1, 1)$, yet only the first belongs to $L$. Consequently a checker restricted to the current action tuple cannot recognize $K$ throughout the ambient domain.

The executor supplies a second check on the design. Its final event $e$ requires receipt of the issuer's token, giving $t < e$ in the completed witness. The legal joint policy $(\pi_1^*, \pi_2^*)$ at $(0, 0)$ belongs to $K$ and exhibits such an execution. The type $D(1)$ specifies the token interface. To establish that the issuer actually produced $t$ for this execution, follow the causal predecessors in the witness. The checker thus has evidence for its decision, and the executor has evidence that the decision reached it.

**From completed communication to continuing agency.** A completed exchange now supports two later decisions. Recipients may base a new response on the joint signal they received, and the executor may use its token for the guarded action. To analyze a subsequent round we must know which signals and permissions persist and which actions remain enabled. The residual strategy and the remaining policy supply the two parts of that continuation.

# Part V
# Continuing agency and exactness

## 12 Continuing Agents and Temporal Agency

After one round, what must a controller remember in order to act in the next? Its residual policy specifies a response to each future observation. Its residual strategy records which future interactions the executed history still allows. We use the former to identify the least number of reachable states needed for a finite-state implementation. We use the latter to test whether a visible description preserves the available moves. Temporal fixed points express what a coalition can enforce across rounds. In finite public trees, backward induction also carries the accumulated charge into each continuation.

### 12.1 Fixed points in assertion fibres and their transport

The assertion fibres are complete lattices, so the fixed-point construction of Tarski (1955) applies to monotone modal operators. We write $\mu$ for the least and $\nu$ for the greatest fixed predicate. These operations express distinct temporal tasks: least fixed points capture progress towards a goal, while greatest fixed points capture its persistence. Their transport must respect the witnessed successor relations on which the operators depend.

**Construction 12.1** (Predicate fixed points)**.**

(1) **Form the two fixed points.** For a monotone $F : \mathsf{Pred}(X) \to \mathsf{Pred}(X)$, put

$$\mu F = \bigwedge\{P : F(P) \leq P\}, \qquad \nu F = \bigvee\{P : P \leq F(P)\}.$$

(2) **Compute joins and meets.** The set-indexed joins and meets are pointwise existential and universal quantification in the chosen predicate doctrine.

**Theorem 12.2** (Predicate induction and coinduction; recalled)**.** *$\mu F$ and $\nu F$ are respectively the least and greatest fixed points. They satisfy $F(P) \leq P \Rightarrow \mu F \leq P$ and $P \leq F(P) \Rightarrow P \leq \nu F$. The same holds on finite products of predicate lattices.*

*Reference.*

(1) **Fixed-point argument.** Apply the complete-lattice fixed-point theorem of Tarski (1955). The indicated products and pointwise predicate lattices are complete.

□

**Definition 12.3** (Positive modal fixed-point language)**.** Fix a state set $X$, atomic predicates $A_p \hookrightarrow X$, relations $R_a \hookrightarrow X^2$, and predicate variables $Z$ drawn from a set $\mathsf{Var}$. Formulas have the grammar

$$\varphi ::= p \mid Z \mid \top \mid \bot \mid (\varphi \wedge \varphi) \mid (\varphi \vee \varphi) \mid (\varphi \Rightarrow \varphi) \mid [R_a]\varphi \mid \langle R_a\rangle\varphi \mid \mu Z.\varphi \mid \nu Z.\varphi.$$

An environment $\eta : \mathsf{Var} \to \mathsf{Pred}(X)$ gives the interpretation $[\![\varphi]\!]_{X,\eta} \in \mathsf{Pred}(X)$ by interpreting atoms as $A_p$, variables as $\eta(Z)$, and connectives and modalities by the already-defined operations. A bound variable must occur **positively** in its binder: antecedents of implication reverse polarity, consequents preserve it, and the other constructors preserve polarity. Hence $F_{\varphi,\eta}(U) = [\![\varphi]\!]_{X,\eta[Z\mapsto U]}$ is monotone, and the binder clauses are

$$[\![\mu Z.\varphi]\!]_{X,\eta} = \mu F_{\varphi,\eta}, \qquad [\![\nu Z.\varphi]\!]_{X,\eta} = \nu F_{\varphi,\eta}.$$

The two fixed points are those of Construction 12.1. This sign calculation is for the fixed relations in the displayed grammar. The assertion language with formula-valued public announcements uses the guarded expansion and formation derivation of Lemma A.5.

**Lemma 12.4** (Transport across a retraction)**.** *Let $\rho : L \to M$ and $J : M \to L$ be monotone maps between complete lattices with $\rho J = \mathrm{id}_M$. For monotone $F : M \to M$, put $\widehat{F} = JF\rho$. Then*

$$\mu\widehat{F} = J\mu F, \qquad \nu\widehat{F} = J\nu F.$$

*Proof.*

(1) **Fixed-point argument.** If $Fx = x$, then $\widehat{F}(Jx) = Jx$. If $\widehat{F}y = y$, then $F(\rho y) = \rho y$ and $J\rho y = y$. Thus $J$ and $\rho$ restrict to inverse order isomorphisms between the two fixed-point posets. They preserve their least and greatest elements.

□

## 12.2 Residual strategies and continuing agents

To interpret action after a history, we must retain both the strategy's remaining event structure and the owner's contingent responses at later decision positions. Pointed residuals provide the operational data. Restriction of a complete policy gives the corresponding future response family, while epistemic action models record information from actually completed rounds. The resulting continuation interface connects temporal objectives to the same control and observation data already used for one-stage agency.

Each completed public round can be read as an enabled epistemic action event from Section 9.1.3. Its state map records the concurrent transition, while its observation records the information actually delivered at completion. Intermediate histories remain represented by operational configurations. Thus successive public updates do not impose a total schedule on events that were independent within a round.

A continuing state can carry commitments or goals as ordinary records. Their persistence, discharge, and revision require explicit update maps; current preference selection alone does not determine such transitions. This is the point of contact with intention accounts such as Cohen and Levesque (1990). Here continued agency is established for the displayed record and residual operations rather than by assigning an intention law to every strategy.

### 12.2.1 Complete policies and their residual controllers

Two observation histories can share a controller state if they prescribe the same response to every possible future observation word. Thus the memory needed to implement a complete local policy is measured by the number of distinct remaining response functions. The coalgebraic construction, in the sense of Rutten (2000), makes this count exact. After each input the controller emits the prescribed action and moves to the state representing the policy that remains.

**Definition 12.5** (Local response controller)**.** For a sequential local observation interface $O$ and action interface $M$, a controller is a coalgebra

$$\delta : S \longrightarrow (M \times S)^O.$$

Write $\delta(s)(o) = (a(s,o), n(s,o))$. A controller map $f : S \to S'$ preserves both components: $a'(f(s),o) = a(s,o)$ and $n'(f(s),o) = f(n(s,o))$. A pointed controller also specifies its initial state $s_0$.

Let $O^+$ and $O^*$ denote nonempty and arbitrary finite words, respectively. A complete local policy is an element $b \in \mathcal{B} = M^{O^+}$: its value at a word is the action after the final observation of that word. Set

$$\zeta : \mathcal{B} \to (M \times \mathcal{B})^O, \qquad \zeta(b)(o) = (b(o), b_o), \qquad b_o(v) = b(ov).$$

**Theorem 12.6** (Final behaviour and the generated residual controller)**.** *The coalgebra $(\mathcal{B}, \zeta)$ is final. For a complete policy $b$, put*

$$b_u(v) = b(uv) \quad (u \in O^*,\ v \in O^+), \qquad S_b = \{b_u : u \in O^*\} \subseteq \mathcal{B}.$$

*It is a pointed subcoalgebra with initial state $b = b_\varnothing$. Every pointed controller implementing $b$ has a unique surjective pointed controller map from its reachable states onto $S_b$. Consequently $b$ has a finite-state implementation if and only if $S_b$ is finite. In that case $|S_b|$ is the least possible number of reachable states of an implementation.*

A map of controllers must preserve two things at each observed input. Reading either way around the square gives the same action and corresponding successor state.

$$\begin{array}{ccc} S & \xrightarrow{\ \zeta\ } & F(S) \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle Ff} \\ T & \xrightarrow[\ \theta\ ]{} & F(T) \end{array}$$

*Proof.*

(1) **Fixed-point argument.** For a controller $\delta$, define its behaviour recursively by

$$\mathrm{beh}_\delta(s)(o) = a(s,o), \qquad \mathrm{beh}_\delta(s)(ov) = \mathrm{beh}_\delta(n(s,o))(v) \quad (v \in O^+).$$

Induction on word length defines a unique function $S \to M^{O^+}$. The two equations assert exactly that this function is a coalgebra map into $(\mathcal{B}, \zeta)$.

(2) **Reverse implication.** Conversely, every such map must satisfy those equations; induction proves uniqueness. This proves finality, including empty carriers whenever the displayed coalgebra exists.

(3) **Closure of controller derivatives.** For a derivative $b_u$, its output on $o$ is $b(uo)$ and its next behaviour is $b_{uo}$, which lies in $S_b$. Hence restriction of $\zeta$ is well defined. Its behaviour map is its inclusion in $\mathcal{B}$, by the same recursion.

(4) **Construction of the comparison.** Let $\delta$ with initial state $s_0$ implement $b$, so that $\mathrm{beh}_\delta(s_0) = b$. Define $s_\varnothing = s_0$ and $s_{uo} = n(s_u, o)$. The reachable carrier is $\{s_u : u \in O^*\}$.

(5) **Lifting along the surjection.** Induction gives $\mathrm{beh}_\delta(s_u) = b_u$. The restricted behaviour map therefore lands in $S_b$, is surjective, and preserves initial state, outputs, and transitions. A pointed controller map must send $s_u$ to $b_u$ by induction on $u$, so it is unique, even when several words reach the same state.

(6) **Verification of the universal property.** Surjectivity implies that a finite reachable implementation has finite $S_b$ and at least its cardinality. Conversely, finite $S_b$ is itself a pointed implementation with that many states.

□

The residual controller remembers exactly the distinctions needed for future local input–output behaviour. A permission, a preference, or a private observation may distinguish states with identical future responses. To use the controller quotient for one of those purposes, the corresponding test must also agree on the merged states.

**Proposition 12.7** (Concurrent realization of a finite controller step)**.** *For finite $S, O, M$, the uncurried response function $\delta : S \times O \to M \times S$ has the one-step realization*

$$D(S) \otimes D(O) \xrightarrow{\mathrm{col}} D(S \times O) \xrightarrow{\widehat{\delta}} D(M \times S) \xrightarrow{\mathrm{dist}} D(M) \otimes D(S).$$

*Connecting the returned state to the next local step realizes, for every finite observation word, the output sequence of the pointed controller. If the controller implements $b$, its final output on a nonempty word $u$ is $b(u)$.*

*Proof.*

(1) **Fixed-point argument.** The finite-data collector and distributor and Proposition 1.16 realize the displayed ordinary function on value inputs. Thus the value pair $(s, o)$ yields exactly $(a(s, o), n(s, o))$. Induction on the number of connected steps shows that the state after $u$ is $s_u$ and that the output at its last step is $\mathrm{beh}_\delta(s_0)(u)$.

(2) **Observation and knowledge.** The implementation equation makes this $b(u)$. Each finite composite is an arrow of the existing concurrent semantics. The collector waits for the local state and observation; its synchronization is part of that arrow and is not an isomorphism identifying $D(S \times O)$ with $D(S) \otimes D(O)$.

□

The words used in the controller are sequential observations at one specified local interface. Concurrent events belonging to other interfaces retain their own partial orders, so the implementation does not choose a global interleaving. The constructed realization covers each finite connected segment, not an unspecified recursive process beyond the declared local protocol.

Petri games provide another setting for causally informed strategies of distributed processes (Finkbeiner and Olderog, 2017). The comparison here concerns implementation: which complete local policies have controllers respecting the observation available at each step? The residual coalgebra answers that question without conflating global concurrency with the chosen local sequence of observations.

### 12.2.2 Maintaining and eventually achieving collective goals

**Definition 12.8** (Public states, histories, and winning regions)**.** For the finite public-state interaction specified below, let $W$ be the set of publicly observed round states, $A_i(w)$ the finite nonempty set of agent $i$'s available actions at $w$, and $\delta_w : \prod_i A_i(w) \to W$ its deterministic transition. The **history carrier** is the set of finite legal histories

$$\mathsf{Hist} = \coprod_{n \geq 0} \{(w_0, a^0, w_1, \ldots, a^{n-1}, w_n) : \delta_{w_k}(a^k) = w_{k+1} \ (k < n)\},$$

with last : $\mathsf{Hist} \to W$. A **history-dependent pure policy** of owner $i$ is a section of the pulled-back action bundle, equivalently a family $\pi_i(h) \in A_i(\mathrm{last}(h))$ for every $h \in \mathsf{Hist}$. Let $\mathsf{Play}$ be the carrier of infinite legal transition-consistent action/state sequences. For initial $w$ and a complete policy profile $\pi$, deterministic recursion gives $\mathrm{play}(w, \pi) \in \mathsf{Play}$. For a **temporal objective** $\Phi \hookrightarrow \mathsf{Play}$, the **winning region** of coalition $C$ is

$$\mathrm{Win}_C(\Phi) = \{w \in W : \exists \pi_C \ \forall \pi_{-C},\ \mathrm{play}(w, \pi_C, \pi_{-C}) \in \Phi\} \hookrightarrow W.$$

All agents observe the completed public history, not their opponents' current simultaneous choices. The predecessor construction below derives this ability using dependent action bundles.

Temporal ability requires more than one-stage effectivity: a coalition must choose at successive public histories while outsiders may respond at each stage. The predecessor operator uses dependent quantifiers in the existing predicate doctrine. The ordinary classical **Set** instance supplies the duality between failed coalition ability and a counterpolicy, expressed by $\neg\exists a_C \, \forall a_{-C} P \Rightarrow \forall a_C \, \exists a_{-C} \, \neg P$. The fixed-point construction below retains the quantifier order when moving from one round to repeated play.

**Construction 12.9** (Finite public-state interaction and controllable predecessor)**.**

(1) **Specify the public interaction.** A finite public-state interaction has a finite state set $W$, a finite nonempty action set $A_i(w)$ for each agent at each state, and a transition function

$$\delta_w : \prod_i A_i(w) \to W.$$

Players observe the finite public history before each simultaneous round.

(2) **Form complete history policies.** A coalition policy assigns its action tuple to each such history. Opponents have policies of the same kind. This is an economic policy space in the preceding sense; one round is a causal decision protocol with concurrent action inputs and a deterministic transition rule. Finite prefixes are obtained by composition of these rounds, retaining the current state as the interface.

(3) **Display the state projections.** To expose the categorical structure when action sets depend on the current state, form the action bundles

$$E_C = \{(w, a_C) : a_C \in \prod_{i \in C} A_i(w)\},$$
$$E = \{(w, a_C, a_{-C}) : a_i \in A_i(w) \text{ for all } i\}.$$

Let

$$E \xrightarrow{\pi_C} E_C \xrightarrow{\pi_W} W$$

forget first the outsider actions and then the coalition actions, and let $\delta : E \to W$ be the transition map.

(4) **Construct the controllable predecessor.** For a state predicate $X$, define

$$\mathrm{Pre}_C(X) = \exists_{\pi_W} \, \forall_{\pi_C} \, \delta^* X. \tag{12.1}$$

Pointwise this is exactly

$$\mathrm{Pre}_C(X)(w) \quad \Longleftrightarrow \quad \exists a_C \ \forall a_{-C},\ X(\delta_w(a_C, a_{-C})).$$

The operator is monotone. It is the state-indexed version of Equation (7.13) and is constructed entirely from reindexing and the two quantifier adjoints of the assertion doctrine.

**Proposition 12.10** (Epistemic product update of a public concurrent round)**.** *For the finite public-state interaction above, let the action-occurrence set be the disjoint union*

$$E^{\mathrm{round}} = \coprod_{w \in W} \prod_{i} A_i(w),$$

*so an occurrence is a tagged pair $(w, a)$. Give it enabling predicate $P(u, (w, a)) \iff u = w$. Take equality as the prior public-state relation and as the action relation representing the publicly revealed completed tuple. Then its product-update carrier is canonically the original joint-action bundle $E$ used to define Equation (12.1), with the transition postcondition*

$$t(u, (u, a)) = \delta_u(a).$$

*Under this identification the controllable predecessor is $\exists_{\pi_W} \forall_{\pi_C} t^* X$. This action-model presentation uses a completed joint action as one public update, while the underlying operational realization uses the independent concurrent choice events and the collector's synchronization barrier.*

*Proof.*

(1) **Monoidal comparison.** Comprehension consists precisely of the pairs $(u, (u, a))$, so forgetting the redundant first copy of $u$ is a bijection with the original action bundle $E$. Both action and old-state relations are equality, hence their restricted product is equality on the updated carrier. The postcondition is the original transition map.

(2) **Ownerwise alternatives.** The quantifier expression is Equation (12.1) under the carrier bijection. The circuit of Theorem 10.4 realizes each action tuple by independent owner-controlled response protocols, followed by the finite collector and transition response; no causal order between simultaneous decisions is added.

□

**Definition 12.11** (Temporal objectives)**.** Let $\mathsf{Play}$ be the carrier of infinite legal plays $\lambda = (w_0, a^0, w_1, a^1, \ldots)$ of the public-state interaction. For state predicates $P, Q \hookrightarrow W$, define the play predicates

$$\begin{aligned} [\![P \,\mathsf{U}\, Q]\!] &= \{\lambda : \exists n \in \mathbb{N},\ w_n \in Q \ \wedge\ \forall k < n\ (w_k \in P)\}, \\ [\![\mathsf{G}P]\!] &= \{\lambda : \forall n \in \mathbb{N},\ w_n \in P\}. \end{aligned}$$

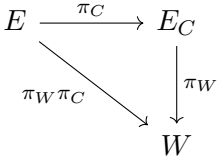
The first requires reaching $Q$ after finitely many rounds while maintaining $P$ beforehand; the second requires $P$ at every state. The coalition-enforcement predicate for either objective is $\mathrm{Win}_C([\![\cdot]\!]) \hookrightarrow W$ of Definition 12.8, quantifying one coalition policy against every outsider policy.

**Theorem 12.12** (Temporal ability and fixed points)**.** *For a finite public-state interaction,*

$$\begin{aligned} \mathrm{Win}_C(P \,\mathsf{U}\, Q) &= \mu X\,(Q \cup (P \cap \mathrm{Pre}_C(X))), \\ \mathrm{Win}_C(\mathsf{G}P) &= \nu X\,(P \cap \mathrm{Pre}_C(X)). \end{aligned}$$

*Every state in either winning set has a policy depending only on the current state. For the until objective, the construction also gives a finite bound on the time to reach $Q$.*

The coalition chooses first in the logical test. A state belongs to the controllable predecessor when one coalition action sends every outsider response into the target set.




*Proof.*

(1) **Fixed-point argument.** For until, iterate $X_0 = \varnothing$ and $X_{n+1} = Q \cup (P \cap \mathrm{Pre}_C(X_n))$. Because $W$ is finite this increasing sequence reaches its least fixed point $L$ after finitely many steps. Give each state in $L$ its least positive index of entry.

(2) **Witness and legality comparison.** States of index 1 lie in $Q$. At every other state take an action tuple witnessing predecessor membership in the preceding stage. There are only finitely many such states, so these finitely many witnesses assemble by finite recursion into a state-dependent coalition policy.

(3) **Ownerwise alternatives.** Against every outsider action it moves to a state of smaller index. Until $Q$ is reached, the current state lies in $P$. The index decreases at each step, so this state-dependent policy reaches $Q$ in fewer steps than the starting index.

(4) **Ownerwise alternatives: second comparison.** To prove the converse, fix any coalition policy at a state outside $L$. Such a state is not in $Q$. If it is outside $P$, the objective has already failed.

(5) **Fibrewise calculation.** At a state in $P \setminus L$, fixed-point equality implies $w \notin \mathrm{Pre}_C(L)$. Expanding the predecessor quantifiers and using the classical finite specialization fixed above gives

$$\forall a_C \; \exists a_{-C}, \qquad \delta_w(a_C, a_{-C}) \notin L.$$

For the finitely many state/action pairs, assemble one such outsider response for each pair by finite recursion.

(6) **Ownerwise alternatives.** Against the fixed coalition policy, this gives an outsider policy on public histories: evaluate that fixed policy at the history and use the selected response. It keeps the play outside $L$ until $P$ is left, if that happens. The play never reaches $Q$; in either case the until objective fails. Thus no state outside $L$ is winning.

(7) **Policy-level comparison.** For safety let $S = \nu X(P \cap \mathrm{Pre}_C(X))$. For the finitely many states in $S$, assemble predecessor witnesses that stay in $S$ into a state-dependent policy by finite recursion. Since $S \subseteq P$, this policy enforces safety.

(8) **Reverse implication.** Conversely let $Y$ be the set of states from which some policy enforces safety. At $w \in Y$, the first coalition action must send every outsider completion to another state of $Y$. Indeed the winning continuation after that first history, with its prefix fixed as a parameter, is a valid policy from the successor state.

(9) **Coinduction and bisimulation.** Also $w \in P$. Hence $Y \subseteq P \cap \mathrm{Pre}_C(Y)$. Coinduction gives $Y \subseteq S$ and proves the converse.

□

The counterpolicy constructed in the until argument responds to a fixed coalition policy: the conclusion is $\forall\sigma_C \exists\sigma_{-C}$, not $\exists\sigma_{-C} \forall\sigma_C$. Both policies use completed public histories as inputs. Exchanging these quantifiers would claim a single outsider policy defeating all coalition policies, which is not what the construction proves.

Consider a single round in which two agents simultaneously choose bits. Whichever bit the first agent chooses, the second has a matching and a nonmatching response. The same holds with the roles reversed. Neither agent can force equality or force inequality. Failure of one coalition to enforce a goal therefore leaves open whether its opponent can enforce the complement.

For a publicly metered finite interaction with natural round charges and cap $b$, a finite monitored state is $(w, n)$ with $n \in \{0, \dots, b, \mathsf{over}\}$. Start at zero, add the charge of each completed transition, and send a sum exceeding $b$ to the absorbing meter value $\mathsf{over}$. Induction on rounds identifies $n$ with the accumulated cost up to saturation. Apply Theorem 12.12 on this finite product to objectives testing the public budget. A private meter remains on its witness carrier; publicly revealing it would change the available history-dependent policies.

**Corollary 12.13** (Translation of alternating-time state formulas)**.** *On these interactions, the state language generated by propositional connectives and the operators $\langle\langle C\rangle\rangle\mathsf{X}$, $\langle\langle C\rangle\rangle\mathsf{U}$, and $\langle\langle C\rangle\rangle\mathsf{G}$ has an exact interpretation in the assertion extension: next is $\mathrm{Pre}_C$, until is the displayed least fixed point, and always is the displayed greatest fixed point.*

*Proof.*

(1) **Fixed-point argument.** Induct on state formulas. Atomic and propositional cases are the predicate interpretation. The next case is the definition of controllable predecessor.

(2) **Ownerwise alternatives.** Once the immediate subformulas have been interpreted as state predicates, Theorem 12.12 supplies the until and always cases. This also allows nested coalition operators, since their interpretations are again state predicates.

□

This gives the specified connection to temporal and alternating-time reasoning (Alur, Henzinger, and Kupferman, 2002). The linear semantics constructs and composes the rounds; ownerwise policy quantifiers specify control; and least and greatest fixed points express progress and persistence. The resulting policies and counterpolicies are exhibited by the finite-history construction rather than inferred from the modal notation alone.

### 12.2.3 What remains after a history

**Definition 12.14** (Residual games, continuation policies, and off-path histories)**.** For a receptive courteous strategy $s : S \to A^{\perp} \parallel B$ and a finite source configuration $x \in \mathrm{Conf}(S)$ with $s(x) = a_x \amalg b_x$, its **pointed operational residual** is the restriction

$$s/x : S/x \longrightarrow (A^{\perp} \parallel B)/s(x) \cong (A/a_x)^{\perp} \parallel (B/b_x).$$

It records the remaining events of that particular strategy after the actually exhibited source history $x$; its validity is proved in Proposition 12.18.

For a complete-policy game with a total family of histories $\mathsf{Hist}$, a **continuation game** or **subgame** at $h \in \mathsf{Hist}$ is the owned open arena $G_h = (\vec{\Sigma}^h, g_h) : A_h \to B_h$ together with the restriction

$$r_h : \Sigma \to \Sigma^h, \qquad g_h : \Sigma^h \to \mathcal{C}(A_h, B_h), \qquad g^h = g_h r_h.$$

A continuation policy is an element of $\Sigma_i^h$; its restriction from a complete policy is $r_{h,i}(\sigma_i)$. If $\mathsf{Reach}(\sigma) \subseteq \mathsf{Hist}$ denotes the histories on the play of $\sigma$, then $\mathsf{Off}(\sigma) = \mathsf{Hist} \setminus \mathsf{Reach}(\sigma)$ is the **off-path history** set. A subtree game is defined even at such a history; it is not asserted to be the residual of the particular strategy realized by $\sigma$. The comparison at reached histories is constructed below.

**Construction 12.15** (The continuing agent at a history)**.**

(1) **Specify the residual agent.** For the continuation presentation $G_h$ and ownerwise restriction $r_h$, the continuing agent of identity $i$ has interface

$$\mathsf{Ag}_i(G_h) = \big(i, \Sigma_i^h, \varepsilon_{\Sigma_i^h}\big), \qquad r_{h,i} : \Sigma_i \to \Sigma_i^h.$$

Its situated observations are the displayed observation terms on the history or continuation carrier. Its comparison is $P_{i,h}$ on the continuation-arrow sort.

(2) **Transport its comparison.** Pulling that comparison back along $g_h$ and then $r_h$ gives the complete-policy comparison

$$(r_h \times r_h)^*(g_h \times g_h)^* P_{i,h}.$$

Functoriality of inverse image identifies this with feedback through $g^h = g_h r_h$. Thus the history changes the agent's continuation interface while retaining its identity and the complete-policy restriction map. Pointed operational residuals supply the exhibited implementation at reached histories; the full continuation family retains the off-path interfaces used by sequential rationality.

A visible history says what an observer has seen. To compute what the system can do next, we may also need to know which hidden events have occurred. The pointed history retains the source configuration needed to construct that residual.

Write $r_h : \Sigma \to \Sigma^h$ for restriction of complete policies to the future of a history, and $g_h : \Sigma^h \to \mathcal{C}(A_h, B_h)$ for residual play. The history-indexed open play map is $g^h = g_h r_h$. For every continuation comparison $P_h$, its complete-policy feedback is

$$(g^h)^\sharp P_h = (r_h \times r_h)^*(g_h \times g_h)^* P_h.$$

The equality is the functorial action of coplay on residual realizations. The operational construction first produces continuations at pointed histories; the finite complete-policy trees then organize all continuation choices, including histories not reached in the current play. This retains the alternatives needed for subgame-perfect evaluation.

**Construction 12.16** (Residual event structure and residual strategy)**.**

(1) **Restrict the event structure.** For a finite configuration $x$ of an event structure $E$, define its residual $E/x$ as follows. Its events are the $e \notin x$ such that $x \cup \{e\} \in \mathrm{Con}_E$; its order is the restriction of $\leq_E$; and its finite consistency relation is

$$Z \in \mathrm{Con}_{E/x} \quad \Longleftrightarrow \quad x \cup Z \in \mathrm{Con}_E\,.$$

(2) **Inherit polarities.** Residual polarities are inherited.

**Lemma 12.17.** *The construction $E/x$ is an event structure, and for finite sets of residual events $z$,*

$$z \in \mathrm{Conf}(E/x) \quad \Longleftrightarrow \quad x \cup z \in \mathrm{Conf}(E).$$

*If $x \subseteq y$ are configurations, there is a canonical isomorphism*

$$(E/x)/(y \setminus x) \cong E/y.$$

*Proof.*

(1) **Causal execution.** Finite pasts and the consistency axioms are inherited: adjoining a cause of a consistent event preserves consistency with $x$. Every event in a residual past is compatible with $x$, by the same axiom. If $z$ is a residual configuration, the causes of an event of $z$ either lie in $x$ or are residual causes and lie in $z$.

(2) **Residual configuration equivalence.** Thus $x \cup z$ is down-closed, and it is consistent by definition. Conversely, down-closure and consistency of $x \cup z$ give the corresponding properties in the residual.

(3) **Causal execution.** The events of either iterated residual are exactly the events outside $y$ compatible with $y$. On these events the orders are both the restriction of the original order, and consistency on both sides says that adjoining $y$ is consistent in $E$. The identity on remaining events is the asserted isomorphism.

□

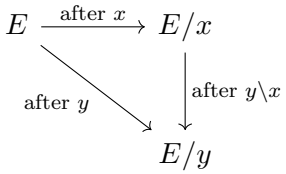



Figure 6: Continuation is independent of whether a history is exposed in one step or in successive prefixes. The arrows denote residual operations.

**Proposition 12.18** (Residual strategies)**.** *If $s : S \to A$ is a receptive courteous strategy and $x \in \mathrm{Conf}(S)$, restriction of the labelling map gives a receptive courteous strategy*

$$s/x : S/x \longrightarrow A/s(x).$$

*Proof.*

(1) **Causal execution.** First the restricted map is well defined. Let $e$ be an event of $S/x$. Then $x \cup \{e\}$ is consistent.

(2) **Causal execution: second comparison.** Adjoining the finitely many causes of $e$ one at a time shows that $x \cup \downarrow e$ is a finite configuration of $S$. Its image under $s$ is therefore a configuration of $A$. Local injectivity on this configuration implies $s(e) \notin s(x)$, while $s(x) \cup \{s(e)\}$ is consistent. Hence $s(e)$ is an event of $A/s(x)$.

(3) **Causal execution.** If $z \in \mathrm{Conf}(S/x)$, then $x \cup z \in \mathrm{Conf}(S)$ by Lemma 12.17. Thus $s(x \cup z)$ is a configuration of $A$. Local injectivity separates the labels in $s(x)$ from those in $s(z)$, so $s(z)$ is a configuration of $A/s(x)$. Local injectivity of $s/x$ is inherited from $s$.

(4) **Secured operational interaction.** For receptivity, let $z$ be a residual configuration and let $a$ be a negative event enabled after $(s/x)(z)$ in $A/s(x)$. Then

$$s(x) \cup s(z) \cup \{a\}$$

is a configuration of $A$. Receptivity of $s$ at $x \cup z$ gives a unique event $e$ with $s(e) = a$ and $x \cup z \cup \{e\} \in \mathrm{Conf}(S)$.

(5) **Lifting through the comparison.** Since $a \notin s(x)$, local injectivity implies $e \notin x$; hence $e$ lies in $S/x$ and is the unique residual lift.

(6) **Secured operational interaction.** For courtesy, suppose $e \lessdot_{S/x} e'$. If some $d$ satisfied $e < d < e'$ in $S$, then $d \notin x$ because $x$ is down-closed while $e \notin x$. Since $e'$ is compatible with $x$ and $d \leq e'$, closure of consistency under adjoining causes makes $d$ compatible with $x$, so $d$ would be an event of $S/x$.

(7) **Secured operational interaction: completion.** This contradicts immediacy there. Therefore $e \lessdot_S e'$. Whenever courtesy for $s$ requires $s(e) \lessdot_A s(e')$, that immediate edge also remains immediate in the residual arena: an event strictly between the two residual labels would already be an event strictly between them in $A$. Hence the courtesy condition restricts to the residual strategy.

□

The source history $x$ cannot be discarded: different source configurations may have the same visible history but distinct future interactions. Residualization also does not claim that every arbitrary future policy extends to the original complete policy space. It constructs the continuation of a specified strategy and the exact restriction maps that are available in the presentation.

**Construction 12.19** (The dependent object of pointed operational histories)**.**

(1) **Form pointed history classes.** For an open arena $G = (\vec{\Sigma}, g) : A \to B$, take all triples

$$(\sigma, s : S \to A^{\perp} \parallel B, x), \qquad [s] = g(\sigma), \quad x \in \mathrm{Conf}(S),$$

using the set of countable strategy presentations on subsets of $\mathbb{N}$ from Convention 1.29. Quotient by strategy isomorphisms preserving $\sigma$ and carrying the marked configuration to the marked configuration. Write $H_G$ for this quotient.

(2) **Project and residualize.** It has canonical projections to $\Sigma$ and to $\mathrm{Conf}(A) \times \mathrm{Conf}(B)$, with the latter defined by $s[x] = a \amalg b$. Residualization is a term in the pulled-back dependent arrow family:

$$[\sigma, s, x] \longmapsto [s/x] \in \mathcal{C}(A/a, B/b).$$

Indeed an isomorphism $f : S \cong S'$ carrying $x$ to $x'$ restricts to an isomorphism $S/x \cong S'/x'$ over the same residual boundary. Hence this term is independent of the representative. Proposition 12.18 supplies its strategy conditions.

(3) **Compare successive residuals.** The isomorphisms $(S/x)/(y \setminus x) \cong S/y$ of Lemma 12.17 are identities on remaining events, so their iterated comparison maps satisfy coherence. No representative strategy is selected for each $\sigma$.

The projection $H_G \to \Sigma$ retains histories actually realized by each strategy. It need not be a product with a common history set. Complete-policy trees additionally specify choices at unreached nodes; their subtree construction below supplies a total continuation family over $\Sigma \times \mathsf{Hist}$.

**Construction 12.20** (The dependent type of residual realizations)**.**

(1) **Form the indexed interfaces.** For a full continuation family, as constructed below for finite complete-policy trees, let $\mathsf{Hist}$ be its history set, and write $G_h : A_h \to B_h$ for the residual arena, with policy restriction $r_h : \Sigma \to \Sigma^{G_h}$. The families $\mathbf{A}_h = yA_h$ and $\mathbf{B}_h = yB_h$ are objects of $\mathsf{L}(\mathsf{Hist})$.

(2) **Recover the ordinary value family.** Their ordinary value family is

$$\mathsf{Val}_{\mathsf{Hist}}[\mathbf{A},\mathbf{B}]_D \cong (\mathcal{C}(A_h,B_h))_{h\in\mathsf{Hist}}.$$

Its context extension is exactly $\coprod_h \mathcal{C}(A_h,B_h) \to \mathsf{Hist}$. Thus the dependent residual interfaces and the existing continuation carrier are the linear and ordinary parts of the same constructed model.

(3) **Interpret the realization.** The realization section sends $(\sigma,h)$ to $(h, g_h r_h \sigma)$; comparison terms take values in the corresponding dependent family of squares.

(4) **Quantify over continuations.** Write $\mathsf{NE}_{G_h}$ for the residual comparison predicate and form the comprehension

$$E = \{(\sigma,h) : r_h\sigma \in \mathsf{NE}_{G_h}\} \hookrightarrow \Sigma \times \mathsf{Hist}.$$

For $\pi : \Sigma \times \mathsf{Hist} \to \Sigma$, universal continuation equilibrium is the internal predicate $\forall_\pi E$. The quantifier formula proves that it holds exactly when every residual profile is an equilibrium. The history type contains all histories, so the formula includes histories not reached by the reference profile. The subsequent finite decision and concurrent-round constructions specify these residual games and their solution predicates explicitly.

### 12.2.4 Operational action updates and pointed-history transport

The epistemic action-model construction can now obtain its enabling data directly from event-structure steps. The updated states are pointed transitions, so they retain the causal prerequisites and consistency of the move as well as its visible label. Keeping the source configuration prevents two different internal histories from being identified merely because they expose the same event.

**Construction 12.21** (An action model generated by a concurrent strategy)**.**

(1) **Form the enabled action family.** For a receptive courteous strategy $s : S \to A^\perp \parallel B$, put $W_s = \mathrm{Conf}(S)$ and $E_s = |S|$. Use the enabling predicate

$$P_s(x,e) \quad\Longleftrightarrow\quad e \notin x \ \wedge\ x \cup \{e\} \in \mathrm{Conf}(S).$$

Its comprehension is the enabled-event span $\mathsf{Step}(S)$ of Construction 1.2.

(2) **Specify its target.** The state map after an action is the target

$$t_s : \mathsf{Step}(S) \longrightarrow W_s, \qquad (x,e) \longmapsto x \cup \{e\}.$$

For $x \in W_s$, write $s(x) = a_x \amalg b_x$ and form the representable dependent boundary families $\mathbf{A}_x = y(A/a_x)$ and $\mathbf{B}_x = y(B/b_x)$.

(3) **Realize the residual.** Define the pointed residual section

$$\rho_s : 1_{W_s} \longrightarrow \mathsf{Val}_{W_s}[\mathbf{A},\mathbf{B}]_D, \qquad \rho_s(x) = [s/x].$$

Proposition 12.18 gives the component's strategy conditions, and pointwise Yoneda gives its displayed type.

(4) **Update a history observable.** For a history observable $v : W_s \to Z$, define its operational postcondition by $v^+ = vt_s$. No choice of total order on independent events is involved. Action uncertainty relations will be constructed from the actual observation data below.

**Theorem 12.22** (Operational realization of one-step product update)**.** *The carrier of the semantic action-model product update determined by $P_s$ is canonically*

$$W_s \odot (E_s, P_s) \cong \mathsf{Step}(S). \tag{12.2}$$

*The residual realization over this carrier is*

$$(x, e) \longmapsto [s/(x \cup \{e\})] \in \mathcal{C}(A/a_{x,e}, B/b_{x,e}), \tag{12.3}$$

*where $s(x \cup \{e\}) = a_{x,e} \amalg b_{x,e}$. It is the reindexing $t_s^* \rho_s$ of the pointed residual family. For an enabled $e$ after $x$ there is also a canonical residual comparison*

$$(s/x)/\{e\} \cong s/(x \cup \{e\}).$$

*All comparisons respect isomorphisms of strategy presentations.*

*Proof.*

(1) **Equality of the two constructions.** Both sides of the carrier bijection consist of exactly the pairs $(x, e)$ satisfying $P_s(x, e)$, and their source and action projections are the same. The target $t_s$ is the configuration reached by that step. Proposition 12.18 constructs the residual arrow at $x \cup \{e\}$; the iterated-residual isomorphism is Lemma 12.17.

(2) **Secured operational interaction.** The corresponding dependent section is Construction 12.20 on the appropriate pointed history base, and its reindexing is Proposition 1.43. Every construction is by restriction of labels, configurations, and causal order, so it commutes with source isomorphisms preserving marked histories.

□

A witnessed event step determines the corresponding epistemic action transition.

$$\begin{array}{ccc} \mathsf{Step}(S) & \xrightarrow{\text{record}} & U \\ \Big\downarrow{\scriptstyle p} & & \Big\downarrow{\scriptstyle \pi_W} \\ \mathrm{Conf}(S) & \xrightarrow[\text{state}]{} & W \end{array}$$

**Construction 12.23** (Action observations and their effective quotients)**.**

(1) **Form recipient observations.** For each owner $i$, take its already designated observation on source histories $o_i : W_s \to O_i$. If the report of the new event is a map $b_i : E_s \to L_i$, construct the retained observation

$$o_i^+ : \mathsf{Step}(S) \to O_i \times L_i, \qquad o_i^+(x, e) = (o_i(x), b_i(e)).$$

(2) **Calculate the updated kernel.** Use $R_i = \ker(o_i)$ and $S_i = \ker(b_i)$ in the action model. Then the product-updated relation is $\ker(o_i^+)$ by Proposition 9.22.

(3) **Retain a history-dependent report.** A protocol may instead supply a history-dependent fresh observation $\omega_i : \mathsf{Step}(S) \to L_i$. To retain both the preceding information and this new observation, construct

$$o_i^+ = \langle o_i p, \omega_i \rangle : \mathsf{Step}(S) \to O_i \times L_i.$$

This too has an exact action-model presentation: use the *pointed-action set* $E_s^\bullet = \mathsf{Step}(S)$, make an action $\alpha = (x, e)$ enabled exactly at its source $x$, and put $S_i^\bullet = \ker(\omega_i)$ on pointed actions. Product update then has carrier canonically isomorphic to $\mathsf{Step}(S)$, with epistemic relation $\ker\langle o_i p, \omega_i \rangle$.

**Proposition 12.24** (Pointed action presentation of retained observations)**.** *For the history-dependent construction above, let*

$$P^\bullet(w,\alpha) \quad \Longleftrightarrow \quad w = p(\alpha), \qquad \alpha \in E_s^\bullet.$$

*Its action-model update and resulting kernel relations identify with* $\mathsf{Step}(S)$ *and* $\ker(o_i^+)$*, respectively. State-dependent action observations are constructed from semantic precondition predicates and effective observation quotients.*

*Proof.*

(1) **Monoidal comparison.** The comprehension of $P^\bullet$ is the graph of $p : E_s^\bullet \to W_s$, canonically isomorphic to $E_s^\bullet$. Under that isomorphism the old-world projection is $p$, while the new-action projection is the identity. The product-updated relation is

$$(p \times p)^* \ker(o_i) \wedge \ker(\omega_i) = \ker\langle o_i p, \omega_i \rangle.$$

The latter kernel has its effective image factorization as in Corollary 6.6.

□

**Construction 12.25** (Owner-parameterized pointed action families)**.**

(1) **Form the classes of pointed steps.** For an owned interaction $G = (\vec{\Sigma}, g)$, use the pointed-history carrier $H_G$ of Construction 12.19. Form the set of classes

$$\mathsf{Step}(G) = \{[\sigma, s, x, e] : [s] = g(\sigma),\ x \in \mathrm{Conf}(S),\ e \notin x, x \cup \{e\} \in \mathrm{Conf}(S)\},$$

where strategy isomorphisms preserve $\sigma$, $x$, and the marked event $e$.

(2) **Compute endpoints and residuals.** Its source and target maps send a class to $[\sigma, s, x]$ and $[\sigma, s, x \cup \{e\}]$, respectively. The post-step residual is the class of $s/(x \cup \{e\})$.

(3) **Reindex the family.** For $r : G \Rightarrow H$, the realization equation $g = h\Pi_N r$ sends these classes to the corresponding classes for $H$, changing only their policy index by $\Pi_N r$. These maps commute with both endpoints and the residual realization. Pulling back the reference along $W \to \Sigma^G$ retains the whole fibre of pointed operational histories; it does not restrict the alternative policy sets.

**Proposition 12.26** (Causal interchange of iterated action updates)**.** *Let* $\mathsf{Path}(S)$ *be the set of pairs consisting of a finite initial configuration* $x$ *and a finite sequence* $(e_1, \ldots, e_n)$ *of events successively enabled from* $x$ *(including* $n = 0$*). Let* $\equiv$ *be generated by exchanging two adjacent occurrences whenever they are concurrent and both orders yield configurations. Then endpoint assignment induces*

$$\mathsf{Path}(S)/{\equiv} \ \cong \{(x, y) \in \mathrm{Conf}(S)^2 : x \subseteq y\}. \tag{12.4}$$

*Residualization after a path depends only on its class, with resulting arrow* $s/y$*. This quotient removes arbitrary linear readings of independence but does not identify epistemic records that genuinely observe their execution order: such records remain on the path carrier until their effective descent is established.*

*Proof.*

(1) **Causal execution.** A sequence yields $y = x \cup \{e_1, \ldots, e_n\}$, unchanged by an interchange. Conversely, given $x \subseteq y$, any linear extension of the finite causal order on $y \setminus x$ provides a sequence of enabled events, since $y$ is consistent and down-closed. Any two linear extensions are connected by swaps of adjacent incomparable occurrences; both swapped steps lie within $y$ and are compatible.

(2) **Coherence of iterated residuals.** This proves the bijection. Iterated residuals are canonically isomorphic to $s/y$ by Lemma 12.17; their comparison maps are identities on the remaining events and hence coherent.

□

### 12.2.5 Secured visible actions, hiding, and epistemic witnesses

The comparison of communication update with concurrent composition has to be made before hiding removes private events. Secured synchronization (Castellan et al., 2017) retains the causal witness, while epistemic product update (Baltag, Moss, and Solecki, 1998) retains the observed event. We construct their correspondence and then identify the additional information required to descend it to visible executions. One visible event can depend on several hidden histories, so forgetting those witnesses too early can change knowledge.

**Construction 12.27** (Visible projection and causal lifts)**.**

(1) **Restrict to visible events.** Let $U$ be an event structure with a designated set $V \subseteq |U|$ of visible events. Write $H = U{\downarrow}V$ for the event structure with event set $V$, the restricted causal order, and consistency inherited from $U$.

(2) **Lift a visible history.** For $z \in \mathrm{Conf}(H)$ put

$$\widehat{z} = \bigcup_{v \in z} {\downarrow_U}\, v. \tag{12.5}$$

The visible projection of a configuration is $\mathrm{vis}(X) = X \cap V$; its fibre is the dependent witness set

$$K_z = \{X \in \mathrm{Conf}(U) : X \cap V = z\}. \tag{12.6}$$

**Lemma 12.28** (Causal lifting and residualization across hiding)**.** *For every $z \in \mathrm{Conf}(H)$, $\widehat{z}$ is a configuration of $U$, is the least such configuration with visible projection $z$, and satisfies $\widehat{z} \cap V = z$. In particular* $\mathrm{vis} : \mathrm{Conf}(U) \twoheadrightarrow \mathrm{Conf}(H)$ *is a surjection with canonical section $z \mapsto \widehat{z}$. Put $V_z = V \cap |U/\widehat{z}|$, the visible events that remain in the residual. There is a canonical isomorphism*

$$(U{\downarrow}V)/z \;\cong (U/\widehat{z}){\downarrow}V_z. \tag{12.7}$$

*For $z \subseteq z'$ the residual extension corresponds to the finite configuration $\widehat{z'} \setminus \widehat{z}$ of $U/\widehat{z}$.*

*Proof.*

(1) **Causal execution.** Consistency of $z$ and repeated adjoining of causal predecessors show that $\widehat{z}$ is consistent. It is finite because every principal past is finite, and it is down-closed by construction. Any configuration with visible part $z$ contains all these predecessors.

(2) **Lifting along the surjection.** Since $z$ is down-closed in the restricted visible order, $\widehat{z} \cap V = z$. Hence $\widehat{z}$ is the least lift and the visible projection is surjective.

(3) **Causal execution.** An event $v \in V \setminus z$ belongs to the left residual exactly when $z \cup \{v\}$ is consistent in $U$. Closure of consistency under adjoining causes makes this equivalent to $\widehat{z} \cup \{v\} \in \mathrm{Con}_U$, which is precisely membership in the right residual.

(4) **Causal execution: completion.** For every finite set $Z$ of these remaining visible events, the same cause-extension argument gives

$$z \cup Z \in \mathrm{Con}_U \quad\Longleftrightarrow\quad \widehat{z} \cup Z \in \mathrm{Con}_U\,.$$

Both residuals inherit the same restricted causal order, proving the isomorphism. Monotonicity of $z \mapsto \widehat{z}$ and Lemma 12.17 give the final comparison.

□

**Proposition 12.29** (Secured interaction at a pointed history)**.** *Let $U = S \circledast_B T$ be the secured interaction of $s : S \to A^{\perp} \parallel B$ and $t : T \to B^{\perp} \parallel C$. For $X \in \mathrm{Conf}(U)$, write $x$ and $y$ for its projections to $S$ and $T$, and $b$ for their common synchronized $B$-history. There is a canonical event-structure isomorphism, over the residual three-interface arena,*

$$U/X \;\cong\; (S/x) \circledast_{B/b} (T/y). \tag{12.8}$$

*It is natural under marked secured-history isomorphisms and compatible with further residualization.*

*Proof.*

(1) **Secured matching of interaction configurations.** By Theorem 1.7, configurations of the secured interaction are secured matchings of the augmented component maps. The configuration $X$ determines a fixed secured matching of the component histories $x, y$, including the private interface events. A configuration extending $X$ gives a matching of the additional residual configurations of $S/x$ and $T/y$, with the already matched $B$-labels removed.

(2) **Reverse implication.** Conversely, the union of this fixed matching with a matching of the residual component configurations is a matching of the original augmented maps. The causality relation has no edges from a new pair to an old pair, because $x, y$ are down-closed. Consequently a cycle in the union exists exactly when a cycle exists in the residual matching; security is preserved and reflected.

(3) **Secured operational interaction.** These constructions are inverse and inclusion-preserving on the secured-matching configuration posets. Taking their prime secured histories identifies the corresponding event structures, including causality, consistency, and boundary projections. The identification uses the same fixed prefix under further residualization, so the comparison maps compose by the identities of Lemma 12.17.

□

**Construction 12.30** (Secured weak-step relation)**.**

(1) **Retain internal step witnesses.** For the visible projection above construct

$$\begin{aligned}\mathsf{WStep}(U, V) = \{(X, Y) : X, Y \in \mathrm{Conf}(U),\\ X \subseteq Y, \quad \mathrm{vis}(Y) = \mathrm{vis}(X) \cup \{v\} \text{ for one } v \in V \setminus \mathrm{vis}(X)\}.\end{aligned}$$

The two projections to $\mathrm{Conf}(U)$ are the source and post-step internal history.

(2) **Project to the visible action.** The comparison with the visible action-update carrier is

$$\Theta : \mathsf{WStep}(U, V) \to \mathsf{Step}(H), \qquad (X, Y) \longmapsto (\mathrm{vis}(X), v), \tag{12.9}$$

where $v$ is the unique newly visible event. Each fibre $\Theta^{-1}(z, v)$ is a dependent set of secured witnesses realizing the visible action.

**Theorem 12.31** (Secured interaction, epistemic update, and hiding)**.** *Let $U = S \circledast_B T$ be the secured interaction of two concurrent strategies and let $V$ consist of their exterior visible events. The source event structure of their composite is $H = U{\downarrow}V$. The action-model carrier*

*generated from $H$ is* $\mathsf{Step}(H)$*. The comparison* $\Theta : \mathsf{WStep}(U,V) \to \mathsf{Step}(H)$ *is surjective, with canonical section*

$$(z, v) \longmapsto (\widehat{z}, \widehat{z \cup \{v\}}). \tag{12.10}$$

*For every visible step $(z, v)$, residualizing after the visible update is canonically isomorphic to the visible part of the secured residual:*

$$H/(z \cup \{v\}) \;\cong (U/\widehat{z \cup \{v\}})\downarrow(V \cap |U/\widehat{z \cup \{v\}}|). \tag{12.11}$$

*The visible residual strategy has the same external labels on both sides. If $x, y$ are the component projections of $\widehat{z \cup \{v\}}$, then by Proposition 12.29 this residual strategy is also the concurrent composite of the residual component strategies $s/x$ and $t/y$, with the common synchronized $B$-past removed. The comparison and the residual isomorphisms are invariant under arena-preserving isomorphisms of the two component strategies.*

*Proof.*

(1) **Causal execution.** Every $(X, Y)$ in the weak-step carrier projects to a visible source configuration followed by exactly one enabled visible event, so $\Theta$ is well-defined.

(2) **Reverse implication.** Conversely, Lemma 12.28 constructs the two configurations in Equation (12.10); the first is included in the second, and their visible parts differ by precisely $v$. This is a section and proves surjectivity. Apply Equation (12.7) at $z \cup \{v\}$ to obtain Equation (12.11).

(3) **Secured operational interaction.** The hidden composite's source is the visible restriction of $U$, and its label map is the restriction to the exterior arena. Therefore the identity on visible remaining events also preserves their arena labels. Proposition 12.29 identifies the secured residual at the canonical lift with interaction of the component residual strategies.

(4) **Secured operational interaction: completion.** Hiding those matched internal events gives precisely the displayed exterior residual. The causal-closure construction commutes with isomorphisms of the secured interaction and with the canonical source comparisons of its pullback presentation.

□

*Remark* 12.32 (Internal histories are not erased by epistemic hiding). The configuration projection $\mathrm{vis} : \mathrm{Conf}(U) \to \mathrm{Conf}(H)$ preserves the secured weak-step relation of Construction 12.30 and the visible enabled-step relation, but it need not be bounded: a visible step need not lift from each internal history over its source. The surjection $\Theta$ compares the corresponding step carriers; its surjectivity asserts only that every visible step has some witness. An internal history may include a hidden occurrence that disables a visible move, even though that move remains available at its coarse visible projection. For example, take a hidden event $h$ conflicting with a visible event $v$, both initially enabled. At internal history $\{h\}$, the event $v$ is impossible, while the empty visible history still permits $v$. The minimal section in Equation (12.10) and the complete witness fibres $K_z$ are therefore both essential. We do not identify arbitrary private observations of $K_z$ with predicates on visible histories.

**Proposition 12.33** (Exact epistemic descent of secured weak steps)**.** *For an owner $i$, let $\omega_i : \mathsf{WStep}(U,V) \to O_i$ be a specified observation of secured weak steps. The following are equivalent:*

(i) *$\omega_i$ is constant on the fibres of $\Theta$;*

(ii) *its two pullbacks to the kernel pair of* $\Theta$ *agree;*

(iii) *there is a unique* $\bar{\omega}_i : \mathsf{Step}(H) \to O_i$ *with* $\omega_i = \bar{\omega}_i \Theta$.

*Under these equivalent descriptions,* $\Theta$ *is a surjective bounded morphism from* $\ker(\omega_i)$ *to* $\ker(\bar{\omega}_i)$. *For a finite family of owners satisfying the same descent test, it preserves and reflects all modal fixed-point predicates with descended atomic tests. Observations that do not descend remain on the dependent witness carrier* $\Theta^{-1}(z, v)$ *and have their own epistemic update there.*

*Proof.*

(1) **Descent along the kernel pair.** The equivalence is effective descent along the surjection $\Theta$, using Proposition 5.34. Since $\ker(\omega_i) = (\Theta \times \Theta)^* \ker(\bar{\omega}_i)$, forward preservation of the kernel relation follows immediately. For a reduced successor $z'$ of $\Theta(X, Y)$ with the same $\bar{\omega}_i$-value, surjectivity of $\Theta$ supplies any preimage of $z'$.

(2) **Descent along the kernel pair: completion.** Its $\omega_i$-value equals that of $(X, Y)$, so it supplies the bounded-morphism back lift. Theorem 9.29 then gives the modal fixed-point assertion. For non-descending observations, the witness family is the ordinary fibre of $\Theta$; no image-level observation term is falsely asserted.

□

The results give two complementary readings. The visible action model is constructed from the hidden composite strategy $H$ through its least causal lift; the internal action model retains every secured history and its private observations. Their comparison identifies which information descends to visible histories and which must remain with the witnesses. Hiding alone is not an epistemic quotient.

### 12.2.6 Canonical exact refinements of visible information

Two internal histories may look identical while allowing different next actions. In the preceding example a hidden occurrence disables a visible move that was initially possible. Recording the private observations separates the histories those observations distinguish. To use the result for temporal reasoning, every transition attributed to a visible state must also be available from each internal source that the state represents. The next refinement enforces this requirement for the specified relations.

**Proposition 12.34** (The coarsest observation-retaining refinement)**.** *Let* $q : X \twoheadrightarrow Y$ *be a surjection and* $o_i : X \to O_i$ *specified observations. The image factorization*

$$X \xrightarrow{r_0} Z_0 := \mathrm{im}\langle q, (o_i)_{i \in N} \rangle \hookrightarrow Y \times \prod_i O_i$$

*is the coarsest quotient through which* $q$ *and every* $o_i$ *factor. Moreover* $r_0$ *is a surjective bounded morphism from each* $\ker(o_i)$ *to* $\ker(\bar{o}_i)$ *on* $Z_0$.

*Proof.*

(1) **Descent along the kernel pair.** Its kernel pair is $\ker(q) \cap \bigcap_i \ker(o_i)$. If a quotient $s : X \twoheadrightarrow Z$ retains all these maps, then $\ker(s) \subseteq \ker(r_0)$, giving the unique factor $h : Z \twoheadrightarrow Z_0$ with $r_0 = hs$. Forward preservation of observation kernels is immediate.

(2) **Lifting along the surjection.** For a back lift from $r_0(x)$ to $z$ with the same $i$-observation, choose a preimage $x'$ of $z$. Then $o_i(x') = o_i(x)$, as required.

□

The image quotient alone can still fail to lift visible transitions. To retain both observations and transitions, take a set $\mathcal{A}$ of relation labels with $R_a \subseteq X^2$ for $a \in \mathcal{A}$, together with the specified observation and test maps $t_\lambda : X \to V_\lambda$. The greatest-bisimulation construction identifies histories only when they are indistinguishable by all of these designated interfaces.

$$\ell = \langle q, (o_i)_i, (t_\lambda)_\lambda \rangle : X \to L, \qquad L = Y \times \prod_i O_i \times \prod_\lambda V_\lambda.$$

An atomic predicate can be retained by its characteristic map to 2.

**Construction 12.35** (The bisimulation predicate)**.**

(1) **Match successors in both directions.** For $B \subseteq X^2$ define $\mathcal{F}(B)$ by

$$\begin{aligned}(x, x') \in \mathcal{F}(B) \quad \iff \quad & \ell(x) = \ell(x') \text{ and, for each } a \in \mathcal{A}, \\ & \forall z\, (xR_a z \Rightarrow \exists z'\, (x'R_a z' \wedge zBz')), \\ & \forall z'\, (x'R_a z' \Rightarrow \exists z\, (xR_a z \wedge zBz')).\end{aligned}$$

The last two lines are simultaneous clauses.

(2) **Take the greatest fixed point.** The map $\mathcal{F}$ is monotone. Define $B_* := \nu\mathcal{F}$ in the existing predicate lattice $\mathcal{P}(X^2)$.

**Definition 12.36** (Exact quotient refinements over retained data)**.** Let $\mathbf{Q}_\ell(X)$ have objects consisting of a relation frame $(Z, (\bar{R}_a))$, a surjective bounded map $s : X \to Z$, and a map $\bar{\ell} : Z \to L$ with $\ell = \bar{\ell}s$. An arrow from $(s_1, \bar{\ell}_1)$ to $(s_2, \bar{\ell}_2)$ is a bounded map $h : Z_1 \to Z_2$ satisfying $s_2 = hs_1$. It is automatically surjective, and $\bar{\ell}_1 = \bar{\ell}_2 h$ follows by cancellation of the surjection $s_1$. Identities and compositions are inherited from frame maps; successive back lifts prove boundedness of composition. Thus the category retains the visible map, observations, and tests as well as the selected relations.

**Theorem 12.37** (Coarsest exact quotient refining the visible quotient)**.** *The relation $B_*$ is an equivalence relation. Let $r : X \twoheadrightarrow Z = X/B_*$ and give $Z$ the direct-image relations*

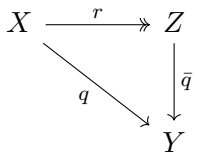
$$u\bar{R}_a v \quad \iff \quad \exists x, x'\, (r(x) = u,\ r(x') = v,\ xR_a x').$$

*Then:*

1. *$q$, all observations, and all retained tests factor uniquely through $r$.*
2. *$r$ is a surjective bounded morphism for every $R_a$.*
3. *For every other surjective bounded morphism $s : X \to Z'$ through which $\ell$ factors, there is a unique surjective bounded morphism $h : Z' \to Z$ satisfying $r = hs$.*

*Thus $(r, \bar{\ell})$ is terminal in $\mathbf{Q}_\ell(X)$. It preserves and reflects the modal fixed-point predicates built from the retained atoms.*

The retained visible observation factors through the greatest-bisimulation quotient.




*Proof.*

(1) **Coinduction and bisimulation.** A relation $B$ with $B \subseteq \mathcal{F}(B)$ is a label-preserving bisimulation. The diagonal is such a relation. The converse of such a relation is again such a relation.

(2) **Witness and legality comparison.** The relational composite of two such relations is also one: to match a successor, first use the witness for the first relation and then the witness for the second; the reverse direction is identical, and label equalities compose. Greatestness therefore gives

$$\Delta_X \subseteq B_*, \qquad B_*^{\mathrm{op}} \subseteq B_*, \qquad B_*; B_* \subseteq B_*.$$

These are reflexivity, symmetry, and transitivity. Since $B_* \subseteq \ker(\ell)$, effective quotient gives the factorizations.

(3) **Lifting through the comparison.** Forward preservation of relations by $r$ is built into $\bar{R}_a$. For back lifting suppose $r(x)\bar{R}_a v$. By definition choose $yR_a z$ with $r(y) = r(x)$ and $r(z) = v$.

(4) **Coinduction and bisimulation.** As $xB_*y$, the bisimulation clause gives $xR_az'$ with $z'B_*z$. Then $r(z') = v$.

(5) **Universal property of the bounded quotient.** For the universal property, $\ker(s)$ preserves labels. We construct $h$ from the quotient universal property and verify both bounded clauses, including the existence of a set map and both bounded-morphism clauses. If $s(x) = s(y)$ and $xR_az$, forward preservation gives $s(y)R_a^{Z'}s(z)$; back lifting along $s$ gives a matching $yR_az'$ with $s(z') = s(z)$.

(6) **Lifting along the surjection.** Interchanging $x, y$ gives the other clause. Thus $\ker(s)$ is a label-preserving bisimulation and is contained in $B_*$. Quotient universality gives the unique surjective map $h : Z' \twoheadrightarrow Z$ with $hs = r$.

(7) **Lifting along the surjection: second comparison.** Its compatibility with the retained labels is forced by the equality $\bar{\ell}_s s = \ell = \bar{\ell}_r r$ and surjectivity of $s$.

(8) **Lifting through the comparison.** To see that $h$ preserves a relation $uR_a^{Z'}v$, take $s(x) = u$ and lift to $xR_ax'$ with $s(x') = v$. Forward preservation by $r$ gives $h(u)\bar{R}_a h(v)$. For the back condition, if $h(u)\bar{R}_a v$, take $s(x) = u$ and use boundedness of $r$ to obtain $xR_ax'$ with $r(x') = v$.

(9) **Choice of representatives and quotients.** Put $u' = s(x')$. Then $uR_a^{Z'}u'$ and $h(u') = v$. Modal fixed-point preservation and reflection are exactly Theorem 9.29, since $r$ is surjective and the atoms descend.

□

**Corollary 12.38** (Compatibility with knowledge and action update)**.** *If $R_i = \ker(o_i)$ is among the retained relations, then $\bar{R}_i = \ker(\bar{o}_i)$. If the only relations are these observation kernels, then $B_* = \ker(\ell)$, so the theorem reduces to the image construction, with any additional tests included in that image.*

*For an action set $E$ with action relations $(S_a)_{a\in\mathcal{A}}$, an arbitrary enabling subobject $P \subseteq X \times E$ is retained by including among the tests the function*

$$t_P : X \to \mathcal{P}(E), \qquad t_P(x) = \{e : (x, e) \in P\}.$$

*Its factor $\bar{t}_P : Z \to \mathcal{P}(E)$ defines $\bar{P}(z, e) \iff e \in \bar{t}_P(z)$, and $P = (r \times 1_E)^*\bar{P}$. The resulting map on product updates $\{P\} \to \{\bar{P}\}$, $(x, e) \mapsto (r(x), e)$, is again surjective and bounded for*

*the updated relations. Hence modal fixed-point adequacy continues through this update and through finite successions for which the enabling data are retained at each stage.*

*Proof.*

(1) **Observation and knowledge.** Equality of quotient observations is equality of the original observations of any representatives, which is exactly $R_i$; this proves the first claim.

(2) **Coinduction and bisimulation.** For the second, if $\ell(x) = \ell(x')$, every $R_i$-successor of $x$ is also an $R_i$-successor of $x'$. It matches itself under $\ker(\ell)$. Thus this entire kernel is a bisimulation.

(3) **Lifting along the surjection.** The added test makes $P$ constant on $r$-fibres for each event. Therefore the updated map is well-defined and surjective. A target successor $(v, f)$ of $(r(x), e)$ gives $r(x)\bar{R}_a v$, $eS_a f$, and enabled $(v, f)$.

(4) **Lifting along the surjection: completion.** Boundedness supplies $xR_a x'$ with $r(x') = v$; descent of enablement makes $(x', f)$ enabled. This is the required back lift. Forward preservation is immediate. Iterate this argument for the last claim.

□

Coalgebraic behavioural equivalence (Rutten, 2000) supplies the underlying bisimulation construction. Here the refinement must preserve the visible map, the declared private observations, and the specified epistemic and secured transitions together. The resulting quotient supports reasoning with these tests while leaving other details on the retained history.

For the example in Remark 12.32, take $X = \mathrm{Conf}(U)$ and $q = \mathrm{vis}$, with the secured weak-step relation and the chosen history observations. The empty history and the hidden history $\{h\}$ have the same visible projection, but only the former can match the visible $v$-step. They cannot be $B_*$-equivalent. The refined carrier therefore retains exactly the distinction that the original visible projection loses. The map from the refined quotient to visible configurations still preserves the visible-step relation: each quotient edge is represented by a secured weak step, whose visible image is an enabled step. For the map $\Theta$ between step carriers, apply the theorem on those carriers with their explicitly specified relations; do not identify it with the configuration projection.

**Corollary 12.39** (Comparison of retained data and invariance under isomorphism)**.** *Adding retained tests to $\ell$ gives a canonical surjective bounded map from the more discriminating exact quotient to the previous one. An isomorphism of the source relation frames that intertwines all retained data induces a unique isomorphism of their exact quotients. These comparison maps compose coherently.*

*Proof.*

(1) **Coinduction and bisimulation.** The quotient retaining more tests is itself an exact quotient retaining the original tests. The terminal property of Theorem 12.37 supplies the comparison. An intertwining isomorphism transports label-preserving bisimulations in both directions, hence transports their greatest relation exactly.

(2) **Lifting along the surjection.** It therefore induces inverse quotient maps. All composition and identity equations follow from uniqueness after precomposition with the relevant surjective quotient map.

□

**Construction 12.40** (Dependent operational witnesses over the refined quotient)**.**

(1) **Form the fibre of the quotient.** For $r : X \twoheadrightarrow Z = X/B_*$, retain the ordinary witness family $K_z = r^{-1}(z)$.

(2) **Recover the total witness context.** The comparison

$$X \xrightarrow{\cong} Z.K, \qquad x \longmapsto (r(x), x),$$

has inverse $(z, x) \mapsto x$. Thus every ordinary or linear family over $X$, including the pointed residual arenas and their realized strategies, is transported exactly to the total witness context $Z.K$. The refined state quotient carries the preserved modal predicates; its dependent witnesses retain the individual operational residuals. This is an isomorphism of contexts, so all reindexing and realization comparisons follow from Proposition 1.43.

The quotient concerns the chosen visible and modal tests, not an identification of residual linear arenas. Complete linear data remain available on the underlying witness family; Proposition 6.21 then specifies when an interface itself descends. This separates exact logical abstraction from isomorphism of all operational structure.

### 12.2.7 Finite perfect-information decisions

**Definition 12.41** (Finite perfect-information tree and complete policies)**.** A **finite perfect-information tree** is a finite prefix-closed set $H$ of histories with root $\varepsilon$, a set $H_{\mathrm{nt}} \subseteq H$ of nonterminal histories, and leaf set $Z = H \setminus H_{\mathrm{nt}}$. Each $h \in H_{\mathrm{nt}}$ has an owner $i(h) \in N$ and a finite nonempty successor set $A(h)$, with the extension $ha \in H$ for each $a \in A(h)$ giving precisely the children of $h$. Each owner evaluates a measured terminal record by $e_i : T_C Z \to \mathbb{K}$. The ordinary leaf-payoff case $u_i : Z \to \mathbb{K}$ is $e_i = u_i \epsilon_Z$. A public edge $h \to ha$ may carry a charge $\delta_h(a) \in C$; zero on every edge gives the ordinary tree. A **complete policy** of owner $i$ is an element of

$$\Sigma_i = \prod_{h \in H_{\mathrm{nt}} :\, i(h) = i} A(h), \qquad \Sigma = \prod_i \Sigma_i,$$

including its actions at nodes not on the realized path. Following the prescribed successor from the root defines play : $\Sigma \to Z$. The **subtree** $T_h$ is rooted at a history $h$ and has as nodes the extensions $H_h = \{k \in H : h \preceq k\}$, with inherited ownership, action sets, edge charges, and leaf labels with their evaluators. Its policy restrictions are the coordinate projections $r_h : \Sigma \to \Sigma^h$ constructed below. An agent observes its current history and therefore knows its decision node; perfect information says nothing about shared preferences.

**Construction 12.42** (Retained-history realization)**.**

(1) **Form the history arena.** There is a fixed-polarity event-structure realization of the complete history. Let $\mathsf{Hist}(T)$ have the nonempty tree histories as events, ordered by prefix extension, with finite consistent sets precisely the chains, and give every event positive polarity.

(2) **Realize the policy path.** For a profile $\sigma$, let $P_\sigma$ be the chain of nonempty prefixes of play$(\sigma)$ and let $S_\sigma$ be the induced chain event structure. Inclusion gives

$$t_\sigma : S_\sigma \longrightarrow \mathsf{Hist}(T),$$

regarded as a strategy $I \to \mathsf{Hist}(T)$. Receptivity is vacuous because the arena has no negative events. Every immediate dependency in $S_\sigma$ is the corresponding prefix dependency in the arena, so courtesy holds. Hence

$$\sigma \longmapsto t_\sigma$$

is an owned family on one common polarized arena; no event changes polarity with the identity of the agent who owns its decision.

**Construction 12.43** (Leaf observation, including the terminal root)**.**

(1) **Define the terminal and nonterminal reports.** If the root is a leaf $z$, then $\mathsf{Hist}(T) = I$, its unique complete-policy profile has $t_\sigma = \mathrm{id}_I$, and put $r = \mathsf{pt}(z) : I \to D(\{z\})$. For a nonterminal root every leaf is a nonempty history. Define $r : \mathsf{Hist}(T) \to D(Z)$ to have one negative input event $\bar{h}$ for every history $h$ and one positive output $o_z$ for every leaf $z$. The negative events inherit the tree order. Add the dependency $\bar{z} < o_z$.

(2) **Generate the event structure.** A finite set is consistent precisely when its negative history events form a chain, it contains at most one output, and, if it contains $o_z$, every negative history in the set is a prefix of $z$. These clauses make the source an event structure: adjoining a cause of $o_z$ stays on the same prefix chain, while inherited causes of negative history events remain on that chain.

(3) **Verify the labelling.** The labelling sends $\bar{h}$ to the input occurrence of $h$ and $o_z$ to the output value $z$. A source configuration containing $o_z$ contains $\bar{z}$ and all its prefixes, so its image is a configuration of $\mathsf{Hist}(T)^\perp \parallel D(Z)$; local injectivity is immediate. Thus the labelling is a total map. Every enabled negative history extension is accepted exactly once. The only dependency not already present on the input arena is $\bar{z} < o_z$, which is negative-to-positive. Thus $r$ is receptive and courteous.

(4) **Compose with a realized history.** Synchronizing $r$ with $t_\sigma$ exposes exactly the prefix chain $P_\sigma$; only the leaf $\mathrm{play}(\sigma)$ occurs on that chain. Hiding the history events therefore gives

$$r \circ t_\sigma \cong \mathsf{pt}_Z(\mathrm{play}(\sigma)) : I \to D(Z).$$

Thus the retained-history and outcome-only presentations are related by an explicit strategy with the interface polarities fixed.

(5) **Compare the staged circuit.** The staged public-state circuit constructed below gives a decentralized factorization of the same play function. When specialized to a perfect-information tree, every nonowner has a singleton action at the current node. To compare its flat outcome interface with the prefix-ordered history arena, we now construct an explicit strategy. Merely copying public state values into independent data ports would not itself impose the history arena's prefix order.

**Construction 12.44** (Expansion of a leaf into its prefix history)**.**

(1) **Specify the leaf branches.** Let $Z$ be the leaves of the finite tree $T$. For $z \in Z$, write $P_z$ for its chain of nonempty prefixes and let $t_z : I \to \mathsf{Hist}(T)$ be inclusion of that chain. Construct

$$e_T : D(Z) \longrightarrow \mathsf{Hist}(T)$$

as follows.

(2) **Build the prefix histories.** For each $z$ its source has a negative event $n_z$ and a positive event $p_{z,h}$ for every $h \in P_z$. Within this branch, $n_z$ precedes every positive

event and the positive events are ordered by prefix. A finite set is consistent exactly when it is contained in one branch; events in distinct branches conflict.

(3) **Label the expansion.** The labelling sends $n_z$ to the input value $z$ and $p_{z,h}$ to the history event $h$. If the root is terminal, $P_z$ is empty, so the unique branch has only its negative input and $e_T : D(\{z\}) \to I$ is disposal.

**Proposition 12.45** (Comparison with the retained-history realization)**.** *The displayed map $e_T$ is a receptive courteous strategy. For every leaf $z$ and complete policy profile $\sigma$,*

$$e_T \circ \mathsf{pt}_Z(z) \cong t_z, \qquad e_T \circ \mathsf{pt}_Z(\mathrm{play}(\sigma)) \cong t_\sigma.$$

*These comparisons include the terminal-root case.*

*Proof.*

(1) **Causal execution.** The source is an event structure: each branch is a finite chain and its causes remain in that branch, so the consistency axioms hold. A nonempty configuration contains one $n_z$ and an initial segment of the prefix chain $P_z$. Its image is consequently a configuration of $D(Z)^\perp \parallel \mathsf{Hist}(T)$, and the labels are injective on that configuration.

(2) **Local injectivity across conflicting branches.** Repeated history labels on different branches cause no failure of local injectivity because those branches conflict.

(3) **Fibrewise calculation.** At the empty configuration every negative input $z$ has the unique lift $n_z$. Once an input has occurred, the other negative inputs conflict with it already in $D(Z)^\perp$, so there are no further enabled negative extensions.

(4) **Receptivity and courtesy of prefixes.** This proves receptivity. The only new immediate dependency is from $n_z$ to the first positive prefix event. All positive-to-positive immediate dependencies are the corresponding prefix edges of $\mathsf{Hist}(T)$. Hence courtesy holds.

(5) **Causal execution.** Synchronizing with $\mathsf{pt}_Z(z)$ selects the $z$-branch and hides its input event. The remaining labelled source is exactly the chain $P_z$, giving the first isomorphism. Substituting the leaf $\mathrm{play}(\sigma)$ gives the second.

(6) **Causal execution: completion.** For a terminal root there are no remaining events, and both sides are the unique arrow $I \to I$.

□

The comparison recovers prefix-ordered records from terminal history values, giving equality for the constructed closed strategies. It does not identify arbitrary strategies on a flat data arena with arbitrary strategies on a history arena. The causal record retained by the protocol is what makes the reconstruction possible.

### 12.2.8 Backward induction, with all ties retained

A choice can change rank after an earlier round has consumed part of a budget. To evaluate a child on a public tree, we therefore need both its remaining play and the charge inherited along its own path. Backward induction must make this calculation even for a child the proposed policy never visits, because an alternative policy can lead there.

**Construction 12.46** (Measured tree play and translated continuations)**.**

(1) **Meter a completed public edge.** In either finite tree presentation, label a designated transition-to-child record by $ha$ and weight this class by $\delta_h(a)$. One realization uses the finite next-history response and its receipt on a recorder port; give all other classes weight zero. The same circuit can use the increments of Proposition 8.7 when its total for a round descends to the public edge. For the displayed child meter the charge is determined by $ha$ on every completing witness. Compute $c_\epsilon = 0$ and $c_{ha} = c_h + \delta_h(a)$ along every legal history, including ones off the selected path. A hidden charge that varies over this public history remains on the witness fibre.

(2) **Return measured play.** At a leaf set $\widehat{P}_z(*) = (0, z)$. Fix a subtree profile $s$ with current action $a$ and child profile $s^{ha}$. Put

$$S_d(b, z) = (d + b, z), \qquad \widehat{P}_h(s) = S_{\delta_h(a)}(\widehat{P}_{ha}(s^{ha})).$$

Induction on height gives $\epsilon_Z \widehat{P}_h(s) = \mathrm{play}_h(s)$ (with the original terminal history as leaf label). Its first projection sums the charges on the remaining path. These are the class increments of the retained circuit; the price is read after measurement.

(3) **Translate the owner evaluation.** Define $e_i^c = e_i S_c$ and $V_i^{h,c}(s) = e_i^c(\widehat{P}_h(s))$. Since $S_0 = 1$ and $S_c S_d = S_{c+d}$, one has $(e_i^c)^d = e_i^{c+d}$. At $d = \delta_h(a)$ the equality is the commuting square

$$\begin{array}{ccc} \Sigma^{ha} & \xrightarrow{\widehat{P}_{ha}} & T_C Z \\ {\scriptstyle S_d \widehat{P}_{ha}}\downarrow & & \downarrow{\scriptstyle e_i^{c+d}} \\ T_C Z & \xrightarrow[e_i^c]{} & \mathbb{K} \end{array}$$

Thus $V_i^{h,c}(s) = V_i^{ha,c+\delta_h(a)}(s^{ha})$. For an additive price, $e_i^c(b, z) = e_i(b, z) - \ell_i(c)$, so the prefix leaves every ranking inside that subtree unchanged. A nonlinear owner may rank continuations differently: give a safe action reward 1 and extra charge 0, and a fast action reward 2 and extra charge 1. Deduct 3 when total charge exceeds 1. The fast action wins at incoming charge zero and the safe action wins at incoming charge one.

**Definition 12.47** (Backward induction and subgame-perfect equilibrium)**.** For a finite perfect-information tree write $\Sigma^h$ for its subtree profile carrier. For incoming charge $c$ set $\mathsf{BI}(z; c) = \{*\}$ at a leaf. At a nonterminal $h$ with owner $i = i(h)$ use $\Sigma^h \cong A(h) \times \prod_b \Sigma^{hb}$ and put

$$\begin{aligned} (a, (s^{hb})_b) \in \mathsf{BI}(h; c) \quad \Longleftrightarrow \quad & s^{hb} \in \mathsf{BI}(hb; c + \delta_h(b)) \quad (\forall b), \\ & V_i^{ha,c+\delta_h(a)}(s^{ha}) = \max_{b \in A(h)} V_i^{hb,c+\delta_h(b)}(s^{hb}). \end{aligned}$$

The two right-hand clauses hold jointly. Write $\mathsf{NE}_{G_h}(c)$ for Nash in the subtree using $V_i^{h,c}$ for each owner, and $\mathsf{SPE}(h; c)$ for the subtree profiles that are Nash in every descendant subtree at the charge obtained by extending $(h, c)$ along its unique path. At the root use $c = 0$ and abbreviate $\mathsf{BI}(\epsilon; 0)$ and $\mathsf{SPE}(\epsilon; 0)$ by $\mathsf{BI}(\epsilon)$ and $\mathsf{SPE}$. The state-indexed conjunction is (12.12). Theorem 12.50 identifies these two root predicates.

This rules out a prescribed future action that ceases to be optimal when its decision is actually reached.

These backward-induction characterizations concern finite trees. For exact rational leaf payoffs the finite maximum is computed by rational comparison. The same choice exists

for classical real payoffs by the usual linear-order argument. The categorical construction supplies all finite continuation choices; ordinary maximization selects an optimal continuation while retaining all ties in the resulting policy set.

**Construction 12.48** (Subtree policies, restriction, and reference-based extension)**.**

(1) **Form the subtree policy spaces.** For each history $h$, let $T_h$ be the rooted subtree whose nodes are the histories extending $h$. Construct its ownerwise policy spaces by

$$\Sigma_i^h = \prod_{\substack{k \succeq h \\ i(k)=i}} A(k),$$

where the product ranges over the nonterminal nodes of $T_h$ owned by $i$. The canonical product projections give restriction maps

$$r_{h,i} : \Sigma_i \longrightarrow \Sigma_i^h, \qquad r_h = \prod_i r_{h,i} : \Sigma \longrightarrow \Sigma^h.$$

Applying the retained-history and outcome construction above to $T_h$ gives an owned continuation game $G_h$ on the parameter family $\vec{\Sigma}^h$. Thus "the game after $h$" is a constructed subtree game, with continuation data carried by its dependent policy family.

(2) **Extend a deviation.** Every residual alternative has an explicit extension to a complete policy relative to a reference profile. For $\sigma_i \in \Sigma_i$, define

$$\mathrm{ext}_{h,\sigma_i} : \Sigma_i^h \longrightarrow \Sigma_i$$

coordinatewise by

$$\mathrm{ext}_{h,\sigma_i}(\tau_i)(k) = \begin{cases} \tau_i(k), & k \succeq h, \\ \sigma_i(k), & k \not\succeq h. \end{cases}$$

Then

$$r_{h,i}\,\mathrm{ext}_{h,\sigma_i} = \mathrm{id}_{\Sigma_i^h}.$$

No global section of $r_{h,i}$ is selected once and for all; the displayed extension is constructed from the reference policy whose off-subtree coordinates are to be retained.

(3) **Identify the residual along a realized history.** When $h$ lies on the realized history of a profile $\sigma$, this subtree construction agrees with the operational residual. Regard $T_h$ as rooted at $h$; the nonempty histories of $T_h$ are the nonempty suffixes after $h$ and therefore correspond to the strict extensions of $h$ in $T$. If $P_h$ is the prefix configuration containing all nonempty prefixes through $h$, deletion of the executed prefix gives canonical isomorphisms

$$\mathsf{Hist}(T)/P_h \cong \mathsf{Hist}(T_h), \qquad t_\sigma/P_h \cong t^h_{r_h(\sigma)}.$$

The first sends a strict extension of $h$ to its relative suffix in the rooted subtree, and the second restricts that map to the realized continuation chain; order, chain consistency, polarities, and labels are inherited. Off-path subgames are still defined by the subtree construction even though they are not residuals of the particular realized strategy.

**Definition 12.49** (Subgame perfection)**.** A profile is **subgame perfect** exactly when every one of these reindexed Nash predicates holds. In the assertion doctrine,

$$\mathsf{SPE} = \bigwedge_h r_h^* \mathsf{NE}_{G_h}(c_h). \tag{12.12}$$

Subgame perfection is the conjunction of equilibrium predicates over all constructed continuation games. The backward-induction theorems below determine this predicate for finite causal trees.

**Theorem 12.50.** *At every incoming charge $c$, $\mathsf{BI}(h;c) = \mathsf{SPE}(h;c)$.*

*For every finite perfect-information tree, all pure subgame-perfect profiles are generated by the following recursion. At a leaf there is the unique empty policy profile. At a node owned by $i$, choose a subgame-perfect profile in every child subtree, and choose any child whose resulting payoff to $i$ is maximal, using $c + \delta_h(b)$ at each child $hb$. Retain the chosen profiles in all child subtrees, including those not selected at the node. In particular a pure subgame-perfect profile exists.*

*Proof.*

(1) **Fixed-point argument.** Proceed by induction on tree height. The leaf case is immediate. At a nonterminal root, let the selected child be $a$ and the chosen subgame-perfect continuation in child $b$ be $\sigma^b$.

(2) **Strategic comparison.** If a deviating agent $j$ is not the root's owner $i$, the selected child does not change. The deviation cannot increase its payoff within child $a$, because $\sigma^a$ is a Nash equilibrium there. If the deviator is $i$, it can choose another child $b$ and also change its entire policy inside that child.

(3) **Strategic comparison: second comparison.** The Nash property of $\sigma^b$ bounds the resulting payoff by $V_i^{hb,c+\delta_h(b)}(\sigma^b)$. This is the parent's evaluation of choosing $b$ with the displayed continuation by Construction 12.46. Maximization over the same indexed values bounds it by the current payoff. Thus no agent has a profitable complete-policy deviation at the root. All proper subgames already have the required property by induction.

(4) **Reverse implication.** Conversely, a subgame-perfect profile restricts to a subgame-perfect profile in every child. Its chosen child must maximize the root owner's payoff among those fixed continuations: otherwise changing only the root decision is profitable. This is exactly one of the profiles in the stated recursion.

(5) **Finite maxima and continuation witnesses.** Finally each finite nonempty set of child payoffs has a maximum, and the finitely many continuation witnesses are assembled by induction over the child set. No choice over an infinite family is involved. This proves existence.

□

The later node may belong to the same owner as an earlier one. Accordingly, optimality is tested against changes to that owner's complete contingent policy rather than one local action at a time. The recursion therefore avoids the componentwise-equilibrium error identified in Section 5.6. The construction relates to selection-function accounts of backward induction (Escardó and Oliva, 2010), but preserves the explicitly concurrent structure of completed stages.

*Remark* 12.51 (Finite and infinite continuation). The continuation arguments here use finite trees and their residual games. For a coalgebraic treatment of subgame-perfect equilibria in infinite games without discounting, see Abramsky and Winschel (2017). Such an infinite-game analysis is separate from the finite backward decompositions proved here.

### 12.2.9 Subgame perfection with concurrent public rounds

**Definition 12.52** (Finite public game tree with simultaneous rounds)**.** A **finite public game tree** has a finite prefix-closed history set $H$, its nonterminal subset $H_{\mathrm{nt}}$, leaves

$Z = H \setminus H_{\mathrm{nt}}$, and, at every $h \in H_{\mathrm{nt}}$, finite nonempty action sets $A_i(h)$ for all agents. Its dependent joint-action carrier and successor map are

$$\mathsf{Act} = \coprod_{h \in H_{\mathrm{nt}}} \prod_{i \in N} A_i(h) \xrightarrow{\mathrm{succ}} H, \qquad \mathrm{succ}(h, a) = ha.$$

The $ha$ are precisely the distinct children of $h$. Each owner has a terminal evaluator $e_i : T_C Z \to \mathbb{K}$. Give its public edges the meter of Construction 12.46, using this round's joint action for $a$. An unpriced leaf payoff $u_i : Z \to \mathbb{K}$ is evaluated by $u_i \epsilon_Z$. Complete owner policies are

$$\Sigma_i = \prod_{h \in H_{\mathrm{nt}}} A_i(h), \qquad \Sigma = \prod_i \Sigma_i.$$

All agents see the public history $h$ before choosing their current components of $a$, but no agent sees the others' current choices before making its own. The joint tuple is revealed only when the child $ha$ is reached. The **stage game** at $h$ is the one-round normal-form game with its payoffs determined by a specified complete family of child continuation profiles, as typed in Definition 12.55. Its Nash set need not be nonempty. The one-round interaction is a concurrent reporting protocol followed by the continuation interface.

The following construction gives a fixed interface for the entire tree.

**Construction 12.53** (Concurrent circuit for a finite public tree)**.**

(1) **Define layers of public states.** Let its height be $d$. At layer $k$, let $H_k$ contain its histories of depth $k$, together with absorbing copies of leaves reached earlier and an error symbol $\perp$. A leaf keeps its label in every subsequent layer.

(2) **Realize the owned decisions.** For agent $i$ take the layer policy set

$$S_i^k = \prod_{\substack{h \in H_k \\ h \text{ nonterminal}}} A_i(h)$$

and the finite response set

$$B_i^k = \{(h, a) : h \in H_k \text{ nonterminal},\ a \in A_i(h)\} \amalg \{\mathsf{idle}\}.$$

The policy $s_i^k$ realizes the function $f_{s_i^k} : H_k \to B_i^k$ which sends a nonterminal $h$ to $(h, s_i^k(h))$ and a leaf or $\perp$ to $\mathsf{idle}$. The map

$$\iota_i^k : S_i^k \longrightarrow (B_i^k)^{H_k}, \qquad s_i^k \longmapsto f_{s_i^k},$$

is injective, since the value of $f_{s_i^k}$ at every nonterminal history recovers the chosen action there. The resulting singly owned response family is therefore the cartesian reindexing

$$(\iota_i^k)^* \mathsf{Dec}_i(H_k, B_i^k)$$

of the full finite decision presentation. In particular it belongs to $\mathbf{DecArena}_N(\mathcal{C})$ by construction.

(3) **Collect concurrent responses.** Use the iterated data comonoid to copy the public $H_k$ input to all agents and to one transition record; if layer-labelled data records are desired, retain one additional copied value at each layer as well. Apply the independent singly owned response families and collect their responses together with the transition copy of the public state. The next-state function sends a tuple of correctly tagged responses at

$h$ to the child $ha$, preserves a leaf when all responses are idle, and sends every other tuple to $\bot$. Set $\bot$ absorbing. This is a total finite function, so its response strategy exists by Proposition 1.16. Every remaining box in the layer is nonstrategic wiring, copying, collecting, or one of the displayed reindexed decision families. Hence each layer, and therefore the complete finite tree circuit, is an open arena in $\mathbf{DecArena}_N(\mathcal{C})$.

(4) **Compose the layers.** Start with the root state and compose the $d$ layers. Induction on the layers proves that $\bot$ is never reached by an available policy: all responses to a genuine nonterminal state have its tag, and all responses to a retained leaf are idle. The same induction shows that the state at each layer is the prescribed public history or the already reached leaf. The optional copied values record the successive public histories in separate data ports; hiding them leaves the leaf-only realization.

(5) **Project to a leaf.** After the last layer, the reachable output states lie in the leaf set $Z \subseteq H_d$. Define the nonstrategic leaf projection $p_Z : D(H_d) \to D(Z)$ as follows. For every $h \in H_d$ it has one negative input $n_h$; for every $z \in Z$ it has one positive output $o_z$ with $n_z < o_z$. Distinct input branches conflict, and an input outside $Z$ has no positive continuation. This strategy is total as an event-structure map and receptive to every negative input, but is value-partial on the unreachable complement $H_d \setminus Z$. Its only added immediate dependencies are negative-to-positive, so it is courteous. The preceding reachability induction ensures that its composite with the circuit is value-total on all policy profiles. It satisfies $p_Z \circ \mathsf{pt}_{H_d}(z) = \mathsf{pt}_Z(z)$.

(6) **Compare the two realizations.** Composing the layer circuit with $\mathbf{j}_N(p_Z)$ gives, within $\mathbf{DecArena}_N(\mathcal{C})$, the leaf-only family $\ell_\sigma : I \to D(Z)$. The finite-function response and collector equations identify each internal value with its uniquely prescribed value on that profile; after hiding, the sole output event is its leaf. Hence

$$\ell_\sigma \cong \mathsf{pt}_Z(\mathrm{play}(\sigma)).$$

Regard each completed public round as one tree step when forming $\mathsf{Hist}(T)$. Applying the explicit comparison of Proposition 12.45 now gives

$$e_T \circ \ell_\sigma \cong t_\sigma.$$

Thus the circuit and the retained-history presentation are related by a constructed operational strategy, not by identifying independent copied data ports with prefix-ordered history events. Its ownerwise parameter product $\prod_k S_i^k$ is canonically the complete contingent policy space, with no choices added at leaves or at the error state. The error symbol merely makes the transition functions total on unreachable tuples; it creates no additional strategic option. Within each round the response components are tensored, so the agents' simultaneous choices remain causally independent.

**Proposition 12.54** (Public trees as iterated epistemic action updates)**.** *For the finite public tree of Construction 12.53, let $W_k$ consist of its valid public histories at layer $k$, including absorbing copies of earlier leaves but excluding the circuit's unreachable error symbol. At a nonterminal $h \in W_k$ put $A(h) = \prod_i A_i(h)$; at a leaf put $A(h) = 1$. Form the tagged action set $E_k = \coprod_{h \in W_k} A(h)$ and enable $(h, a)$ only at $h$. Take the publicly observed completed action as the action label. The semantic product-update carrier is then canonically $\coprod_{h \in W_k} A(h)$, and its deterministic successor map is a bijection with $W_{k+1}$. Iterating these updates from the root recovers the full set of reachable complete public histories, including histories off the path of any particular policy profile. On the product with the complete-policy context $\Sigma$, the residual realization at the successor history $ha$ is $g_{ha} r_{ha}$, with $r_{ha} = r_{h,ha} r_h$ for the subtree*

*restriction maps. Thus history substitution and owner-policy restriction act on their distinct indices and commute in the constructed continuation family.*

*Proof.*

(1) **Lifting along the surjection.** The enabling comprehension pairs $h$ with a single tagged legal action $(h, a)$, so it is isomorphic to $E_k$. A child is uniquely the history $ha$, and an absorbing leaf has its sole idle successor. Hence the successor function is injective and surjective onto $W_{k+1}$.

(2) **Fixed-point argument.** Induction on layers gives the iterated carrier correspondence. The underlying concurrent circuit realizes the same child map by the finite response and collector equations in Construction 12.53. The residual fibre after $h$ is the constructed subtree game.

(3) **Policy-level comparison.** Its policy restriction to $ha$ is the product projection $r_{h,ha}$; composing it with $r_h$ gives $r_{ha}$, by equality of the selected policy coordinates. Proposition 1.43 applies separately to the history-indexed linear family and shows that the resulting component is $g_{ha} r_{ha}$. This keeps off-path fibres: no update is restricted to the actual reference policy's unique path.

□

**Definition 12.55** (Stage game induced by continuation profiles)**.** At a nonterminal public history $h$, put $A(h) = \prod_{i \in N} A_i(h)$. A specified family of complete child-continuation profiles has type

$$s = (s^{ha})_{a \in A(h)} \in \prod_{a \in A(h)} \Sigma^{ha}.$$

For incoming charge $c$, the **stage game at** $(h, c, s)$ has utilities

$$U_i^{h,c,s}(a) = e_i^{c + \delta_h(a)}\big(\widehat{P}_{ha}(s^{ha})\big).$$

In the game started at the root, the incoming charge at $h$ is $c_h$. With $e_i = u_i \epsilon_Z$, the expression is $u_i(\mathrm{play}_{ha}(s^{ha}))$, recovering the unpriced stage game. Each current action is tested against the same full family of child profiles; the incoming charge for its selected child is recomputed from that action.

**Theorem 12.56** (Concurrent backward decomposition)**.** *A profile at $(h, c)$ is subgame perfect exactly when every child continuation is subgame perfect at its own incoming charge and the current joint action is a pure Nash equilibrium of the stage game. All complete profiles satisfying the criterion are given by*

$$\mathrm{SPE}(h; c) \cong \coprod_{s \in \prod_{b \in A(h)} \mathrm{SPE}(hb; c + \delta_h(b))} \mathrm{NE}\big((A_i(h))_i, (U_i^{h,c,s})_i\big).$$

*At a leaf use the singleton empty profile. The indexing retains all child policies, including those off the current path.*

*Proof.*

(1) **Ownerwise alternatives.** Subgame perfection implies the property in every child. A change only to agent $i$'s current action is among its admissible complete policy deviations; holding its child policies fixed gives exactly the stage-game inequality. This proves necessity.

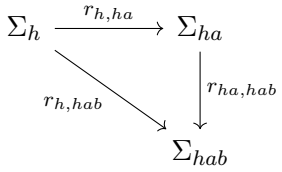
(2) **Forward implication.** For sufficiency, take a complete deviation of agent $i$. Its new current action, combined with the other agents' unchanged actions, selects a child $hb$. In that child the deviator may also change its entire continuation.

(3) **Strategic comparison.** The Nash property of the chosen child profile at $(hb, c+\delta_h(b))$ bounds the deviator's payoff by $U_i^{h,c,s}(b)$. Indeed its measured suffix result is evaluated by $e_i^{c+\delta_h(b)}$ on both sides of the child Nash inequality. By the square in Construction 12.46, the left side is the parent's value of the full deviation. Stage Nash bounds the right side by $U_i^{h,c,s}(a)$ for the original joint action $a$. Thus the complete deviation is not profitable at $h$.

(4) **Fixed-point argument.** Every proper subgame already has the required property. Induction on height proves the theorem and the displayed decomposition.

□

Child-continuation policy restriction composes with deeper restriction.




A simultaneous stage need not have a pure Nash equilibrium, so the resulting subgame-perfect set may be empty. When each stage has only one nontrivial owner, its Nash condition becomes the nonempty argmax condition and the construction recovers Theorem 12.50. The same continuation restriction maps cover both cases, distinguishing finite perfect-information choice from simultaneous concurrent rounds.

**Corollary 12.57** (One-round deviations suffice)**.** *For a finite public game tree, a profile is subgame perfect iff, at every history, no agent can improve by changing only its action in that round and retaining its original continuation everywhere, evaluating each reached child at its inherited charge.*

*Proof.*

(1) **Fixed-point argument.** The forward implication is immediate. For the reverse, induct upward from the leaves. The induction gives subgame perfection of every child profile, while the one-round inequalities are the stage Nash condition. Theorem 12.56 proves the inductive step.

□

# 13 Epistemic Rationality and Strategic Dynamics

The previous section tests a complete policy after every public history. We now ask what gives an owner a reason to change its policy. The gain from a deviation determines an improvement relation. To study adjustment over time we specify that relation as the step relation and construct a termination rank when it is acyclic. Beliefs introduce another comparison, since an owner evaluates a response against the opponent policies it considers possible. Truthful reporting provides a final test in which the true type stays fixed while the report varies.

## 13.1 Improvement dynamics and termination

Evaluating both ends of a deviation gives its signed gain. The deviations with positive gain form a directed graph, whose sinks are precisely the Nash equilibria. On a finite nonempty carrier, acyclicity supplies an integer rank that decreases at each improvement step. The termination argument applies to a process that actually takes such steps.

**Notation 13.1** (Utilities on profiles)**.** In the numerical results below, $u_i : \Sigma \to \mathbb{K}$ denotes the profile value descended from $e_i r$ by Proposition 5.43 on the full completed profile carrier. The elementary strategic-form input is included by taking $K = \Sigma$, $r(\sigma) = (0, \sigma)$ and $e_i(c, \sigma) = u_i(\sigma)$. When the value factors through the concurrent arrow this also gives $u_i = k_i g$. On a proper legal completed carrier, deviations use its restricted span; the full-product elimination results below concern their stated product games. A utility evaluated on a typed outcome will instead have its domain displayed explicitly, as in the reporting contexts of Section 13.3.

**Construction 13.2** (The comprehension of profitable alternatives)**.**

(1) **Measure the gain of a deviation.** On the dependent deviation context $p_1 : D_i \to \Sigma$, put $\gamma_i = u_i p_2 - u_i p_1$ and take the comprehension $J_i = \{\gamma_i > 0\} \hookrightarrow D_i$. Its paired endpoint map is jointly monic and presents $\mathsf{PDev}_i \subseteq \Sigma^2$.

(2) **Recover the equilibrium predicate.** Hence in the assertion fibre over $1_G$,

$$\mathsf{NE}_G = \bigwedge_i \neg \exists_{p_1 : J_i \to \Sigma} \top.$$

Indeed the existential image consists exactly of profiles with a positive-gain alternative. The term $\gamma_i$ is retained before comprehension, so the construction preserves the magnitudes used by potential and integrability arguments. For a completed profile with descended exact net value as in Construction 2.8 and Proposition 5.43, choose endpoint witnesses $k, k'$ for a deviation. With $e_i(c, z) = u_i^0(z) - \ell_i(c)$ its gain is

$$\gamma_i = [u_i^0(a(k')) - u_i^0(a(k))] - [\ell_i(m(k')) - \ell_i(m(k))].$$

Value descent makes this difference independent of the endpoint witnesses. Both differences are taken in the signed payoff group.

### 13.1.1 Profitable changes and termination

**Definition 13.3** (Improvement paths and ordinal potentials)**.** Put $P = \bigcup_{i \in N} \mathsf{PDev}_i \hookrightarrow \Sigma^2$, the directed relation of strictly profitable unilateral changes. A **finite improvement path** of length $n$ is a tuple $(\sigma_0, \ldots, \sigma_n) \in \Sigma^{n+1}$ with $\sigma_k P \sigma_{k+1}$ for every $k < n$. An **infinite improvement path** is a map $\sigma : \mathbb{N} \to \Sigma$ with $\sigma(k) P \sigma(k+1)$ for every $k$. A path is **maximal** when it is infinite or its final vertex has no outgoing $P$-edge.

A **generalized ordinal potential** is a map $\Phi : \Sigma \to \mathbb{K}$ satisfying

$$\sigma P \tau \Longrightarrow \Phi(\sigma) < \Phi(\tau).$$

In the numerical setting, an **exact potential** additionally satisfies $\Phi(\tau) - \Phi(\sigma) = u_i(\tau) - u_i(\sigma)$ for every $(\sigma, \tau) \in D_i$ and every owner $i$. Exact potentials can be analyzed through the same gain interface; neither kind of potential is identified with welfare or an adjustment scheduler.

**Lemma 13.4** (Exact potential in a cancellative value monoid)**.** *For values $u_i, \Phi : \Sigma \to M$ in a commutative cancellative monoid $M$, the signed gains of $u_i$ and $\Phi$ agree in $\mathrm{Gr}(M)$ on the owner-$i$ deviations exactly when*

$$\Phi(\tau) + u_i(\sigma) = \Phi(\sigma) + u_i(\tau) \qquad ((\sigma, \tau) \in D_i).$$

*Proof.*

(1) **Construct signed differences.** On $M^2$ put $(a,b) \sim (c,d)$ when $a+d=b+c$. For transitivity add the two balance equations and cancel the common intermediate terms. Reflexivity and symmetry follow from commutativity. Componentwise addition descends to the quotient, $[(0,0)]$ is its unit and $[(b,a)]$ is the inverse of $[(a,b)]$. Every monoid map $f$ from $M$ to an abelian group extends uniquely by $[(a,b)] \mapsto f(a) - f(b)$, giving the group completion.

(2) **Compare deviations.** Represent the gain of $u_i$ by $[(u_i(\tau), u_i(\sigma))]$. The equivalence relation identifies its sum along two successive deviations with the gain between their outer endpoints. Equality with the corresponding potential gain is precisely the displayed balance equation.

(3) **Recover exact numerical gains.** For $M = R_+$ the completion is $R_\pm$. For signed values, adding the two initial values to the difference equation gives the balance equation, and cancellation recovers the difference equation. Strict improvement is still read from the owner's comparison on its value carrier.

□

**Corollary 13.5** (A shared execution price and exact potentials)**.** *Let $\Phi$ be an exact potential for intrinsic profile values $u_i^0$. For a common descended scalar deduction $b : \Sigma \to \mathbb{K}$, the measured utilities $u_i = u_i^0 - b$ have exact potential $\Phi - b$.*

*Proof.*

(1) **Compute a measured gain.** For an $i$-deviation $(\sigma, \tau)$, the change in $u_i$ equals the change in $u_i^0$ minus $b(\tau) - b(\sigma)$. The intrinsic change is $\Phi(\tau) - \Phi(\sigma)$ by exactness. Rearrangement gives exactly $(\Phi - b)(\tau) - (\Phi - b)(\sigma)$.

(2) **Identify the deduction.** For an additive price $\ell$, a charge whose price descends gives $b(qk) = \ell(m(k))$. Thus the calculation uses subtraction in the ordered payoff group and only addition in the charge monoid. For distinct owner prices the exactness test uses the corresponding owner's deduction on each deviation, as in Lemma 13.4.

□

**Theorem 13.6** (Finite improvement and equilibrium)**.** *Let the profile space $\Sigma$ be finite and nonempty. The following are equivalent: $P$ has no directed cycle; every improvement path is finite; and an integer-valued generalized ordinal potential exists. Under these conditions every maximal improvement path ends at a Nash equilibrium, and at least one such equilibrium exists. A potential can be constructed with values in $\{0, \dots, |\Sigma| - 1\}$.*

*Proof.*

(1) **Infinite paths and directed cycles.** A directed cycle can be repeated to make an infinite path. Conversely an infinite path in a finite graph repeats a vertex and therefore contains a directed cycle. An increasing potential excludes such a cycle, since returning to its starting vertex would strictly increase the same potential value.

(2) **Construction of a ranking function.** If the graph is acyclic, define $h(\sigma)$ to be the largest length of a directed path ending at $\sigma$. Every such path is simple, so this integer exists and is at most $|\Sigma| - 1$. For an edge $\sigma P \tau$, a longest path to $\sigma$ cannot already contain $\tau$, since that would give a cycle.

(3) **Strategic comparison.** Appending the edge proves $h(\tau) \geq h(\sigma) + 1$. Thus $h$ is the required potential. Every maximal path is finite and ends at a vertex with no profitable deviation, exactly Equation (5.6).

(4) **Strategic comparison: completion.** For existence, take a profile on which the integer $h$ is maximal. It cannot have an outgoing improvement edge, since every such edge raises $h$ by at least one. Thus it is a Nash equilibrium.

□

An arrow records a profitable deviation. A profile satisfies the equilibrium test exactly when it has no outgoing arrow.

$$\begin{array}{ccc} \mathsf{PDev}_i & \hookrightarrow & L \times L \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle \pi_1} \\ L & = & L \end{array}$$

This is also a logical termination certificate. The integer $\rho = |\Sigma| - 1 - h$ satisfies

$$P(\sigma, \tau) \Longrightarrow \rho(\tau) < \rho(\sigma).$$

The rank is a certificate for paths that actually take profitable steps. A process permitted to idle at a non-equilibrium state can still run forever; no fairness principle turns arbitrary execution into an improvement path. The termination result is therefore about the displayed improvement relation, not every operational transition.

## 13.2 Epistemic rationality and elimination

Situated rationality combines the realized profile $s : W \to \Sigma^G$, supplied comparison feedback, and the owners' possibility relations. An agent's response is tested against the opponents it considers possible, which need not be the actual opponents. To connect such a predicate to elimination of strategies, we must keep the rectangular closure of these possibilities explicit and retain the witnesses used by best-response and dominance tests.

Public restriction uses the comprehension update of Section 9.1. Iterated elimination applies that operation repeatedly, with the rationality predicate evaluated afresh in the surviving model. Following van Benthem and Gheerbrant (2010); van Benthem, Pacuit, and Roy (2011), we compare the policies that survive with the policies compatible with common belief in rationality.

### 13.2.1 Belief, common belief, and survival under elimination

**Definition 13.7** (Pointwise rationality and rectangular closure)**.** For $S \hookrightarrow \Sigma = \prod_{j \in N} \Sigma_j$, put

$$\mathrm{Rect}(S) = (\pi_j[S])_{j \in N}, \qquad \mathrm{Prod}(\mathrm{Rect}(S)) = \prod_{j \in N} \pi_j[S] \hookrightarrow \Sigma.$$

The latter is the **rectangular closure**: it recombines the coordinates of represented profiles independently. Given a possibility model $s : W \to \Sigma$ with possibility fibres $P_i(w) = \{w' : wR_iw'\}$, form

$$Q_i(w) = \prod_{j \in N} s_j[P_i(w)] \hookrightarrow \Sigma.$$

For numerical utility maps $u_i : \Sigma \to \mathbb{K}$, owner $i$ is **pointwise rational** at $w$ precisely when

$$\exists t \in Q_i(w)\ \forall a_i \in \Sigma_i, \qquad u_i(s_i(w), t_{-i}) \geq u_i(a_i, t_{-i}).$$

Thus rationality uses a possible pure opponent tuple, not expected utility or simultaneous optimality against every possible tuple. The universal property of Rect and the internally typed rationality predicate are given below.

Pointwise rationality asks whether a policy is a pure best response to some opponent profile considered possible. That differs from expected-utility maximization and from one response optimal at every possible opponent profile. In particular, the profiles $(a, b)$ and $(a', b')$ generate the rectangular opponent set $\{a, a'\} \times \{b, b'\}$, including the crossed combinations. The rectangular opponent set below uses that rectangular closure so that individual beliefs are not silently treated as a correlated joint opponent policy.

Fix a possibility model as in Section A.2.14, over a $\mathbb{K}$-valued strategic game. The rectangular closure used below has a universal property.

**Proposition 13.8** (Rectangular closure)**.** *For $S \subseteq \Sigma = \prod_j \Sigma_j$ put*

$$\mathrm{Rect}(S) = (\pi_j[S])_{j \in N}, \qquad \mathrm{Prod}(L) = \prod_j L_j.$$

*With rectangular restrictions ordered coordinatewise,*

$$\mathrm{Rect}(S) \leq L \quad \Longleftrightarrow \quad S \subseteq \mathrm{Prod}(L). \tag{13.1}$$

*Hence* $\mathrm{Rect} \dashv \mathrm{Prod}$*, and* $\mathrm{Prod}(\mathrm{Rect}(S))$ *is the closure of $S$ under independent recombination of its represented owner coordinates.*

*Proof.*

(1) **Adjunction and transposition.** The inequality $\mathrm{Rect}(S) \leq L$ says $\pi_j[S] \subseteq L_j$ for every owner $j$. This holds exactly when every $s \in S$ has each coordinate in $L_j$, namely when $S \subseteq \prod_j L_j$. The adjunction follows.

(2) **Rectangular-closure unit.** Its unit $S \subseteq \mathrm{Prod}(\mathrm{Rect}(S))$ is the asserted rectangular closure.

□

**Corollary 13.9** (Rectangular algebras)**.** *The endomap* $\mathrm{Prod}\,\mathrm{Rect}$ *is an idempotent monad on* $\mathsf{Pred}(\Sigma)$*. Its algebra poset consists of the subsets that are products of coordinate subsets, ordered by inclusion.*

*Proof.*

(1) **Adjunction and transposition.** The adjunction gives an extensive monotone map, and the two triangle inequalities give idempotence. Its fixed subsets are exactly the image of Prod, because any right-adjoint value is fixed by the induced posetal monad and every fixed value has that form.

□

**Definition 13.10** (Pointwise rationality)**.** For agent $i$ define

$$Q_i(w) = \mathrm{Prod}(\mathrm{Rect}(s[P_i(w)])) = \prod_{j \in N} s_j[P_i(w)].$$

Rectangularity is the closure under independent recombination of strategy coordinates represented at states related to $w$. Player $i$ is **pointwise rational** at $w$ when some $t \in Q_i(w)$ makes its current strategy a global best response:

$$\exists t \in Q_i(w)\ \forall a_i \in \Sigma_i, \qquad u_i(s_i(w), t_{-i}) \geq u_i(a_i, t_{-i}). \tag{13.2}$$

This is rationality relative to a possible pure opponent profile, not expected utility relative to a probability distribution. Let Rat be the states where every agent is pointwise rational and put

$$E(X) = \bigcap_i B_i X.$$

**Proposition 13.11** (Pointwise rationality after private action update)**.** *Let $s : W \to \Sigma$ be the world realization of the qualitative possibility model, and apply an epistemic action model $\mathcal{E} = (E, (S_i), P)$. Use the unchanged reference-profile term $sp : U \to \Sigma$. At $(w, e) \in U$, define the old-world projection of the updated possibility set by*

$$V_i^{\mathcal{E}}(w, e) = \{w' \in W : \exists f \in E,\ R_i(w, w') \wedge S_i(e, f) \wedge P_f(w')\}.$$

*The updated pointwise-rationality predicate of owner $i$ is exactly*

$$\exists t \in \prod_{j \in N} s_j[V_i^{\mathcal{E}}(w, e)]\ \forall b_i \in \Sigma_i, \qquad u_i(s_i(w), t_{-i}) \geq u_i(b_i, t_{-i}). \tag{13.3}$$

*Thus the same selection predicate is tested against the product-updated possibility fibre, which can change under private action information even when the economic policy and utility maps do not change.*

*Proof.*

(1) **Policy-level comparison.** Equation (9.9) says that the updated possible states at $(w, e)$ are exactly the enabled pairs $(w', f)$ with $R_i(w, w')$ and $S_i(e, f)$. Their image under the unchanged profile map $sp$ is the set of profiles $s(w')$ with $w' \in V_i^{\mathcal{E}}(w, e)$. Taking rectangular closure gives the product in Equation (13.3), as in Definition 13.10.

(2) **Strategic comparison.** The global best-response inequality uses the same current policy $s_i(w)$ and original alternatives $\Sigma_i$, proving the formula even when the updated possibility fibre is empty.

□

**Construction 13.12** (Rationality and common belief)**.**

(1) **Form the greatest fixed point.**

$$C = \nu X(\mathrm{Rat} \wedge E(X)) = \mathrm{Rat} \wedge \mathsf{CB}(\mathrm{Rat}). \tag{13.4}$$

(2) **Identify the common-belief predicate.** The equality is Proposition 6.10.

By Theorem 12.2, $C \subseteq \mathrm{Rat}$ and $P_i(w) \subseteq C$ whenever $w \in C$. Equivalently, $C$ consists of states that are rational themselves and whose states reachable by every finite positive-length chain of supplied belief relations are rational. The factorization above makes explicit that no factivity axiom has been hidden in the word "belief."

**Definition 13.13** (Local strict dominance and iterated elimination)**.** For a restriction $L = (L_j)_{j \in N}$ with $L_j \subseteq \Sigma_j$, a policy $a_i \in L_i$ is **strictly dominated within** $L$ if there is $b_i \in L_i$ such that the opponent product $\prod_{j \neq i} L_j$ is inhabited and

$$u_i(b_i, a_{-i}) > u_i(a_i, a_{-i}) \quad \text{for every } a_{-i} \in \prod_{j \neq i} L_j.$$

**Iterated elimination** repeatedly removes such dominated policies, using the remaining policy sets at each step and intersections at limits.

The tested policy and its possible dominator are both drawn from the current restriction. The explicit opponent witness also prevents vacuous strict dominance when the opponent carrier is empty. This is a different test from being a best response to at least one possible opponent profile; changing their quantifiers changes the resulting elimination operator.

**Construction 13.14** (Pure strict-dominance elimination)**.**

(1) **Iterate strict-dominance elimination.** Define iterated elimination of locally strictly dominated pure strategies by $L_i^0 = \Sigma_i$ and

$$\begin{aligned} L_i^{\alpha+1} = \{a_i \in L_i^\alpha : \text{there is no } b_i \in L_i^\alpha \text{ such that} \\ (\exists a_{-i} \in \prod_{j \neq i} L_j^\alpha) \wedge \\ \forall a_{-i} \in \prod_{j \neq i} L_j^\alpha,\ u_i(b_i, a_{-i}) > u_i(a_i, a_{-i})\}, \end{aligned}$$

with intersections at limit ordinals.

(2) **Test opponents in the restriction.** The explicit witness of an opponent profile prevents strict dominance from becoming vacuously true after a restriction has no realizable opponent tuple. On every inhabited rectangular restriction this is exactly the ordinary definition. Each stage is determined extensionally by the preceding restriction; no strategy is selected for deletion.

(3) **Read off stabilization.** For finite games the sequence stabilizes after finitely many strict deletions. The survival theorem below only uses transfinite induction over these recursively defined stages, so no stabilization claim for arbitrary strategy sets is needed.

**Theorem 13.15.** *Every strategy played at a state in $C$ survives every stage of this elimination procedure.*

*Proof.*

(1) **Construction of the comparison.** Put $S_i = s_i[C]$. We prove $S_i \subseteq L_i^\alpha$ for every $i$ by transfinite induction. The initial and limit cases are immediate.

(2) **Fixed-point argument.** For the successor step, take $a_i = s_i(w)$ with $w \in C$. Pointwise rationality gives $t \in Q_i(w)$ satisfying Equation (13.2). Since $P_i(w) \subseteq C$, each coordinate $t_j$ belongs to $S_j$, hence by induction to $L_j^\alpha$.

(3) **Strategic comparison.** If a surviving $b_i$ strictly dominated $a_i$ against every surviving opponent profile, then in particular

$$u_i(b_i, t_{-i}) > u_i(a_i, t_{-i}),$$

contradicting the global best-response inequality, which quantifies over every original $b_i$. Thus $a_i$ is not removed at the successor stage.

□

This reconstructs a qualitative rationality-to-survival comparison of the kind considered in Apt and Zvesper (2010). Global best response and local strict dominance have deliberately different quantifier patterns, and the theorem relies on those exact definitions. The resulting implication uses the pointwise rationality predicate of (13.2), not an expected-payoff or arbitrary best-response variant.

Epistemic conditions leading to Nash equilibrium are studied in Aumann and Brandenburger (1995). Here the comparison keeps actual and possible outsiders distinct at every step, so the epistemic assumptions are represented by the supplied possibility relation rather than folded into the outcome comparison.

### 13.2.2 Common belief under public restriction

**Construction 13.16** (Relativized common belief)**.**

(1) **Restrict the belief relation.** For $a : U \hookrightarrow W$ put $A = \exists_a \top$, $E_W(X) = \bigwedge_i [R_i]X$, and $E_U(X) = \bigwedge_i [R_i|_U]X$.

(2) **Lift a predicate from the restriction.** For $Q$ on $U$, put $\widehat{Q} = \forall_a Q$ and

$$\mathsf{RCB}_{A,\widehat{Q}} = \nu X \, (A \Rightarrow E_W(A \Rightarrow (\widehat{Q} \wedge X))).$$

**Theorem 13.17** (Common belief after restriction)**.**

$$[!a]\mathsf{CB}_U(Q) = \mathsf{RCB}_{A,\widehat{Q}}. \tag{13.5}$$

*Proof.*

(1) **Construction of the comparison.** Let $\rho = a^*$, $J = \forall_a$, and $j_A = J\rho$. The public-update equation gives $JE_U(X) = j_A E_W(JX)$. For $F(X) = E_U(Q \wedge X)$, the operator $JF\rho$ is therefore

$$Y \longmapsto j_A E_W(JQ \wedge j_A Y) = j_A E_W(j_A(\widehat{Q} \wedge Y)),$$

since $\widehat{Q}$ is fixed by the meet-preserving nucleus $j_A$. Apply Corollary 9.10 to $\nu F$.

□

Restriction of the epistemic model is obtained by intersecting each accessibility relation with the announced subspace.

$$\begin{array}{ccc} U \times U & \hookrightarrow & W \times W \\ {\scriptstyle \pi_1}\downarrow & & \downarrow{\scriptstyle \pi_1} \\ U & \xhookrightarrow[m]{} & W \end{array}$$

### 13.2.3 An exact epistemic characterization

**Definition 13.18** (Global pure best responses and point rationalizability)**.** A **point-rationalizable** policy survives repeated removal of policies that cannot be a best response to any remaining pure opponent tuple. In the global operator defined here, a candidate is tested against all of the agent's original alternatives; only the possible opponents are restricted at each step. This distinction from local dominance elimination determines the

exact epistemic characterization. No distribution over opponents is used in this notion of rationalizability.

The preceding survival theorem can be strengthened to an exact characterization when the elimination operator matches the rationality predicate. For a finite game, let a restriction be a tuple $L = (L_i)_{i \in N}$ with $L_i \subseteq \Sigma_i$. Define

$$\mathcal{B}_i(L) = \{a_i \in \Sigma_i : \exists t_{-i} \in \prod_{j \neq i} L_j \ \forall b_i \in \Sigma_i, \quad u_i(a_i, t_{-i}) \geq u_i(b_i, t_{-i})\}.$$

The opponent witness ranges over the current restriction, but the comparison ranges over every original alternative of the agent. Throughout this paper, "point rationalizability" refers to the greatest fixed point of this displayed global-best-response operator with pure opponent witnesses. The operator $\mathcal{B} = (\mathcal{B}_i)_i$ is monotone on the product lattice $\prod_i \mathcal{P}(\Sigma_i)$. Call its greatest fixed point $R = \nu\mathcal{B}$ the point-rationalizable restriction.

**Construction 13.19** (Descending best-response iteration)**.**

(1) **Iterate best response.** Starting at the full restriction, put

$$L^0 = \Sigma, \qquad L^{n+1} = \mathcal{B}(L^n).$$

The sequence is descending: $\mathcal{B}(\Sigma) \subseteq \Sigma$, and if $L^n \subseteq L^{n-1}$ then monotonicity gives $L^{n+1} = \mathcal{B}(L^n) \subseteq \mathcal{B}(L^{n-1}) = L^n$.

(2) **Identify the greatest fixed point.** The product lattice is finite, so the sequence stabilizes at some $L^*$ with $L^* = \mathcal{B}(L^*)$. If $F = \mathcal{B}(F)$ is any fixed point, then $F \subseteq L^0$, and induction gives $F = \mathcal{B}(F) \subseteq \mathcal{B}(L^n) = L^{n+1}$. Hence $F \subseteq L^*$, so $L^*$ is the greatest fixed point $R = \nu\mathcal{B}$. No separate nonemptiness premise is needed. The fixed-point equation itself supplies exactly the extension property required by the epistemic construction.

**Lemma 13.20** (Surviving strategies extend to surviving profiles)**.** *Let $R = \mathcal{B}(R)$. For every owner $i$ and every $a_i \in R_i$, there is a profile $w \in \prod_j R_j$ with $w_i = a_i$. Consequently, if any surviving individual strategy exists then the full product $\prod_j R_j$ is nonempty.*

*Proof.*

(1) **Strategic comparison.** Since $a_i \in R_i = \mathcal{B}_i(R)$, the definition of $\mathcal{B}_i$ provides a tuple

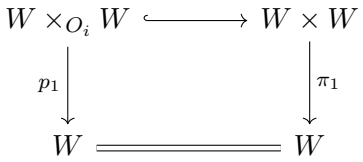
$$t_{-i} \in \prod_{j \neq i} R_j$$

against which $a_i$ is a global best response. The profile $w = (a_i, t_{-i})$ lies in $\prod_j R_j$ and has the required $i$th coordinate. No selection from a family of nonempty sets is involved: the needed opponent tuple is already the witness carried by membership in the fixed point.

□

**Theorem 13.21** (Rationality with common belief and point rationalizability)**.** *In every possibility model satisfying the definitions of Section 13.2, every strategy played at a state satisfying pointwise rationality together with common belief of pointwise rationality belongs to $R$. Equivalently, the conclusion holds at every state in the predicate $C$ of Equation (13.4). Conversely every profile in $\prod_i R_i$ occurs in a finite epistemic model in which agents know their own strategies and pointwise rationality is common knowledge at every state. Thus the surviving strategies have an exact epistemic realization.*

Pointwise rationality uses the actual observation kernel pair on epistemic states.



*Proof.*

(1) **Witness and legality comparison.** For the forward direction, let $C$ be the rationality-and-common-belief predicate constructed above and put $S_i = s_i[C]$. For $a_i = s_i(w)$ with $w \in C$, rationality supplies an opponent witness drawn from $s_j[P_i(w)]$ for each $j \neq i$. Possibility closure gives $s_j[P_i(w)] \subseteq S_j$.

(2) **Coinduction and bisimulation.** The rationality inequality is global in the agent's alternatives, so $a_i \in \mathcal{B}_i(S)$. Therefore $S \subseteq \mathcal{B}(S)$. Coinduction in the product lattice gives $S \subseteq R$.

(3) **Reverse implication.** For the converse take the finite state space

$$W = \prod_i R_i.$$

It may be empty only when there is no surviving individual strategy to represent, by Lemma 13.20. Let the strategy map be inclusion into $\Sigma$ and define $o_i(w) = w_i$.

(4) **Choice of representatives and quotients.** Take the epistemic relation to be the constructed kernel pair

$$R_i^{\text{ep}} = \ker(o_i),$$

so its possibility fiber is

$$P_i(w) = \{t \in W : t_i = w_i\}.$$

Player $i$ therefore knows its own strategy. Since $R_i = \mathcal{B}_i(R)$, the current strategy $w_i$ is a global best response to some tuple in $R_{-i}$. Combining that tuple with $w_i$ gives a state in $P_i(w)$ and hence a witness for Equation (13.2).

(5) **Common knowledge of rationality.** Thus all agents are rational at every state. Every epistemically accessible state is again in $W$, so rationality is common knowledge everywhere. Every profile in $W$ is represented by construction.

(6) **Policy-level comparison.** Lemma 13.20 shows in addition that every surviving individual strategy occurs as a coordinate of one of those profiles. Thus the same constructed model realizes all surviving behavior without a separate nonemptiness premise.

□

The converse constructs a finite state space and observation maps in which every surviving profile satisfies the specified common-knowledge rationality condition. It also identifies the relevant elimination operator: global never-best-response elimination with pure opponent witnesses. This is the exact characterization compared with Apt and Zvesper (2010), not a claim about every iterative deletion procedure.

**Example 13.22** (Pure best responses versus pure strict dominance)**.** Pure-witness rationalizability need not coincide with elimination of pure strictly dominated actions. For example,

give agent 1 the rows $a, b, c$, agent 2 the columns $L, R$, and utilities

| | $L$ | $R$ |
|---|---|---|
| $a$ | 2 | 0 |
| $b$ | 0 | 2 |
| $c$ | 11/10 | 11/10 |

for agent 1; give agent 2 utility zero everywhere. No row strictly dominates $c$, but $c$ is not a best response to either pure column and is removed by $\mathcal{B}$. This exact difference is why the development has not called all these elimination procedures one notion of rationality. Probability distributions over opponent choices give a further best-response operator in probabilistic models of games.

The inclusion of a public comprehension has the quantifier adjoints of Theorem 5.22. When an update also changes the policy carriers, the counterfactual relation must be transported by Construction 5.64. The rectangular-image and possibility comparisons therefore occur at the correct updated strategic context, rather than merely at its reference world.

### 13.2.4 Re-evaluation and repeated public restriction

Iterated public announcements are successive comprehensions of the epistemic state carrier, as in Plaza (2007). The relevant rationality or survival predicate is re-evaluated after each restriction. Keeping this change of model explicit distinguishes public elimination from simply repeating a predicate on the original state space.

**Proposition 13.23** (Finite stabilization of re-evaluated announcements)**.** *For every submodel carrier $A \subseteq W$, let $\Phi_A \subseteq A$ be a specified updated goal predicate, for example the denotation of a fixed epistemic formula in $M|_A$. Define*

$$F_\Phi(A) = \Phi_A \quad \textit{viewed as a subset of } W, \qquad A_0 = W, \quad A_{n+1} = F_\Phi(A_n).$$

*If $W$ is finite, this chain stabilizes after at most $|W|$ strict deletions. At the stable carrier the goal holds everywhere. If $F_\Phi$ is monotone, the stable carrier is its greatest fixed point. Without monotonicity that last conclusion need not hold.*

*Proof.*

(1) **Fixed-point argument.** By construction $F_\Phi(A) \subseteq A$. Every strict step removes at least one element; thus there are at most $|W|$ strict steps. Equality at one step is the fixed-point equation.

(2) **Fixed-point argument: second comparison.** If $F_\Phi$ is monotone, every fixed point $B$ is contained in $A_0$ and induction gives $B = F_\Phi(B) \subseteq F_\Phi(A_n) = A_{n+1}$.

(3) **Greatest fixed point and counterexample.** This proves greatestness. For failure without monotonicity, use the two-world model of Example 9.8 and $\Phi_A = [\![Kp]\!]_{M|_A}$. Then $F_\Phi(W) = \varnothing$ while $F_\Phi(\{x\}) = \{x\}$.

□

The operator $\mathcal{B}$ records one such elimination step, and $\nu\mathcal{B}$ denotes the states surviving every finite iteration. Its greatest-fixed-point meaning is justified by the descending comprehensions, not by treating a public update as an ordinary reference substitution.

### 13.2.5 Public elimination as an epistemic update

For each current policy carrier, the institutional test agent of Construction 11.10 uses the current elimination predicate as its observable test. By Proposition 11.11, success of that public test realizes the next comprehension carrier. Thus the fixed-point elimination sequence also has an agent-labelled execution at each step; its predicate is evaluated anew on the surviving carrier.

**Construction 13.24** (Successive rationality announcements)**.**

(1) **Initialize the epistemic model.** Begin with state space $W_0 = \prod_i \Sigma_i$ and observations $o_i(s) = s_i$.

(2) **Announce rationality at each stage.** At a stage with $W_n = \prod_i L_i^n$, announce the assertion

$$\mathrm{Rat}_n(s) \quad \Longleftrightarrow \quad \forall i\ \exists t_{-i} \in L_{-i}^n\ \forall b_i \in \Sigma_i,\ u_i(s_i, t_{-i}) \geq u_i(b_i, t_{-i}).$$

This is pointwise rationality in the restricted observation model: the agent's own action is known, while every surviving opponent tuple is possible.

(3) **Restrict the model.** Public announcement restricts the state space and its observation equivalence relations as in Equation (9.5).

**Proposition 13.25** (Update equals elimination)**.** *Repeated announcements of these current rationality predicates give*

$$W_{n+1} = \prod_i \mathcal{B}_i(L^n).$$

*They stabilize at the canonical point-rationalizable epistemic model with state space* $\prod_i R_i$.

*Proof.*

(1) **Descent along the kernel pair.** Within the current product domain, the announcement says independently for each coordinate that $s_i \in \mathcal{B}_i(L^n)$. Thus the new domain is $\prod_i (L_i^n \cap \mathcal{B}_i(L^n))$. Along the descending iteration already proved, $\mathcal{B}_i(L^n) \subseteq L_i^n$, giving the displayed equality.

(2) **Observation and knowledge.** Observation relations remain equality of the appropriate coordinate, restricted to the new domain. Each announcement is the singleton action model with enabling predicate $\mathrm{Rat}_n$ and a trivial action relation; their iterated product update is canonically the successive comprehension by Theorem 9.25. Finite stabilization of the restrictions therefore proves stabilization of the models as well.

□

An elimination step is the pullback of the surviving-strategy inclusion along the profile map.

$$\begin{array}{ccc} W_{n+1} & \hookrightarrow & W_n \\ {\scriptstyle s_{n+1}}\downarrow & & \downarrow{\scriptstyle s_n} \\ \Sigma_{n+1} & \hookrightarrow & \Sigma_n \end{array}$$

The resulting public updates realize the successive eliminations through the announcement and product-update semantics, while retaining the rationality predicate appropriate to each restricted game. This explains the link with the elimination result: repeatedly announcing one fixed subset of the original model would not, in general, implement the same sequence.

**Proposition 13.26** (Actual-opponent rationality)**.** *At a pointwise rational state $w$, if every state considered possible by $i$ has opponents' tuple $s_{-i}(w)$, for every $i$, then $s(w)$ is Nash. Conversely every Nash profile occurs in a one-state observation model with these properties.*

*Proof.*

(1) **Observation and knowledge.** The rationality witness has the actual opponents' tuple, so its inequalities are the Nash inequalities. Conversely take a singleton state with the given profile and the unique observation maps. Its only possible tuple is the given profile, and its Nash inequalities establish pointwise rationality.

□

At a fixed price, an opponent tuple that witnesses a global best response still witnesses it when the set of possible opponents expands. All the alternatives against that tuple have the same measured values as before, by Proposition 5.43. This gives the monotonicity step in Theorem 13.21. Changing the price can change the winning comparison. Revealing the charge to the owner would additionally require a signal and its observation map.

## 13.3 Reporting agents, incentive compatibility and truthfulness

A checker that accepts a report has received a message. What would give the sender an incentive to report its true type? We model the types and messages as ordinary data in $\mathsf{T}$ and their transmission and receipt as linear processes in $\mathsf{L}$. Incentive compatibility compares the records produced by truthful and alternative messages at the same true type.

### 13.3.1 Type-indexed reporting interfaces

**Definition 13.27** (Economic types and type-dependent valuations)**.** An **economic type** of owner $i$ is a point $t_i \in T_i$ of its specified ordinary type carrier. For an outcome or allocation carrier $Z$, a **type-dependent valuation** has type

$$v_i : T_i \times Z \longrightarrow \mathbb{K}, \qquad (t_i, z) \longmapsto v_i(t_i, z).$$

If an outcome is presented by a resource certificate in $K$ with allocation map $a : K \to Z$, the corresponding evaluation of a certificate is the composite $T_i \times K \xrightarrow{1\times a} T_i \times Z \xrightarrow{v_i} \mathbb{K}$; a certificate need not be identified with its allocation. The true type is held fixed when reports are varied. Economic types are ordinary parameters of valuation, distinct from resource types and from dependent semantic types.

**Definition 13.28** (Messages, reports, and truthful reporting)**.** For an economic type carrier $T_i$ and an allowed message carrier $M_i$, a **message** is $m_i \in M_i$. A **report** is a message interpreted by the institution as a declaration of information about type. A **reporting policy** has type

$$s_i : T_i \longrightarrow M_i;$$

its value $s_i(t_i)$ is the message sent at true type $t_i$. In a **direct mechanism** $M_i = T_i$, and the **truthful reporting policy** is $\mathrm{tr}_i = \mathrm{id}_{T_i} : T_i \to T_i$. For finite carriers, its concurrent response realization has type $\widehat{s_i} : D(T_i) \to D(M_i)$. A general mechanism may instead use messages that do not denote types; no probabilistic inference from a message to a true type is implied.

Each reporting policy is a type-indexed function $s_i : T_i \to M_i$. It chooses the message associated with each possible type, including types not realized in the current execution. No prior or probabilistic inference from a message to a true type is introduced by the protocol.

**Construction 13.29** (Concurrent reporting protocol)**.**

(1) **Wire the reporting components.** For finite message sets, a reporting rule has a direct concurrent realization even when its ambient outcome set is infinite. Tensor the agents' message-selection components and collect the chosen messages.

(2) **Collect the profile of messages.** The collector has one negative event for each possible input on each component; distinct values on a component conflict. For each full tuple it has an output event labelled by that tuple, dependent on all its input events. An input can occur as soon as its component is ready, in any order. Its only new dependencies are negative-to-positive, and it accepts every enabled input, so it is a receptive courteous strategy.

(3) **Compute the reported outcome.** If $m : \prod_i M_i \to R$, $R = T_C Z$, is the measured rule, construct its finite image $F_m = m[\prod_i M_i]$ and corestrict it to $\bar{m} : \prod_i M_i \to F_m$. Proposition 1.16 realizes $\widehat{\bar{m}}$ on the finite outcome arena $D(F_m)$. Thus the construction preserves concurrency of reporting while explicitly waiting for the data the rule needs; no operational presentation of unused elements of $R$ is required. The token labels a declared measured result. If the reporting circuit itself is separately metered, its physical charge is computed on the corresponding secured witness.

### 13.3.2 Dominant strategies and direct mechanisms

**Definition 13.30** (Mechanism, selected dominance, and direct truthfulness)**.** A reporting strategy is **dominant** when, for every true type and every outsider message tuple, its prescribed message is at least as good as every alternative message. The weak inequality permits ties. A direct mechanism is **dominant-strategy incentive compatible** (DSIC) when truthful reporting has this property. Nash equilibrium fixes the outsiders' actual policies in its comparison; dominance requires the comparison for every outsider message tuple. A dominant truthful strategy can coexist with other equilibrium reports.

Fix finite type and message sets $T_i, M_i$, put $M = \prod_i M_i$, and use the measured result sort $R = T_C Z$. A reporting rule is $m : M \to R$ and the numerical evaluator is $e_i : T_i \times R \to \mathbb{K}$. The ordinary outcome rule is $\epsilon_Z m$. Conversely an unpriced rule $m_0 : M \to Z$ and utility $u_i^0$ give $m = \eta_Z^T m_0$, $e_i(t,(c,z)) = u_i^0(t,z)$. This embeds the original reporting interface and recovers its inequalities. Only the finite image of $m$ need be represented by the finite-data protocol. Its selected strategy is a family $s_i : T_i \to M_i$. The selection is dominant if

$$\forall i, \theta_i, m_i, m_{-i}, \qquad e_i\big(\theta_i, m(s_i(\theta_i), m_{-i})\big) \geq e_i\big(\theta_i, m(m_i, m_{-i})\big). \tag{13.6}$$

The quantifiers range over types and messages without a prior distribution.

A direct rule $r : \prod_i T_i \to R$ is **dominant-strategy incentive compatible** (DSIC) if truthful reporting is dominant. In the assertion language this is the formula

$$\forall i, \theta_i, \widehat{\theta}_i, \widehat{\theta}_{-i}, \qquad e_i\big(\theta_i, r(\theta_i, \widehat{\theta}_{-i})\big) \geq e_i\big(\theta_i, r(\widehat{\theta}_i, \widehat{\theta}_{-i})\big).$$

It is an explicit comparison of two realizations of the same typed reporting protocol. The report changes; the true type in the utility comparison does not.

**Construction 13.31** (The dependent type of dominant selections)**.**

(1) **Construct the dominant-message fibres.** For each owner, form the comprehension

$$\mathsf{Dom}_i(m) = \{(t,a) \in T_i \times M_i : \quad \forall \beta \in M_{-i}\, \forall b \in M_i, \\ e_i(t, m(a,\beta)) \geq e_i(t, m(b,\beta))\}.$$

with projection $d_i : \mathsf{Dom}_i(m) \to T_i$. Its section set is $\{s_i : T_i \to \mathsf{Dom}_i(m) : d_i s_i = \mathrm{id}_{T_i}\}$.

(2) **Select a message at every type.** Define the dependent product

$$\mathsf{Sel}(m) = \prod_i \mathsf{Sect}(d_i) \cong \prod_i \prod_{t \in T_i} \mathsf{Dom}_i(m)_t.$$

(3) **Verify incentive compatibility.** Evaluation gives the selected messages uniformly over $\mathsf{Sel}(m)$; write $s_i(t) \in M_i$ for the message component. Its defining inequality is exactly Equation (13.6). Substituting $b = s_i(\widehat{t})$ and $\beta = s_{-i}(\widehat{\theta}_{-i})$ proves DSIC of $r_s = m \prod_i s_i$. Thus a selected implementation is precisely an element of this constructed witness type, and directization is defined uniformly on that type.

(4) **Identify empty fibres.** The type can be empty. In the two-player bit-matching game, agent 1 receives 1 for equal bits and 0 otherwise. Message 0 fails to dominate against opponent message 1, and message 1 fails against opponent message 0. With a singleton true-type set, $\mathsf{Dom}_1(m)$ is empty. In a concrete truthful mechanism, its incentive proof exhibits the identity report map as an inhabitant of this predicate.

**Proposition 13.32** (Dominance as uniform agent selection)**.** *For a true type $t \in T_i$ and fixed outsider messages $\beta$, put*

$$q_{i,t,\beta}(a)(b) \quad \Longleftrightarrow \quad e_i(t, m(a,\beta)) < e_i(t, m(b,\beta)).$$

*Then selected dominance is $\forall i, t, \beta\ \varepsilon_{M_i}(s_i(t), q_{i,t,\beta})$. If $r = m \prod_i s_i$, the direct mechanism's feedback is the pullback of the mechanism's feedback along the selected report maps in both comparison arguments. At the measured sort this equality retains both the outcome and its charge. For a witness family $p_m : K_m \to M$ with record $r_m : K_m \to T_C Z$, directization along $s : T = \prod_i T_i \to M$ is given by*

$$\begin{array}{ccccc} T \times_M K_m & \xrightarrow{\pi_2} & K_m & \xrightarrow{r_m} & T_C Z \\ {\scriptstyle \pi_1}\downarrow & \lrcorner & \downarrow{\scriptstyle p_m} & & \\ T & \xrightarrow[s]{} & M & & \end{array}$$

*and record $r_m \pi_2$.*

*Proof.*

(1) **Strategic comparison.** Expanding selection gives the absence of every strictly improving message. In the linear order of $\mathbb{K}$ this is exactly Equation (13.6). For directization the play equation is $r = m \prod_i s_i$; applying Theorem 5.11 gives its two-argument feedback equation.

(2) **Direct-message alternative comparison.** Hence every alternative direct report is compared through its exhibited message, with the true type fixed in the continuation.

(3) **Lift each direct report witness.** The fibre at a reported tuple $t$ consists of $(t, k)$ with $p_m(k) = s(t)$. Projection to $k$ is a bijection with inverse $k \mapsto (t, k)$. Hence the record, its evaluation at the fixed true type, and the two quantifiers on witness pairs transport through these bijections. A separately executed adapter $f : T \to T_C M$ instead realizes $m^\dagger f$. For $f(t) = (d, s(t))$ its result is $S_d(m(s(t)))$. A report-dependent $d$ can change the comparison: even two reports with identical original payoff can acquire different net values. The pulled-back DSIC proof applies to the inherited measured rule; a separately priced adapter is tested on its composite record.

□

### 13.3.3 Dominance and direct truthfulness in the assertion doctrine

**Proposition 13.33** (Dominance in the internal reporting language)**.** *For owner $i$, put $\Gamma_i = T_i \times M_{-i}$ and let $\rho_i : \Gamma_i \times M_i \to \Gamma_i$ project the alternative message. Define the bad-alternative predicate*

$$B_i(t, \beta, a) \iff e_i(t, m(s_i(t), \beta)) < e_i(t, m(a, \beta)).$$

*Selected dominance is $\bigwedge_i \forall_{!:\Gamma_i \to 1} \forall_{\rho_i} \neg B_i$. Substituting $a = s_i(\widehat{t})$ and $\beta = s_{-i}(\widehat{\beta})$ gives direct truthfulness of $r = m \prod_i s_i$.*

*Proof.*

(1) **Adjunction and transposition.** The adjoint formulas expand the displayed predicate to all true types, all outsider messages, and all alternative messages, with no improving alternative. This is Equation (13.6). Universal elimination substitutes the two displayed report terms, leaving the true type $t$ fixed in both values.

(2) **Lifting along the surjection.** The resulting inequality is exactly DSIC for $r$. The terms take values in constant message types; neither surjectivity nor a choice of representatives is used.

□

Truthful reporting has now been checked against every outsider message and every alternative report. The remaining question is how the direct reporting rule relates to other implementations that realize the same measured rule. The selected message policies themselves supply the comparison maps.

**Construction 13.34** (The category of selected implementations)**.**

(1) **Fix the selected outcome rule.** Fix the type carriers $T_i$, measured result carrier $R = T_C Z$, and evaluators $e_i : T_i \times R \to \mathbb{K}$ of the reporting interface. For a selected dominant implementation $(M, m, s)$, put $r = m \circ \prod_i s_i$.

(2) **Specify the objects and arrows.** Define $\mathbf{Impl}_r$ to have as objects the selected dominant implementations $(M', m', s')$ of this same rule: each $M'_i$ is a finite message carrier, $s'_i : T_i \to M'_i$ is dominant for $m'$, and $m' \circ \prod_i s'_i = r$. A morphism

$$f : (M', m', s') \longrightarrow (M'', m'', s'')$$

is an ownerwise family $f_i : M'_i \to M''_i$ satisfying

$$m'' \circ \prod_i f_i = m', \qquad f_i \circ s'_i = s''_i.$$

(3) **Define composition.** Identities are coordinatewise identity maps, and composition is coordinatewise function composition. Substitution in the two displayed equations proves closure under composition; associativity and the unit laws follow from those of functions.

**Proposition 13.35** (Initiality of the truthful direct implementation)**.** *The truthful direct implementation $(T, r, \mathrm{id})$ is initial in $\mathbf{Impl}_r$. Its unique morphism to $(M', m', s')$ is the selected policy family $(s'_i)_i$.*

*Proof.*

(1) **Construct the object and its arrow.** By Proposition 13.33, $r = m \prod_i s_i$ is DSIC when evaluated at the true type, so $(T, r, \mathrm{id})$ is an object. For any $(M', m', s')$ of $\mathbf{Impl}_r$, $m' \prod_i s'_i = r$ is equality in $T_C Z$ and $s'_i \,\mathrm{id} = s'_i$. Thus $s'$ is a morphism.

(2) **Prove uniqueness and recover the ordinary comparison.** Any morphism $f$ from the direct object satisfies $f_i \,\mathrm{id} = s'_i$. Therefore $f = s'$. Erasure of the record equation gives the original outcome equation. For an ordinary fixed rule, lifting every implementation by $\eta^T$ preserves and reflects its morphism equations since $\epsilon\eta^T = 1$. This recovers the unpriced initiality statement on the zero-charge presentations.

□

**Proposition 13.36** (Reporting agents as observation-based response interfaces)**.** *At a reporting interface where agent $i$ observes its economic type $t_i$, the complete reporting carrier is $M_i^{T_i}$. Its policy $s_i$ realizes the response $t_i \mapsto s_i(t_i)$, and for finite carriers its concurrent realization is $\widehat{s_i}$. Connecting the reporting agents to the institutional outcome rule gives the reporting protocol above. Uniform selection over true types and outsider reports is the dominance predicate of Proposition 13.32.*

*Proof.*

(1) **Choice of representatives and quotients.** Take the observation quotient to be the identity of $T_i$ in the response interface of Proposition 6.7. Its policies are exactly the maps $T_i \to M_i$, and the finite decision generator realizes each by its response strategy. Tensor and the collector connect these responses to the same rule.

(2) **Ownerwise alternatives.** The selection comparison keeps $t_i$ fixed and varies only the report; quantification over all true types and outsider reports is exactly the displayed dominance construction. The internal identification is proved in Proposition 13.32.

□

# 14 Reconstruction and Exact Abstraction of Agent Systems

Two controllers can agree on their current action yet permit different interventions tomorrow. Whether we may identify them depends on the questions the reduced model must answer. We first reconstruct the declared owners, policies, observations, permissions, and execution witnesses from the displayed presentation. We then construct quotients for specified tests and prove which strategic and epistemic alternatives lift at a given source. Finally we distinguish what the fixed concurrent semantics can observe from which equations are derivable in the calculus.

## 14.1 Reconstructing the complete agent presentation

The fibred semantics retains a structured agent rather than merely its truth predicates: owner coordinates, concurrent realizations, comparisons, observations, legal alternatives, continuations, and execution witnesses. We first reconstruct this displayed data and its composition. Exact reductions will then retain the tests and source-fixed alternative lifts on which strategic and epistemic assertions depend. This also recovers the policy witnesses of the communication comparison in Section 11.4 and the observations and transitions of the visible refinement.

### 14.1.1 Reconstruction of the structured agent presentation

Reconstruction concerns the displayed economic presentation. Ownership, realization, observation, and comparison data must be recovered together; a collection of truth predicates alone would not identify who controls a given deviation. The constructions below exhibit inverse maps between the declared presentation and its fibred semantic record.

**Theorem 14.1** (Equivalence of agent presentations and fibred economic presentations)**.** *Fix the operational category $\mathcal{C}$ of Section 1.1. Its owner-indexed presentations and the constructed fibred open-arena presentations are equivalent, with inverse constructions in indexed-family notation and canonical equivalences in slice notation. The fibred presentation retains*

$$\vec{\Sigma}^G, \quad \mathsf{Assn}_G \longrightarrow \mathsf{St}(G), \quad \Delta_G, \quad \widehat{g} : 1_G \longrightarrow \Delta_G \mathcal{C}(A, B),$$

*the designated owner projections and deviation contexts, and canonical coplay. It retains every ownerwise reparametrisation, including its deviation comparison maps, and commutes with sequential and parallel composition.*

*Equivalently, by Proposition 2.13, these are the agent presentations with their designated interfaces. The correspondence recovers the complete-policy composition law and the control regroupings.*

*For a measured execution the retained family consists of $p : K \to \Sigma$ with $r = \langle m, a \rangle : K \to T_C Z$ and the evaluators $e_i : T_C Z \to V_i$. The projections of $r$ recover $m, a$; the composite $e_i r$ recovers the evaluated utility. Comprehension of $B(m(k))$ reconstructs the budget witnesses before taking their endpoint image. Retaining $e_i r$ alone preserves that evaluation; changing its price or budget can distinguish witnesses with the same retained value.*

*The correspondence extends to evaluated, observed, legal, history-indexed, certified, and epistemically action-decorated games by retaining their evaluation terms, observation and action-relation terms, enabling comprehensions, updated observable terms, dependent continuation families, and certificate carriers with their structure maps. Public inspection, the full dependent announcement games and their complete-policy realizations, and the exact visible refinements are recovered from these retained data. Every interpreted formula of the agency language has the same denotation. Player selection, best response, Nash equilibrium, coalition stability, continuation equilibrium, and truthful implementation are preserved and reflected on their original carriers.*

The proof is given in Appendix B.

The recovered object includes the full declared presentation. The hyperdoctrine alone would not determine ownership or concurrent play, since distinct realizations can share one profile carrier. Keeping the underlying arena and its distinguished terms makes the reconstruction reversible. The operational arrow remains taken up to arena-preserving strategy isomorphism; the reductions to follow are additional quotients relative to explicitly chosen economic tests.

**Corollary 14.2** (Recovery from the completed dependent semantics)**.** *Embedding an original game and its dependent arena families into the completed model preserves and reflects its concurrent realizations and all the economic predicates in Theorem 14.1.*

*Proof.*

(1) **Representation and Yoneda decoding.** For each representable boundary, recover a realized map $\alpha : yA \to yB$ as $\alpha_c(u)u^{-1}$ for any local isomorphism $u : Jc \to A$, by Theorem 1.31. This is inverse to embedding and commutes with composition, tensor, and parameter substitution.

(2) **Predicate calculation.** Ordinary carriers and their maps are recovered by $\mathsf{ValLin} \cong 1$; the subterminal fragment, its quantifiers, and its complete predicate lattices are the same ones. Apply the preceding reconstruction proof to these recovered maps. The argument applies fibrewise to dependent residual arenas and retains the specified certificate objects and their realization maps.

□

**Corollary 14.3** (Recovery of action-updated concurrent realizations)**.** *For an original concurrent interaction with a specified epistemic action model, embedding its boundaries by Yoneda preserves the action enablement comprehension and all ordinary updated epistemic relations. For an operationally generated action, its post-action residual is recovered pointwise from the updated dependent linear internal-hom value family. Secured weak-step witnesses and their observable predicates are recovered on their own dependent carriers.*

*Proof.*

(1) **Representation and Yoneda decoding.** The epistemic update data and their pullbacks are objects and arrows of $\mathsf{T}$, unchanged by the embedding. Yoneda's full faithfulness recovers each residual strategy from its linear representative; Proposition 1.43 commutes this identification with action-target reindexing.

(2) **Observation and knowledge.** The witness carrier of Construction 12.30 is ordinary and is retained before any existential image or observational descent.

□

**Corollary 14.4** (Reconstruction of agents and their communication witnesses)**.** *The reconstruction equivalence recovers the agent interfaces of Definition 2.12, their complete-policy composition, regrouped control, legal choice fibres, observations, and continuation interfaces. For the complete-policy announcement presentation it also recovers* $\mathsf{AnnPol}_i$*, the source* $\mathcal{X}$*, the comparison* $F : \mathcal{X} \to \mathcal{U}$*, and the dependent family of policy witnesses and concurrent realizations over* $\mathcal{U}$*. The retained modal, fixed-point, dynamic, selection, and coalition predicates have the comparison equations established in Section 9.1.*

The proof is given in Appendix B.

**Corollary 14.5** (Reconstruction of the enriched agent calculus)**.** *Retain, in addition to the preceding agent presentation, its displayed delegation maps, feasibility predicates, institutional execution spans with realizers and permissions, and local controller data whenever these are part of the presentation. The reconstruction of these ordinary carriers and maps and of their linear realization values also reconstructs the following derived objects and comparisons:*

1. *substituted agent processes, their implemented-policy images, and the deviation comparison of Theorem 4.12;*
2. *informed legal policy carriers and the decentralized-to-pooled comparison of Proposition 7.13;*
3. *composites of witnessed institutions, their permission predicates, and their box and diamond operations;*
4. *the behaviour of each retained local controller and the residual controller generated by each retained complete local policy.*

*The identifications are canonical up to the same carrier isomorphisms as the reconstructed presentation.*

The proof is given in Appendix B.

The extra displayed data are genuine parts of the enriched presentation: they are not claimed to be recoverable from a bare concurrent arrow. Reconstruction recovers what the presentation retains and the universal constructions derived from it. All these constructions can also be made fibrewise over an ordinary base. Reindexing then selects the same fibrewise products, images, sections, pullbacks, and controller equations, so the displayed comparisons commute with base substitution.

### 14.1.2 Strategic reduction relative to semantic tests

**Definition 14.6** (Semantic tests and contextual policy equivalence)**.** A **semantic-test family** on $\Sigma = \prod_{i\in N} \Sigma_i$ consists of set-indexed functional tests $T_\lambda : \Sigma \to X_\lambda$ $(\lambda \in \Lambda)$ and predicate tests $P_\mu \hookrightarrow \Sigma$ $(\mu \in M)$. For owner $i$, transpose these along $\Sigma \cong \Sigma_{-i} \times \Sigma_i$ into

$$\kappa_i^{T,P} : \Sigma_i \longrightarrow \left(\prod_{\lambda\in\Lambda} X_\lambda^{\Sigma_{-i}}\right) \times \left(\prod_{\mu\in M} \mathsf{Pow}(\Sigma_{-i})\right),$$
$$\kappa_i^{T,P}(a_i) = \Big((\sigma_{-i} \mapsto T_\lambda(\sigma_{-i}, a_i))_\lambda, (\{\sigma_{-i} : P_\mu(\sigma_{-i}, a_i)\})_\mu\Big).$$

Here $\mathsf{Pow}(X)$ is the power object representing predicates on $X$. Two policies are **contextually equivalent** for the tests precisely when

$$a_i \approx_i^{T,P} b_i \quad\Longleftrightarrow\quad \kappa_i^{T,P}(a_i) = \kappa_i^{T,P}(b_i).$$

Thus $\approx_i^{T,P}$ is the kernel pair of the contextual-test map. Its image factorization gives the exact test-induced quotient below; equivalence compares policies against every fixed outsider profile.

Choosing a test family gives a precise meaning to agreement of behaviour. A test of the current payoff can identify two policies that a later permission check, observation, or continuation distinguishes. The quotient below retains each distinction that some chosen test detects in an opponent context.

A strategic abstraction should forget a difference in policies only when the designated assertions cannot detect it. Some of those tests are functions recording observable outputs; others are predicates recording comparisons or permissions. The following image factorization reduces both kinds of test simultaneously, making the meaning of the resulting equivalence explicit.

**Construction 14.7** (Contextual test quotient and its comparison category)**.**

(1) **Specify the retained tests.** Let

$$T_\lambda : \Sigma = \prod_i \Sigma_i \longrightarrow X_\lambda \qquad (\lambda \in \Lambda)$$

be a set-indexed family of functional semantic tests, and let

$$P_\mu \hookrightarrow \Sigma \qquad (\mu \in M)$$

be a set-indexed family of predicate tests. A predicate is treated by pullback to owner contexts and transposition along the power-object representation. Thus the construction remains valid in the intuitionistic reading of the assertion doctrine.

We use the predicate-descent form of Proposition 5.34, placed beside functional descent above.

(2) **Represent their contexts.** Because **Set** is Cartesian closed and its subobject doctrine has power objects, the contextual tests can be assembled without replacing a predicate by a chosen Boolean-valued characteristic function. Write $\mathsf{Pow}(X)$ for the power object representing subobjects of $X$. For each owner $i$, exponential transposition gives

$$\widehat{T}_{\lambda,i} : \Sigma_i \longrightarrow X_\lambda^{\Sigma_{-i}}, \qquad \widehat{T}_{\lambda,i}(a_i)(\sigma_{-i}) = T_\lambda(\sigma_{-i}, a_i),$$

while power-object transposition of $P_\mu \hookrightarrow \Sigma_{-i} \times \Sigma_i$ gives

$$\widehat{P}_{\mu,i} : \Sigma_i \longrightarrow \mathsf{Pow}(\Sigma_{-i}),$$

characterized by

$$\sigma_{-i} \in \widehat{P}_{\mu,i}(a_i) \quad \Longleftrightarrow \quad P_\mu(\sigma_{-i}, a_i).$$

(3) **Form the contextual image.** Define the canonical contextual-test map

$$\kappa_i^{T,P} = \left\langle (\widehat{T}_{\lambda,i})_{\lambda\in\Lambda}, (\widehat{P}_{\mu,i})_{\mu\in M} \right\rangle : \Sigma_i \longrightarrow \left( \prod_{\lambda\in\Lambda} X_\lambda^{\Sigma_{-i}} \right) \times \left( \prod_{\mu\in M} \mathsf{Pow}(\Sigma_{-i}) \right). \tag{14.1}$$

Take its image factorization in **Set**,

$$\Sigma_i \xrightarrow{q_i^{T,P}} \bar{\Sigma}_i^{T,P} \hookrightarrow \left( \prod_{\lambda\in\Lambda} X_\lambda^{\Sigma_{-i}} \right) \times \left( \prod_{\mu\in M} \mathsf{Pow}(\Sigma_{-i}) \right). \tag{14.2}$$

Thus $\bar{\Sigma}_i^{T,P}$ is the image of the map that records exactly the functional behavior and predicate behavior visible in every opponent context.

(4) **Identify its fibres.** Elementwise,

$$\begin{aligned} q_i^{T,P}(a_i) = q_i^{T,P}(b_i) \quad \Longleftrightarrow \quad & \forall\sigma_{-i}\ \forall\lambda\in\Lambda, \quad T_\lambda(\sigma_{-i}, a_i) = T_\lambda(\sigma_{-i}, b_i), \\ & \forall\sigma_{-i}\ \forall\mu\in M, \quad P_\mu(\sigma_{-i}, a_i) \leftrightarrow P_\mu(\sigma_{-i}, b_i). \end{aligned} \tag{14.3}$$

Writing $\mathcal{K}_i$ for the displayed target of $\kappa_i^{T,P}$, the image construction and its terminal factorization have the forms

$$\begin{array}{ccc} \Sigma_i & \xrightarrow{q_i^{T,P}} & \bar{\Sigma}_i^{T,P} \\ & \searrow^{\kappa_i^{T,P}} & \downarrow \\ & & \mathcal{K}_i \end{array} \qquad\qquad \begin{array}{ccc} \Sigma_i & \xrightarrow{f_i} & \Theta_i \\ & \searrow_{q_i^{T,P}} & \downarrow \bar{f}_i \\ & & \bar{\Sigma}_i^{T,P}. \end{array}$$

For utility at fixed prices, include the descended maps $\bar{v}_i$ among the functional tests. To permit reevaluation at other prices or recovery of the charge, include the full descended record instead. A record defined only on $K$ stays on that dependent witness family with its projection and chosen tests: the displayed profile test map takes functions on profiles. The dashed factorization is proved in the theorem below. No representatives of equivalence classes are chosen: the reduced policy set is the image object of the displayed universal map.

(5) **Construct the comparison category.** To state the universal property, let $\mathsf{Quot}_{T,P}(\vec{\Sigma})$ have as objects ownerwise surjections

$$f : \vec{\Sigma} \twoheadrightarrow \vec{\Theta}$$

through which every $T_\lambda$ factors and along which every $P_\mu$ descends: there are maps $\widetilde{T}_\lambda$ and predicates $\widetilde{P}_\mu$ with

$$T_\lambda = \widetilde{T}_\lambda \prod_i f_i, \qquad P_\mu = \left(\prod_i f_i\right)^* \widetilde{P}_\mu.$$

A morphism $h : f \to f'$ is an ownerwise map between quotient parameter objects satisfying $f' = hf$. Surjectivity of $f$ makes the descended maps and predicates commute with $h$ automatically.

**Theorem 14.8** (Test-induced strategic reduction)**.** *Let $q^{T,P} = \prod_i q_i^{T,P}$. Every functional test descends uniquely to*

$$\bar{T}_\lambda : \prod_i \bar{\Sigma}_i^{T,P} \longrightarrow X_\lambda, \qquad T_\lambda = \bar{T}_\lambda q^{T,P},$$

*and every predicate test descends uniquely to*

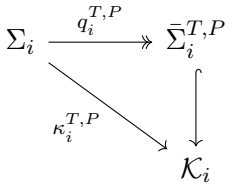
$$\bar{P}_\mu \hookrightarrow \prod_i \bar{\Sigma}_i^{T,P}, \qquad P_\mu = (q^{T,P})^* \bar{P}_\mu.$$

*The reduced family has no distinct policies of one owner with the same values under every functional test and the same truth values under every predicate test against every reduced opponent profile. Moreover, if $f_i : \Sigma_i \twoheadrightarrow \Theta_i$ are ownerwise surjections through which every $T_\lambda$ factors and along which every $P_\mu$ descends, then there is a unique ownerwise map*

$$\bar{f} : \vec{\Theta} \longrightarrow \bar{\vec{\Sigma}}^{T,P} \qquad \textit{with} \qquad q^{T,P} = \bar{f} f.$$

*Hence $q^{T,P}$ is terminal in $\mathsf{Quot}_{T,P}(\vec{\Sigma})$: it is the coarsest ownerwise effective quotient preserving all chosen functional and predicate tests.*

The contextual-test map admits its canonical image factorization.




*Proof.*

(1) **Lifting along the surjection.** Each $q_i^{T,P}$ is surjective by image factorization, so their finite product $q^{T,P}$ is surjective by Proposition 5.34. Suppose two profiles have the same $q^{T,P}$-image. Replace their finitely many owner coordinates one at a time.

(2) **Fibrewise calculation.** At the $i$th replacement, Equation (14.3) says, against the currently fixed opponent profile, that every $T_\lambda$ retains its value and every $P_\mu$ retains its truth value. Hence every $T_\lambda$ is constant on the fibers of $q^{T,P}$, while every $P_\mu$ is fiber-invariant.

(3) **Predicate calculation.** Functional descent follows from Proposition 5.34 and predicate descent from Proposition 5.34.

(4) **Separation by contextual tests.** For the separation statement, write $\bar{\Sigma}_{-i} = \prod_{j \neq i} \bar{\Sigma}_j^{T,P}$ and

$$\mathcal{K}_i = \left( \prod_{\lambda} X_{\lambda}^{\Sigma_{-i}} \right) \times \left( \prod_{\mu} \mathsf{Pow}(\Sigma_{-i}) \right)$$

for the original contextual-test object, and let

$$\bar{\mathcal{K}}_i = \left( \prod_{\lambda} X_{\lambda}^{\bar{\Sigma}_{-i}} \right) \times \left( \prod_{\mu} \mathsf{Pow}(\bar{\Sigma}_{-i}) \right)$$

be its reduced analogue.

(5) **Conclusion: second comparison.** The quotient map $q_{-i} : \Sigma_{-i} \twoheadrightarrow \bar{\Sigma}_{-i}$ induces a canonical map

$$\theta_i : \bar{\mathcal{K}}_i \longrightarrow \mathcal{K}_i$$

by precomposition with $q_{-i}$ on every functional component and inverse image along $q_{-i}$ on every predicate component. Let

$$\bar{\kappa}_i : \bar{\Sigma}_i \longrightarrow \bar{\mathcal{K}}_i$$

be the contextual-test map formed from the descended tests.

(6) **Lifting along the surjection.** By construction,

$$\kappa_i^{T,P} = \theta_i \bar{\kappa}_i q_i^{T,P}.$$

If

$$\kappa_i^{T,P} = m_i q_i^{T,P}$$

is its image factorization, epimorphicity of $q_i^{T,P}$ gives

$$m_i = \theta_i \bar{\kappa}_i.$$

Since the image inclusion $m_i$ is monic, $\bar{\kappa}_i$ is monic. Hence two reduced policies with the same values under all reduced functional tests and the same reduced contextual predicate behavior are equal. No representatives of the reduced policies are selected in this argument.

(7) **Policy-level comparison.** For the universal property, suppose $f_i(a_i) = f_i(b_i)$. Against every original opponent profile the two full profiles have the same $f$-image. Factorization of the functional tests and descent of the predicate tests then give both equalities in Equation (14.3).

(8) **Factorization of the representation map.** Thus $q_i^{T,P}$ equalizes the kernel pair of $f_i$. Since $f_i$ is the coequalizer of that kernel pair, there is a unique $\bar{f}_i : \Theta_i \to \bar{\Sigma}_i^{T,P}$ with $q_i^{T,P} = \bar{f}_i f_i$. Taking the ownerwise product gives $\bar{f}$.

□

**Corollary 14.9** (Canonical strategic reduction)**.** *Take the single functional test*

$$T = g : \Sigma \longrightarrow \mathcal{C}(A, B)$$

*and no predicate tests. Then owner $i$'s contextual-behavior map is*

$$\kappa_i^g : \Sigma_i \longrightarrow \mathcal{C}(A, B)^{\Sigma_{-i}}, \qquad a_i \longmapsto \big(\sigma_{-i} \mapsto g_{(\sigma_{-i}, a_i)}\big),$$

*and its image gives the canonical behavioral reduction $\bar{G}$. For every coalition, the quotient map is a bounded morphism of deviation relations. For every ownerwise surjective realization-preserving 2-cell $f : G \Rightarrow H$, there is a unique realization-preserving 2-cell $\bar{f} : H \Rightarrow \bar{G}$ with $q = \bar{f} f$.*

*Proof.*

(1) **Descent along the kernel pair.** The realization descends by Theorem 14.8. Write the realizations of $G, H, \bar{G}$ as $g, h, \bar{g}$, respectively. The universal property supplies a unique ownerwise parameter map $\bar{f}$ with $q = \bar{f} f$.

(2) **Lifting along the surjection.** Since $f$ is realization-preserving and $q$ is the quotient presentation map,

$$hf = g = \bar{g}q = \bar{g}\,\bar{f}\,f.$$

The total profile map $f = \prod_i f_i$ is surjective by Proposition 5.34, hence is an epimorphism in **Set**. Cancelling it gives

$$h = \bar{g}\,\bar{f},$$

so $\bar{f} : H \Rightarrow \bar{G}$ is itself realization-preserving. Uniqueness as a 2-cell follows from uniqueness of the ownerwise parameter map.

(3) **Descent along the kernel pair.** Every coalition alternative descends to the quotient. To lift a reduced alternative, keep the original outsider coordinates and lift the finite tuple of coalition coordinates along $\prod_{i \in C} q_i$. This product is an effective quotient by Proposition 5.34, so such a lift exists without choosing sections.

(4) **Lifting through the comparison.** Realizations and every comparison or full-product outcome predicate that factors through realization have the same truth value before and after this lifting. Thus the quotient has the claimed lifting property, and Theorem 9.29 applies to all descended atoms.

□

**Proposition 14.10** (Reduction and composition)**.** *Let $q_G : G \Rightarrow \bar{G}$ and $q_H : H \Rightarrow \bar{H}$ be canonical behavioral reductions. There is a unique ownerwise surjective, realization-preserving comparison*

$$\bar{H} \circ \bar{G} \longrightarrow \overline{H \circ G}$$

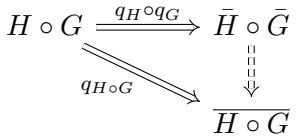
*whose composite with $q_H \circ q_G$ is the canonical reduction of $H \circ G$. The analogous comparison exists for tensor. Such a comparison need not be invertible.*

*Proof.*

(1) **Lifting along the surjection.** The ownerwise products of the two quotient maps are surjective and preserve the composite realization. Terminality of the composite's test quotient supplies the unique comparison. It is surjective at each owner because its composite with a surjection is the surjective canonical quotient.

(2) **Ownerwise alternatives.** For noninvertibility, let one owner choose 0 or 1 in $G : I \to D(\{0,1\})$, and let $H$ be the nonstrategic response induced by the constant function $\{0,1\} \to 1$. The reduction of $G$ retains two policies, whereas every composite realization is the same value state in $D(1)$ and its behavioral reduction has one policy. Thus the comparison has a two-element domain and a singleton codomain at that owner.

□




### 14.1.3 The reflection and reduced composition

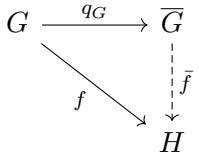
**Definition 14.11** (Surjective presentation category)**.** For fixed $A, B$, let $\mathsf{Pres}_{\mathrm{epi}}(A, B)$ have open arenas as objects and ownerwise surjective realization-preserving 2-cells as morphisms. Let $\mathsf{Red}(A, B)$ be its full subcategory on presentations whose contextual-behavior maps $\kappa_i^g$ are injective. Write $I$ for its inclusion.

**Theorem 14.12** (Strategic reduction is a reflection)**.** *Behavioral reduction defines a functor*

$$\mathrm{Red} : \mathsf{Pres}_{\mathrm{epi}}(A, B) \longrightarrow \mathsf{Red}(A, B), \qquad \mathrm{Red} \dashv I.$$

*Its unit is $q_G : G \to \bar{G}$. For every morphism $f : G \to H$, $\mathrm{Red}(f) : \bar{G} \to \bar{H}$ is an isomorphism.*

The quotient comparison satisfies the universal property of a reflection.




*Proof.*

(1) **Lifting along the surjection.** Write $g = h \prod_i f_i$. For two $i$-policies $a, b$, one has

$$a \sim_G b \quad \Longleftrightarrow \quad f_i(a) \sim_H f_i(b).$$

For the forward implication, lift every $H$-opponent profile along the surjection $\prod_{j \neq i} f_j$ and use the realization equation.

(2) **Reverse implication.** For the reverse implication substitute the image of each $G$-opponent profile. Hence $q_{H,i} f_i$ descends uniquely through $q_{G,i}$ to a bijection $\bar{f}_i$. The descended realization equation follows by cancelling the surjective total parameter map.

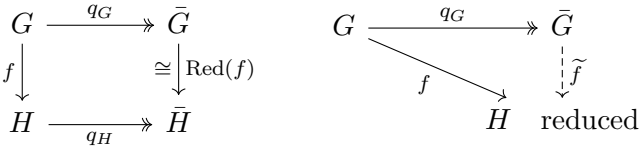
(3) **Lifting along the surjection.** The equations $\bar{f} q_G = q_H f$ prove identity and composition preservation by epimorphic cancellation.

(4) **Lifting along the surjection: second comparison.** If $H$ is reduced, $a \sim_G b$ implies $f_i(a) = f_i(b)$. Thus $f$ factors uniquely as $\widetilde{f} q_G$ with $\widetilde{f} : \bar{G} \to H$. It is surjective because $f$ is, and preserves realization by cancelling $q_G$.

(5) **Reverse implication.** Conversely every such $\widetilde{f}$ gives a morphism by precomposition with $q_G$. These inverse operations are natural and give the adjunction. Reduction is reduced by Theorem 14.8, so the stated codomain is correct.

(6) **Policy-level comparison.** The bijections constructed in the first paragraph prove the last assertion, including the empty-policy cases.

□




**Construction 14.13** (Composition after reduction)**.**

(1) **Reduce the composite.** For reduced $G : A \to B$ and $H : B \to C$, put

$$H \star G = \mathrm{Red}(H \circ G), \qquad H \mathbin{\widehat{\otimes}} G = \mathrm{Red}(H \otimes G).$$

(2) **Act on maps and identities.** On 2-cells apply the original horizontal composition or tensor and then the reduction functor. Retain the original arena objects and the reduced identity presentations.

**Theorem 14.14** (Coherence of reduced composition)**.** *The preceding construction is a symmetric monoidal bicategory, with hom-categories* $\mathsf{Red}(A, B)$*. Its associators, unitors, and monoidal coherence maps are the canonical comparisons induced from those of open arenas. In particular,*

$$\mathrm{Red}(\bar{H} \circ \bar{G}) \cong \mathrm{Red}(H \circ G),$$

*although the unreduced comparison of Proposition 14.10 need not be invertible.*

*Proof.*

(1) **Composition.** Composition and tensor of ownerwise surjections are ownerwise surjections. The preceding theorem therefore sends every quotient map inserted in a composite to an isomorphism after reduction. For example, reduce the map $q_H \circ q_G : H \circ G \to \bar{H} \circ \bar{G}$; its image is the displayed isomorphism, with the orientation reversed if necessary.

(2) **Monoidal comparison.** For three arrows, both $(K \star H) \star G$ and $K \star (H \star G)$ are consequently canonically isomorphic to the reduction of an unreduced triple composite. Transport its original associator through these isomorphisms. Define the unitors and tensor comparisons in the same way. Naturality follows from the naturality of $q$ and of the original structural maps.

(3) **Lifting along the surjection.** For each pentagon, triangle, symmetry, or monoidal interchange diagram, precompose the two candidate composites with the product of the unreduced parameter quotients. This is surjective at every owner. The resulting diagram is the corresponding coherence diagram in $\mathbf{Arena}_N(\mathcal{C})$, followed by the unique reduction map, and therefore commutes.

(4) **Lifting along the surjection: completion.** Cancel the surjections ownerwise. This proves all the asserted coherence equations and their compatibility, without choosing representatives of reduced policies.

□

*Remark* 14.15 (The morphisms used by the reflection)*.* The proof uses the lifting of opponent contexts along the displayed surjections. Arbitrary parameter maps remain available in the original open-arena bicategory; the reflection just proved is on its surjective presentation categories. A change that introduces new opponent policies is represented there before any new contextual reduction is performed.

*Remark* 14.16 (Strategy reduction and epistemic state spaces). For $s : W \to \Sigma$, behavioral reduction replaces it by $qs : W \to \bar{\Sigma}$ and preserves $gs = \bar{g}qs$. It need not identify states of $W$. Observations and belief relations on $W$ may be retained unchanged. If a separate state quotient is taken, its relations and atoms are subject to the bounded-morphism theorem independently. When action enablement descends along such a bounded state quotient, the updated state quotient is again bounded by Corollary 9.27; this is not inferred from equality of play maps alone.

### 14.1.4 Reduction of dependent legal agency

The authorization example gives two policies with the same immediate output and different legal status. A quotient that merged them could give the regulated system a deviation it cannot execute. We therefore descend the legal domain and its dependent alternative fibres along with the outcome map.

**Construction 14.17** (Legality-sensitive strategic reduction)**.**

(1) **Form the ownerwise quotient.** The predicate-sensitive strategic reduction now gives the reduction that is appropriate when institutional legality must survive abstraction. Apply Theorem 14.8 to the functional test

$$g : \Sigma \to \mathcal{C}(A, B)$$

and the predicate test $L \hookrightarrow \Sigma$. Write the resulting ownerwise quotient as

$$r_i : \Sigma_i \twoheadrightarrow \bar{\Sigma}_i, \qquad r = \prod_i r_i : \Sigma \twoheadrightarrow \bar{\Sigma}.$$

Then there are a unique realization $\bar{g}$ and a unique descended legal subobject $\bar{L} \hookrightarrow \bar{\Sigma}$ such that

$$g = \bar{g}r, \qquad L = r^*\bar{L}. \tag{14.4}$$

(2) **Restrict to legal profiles.** Let

$$r_L : L \twoheadrightarrow \bar{L}$$

be the restriction of $r$. It is surjective: every point of $\bar{L}$ has an $r$-preimage by surjectivity, and Equation (14.4) places every such preimage in $L$.

(3) **Descend the coordinate maps.** Construct the realizable-coordinate functor $\bar{L}_{(-)}$ of the reduced legal domain in the same way as Proposition 7.17. For every $S \subseteq N$, the composite $\bar{e}_S r_L : L \to \bar{L}_S$ is constant on the fibers of $e_S$, because equal original $S$-coordinates have equal reduced $S$-coordinates. Effective descent therefore gives a unique map

$$r^L_S : L_S \longrightarrow \bar{L}_S, \qquad r^L_S e_S = \bar{e}_S r_L. \tag{14.5}$$

It is surjective: for $\bar{s} \in \bar{L}_S$, take $\bar{\ell} \in \bar{L}$ with $\bar{e}_S(\bar{\ell}) = \bar{s}$, lift $\bar{\ell}$ along $r_L$ to $\ell \in L$, and then $r^L_S(e_S(\ell)) = \bar{s}$.

(4) **Verify their compatibility.** These quotient maps are compatible with every coordinate restriction. If $S \subseteq T$, then

$$\bar{\rho}^T_S r^L_T = r^L_S \rho^T_S. \tag{14.6}$$

Indeed both sides become $\bar{e}_S r_L$ after precomposition with the surjection $e_T$. Hence

$$r^L_{(-)} : L_{(-)} \Longrightarrow \bar{L}_{(-)}$$

is a levelwise surjective natural transformation of realizable-coordinate functors.

**Theorem 14.18** (Exact descent of legal alternatives)**.** *For every $S \subseteq N$, the commutative square*

$$\begin{array}{ccc} L & \xrightarrow{r_L} & \bar{L} \\ {\scriptstyle e_S}\downarrow & & \downarrow{\scriptstyle \bar{e}_S} \\ L_S & \xrightarrow[r_S^L]{} & \bar{L}_S \end{array}$$

*has a surjective comparison map*

$$L \twoheadrightarrow L_S \times_{\bar{L}_S} \bar{L}, \qquad \ell \longmapsto (e_S(\ell), r_L(\ell)). \tag{14.7}$$

*Thus, in* **Set***, each displayed square is a regular pushout of surjections. Consequently, for every coalition $C$, $r_L$ is a surjective bounded morphism from the legal deviation relation $D_C^L$ to $D_C^{\bar{L}}$, and the counterfactual realizations commute with the induced map of deviation objects. Legal Nash equilibrium is therefore preserved and reflected by $r_L$; the same is true of every modal fixed-point assertion built from legal deviation relations and atomic predicates that descend through $r_L$.*

Exact legal reduction is a Cartesian square of legal profiles over their policy spaces.

$$\begin{array}{ccc} L & \xrightarrow{r_L} & \bar{L} \\ \downarrow & & \downarrow \\ \Sigma & \xrightarrow[r]{} & \bar{\Sigma} \end{array}$$

*Proof.*

(1) **Choice of representatives and quotients.** Take a compatible pair

$$(s, \bar{\ell}) \in L_S \times_{\bar{L}_S} \bar{L}.$$

Surjectivity of $e_S$ gives $\ell_0 \in L$ with $e_S(\ell_0) = s$, and surjectivity of $r_L$ gives $\ell_1 \in L$ with $r_L(\ell_1) = \bar{\ell}$.

(2) **Witness and legality comparison.** Compatibility says

$$r_S^L(e_S(\ell_0)) = \bar{e}_S(r_L(\ell_1)).$$

Using Equation (14.5) on the left gives

$$\bar{e}_S(r_L(\ell_0)) = \bar{e}_S(r_L(\ell_1)).$$

Since $\bar{L}_S \hookrightarrow \bar{\Sigma}_S$ is monic, the reduced $S$-coordinates of $\ell_0$ and $\ell_1$ coincide. Form the ambient profile $\tau \in \Sigma$ by taking its $S$-coordinates from $\ell_0$ and its coordinates outside $S$ from $\ell_1$.

(3) **Ownerwise alternatives.** Ownerwise equality of the reduced $S$-coordinates gives

$$r(\tau) = r(\ell_1) = \bar{\ell}.$$

Because $L = r^*\bar{L}$ and $\bar{\ell} \in \bar{L}$, the profile $\tau$ belongs to $L$. It also satisfies $e_S(\tau) = s$. Hence $\tau$ maps to the chosen compatible pair, proving surjectivity of Equation (14.7).

(4) **Lifting through the comparison.** For the bounded-morphism lifting property take $S = -C$. If

$$r_L(\ell)\ D_C^{\bar{L}}\ \bar{\tau},$$

then $\bar{e}_{-C}(r_L\ell) = \bar{e}_{-C}(\bar{\tau})$. By Equation (14.5), the pair $(e_{-C}(\ell), \bar{\tau})$ lies in $L_{-C} \times_{\bar{L}_{-C}} \bar{L}$.

(5) **Verification of the universal property.** Surjectivity of the comparison map supplies $\tau \in L$ with

$$e_{-C}(\tau) = e_{-C}(\ell), \qquad r_L(\tau) = \bar{\tau}.$$

Thus $\ell D_C^L \tau$, giving the required lift. Forward preservation follows from the commutative square.

(6) **Strategic comparison.** The induced map on legal deviation objects sends $(\ell, \tau)$ to $(r_L\ell, r_L\tau)$. Equation (14.4) gives

$$\delta_C^{\bar{G},\bar{L}}(r_L\ell, r_L\tau) = \delta_C^{G,L}(\ell, \tau).$$

Hence profitable legal alternatives are preserved and lifted with the same two realized interactions. Equation (7.8) is therefore preserved and reflected.

(7) **Fixed-point argument.** The fixed-point statement is the bounded-morphism argument of Theorem 9.29, applied to the legal state sort $L$ and its legal deviation relations.

□

**Theorem 14.19** (Legal ability under strategic reduction)**.** *Let $r_L : L \twoheadrightarrow \bar{L}$ be the legality-sensitive reduction of Theorem 14.18, and suppose an outcome map $q : L \to Z$ descends to $\bar{q} : \bar{L} \to Z$, so $q = \bar{q} r_L$. Then for every coalition $C$ and predicate $P \hookrightarrow Z$,*

$$[C]_L P \quad \Longleftrightarrow \quad [C]_{\bar{L}} P.$$

*Thus the legality-sensitive strategic quotient preserves and reflects coalition ability as well as legal strategic stability.*

*Proof.*

(1) **Choice of representatives and quotients.** Put

$$A = \forall_{e_C : L \to L_C} q^* P, \qquad \bar{A} = \forall_{\bar{e}_C : \bar{L} \to \bar{L}_C} \bar{q}^* P.$$

We first prove

$$A = (r_C^L)^* \bar{A}.$$

Suppose $A(c)$ and let $\bar{\ell} \in \bar{L}$ satisfy $\bar{e}_C(\bar{\ell}) = r_C^L(c)$. By the surjective comparison map in Theorem 14.18, there is $\ell \in L$ with $e_C(\ell) = c$ and $r_L(\ell) = \bar{\ell}$. Hence $P(q(\ell))$, and therefore $P(\bar{q}(\bar{\ell}))$.

(2) **Reverse implication.** Conversely, suppose $\bar{A}(r_C^L(c))$ and let $\ell \in L$ satisfy $e_C(\ell) = c$. Then

$$\bar{e}_C(r_L\ell) = r_C^L(c),$$

so $P(\bar{q}(r_L\ell))$, equivalently $P(q(\ell))$. This proves the predicate equality.

(3) **Preservation of coalition effectivity.** Finally $r_C^L$ is surjective, so existential quantification over $L_C$ is preserved and reflected:

$$\exists_{!:L_C \to 1} A \quad \Longleftrightarrow \quad \exists_{!:\bar{L}_C \to 1} \bar{A}.$$

Using Equation (7.12) on both sides gives the result.

□

In particular, an observable outcome map of the form $q = o \circ g \circ m$ descends automatically: if $\bar{m} : \bar{L} \hookrightarrow \bar{\Sigma}$ is the descended legal inclusion, then

$$\bar{q} = o \circ \bar{g} \circ \bar{m}$$

satisfies $q = \bar{q} r_L$ by Equation (14.4).

### 14.1.5 Continuation-sensitive open-game reduction

**Definition 14.20** (Continuation tests)**.** For a complete history-indexed continuation family with history set $\mathsf{Hist}$, use its restriction and realization maps

$$r_h : \Sigma \to \Sigma^h, \qquad g_h : \Sigma^h \to \mathcal{C}(A_h, B_h), \qquad g^h = g_h r_h : \Sigma \to \mathcal{C}(A_h, B_h) \quad (h \in \mathsf{Hist}).$$

A **functional continuation test** is the map $g^h$, which observes the continuation of a complete policy after $h$, whether reached or off-path. The family $(g^h)_{h \in \mathsf{Hist}}$ induces the ownerwise contextual equivalence

$$a_i \approx_i^{\mathrm{cont}} b_i \quad \Longleftrightarrow \quad \forall h \in \mathsf{Hist}\ \forall \sigma_{-i} \in \Sigma_{-i}, \qquad g^h(a_i, \sigma_{-i}) = g^h(b_i, \sigma_{-i}).$$

Comparison predicates $P_h \hookrightarrow \mathcal{C}(A_h, B_h)^2$ are transported along the same maps; identical continuation realizations give identical comparison tests. The superscript on $\approx_i^{\mathrm{cont}}$ is suppressed in the continuation-quotient construction below. Additional factual or certificate observations are retained by including them in the general semantic-test family.

A root outcome records what the selected policies do along the realized play. A test of subgame perfection can also ask what they prescribe after a history that play never reaches. To preserve that test, the policy quotient must commute with restriction to the relevant continuations. The example below exhibits the distinction that a root test misses.

**Example 14.21** (Root behavior does not determine sequential rationality)**.** Root-outcome equivalence alone does not preserve sequential rationality. Consider a one-player tree: the root can stop for payoff 2, or continue to a choice between payoffs 1 and 0. The complete policies "stop, then choose 1" and "stop, then choose 0" have the same root behavior and are both root Nash equilibria. Only the first is subgame perfect. The canonical root strategic quotient therefore identifies policies that the continuation predicates in Equation (12.12) still need to distinguish.

**Construction 14.22** (Continuation test quotient)**.**

(1) **Evaluate every continuation.** For each history $h$, let

$$g^h : \Sigma \longrightarrow \mathcal{C}(A_h, B_h)$$

be the realization of the residual game after restricting complete policies. Use the family $(g^h)_h$ as the semantic tests of Theorem 14.8. This is the family in Definition 14.20.

(2) **Identify the policy equivalence.** The resulting ownerwise equivalence is

$$a_i \approx_i b_i \quad \Longleftrightarrow \quad \forall h\ \forall s_{-i}, \qquad g^h(a_i, s_{-i}) = g^h(b_i, s_{-i}).$$

**Lemma 14.23** (Coherent reduction of continuation games)**.** *For either finite complete-policy tree construction, order histories by extension and write $r_{hk,i} : \Sigma_i^h \to \Sigma_i^k$ for coordinate restriction when $h \preceq k$. At each history define*

$$a_i \approx_i^h b_i \quad \Longleftrightarrow \quad \forall k \succeq h \; \forall \tau_{-i} \in \Sigma_{-i}^h, \qquad g_k\big(r_{hk}(a_i, \tau_{-i})\big) = g_k\big(r_{hk}(b_i, \tau_{-i})\big).$$

*Let $q_i^h : \Sigma_i^h \twoheadrightarrow \bar{\Sigma}_i^h$ be its quotient. There are unique surjective maps $\bar{r}_{hk,i} : \bar{\Sigma}_i^h \to \bar{\Sigma}_i^k$ satisfying*

$$\bar{r}_{hk,i} q_i^h = q_i^k r_{hk,i}. \tag{14.8}$$

*They satisfy $\bar{r}_{hh,i} = 1$ and $\bar{r}_{k\ell,i}\bar{r}_{hk,i} = \bar{r}_{h\ell,i}$. Every $g_h$ descends uniquely to $\bar{g}_h$ with $g_h = \bar{g}_h q^h$, where $q^h = \prod_i q_i^h$. At the root, this is the continuation test quotient above.*

*Proof.*

(1) **Lifting along the surjection.** The coordinate projections satisfy $r_{hh,i} = 1$ and $r_{k\ell,i} r_{hk,i} = r_{h\ell,i}$. They are surjective: the finitely many omitted coordinates have nonempty action sets, so finite induction supplies their product, and adjoining those coordinates extends any subtree policy. Equivalently, the paper's reference-based extension gives a section after fixing the reference policy. Neither argument selects sections globally.

(2) **Lifting along the surjection: second comparison.** Suppose $a_i \approx_i^h b_i$. For $\ell \succeq k \succeq h$ and an opponent profile $\eta_{-i} \in \Sigma_{-i}^k$, lift $\eta_{-i}$ to $\tau_{-i} \in \Sigma_{-i}^h$.

(3) **Stability under continuation restriction.** Restriction composition gives

$$\begin{aligned} g_\ell\big(r_{k\ell}(r_{hk,i} a_i, \eta_{-i})\big) &= g_\ell\big(r_{h\ell}(a_i, \tau_{-i})\big) \\ &= g_\ell\big(r_{h\ell}(b_i, \tau_{-i})\big) \\ &= g_\ell\big(r_{k\ell}(r_{hk,i} b_i, \eta_{-i})\big). \end{aligned}$$

Thus $r_{hk,i} a_i \approx_i^k r_{hk,i} b_i$, and the quotient universal property supplies the unique map in Equation (14.8).

(4) **Lifting along the surjection.** It is surjective because $q_i^k r_{hk,i}$ is surjective. Moreover,

$$\bar{r}_{k\ell,i}\bar{r}_{hk,i} q_i^h = q_i^\ell r_{k\ell,i} r_{hk,i} = q_i^\ell r_{h\ell,i} = \bar{r}_{h\ell,i} q_i^h.$$

Cancelling the epimorphism $q_i^h$ proves composition coherence; the identity law follows in the same way. The test at $h$ occurs in the defining family.

(5) **Ownerwise alternatives.** Replacing the finitely many owner coordinates successively therefore proves that $g_h$ is constant on the fibres of $q^h$, giving its unique descent. The root tests are exactly $(g^k)_k$.

□

**Theorem 14.24** (Continuation-sensitive reduction)**.** *The test-induced quotient for the family $(g^h)_h$ is the coarsest ownerwise quotient preserving all continuation realizations and has the universal factorization property of Theorem 14.8. For measured games retain, at each subtree $h$, the tuple consisting of $g_h$ and $(V_i^{h,c_h})_i$ in the test family and in its induced subtree quotients. This is the corresponding coarsest quotient for the enlarged tests. To vary the initial charge retain all desired translations, or retain the full measured suffix outcome from which they are computed. The assertion below uses the quotients for the chosen tests. In the measured instance $\mathsf{NE}_{G_h}$ and $\mathsf{NE}_{\bar{G}_h}$ below use the values at $c_h$ and their descents, respectively.*

*For the finite complete-policy trees, the reduced games and restriction maps of Lemma 14.23 preserve and reflect subgame-perfect profiles. Writing $o$ for the root,*

$$\mathsf{SPE}_G = (q^o)^* \left( \bigwedge_h \bar{r}_{oh}^* \mathsf{NE}_{\bar{G}_h} \right). \tag{14.9}$$

*Proof.*

(1) **Reduced continuations and test descent.** Theorem 14.8 gives descent and the universal property. Lemma 14.23 constructs the reduced continuation diagram. For the enlarged tuple of tests the proof of that lemma applies to each component: lift an opponent profile from a child, restrict it along the same composite restriction, and evaluate the same descendant test. The value components therefore descend at their inherited charges, and the quotient restriction squares commute by epimorphic cancellation. For each $h$, an original unilateral alternative maps to a reduced alternative.

(2) **Reverse implication.** Conversely any reduced unilateral alternative lifts through $q_i^h$, while the reference opponents remain fixed. Since $g_h = \bar{g}_h q^h$, both realized interactions in the comparison are preserved. Hence, for each continuation comparison,

$$\mathsf{NE}_{G_h} = (q^h)^* \mathsf{NE}_{\bar{G}_h}.$$

The product of the ownerwise descent squares gives $q^h r_{oh} = \bar{r}_{oh} q^o$.

(3) **Functoriality and substitution.** Functoriality of reindexing and its preservation of meets therefore yield

$$\bigwedge_h r_{oh}^* \mathsf{NE}_{G_h} = (q^o)^* \left( \bigwedge_h \bar{r}_{oh}^* \mathsf{NE}_{\bar{G}_h} \right),$$

which is Equation (14.9).

□

Reducing a policy and then restricting it to a later history gives the same reduced continuation as restricting it first. This is the equality needed to transport a subgame test.

$$\begin{array}{ccc} \Sigma_i^h & \xrightarrow{q_i^h} & \bar{\Sigma}_i^h \\ {\scriptstyle r_{hk,i}}\downarrow & & \downarrow{\scriptstyle \bar{r}_{hk,i}} \\ \Sigma_i^k & \xrightarrow[q_i^k]{} & \bar{\Sigma}_i^k \end{array}$$

Thus the two strategic reductions in the paper are not different abstraction mechanisms. They are the same test-induced quotient with different observation families: root realization for ordinary strategic stability, and all residual realizations for sequential stability.

$$\begin{array}{ccc} \Sigma_i^h & \xrightarrow{\mathrm{ext}_{h,\sigma_i}} & \Sigma_i \\ & \searrow & \downarrow{\scriptstyle r_{h,i}} \\ & & \Sigma_i^h \end{array} \qquad \begin{array}{ccc} \Sigma & \xrightarrow{q} & \bar{\Sigma} \\ & {\scriptstyle g^h}\searrow & \downarrow{\scriptstyle \bar{g}^h} \\ & & \mathcal{C}(A_h, B_h) \end{array}$$

## 14.2 Soundness and classifying-model completeness

### 14.2.1 The two-layer soundness theorem

The soundness statement connects the four levels already constructed: the ordinary judgments of Section A.1.1, the displayed agent operations, the finite protocols of Section 10.2, and the contextual assertions in $\mathcal{T}(W)$, $\mathcal{L}(W)$, and $\mathcal{P}(W)$. Ordinary equality is interpreted through diagonal comprehension, while linear variables remain governed by tensor-context splitting. Agent abstraction is internal-hom transposition; observed responses are effective coequalizer factors; legal policies are dependent universal images; and institutional composition matches witnesses by pullback. Primitive comparisons, permissions, and undelivered observations are interpreted by their declared typed data, not inferred from the operational arrows. The inference rules and their proof obligations are displayed in the appendices.

**Theorem 14.25** (Soundness of the two-layer agency calculus)**.** *The completed true-concurrent semantics of Section 1 interprets every well-formed type, term, display, and contextual assertion of this calculus. Derivable term equations are equalities of morphisms; derivable assertion entailments are inclusions of predicates. Interpretation preserves ordinary and linear substitution, agent composition, owner regrouping, delegation, finite protocol execution, perspective, witnessed institutional composition, and epistemic update.*

The proof is given in Appendix B.

The two semantic layers remain distinct: linear terms retain resources and execution witnesses, whereas assertions retain truth values. Soundness does not identify their units, products, or notions of equality.

### 14.2.2 Reconstruction of the displayed economic model

**Theorem 14.26** (Exact reconstruction)**.** *Encoding a displayed owned presentation on original arenas as an agent state and decoding it by Theorem 2.18 recovers that presentation. The construction extends to situated presentations, regroupings, delegation maps, witnessed institutions, and complete finite communication presentations, retaining every displayed ordinary carrier and map. Both round trips are identity up to their canonical isomorphisms.*

Agent reconstruction is the inverse of policy-indexed currying and Yoneda decoding.

$$\begin{array}{ccc} \mathbf{Set}(\Sigma, \mathcal{C}(A,B)) & \xrightarrow{\text{encode}} & \mathcal{V}(\mathsf{I}, \mathsf{Ag}_{\vec{\Sigma}}(yA, yB)) \\ & \searrow^{\text{id}} & \downarrow{\scriptstyle \text{decode}} \\ & & \mathbf{Set}(\Sigma, \mathcal{C}(A,B)) \end{array}$$

The proof is given in Appendix B.

Exact reconstruction depends on retaining the ownership signature and execution witnesses as displayed data. The underlying linear agent object does not recover who controls its policy factors, and the predicate doctrine cannot recover a hidden execution from its truth value. The reconstruction theorem recovers precisely the data that were retained.

| Type-theoretic construction | Recovered semantic construction |
|---|---|
| Agent state | Owned complete-policy realization |
| Linear sequential and parallel operations | Composition and tensor of concurrent realizers |
| Ownerwise parameter substitution | Realization-preserving policy maps |
| Regrouping and delegation | Collective control and implemented parent policies |
| Linear dependent products | Families of perspective-dependent resources |
| Predicate observation box | Knowledge from observation fibres |
| Uniform policy section | Observation-based legality at every compatible world |
| Session interaction witness | Secured communication history |
| Institutional span composition | Matching records, realizers, and permissions |
| Public comprehension | Restriction of both epistemic endpoints |
| Complete communication policies | Truthful reporting and delivered private/public observations |

### 14.2.3 The classifying model of the first layer

Fix a typed primitive signature $\mathcal{K}$ for ordinary sorts and operations, linear boundary sorts, and linear generators. All primitive declarations have specified arities. The first-layer theory $\mathbb{T}_{\mathrm{Ag}}(\mathcal{K})$ is generated by the rules of Section A.1.1 and Section A.1.1; agent types, composition, regrouping, and delegation are the displayed definitions above. The measured witness package and its explicit charge tag in Section A.1.2 expand into dependent linear sum, tensor and $\mathsf{Lin}$ terms over the declared ordinary maps. The budget modalities belong to the assertion and dependent-family interpretations. Finite data use the exhibited generators of Construction 10.7. Finite sessions have the guarded prefix interpretation of Lemma 10.14 and its factorization through fixed session wiring in $\mathbf{j}_N(\mathcal{C})$. A primitive signature declares operation symbols with typed arities; it does not silently assert that unrelated symbols obey equations proved only in one semantic interpretation. Consequently the following completeness statement concerns the equational first-layer theory, not operational progress or equality in one fixed concurrent model.

A split indexed linear–nonlinear model consists of an ordinary comprehension model for the stated extensional dependent type constructors (with equality comprehensions given by pullbacks of diagonals), an indexed symmetric monoidal closed linear category with additives and dependent linear sums and products, and a substitution-stable adjunction $\mathsf{Lin} \dashv \mathsf{Val}$ with strong monoidal $\mathsf{Lin}$. Its indexed adjoints obey the base-change and Frobenius comparisons expressed by the rules. A model of $\mathcal{K}$ interprets its declared constants at their arities. These are the semantic structures specified by the type constructors, rather than an identification of all ordinary contexts with sets.

**Construction 14.27** (Syntactic indexed model)**.**

(1) **Construct the ordinary base.** Let $\mathcal{B}_{\mathrm{syn}}$ have ordinary contexts as objects and ordinary substitutions modulo derivable equality as arrows. Its terminal object is the empty context and its comprehension is context extension.

(2) **Construct the linear fibres.** Over $\Gamma$, let $\mathcal{L}_{\mathrm{syn}}(\Gamma)$ have linear types modulo conversion as objects and arrows $A \to B$ represented by

$$\Gamma; x : A \vdash t : B$$

modulo derivable equality and bound-variable renaming. Composition is linear substitution.

(3) **Define reindexing.** Reindexing substitutes ordinary parameters in types and terms. The ordinary type fibre is defined similarly using ordinary terms.

**Proposition 14.28** (Diagonal construction of extensional equality)**.** *In an ordinary extensional comprehension model, equality between sections $a, b : \Gamma \to \Gamma.X$ is the diagonal pullback*

$$\begin{array}{ccc} \Gamma.\,\mathrm{Eq}_X(a,b) & \longrightarrow & \Gamma.X \\ \downarrow e & & \downarrow \Delta_X \\ \Gamma & \xrightarrow{\langle a,b\rangle} & \Gamma.X \times_\Gamma \Gamma.X. \end{array}$$

*A term of this equality family yields the equation $a = b$; two terms of the same equality family are equal. These properties are stable under ordinary substitution. The corresponding extensional equality conversion in the syntax* presents *this universal property: the categorical argument proves its soundness, not a derivation of equality reflection from weaker syntactic constructors. It is part of the chosen* ordinary *foundation, not a rule for agents or linear resources, and it does not follow from intensional identity types or from dependent sums and products alone.*

*Proof.*

(1) **Extensional equality elimination.** The diagonal and its pullback $e$ are monomorphisms. The pullback equations give $ae = be$. A term $p$ of the equality family is a section $\bar{p}$ of $e$, so $a = ae\bar{p} = be\bar{p} = b$.

(2) **Functoriality and substitution.** For two such sections $p, q$, monicity gives $p = q$ from $ep = eq = \mathrm{id}$. Pullback stability of diagonals proves substitution stability. This proof derives the categorical consequences of the chosen extensional equality interpretation; it does not identify intensional equality with extensional conversion.

□

**Lemma 14.29** (General-map dependent adjoints in the syntax)**.** *For every definable ordinary substitution $f : V \to W$, its ordinary equality fibres together with the linear dependent sum and product along comprehension construct left and right adjoints to reindexing on the syntactic linear fibres. They obey Beck–Chevalley for pullback squares and Frobenius for tensor.*

The proof is given in Appendix B.

**Lemma 14.30** (Equality-fibre base change and Frobenius coherence)**.** *For a pullback of definable ordinary substitutions*

$$\begin{array}{ccc} V' & \xrightarrow{a} & V \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ W' & \xrightarrow[b]{} & W, \end{array}$$

*there is a substitution-natural equivalence of equality fibres*

$$\chi_{w'} : \mathrm{Fib}_{f'}(w') \xrightarrow{\cong} \mathrm{Fib}_f(b(w')).$$

*Its induced comparison maps give*

$$b^*\Sigma^\ell_f \cong \Sigma^\ell_{f'} a^*, \qquad b^*\Pi^\ell_f \cong \Pi^\ell_{f'} a^*.$$

*The comparisons compose coherently under iterated pullbacks, and intertwine the Frobenius maps $\Sigma^\ell_f(f^*B \otimes A) \cong B \otimes \Sigma^\ell_f A$.*

The proof is given in Appendix B.

**Lemma 14.31** (The syntax carries the required structure)**.** *The construction is a split indexed linear–nonlinear model of* $\mathbb{T}_{\mathrm{Ag}}(\mathcal{K})$.

The proof is given in Appendix B.

**Interpretations and their comparisons.** Fix a target model and an interpretation of the typed signature $\mathcal{K}$. Strict interpretations preserve the chosen operations and substitution on the nose. For the weak formulation, use functors on contexts and functors on the ordinary and linear fibres over that base functor. Their specified invertible comparisons transport reindexing, comprehension, the monoidal and closed structures, the additives, the indexed adjoints and the $\mathsf{Lin}, \mathsf{Val}$ adjunction. Preservation of a universal construction includes its defining arrows (projections, injections, evaluation, or unit and counit). Coherence compares identity and composite reindexing and the substitution comparisons of the constructors, including Beck–Chevalley and Frobenius. At a primitive declaration the identification with its fixed interpretation is part of the data, over the already interpreted arity. This is the usual distinction between strict and pseudo interpretations in a category with families; compare the ordinary fragment in Castellan, Clairambault, and Dybjer (2017).

In the category of such interpretations a structural 2-cell $\theta : F \Rightarrow G$ consists of a natural isomorphism on contexts and natural fibre isomorphisms

$$\theta_{\Gamma,A} : F_\Gamma A \longrightarrow \theta_\Gamma^* G_\Gamma A$$

for ordinary and linear types. These commute with substitutions, with the specified structure comparisons and with the identifications of primitive declarations. In particular, on an arrow $t : A \to B$ the fibre naturality square is $\theta_{\Gamma,B} F(t) = (\theta_\Gamma^* G(t))\theta_{\Gamma,A}$. Composition uses composition on contexts and reindexing on the fibres. Thus uniqueness up to coherent comparison is a statement in a definite category of interpretations over the fixed signature.

**Theorem 14.32** (Initiality and equational completeness)**.** *Every interpretation of* $\mathcal{K}$ *in a split indexed linear–nonlinear model extends uniquely, up to its specified coherent structure comparisons, to an interpretation of the first-layer syntax. For well-typed terms* $t, u$*,*

$$\mathbb{T}_{\mathrm{Ag}}(\mathcal{K}) \vdash t = u \quad \Longleftrightarrow \quad [\![t]\!]_{\mathcal{M}} = [\![u]\!]_{\mathcal{M}} \text{ in every such model } \mathcal{M}.$$

*The same assertion holds for its derived agent operations.*

The proof is given in Appendix B.

### 14.2.4 Operational and modal completeness in the fixed semantics

The classifying-model result concerns derivability relative to all split indexed linear–nonlinear models. The fixed concurrent model admits stronger comparison results of a different kind: original arenas separate its open agent morphisms, while its retained transition relations admit a logical characterization by modal tests. These statements concern equality and expressivity in the specified semantics, rather than derivability of every fixed-model equation in the freely presented syntax.

**Theorem 14.33** (Original-arena probes separate open agent morphisms)**.** *For original arenas* $A, B$ *and an ordinary complete-policy carrier* $\Sigma$*, put* $H_\Sigma(A,B) = [\mathsf{Lin}\Sigma \otimes yA, yB] \in \mathcal{V}$*. For every* $K \in \mathcal{C}_0$ *there is a natural bijection*

$$\mathcal{V}(y(JK), H_\Sigma(A,B)) \cong \mathbf{Set}(\Sigma, \mathcal{C}(JK \otimes A, B)). \tag{14.10}$$

*For any* $T \in \mathcal{V}$ *and* $f, g : T \to H_\Sigma(A,B)$*, the following are equivalent:*

(i) *$f = g$ in the fixed presheaf category;*

(ii) *for every $K \in \mathcal{C}_0$, every $z : y(JK) \to T$, and every $\sigma \in \Sigma$, the concurrent arrows $JK \otimes A \to B$ decoded from $fz$ and $gz$ at $\sigma$ are equal in $\mathcal{C}$.*

*The comparison is stable under ordinary parameter substitution, by applying it in each indexed fibre. For $T = yR$, it reduces to pointwise equality of the families of actual concurrent arrows $R \otimes A \to B$.*

*Proof.*

(1) **Closedness, copowers, and Yoneda decoding.** By closedness, the copower description of $\mathsf{Lin}$, and strong monoidality of restricted Yoneda, there are natural isomorphisms

$$\begin{aligned}\mathcal{V}(y(JK), H_\Sigma(A,B)) &\cong \mathcal{V}(y(JK) \otimes \mathsf{Lin}\Sigma \otimes yA, yB)\\ &\cong \prod_{\sigma\in\Sigma} \mathcal{V}(y(JK \otimes A), yB)\\ &\cong \prod_{\sigma\in\Sigma} \mathcal{C}(JK \otimes A, B).\end{aligned}$$

The second isomorphism distributes Day tensor over the copower and identifies the resulting representables; the third is restricted Yoneda. Each comparison is natural in the arena probe and in the policy coordinate.

(2) **Separation by original arenas.** Yoneda identifies a map $z : y(JK) \to T$ with an element of $T(K)$. Two natural transformations $f, g : T \to H_\Sigma(A,B)$ are equal exactly when their component maps agree on every such element, for every $K \in \mathcal{C}_0$. The displayed bijection identifies this equality componentwise with equality of the decoded concurrent arrows at all $\sigma \in \Sigma$, proving both implications.

(3) **Representable contexts and substitution.** For $T = yR$, choose $K \in \mathcal{C}_0$ and an isomorphism $JK \cong R$; evaluation at the corresponding Yoneda element reduces the comparison to equality of the families $R \otimes A \to B$. In a dependent fibre all comparisons are pointwise, so ordinary reindexing preserves them. Ownerwise reparametrization simply precomposes the displayed policy family.

□

**Corollary 14.34** (Effective semantic equality of finite causal agents)**.** *Consider two closed agent states with the same displayed finite policy carrier $\Sigma$ and boundaries $A, B$, generated by the explicitly presented finite causal protocols of Theorems 10.4 and 10.10. Their denotations in $\mathcal{V}$ are equal if and only if, for every $\sigma \in \Sigma$, their realized finite strategies are isomorphic over $A^\perp \parallel B$. Equality of these explicitly encoded finite interpretations is decidable: a successful comparison supplies an arena-preserving event-structure isomorphism for every policy coordinate. The procedure also tests an exhibited finite ownerwise reparametrization.*

*Proof.*

(1) **Recovery of concurrent arrows.** For closed agent states, Theorem 2.18 and restricted Yoneda identify semantic equality with equality of the realized arrows in $\mathcal{C}(A,B)$ at each complete policy. By the definition of $\mathcal{C}$, equality of two such arrows is precisely isomorphism of their strategy event structures over the common polarized arrow arena.

(2) **Finite comparison procedure.** In the finite causal protocols, the policy sets, event carriers, event labels, orders, and consistency tables are finite explicitly specified data. For each policy, enumerate the finitely many bijections between the two event carriers

and test preservation of order, consistency, polarity, and arena labelling. The procedure returns an isomorphism exactly when one exists and terminates over the finite policy carrier.

(3) **Ownerwise reparametrization.** For a displayed finite map $r : \Theta \to \Sigma$, compare the source arrow at $\theta$ with the target arrow at $r(\theta)$. This is the defining realization equation for reparametrization and again requires finitely many comparisons.

□

Let $X$, $\ell : X \to L$, and $(R_a \subseteq X^2)_{a\in\mathcal{A}}$ be the retained carrier, observable data, and designated relations of Construction 12.35. Write $B_* = \nu\mathcal{F}$ for its greatest retained-label bisimulation. The predicates $P_v = \ell^{-1}\{v\} \hookrightarrow X$, $v \in \ell[X]$, are inverse images of ordinary equality predicates in the already constructed doctrine. Let $\mathsf{ML}_{\ell,R}$ be the Boolean *semantic test language* built from these equality-fibre tests, $\top$, Boolean operations, and $\langle a\rangle Q = \exists_{p_1} p_2^* Q$ for $R_a$. As usual, $[a]Q = \neg\langle a\rangle\neg Q$. Each $P_v$ may be regarded as a parameterized equality test at the retained value $v$; a closed syntactic formula requires a term naming that value. This distinction will be maintained in the expressive-completeness result.

**Theorem 14.35** (Modal characterization of the exact retained refinement)**.** *If each successor set $\{y : xR_ay\}$ is finite, then, for $x, y \in X$,*

$$x\, B_*\, y \quad\Longleftrightarrow\quad (\forall\varphi \in \mathsf{ML}_{\ell,R})\,(x \models \varphi \Longleftrightarrow y \models \varphi).$$

*Thus the coarsest exact quotient of Theorem 12.37 identifies precisely the states indistinguishable by these retained Boolean modal tests. This is the Hennessy–Milner separation argument (Hennessy and Milner, 1980, 1985), applied to the paper's displayed observation, strategic, and execution relations.*

*Proof.*

(1) **Invariance under retained bisimulation.** The relation $B_*$ preserves $\ell$ and matches $R_a$-successors in both directions. Therefore each equality-fibre test $P_v$ is invariant. Boolean operations preserve invariance, and the two successor clauses show that the diamond (hence also the box) preserves it. Induction on modal tests proves the forward implication.

(2) **Separating an unmatched successor.** Let $M$ relate states satisfying exactly the same tests. The tests $P_v$ give $M \subseteq \ker(\ell)$. Suppose $xMy$ and $xR_ax'$, but no $a$-successor $y'$ of $y$ satisfies $x'My'$. For each of the finitely many such $y'$ choose a modal test $\varphi_{y'}$ true at $x'$ and false at $y'$. Boolean negation permits this orientation. Then

$$\psi = \bigwedge_{yR_ay'} \varphi_{y'}$$

is true at $x'$ and false at every $a$-successor of $y$; the empty conjunction is $\top$. Thus $x \models \langle a\rangle\psi$ but $y \not\models \langle a\rangle\psi$, contradicting $xMy$.

(3) **Coinduction.** Interchanging $x, y$ gives the reverse successor clause. Consequently $M \subseteq \mathcal{F}(M)$. Coinduction for the greatest fixed point gives $M \subseteq B_*$, completing the converse.

□

The characteristic formulas below belong to the semantic language of retained equality tests: $P_{\ell(x)}$ is available as a predicate interpreting equality with the displayed label $\ell(x)$.

Their finite expressivity is relative to those named semantic tests. A separately presented first-layer signature denotes such a formula only when it contains a term naming the corresponding retained label; no general surjectivity from freely generated terms onto semantic labels is used.

**Theorem 14.36** (Finite-depth characteristic formulas and full expressivity)**.** *Suppose $X$ and $\mathcal{A}$ are finite. Define*

$$E_0 = \ker(\ell), \qquad E_{n+1} = \mathcal{F}(E_n).$$

*If $X$ is nonempty, this sequence stabilizes at $B_*$ after at most $|X| - 1$ strict refinement steps. In particular, $B_*$ is definable without a fixed-point binder by finitely many instances of ordinary equality, the designated relation tests, and their quantifier adjoints.*

*More explicitly, for each $x \in X$ put*

$$\chi_x^0 = P_{\ell(x)},$$

$$\chi_x^{n+1} = P_{\ell(x)} \ \wedge \bigwedge_{a \in \mathcal{A}} \left( \bigwedge_{xR_a z} \langle a \rangle \chi_z^n \ \wedge \ [a] \Big( \bigvee_{xR_a z} \chi_z^n \Big) \right).$$

*Empty conjunctions and disjunctions mean $\top$ and $\bot$. For all $x, y, n$,*

$$y \models \chi_x^n \quad \Longleftrightarrow \quad (x, y) \in E_n.$$

*If $m$ is the first stable index, then $\chi_x^m$ defines the $B_*$-class of $x$. Every $B_*$-invariant predicate $P \hookrightarrow X$ is consequently denoted by the finite modal expression $\bigvee_{x \in P} \chi_x^m$. In particular, evaluation induces an isomorphism of Boolean algebras*

$$\mathsf{ML}_{\ell,R}/{\equiv_X} \ \cong \ \mathsf{Pred}(X/B_*),$$

*where $\equiv_X$ denotes equality of denotations in this fixed finite model. This is expressive completeness of the semantic test language, not proof-theoretic completeness of the freely generated first-layer syntax. For $X = \varnothing$ the analogous statement holds with the unique predicate and $\bot$.*

*Proof.*

(1) **Finite bisimulation refinement.** For an equivalence relation $E$, let $[z]_E$ denote its equivalence class. Then $\mathcal{F}(E)$ is the kernel equivalence of the map

$$x \longmapsto \left(\ell(x), \left(\{[z]_E : xR_a z\}\right)_{a \in \mathcal{A}}\right).$$

Indeed, equality of the reached class sets expresses both successor matching clauses. Hence $\mathcal{F}(E)$ is an equivalence relation. Since $E_1 \subseteq E_0$ and $\mathcal{F}$ is monotone, the sequence $E_n$ descends. On nonempty finite $X$ every strict refinement splits at least one class, so there are at most $|X| - 1$ strict refinements.

(2) **Identification with the greatest fixed point.** If $E_m = E_{m+1}$, then $E_m$ is a fixed point and therefore $E_m \subseteq B_*$. Conversely $B_* \subseteq E_0$ by retained-label preservation, and monotonicity inductively gives $B_* \subseteq E_n$ for all $n$. Hence $E_m = B_*$. An explicit first-order definition is obtained recursively from $E_0(x, y) \equiv \ell(x) = \ell(y)$ by

$$\begin{aligned} E_{n+1}(x, y) \ \equiv \ & \ell(x) = \ell(y) \ \wedge \\ & \bigwedge_{a \in \mathcal{A}} \big[ \forall x' \, (xR_a x' \Rightarrow \exists y' \, (yR_a y' \wedge E_n(x', y'))) \\ & \qquad \wedge \, \forall y' \, (yR_a y' \Rightarrow \exists x' \, (xR_a x' \wedge E_n(x', y'))) \big]. \end{aligned}$$

The construction uses the existing equality, predicate, and quantifier operations and gives $\mathcal{F}(E_n)$ at each stage.

(3) **Characteristic modal tests.** We prove $y \models \chi_x^n \iff (x,y) \in E_n$ by induction on $n$. At $n = 0$ it says precisely $\ell(x) = \ell(y)$. For the successor case, the diamond conjunction states that every $a$-successor of $x$ has an $E_n$-matching successor of $y$; the box states that every $a$-successor of $y$ has an $E_n$-matching successor of $x$. These are exactly the clauses defining $E_{n+1}$. At stabilization, $\chi_x^m$ therefore denotes the $B_*$-class of $x$.

(4) **Classification of invariant predicates.** A predicate $P \subseteq X$ is $B_*$-invariant precisely when it is a union of $B_*$-classes. Therefore $\bigvee_{x \in P} \chi_x^m$ denotes $P$. All modal tests are $B_*$-invariant by the previous theorem, so their interpretations descend to predicates on $X/B_*$. Passing to tests modulo equality of denotations gives an injective Boolean-algebra map to $\mathsf{Pred}(X/B_*)$, and the characteristic formulas make it surjective. When $X = \varnothing$, both Boolean algebras have one element, represented by the denotation of $\bot$.

□

**Corollary 14.37** (Finite-model elimination of modal fixed points)**.** *On a fixed finite retained relation frame, every modal fixed-point predicate built from the retained atoms and relations has the same denotation as a finite Boolean modal expression without $\mu$ or $\nu$. For any monotone $F : \mathsf{Pred}(X) \to \mathsf{Pred}(X)$ given by the displayed assertion constructors, its least and greatest fixed points are also obtained by iteration from $\bot$ and $\top$, respectively, after at most $|X|$ strict changes.*

*Proof.*

(1) **Descent of modal fixed-point predicates.** By Theorem 12.37, the quotient by $B_*$ is surjective and bounded for each retained relation, and the designated atoms descend. The modal fixed-point transport theorem 9.29 consequently makes every such fixed-point denotation $B_*$-invariant. The characteristic-formula theorem then expresses it by a finite Boolean modal test without fixed-point binders.

(2) **Finite monotone iteration.** For any monotone $F$ on $\mathsf{Pred}(X)$, the iteration $\bot \leq F(\bot) \leq F^2(\bot) \leq \cdots$ is increasing, while $\top \geq F(\top) \geq F^2(\top) \geq \cdots$ is decreasing. Each strict change adds or removes at least one state, so a chain stabilizes after at most $|X|$ strict changes. At stabilization the value is a fixed point; induction and coinduction identify it with $\mu F$ and $\nu F$, respectively.

□

*Remark* 14.38 (Why the fixed-model result is not unrestricted syntactic completeness). These results separate all denotationally distinct open agent maps and characterize every finite-state bisimulation-invariant predicate, but they do not infer definitional equality of arbitrary primitive terms from equality in the fixed concurrent model. In the interpreted finite-data signature of Construction 10.7, the primitive name for the response $\widehat{\mathrm{id}_X} : D(X) \to D(X)$ denotes copycat, which is also the interpretation of the structural identity. When that name is freely declared, the existing rules need not equate its term with the identity term. Indeed, the split indexed linear–nonlinear model with both ordinary and linear fibres $\mathbf{Set}^{\Gamma_{\mathrm{disc}}}$, Cartesian tensor, and $\mathsf{Lin} = \mathsf{Val} = 1$ may interpret $d_X$ as a two-element set and the free response name as a nonidentity endomap. Soundness then rules out derivability of their equation in the free theory. The modal Boolean-algebra isomorphism above is likewise modulo *equality of denotations in the specified finite model*, not an unproved assertion about equations derivable in all indexed models. In particular, the concrete assertion fibres are Boolean whereas the listed logical inference rules are formulated through Heyting operations; a proof-theoretic completeness claim would also need to derive the corresponding fixed-model Boolean validities from the stated rules.

### 14.2.5 The different meanings of completeness

Several completeness results must be distinguished. The representation theorem 2.18 identifies semantic agent states with owned concurrent realizations. The finite causal protocol theorem 10.10 gives definability for that generated fragment; Theorem 10.23 retains its finite session realizations and witnesses. The classifying-model theorem 14.32 establishes derivability from equality in all the designated first-layer models. In contrast, Theorem 14.33 separates arrows in the single fixed presheaf model, and Theorems 14.35 and 14.36 characterize modal indistinguishability and finite-state expressivity relative to the retained tests. Neither conclusion equates all fixed-model denotational equalities with derivable syntactic equations.

Classifying-model completeness ranges over a class of indexed linear–nonlinear models. Definability and equality reflection in a fixed model are different questions. For instance, two unrelated primitive scalar constants can remain distinct in the free syntax while both necessarily denote the unique element of $\mathcal{C}(I, I)$. Remark 14.38 exhibits the same obstruction using a named finite response and copycat. The operational separation and modal results characterize fixed-model semantic equality at their specified interfaces; they do not prove derivability of those equalities. Full completeness in particular causal-invariant linear fragments (Castellan and Yoshida, 2019) is not transferred here to all receptive courteous strategies.

We have two ways to inspect a proposed reduction. Operational tests distinguish the concurrent arrows, while the chosen modal tests distinguish the retained alternatives on a finite frame. The authorization protocol illustrates the practical difference. A guard certifies that the issuer's event preceded execution. A judgement about whether the issue was permitted uses the institutional rule as well.

## Conclusion

The authorization checker at the beginning of the paper faced two policies that agreed on the current action. Their different prescriptions at another input made one joint profile legal and the other illegal. The construction identifies the extra evidence the checker needs, routes it to the checker alone, and traces the authorization token to the final execution. It therefore answers three oversight questions separately. The contents of the policy determine whether the rule applies. The observation interface determines whether the checker can decide. The execution witness determines whether this action used the issued authorization.

For a cybernetic system, a related design choice is what to keep when replacing a controller by a smaller one. If the required behaviour is its future local response, histories can be merged exactly when every observation continuation elicits the same responses. Their distinct residuals give the minimum reachable state count in the finite case. If the reduced system is also supposed to support reasoning about interventions, its states must preserve the relevant alternatives from each source. A hidden event that disables a visible move is enough to separate these two design questions.

The formal results specify the scope of both readings. Composition preserves the designated owner policies and their concurrent realizations. The calculus is sound, and validity across its classifying-model semantics characterizes first-layer derivability. Within the fixed semantics, original concurrent arenas separate open-agent arrows. On finite retained frames, modal tests characterize bisimulation and characteristic formulas describe its classes.

**Scope and interfaces.** For an application, the choice of interface determines what can be certified. Yoneda recovers the original concurrent arrows. The owner factors say who

controls a deviation, and the legal and observation data determine which responses can be used. The execution span retains the witness on which a later action may depend. The comparisons in Theorems 11.18, 14.1 and 14.26 specify which of these data survive each abstraction. In the finite checker, for instance, the extra policy entries are delivered by construction. Applying the same rule to another implementation requires identifying what that implementation can actually supply.

*Remark* 14.39 (Retained economic interfaces). The linear semantics determines how exhibited interactions compose. Ownership, observations, comparison predicates, technologies, and institutional rules are explicitly typed economic data. Concurrent strategies determine event enablement and residualization; observation wires and retained pointed histories determine the displayed action uncertainty. The resulting predicates express equilibrium, ability, rationalizability, feasibility, coalition blocking, and truthful implementation for the specified economic presentation. The Cartesian-frame, local-state, and choice-cell comparisons identify their respective interfaces. Persistent intentions and corporate judgements can be represented through explicit commitment and aggregation data retained by the same presentation.

The cost construction gives a further reason to keep a completed witness. A matched delivery contributes one class to an additive charge, while its place in the causal order can also contribute to the critical-path duration. Applying a budget to these witnesses selects the executions the institution admits. Applying a price to them changes the gains from deviation. In the two-bit example, the joint affirmative action remains an equilibrium at the threshold price of one and ceases to be one above it. In a continuation game the writer carries the accrued charge into each child, so alternatives are compared from the same prior history. Projecting away the charge recovers outcome-only play. The same measured record can therefore serve an audit, an incentive comparison, or a decision about a later action, with a different test in each case.

*Remark* 14.40 (Probability and intensional interfaces). The policy sets and private policy records used here are semantic values; their finite realizations are extensional response protocols. Models of open-source games additionally represent program descriptions, permitting self-reference and inspection of another agent's strategy description. Interactive diagonalization is studied by Abramsky and Zvesper (2010), and provability-based equilibria by Barasz et al. (2014). Rational synthesis, as studied by Fisman, Kupferman, and Lustig (2009), asks for implementations satisfying strategic specifications.

The present theory uses pure policies and qualitative information. Probabilistic extensions require distributions and chance kernels, as in Bayesian games (Harsanyi, 1967) and stochastic games (Shapley, 1953). The disclosure and action-model constructions here have the formula semantics and reduction laws of Appendix A.2. Observational distinctions that do not descend to the original visible quotient are retained on the exact refinement and its dependent operational witness family.

The completion, linearization, and dependent adjoints are set-theoretic categorical constructions, and least and greatest fixed points are order-theoretic operations on predicates. Finite path maxima construct the potentials of Theorem 13.6; directization constructs $r_s = m \prod_i s_i$ with the universal comparison of Proposition 13.35.

For a proposed oversight mechanism the first design decision is therefore what evidence reaches each decision point. For a controller abstraction it is which future responses or interventions the abstraction must preserve. In both cases the mathematical work starts once the interface is specified. One can then exhibit a decision protocol, compare its complete alternatives, and trace the event on which a later action depends.

# A Natural-Deduction Presentation of the Logic of Agency

The main text constructs policies, their causal realizations, and the assertions an observer can make about them. This appendix gives the inference rules for reasoning at each of those three levels. In particular the context of a judgement shows which ordinary parameters can be reused and which linear occurrences an execution must account for. Appendix B proves soundness and the classifying-model result for the rules.

## A.1 Ordinary and linear agent type theory

The proof-theoretic fragment is interpreted through the linear–nonlinear adjunction of Benton (1995) and the indexed dependent setting of Vákár (2015). The rule tables preserve the distinction between ordinary variables, which may be reused, and linear variables, which must be accounted for by context splitting.

The ordinary and linear judgements below are interpreted in the indexed categories of the main text. In a rule with several premises, the contexts show how the linear resources are divided. This is the information used in the soundness induction and in deriving the displayed operations on agents.

### A.1.1 Judgments of the agent calculus

**Ordinary parameters and linear resources** The ordinary fragment is extensional dependent type theory with finite products and coproducts, dependent sums and products, and equality. Ordinary types and terms have judgments

$$\Gamma \vdash X\ \mathsf{Type}, \qquad \Gamma \vdash u : X.$$

An ordinary context is built by comprehension $\Gamma, x : X$. Linear families may depend on ordinary variables:

$$\Gamma \vdash A\ \mathsf{LType}, \qquad \Gamma; \Delta \vdash t : A.$$

Here $\Delta = x_1 : A_1, \dots, x_n : A_n$ is a linear context whose families are all indexed by $\Gamma$. Each $A_k$ is indexed by the ordinary context. A later linear type does not depend on a preceding linear variable. Ordinary dependence is therefore available in type formation, while resource occurrences remain governed by tensor-context use.

The linear type constructors are

$$\mathsf{I}, \quad A \otimes B, \quad A \multimap B, \quad A \,\&\, B, \quad \top, \quad A \oplus B, \quad \mathbf{0}, \quad \Sigma^{\ell}_{x} A, \quad \Pi^{\ell}_{x} A, \quad \mathsf{Lin} X.$$

The ordinary family $\mathsf{Val}A$ records monoidal states of the linear family $A$. The adjunction $\mathsf{Lin} \dashv \mathsf{Val}$ connects the two kinds of judgments, and $!A$ abbreviates $\mathsf{LinVal}A$, the resulting exponential. These rules do not supply an unrestricted contraction on arbitrary linear terms.

**The multiplicative and adjunction rules** The essential multiplicative rules are

$$\frac{}{\Gamma; x : A \vdash x : A}, \qquad \frac{\Gamma; \Delta \vdash t : A \quad \Gamma; \Theta \vdash u : B}{\Gamma; \Delta, \Theta \vdash t \otimes u : A \otimes B},$$

$$\frac{\Gamma; \Delta, x : A \vdash t : B}{\Gamma; \Delta \vdash \lambda x.t : A \multimap B}, \qquad \frac{\Gamma; \Delta \vdash f : A \multimap B \quad \Gamma; \Theta \vdash t : A}{\Gamma; \Delta, \Theta \vdash f(t) : B}.$$

Each multiplicative rule partitions its linear premises into disjoint contexts. Tensor elimination consumes its argument and binds both components, while the unit is introduced only

with the appropriate empty linear context. These conditions implement the resource-sensitive meaning of the connectives.

Ordinary terms are introduced into $\mathsf{Lin}X$ by

$$\frac{\Gamma \vdash u : X}{\Gamma; \varnothing \vdash \mathsf{return}(u) : \mathsf{Lin}X}.$$

Elimination of $\mathsf{Lin}X$ binds an ordinary $x : X$ inside the continuation:

$$\frac{\Gamma; \Delta \vdash s : \mathsf{Lin}X \quad \Gamma, x : X; \Theta \vdash t : A}{\Gamma; \Delta, \Theta \vdash \mathsf{let}\ \mathsf{return}(x) = s\ \mathsf{in}\ t : A}. \tag{A.1}$$

The displayed instance has $\Theta$ and $A$ pulled back from $\Gamma$. Dependent result types are expressed by the indexed sum $\Sigma^{\ell}$. Elimination binds an ordinary parameter that may be reused in its continuation, while the supplied linear term $s$ is consumed exactly once.

The value rules are

$$\frac{\Gamma; \varnothing \vdash t : A}{\Gamma \vdash \mathsf{thunk}(t) : \mathsf{Val}A}, \qquad \frac{\Gamma \vdash v : \mathsf{Val}A}{\Gamma; \varnothing \vdash \mathsf{force}(v) : A}.$$

The value rules express the hom-set adjunction by their computation and uniqueness laws. The introduction rule requires an empty linear context: a term depending on a linear resource cannot thereby be turned into reusable ordinary information.

Definitional equality consists of the displayed $\beta$ and $\eta$ equations for tensor, unit, internal hom, additives, dependent sums and products, and the adjunction. Its commuting conversions make eliminators natural in their continuations. The soundness proof checks each family of equations against the corresponding categorical universal property.

**Indexed linear abstraction** For $\Gamma \vdash X\ \mathsf{Type}$ and $\Gamma, x : X \vdash A\ \mathsf{LType}$, the dependent product introduction rule is

$$\frac{\Gamma, x : X; p^{*}\Delta \vdash t : A}{\Gamma; \Delta \vdash \Lambda x.t : \Pi^{\ell}_{x} A}, \qquad p : \Gamma.X \to \Gamma.$$

Application selects one ordinary index from the dependent product. The same linear resource may be mentioned in alternative components of an additive product, but it cannot be extracted twice as simultaneous linear occurrences. The distinction is essential for the agent abstraction derived from the dependent product.

**Lemma A.1** (Substitution)**.** *Ordinary substitution preserves all formation, typing, and equality judgments. If*

$$\Gamma; \Delta \vdash s : A, \qquad \Gamma; \Theta, x : A \vdash t : B,$$

*then* $\Gamma; \Theta, \Delta \vdash t[s/x] : B$*. The substitution uses the resources of* $\Delta$ *once in the multiplicative sense and preserves definitional equality.*

*Proof.*

(1) **Fixed-point argument.** Proceed by induction on the derivation into which substitution is made. For a variable, substitute $s$ if it is $x$, and retain the variable otherwise. At a multiplicative rule, $x$ belongs to exactly one premise; apply the induction to that premise and retain the other disjoint context.

(2) **Fibrewise calculation.** At an additive introduction or case rule, the same context is present in the alternative premises, so substitute in each alternative, retaining one shared context in the conclusion. Under an ordinary binder, rename its variable to avoid capture and pull $s$ back along comprehension. Under a linear binder, rename that variable and apply the induction to the body.

(3) **Fixed-point argument.** The Val introduction case has empty linear context, so it cannot contain a free $x$. The Lin eliminator uses the two induction steps for its split premises. These exhaust the constructors.

(4) **Functoriality and substitution.** Each defining equation is stable under the same capture-avoiding substitutions, proving the equality claim.

□

**Corollary A.2** (Preservation for the core reductions)**.** *Every well-typed $\beta$ reduction and every typed commuting conversion preserves its judgment.*

*Proof.*

(1) **Secured operational interaction.** The internal-hom and dependent elimination cases are instances of Lemma A.1. Tensor and unit reduction consume precisely the components introduced by their matching constructors. Additive reduction selects one branch with its declared context.

(2) **Adjunction and transposition.** Adjunction reduction substitutes the ordinary value supplied by return or removes a matching thunk–force pair. Commuting conversions rearrange the same context splits and bindings.

□

**Example A.3** (Linear occurrences and ordinary descriptions)**.** Given $a : R \multimap A$, $b : R \multimap B$, and one linear occurrence $x : R$, the expression $a(x) \otimes b(x)$ is not derivable: tensor introduction would place $x$ in both split premises. An exhibited map $R \to R \otimes R$, or a supplied input $R \otimes R$, makes the needed resource available. An ordinary identifier of $R$ supplies neither. There is an important distinction for an actual value $v : \mathsf{Val}R$: ordinary contraction together with the value and tensor rules derives

$$\Gamma, v : \mathsf{Val}R; \varnothing \vdash \mathsf{force}(v) \otimes \mathsf{force}(v) : R \otimes R.$$

This constructs two states from a reusable ordinary state description; it does not contract a linear variable. Consequently linear typing alone does not establish unforgeability of an authorization token. The causally token-guarded protocol of Lemma 10.8 provides the operational dependency needed in the finite-data fragment.

### Complete rule tables and formation conventions

**Binding and formation conventions** The rules are schematic under ordinary substitution and type conversion. Bound variables are chosen fresh and substitutions avoid capture. Multiplicative context concatenation requires disjoint variable sets; a rule over $\Gamma, x : X$ pulls back any context or target originally over $\Gamma$ along its projection. A resource may appear in alternative additive premises, but this does not allow simultaneous duplication. These are the binding conventions used in the later soundness induction.

Ordinary formation is generated from the declared sorts by $1, 0, X + Y, \Sigma_{x:X} Y, \Pi_{x:X} Y$, and extensional equality. The ordinary rules are variable, weakening, contraction, exchange, substitution, unit introduction, empty elimination, sum injections and case, dependent pairing and projections, abstraction and application, and equality reflexivity and substitution. Their $\beta\eta$ equations are the evaluation and uniqueness equations for these constructors. Ordinary equality is *extensional*: the ordinary foundation presents equality by comprehension of the diagonal. Its eliminator and conversion rule are those of this extensional ordinary type

theory. Semantically, a section of a diagonal pullback forces its two compared sections to be equal, and monicity makes its equality witness unique. The proof is given in Proposition 14.28. This is a genuine feature of the selected ordinary foundation, not a consequence of products, sums, or dependent products and not an additional law of agency. No equality reflection for arbitrary linear resources is introduced.

The mixed formation rules are

$$\frac{\Gamma \vdash X\ \mathsf{Type}}{\Gamma \vdash \mathsf{Lin}X\ \mathsf{LType}}, \qquad \frac{\Gamma \vdash A\ \mathsf{LType}}{\Gamma \vdash \mathsf{Val}A\ \mathsf{Type}},$$

$$\frac{\Gamma \vdash A, B\ \mathsf{LType}}{\Gamma \vdash A \otimes B,\ A \multimap B,\ A \,\&\, B,\ A \oplus B\ \mathsf{LType}},$$

$$\frac{\Gamma \vdash X\ \mathsf{Type} \qquad \Gamma, x : X \vdash A\ \mathsf{LType}}{\Gamma \vdash \Sigma^{\ell}_{x} A,\ \Pi^{\ell}_{x} A\ \mathsf{LType}}.$$

The nullary constructors are $\mathsf{I}, \top, \mathbf{0}$. For every displayed signature $(\vec{\Sigma}; A, B)$, $\mathsf{Ag}_{\vec{\Sigma}}(A, B)$ abbreviates $[\mathsf{Lin}(\prod_i \Sigma_i) \otimes A, B]$ with that signature retained.

**The remaining linear introduction and elimination rules** Tensor elimination and the multiplicative unit are

$$\frac{\Gamma; \Delta \vdash s : A \otimes B \qquad \Gamma; \Theta, x : A, y : B \vdash t : C}{\Gamma; \Delta, \Theta \vdash \mathsf{let}\ x \otimes y = s\ \mathsf{in}\ t : C},$$

$$\frac{}{\Gamma; \varnothing \vdash * : \mathsf{I}}, \qquad \frac{\Gamma; \Delta \vdash s : \mathsf{I} \qquad \Gamma; \Theta \vdash t : C}{\Gamma; \Delta, \Theta \vdash \mathsf{let}\ * = s\ \mathsf{in}\ t : C}.$$

Additive products have

$$\frac{\Gamma; \Delta \vdash t : A \qquad \Gamma; \Delta \vdash u : B}{\Gamma; \Delta \vdash \langle t, u \rangle : A \,\&\, B}, \qquad \frac{\Gamma; \Delta \vdash t : A \,\&\, B}{\Gamma; \Delta \vdash \pi_1 t : A},$$

and the analogous second projection. The nullary additive product has a unique term $\Gamma; \Delta \vdash \langle\rangle : \top$ at every linear context. This is the terminal-object rule, not weakening into an arbitrary target.

For additive sums, either injection uses its premise's context, and

$$\frac{\Gamma; \Delta \vdash s : A \oplus B \qquad \Gamma; \Theta, x : A \vdash t : C \qquad \Gamma; \Theta, y : B \vdash u : C}{\Gamma; \Delta, \Theta \vdash \mathsf{case}\ s\ \mathsf{of}\ (x.t \mid y.u) : C}.$$

A term of $\mathbf{0}$ eliminates into any target, with an additional linear context allowed in that impossible branch. Semantically this is the unique map $0 \otimes X \to C$, using $0 \otimes X \cong 0$.

For dependent linear sums, with $p : \Gamma.X \to \Gamma$,

$$\frac{\Gamma \vdash u : X \qquad \Gamma; \Delta \vdash t : A[u/x]}{\Gamma; \Delta \vdash (u, t) : \Sigma^{\ell}_{x} A},$$

$$\frac{\Gamma; \Delta \vdash s : \Sigma^{\ell}_{x} A \qquad \Gamma, x : X; p^{*}\Theta, z : A \vdash t : p^{*}B}{\Gamma; \Delta, \Theta \vdash \mathsf{let}\ (x, z) = s\ \mathsf{in}\ t : B}.$$

Dependent product abstraction is in Section A.1.1; its application is

$$\frac{\Gamma; \Delta \vdash f : \Pi^{\ell}_{x} A \qquad \Gamma \vdash u : X}{\Gamma; \Delta \vdash f[u] : A[u/x]}.$$

The ordinary–linear rules are those of Section A.1.1. Each primitive linear generator consumes the tensor of its declared input resources, with the indicated context split. No additional copying rule is attached to a primitive generator.

The defining equations include

$$\begin{aligned} (\lambda x.t)(u) &= t[u/x], & \lambda x.f(x) &= f, \\ \pi_1\langle t, u\rangle &= t, & \langle \pi_1 s, \pi_2 s\rangle &= s, \\ (\Lambda x.t)[u] &= t[u/x], & \Lambda x.f[x] &= f, \\ \mathsf{force}(\mathsf{thunk}(t)) &= t, & \mathsf{thunk}(\mathsf{force}(v)) &= v. \end{aligned}$$

Tensor, dependent-sum, additive-sum, unit, and $\mathsf{Lin}$ computation laws identify elimination immediately following the corresponding introduction. Their uniqueness counterparts say that a value is determined by its projections or case components; in particular, maps out of sums are determined by all branches. For the mixed ordinary–linear eliminator, the fundamental equations (with the context conditions of (A.1)) include

$$\begin{aligned} &\mathsf{let}\ \mathsf{return}(x) = \mathsf{return}(u)\ \mathsf{in}\ t = t[u/x], \\ &\mathsf{let}\ \mathsf{return}(x) = z\ \mathsf{in}\ \mathsf{return}(x) = z \qquad (z : \mathsf{Lin}X). \end{aligned}$$

The remaining commuting conversions express naturality of the eliminators in their linear continuations. If $E_s(t)$ eliminates a constructor whose parameter is $s$ and continuation is $t$, and $k$ follows the result, then eliminating into the composite continuation agrees with eliminating first and composing with $k$, with identical context splitting. These equations include the coproduct and $\mathsf{Lin}$ commuting conversions used by the classifying-model construction. Each equation is checked under the same typed substitution rules. The value equations retain the empty linear context required by $\mathsf{thunk}$. This completes the formal first-layer equations used by the classifying model.

### A.1.2 Measured witnesses in the dependent linear calculus

The data of Construction 2.8 have an ordinary indexed presentation. Over $\Gamma, \sigma : \Sigma$ declare the completed witness family $K_\sigma$. In context $\Gamma, \sigma : \Sigma, k : K_\sigma$ the outcome $a_\sigma(k) : Z$ and the charge $m_\sigma(k) : C$ are ordinary terms. The measured-record term and its projections are

$$r_\sigma(k) = \langle m_\sigma(k), a_\sigma(k)\rangle : T_C Z, \qquad \pi_1 r_\sigma(k) = m_\sigma(k), \quad \pi_2 r_\sigma(k) = a_\sigma(k).$$

Here $(C, +, 0)$ interprets the selected charge monoid. In a finite secured execution the term $m$ evaluates the multiset of matched classes in Construction 8.6. It reads the ordinary execution witness, which is why a derivation using the charge must retain that witness even when the linear strategy has been composed and hidden.

**Retaining an execution and its linear output** Abbreviate $\Gamma, \sigma : \Sigma$ by $\Gamma'$ and let $\Gamma', k : K_\sigma \vdash A(k)\ \mathsf{LType}$. Define $\mathsf{Run}_K(A) = \Sigma^{\ell}_{k:K_\sigma} A(k)$. For a linear family $D$ over $\Gamma'$ the introduction and elimination rules specialize to

$$\frac{\Gamma' \vdash h : K_\sigma \qquad \Gamma'; \Delta \vdash t : A(h)}{\Gamma'; \Delta \vdash (h, t) : \mathsf{Run}_K(A)},$$

$$\frac{\Gamma'; \Delta \vdash u : \mathsf{Run}_K(A) \qquad \Gamma', k : K_\sigma; p^*\Theta, x : A(k) \vdash v : p^*D}{\Gamma'; \Delta, \Theta \vdash \mathsf{let}\ (k, x) = u\ \mathsf{in}\ v : D}, \qquad p : \Gamma'.K_\sigma \to \Gamma'.$$

Eliminating $(h, t)$ gives $v[h/k, t/x]$. Repacking $(k, x)$ after opening $u$ gives $u$. In the elimination branch $a_\sigma(k)$ and $m_\sigma(k)$ may each be used in an ordinary comparison, while the output $x$ occupies one linear slot.

**An explicit cost tag** For a linear output $A$ over a fixed ordinary context put $\mathsf{Log}_C(A) = \mathsf{Lin}C \otimes A$. Write $\mathsf{tag}_C(c,t) = \mathsf{return}(c) \otimes t$. Its introduction uses an ordinary term $c : C$ and exactly the linear context of $t : A$. A completed witness $h$ can therefore accompany its output with the tag $m_\sigma(h)$. Opening a tag is tensor elimination followed by $\mathsf{Lin}$ elimination on its first component. A homomorphism $j : C \to C'$ induces $j_* = \mathsf{Lin}(j) \otimes 1_A$; adding an ordinary charge $d : C$ induces $\mathsf{step}^d = \mathsf{Lin}(c \mapsto d + c) \otimes 1_A$. Erasure is induced by $C \to 1$ and the unit comparison $\mathsf{Lin}1 \otimes A \cong A$. On a displayed tag these maps give

$$\begin{aligned} j_*\mathsf{tag}_C(c,t) &= \mathsf{tag}_{C'}(j(c),t), \\ \mathsf{step}^d\mathsf{tag}_C(c,t) &= \mathsf{tag}_C(d+c,t), \\ \mathsf{erase}_C\mathsf{tag}_C(c,t) &= t. \end{aligned}$$

The zero and addition laws for $\mathsf{step}$, the equation $j_*\mathsf{step}^d = \mathsf{step}^{j(d)}j_*$, and $\mathsf{erase}_{C'}j_* = \mathsf{erase}_C$ follow by functoriality of $\mathsf{Lin}$ and the corresponding ordinary maps. Reindexing substitutes the ordinary parameters in each formula. These are derived operations of the existing type theory. Their concrete equations use the indicated equalities of the interpreted ordinary maps. The tag accompanies a completed output; the fibre $K_\sigma$ separately records whether any such output was obtained.

**Proposition A.4** (Linear interpretation of the cost writer)**.** *In each linear fibre $L(A) = \mathsf{Log}_C(A) = \mathsf{Lin}C \otimes A$ has the writer monad structure induced by the ordinary cost monoid. There is a natural comparison $\chi_X : \mathsf{Lin}(T_C X) \cong L(\mathsf{Lin}X)$ intertwining both units and multiplications. The constructions commute with ordinary base substitution.*

*Proof.*

(1) **Transport the cost operations.** Strong monoidality gives $u : \mathsf{I} \cong \mathsf{Lin}1 \xrightarrow{\mathsf{Lin}(0)} \mathsf{Lin}C$ and $v : \mathsf{Lin}C \otimes \mathsf{Lin}C \cong \mathsf{Lin}(C \times C) \xrightarrow{\mathsf{Lin}(+)} \mathsf{Lin}C$. Put $\eta^L_A = (u \otimes 1_A)\lambda_A^{-1}$ and $\mu^L_A = (v \otimes 1_A)\alpha^{-1}_{\mathsf{Lin}C,\mathsf{Lin}C,A}$, with $\alpha : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$. The unit laws follow from the two unit laws for $v$, and both routes from $L^3A$ use the same associativity equation for $v$ after coherent rebracketing. These are equalities of open arrows.

(2) **Verify the comparison.** Take $\chi_X$ to be the strong monoidal comparison $\mathsf{Lin}(C \times X) \cong \mathsf{Lin}C \otimes \mathsf{Lin}X$. Then $\chi_X \mathsf{Lin}(\eta^T_X) = \eta^L_{\mathsf{Lin}X}$ by unit coherence, and

$$\begin{array}{ccc} \mathsf{Lin}(T_C^2 X) & \xrightarrow{L(\chi_X)\,\chi_{T_C X}} & L^2(\mathsf{Lin}X) \\ {\scriptstyle \mathsf{Lin}(\mu^T_X)}\downarrow & & \downarrow{\scriptstyle \mu^L_{\mathsf{Lin}X}} \\ \mathsf{Lin}(T_C X) & \xrightarrow[\chi_X]{} & L(\mathsf{Lin}X) \end{array}$$

Both routes apply $\mathsf{Lin}(+)$ to the two charge factors and identity to the result. Naturality of the monoidal comparison proves this equation. The indexed adjunction comparisons reindex $u, v, \chi$ and their diagrams fibrewise.

(3) **Compute the term operations.** Unit tags a linear input with the ordinary zero. Multiplication opens the outer and inner tags, then returns one tag with the sum and the original linear output. Tensor and $\mathsf{Lin}$ elimination consume that output once. Applying $\mathsf{Lin}$ to $c \mapsto d + c$, to $j$, or to $C \to 1$ recovers the step, regrading and erasure calculations above. In the free first layer, equalities of compound tags use the corresponding derivable equalities of their ordinary charge terms; for a concrete interpreted monoid the displayed laws hold in its interpretation.

□

## A.2 Strategic, epistemic, institutional, and temporal assertions

The assertion calculus is generated from the quantifier adjunctions in Jacobs (1999) and the complete-lattice fixed-point construction of Tarski (1955). The rules below record which modalities are derived from which maps, so that ordinary logical reasoning and linear execution remain separate judgments.

There are two coordinated languages: assertions over ordinary state contexts and linear terms over those contexts. For a situated agent with complete policy carrier $\Sigma$, legal carrier $L$, reference map $s : W \to L$, observation quotients $o_i : W \twoheadrightarrow O_i$, and comparison predicates $P_i$, the clauses of the predicate language are interpreted in $\mathsf{Pred}(W)$ using exactly those carriers and their typed relations. Its formulas therefore inherit the available alternatives from the declared game rather than treating control as an untyped modal primitive.

### A.2.1 Assertion and strategic judgments

Write $W \mid \Phi \vdash \varphi$ when $\bigcap_{\psi\in\Phi}[\![\psi]\!] \subseteq [\![\varphi]\!]$. Ordinary equality, conjunction, disjunction, implication, quantifiers, and substitution have the extensional subobject interpretation. For a coalition $C$ and the kernel pair $D_C^L = L \times_{L_{-C}} L$, the alternatives from a reference are its fibres under the first projection. The existential and universal counterfactual operations are

$$\langle C\rangle Q = \exists_{p_1} p_2^* Q, \qquad [C]Q = \forall_{p_1} p_2^* Q.$$

Selection and best response are obtained by pulling back strict-improvement comparisons along paired concurrent realizations. The predicate $\mathsf{NE}$ universally excludes profitable unilateral alternatives. Coalition effectivity uses a different order of quantifiers: it chooses a realizable coalition coordinate and tests every legal completion. The formal rules retain that distinction.

### A.2.2 Knowledge, perspective, and common knowledge

For $o_i : W \twoheadrightarrow O_i$ put

$$K_i P = o_i^* \forall_{o_i} P, \qquad D_C P = q_C^* \forall_{q_C} P,$$

where $q_C$ is the effective image of the tuple of the members' observations. Let $R_C^*$ be the finite-path equivalence closure of $\bigcup_{i\in C} \ker(o_i)$. Then

$$\mathsf{CK}_C P = [R_C^*]P = \nu X.\left(P \wedge \bigwedge_{i\in C} K_i X\right).$$

The knowledge rules follow from the observation quotient and its kernel pair. The proof-relevant linear operation $K_{o_i}^{\ell} = o_i^* \Pi_{o_i}^{\ell}$ is separately interpreted as a comonad. Its predicate reflection yields the stated knowledge operator, without identifying their witness products.

### A.2.3 Institutional and epistemic action formulas

For a witnessed span $X \xleftarrow{s} E \xrightarrow{t} Y$, the dynamic clauses are

$$\langle E\rangle Q = \exists_s t^* Q, \qquad [E]Q = \forall_s t^* Q.$$

A public announcement of $\varphi$ has updated carrier $j : U = \{w : \varphi(w)\} \hookrightarrow W$ with relations $(j \times j)^* R_i$ and box $[!\varphi]\psi = \forall_j [\![\psi]\!]_{M|\varphi}$. A general epistemic action $(E, (S_i), \mathsf{pre})$ has updated carrier $U = \{(w, e) : \mathsf{pre}(w, e)\}$ and relation $(w, e)R_i^+(v, f)$ exactly when $wR_i v$ and $eS_i f$. All postformulas are evaluated in the updated model.

### A.2.4 Fixpoint and temporal rules

Least and greatest fixed points, denoted by $\mu$ and $\nu$, are evaluated in the complete predicate lattice of the current context. Induction uses the least prefixpoint property, while coinduction uses the greatest postfixpoint property. Temporal maintenance and achievement apply these operations to the controllable predecessor supplied by the finite public transition game. The formulas do not create transition witnesses; those are supplied by the operational construction.

### A.2.5 Formal reductions and soundness obligations

Public announcement, epistemic action, substitution, and modal reduction are consequences of comprehension, pullback, and the dependent adjunctions. The displayed equations fix the interpretation of these operations for the soundness proof; no independent reduction principle is added at the level of concurrent arrows.

$$[!\varphi]K_i\psi \leftrightarrow (\varphi \Rightarrow K_i[!\varphi]\psi), \qquad [!\varphi][!\psi]\chi \leftrightarrow [!(\varphi \wedge [!\varphi]\psi)]\chi.$$

Each assertion rule is sound when it denotes an inclusion of subobjects, and each typed term equation is sound when it denotes equality of arrows. Appendix B verifies these obligations by induction on derivations.

### A.2.6 Types, formulas, and contexts

Fix a displayed agent process $G = (\vec{\Sigma}, g) : A \to B$, a legal comprehension $m : L \hookrightarrow \Sigma$, and a reference context $s : W \to L$. Its sorts include ordinary state and policy families, linear families over those states, the concurrent arrow carrier $\mathcal{C}(A, B)$, and dependent witness families. An ordinary term is interpreted as a section of an ordinary family over its declared context; a linear term is interpreted as a morphism from the tensor of its linear context. A formula in context $W$ denotes a subobject $[\![\varphi]\!] \hookrightarrow W$. For each ordinary context map $f : V \to W$ there is a substitution operation $f^* : \mathsf{Pred}(W) \to \mathsf{Pred}(V)$.

The logical grammar may be given schematically by

$$\begin{aligned}\varphi, \psi ::= {} & X \mid \top \mid \bot \mid P(t_1, \ldots, t_n) \mid t = u \mid (\varphi \wedge \psi) \mid (\varphi \vee \psi) \mid (\varphi \Rightarrow \psi) \\ & \mid \exists_f \varphi \mid \forall_f \varphi \mid \langle C \rangle \varphi \mid [C]\varphi \mid K_i\varphi \mid \mathsf{DK}_C\varphi \mid \mathsf{CK}_C\varphi \\ & \mid \langle E \rangle \varphi \mid [E]\varphi \mid [!\varphi]\psi \mid [\mathcal{A}, e]\psi \mid \mu X.\varphi(X) \mid \nu X.\varphi(X).\end{aligned}$$

Each predicate atom is typed at a declared ordinary carrier. The coalition variable $C$ refers to actual owner coordinates; $E$ is an exhibited witness span; and $(\mathcal{A}, e)$ is an epistemic event with enabling data. Each predicate variable has a state sort. For the extended grammar, binder formation checks the variance of its expanded body as follows.

**Expansion over a restricted state carrier.** For $j_D : D \hookrightarrow W$, write $T_D(\psi)$ for the ambient predicate supported on $D$ that expresses truth of $\psi$ in $M|_D$. Free world predicates are restricted along $j_D$. For a fixed factual atom $p$ and a variable $Y$, the clauses are $T_D(p) = D \wedge p$ and $T_D(Y) = D \wedge Y$. Updated atoms use their specified postcondition term at the updated sort. Put $T_D(\top) = D$, $T_D(\bot) = \bot$, distribute $T_D$ over meets and joins, and put

$$T_D(\theta \Rightarrow \psi) = D \wedge \big(T_D(\theta) \Rightarrow T_D(\psi)\big).$$

For a fixed accessibility relation $R$ on $W$, and for a public announcement, the clauses are

$$\begin{aligned} T_D([R]\psi) &= D \wedge [R](D \Rightarrow T_D(\psi)), \\ T_D(\langle R\rangle\psi) &= D \wedge \langle R\rangle T_D(\psi), \\ T_D([!\theta]\psi) &= D \wedge (A \Rightarrow T_A(\psi)), \qquad A = T_D(\theta). \end{aligned} \tag{A.2}$$

Here $A$ is already supported on $D$. Thus successive announcements compute their guards in the appropriate restricted model.

More generally the expansion carries a domain comprehension for *each* occurring sort. For an actual span from sort $U$ to sort $V$, let $D_U, D_V$ be those domains, and write $G(k)$ for membership in the displayed witness comprehension in the current model, including its endpoint conditions. The guarded box at $u$ is

$$D_U(u) \ \wedge \ \forall k \, \big((G(k) \wedge s(k) = u) \Rightarrow T_{D_V}(\psi)(t(k))\big).$$

The diamond replaces the universal implication by existential conjunction. Graph spans and their converses give reindexing and the ordinary quantifiers. This construction uses the witnesses of the declared span: for instance a change of reference restricts the source of a legal deviation span, whereas a public epistemic restriction restricts both ends of an observation relation. Knowledge uses that restricted relation, distributed knowledge the intersection of the member relations, and common knowledge its defining greatest-fixed-point expression with the restricted individual boxes. For an epistemic action with precondition $P(w, e)$ in the current model, its updated domain in $W \times E$ is $D'(w, e) = D(w) \wedge P(w, e)$. At a designated event $e$, the pointed box expands to $D(w) \wedge (P(w, e) \Rightarrow T_{D'}(\psi)(w, e))$. The relation on the new carrier is the restricted product relation, and its atoms and terms use the declared updated values. An inherited world variable is pulled back along the old-world projection when it is used at this new sort. All these guards, including preconditions formed from formulas, are expanded before deciding the variance of a bound variable.

**Formation at a binder.** For $\kappa \in \{\mu, \nu\}$, rename $Y$ fresh for the ambient domain parameters. With $P$ an ambient predicate at the binder's sort, form the expression $F(P) = T_D(\psi)[Y := P]$; the occurrence clause for $Y$ already restricts $P$ to $D$. The binder is formed by deriving

$$P \subseteq P' \ \vdash \ F(P) \subseteq F(P') \tag{A.3}$$

at two fresh predicate variables, with the other parameters fixed. The proof of this variance judgment is a premise of binder formation. It uses the already formed proper subexpressions, so the induction on derivations for soundness applies to it before interpreting this binder. Then $T_D(\kappa Y.\psi) = \kappa P.F(P)$. This is a derivation in the existing predicate order. In the usual positive fragment it is constructed by the sign calculation below. A derivably equivalent expanded body can be used by equality substitution. In particular an occurrence inside the formula specifying an announcement domain does not acquire a sign merely from the fact that the outer operator is a box.

**Lemma A.5** (Relativization and sound binder formation)**.** *For each well-formed expression of the extended grammar, $T_D(\psi)$ is the extension by false of its interpretation on $D$. Every binder formed by* (A.3) *denotes the corresponding least or greatest fixed predicate in that fibre. The construction commutes with substitution of external parameters.*

*Proof.* Proceed from the proper subexpressions toward the root. For predicates on $D$, extension by false is the image along $j_D$ and identifies $\mathsf{Pred}(D)$ with the interval $\downarrow D$ in $\mathsf{Pred}(W)$. Relative implication on this interval is $D \wedge (U \Rightarrow V)$. Quantifying the actual

witness $k$ in the guarded span formula gives exactly the restricted span modality at $u$. For (A.2), a point of $D$ outside $A$ satisfies the box vacuously. At a point of $A$ the updated model is the restriction to $A$, so the induction hypothesis for $\psi$ gives $T_A(\psi)$. Both sides have empty support outside $D$. The pointed action case makes the same calculation at $(w, e)$. This proves the nonbinding clauses.

For the sign calculation, an occurrence of a variable is increasing in itself, and a fixed parameter is constant. Meets, joins, inverse image and both quantifiers on fixed maps preserve inequalities. Implication reverses the first inequality and preserves the second. These facts prove the positive and negative variance judgments simultaneously by induction after domains and witness guards have been expanded. At an inner binder, if $H(P, Y) \subseteq H(P', Y)$ for every $Y$ and both inner bodies are monotone in $Y$, then $\mu_Y H(P', Y)$ is a pre-fixed point for $H(P, -)$ and $\nu_Y H(P, Y)$ is a post-fixed point for $H(P', -)$. Induction and coinduction give the two parameter inequalities for $\mu$ and $\nu$. Reversing the parameter inclusion gives the negative case. This constructs (A.3) for positive expanded bodies, including ones with nested binders.

At the current binder $F$ is monotone by its formation derivation. Writing $L = \exists_{j_D}$ and $\rho = j_D^*$, the induction hypothesis on the body gives $F = LF_D\rho$, where $F_D$ is its operator on $D$. Since $\rho L = 1$, $F_D = \rho F L$ is monotone too. An $F$-fixed predicate is supported on $D$. The maps $\rho, L$ therefore identify its fixed predicates in both directions with those of $F_D$, preserving inclusion. Their least and greatest elements correspond by Lemma 12.4. This proves the binder clause without choosing a fixed point of a nonmonotone operator. Every step is fibrewise in the external parameters, so reindexing them gives the same expansion and the same fibrewise least or greatest predicate. □

For example, on a nonempty singleton $W$ the expanded body $[!Y]\bot$ is $W \setminus Y$. It interchanges $\varnothing$ and $W$, so (A.3) fails and there is no fixed predicate. An announcement guard has no uniform negative variance either: take two mutually accessible worlds $w, v$ with $p$ true only at $v$. At $w$, $[!Y]\langle R\rangle p$ has truth values true, false, true for $Y = \varnothing, \{w\}, \{w, v\}$, respectively. A variable may nevertheless occur in a guard of a well-formed binder. For example, in the Boolean state fibre the body $Q \vee \neg[!Y]\neg\langle R\rangle p$ expands on $W$ to

$$Q \ \vee \ (Y \wedge \langle R\rangle(Y \wedge p)).$$

Indeed, the announcement diamond requires membership in $Y$ and the inner possibility needs an $R$-successor in $Y \cap p$. At $Y \subseteq Y'$ that same successor witnesses the required inclusion, so both fixed-point binders have the stated formation derivation. For the fixed-relation language of Definition 12.3 take $D = W$; the expansion then gives its ordinary positivity derivation.

For a state context $s : W \to \Sigma$, logical equality is the diagonal of $W \times_\Sigma W$. If a state is represented by a term $t : V \to W$, an assertion is tested by the inverse image $t^*[\![\varphi]\!]$. In particular, assertions about ownership, legality, or observations are not forced to factor through the realized concurrent arrow: they may use the full displayed state record.

### A.2.7 Predicate inference from adjunctions

The order structure on each predicate fibre supplies the inference relation $\Phi \vdash_W \varphi$ by inclusion. The ordinary connective rules are those of its Heyting algebra, so for example

$$R \wedge P \leq Q \quad \Longleftrightarrow \quad R \leq (P \Rightarrow Q).$$

The dependent quantifier introduction and elimination rules are the following equivalences, applicable to every ordinary context map $f : V \to W$:

$$\boxed{\exists_f P \leq Q \iff P \leq f^*Q, \qquad f^*Q \leq P \iff Q \leq \forall_f P.}$$

The rules for conjunction, implication, and quantification follow from the Heyting structure and adjunctions. Their stability under ordinary substitution is the Beck–Chevalley comparison. In particular, for the following pullback square, the two directions of quantifier transport are fixed by the same fibre bijection.

$$\begin{array}{ccc} V' & \xrightarrow{a} & V \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ W' & \xrightarrow[b]{} & W \end{array}$$

one has

$$b^*\exists_f \cong \exists_{f'}a^*, \qquad b^*\forall_f \cong \forall_{f'}a^*, \qquad \exists_f(P \wedge f^*Q) = \exists_f P \wedge Q.$$

The dependent sum and product underlie ordinary existential and universal quantification, with subterminal reflection converting witness-bearing sums into truth values. The assertion rules therefore reuse the indexed type operations rather than duplicating their semantic structure.

### A.2.8 Counterfactual and selection rules

Let $\pi_{-C} : \Sigma \to \Sigma_{-C}$ be the outsider projection. Its kernel pair $D_C = \Sigma \times_{\Sigma_{-C}} \Sigma$ has source and target $p_1, p_2$. The relation modalities are

$$\begin{aligned} \langle C\rangle P(\sigma) &\iff \exists\tau\,[\pi_{-C}(\tau) = \pi_{-C}(\sigma) \wedge P(\tau)], \\ [C]P(\sigma) &\iff \forall\tau\,[\pi_{-C}(\tau) = \pi_{-C}(\sigma) \Rightarrow P(\tau)]. \end{aligned}$$

They are monotone, preserve the respective finite joins and meets, and satisfy $\langle\varnothing\rangle P = [\varnothing]P = P$ on an unrestricted profile carrier. On unrestricted products the relation identity $D_C; D_D = D_{C\cup D}$ gives

$$\langle C\rangle\langle D\rangle P = \langle C \cup D\rangle P, \qquad [C][D]P = [C \cup D]P.$$

With a legal carrier, replace the full product kernel pair by $D_C^L = L \times_{L_{-C}} L$. Sequentially composed legal changes must retain an intermediate profile lying in the same legal subobject. The unrestricted identity for coalitional composition need not hold without that witness.

For each owner $i$, the paired realization is $\delta_i = (gp_1, gp_2) : D_i \to \mathcal{C}(A,B)^2$. If $P_i \hookrightarrow \mathcal{C}(A,B)^2$ compares the second interaction favorably with the first, define the strict-improvement test $\delta_i^* P_i$. The selected legal best-response predicate says that no improvement exists against the fixed outsider profile:

$$\mathsf{BR}_i(\sigma, \sigma_i) \Longleftrightarrow \forall\tau\,(\sigma D_i^L \tau \Rightarrow \neg P_i(g\sigma, g\tau)).$$

Equilibrium conjoins the ownerwise absence-of-improvement tests. Under reparametrization its exact transport depends on lifting the relevant deviations at a fixed reference, as well as preserving both realized endpoints. Numerical utilities give a particular comparison presentation; they do not alter this logical rule.

### A.2.9 Epistemic and informed-ability rules

An observation quotient $o_i : W \twoheadrightarrow O_i$ induces the relation $R_i = \ker(o_i)$. The knowledge operator has the explicit universal-fibre interpretation

$$K_i P(w) \iff \forall v \in W\,[o_i(v) = o_i(w) \Rightarrow P(v)].$$

It preserves truth and conjunction, and the defining equivalence relation gives

$$K_i(P \Rightarrow Q) \leq (K_i P \Rightarrow K_i Q), \quad K_i P \leq P, \quad K_i P = K_i K_i P, \quad \neg K_i P \leq K_i \neg K_i P.$$

These are the normality, factivity, and introspection properties of an observation equivalence relation. The linear perspective $K^{\ell}_{o_i} = o_i^* \Pi^{\ell}_{o_i}$ instead carries proof-relevant witness products, satisfying its comonad and lax-monoidal laws. The predicate laws are not read as equations of arbitrary linear resources.

For a nonempty coalition $C$ the joint quotient $q_C$ is the image of $\langle o_i \rangle_{i \in C}$, and $\mathsf{DK}_C P = q_C^* \forall_{q_C} P$. The common-knowledge operator is the greatest fixed point

$$\mathsf{CK}_C P = \nu X. \left( P \wedge \bigwedge_{i \in C} K_i X \right).$$

Its unfolding law and its coinduction rule have the forms

$$\mathsf{CK}_C P = P \wedge \bigwedge_{i \in C} K_i \mathsf{CK}_C P, \qquad \frac{X \leq P \quad X \leq K_i X \ (i \in C)}{X \leq \mathsf{CK}_C P}.$$

The greatest fixed point is computed using finite paths in the union of the owners' observation relations. The joint quotient classifies distributed knowledge but does not by itself deliver a usable pooled signal. Section 11 supplies its communication realization.

For a worldwise feasibility or permission predicate $F \hookrightarrow W \times M$, an informed legal policy is a section of $\forall_{o \times 1_M} F$ over $O$. The judgement $\pi : \mathsf{Pol}_o(F)$ is equivalent to the uniformly quantified permission formula $\forall w \, F(w, \pi(o(w)))$. An individual or coalition guarantee uses existential quantification over a realizable choice coordinate, followed by universal quantification over its actual legal completions. A separate existence conjunct is retained for executable guarantees while a vacuous universal assertion.

### A.2.10 Dynamic execution and public restriction

For a witnessed span $X \xleftarrow{s} E \xrightarrow{t} Y$, put

$$\langle E \rangle P = \exists_s t^* P, \qquad [E] P = \forall_s t^* P.$$

The composition of witnessed spans by pullback gives, for a second span $F$,

$$[F \circ E] P = [E][F] P, \qquad \langle F \circ E \rangle P = \langle E \rangle \langle F \rangle P.$$

Identity spans induce identity modalities and pullback composition supplies their sequential laws. The proof-relevant versions use $\Sigma^{\ell}_s t^*$ and $\Pi^{\ell}_s t^*$, so two distinct witnesses sharing one endpoint pair still contribute separate summands or factors. Restricting $E$ to its permission predicate gives the permitted modalities before either construction is applied.

**Budget judgments on completed witnesses** For $X \xleftarrow{s} E_{\mathrm{ret}} \xrightarrow{t} Y$ as in Proposition 8.8, let $E_x$ be the dependent family of completed witnesses with source $x$. In context $x : X, k : E_x$ there are ordinary terms $t(k) : Y$ and $m(k) : C$. Given $c : C \mid B(c)\ \mathsf{Prop}$ and $y : Y \mid P(y)\ \mathsf{Prop}$, the two budget judgments at $x$ expand to

$$(\langle E \rangle^B P)(x) \equiv \exists k : E_x \, [B(m(k)) \wedge P(t(k))],$$
$$([E]^B P)(x) \equiv \forall k : E_x \, [B(m(k)) \Rightarrow P(t(k))].$$

A diamond introduction presents a witness $k$ and derives the two conjuncts. Elimination opens this existential and its two conjuncts, with the fresh witness absent from the conclusion. A box introduction checks $P(t(k))$ for a fresh $k : E_x$ satisfying $B(m(k))$; elimination instantiates it at the supplied completed witness and its budget proof. All four are the ordinary quantifier rules after expansion.

For a linear postcondition $y : Y \vdash A(y)\ \mathsf{LType}$, form the comprehension $E_{x,B} = \{k : E_x \mid B(m(k))\}$. With $|h|$ the underlying witness of $h : E_{x,B}$, the two linear types are

$$\Sigma^{\ell}_{h:E_{x,B}} A(t(|h|)), \qquad \Pi^{\ell}_{h:E_{x,B}} A(t(|h|)).$$

Their introduction, opening, abstraction and application are the dependent linear rules above. For regrading $j$ the comprehensions defined by $B'(j(m(k)))$ and $(j^*B')(m(k))$ are canonically identical over $E_x$. The induced comparison on each dependent sum or product is the identity on the indexed branches in the family presentation. This is the typed form of Proposition 8.8.

For a public announcement $\varphi$ interpreted in model $M$, construct its comprehensional inclusion $j : U \hookrightarrow W$ and restrict every accessibility relation on both endpoints. The box is

$$[\![[!\varphi]\psi]\!]_M = \forall_j [\![\psi]\!]_{M|\varphi}.$$

The postcondition belongs to the restricted epistemic model, not to the original state space. The displayed announcement reductions follow from the dependent-product and comprehension equations and must therefore use the updated accessibility relation.

$$\begin{aligned}
[!\varphi]p &\leftrightarrow (\varphi \Rightarrow p),\\
[!\varphi](\psi \wedge \chi) &\leftrightarrow [!\varphi]\psi \wedge [!\varphi]\chi,\\
[!\varphi]\neg\psi &\leftrightarrow (\varphi \Rightarrow \neg[!\varphi]\psi),\\
[!\varphi]K_i\psi &\leftrightarrow (\varphi \Rightarrow K_i[!\varphi]\psi),\\
[!\varphi][!\psi]\chi &\leftrightarrow [!(\varphi \wedge [!\varphi]\psi)]\chi.
\end{aligned}$$

At a removed state the first announcement makes the universal condition vacuous. At a retained state, a knowledge reduction ranges only over accessible worlds that also survive the restriction. This is why updating both endpoints of the relation differs from substituting the reference state alone.

### A.2.11 General epistemic events and their reduction

For a finite event set $E$, relations $S_i$, and enabling predicate $\mathsf{pre} \hookrightarrow W \times E$, the product-update model has carrier $U = \{(w, e) : \mathsf{pre}(w, e)\}$ with

$$(w, e)R^+_i(v, f) \iff wR_iv \wedge eS_if.$$

For a designated event $e$ let $j_e : W_e \hookrightarrow W$ be its precondition comprehension and $k_e : W_e \to U$ its insertion. Define

$$[\mathcal{A}, e]\psi = \forall_{j_e} k_e^* [\![\psi]\!]_{M\otimes\mathcal{A}}.$$

The derived finite-event knowledge reduction is

$$[\mathcal{A}, e]K_i\psi \leftrightarrow \left(\mathsf{pre}_e \Rightarrow \bigwedge_{f:eS_if} K_i[\mathcal{A}, f]\psi\right).$$

The finite-event knowledge reduction quantifies over enabled, epistemically compatible event alternatives through guarded boxes. Successive product updates retain their event pairs and the conjunction of enabling predicates. The canonical reassociation of these pairs proves compatibility with repeated updates.

### A.2.12 Temporal induction and rationality

For a finite public transition game with controllable predecessor $\mathrm{Pre}_C$, the formulas for maintenance and bounded-until ability are respectively

$$\nu X.(P \wedge \mathrm{Pre}_C(X)), \qquad \mu X.(Q \vee (P \wedge \mathrm{Pre}_C(X))).$$

For a monotone operator $F$ on a predicate lattice, its fixed-point inference principles are

$$\frac{F(P) \leq P}{\mu F \leq P}, \qquad \frac{P \leq F(P)}{P \leq \nu F},$$

The unfolding equations $F(\mu F) = \mu F$ and $F(\nu F) = \nu F$ express the least and greatest fixed points. Finite-state strategy extraction is a separate operational theorem: its witnesses come from the displayed predecessor construction. Likewise, a profitable-deviation relation becomes a temporal step only when the corresponding transition is specified.

### A.2.13 Compositionality and exact transport of formulas

The recursive interpretation of assertions commutes with ordinary substitution by Beck–Chevalley and reindexing of the designated maps. For semantic arrow tests, one has

$$\mathrm{Test}_Q(mtu) = u^* \,\mathrm{Test}_Q(mt) = \mathrm{Test}_{m^*Q}(tu)$$

For a wiring map $m$, a realized-arrow tuple $t$, and a reference substitution $u$, semantic testing commutes with composition. Relation modalities also require their alternatives to be preserved: a comparison map must send related pairs forward and lift each abstract successor at an image state. Under these two conditions the specified modal fixed points transport exactly, rather than merely one-way.

The soundness induction sends an assertion entailment $W \mid \Phi \vdash \varphi$ to inclusion of subobjects and each linear equation to equality of its denotations. The cases use the universal properties established in the main text and this appendix: adjunction and Beck–Chevalley for quantifiers, kernel pairs for alternatives, comprehension for updates, and fixed-point induction and coinduction for temporal operators.

### A.2.14 The formal logic of agency

The agent calculus supplies control, observation, and state contexts with their intended carriers. Relation spans turn these structures into predicate modalities: owner projections describe controlled changes, observation kernels describe indistinguishable states, and witnessed executions describe possible action. The different meanings arise from their displayed relations, not from different untyped modal operators.

**The carriers of agent assertions** Assertions must be interpreted over their correct carrier. Legal profile predicates lie in $\mathsf{Pred}(L)$, whereas situated epistemic assertions lie in $\mathsf{Pred}(W)$. For a reference $s : W \to L$, the corresponding lifted deviation relation has a source on the situated-state carrier and a target in the legal profile space, with type

$$W \xleftarrow{d_C} W \times_{L,p_1} D_C^L \xrightarrow{q_C} L.$$

Its box therefore sends $\mathsf{Pred}(L)$ into $\mathsf{Pred}(W)$. The knowledge relation $\ker(o_i)$ instead relates states within $W$ and induces an endomap of $\mathsf{Pred}(W)$. Fixed-point expressions require compatible state sorts, which the typed span calculus enforces.

# B Soundness, Reconstruction, and Equational Completeness

Reconstruction starts from the data displayed in the indexed semantics and recovers the corresponding agent presentation. Soundness goes the other way, interpreting the rules of the calculus in that semantics. For equational completeness we build the syntactic classifying model. This last argument concerns validity across the designated class of models. The separate tests for equality in the fixed concurrent model are proved in the main text. We use the indexed semantics of Jacobs (1999); Vákár (2015).

## B.1 Reconstruction of semantic presentations

*Proof of Theorem 14.1.*

(1) **Construction of the state-context tower.** From $\vec{\Sigma}^G$ and $g$ construct the tower of Theorem 5.3, its constant types, and the section $\widehat{g}(\sigma) = (\sigma, g_\sigma)$.

(2) **Reverse implication.** Conversely retain the displayed owner carriers and recover $g = \pi_2 \widehat{g}$. The canonical backward map is the inverse-image operation along that same $g$, so forgetting and rebuilding it changes nothing. The family and slice presentations correspond by the canonical maps between a displayed domain and the disjoint union of its fibres; these preserve predicates and all defining maps.

(3) **Lifting through the comparison.** An ownerwise arrow is recovered as the same family $r_i$. Its equation $g = h\bar{r}$ is the equation identifying the pulled-back realization section. Its cartesian context lift is the displayed pullback and its deviation comparison is forced by the two endpoint maps.

(4) **Recovery of original concurrent arrows.** Thus arrows and their composition are recovered as well. Coherence with wiring is Theorem 3.15; joint alternative contexts agree by Theorem 4.4.

(5) **Witness and legality comparison.** For each additional datum retain its carrier and its map, interpreted as a term using constant or dependent types. Comprehension recovers the same subset, and a history-indexed type keeps all its fibres. Certificate objects and their arrows remain the specified witnesses, with their individual identities and causal histories.

(6) **Monoidal comparison.** Action-model product updates are the same enabling comprehensions and relation pullbacks by Theorem 9.16; their composition comparisons are those of Theorem 9.25. The pointed operational update and its linear residual section are recovered by Theorem 12.22.

(7) **Descent along the kernel pair.** For a hidden composite the secured witness carrier and visible comparison are retained as in Theorem 12.31, without identifying non-descending private observations. Image factorization recovers pooled signals, and intersection with their kernels recovers public inspection.

(8) **Observation and knowledge.** The actual observation images, their power objects, and the truthful-message comprehensions reconstruct the announcement game of Construction 11.12; its effectivity uses the same dependent quantifiers. For an exact visible refinement, the retained labels and source relations define the same monotone endomap on $\mathcal{P}(X^2)$, so its greatest fixed point and effective quotient are recovered.

(9) **Causal execution.** Corollary 12.39 supplies the comparison under isomorphic presentations, and Construction 12.40 retains all dependent residual families. For a measured

presentation, the ordinary terms $m, a$ reconstruct $r$ by pairing; conversely the two projections of $r$ recover both terms. Composition with the retained $e_i$ reconstructs $e_i r$; effectiveness of the image quotient reconstructs its unique descended value when it exists. At a prefix, the retained edge increments reconstruct $c_h$ and $e_i^{c_h}$. For a charge regrading $j$, $e_i' T_j r = e_i r$ is precisely agreement of the two evaluations on all retained records. Pullback of the budget reconstructs the same restricted witnesses and realizers by Proposition 8.8.

(10) **Fixed-point argument.** Atomic predicates have the same pullbacks by Equation (5.1). The quantifier formulas of Theorem 5.22 range over the same fibres, so structural induction proves equality for first-order operations and the relation modalities built from them. The two interpretations recover identical domain comprehensions and guarded span maps. By Lemma A.5, the expanded binder bodies therefore give the same monotone endomap on the recovered predicate fibre. Its least and greatest fixed points agree in the two presentations.

(11) **Strategic comparison.** Selection and best response agree by Proposition 5.38; legal selection agrees by Proposition 7.3. All the remaining predicates are the specified combinations of these operations and retained terms, proving preservation and reflection.

□

*Proof of Corollary 14.4.*

(1) **Ownerwise alternatives.** An agent interface is a record of retained owner carriers and the canonical selection predicate; its contextual realization is a rearrangement of the retained realization map. Products reconstruct complete-policy composition and control regrouping, while comprehension and the outsider projection reconstruct the legal choice fibres. The retained observation and comparison terms give its situated and evaluated interfaces. History-indexed families retain every continuation carrier and restriction map.

(2) **Observation and knowledge.** For communication, the retained observation image $\Omega_i$ determines the power object $\mathcal{P}(\Omega_i)$. Exponentiation and the truthful-policy comprehension reconstruct $\mathsf{AnnPol}_i$ and its evaluation. Products therefore reconstruct $\mathcal{X}$ and $F$, including the private policy records.

(3) **Lifting through the comparison.** The fibre family $K_u = F^{-1}(u)$ recovers $\mathcal{X}$ through the canonical bijection $\coprod_u K_u \cong \mathcal{X}$; its realization term is retained as in Construction 11.23. The maps in the lifting proof are the explicit extension policies, so the bounded comparison is reconstructed as well. Its logical consequences follow from Theorem 11.18 and its corollaries.

(4) **Policy-level comparison.** This retains the complete policy presentation together with its logical comparison to the message presentation.

□

*Proof of Corollary 14.5.*

(1) **Lifting through the comparison.** The retained implementation maps give the same coordinate products, composites, and image factorizations, hence the same substituted realizations and legal inverse images. The lifting proof uses precisely the fibres of those image maps, so is transported along their canonical bijections.

(2) **Adjunction and transposition.** The observation and feasibility maps reconstruct $\forall_{o\times \mathrm{id}} F$ through its adjoint characterization. Sections, exponentials, and inverse images then reconstruct the informed and decentralized legal policy carriers. A bijection of the displayed data preserves both evaluation and the descent equations defining the image of $J_C$.

(3) **Witness and legality comparison.** An institutional composite is a pullback with its composite realization term and an intersection of inverse-image permission predicates. These are reconstructed from the retained endpoints, realizers, and permission inclusions. The dependent quantifiers on its source and target consequently reconstruct its modalities.

(4) **Representation and Yoneda decoding.** Linear realizers on representable boundaries are recovered by Yoneda as in the preceding reconstruction results.

(5) **Transport of controller behaviour.** Finally controller behaviour is uniquely determined by its output and next-state equations. Transport of a controller therefore transports that behaviour by induction on observation words. For a retained complete policy, its derivatives and their image $S_b$ are determined by word concatenation and evaluation.

(6) **Observation and knowledge.** Bijections of the observation and action carriers induce bijections of words and behaviours, taking derivatives to derivatives and preserving the pointed residual controller.

□

## B.2 Soundness and reconstruction of the agency calculus

*Proof of Theorem 14.25.*

(1) **Adjunction and transposition.** Formation and the core terms are interpreted by Proposition 2.20. Agent operations are composites of evaluation, tensor, currying, and ordinary parameter maps, so Theorem 3.2 and Proposition 4.16 validate their equations. Regrouping uses the product bijections and kernel-pair comparison of Theorem 4.15.

(2) **Observation and knowledge.** Decision protocols are interpreted by Theorem 10.10, with non-anticipation and stagewise observations proved in Theorem 10.11. The acceptance domains of protocol-generated situated contexts are the image and dependent quantifier constructions of Proposition 5.72.

(3) **Witness and legality comparison.** Finite endpoint terms and their reductions are interpreted by Lemma 10.14, Proposition 10.22, and Theorem 10.23, retaining execution witnesses when the external interface is hidden. Lemma 10.25 supplies the explicit physical output for a retained transcript.

(4) **Causal execution.** Lemma 10.16, Lemma 10.20, and Corollary 10.24 cover physical private-input occurrences, their status-indexed residuals, and their causal constraints.

(5) **Adjunction and transposition.** For contextual terms, Theorem 1.46 validates the dependent adjoints and their substitution laws. Theorem 2.15 constructs the agent abstraction as the corresponding dependent closed product, constructed using the linear–nonlinear adjunction. Predicate introduction and elimination rules are the corresponding order adjunctions in $\mathrm{Sub}(W)$.

(6) **Measured witnesses and budget rules.** The packages of Section A.1.2 are dependent linear sums. Their opening and repacking equations are the sum equations. The unit, multiplication and ordinary-to-linear comparison are the arrows of Proposition A.4; their unit and associativity equations follow there on arbitrary linear contexts. The tagged output uses tensor introduction and $\mathsf{Lin}$, so its elimination uses the declared resource once. Functoriality carries the given ordinary charge maps to the displayed regrading and erasure maps. By Proposition 8.8, restricting the completed witness family gives exactly the quantifier domains of the budget rules, points and linear branches alike.

(7) **Adjunction and transposition: second comparison.** Perspective uses (6.3); its coherence and epistemic laws are Theorem 6.30 and Proposition 6.31. Uniform policies and comparison feedback are subobjects defined by the proved graph adjunction and inverse-image equations. Hence their rules are sound comprehensions and quantifications.

(8) **Witness and legality comparison.** For dynamic rules, Theorems 8.4 and 8.10 validate composition and its modalities. Theorems 9.30 and 9.31, followed by Proposition 9.34, validate public and general epistemic updates.

(9) **Observation and knowledge.** Proposition 9.33 distinguishes a logical public comprehension from the exact tests implementable using an actual finite observation wire.

(10) **Observation and knowledge: second comparison.** Construction 10.26 constructs an implemented epistemic action and its event-observation kernels from the image of actual recipient records in retained witnesses, and Lemma 11.26 supplies the synchronization barrier for completed pooling. Proposition 8.9 constructs the separate realized map of witnessed execution.

(11) **Coinduction and bisimulation.** At a binder of the contextual grammar, Lemma A.5 interprets the expanded body on its actual state domain and derives monotonicity by (A.3). The predicate fixed-point theorem supplies unfolding, induction and coinduction there. Common-knowledge unfolding and coinduction also have the finite-path proof of Proposition 6.33. Induction on a complete derivation therefore establishes each judgment and equation, with the same substitution comparisons at every constructor.

□

*Proof of Theorem 14.26.*

(1) **Representation and Yoneda decoding.** Theorem 2.19 proves the mutually inverse encoding and decoding functors on owned presentations and their ownerwise morphisms. Concretely, encoding copairs the arrows $yg_\sigma$ and curries their map $\mathsf{Lin}\Sigma \otimes yA \to yB$, retaining the owner display. Decoding evaluates at each $\sigma$ and applies Yoneda full faithfulness to recover $g_\sigma : A \to B$.

(2) **Frobenius and coproduct comparison.** Coproduct and internal-hom $\beta\eta$ give both inverse equations, while naturality proves the reparametrization and composition comparisons.

(3) **Descent along the kernel pair.** The owner family is retained, hence the original control projections and their kernel pairs are recovered. The legal subobject, reference map, observations, and comparisons are retained ordinary data.

(4) **Observation and knowledge.** For protocol-generated data, Theorem 10.11 reconstructs the decision-local observation from its actual causal input tuple, while Proposition 5.72 reconstructs acceptance images and universally legal profile domains from the

retained record test. Neither construction substitutes a future record for a decision's local observation.

(5) **Adjunction and transposition.** Their derived legal choices, perspective operations, selection predicates, and modalities are reconstructed by the same pullbacks and adjoints. Regrouping and delegation retain their coordinate maps, so their executions decode to the original precompositions.

(6) **Representation and Yoneda decoding.** For witnessed institutions the map $\rho : E \to \mathcal{C}(A, B)$ is equivalent, pointwise over $E$, to the family morphism $\rho^\sharp : s^*\underline{yA} \to t^*\underline{yB}$ of Proposition 8.9. Encoding and decoding each component by Yoneda are inverse and natural in maps of witnesses. Endpoints and permission remain displayed, while composition uses the same pullback and the same composite realizer.

(7) **Policy-level comparison.** For a finite session the *displayed* session type and endpoint policies reconstruct the two pruned trees; their secured-matching synchronization reconstructs the transcript witness. The witness is retained, not inferred from the hidden scalar, and all off-path coordinates remain in the policy family.

(8) **Secured operational interaction.** For implemented action updates, the retained recipient-port records reconstruct $d_i$, their tuple image $E$, and the derived event relation $\ker(b_i)$ as in Construction 10.26; the original completed matching witnesses remain available above the image. For truthful communication, the retained $\Omega_i$ reconstructs $\mathcal{P}(\Omega_i)$, the complete-policy subtype, evaluation, and the explicit extension map.

(9) **Observation and knowledge.** These observations cover all the displayed fields, and each comparison commutes with them by its construction.

□

### B.3 The classifying model and equational completeness

*Proof of Lemma 14.29.*

(1) **Secured operational interaction.** Encode each finite ordinary context telescope by its iterated dependent sum, retaining the canonical packing and unpacking substitutions. Thus $f$ may be written as a map of the corresponding ordinary total types.

(2) **Functoriality and substitution.** Over $w : W$ form its equality fibre

$$\mathrm{Fib}_f(w) = \sum_{v:V} \mathrm{Id}_W(f(v), w).$$

The maps $v \mapsto (f(v), v, \mathrm{refl}_{f(v)})$ and $(w, v, p) \mapsto v$ give inverse context substitutions between $V$ and $W.\,\mathrm{Fib}_f$; the reverse composite uses the diagonal equality property of Proposition 14.28 to identify $w = f(v)$ and its equality witness with reflexivity.

(3) **Dependent sum and product expressions.** For a linear family $A$ over $V$ put

$$(\Sigma^\ell_f A)(w) = \Sigma^\ell_{(v,p):\mathrm{Fib}_f(w)} A(v), \qquad (\Pi^\ell_f A)(w) = \Pi^\ell_{(v,p):\mathrm{Fib}_f(w)} A(v).$$

These are expressions of the existing dependent constructors, not new quantifier symbols. Given $\alpha : \Sigma^\ell_f A \to B$, define $\beta_v : A(v) \to B(f(v))$ by applying $\alpha$ at the fibre witness $(v, \mathrm{refl})$.

(4) **Reverse implication.** Conversely, for $\beta : A \to f^*B$, map an element $((v,p),a)$ to the transport along $p$ of $\beta_v(a)$. Extensional equality elimination and the dependent-sum $\beta\eta$ equations make the two operations inverse.

(5) **Adjunction and transposition.** For the right adjunction, evaluate a map $\alpha : B \to \Pi^\ell_f A$ at $(v, \mathrm{refl})$ to obtain $\beta_v : B(f(v)) \to A(v)$.

(6) **Reverse implication.** Conversely, $\beta : f^*B \to A$ determines the dependent abstraction

$$\alpha_w(b) = \Lambda(v,p).\beta_v(p_*^{-1}b).$$

The equality-transport and dependent-product $\beta\eta$ equations again prove the two inverse equations. Thus

$$\Sigma^\ell_f \dashv f^* \dashv \Pi^\ell_f.$$

For a pullback of ordinary substitutions, its two equality-fibre presentations are equivalent by inserting or deleting the redundant pullback coordinate.

(7) **Base-change comparison.** Substituting that equivalence into the displayed sum and product constructs both Beck–Chevalley maps and their inverses. Every iterated comparison has the same projections to the original fibre witnesses, proving its cocycle by dependent $\eta$. The explicit comparison maps and the two inverse substitutions are verified in Lemma 14.30; in particular their composite-substitution equations hold at arbitrary linear contexts.

(8) **Linear Frobenius comparison maps.** Finally, the map $\Sigma^\ell_f(f^*B \otimes A) \to B \otimes \Sigma^\ell_f A$ eliminates the fibre witness and moves the independent $B$-factor outside its dependent sum. Its inverse eliminates the sum in the right tensor factor and pairs $B$ with the selected fibre component. Tensor and dependent-sum $\beta\eta$ make the composites identities, giving Frobenius and its substitution coherence.

□

*Proof of Lemma 14.30.*

(1) **Construction and inverse of the fibre comparison.** In the equality-fibre presentation a point of $\mathrm{Fib}_{f'}(w')$ is a pair $(v', p')$ with $p' : f'(v') = w'$. Define

$$\chi_{w'}(v', p') = (a(v'), b_*(p')),$$

where the second component is the equality $f(a(v')) = b(w')$ obtained by transporting $p'$ through the commuting square. Conversely, a pair $(v, p : f(v) = b(w'))$ defines the point $v' = (w', v, p)$ of the ordinary pullback comprehension, with $f'(v') = w'$ by the first projection; send it to $(v', \mathrm{refl}_{w'})$. The pullback's defining property and extensional equality elimination identify the two composites with the identity substitutions, including their equality witnesses.

(2) **Dependent sums, products, and naturality.** After substitution along $b$, the fibre of $\Sigma^\ell_f A$ at $w'$ is $\Sigma^\ell_{(v,p):\mathrm{Fib}_f(b(w'))} A_v$. The fibre of $\Sigma^\ell_{f'} a^* A$ is the analogous dependent sum indexed by $\mathrm{Fib}_{f'}(w')$. Reindexing the summands along $\chi_{w'}$ and its inverse gives the two comparison maps; the same argument using dependent products gives the right-adjoint comparison. Their inverse equations follow from the $\Sigma^\ell$ and $\Pi^\ell$ uniqueness laws. No particular representative of an equality class is selected.

(3) **Iterated base change.** For two composable pullback squares, either route sends a fibre witness to its projection in the original context $V$ together with the equality obtained by successively applying the two base substitutions. Functoriality of equality transport identifies this with the direct transported equality. By the uniqueness laws of the ordinary pullback and the linear dependent sums and products, the associated comparisons commute on arbitrary linear terms, and not only on closed states.

(4) **Frobenius comparison and compatibility.** At $w : W$, the Frobenius map is the copower-distribution comparison

$$\Sigma^{\ell}_{(v,p):\mathrm{Fib}_f(w)}(B_w \otimes A_v) \longrightarrow B_w \otimes \Sigma^{\ell}_{(v,p):\mathrm{Fib}_f(w)} A_v.$$

On a summand indexed by $(v, p)$, it sends $(b, a)$ to $(b, \mathrm{in}_{(v,p)}\, a)$; its inverse eliminates the dependent sum in the right-hand tensor factor and retains the same $b$. The tensor and dependent-sum $\beta\eta$ laws prove the inverse equations. Both base-change routes use $\chi_{w'}$ to transport exactly the same fibre witness and leave the independent $B$-term unchanged. Thus Frobenius commutes with Beck–Chevalley, including its composite-substitution coherence.

□

*Proof of Lemma 14.31.*

(1) **Monoidal comparison.** The substitution lemma makes composition well-defined and associative; variables give identities. Ordinary substitution gives functorial reindexing and comprehension. Tensor on objects is the type constructor; on arrows it is tensor introduction after tensor elimination on the input.

(2) **Secured operational interaction.** Its associators, symmetry, and unitors are the corresponding resource rearrangements. Their equations follow from tensor and unit $\beta\eta$.

(3) **Frobenius and coproduct comparison.** Currying and application give mutually inverse bijections

$$\mathcal{L}_{\mathrm{syn}}(\Gamma)(A \otimes B, C) \cong \mathcal{L}_{\mathrm{syn}}(\Gamma)(A, B \multimap C).$$

Additive pairing, projections, injections, and case analysis prove the product and coproduct universal properties, including their nullary cases.

(4) **Adjunction and transposition.** Dependent abstraction and application prove the right-adjoint bijection for comprehension projections; dependent pairing and elimination prove the left-adjoint bijection. The uniqueness equations supply naturality. Lemma 14.29 constructs their extension to general ordinary substitutions. The equality-fibre comparison of Lemma 14.30 verifies Beck–Chevalley, Frobenius, and the two composite-substitution coherences, so these constructions are coherent on open terms and not merely fibrewise on closed values.

(5) **Term-level linear–nonlinear adjunction.** Finally, the mutually inverse maps for $\mathsf{Lin} \dashv \mathsf{Val}$ send a linear term $z : \mathsf{Lin}X \vdash t : A$ to $x : X \vdash \mathsf{thunk}(t[\mathsf{return}(x)/z]) : \mathsf{Val}A$, and send an ordinary term $x : X \vdash v : \mathsf{Val}A$ to $z : \mathsf{Lin}X \vdash \mathsf{let}\ \mathsf{return}(x) = z\ \mathsf{in}\ \mathsf{force}(v) : A$. Their composites are the adjunction $\beta\eta$ equations.

(6) **Monoidal comparison.** For strong monoidality, the comparison $\mathsf{Lin}X \otimes \mathsf{Lin}Y \to \mathsf{Lin}(X \times Y)$ eliminates the two linearized ordinary values and returns their pair. Its inverse eliminates the ordinary pair $p : X \times Y$ and forms $\mathsf{return}(\pi_1 p) \otimes \mathsf{return}(\pi_2 p)$. Their composites are identities by the $\mathsf{Lin}$, tensor, and ordinary product $\beta\eta$ equations.

(7) **Monoidal comparison: completion.** The same computation gives $\mathsf{Lin1} \cong \mathsf{I}$ and the associativity, unit, and symmetry coherence of this strong monoidal structure. All constructions are stable under ordinary substitution. The declared constants interpret themselves.

□

*Proof of Theorem 14.32.*

(1) **Existence and strict interpretation.** Interpret contexts by successive comprehension, types by the chosen constructors, terms by introduction and elimination and substitutions by tuples. In a strictly stable presentation these clauses determine the interpretation on raw syntax. Identity and composition of substitutions follow by substitution into the components of a tuple. Conversion follows from the computation and uniqueness laws of the chosen constructions, including the diagonal-pullback interpretation of extensional equality. Consequently the interpretation descends to derivable-equality classes.

Here is an explicit way to use the recursion when substitution for a constructor is supplied by coherent isomorphisms. In either indexed fibre over $\Gamma$, present a family by objects $A_f$ over $\Delta$ for every $f : \Delta \to \Gamma$, with coherent isomorphisms $u^*A_f \cong A_{fu}$; arrows are compatible families of fibre arrows. Evaluation at $1_\Gamma$ is an equivalence. An inverse sends $A$ to $(f^*A)_f$, and the structure isomorphism at $(1_\Gamma, f)$ supplies its comparison with any presented family. Reindexing this presentation amounts to precomposing the subscript and is strictly functorial. Use $\Gamma.A_{1_\Gamma}$ as comprehension. The map $\bar{f} : \Delta.A_f \to \Gamma.A_{1_\Gamma}$ pairs $f$ after projection with the generic term transported through $A_f \cong f^*A_{1_\Gamma}$. For $u : \Theta \to \Delta$, both $\overline{fu}$ and the composite of the two extension maps have the same projection and transported generic term by the cocycle law; comprehension makes them equal. Evaluate each nonbinding constructor at $f$ on its $f$-components. For a binder evaluate the body at $\bar{f}$ and use the chosen dependent constructor along $\Delta.A_f \to \Delta$. The substitution and Beck–Chevalley isomorphisms give the family comparisons; their composition law is precisely the pasting law for those comparisons. The identity for the extension maps just proved makes reindexing of the resulting expressions strict. Fibre arrows, units, counits and their equations are interpreted componentwise. This gives the recursive interpretation in the family presentation; evaluation at identities gives its weak interpretation in the original model. As evaluation is faithful on each fibre, the interpretations give the same judgments of arrow equality after the displayed type identifications.

(2) **Reverse implication.** For the converse, use the syntactic model of Lemma 14.31. Interpretation there returns precisely the term's own derivable-equality class. Thus, if two interpretations coincide in every model, they coincide in this model and are related by the derivable-equality judgment.

(3) **Monoidal comparison.** This applies to open terms because the hom-set of the syntactic fibre is *defined* by their equality classes, not by evaluation at global points. Open linear contexts are represented by their tensor and unpacked by tensor elimination; its $\beta\eta$ equations identify this encoding with the original judgment.

(4) **Conservativity of displayed agent types.** The displayed agent judgment is conservative by Definition 2.14: erasure recovers the underlying term and the tag is recovered unchanged. Agent operations are constructed terms with specified displayed maps, so the same syntactic-model argument applies to their typed equations. In split strict presentations the interpretation is unique on the nose.

(5) **The comparison between weak interpretations.** Let $F, G$ be interpretations in the fixed target. We construct $\theta : F \Rightarrow G$ simultaneously on contexts and types and verify naturality on substitutions and terms. At the empty context take the unique isomorphism of the two terminal objects. At a primitive take the composite of its two prescribed identifications, transporting its arity using the components already constructed. At $\Gamma.A$ use the two comprehension comparisons to identify the source and target as context extensions. Pair the map on $\Gamma$ after projection with the generic term transported by $\theta_{\Gamma,A}$. Its inverse uses the inverse components; projection and the generic term determine both composite identities.

For a type constructor $c$ write $\phi^F_c : F(c(\vec{A})) \cong c(F\vec{A})$ for its structure comparison, with reindexing into the common context understood. The component at a composite is forced to be

$$\theta_{c(\vec{A})} = (\phi^G_c)^{-1}\, c_{\#}(\theta_{\vec{A}})\, \phi^F_c .$$

Here $c_{\#}$ transports isomorphisms through the displayed constructor. For tensor it tensors the two components. For internal hom it precomposes with the inverse of the input component and postcomposes with the output component. For a dependent sum or product, first transport its index by the extension isomorphism above, use the body component over that isomorphism, and apply the dependent functor. The change of index is the Beck–Chevalley isomorphism for the resulting square of projections. Ordinary binders use the same square. Products and coproducts use their projection and injection characterizations, equality uses the diagonal pullback, and $\mathsf{Lin}, \mathsf{Val}$ apply the indicated functors to the component. All these maps are invertible by the same construction on the inverse components.

(6) **Substitution, terms and conversion.** For $A[\sigma]$ use the reindexing comparisons of $F, G$ and the pullback of $\theta_A$ along $F\sigma$, using the base naturality equation $\theta_\Gamma F\sigma = G\sigma\, \theta_\Delta$ for $\sigma : \Delta \to \Gamma$. For a tuple $(\sigma, t)$ this equation follows by projecting to the base and to the generic term, which gives the equations for $\sigma$ and $t$. Identity substitutions give identity comparisons. For composable substitutions their comparison is the composite one by the reindexing cocycle law. Substituting into a constructor gives the same component by naturality of its structure comparison; for a dependent binder the square for two substitutions is the pasting of the two extension squares. The two Beck–Chevalley maps agree by pasting, and Frobenius commutes with the same base change by Lemma 14.30 in the syntax and by its interpreting comparison in the target.

For terms, context projection gives the variable case and the identifications of the signature give the generator case. Pairing commutes with $\theta$ because its two projections do; copairing commutes because its restrictions to the injections do. For a transposed term, undo the transposition on both sides of the naturality square. Compatibility of evaluation (or of the adjunction counit) identifies the resulting square with the square for the body. Bijectivity of transposition gives the required equality. This verifies ordinary and linear abstraction, the indexed binders, and the mixed adjunction. Tensor and unit introduction use the monoidal comparison; elimination follows by precomposing with their context-unpacking isomorphism. These verifications apply also in a context with further linear inputs, using the same tensor rearrangement on both sides. Equality reflection equates the two base substitutions because a section of the diagonal pullback equates their interpretations. Proof uniqueness there is monicity. The computation, uniqueness and commuting conversions are respected by the preceding projection, injection and transposition calculations. Congruence, symmetry and transitivity preserve the commuting squares. This simultaneous induction therefore descends to the equality

classes used in the syntactic model.

(7) **Uniqueness.** Any structural 2-cell has the stated components on the terminal context and on each primitive. Its comprehension square forces the component on each extended context, and its square for $c$ forces the displayed formula on each constructed type. Its reindexing square forces the remaining substitution instances. Induction gives equality with $\theta$ on every component. In particular the unique self-comparison is the identity and comparison composites are the unique comparisons with their endpoints. The category of interpretations over the fixed target and fixed signature interpretation is inhabited and has exactly one arrow between every pair of objects. It is therefore equivalent to the terminal category, which proves bi-initiality for the specified weak interpretations.

□

## References


Samson Abramsky and Jonathan Zvesper (2010). *From Lawvere to Brandenburger–Keisler: Interactive Forms of Diagonalization and Self-Reference*. https://arxiv.org/abs/1006.0992.

Samson Abramsky and Viktor Winschel (2017). *Coalgebraic Analysis of Subgame-perfect Equilibria in Infinite Games without Discounting*. *Mathematical Structures in Computer Science*, **27**(5), 751–761. First published online in 2015. DOI: https://doi.org/10.1017/S0960129515000365.

Thomas Ågotnes and Hans van Ditmarsch (2011). *What Will They Say? Public Announcement Games*. *Synthese*, **179**, 57–85. DOI: https://doi.org/10.1007/s11229-010-9838-8.

Thomas Ågotnes, Natasha Alechina, and Rustam Galimullin (2022). *Logics with Group Announcements and Distributed Knowledge: Completeness and Expressive Power*. *Journal of Logic, Language and Information*, **31**, 141–166. DOI: https://doi.org/10.1007/s10849-022-09355-0.

Thomas Ågotnes, Philippe Balbiani, Hans van Ditmarsch, and Pablo Seban (2010). *Group Announcement Logic*. *Journal of Applied Logic*, **8**(1), 62–81. DOI: https://doi.org/10.1016/j.jal.2008.12.002.

Thomas Ågotnes and Yì N. Wáng (2016). *Resolving Distributed Knowledge*. https://arxiv.org/abs/1606.07515.

Rajeev Alur, Thomas A. Henzinger, and Orna Kupferman (2002). *Alternating-time Temporal Logic*. *Journal of the ACM*, **49**(5), 672–713. DOI: https://doi.org/10.1145/585265.585270.

Krzysztof R. Apt and Jonathan A. Zvesper (2010). *The Role of Monotonicity in the Epistemic Analysis of Strategic Games*. *Games*, **1**(4), 381–394. DOI: https://doi.org/10.3390/g1040381.

Robert Atkey, Bruno Gavranović, Neil Ghani, Clemens Kupke, Jérémy Ledent, and Fredrik Nordvall Forsberg (2021). *Compositional Game Theory, Compositionally*. In *Proceedings of the Third International Conference on Applied Category Theory (ACT 2020)*, Electronic Proceedings in Theoretical Computer Science, vol. 333, pp. 198–214. DOI: https://doi.org/10.4204/EPTCS.333.14.

Robert J. Aumann and Adam Brandenburger (1995). *Epistemic Conditions for Nash Equilibrium*. *Econometrica*, **63**(5), 1161–1180.

Alexandru Baltag, Bob Coecke, and Mehrnoosh Sadrzadeh (2007). *Epistemic Actions as Resources*. *Journal of Logic and Computation*, **17**(3), 555–585. DOI: https://doi.org/10.1093/logcom/exm015.

Alexandru Baltag, Lawrence S. Moss, and Sławomir Solecki (1998). *The Logic of Public Announcements, Common Knowledge, and Private Suspicions*. In *Proceedings of the 7th Conference on Theoretical Aspects of Rationality and Knowledge*.

Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky (2014). *Robust Cooperation in the Prisoner's Dilemma: Program Equilibrium via Provability Logic*. https://arxiv.org/abs/1401.5577.

Andrew Barber, Philippa Gardner, Masahito Hasegawa, and Gordon Plotkin (1998). *From Action Calculi to Linear Logic*. In *Computer Science Logic (CSL 1997)*, Lecture Notes in Computer Science, vol. 1414, pp. 78–97.

Nick Benton (1995). *A Mixed Linear and Non-Linear Logic: Proofs, Terms and Models*. In *Computer Science Logic (CSL 1994)*, Lecture Notes in Computer Science, vol. 933, pp. 121–135, Springer.

Elise Bonzon, Marie-Christine Lagasquie-Schiex, Jérôme Lang, and Bruno Zanuttini (2006). *Boolean Games Revisited*. In *Proceedings of the 17th European Conference on Artificial Intelligence*, pp. 265–269.

Luís Caires and Frank Pfenning (2010). *Session Types as Intuitionistic Linear Propositions*. In *21st International Conference on Concurrency Theory (CONCUR 2010)*, Lecture Notes in Computer Science, vol. 6269, pp. 222–236. DOI: https://doi.org/10.1007/978-3-642-15375-4_16.

Matteo Capucci (2023). *Diegetic Representation of Feedback in Open Games*. In *Proceedings of the Fifth International Conference on Applied Category Theory (ACT 2022)*, Electronic Proceedings in Theoretical Computer Science, vol. 380, pp. 145–158. DOI: https://doi.org/10.4204/EPTCS.380.9.

Matteo Capucci, Bruno Gavranović, Jules Hedges, and Eigil Fjeldgren Rischel (2022a). *Towards Foundations of Categorical Cybernetics*. In *Proceedings of the Fourth International Conference on Applied Category Theory (ACT 2021)*, Electronic Proceedings in Theoretical Computer Science, vol. 372, pp. 235–248. DOI: https://doi.org/10.4204/EPTCS.372.17.

Matteo Capucci, Neil Ghani, Jérémy Ledent, and Fredrik Nordvall Forsberg (2022b). *Translating Extensive Form Games to Open Games with Agency*. In *Proceedings of the Fourth International Conference on Applied Category Theory (ACT 2021)*, Electronic Proceedings in Theoretical Computer Science, vol. 372, pp. 221–234. DOI: https://doi.org/10.4204/EPTCS.372.16.

Marco Carbone, Fabrizio Montesi, Carsten Schürmann, and Nobuko Yoshida (2015). *Multiparty Session Types as Coherence Proofs*. In *26th International Conference on Concurrency Theory (CONCUR 2015)*, LIPIcs, vol. 42, pp. 412–426. DOI: https://doi.org/10.4230/LIPIcs.CONCUR.2015.412.

Marco Carbone, Fabrizio Montesi, Carsten Schürmann, and Nobuko Yoshida (2017). *Multiparty Session Types as Coherence Proofs*. *Acta Informatica*, **54**(3), 243–269. DOI: https://doi.org/10.1007/s00236-016-0285-y.

Simon Castellan and Nobuko Yoshida (2019). *Causality in Linear Logic: Full Completeness and Injectivity*. In *Foundations of Software Science and Computation Structures (FoSSaCS 2019)*, Lecture Notes in Computer Science, pp. 150–168. DOI: https://doi.org/10.1007/978-3-030-17127-8_9.

Simon Castellan, Pierre Clairambault, and Glynn Winskel (2019). *Thin Games with Symmetry and Concurrent Hyland–Ong Games. Logical Methods in Computer Science*, **15**(1), article 18. DOI: https://doi.org/10.23638/LMCS-15(1:18)2019.

Simon Castellan, Pierre Clairambault, and Peter Dybjer (2017). *Undecidability of Equality in the Free Locally Cartesian Closed Category (Extended Version). Logical Methods in Computer Science*, **13**(4), article 22. https://doi.org/10.23638/LMCS-13(4:22)2017.

Simon Castellan, Pierre Clairambault, Silvain Rideau, and Glynn Winskel (2017). *Games and Strategies as Event Structures. Logical Methods in Computer Science*, **13**(3), article 35. DOI: https://doi.org/10.23638/LMCS-13(3:35)2017.

Qian Chen, Chenwei Shi, and Yiyan Wang (2022). *Reasoning about Dependence, Preference and Coalitional Power*. https://arxiv.org/abs/2209.08213.

Philip R. Cohen and Hector J. Levesque (1990). *Intention Is Choice with Commitment. Artificial Intelligence*, **42**(2-3), 213–261. DOI: https://doi.org/10.1016/0004-3702(90)90055-5.

Brian Day (1970). *On Closed Categories of Functors*. In *Reports of the Midwest Category Seminar IV*, Lecture Notes in Mathematics, vol. 137, pp. 1–38, Springer. DOI: https://doi.org/10.1007/BFb0060438.

Martín Escardó and Paulo Oliva (2010). *Selection Functions, Bar Recursion and Backward Induction. Mathematical Structures in Computer Science*, **20**(2), 127–168. DOI: https://doi.org/10.1017/S0960129509990351.

Bernd Finkbeiner and Ernst-Rüdiger Olderog (2017). *Petri Games: Synthesis of Distributed Systems with Causal Memory. Information and Computation*, **253**, 181–203. DOI: https://doi.org/10.1016/j.ic.2016.07.006.

Dana Fisman, Orna Kupferman, and Yoad Lustig (2009). *Rational Synthesis*. https://arxiv.org/abs/0907.3019.

Sabine Frittella, Giuseppe Greco, Alexander Kurz, Alessandra Palmigiano, and Vlasta Sikimić (2016). *Multi-type Display Calculus for Dynamic Epistemic Logic. Journal of Logic and Computation*, **26**(6), 2017–2065. DOI: https://doi.org/10.1093/logcom/exu068.

Rustam Galimullin and Natasha Alechina (2018). *Coalition and Group Announcement Logic*. https://arxiv.org/abs/1810.02769.

Scott Garrabrant, Daniel A. Herrmann, and Josiah Lopez-Wild (2021). *Cartesian Frames*. https://arxiv.org/abs/2109.10996.

Neil Ghani, Jules Hedges, Viktor Winschel, and Philipp Zahn (2018). *Compositional Game Theory*. In *Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*, pp. 472–481. DOI: https://doi.org/10.1145/3209108.3209165.

Jean-Yves Girard (1987). *Linear Logic. Theoretical Computer Science*, **50**, 1–102. DOI: https://doi.org/10.1016/0304-3975(87)90045-4.

Joseph Y. Halpern and Ronald Fagin (1989). *Modelling Knowledge and Action in Distributed Systems. Distributed Computing*, **3**(4), 159–177.

Robert Harper (2016). *Practical Foundations for Programming Languages*. 2nd edition.

Cambridge University Press.

John C. Harsanyi (1967). *Games with Incomplete Information Played by Bayesian Players, Part I. Management Science*, **14**(3), 159–182.

Jules Hedges (2020). *The Game Semantics of Game Theory*. https://arxiv.org/abs/1904.11287.

Matthew Hennessy and Robin Milner (1980). *On Observing Nondeterminism and Concurrency*. In *Automata, Languages and Programming (ICALP 1980)*, Lecture Notes in Computer Science, vol. 85, pp. 299–309, Springer. DOI: https://doi.org/10.1007/3-540-10003-2_79.

Matthew Hennessy and Robin Milner (1985). *Algebraic Laws for Nondeterminism and Concurrency*. *Journal of the ACM*, **32**(1), 137–161. DOI: https://doi.org/10.1145/2455.2460.

John Horty and Nuel Belnap (1995). *The Deliberative STIT: A Study of Action, Omission, Ability, and Obligation*. *Journal of Philosophical Logic*, **24**, 583–644.

Bart Jacobs (1999). *Categorical Logic and Type Theory*. Studies in Logic and the Foundations of Mathematics, vol. 141. Elsevier.

Wojciech Jamroga and Wiebe van der Hoek (2004). *Agents that Know How to Play*. *Fundamenta Informaticae*, **63**(2-3), 185–219.

Andrew J. I. Jones and Marek Sergot (1996). *A Formal Characterisation of Institutionalised Power*. *Journal of the IGPL*, **4**(3), 427–443.

Kohei Kishida (2017). *Categories for Dynamic Epistemic Logic*. In *Proceedings of the Sixteenth Conference on Theoretical Aspects of Rationality and Knowledge (TARK 2017)*, Electronic Proceedings in Theoretical Computer Science, vol. 251, pp. 353–372. DOI: https://doi.org/10.4204/EPTCS.251.26.

N. P. Landsman (2001). *Operator Algebras and Poisson Manifolds Associated to Groupoids*. *Communications in Mathematical Physics*, **222**, 97–116. DOI: https://doi.org/10.1007/s002200100496.

Christian List and Philip Pettit (2011). *Group Agency: The Possibility, Design, and Status of Corporate Agents*. Oxford University Press.

Fabio Mogavero, Aniello Murano, Giuseppe Perelli, and Moshe Y. Vardi (2012). *What Makes ATL* Decidable? A Decidable Fragment of Strategy Logic*. In *23rd International Conference on Concurrency Theory (CONCUR 2012)*, Lecture Notes in Computer Science, vol. 7454, pp. 193–208. DOI: https://doi.org/10.1007/978-3-642-32940-1_15.

Yue Niu, Jonathan Sterling, Harrison Grodin, and Robert Harper (2022). *A Cost-Aware Logical Framework*. *Proceedings of the ACM on Programming Languages*, **6**(POPL), Article 9. DOI: https://doi.org/10.1145/3498670.

Marc Pauly (2002). *A Modal Logic for Coalitional Power in Games*. *Journal of Logic and Computation*, **12**(1), 149–166. DOI: https://doi.org/10.1093/logcom/12.1.149.

Jan Plaza (2007). *Logics of Public Communications*. *Synthese*, **158**(2), 165–179. DOI: https://doi.org/10.1007/s11229-007-9168-7.

Daniele Porello and Nicolas Troquard (2014). *A Resource-Sensitive Logic of Agency*. In *21st European Conference on Artificial Intelligence (ECAI 2014)*, pp. 723–728.

Roy Radner (1962). *Team Decision Problems*. *The Annals of Mathematical Statistics*, **33**(3),

857–881.

Jan J. M. M. Rutten (2000). *Universal Coalgebra: A Theory of Systems*. *Theoretical Computer Science*, **249**(1), 3–80. DOI: https://doi.org/10.1016/S0304-3975(00)00056-6.

Peter Selinger (2011). *A Survey of Graphical Languages for Monoidal Categories*. In *New Structures for Physics*, Lecture Notes in Physics, vol. 813, pp. 289–355. DOI: https://doi.org/10.1007/978-3-642-12821-9_4.

Lloyd S. Shapley (1953). *Stochastic Games*. *Proceedings of the National Academy of Sciences*, **39**(10), 1095–1100. DOI: https://doi.org/10.1073/pnas.39.10.1095.

Alfred Tarski (1955). *A Lattice-Theoretical Fixpoint Theorem and Its Applications*. *Pacific Journal of Mathematics*, **5**, 285–309.

Matthijs Vákár (2015). *A Categorical Semantics for Linear Logical Frameworks*. https://arxiv.org/abs/1501.05016.

Johan van Benthem and Amélie Gheerbrant (2010). *Game Solution, Epistemic Dynamics and Fixed-Point Logics*. *Fundamenta Informaticae*, **100**(1-4), 19–41.

Johan van Benthem, Eric Pacuit, and Olivier Roy (2011). *Toward a Theory of Play: A Logical Perspective on Games and Interaction*. *Games*, **2**(1), 52–86. DOI: https://doi.org/10.3390/g2010052.

Hans van Ditmarsch, Wiebe van der Hoek, Barteld Kooi, and Louwe B. Kuijer (2020). *Arrow Update Synthesis*. *Information and Computation*, **275**, article 104544. DOI: https://doi.org/10.1016/j.ic.2020.104544.

Jan van Eijck, Malvin Gattinger, and Yanjing Wang (2017). *Knowing Values and Public Inspection*. In *Logic and Its Applications*, Lecture Notes in Computer Science, vol. 10119, pp. 77–90. DOI: https://doi.org/10.1007/978-3-662-54069-5_7.

Glynn Winskel (2023). *Making Concurrency Functional*. In *38th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*. DOI: https://doi.org/10.1109/LICS56636.2023.10175727.